

Holographic Extreme and Mean Ratio

The holographic arithmetic–geometric coupling constant.

Conservation and reversibility of information

Oumar Haidara Fall · Rubén Ramos Balsa

Co-authorship without precedence of contribution

Documentary integration and compilation generated by ChatGPT - Astra.

Abstract.

Joint exceptional and temporal selection restricts the domain of 19 446 candidates to 79 through incidence and to a unique K through temporal compatibility. The repertoire, orientations and filters producing this uniqueness are retained explicitly.

The holographic arithmetic–geometric coupling constant is determined through an ordered register K , its rational evaluation and its incidence representations, and a conservation law is proved that permits information recovery through these transformations. With base one thousand and the phase chart fixed, the reading $\kappa = N_K/(1000^{12} - 1)$ preserves all twelve register components exactly. Its cyclic orbit forms a basis and provides stable identification of linear operators: the data are the responses to the complete basis, or one vector response under cyclic covariance. The structure that determines the value thus also participates in reconstructing the operations acting on it.

The construction starts from a formal kernel called Triadic Modular Holography, hereafter HMT. It comprises *All Possible Paths* (APP), an arithmetic path structure on a toroidal support with a Cayley graph and distinct additive and multiplicative evaluations; TRIT, an oriented signature taking values in $\{-1, 0, +1\}$; and the *Topological Phase Kernel* (TPK), which composes selection, transport and memory. Nonadic prolongation preserves the enriched state from which arithmetic coordinates and boundary incidences arise. The register K enters the closure of α together with the HMT coordinates of π, φ, e ; its incidence realization transports the same ordered information to the specified spaces.

The general conservation result is a bilateral identity for two isometries with a common domain and codomain, valid in arbitrary dimension. Integer refinement with carry has an exact inverse on admissible labels and a unitary lift that composes with this identity. In the nonadic specialization, the weights $72/73$ and $1/73$ determine the transformation of observables; the invariants are precisely those commuting with the relative transport. An explicit archive policy transfers the entire norm to infinite memory and yields reconstruction by an adjoint of norm one, together with finite inverses whose bounds are uniform in depth. Two complementary observations recover the centred register with an optimal bound, from which the specified dimensional projector is reconstructed. Conservation extends to every admissible exterior degree, including mixed readout–memory contributions.

The reversible extension relates deformation, holonomy, and action; the electromagnetic and gravitational realizations retain their explicit operators and normalizations. For the declared symplectic protocol with two identical centred pure Gaussian inputs and finitely many modes, the antilinear component of memory determines the reduced symplectic spectrum, retaining all Fock occupations. An elliptic–parabolic loop gives purity $9/11$, spectrum $(9/10)10^{-j}$, and dimensionless entropy $(10/9)\log 10 - \log 9$. Taken together, the results characterize what a transformation preserves, what an observation changes, and which data permit its inversion.

The recoverable direction of K also determines a positive determinant-one flow, preserves positroid strata, and produces dual lattices of fixed covolume with ternary symmetry. Their theta–Mellin readout preserves poles and residues, while spectral variation lies in an entire difference. The generated angular contrast is realized as a Klein coordinate; with orientation retained, the radius recovers the action section and re-expresses the complete electronic formula. The proofs connect this geometry with the bilateral archive and provide quantitative reconstruction bounds.

The emitting basis of this recovery is made explicit through the alphabet of 42 decimal blocks, regional junctions, and an example separating ordered memory from its histogram. Gauge comparison determines the normalization of the multiplicative observable, while closure controls separate orientation, terminal condition, and register position. At the joint boundary, saturation and dual neutrality leave two representatives; the fixed orientation determines the selected section. These results identify the data subsequently received by the conservation and recovery laws.

Keywords: Triadic Modular Holography; Dimensional Mechanics; holographic extreme and mean ratio; arithmetic–geometric coupling; reversible refinement; operator memory; holonomy; exceptional incidence; exterior conservation; bilateral transport; invariant observables; stable recovery; Hamiltonian action; Gaussian covariance; entanglement entropy.

Contents

Introduction: structural determination and information recovery	14
Generative architecture and conservation of information	19
Five holographic principles of construction and conservation	26
Evolutionary conservation of dimensional unity through double holographic projection	31
1 Generation, conservation and realization	32
1.1 Principal thesis and statements	32
1.2 Complete memory and partial observation	33
1.3 A calculable spectral consequence	33
1.4 Order of exposition	33
2 Formal kernel of Triadic Modular Holography	35
2.1 APP: positional support and arithmetic evaluations	35
2.2 TRIT: orientation and quadratic regimes	38
2.3 TPK: topological phase kernel	44
2.4 Accumulated memory and transport resolvents	50
3 Topological Phase Kernel and generation of the seed catalogue	54
3.1 Phase selection, transport and enriched state	54
3.2 Initial conditions and reversible enumeration	55
3.3 Sheet evaluation and arithmetic record	56
3.4 Six windows and decimal block emission	57
3.5 Complete calculation of an emission beginning with 501	60
3.6 Factorisation of the finite census	61
3.7 Oriented ternary reading and preservation of fibres	63
3.8 Dihedral action and the forty-three orbits	64
3.9 Self-contained enumeration and verification algorithm	65
3.10 Preservation of the record and the function of the seeds	65
3.11 Cyclic correlation and preservation of path order	65
4 Development of the formal kernel: initial conditions, geometry, and refinement	67
4.1 Initial states and the emission recurrence	67
4.2 Calendar closure and memory advance	68
4.3 Observable compression and orthogonal conservation	70
4.4 Local development: commutation, Lorentzian signature, and discrete curvature	70
4.5 Cubic completion and two notions of boundary	71
4.6 Cubic incidence and Lorentzian realization	73
4.7 Compatibility of modular and positional reductions	74
4.8 Number as a compatible section and value as a reading	75
4.9 Composition of histories and memory multiplier	75
4.10 Double variance of refinement and conservation of unity	76
4.11 Phase return, Weyl representation, and two readings	77
5 Correlative construction of the continuum: histories, transport, incidence and terminal	78
5.1 Admissible histories and memory by composition	78
5.2 Tree refinement and measure of histories	79
5.3 Sheetwise transport and exact accounting of outputs	79
5.4 Oriented balance and cancellation under refinement	81
5.5 Ternary sheet incidence and rectangular complex	81
5.6 Memory complex and compatibility of incidences	82
5.7 Assembly with terminal paths	83
5.8 Determinants, limit and arithmetic readout	84

5.9	Provenance, verification and scope	85
Coinductive generation		87
6	Coinductive generation of π, φ, e at arbitrary depth	88
6.1	Orbits, regions, and characters of closure, propagation, and self-scaling	88
6.2	Incidences between halves of regional emissions	89
6.3	Finite transport and prolongation boundary	92
6.4	The deep coordinate and integer recovery	93
6.5	Five closure regions and angular compensation	94
6.6	Propagation reader and normalization of the unit	95
6.7	Modal incidence and generation of self-scaling	95
6.8	Iteration of the ratio reader and phase recovery	100
6.9	Exponential realizations of the generated coordinates	102
6.10	Compatible cylinders and arbitrary depth	104
6.11	Mirror involution and synchronization of regions	104
6.12	Sturmian helical suspension	105
6.13	Nonadic connection, holonomy, and monodromic representation	108
6.14	Block clock and capacity clock	109
6.15	Symbolic suspension and aperiodic time crystal	110
6.16	Specular symmetry of the regions and double realization	111
6.17	Temporal transport and tritic exponential composition	114
6.18	Multiplicative irreducibility and return periods	114
Holographic arithmetic–geometric coupling constant		117
7	Regional generation and refinement	118
7.1	Positional emission through rational separation of cells	118
7.2	Regional blocks of six trits at arbitrary depth	119
7.3	Transition registers and reconstruction of linear lifts	120
7.4	Simultaneous refinement in ternary and decimal bases	121
7.5	Selection of the next block by the regional reader	122
7.6	Ternary signature, carry, Witt incidence, and reading clock	123
7.7	Composition theorem for the block, cylinder, and signature	124
7.8	Calendar census and simultaneous compatibility of the regional biographies	125
8	Regional determination of the dodecaphasic descriptor and selection of the coupling register	126
8.1	Regional bands, encoding, and integer signatures	126
8.2	Conversion clock and first zero advance	127
8.3	Oriented construction of the twenty-four-trit descriptor	128
8.4	Functional panel and terminal repertoire	129
8.5	Comparative readers, affine reader, and temporal incidence	133
8.6	Enumeration of cylinders and decimal candidates	134
8.7	Exceptional incidence and heterotypy condition	136
8.8	Regional composition and continuity of the realizations	138
8.9	Executable certification and traceability of the operations	139
9	Dodecaphasic extraction of the arithmetic–geometric coupling constant	140
9.1	Cyclic extension and memory grading	141
9.2	The oriented register and the integral $1 + 5 + 6$ chart	141
9.3	Incidence evaluation preceding the dodecaphasic register	142
9.4	Conjugation of counts and boundary terms	143
9.5	Complete census of cylinders and selection of the terminal boundary	144
9.6	Two readouts of the terminal state	147
9.7	From the channel register to the signed vector	148

9.8	Orbitwise Hadamard transform and integer reconstruction	149
9.9	Cyclic differences and the uniform component	150
9.10	Integral compatibility and determination of the transverse register	151
9.11	Two rational readouts: κ_{36} and κ_{per}	154
9.12	Reversibility of the rational constant and phase change	155
9.13	Periodicity and the cokernel of the transport operator	156
10	Determination of the constant and covariance of its realizations	157
Fine-structure constant		159
11	Derivation of the fine-structure constant	160
11.1	First route: positional determination by integer closure	160
11.2	Second route: analytic determination through vacancies	167
11.3	Compatibility of the two determinations at every depth	169
11.4	Compatibility between the dodecaphasic and analytic representations of α	170
Planck constants and orientation		177
12	Determinantal structure and oriented action sections	178
12.1	Local quotient and determinantal norm	178
12.2	Action character and regional continuation	179
12.3	Positive sections and action per revolution	180
13	Angular readout and channel orientation	181
13.1	Circular charts and covariance of action	181
13.2	Positive channels and reversal of the odd coordinate	182
14	Dodecaphasic germ and recovery of unimodal branches	183
14.1	Normalization and analytic profile	183
14.2	Branchwise inversion and itinerary memory	184
14.3	Scope of the rescaled return	184
Algebraic structure of the electron		185
15	Algebraic structure of the electron and action scale	186
15.1	Center, orientation, and vacancy on a preserved fiber	187
15.2	Commutator norm of the arithmetic projections	191
15.3	Spinor representation of the electronic block and helicity	192
15.4	Transfer between sheets and coefficients of the basal stage	194
15.5	Global torsion and the basal coordinate	195
15.6	Determinantal action scale and decimal valuation	196
15.7	Action sections and dimensionless return quotient	197
15.8	Two readers of the electronic return	200
15.9	Complete electronic composition and normalization	202
15.10	Dimensional realization: energy, mass, and chronological scale	202
15.11	Lorentzian transport and electromagnetic representation	204
Incidence and dimensional realization		205
16	Exceptional prolongation: from dodecaphasic incidence to Moonshine	206
16.1	Compatibility of the Archimedean and incidence realizations	206
16.2	The finite realization and the ternary code	207
16.3	Dodecaphasic flag associated with K and α	210
16.4	Isometric intertwining of the incidence representations	213
16.5	The direction selected by the dodecaphasic register	214
16.6	The Witt star and the generation of M_{12}	215
16.7	The binary residue and the five regions of π	216
16.8	From the ternary code to the Niemeier lattice	218

16.9	Incidence selection of the rootless unimodular neighbor	219
16.10	The vertex operator algebra and the Monster	221
16.11	State-based construction of the twelve cyclotomic fibres	222
16.12	Dodecaphasic alignment with the Witt positions	223
16.13	Ternary orientation of the Leech neighbour	223
16.14	Ternary trace, twisted weight, and order-three extension	224
16.15	Equivalence of the order-two and order-three presentations	225
16.16	Eta functions, Fricke involution, and vacuum normalization	226
16.17	The Fricke correspondence and differential recovery	227
16.18	Compatibility of the level-two and level-five modular readers	228
16.19	Exceptional incidence and coordination of the screens	229
16.20	Graded Moonshine traces and representation of the Monster	230
16.21	Full-support orientations and lattice equivariance	231
17	Finite holonomy, the Schläfli configuration, and E_6 symmetry	232
17.1	Mixed APP curvatures	232
17.2	Central extension and ternary reduction	232
17.3	Weyl–Heisenberg representation and central phase	233
17.4	Action of 10 displacements on the Cayley graph	233
17.5	The root lattice realizing E_6	236
17.6	Stratification of the 729 words	236
17.7	Reversible coordinates for ordered pairs of lines	237
17.8	Necessity of the central phase	237
18	Hilbert-space realization of the nonadic phase and preservation of the unit	239
18.1	Phase fiber, labeled basis, and inner product	239
18.2	Translation and phase: finite Weyl representation	239
18.3	Two lifts and two different normalizations	241
18.4	Inductive closure and tracial realization	241
18.5	Integer ranks and continuous tracial dimension	242
18.6	Polar transport of incidence and preservation of readers	242
19	Genealogical path kernels: finite composition, unitary realization and cylindrical limit	245
19.1	Route category and functorial weight	245
19.2	Construction of the internal phase through the real action character on genealogical routes	246
19.3	Unitary realization on the chronological register 27×27	246
19.4	Cylindrical limit on the space of genealogies	248
19.5	Expressed value and fiber memory	249
20	Action ellipse and determination of the normalized radius	250
20.1	Domain and labelled semiaxes	250
20.2	Parametric phase and oriented area	250
20.3	Shape reciprocity and symplectic orientation	251
20.4	Normalized radius and lattice-readout matrix	252
20.5	Inductive–capacitive realization of the action form	253
21	T-duality: integer coordinates, change of section, and operator equivalence	254
21.1	Stable domain of the interchange	254
21.2	Compact Hamiltonian and self-adjoint domain	254
21.3	Specialization to the radius of the action ellipse	255
21.4	Change of representative and quotient memory	256
21.5	Radius normalization in the string realization	256
21.6	Relationship with the modular transformation	257
22	Dimensional screens and coordinated state reduction	258

22.1	Channels, relations, and effective dimension	258
22.2	Ternary puncturing and syndrome screen	258
22.3	The dodecaphase direction and rank reduction	259
22.4	Lorentzian lattice screen	260
22.5	Joint screen morphism	260
22.6	Coordination with the exceptional realization	261
23	Action on trajectories and eleven-dimensional dynamical realization	262
23.1	Descent of an evolution and sufficiency of the observed datum	262
23.2	Cocycle of directional and arithmetic changes	262
23.3	Extended persistence and worldsheet	263
23.4	Eleven-dimensional realization and graded operators	263
23.5	Compatible compactification and conclusion of the development	264
Information conservation and recovery		265
24	Refinement of histories and conservation of unity	266
24.1	Generation of emissions and realization domain	266
24.2	Refinement on the complete multisection	266
24.3	Correlative prolongation of the five operations	267
24.4	Memory fibres: Gram criterion and conservation of unity	268
24.5	Symmetries and readers on the common domain	269
24.6	Passage-to-the-limit theorem for the observable/memory charge	269
25	Conservation of observables and the complete record	272
25.1	Balance of a conserved observable	272
25.2	Specialization to the nonadic connection	272
25.3	Terminal sector and the unlimited-depth record	273
25.4	Symmetry conservation and chart transport	274
25.5	History refinement and compatibility of balances	275
25.6	Variational specialization: action and orientation	275
26	Recovery of the state and its dynamics	277
26.1	Recoverable representation of transport	277
26.2	Explicit reconstruction and record depth	278
26.3	Naturality of recovery under refinement	279
26.4	Two exact regimes of memory distribution	279
26.5	Composition with Euler's identity and incidence	281
27	Exceptional incidence and transport of forms	282
27.1	Generation of the record and conservation of incidence	282
27.2	K-Witt transport of norms, operators, and balances	283
27.3	Order-three action on the lattice fibres	284
27.4	Refinement of energy and its differential	284
28	Incidence reconstruction and covariance	286
28.1	The complete isometry and its domain	286
28.2	Covariant lifting theorem on the image	286
28.3	Explicit recovery from the incidence record of complements	287
28.4	Natural lifting of refinement with local frames	288
28.5	Direct extension to the twelve A_2 fibres	289
28.6	Exact scope with respect to histories	289
29	Affine holonomy and memory symmetries	291
29.1	Effective composition of memory	291
29.2	Application to the effective record of the observable revolution	292
29.3	Parabolic generator determined by memory	293
29.4	APP curvature and conservation of the central phase	295

29.5	Composition between the three regimes and continuity of the balance	296
29.6	Compatibility with refinement and arbitrary extent	296
29.7	Foundational consequence and scope of the composition	297
30	Variational conservation and oriented action	298
30.1	Symmetry of the elliptic action and Noether charge	298
30.2	Change of form: determined transport and connection	299
30.3	Orientation and boundary of the action	300
30.4	Variational clock charge and balance between reading and memory	300
30.5	Covariant balance of the total tensor	302
31	Conservation of action forms and oriented area in the tritic regimes	303
31.1	Balance between two action forms	303
31.2	Exact accumulation law with variable phase	303
31.3	Extended Euler and common oriented area	304
31.4	The common factor of the exponential reader	305
31.5	Action refinement over the complete emissions	305
32	Explicit reconstruction from complete returns	307
32.1	State domain and bilateral extension	307
32.2	Tritic sampling and reconstruction of every step	307
32.3	Conjugation of transport and preservation of readouts	308
33	Observable return and persistence of the register	309
33.1	Joint recovery from complete returns	309
33.2	Explicit subsequent representation of transport	309
33.3	Numerical convergence and correlations with incidence	310
33.4	Covariance of the memory operator under return	311
33.5	Incidence, refinement, and three regimes	311
34	Reversible coupling and nonstationary transport	312
34.1	Reversible extension of the balance between two forms	312
34.2	Exact composition and memory re-entry	313
34.3	Memory of a nonstationary sequence	313
34.4	Three distinct properties: exhaustion, distinguishability, and stability	314
34.5	Quantitative criterion for complete transfer	314
34.6	Complete incidence realization and changes of regime	315
35	Dimensional reconstruction and exterior conservation	316
35.1	Two memories recover the centred register	316
35.2	Reconstruction of the dimensional projector from two memories	318
35.3	Application to an effective vector of the exceptional construction	319
35.4	Conservation law in every exterior degree	320
35.5	Transport memory and relative holonomy	321
35.6	Symmetry, action, and conserved charge	323
36	Canonical realization of covariance	325
36.1	Canonical covariance and reciprocity of the action ellipse	325
37	Sheet return, areal–volumetric incidence and constitutive orientation	327
37.1	Projected return and restoration of the oriented cover	327
37.2	Descent of incidence from the tritic calendar	327
37.3	Boundary marking and areal–volumetric ratio	328
37.4	Transport of orientation to the constitutive channels	328
38	Discrete polarization and the tritic mode of arithmetic projections	329
38.1	Refinement discrepancy and endpoint convention	329
38.2	Bounded polarization and exact temporal balance	329
38.3	Realization on the charge and time lines	330

38.4	Pairing the additive and multiplicative fibres	331
38.5	Commutator norm and electronic prefactor	332
38.6	Quadratic defect and response of the 90/120 sectors	333
39	Constitutive operator and reconstruction of the vacuum channels	334
39.1	Positive response on two sheets	334
39.2	Four coordinates of a common response	334
39.3	Cubic inversion and angular recovery	335
39.4	Logarithmic coordinates and refinement	336
39.5	Dimensional types of the four evaluations	336
39.6	Reconstruction of the action shape from the constitutive response	336
39.7	Constitutive speed and kinematic realization of the clock	338
40	Charge section and electrical quantization relations	340
40.1	Positive determination of the charge section	340
40.2	Resistance, conductance, flux, and frequency	340
40.3	Action return and transport of the sections	341
Barbero–Immirzi parameter		342
41	Hexadic–octadic incidence and the Barbero–Immirzi functional	343
41.1	The weights precede the pole coefficient	343
41.2	Parity and structural sensitivity	344
42	Longitudinal return, radial reciprocity, and the memory archive	345
42.1	Constitutive restitution and oriented moment	345
42.2	Marked inscription and linear return	346
42.3	Arrival modulus and common coupling	347
42.4	Clock, mass, and dimensional sections	347
42.5	Reduction with a residual and transport of symmetries	348
43	Action, gravity, and thermal information	349
43.1	Dodecaphasic period and circular realization of action	349
44	Area operator and reconstruction of sector information	350
44.1	Tritic regrouping and boundary incidence	350
44.2	Transport of collective incidence modes	350
44.3	Angular character and area operator	351
44.4	State normalization and the meaning of the modular Hamiltonian	352
44.5	Reduced state, Schmidt decomposition, and entanglement entropy	352
45	Genealogical memory, informational reduction, and thermal transduction	353
45.1	Reader, memory, and state recovery	353
45.2	Reversible updating, tritic decision, and erasure	354
45.3	Action, frequency, and decision energy	354
45.4	Dimensional definition of Boltzmann’s constant and coordinate selection	355
45.5	Dimensional entropy and information of the reduced state	356
45.6	Spectral normalization of thermal histories	356
46	Oriented incidence and the gravitational tensor representation	357
46.1	Signed covering and calendar transport	357
46.2	Quadratic readout and the five-component space	358
46.3	Transverse projection and helicities	359
47	Phase return and representation of accumulated memory	360
47.1	Reversal cocycle and change of sheet	360
47.2	Affine representation of the cocycle	361
48	Boundary soldering and energy response	362
48.1	Cubic incidence and the rank-four quotient	362
48.2	Cellular soldering and memory transport	362

48.3	Positive response, uniqueness, and minimal extension	363
48.4	Conservation of density under refinement	364
48.5	Variational realization of the response	365
49	Geometric realization of transport and soldering	366
49.1	Screen, co-tetrad, and connection	366
49.2	The three quadratic realizations of curvature	367
50	Variational response of memory energy	368
50.1	Discrete functional and space of variations	368
50.2	Composition and transport of the response	369
50.3	Oriented inversion and transport of the weight	369
50.4	Compatibility with refinement	370
50.5	Matter realization of the response	371
51	Realization of the current in connection coordinates	372
51.1	Differential on invertible transports	372
51.2	Quadratic representations of the TRIT	372
51.3	Composition of routes and oriented inversion	373
51.4	Evaluation of current components	373
51.5	Insertion into the Cartan equation	374
52	Differential of the minimum boundary extension	375
52.1	Minimizing field and response weight	375
52.2	Full differential of the minimum	376
52.3	Fixed quotient and changes of frame	376
52.4	Transported incidence and connection components	377
53	Spin current and algebraic Cartan–Holst response	378
53.1	Geometric domain and current normalization	378
53.2	Connection variation and Holst inversion	378
53.3	Pointwise reconstruction of torsion and contorsion	379
53.4	Elimination of contorsion and quadratic contribution	380
53.5	Effective metric source and covariant balance	381
54	Torsional elimination and metric response of the minimum extension	382
54.1	Variational composition at a cellular level	382
54.2	Local existence of a stationary branch	383
54.3	Effective action and non-affine contribution	383
54.4	Metric differential of the eliminated action	384
54.5	Exact non-affine checking model	385
55	Spinor current and torsional amplitude of the matter state	386
55.1	Spinor representation and affine matter action	386
55.2	Effective contraction and second-moment coefficient	387
55.3	Finite fermionic realization and second moment of the state	388
55.4	Density, pressure, and conservation of comoving amplitude	389
56	Memory dilution and persistence of the bounce	391
56.1	The a^{-6} dilution law	391
56.2	Homogeneous threshold and transversality	391
56.3	Local persistence under controlled perturbations	392
57	Explicit homogeneous solution and causal completeness	394
57.1	Finite curvature and extension of geodesics	395
58	Residual tensor, trace decomposition, and covariant balance	396
58.1	Chart data and construction of the residual tensor	396
58.2	Canonical decomposition and exchange current	396
58.3	Link to the effective Cartan–Holst source	397

	58.4	Trace sector and acceleration of the selected charts	397
59		Pure bipartite states, reduced entropy, and horizon area laws	399
	59.1	Sheet register and bipartite purification	399
	59.2	Purification of the incidence sectors	399
	59.3	Information equation and area fluctuations	400
	59.4	Modular law and finite area variations	401
	59.5	Action, decision energy, and the area unit	402
	59.6	Cosmological normalization of the horizon	402
	59.7	Kruskal chart and exchange of exterior regions	403
	59.8	Transport of states, observables, and refinements	404
60		Observer, memory, and relational response	406
	60.1	Boundary transport and intrinsic readout	406
	60.2	Boundary identity and cellular interior	407
	60.3	Isometric inscription of the register in the apparatus	408
	60.4	Outcome map and record conditioning	408
	60.5	Accessible algebra and pointer complement	409
	60.6	Factorization of the response through the effective boundary	410
	60.7	Ambivalence weights and self-inertial response	411
	60.8	Connection of the joint state and expressed acceleration	413
61		Bidirectional propagation and the boundary absorption condition	414
	61.1	Route action and register involutions	414
	61.2	Reversal of the realized evolution	415
	61.3	Propagators of the chronological channel of a route	415
	61.4	Causal interchange and decomposition of the response	416
	61.5	Projection of absorbed currents and variation of the action	417
	61.6	Explicit absorbing condition in a chronological window	418
	61.7	Refinement and continuity of the boundary condition	419
62		Action realization and metrological comparison	420
	62.1	Comparison of the reduced Planck constant	420
63		Covariance, normal form, and Gaussian spectrum	421
	63.1	Canonical coordinates and characteristic function	421
	63.2	Symplectic reduction of the positive form	422
	63.3	Calculation of the spectrum of the isotropic density operator	422
	63.4	Preparation and transport of the two covariances	423
64		Physical actions and the spectrum of reduced memory	424
	64.1	Genealogy of the dimensional sections	424
	64.2	Normalization of the Maxwell–Dirac action	425
	64.3	Areal scale of gravitational action and entropy	426
	64.4	Prolongation of memory transport: quantum realization	427
	64.5	Extension to arbitrary finite canonical dimension	430
65		Bilateral memory, relative holonomy, and oriented recovery	432
	65.1	Unitary composition with bilateral archive	432
	65.2	Quarter-turn plane in the dodecaphasic record	432
	65.3	Relative commutator and action recovery	434
	65.4	Dodecaphasic lift and compatible amplification	434
	65.5	Chronological information and energy reading	435
66		Synthesis of the results and consequences	436
	66.1	Conservation, reconstruction, and dimension	436
	66.2	Action and physical consequences	436
	66.3	Memory and quantum spectrum	436

66.4	Foundational content	436
Arithmetic–geometric coupling: bilateral conservation and reversible refinement		437
67	The holographic arithmetic–geometric coupling constant	438
67.1	Object, chart, and production of the register	438
67.2	Periodic encoding, closure, and exact inversion	439
67.3	The complete orbit as a reference for its carrier	440
67.4	Operator identification and relative holonomy	441
67.5	Complete memory preserves the reference and the observables	442
67.6	Exceptional incidence and selected direction	443
67.7	Continuous stability and the need to preserve correlations	444
67.8	Discrete correction before incidence selection	445
67.9	Covariance, refinement, and variational charge	446
67.10	Encoding, operator identification, and incidence stability	448
68	Ternary algebraic norm and optimal recovery radii	449
68.1	A two-memory collection and its Gram matrix	449
68.2	Minimum distances: all integer registers	450
68.3	Finite decoder and parameter duality	452
68.4	The norm on the ternary augmentation module	453
68.5	The complete cycle and the origin of the fixed term	454
68.6	Six received states and conservation of the complete reading	454
68.7	Inversion, charts, and bilateral balance	456
68.8	The alpha datum and collections of integer iterations	457
68.9	Angular reciprocity and the existing regional correction	459
68.10	Angular inverse test on the twelve positions	460
68.11	Result of the composition and scope of the tests	462
69	Conservative refinement of unity and carry memory	463
69.1	From the quadratic norm to cubic completion	463
69.2	Concatenations of the central block and restoration of unity	463
69.3	Affine transport and the positivity domain	464
69.4	Canonical recovery through the residue	465
69.5	Variable iteration and complete finite memory	465
69.6	Block composition and compatibility with the partition	466
69.7	Convergence and quantitative control of the reading	466
69.8	Compatible cylinders and boundary multiplicity	467
69.9	Signed carries and prospective specification	467
69.10	Operational content and transition to amplitudes	468
70	Bilateral conservation and arithmetic–geometric transduction	469
70.1	Bilateral law for two isometric transports	469
70.2	Factorization, recovery, and channel compatibility	470
70.3	Transformation of readers and relative symmetry	470
70.4	Reversible updating with incoming memory	471
70.5	Causal reduction of the bilateral update	471
70.6	Nonstationary archiving and uniform exhaustion of the terminal	473
70.7	Geometric forms and conservation under transport	474
70.8	Channel correlation and an inverse test of a reading	474
70.9	Cyclic frame and transport of responses	475
70.10	Positional reading and scalar precision	475
70.11	Incidence, charge, and exact recovery of the record	476
70.12	Preservation of the conditioning of two memories	477

70.13	Orbital norm and polar normalization of the frame	477
70.14	An exact example of transduction and error correction	478
71	Unitary lift of refinement and transport of readers	479
71.1	Bijection of admissible arithmetic states	479
71.2	Unitarity on label amplitudes	479
71.3	Diagonal readers of proportion and carry	480
71.4	Complete paths at arbitrary depth	480
71.5	Fixed input and exact image	481
71.6	Bilateral coupling of refinement	481
71.7	Law of observables on the two channels	482
71.8	Arithmetic example in dimension one thousand and one	482
71.9	Relation to the generated register and domains of application	483
72	Dimensional unity, relative symmetry, and recovery of structure	484
72.1	Arithmetic conservation, unitarity, and isometric decomposition	484
72.2	Complete memory and the logical contribution to the Landauer bound	484
72.3	Conservation of an observable and its relative symmetry	485
72.4	Role of the register in transduction	485
72.5	Depth, dimensions, and physical realizations	486
72.6	Synthesis of conservation under evolution	486
	Arithmetic formalization of the bilateral balance	487
	Recoverable geometry and reciprocity of coupling	489
73	Positive deformation and lattice duality of the generated register	490
73.1	Coupling direction and unimodular flow	490
73.2	Exact reciprocity of the double circle	490
73.3	Positroid strata and balanced invariants	491
73.4	Covolume, dual lattice, and ternary symmetry	491
74	Split form, integrality, and lattice energy	493
74.1	Arithmetic variation of the Euclidean projection	493
74.2	Integral self-duality for the split-signature form	494
74.3	Positive operator and unitary equivalence of domains	495
75	Theta–Mellin reciprocity and conservation of the polar structure	496
75.1	Periodization of the generated lattice	496
75.2	Meromorphic continuation and functional equation	496
75.3	Stable reconstruction from the bilateral archive	497
75.4	Pell dilation, Catalan moment and oriented return	498
76	Hyperbolic geometry of the angular pair and recovery of action	500
76.1	Radion, action ellipse, and marked charts	500
76.2	Complete substitution of the angular representations	501
76.3	Recovery of Planck’s action with orientation preserved	502
76.4	Intertwining phase and anisotropy	503
76.5	Radial geodesic and differential realization of the direction of K	503
76.6	Lattice lift and rectangular lift	504
77	Electron coordinate, operator scale, and reversibility of readings	505
77.1	Algebraic structure in the complete electronic expression	505
77.2	Separation of scale and shape in a positive mass fibre	506
77.3	Circular compression and identification of transport	507
77.4	Hierarchy of invariants and scope of the proofs	507
	Compatible recovery and collections of the continuum	509
78	Carry and incidence registers and projective reconstruction of histories	510

78.1	Carry, history, incidence and archive	510
78.2	Refinement tree and local weights	511
78.3	Enriched readers and fiber memory	511
78.4	Equivariance of the measure under tree symmetries	511
78.5	Recovery of histories and transport of membership	512
78.6	Connection with the genealogical closure of the continuum	512
79	Genealogical closure and cardinality of the generated continuum	513
79.1	Extension of states and transfinite semantic extension	513
79.2	TPK lift of the ternary fiber and complete cylinder partition	516
79.3	Enumeration by genealogical codes and HMT condensation	530
79.4	Injection of the generated line into the first uncountable ordinal of the genealogical closure	532
79.5	Injection of the first uncountable ordinal into the line and cardinal equivalence	532
79.6	Full quantifier over all internal subsets	533
79.7	Representation of the genealogical closure as $L[h, m_\infty]$	533
79.8	Theorem giving an affirmative resolution of the Continuum Hypothesis	534
79.9	Falsification criteria for the cardinal closure	534
80	Census of residual-emitter signatures	535
81	Conclusions	537
	References	540

Introduction: structural determination and information recovery

The holographic extreme and mean ratio addresses a precise question: how a structure can produce a numerical evaluation, undergo a transformation, and preserve the information required to recognize and invert that transformation. The problem concerns both the formation of the value and the persistence of its relations. An equality between quantities establishes a balance; a recovery law additionally determines which state realizes it, which data identify that state, and which operation restores it. This work develops the latter requirement from an arithmetic construction of paths through to its linear, incidence and physical representations.

The first result organizing the argument is the determination of the holographic arithmetic–geometric coupling constant. Its ordered representation is the dodecaphasic register K ; its positional evaluation is the rational number κ . This reading preserves the register exactly when the base, length and phase origin are fixed. The object is also significant through its operations: its cyclic shifts form a basis, its responses permit operator identification, and its incidence images transport the relations of the initial configuration. Structure, equation and value thus arise as stages of a construction whose recoverable information is determined mathematically.

The second result supplies the general law of that conservation. Two isometric transports with a common domain and codomain produce two components whose weighted balance restores the initial inner product exactly. The nonadic specialization retains a ternary norm of value 73 and determines an observable transformation with weights $72/73$ and $1/73$. A policy of prolongation and archiving turns this local identity into stable recovery at arbitrary depth. The article's thesis thereby acquires a verifiable content: the evaluated value, the preserved state and the observed readout are related by explicit maps and inverse formulae.

The generating kernel and its two realizations

The construction is established in a formal kernel called *Triadic Modular Holography*, hereafter HMT. Its first object is *All Possible Paths*, hereafter APP: an arithmetic path structure on the toroidal support $(\mathbb{Z}/9\mathbb{Z})^2$, whose Cayley graph describes cardinal displacements. Additive and multiplicative evaluations are distinguished on this same support. Preserving the residue together with the quotient retains the evaluated integer and permits the origin of each evaluation to be traced. Addition accumulates; multiplication organizes the second sheet through its own evaluation and transport rules.

The second object, *TRIT*, supplies the oriented signature $\{-1, 0, +1\}$ and distinguishes the local regimes. The extended Euler equation expresses the elliptic, parabolic and hyperbolic regimes in their quadratic algebras through a common exponential law. The third object is the *Topological Phase Kernel*, hereafter TPK. Its action composes selection, transport and memory updating. Route, direction, sheet, residue, quotient, carry and boundary are retained in the relevant enriched state; phase return prolongs its history. The nonadic connection and holonomy belong to this dynamics.

This state gives rise to the joint discrete structure of the continuum and its two realizations: contraction of compatible cylinders determines arithmetic coordinates, while boundary incidence preserves combinatorial and geometric relations. The five correlative constructions of the continuum remain within this same development. The linear spaces, forms and observables used subsequently realize previously generated objects, with their domains and normalizations. The continuity of the argument lies in each operation preserving the relations required by the next.

Completion of unity and reversible refinement

Completing a cell with nine possibilities per coordinate adds one distinguished boundary position. In three coordinates, stratification by the number of boundary positions gives

$$10^3 = 729 + 243 + 27 + 1 = 729 + 271.$$

The interior, intermediate strata and apex compose one normalized unity. In arbitrary finite dimension, the expansion of $(9 + 1)^d$ expresses the same stratification; summing the contributions of the omitted coordinate proves compatibility with projection.

The central block $\begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$ provides a concrete arithmetic antecedent. Its eigenvalues are 9 and 5, and its concatenations satisfy $72 + 27 = 77 + 22 = 99$. Restoring one unit and then refining gives the chain $(72, 27) \mapsto (73, 27) \mapsto (729, 271)$. The first operation incorporates the recorded unit; the second changes scale and redistributes one unit by a zero-sum correction. This explicit sequence relates centre, completion and proportion while retaining the origin of every term.

Refinement with carry introduces an evolution of this distribution. The transformation $(x, y) \mapsto (Bx - c, By + c)$ preserves the sum after multiplication by B . On admissible integer labels, with $0 \leq c < B$, Euclidean division of the second component first recovers c and then the preceding pair. Depth fixes the number of inverse steps. When lifted carries are admitted, their additional quotient belongs to the preserved register. Reversibility therefore depends on precise data and has an explicit recovery operation.

This distinction gives holographic conservation an operational meaning. The normalized reading determines a coordinate; the register preserves the relations permitting state reconstruction or further refinement. The subsequent development makes this correspondence linear: the bijection between admissible labels induces a unitary operator between their bases and permits amplitudes, correlations and observables to be transported.

The arithmetic and incidence roles of K

The register K is obtained from the terminal state through its visits, traces, counts, differences and charge. The evaluation

$$\kappa = \frac{N_K}{1000^{12} - 1}$$

is reversible in the fixed chart: multiplication by the denominator restores the integer, and its decomposition into twelve blocks recovers the complete register, including leading zeroes. The enriched history additionally carries the depth and transport coordinates used by each prolongation. Recovery of each object is formulated on the data that identify it.

In the arithmetic reading, κ enters

$$\alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa.$$

The three regional coordinates and the register arise from the same generating kernel and act in their constructive order. The complete analytic chart of α represents this same pre-coordinate through a simple root and a compatible interval collection. In the incidence reading, the register transports its phase relations to the exceptional configuration. The unity of the object is manifested in the effective composition of its readers, inverses and actions.

The cyclic orbit of K adds an identification result. Its twelve vectors form a basis of the twelve-component linear realization. The responses to this basis determine any linear operator in the class considered; for an operator commuting with the shift, one complete vector response determines the others. The smallest Gram eigenvalue, 589609, provides an exact stability bound. Two complementary memory observations also recover the centred part of the register; the uniform charge completes its reconstruction. The constant thus functions as a coupling object with quantified capacities for evaluation, transport and identification.

Bilateral balance and observable transformation

The integer 73 occurs as the norm of two conjugate elements and as the index of a rank-two lattice inclusion. The identity $10 \cdot 73 = 729 + 1$ relates this ternary norm to decimal completion. The bilateral theorem extends the balance to two isometries B, C between spaces with a common domain and codomain. For $a > 0$, the components $aB - C$ and $(a + 1)B + C$, weighted by $a + 1$ and a , preserve the inner product after the stated normalization. Expansion and cancellation of the cross terms prove the identity in arbitrary dimension. The specialization $a = 8$ retains the nonadic operators and their arithmetic information.

Conservation of the complete state permits the effect of a particular observation to be calculated. If the relative transport R is unitary and G is the observable expressed in the common chart, the diagonal reading transforms

$$\mathcal{T}(G) = \frac{72}{73}G + \frac{1}{73}R^*GR.$$

An observable remains invariant exactly when it commutes with R . The observable transported on the complete space preserves the entire initial reading. Within a single construction, the result specifies both the information that persists and the change introduced by observing it through a particular channel.

Recovery at arbitrary depth

At each step, one component may be prolonged and its complement archived. The finite balance sums the squared norms of the memories and the terminal state. An explicit policy imposes the geometric bound $(729/1241)^N$ on the squared norm of the nonadic terminal operator. In the limit, the entire norm is contained in the infinite archive; this archive is an isometry and its adjoint reconstructs the state with operator norm one.

The first archived blocks likewise admit an exact inverse corrected by their Gram operator, with a stability bound uniform over positive depths. The infinite archive, the finite inverse and the approximation obtained by truncating the adjoint have their own formulae and estimates. In general dynamics with a surviving terminal, its contribution is retained together with memory. The proved policy identifies the case in which that terminal tends to zero.

The unitary lift of refinement composes with bilateral balance. The decimal example carrying $(73, 27, 1)$ to $(729, 271)$ keeps the origin of the carry explicit; comparison with a neighbouring carry determines the exact change in a reading. The evolutionary conservation of unity is thereby expressed through a sequence of recoverable transformations.

Forms, action and quantum observation

Inner-product transport extends to every admissible exterior degree. Areas and volumes also preserve the mixed readout-memory contributions. Exceptional incidence transports these identities on its specified images; variational charges are obtained from a declared action and its symmetries. The electromagnetic and gravitational realizations retain the scales and operators involved in their proofs.

The Gaussian realization shows why preserving the state and knowing a reduced observation are complementary questions. In the declared protocol with two identical centred pure Gaussian inputs and finitely many modes, symplectic transport preserves the global structure, while its reduction acquires a spectrum determined by the antilinear component of memory. The elliptic-parabolic loop studied here produces purity $9/11$, spectrum $(9/10)10^{-j}$, and dimensionless entropy $(10/9)\log 10 - \log 9$, with all Fock-space occupations retained in the calculation.

The argument therefore concludes with a transformation law richer than a scalar equality: the value is determined, the structure involved in its reading is preserved, and the conditions for its recovery are calculated. The inverse formulae, stability bounds and specified realizations jointly express the mathematical meaning of the holographic extreme and mean ratio.

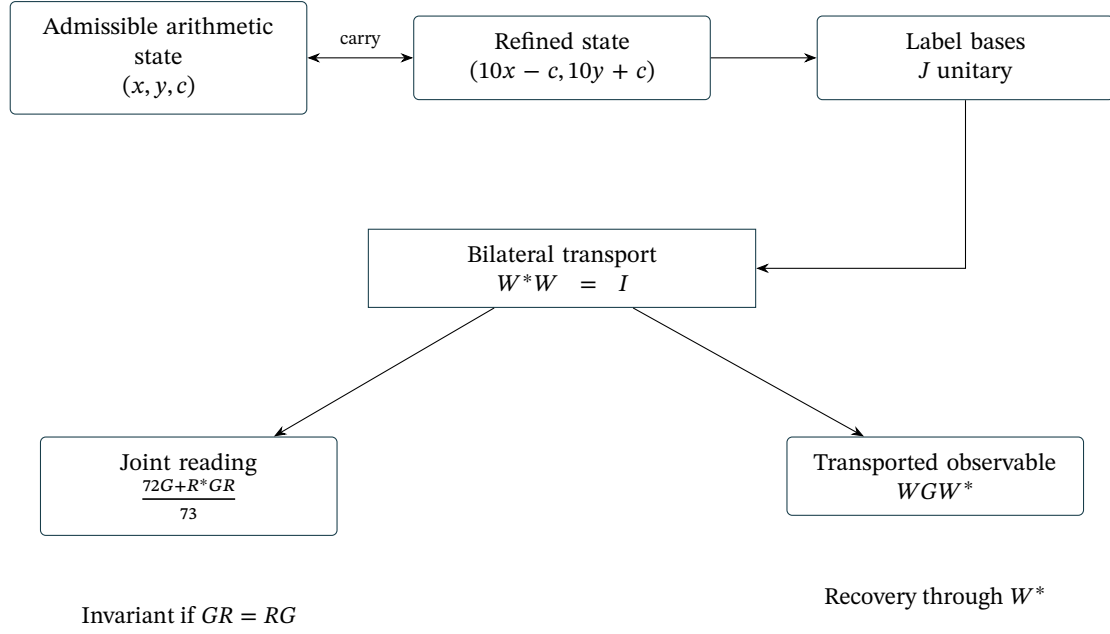


Figure 1: Composition of proved operations. Refinement preserves the normalized sum and allows the label to be recovered. Its lift acts on state amplitudes and preserves the norm. Reading an observable and transporting it in full retain different functions, specified by the two lower formulas.

Section 65 makes the archive contribution to bilateral composition explicit, constructs the quarter-turn plane of the dodecaphasic calendar, and proves oriented action recovery through a relative commutator. Compatible amplification preserves the operator; the detector distinguishes state dilation from energy gain and the ordered archive from operation counting.

Governing results and proof sequence

The opening exposition anticipates the function of the coupling constant; its construction follows the definition of the APP sheets, the TRIT orientation, the TPK operators, and the regional prolongations. Definition 9.1 specifies the ordered record, its positional readout, and the chart in which they are represented. Section 9.6 constructs terminal extraction and distinguishes visible information from the signed information preserved in the state.

The arithmetic function of the record is established in the construction of α . Section 11.4 contains the complete analytic representation, its recurrence in every degree, the convergence of its truncations, and the identification of its root with the generated pre-coordinate. Its dependence on this pre-coordinate is part of the reader's definition. This construction precedes the action scale, the angular coordinates, and the realizations of area and gravitation.

The reconstruction function is expressed through distinct, composable results. Theorem 67.2 proves that the cyclic orbit of the record forms a basis; Theorem 67.4 determines which responses permit recovery of an operator. Theorem 70.1 establishes bilateral conservation for two isometries. Theorem 70.7 constructs the archive of unbounded depth and its continuous inverse. Finally, Theorems 71.1 and 71.6 incorporate arithmetic refinement and calculate the transformation of observables.

These results make the article's hierarchy explicit: generation of the state, extraction of the constant, arithmetic and incidence realizations, conservation of forms, and effective recovery of information. In each case, the reconstruction formulas specify the record they receive; composition preserves the meaning of their domains, scales, and orientations.

Joint selection of the constant is developed before its algebraic reconstruction: orderings of the terminal repertoire produce 19 446 registers; exceptional compatibility retains 79, and temporal compatibility determines a unique register K . The conditions of both filters and their dependence on the enriched history are retained. Section 10 connects this determination with arithmetic closure, incidence representations and operator reconstruction.

Generative architecture and conservation of information

Triadic Modular Holography, abbreviated HMT, locates the origin of the construction in a finite system of positions, operations, and transports. Its prolongation yields compatible histories; the histories yield numerical and incidence readings; these realizations yield the structures with which Dimensional Mechanics operates. The question organizing this article concerns that constructive continuity: what remains when a unit changes scale, when an operation modifies its representation, and when part of the state is transferred to memory.

The content of the unit is specified throughout this progression. In an arithmetic partition, the normalized sum of its components is conserved. In an inner-product space, norms and correlations are conserved. In a geometric representation, forms, directions, and incidences are transported. Each conservation has its own object and equation. Their articulation makes it possible to follow a single construction from the carry of a digit to the recovery of a state distributed across an archive of arbitrary depth.

The following exposition presents this progression in dependency order. The complete definitions and proofs appear later, in the mathematical body. The purpose here is to enable the reader to recognize the role of each operation before studying its general formulation: how a history is produced, what determines a reading, and how information that a visible representation no longer displays can be reconstructed.

From finite positions to operational histories

The construction called APP, for *All Possible Paths*, begins with two axes of nine marks and two operations on their pairs of positions: addition and multiplication. The positions have an identity that is preserved when they are evaluated. The pair (i, j) admits the additive reading $i + j$ and the multiplicative reading ij ; a route also records how that pair was reached, which operation acted, and which orientation the transport followed. The relation between additive accumulation and multiplicative composition is thus situated on a common support.

The modular step preserves both the remainder and the quotient. For a positive evaluation n , the nonadic mark r , lying between one and nine, and the quotient q satisfy

$$n = r + 9q.$$

This equality allows the integer evaluation to be recovered from its enriched modular reading. Two positions may share the same mark while retaining different quotients. For example, $(5, 5)$ and $(8, 2)$ produce the sum ten; their products are twenty-five and sixteen. Both products have mark seven, with quotients two and one, respectively. The operational difference is located in the record accompanying the mark. The definitions in section 2 develop this conservation jointly with the operation sheets and routes.

The oriented ternary unit, called TRIT, distinguishes three regimes by means of the signs $-1, 0, +1$. Its action includes phase selection, the orientation of prolongations, and carry. Balanced ternary division writes an integer as $n = r + 3c$, with $r \in \{-1, 0, +1\}$; the remainder identifies the local state and the quotient preserves what is transported to the next level. Subsequent geometric realizations distinguish the elliptic, parabolic, and hyperbolic regimes. In the temporal reading of the corpus, return with memory, the threshold, and bidirectional opening have specific actions on the state.

The *Topological Phase Kernel*, or topological phase kernel, is abbreviated TPK. It composes tritic selection, transport, and memory updating. A transition modifies positions, sheets, or phase and preserves the data needed to relate the previous state to the next. Its result is an enriched state: it contains residues, quotients, carries, orientation, boundary, and route prefixes. The number that will subsequently be read arises from that operational history. Its genealogy preserves the operations that allow it to be continued and the relations that allow it to be compared with other histories.

This organization gives return a precise meaning. The calendar of one hundred and eight positions is distributed across twelve windows of nine. Completing a nonadic turn returns the corresponding

phase and advances the window record. Returning to a visible position and recovering the enriched state are therefore two distinct events. section 4 formalizes the calendar, holonomy, and the memory increment accompanying their returns.

Contraction of the value and persistence of incidence

A history is prolonged through compatible prefixes. Each prefix preserves the operations performed and delimits its admissible continuations. The joint construction of the continuum maintains graded transport, memory balances, ternary incidences, boundary relations, and determinantal compositions within that same system. These operations are linked through their actions on the same histories and through their compatibility with refinement (section 5). Their correlation is part of the construction of the object.

The Archimedean reading associates cylinders of increasing precision with the prefixes. Their compatible intervals contract and determine a value when their diameters tend to zero. The half-open convention distinguishes boundary representatives during refinement; carry information records the continuations that reach the same endpoint. This reading gives the numerical realization of the generated continuum. The sequence of intervals expresses how much of the value has been determined; the enriched state expresses how it has been determined.

The incidence reading preserves other information: which elements form a block, how a boundary is oriented, which transports relate the prefixes, and which actions preserve those relations. Through the maps developed in section 16, this information is realized in Paley and Hadamard matrices, Witt designs and Steiner systems, Golay codes, lattice constructions, and the Leech lattice. Prolongation to the vertex operator algebra V^h locates the action of the Monster group there. Each step preserves its carrier, its map, and its compatibility conditions.

The relation between number and symmetry thus becomes concrete. The Archimedean value identifies histories under a numerical equivalence; the incidence structure preserves relations between histories and their boundaries. Both readings arise from the same state and fulfil complementary functions. A scalar magnitude may remain fixed while the incidence representation undergoes a transformation. Following both readings makes it possible to study jointly the stabilization of the value and the evolution of the structure realizing it.

The unit is also conserved when distributed among prolongations. The measures of the successor cylinders sum to the measure of the predecessor cylinder. In the associated function spaces, this equality becomes a refinement isometry: a function from the previous level preserves its norm when represented on the new positions. The same principle relates partition, measure, and transport. The proofs of the joint construction specify composition and passage to the limit, so that every new depth remains linked to the preceding ones.

The realization of a measurement adds a concrete operation to this relation between state and reading. The system and the device couple, produce a record, and establish which distinctions become accessible. In an isometric realization of the coupling, the joint description preserves the correlations; the recorded outcome then determines the corresponding update. Evaluating an observable, conditioning the state on an outcome, and preparing the record anew have distinct functions. This sequence makes it possible to follow both the information that reaches the reading and that which remains in the state and the device. Memory intervenes in this operational continuity between an observation and subsequent ones (Sections 26 and 70.6).

The dimensional unit and recoverable carry

The holographic extreme and mean ratio is studied here through the operations that conserve the unit during its extension and refinement. The added position has its own identity. If E_9 contains the nine nonadic classes, its completion is written $E_9 \sqcup \{*\}$. The zero residue class already belongs to E_9 and has the positive representative nine; $*$ denotes the additional boundary state. The tenth possibility thus retains a function distinct from modular reduction. In several coordinates, counting how many

have reached * determines the strata of the completion and makes it possible to follow each stratum's contribution to the unit.

The cubic completion provides a first exact representation. Extending the positions of each coordinate from nine to ten decomposes the cubic unit into the interior, extension strata, and final vertex:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1 = 729 + 271.$$

Dividing by one thousand converts the count into a partition of unit mass. The terms preserve their dimensional provenance: one, two, or three coordinates occupy the added position. The expansion $(9 + 1)^d$ preserves this distinction for every positive integer dimension. Increasing dimension and changing scale are related by coordinate maps and sums of strata (sections 4.5.1 and 69.1).

Carry introduces a finer transformation. Consider the partition $73 + 27 = 100$. Multiplication by ten produces $730 + 270 = 1000$. Transferring one unit from the first component to the second gives

$$(73, 27, 1) \mapsto (10 \cdot 73 - 1, 10 \cdot 27 + 1) = (729, 271).$$

The third datum is the carry. The final sum remains one thousand and the normalized partition remains one. At the same time, the decimal residue of the second component preserves the transfer: from 271 one recovers the remainder one, then $(271 - 1)/10 = 27$, and finally $(729 + 1)/10 = 73$. The result carries with it the data of the operation that produced it.

Theorem 69.4 proves this inversion for the map

$$L_c(x, y) = (Bx - c, By + c), \quad x + y = M,$$

with base B , canonical carry $0 \leq c < B$, and the stated positivity conditions. After several stages, the accumulated record C_n collects the carries as a word in base B , and the components satisfy

$$x_n = B^n x_0 - C_n, \quad y_n = B^n y_0 + C_n.$$

The sum preserves $B^n M$. With base and depth fixed, the final state recovers the complete carry word and the initial partition (corollary 69.6). Depth is part of the reconstruction data, since it also determines the initial positions that contain zeros.

The normalized readings converge with explicit tail bounds. Three levels of description are thereby linked: a reversible integer operation, a sequence of partitions of the unit, and a system of cylinders determining a limit reading. The effective selection of continuations follows the rules of the transport under consideration. The example $(729, 271)$ shows its first transfer; the refinement proofs also describe the memory of all subsequent transfers and the boundary coincidences between expansions (theorems 69.7 and 69.8).

Numerical regions and the coupling register

In the corpus, the coordinates π_{HMT} , φ_{HMT} , and e_{HMT} arise from regional readers on the APP-TRIT-TPK prolongation. The closure region, propagation, and stable relation between modes have their own operators. For example, the calendar of additive and multiplicative modes produces the incidence matrix $F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. Its stable positive direction determines the golden coordinate in the corresponding realization. The generation proofs specify, for each requested depth, how a certified prefix is obtained and how its compatibility with subsequent prefixes is preserved (section 6).

The same terminal state preserves an oriented register of its twelve windows. Extraction of the signed channels, total charge, and Hadamard inversion produce the ordered vector K . The term *holographic arithmetic-geometric coupling constant* denotes that register together with its readers and its fixed phase origin. Its function is to relate an exact arithmetic encoding to the incidence and geometric realizations acting on the same object (definition 9.1).

In the canonical integral representation one obtains

$$K = (234, 543, 140, 729, 659, 824, \\ 621, 58, 914, 794, 146, 601), \quad \sum_{j=1}^{12} K_j = 6263.$$

section 67.1 shows the complete extraction, the congruences of its channels, and the divisibility allowing the Hadamard matrix to be inverted. The order of its components preserves window position. This information intervenes in the digit reading as well as in cyclic operators and incidences.

The total sum serves a reconstruction function. Differences between positions separated by three and four steps jointly determine the relative variations of the register. Adding the same quantity to all twelve components preserves these differences; the sum fixes precisely that common level. In K , the charge $Q(K) = 6263$ restores the uniform component. Differences, charge, and integer compatibilities allow the complete register to be recovered (proposition 9.13). The global relation between positions and the relative organization of their channels are preserved by complementary readings.

In base one thousand, with twelve declared positions, the register is encoded by the repeating fraction

$$\kappa_{\text{ag}} = \frac{234543140729659824621058914794146601}{10^{36} - 1}.$$

Each component occupies three digits; thus, the component fifty-eight is written 058 in the expansion. Multiplying the fraction by its denominator and performing twelve Euclidean divisions recovers the exact register. The periodicity of this encoding describes a fixed reading of the finite register; updating the enriched history, for its part, preserves its own memory dynamics.

The connection with the coordinate α_A is expressed in the closure reader by

$$\alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{ag}}.$$

The three regional coordinates and the register arise from the generated state and act in the constructive order described in equation (67.3). The formula makes their arithmetic coupling visible: the extraction of K preserves part of the structure intervening in that closure. The other realizations of K preserve their respective readers. In this way, a scalar identity, an incidence matrix, and a phase transport can be studied from the same provenance, with their actions clearly distinguished.

A register capable of identifying a transformation

The function of the register can be examined through a concrete operation. Let S be the cyclic shift of its twelve components. The twelve columns $K, SK, \dots, S^{11}K$ form an invertible matrix O_K . The result means that the cyclic orbit of this register reaches all directions in its twelve-component space. The proof computes the spectrum of the Gram matrix $O_K^* O_K$ and gives exact positive bounds for its eigenvalues (theorem 67.2).

Suppose that an operator H respects the cyclic shift, that is, $HS = SH$. When the vector response $y = HK$ is observed, its shifts provide the responses to all columns of O_K . The complete operator is then recovered by

$$H = O_y O_K^{-1}, \quad O_y = (y, Sy, \dots, S^{11}y).$$

The observation contains twelve components, and the commutation hypothesis identifies the class of recoverable transformations. Within that class, a single vector response determines all coefficients of the circular filter. For example, the combination $2I + 3S - S^5$ is identified from its action on K , with the error bounds of Theorem 67.4.

This fact provides an operational application of the structure of the register. Its order and spectrum allow it to be used as a test state for identifying compatible transformations. If the state and its responses are transported by a full-memory isometry, the Gram matrix of the orbit is conserved and identification remains recoverable by the adjoint. Stability follows from computed eigenvalues and explicit inverse operators. The same register participating in arithmetic closure thus acquires a precise function in the recognition of transports.

Normalization also has associated information. The unitary factor of O_K preserves orientation and angles; its Gram matrix preserves the amplitudes needed to reconstruct the original matrix. The pair formed by the two factors recovers O_K (corollary 67.3). This separation allows a normalized representation to be chosen while maintaining the exact scale of the register.

Two transports and bilateral conservation

The next step relates refinement to a law of correlation conservation. Let $\mathcal{B}$ and $\mathcal{C}$ be two isometries transporting the same state space to a common output space. For $a > 0$, form the two operators

$$Q_- = a\mathcal{B} - \mathcal{C}, \quad Q_+ = (a+1)\mathcal{B} + \mathcal{C}.$$

The two combinations bring together a common contribution and a relative contribution with different signs. Summing their norms with the corresponding weights cancels the cross terms:

$$(a+1)\|Q_-v\|^2 + a\|Q_+v\|^2 = (2a+1)(a^2 + a + 1)\|v\|^2.$$

Theorem 70.1 proves this equality for spaces of any dimension and for the stated isometric transports. The same cancellation preserves inner products and admits a formulation for forms preserved by the transports.

In the case $a = 8$, the factor $a^2 + a + 1$ equals seventy-three. The factorization of the full isometry first distributes the norm between two components with weights

$$\frac{72}{73} \quad \text{and} \quad \frac{1}{73}.$$

An orthogonal mixing then acts to produce the normalized channels of Q_- and Q_+ . The weights thus have a precise operational origin: $72 = 8 \cdot 9$ and $73 = 8 \cdot 9 + 1$. Their function is preserved when the dimension or the transports change, within the hypotheses of the bilateral law (section 70.2).

Recovery is seen directly in the output combinations. If $x = Q_-v$ and $y = Q_+v$, then

$$x + y = (2a+1)\mathcal{B}v, \quad ay - (a+1)x = (2a+1)\mathcal{C}v.$$

The two transports of the initial state remain determined by the pair of channels. The adjoint of either isometry then recovers the corresponding state. With full normalization, the operator W_a satisfies $W_a^*W_a = I$: it preserves the norm and provides an exact inverse on its image.

A particular reading may vary while all this information is conserved. When $\mathcal{C} = \mathcal{B}R$, with $\mathcal{B}$ and R unitary, the latter serving as relative transport, the diagonal reading of an observable G in both channels produces

$$\mathcal{J}(G) = \frac{72G + R^*GR}{73} \quad (a = 8).$$

Observables compatible with R remain fixed; the others record the difference between its two actions. This equation distinguishes conservation of the complete state from the response of a particular reader (theorem 70.3).

The same carry, now as a transportable state

The preceding relation applies materially to carry. Each admissible label (x, y, c) is associated with a vector of an orthonormal basis. The integer transformation L_c then induces the operator J taking that basis to the output labels $(Bx - c, By + c)$. Its inverse recovers the carry through the residue. The bijection of labels makes J a unitary transformation (proposition 71.2). Integer values designate the labels; norms correspond to the amplitudes of the vectors representing them.

For base ten and initial sum one hundred, consider the two labels $d_1 = (73, 27, 1)$ and $d_2 = (73, 27, 2)$. Their images are $e_1 = (729, 271)$ and $e_2 = (728, 272)$. Fix the transport R that interchanges these two labels and leaves the others fixed. The composition uses $\mathcal{B} = J$ and $\mathcal{C} = JR$. This choice provides an explicit example within the bilateral theorem, with the transport rule specified before its evaluation.

On the state $|d_1\rangle$, the channels for $a = 8$ are

$$Q_-|d_1\rangle = 8|e_1\rangle - |e_2\rangle, \quad Q_+|d_1\rangle = 9|e_1\rangle + |e_2\rangle.$$

Their squared norms are sixty-five and eighty-two. The exact balance is $9 \cdot 65 + 8 \cdot 82 = 1241 = 17 \cdot 73$, while the sum of the two channels recovers $17|e_1\rangle$. Applying the label inverse recovers the partition $(73, 27)$ and carry one.

The fractional reading of the first component also provides an exact prediction for this declared transport. Evaluating it diagonally in both normalized channels yields

$$\frac{72 \cdot 729 + 728}{73 \cdot 1000} = \frac{729}{1000} - \frac{1}{73000}.$$

The diagonal carry reader gives $(72 \cdot 1 + 2)/73 = 74/73$. The observable transported from the input, by contrast, preserves the eigenvalue one of the original carry. The difference between the two results makes it possible to identify which reading operation has been performed. The state information remains recoverable, and the diagonal reading quantifies how the relative transport intervenes (section 71.8).

The example brings together the two conservations of the article. The arithmetic sum is conserved in the refined labels, and the norm is conserved in their unitary representations. The connection between them arises from the bijection and its lift to orthonormal bases. This composition allows readers to be computed, transport to be inverted, and error to be controlled while keeping the objects on which each law acts identified.

Full memory, symmetries, and arbitrary depth

Bilateral conservation is prolonged stage by stage. At each level, one channel continues the state and the other records a memory complement. If $R_N v$ is the state reaching level N and $m_0, \dots, m_{N-1}$ are the recorded complements, the accumulated balance is

$$\|v\|^2 = \|R_N v\|^2 + \sum_{n=0}^{N-1} \|m_n\|^2.$$

The equality identifies exactly where the initial norm remains. Finite recovery uses the continued state and all available complements; their correlations are preserved by the same joint isometry.

For the archiving policy of Theorem 70.7, the norm of the continued channel has a uniform geometric bound. In the case $a = 8$, its squared norm is bounded at every stage by $729/1241$. As depth increases, the continued component tends to zero and the infinite archive preserves the entire norm. If A_n and D_n denote the continuation and memory blocks, respectively, the state is reconstructed by the convergent series

$$v = \sum_{n \geq 0} R_n^* D_n^* m_n.$$

The inverse operator has norm one. A perturbation of norm ε in the complete archive produces a reconstruction error of norm at most ε . Finite truncations have their own bounds and, when only their memory is used, the finite inverse specified in the same development.

The incidence structure of the register K , its cyclic orbit, and its distinguished directions can be transported through these isometries. The inner products determining angles and Gram matrices are preserved; projectors and readers are transported with their domains; the action of the adjoint recovers the initial representation. Exterior powers transport orientation and preserve Gram determinants, which measure the squares of the corresponding volumes. The dimension of the carrier may be increased, and the geometric information of the image remains specified by the isometry.

The relation to Noether conservation is formulated at a well-defined additional level. For the auxiliary discrete action constructed in proposition 67.12, a collection of generators intertwining the transports produces a conserved charge of the form $p_n^* A_n q_n$. A variational action, its equations, and the symmetry preserving them intervene here. The norm balance and the variational charge are articulated through the common transports, each with its hypotheses and its function in the proof.

This progression establishes an operational relation between unit, number, memory, and geometry. The arithmetic unit admits reversible refinements; the enriched state preserves the record of operations; the numerical and incidence realizations extract correlated properties; bilateral coupling maintains information and correlations during their redistribution. The holographic extreme and mean ratio thus acquires a demonstrative development in which partition, extension, transport, and recovery can be followed through explicit maps. The following pages present those maps, their proofs, and their complete examples.

Five holographic principles of construction and conservation

The *Elements* provide a precise antecedent for presenting a mathematical architecture through declared operations and proved consequences. Book I distinguishes definitions, postulates, and common notions. Its first three postulates allow a segment to be drawn between points, a segment to be extended in a straight line, and a circle to be described with a given centre and radius; the fourth establishes the equality of right angles. The fifth formulates a condition of intersection: two straight lines cut by a transversal meet, when extended, on the side where the interior angles sum to less than two right angles. These are statements about the objects and constructions of that geometry.¹

The extreme and mean ratio appears in Book VI as a relation between a whole and its parts: the whole is to the greater segment as the latter is to the lesser. In a modern algebraic translation, for positive lengths $u > v$, one writes $(u + v)/u = u/v$; hence the ratio $r = u/v$ satisfies $r^2 = r + 1$. This notation explains the transmitted proportion without turning modern algebraic language into Euclid's starting point. In HMT, the self-scaling coordinate is obtained from an incidence produced by the operational calendar; its recognition through the same equation belongs to a subsequent stage.²

The comparison that follows concerns the organization of the construction. HMT declares an arithmetic support, two related sheets, a tritic orientation, and operations that prolong states while preserving their records. A cylinder represents the information fixed by a prefix; a Euclidean line is another object, with another definition and other operations. The five principles in this section express effective relations between holographic objects. Their number provides a recognizable exposition; it does not establish a term-by-term correspondence with the five postulates, a refutation of the fifth, or a replacement for the geometric axioms.

These principles have an *organizing* status: they bring together conditions and theorems that the body of the article constructs and proves. Each first presents its meaning, then its domains and an identity by which it can be checked. The common sequence is

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{enriched state} \longrightarrow \text{joint discrete structure of the continuum.}$$

Numerical, linear, and geometric realizations subsequently read that structure. This placement of the realizations keeps visible the origin of the coefficients and the information preserved by each reading.

Principle I. Relational unity of generation

Each reading is constructed through typed operations on a common support, preserving the relations that allow it to be continued. An APP cell simultaneously admits an additive and a multiplicative reading. The passage to a visible digit records part of both operations; the quotient, orientation, and sheet preserve the conditions of its prolongation. Unity consists in this relational provenance, which allows the same execution to be followed through its different realizations.

The initial domain is D_9^2 , with $D_9 = \{1, \dots, 9\}$. On it act $S(i, j) = i + j$ and $P(i, j) = ij$. For an integer n , the positive residue-and-quotient chart is given by

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n).$$

The lifted APP evaluation preserves $((\rho_9(S), q_9(S)), (\rho_9(P), q_9(P)))$. The identity reconstructs both integer readings; for example, it distinguishes the values 10 and 19, which have the same positive residue and different quotients. The result and its proof are developed in (2.1) and (2.2). The representation by residues 0, ..., 8, used in other charts of the article, preserves its corresponding quotient: changing representatives requires changing both components of the division.

¹Euclid, *Elements*, Book I, Postulates 1–5. Heiberg's Greek text and Richard Fitzpatrick's translation, p. 7: <https://farsid.e.ph.utexas.edu/books/Euclid/Elements.pdf>. They may also be consulted separately in David E. Joyce's edition: <https://mathcs.clarku.edu/~djoyce/elements/bookI/bookI.html>.

²Euclid, *Elements*, Book VI, Definition 3, in Joyce's edition: <https://mathcs.clarku.edu/~djoyce/elements/bookVI/defV13.html>. Fitzpatrick's edition prints this same statement as Definition 2 of Book VI, p. 156. Both editorial numberings are retained here.

TRIT adds the oriented representation of $\mathbb{F}_3$ by $\{-1, 0, +1\}$, with its exact division $n = r + 3c$. The operational selector M_{ph} takes the values $+1, -1, 0$ in the phase classes $\{1, 4, 7\}, \{2, 5, 8\}, \{3, 6, 9\}$, respectively. Cardinal transport has another domain and another function: it moves the cursors on $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. In the enriched state $\hat{X}$, TPK composes selection, transport, and memory updating,

$$U_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Definition (2.12) and the development in sections 2.3.1 and 2.3.2 specify phase, cursors, sheets, directions, residues, quotients, and records. Phase sign and cardinal direction are composed while preserving their distinct types.

Self-scaling provides an example of generation that can be followed without starting from its final value. The calendar $(+1, -1, 0)$ produces, with the rule for changing and preserving sheets, the modal cycle (Σ, Π, Π) . Its three edges determine

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

In its subsequent linear realization, the positive eigenray normalized as $(1, x)^\top$ satisfies $x = \lambda$ and $1 + x = \lambda x$. Thus $x^2 - x - 1 = 0$ is obtained; the positive root is recognized as φ . The theorem following (6.15) proves this identification after constructing the entries of the matrix. The relation to the Euclidean proportion is exact in this self-scaling reading and preserves the distinction between their two procedures of origin.

Principle II. Oriented prolongation and composition memory

An admissible prolongation adds an event to the state and transports the preceding memory through the same operation. The history allows paths ending at the same position or phase to be distinguished. This principle provides a precise way of speaking about continuation: the prefix actually traversed is preserved, and an emission allowed by the current selector and regime is added.

Let X_k be the set of admissible histories of length k , constructed from the seeds of the kernel. If ε is admissible after h , the prolongation $h\varepsilon$ belongs to X_{k+1} and truncation satisfies $\rho_k(h\varepsilon) = h$. The emission label and the preceding sheets are retained, even when two emissions have the same final reading. In a memory chart over a module M , the update has the form $U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon$, with $C_\varepsilon \in \text{End}(M)$ and $\kappa_\varepsilon \in M$. If two segments are composable in that chart, then

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

The proof of proposition 5.1 consists in substituting the first update into the second. Memory is added after the transport corresponding to it; associativity makes the result independent of the subdivision of the path.

The distinction between phase and memory already appears in an elementary reading. For $(w, r) \in \mathbb{Z} \times C_9$, with $\bar{r} \in \{0, \dots, 8\}$, one has

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

The second equality counts exactly one crossing from 8 to 0. Its first coordinate records the advance that the second no longer distinguishes. In the calendar of twelve windows, the exact sequence $0 \rightarrow C_{12} \rightarrow C_{108} \rightarrow C_9 \rightarrow 0$ preserves carry between windows. It does not split: C_{108} has exponent 108, and $C_{12} \times C_9$ has exponent 36. The proof and the carry law are in proposition 9.2.

Phase return thus acquires an operational meaning: it allows readings made at the same position in the calendar to be compared while the state preserves what happened between them. The holonomy Γ_9 is obtained from the enriched dynamics; K_{ph} is its subsequent reduced representation. The integer dodecaphasic register K , which also uses orientation and incidences, likewise has its own construction and domain. Equality of a phase, a residue, or a expressed coordinate does not by itself identify the histories. This is the initial information preserved in passing to refinement.

Principle III. Compatibility of refinement and realization

Refinement consists in adding resolution compatible with the prefix while maintaining truncation relations and the masses of its prolongations. The reading at a finer scale specifies what occurred inside a cylinder; forgetting that detail recovers the previous description. The realization of a value arises when the cylinders of its reading contract compatibly. The history determining those cylinders remains in the domain of the construction.

In the tree of section 5.2, the initial set is finite and nonempty, and each history has a finite, positive number of admissible successors. A positive initial mass summing to one and local probabilities $p(\varepsilon \mid h) > 0$, summing to one at each predecessor, produce

$$\mu(h\varepsilon) = \mu(h)p(\varepsilon \mid h), \quad \sum_{\rho_k(y)=h} \mu(y) = \mu(h).$$

Therefore, in the subsequent realization $H_k = L^2(X_k, \mu_k)$, the lift $(J_k f)(y) = f(\rho_k(y))$ is isometric. Its adjoint is the conditional expectation

$$(E_k g)(h) = \sum_{\rho_k(y)=h} \frac{\mu(y)}{\mu(h)} g(y), \quad E_k J_k = I.$$

Grouping the squared norm by predecessors proves the isometry, and grouping the inner product proves the adjoint formula (proposition 5.2). Lifting a reading and returning it to the preceding level thus form a pair of explicit operations.

The hypotheses of finite branching and nonempty prolongation provide compatible infinite histories. Their cylinders receive the preceding masses, which determine a measure on the cylindrical sigma-algebra. The choice of p belongs to the declared reader: the case of equiprobable successors gives $p = 1/d(h)$, without imposing that measure on any other reading. This construction preserves multiplicities of histories, including those that share a realized value. Prolongability refers to the constructed admissible tree; an additional restriction on routes must explicitly preserve that property.

For the regional numerical reading, each block $b \in \mathbb{F}_3^6$ has an integer index $\nu(b) = \sum_{j=1}^6 b_j 3^{6-j}$, lying between 0 and 728. The reader fixes $N_0 = m$ and produces

$$N_{n+1} = 729N_n + \nu(b_{n+1}), \quad I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right].$$

One has $I_{n+1} \subseteq I_n$ and $\text{diam}(I_n) = 729^{-n}$. The left endpoints form a rational Cauchy sequence; its class first defines the internal realization of the reading. Subsequently identified with the usual completion of the rationals, it corresponds to the unique point of the nested closed intervals. The theorem following (6.31) proves this statement and treats boundary conventions separately. For regional characters, the effective reader of each decimal prefix uses the computable brackets and the proved separation from their boundaries, as specified in corollary 6.18. This arbitrary depth preserves the quantifier “for each finite length”; the existence of a limit by compatibility and contraction is a distinct statement and does not, by itself, provide an algorithm for every abstract history. Histories representing the same point retain their distinct records.

Principle IV. Correlative double reading and frame transport

The numerical and incidence readings arise from the same enriched state and preserve their compatibilities under prolongation and changes of frame. The first specifies a realization through increasingly narrow intervals. The second retains relations between positions, signs, and boundaries. The holographic link consists in their common maps of origin and transport, which allow the information expressed by each to be compared.

Let X_n^{cal} be the domain of admissible prefixes including their initial frame and successive frame transports. The arithmetic reading uses the blocks of that prefix; the incidence reading uses the words $w_j \in \mathbb{F}_3^6$ and the frames f_j . For a matrix $A \in M_6(\mathbb{F}_3)$, the linear reader

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad \gamma_A(w) = (w, wA)$$

acts on words written as rows. Its specialization to the Paley–Witt matrix constructed in section 16.2 produces the ternary code and its incidences. The matrix is obtained in that finite construction; a terminal real number does not select its entries. The order $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ maintains the transformations and memories used before each reading.

Compatibility can be checked on the operator itself. For $f \in \text{GL}_6(\mathbb{F}_3)$, let $A^f = fA f^{-1}$. Then

$$\gamma_{A^f}(w f^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}),$$

since both sides are $(w f^{-1}, w A f^{-1})$. In particular, the reading of the prefix

$$Q_n(x) = (f_j, \gamma_{f_j A w f_j^{-1}}(w_j))_{0 \leq j < n}$$

satisfies $r_{n,m} Q_n = Q_m p_{n,m}$, where $p_{n,m}$ truncates histories and $r_{n,m}$ truncates records. The causality of frame updating guarantees that both procedures retain exactly the same terms. The result proposition 16.1 constructs its passage to the limit. Linear covariance holds for the stated change of frame; additionally preserving a coordinate support or a metric requires that structure to be transported, or frame changes to be restricted to the group preserving it.

The correlative construction of the continuum extends this compatibility to sheet transport, oriented balance, ternary incidence, the rectangular port complex, and cellular memory with its fillings and determinant lines. These aspects are constructed on the same histories, not on five domains chosen afterward. The identities of concatenation, naturality, and boundary ensure their assembly in propositions 5.7 and 5.8 and sections 5.7 and 5.9. The five expository principles of this opening are not a new partition of those five consubstantial constructions. Recovery of a word by the first projection of γ_A , for its part, concerns that word: recovery of the complete enriched history requires the other records distinguishing it to be preserved.

Principle V. Conservation and recovery of the complete record

Recoverable information is conserved jointly in the continuing state, in the records produced, and in their correlations. A visible reading may contract while the complete record preserves an isometric representation of the initial state. This principle allows verifiable reversibility to be formulated: what is stored, in which space it resides, and by which operator it is reconstructed are declared.

The transport of lifted histories in section 5.3 acts between the spaces $H_k = \ell^2(\hat{X}_k)$ by

$$u_k e_{\hat{h}} = \sum_{\varepsilon \text{ admissible}} \sqrt{p(\varepsilon | \hat{h})} e_{\hat{h}\varepsilon}, \quad u_k^* u_k = I.$$

The second equality follows from local normalization and from distinct predecessors having successors with distinct labels. If R_k reads sheet Σ and $Q_k = I - R_k$ reads sheet Π , the operations

$$\begin{aligned} A_k &= R_{k+1} u_k R_k + Q_{k+1} u_k Q_k, \\ \eta_k &= Q_{k+1} u_k R_k + R_{k+1} u_k Q_k \end{aligned} \quad \text{satisfy} \quad A_k^* A_k + \eta_k^* \eta_k = I.$$

The Gram identity in proposition 5.3 is proved by annihilating the products of complementary projectors. It distinguishes remaining on a sheet from changing sheets; the tritic value zero remains an orientation and does not become a third sheet.

Write $F_0 = I$, $F_{n+1} = A_n F_n$. Telescoping the preceding identity gives, for every horizon N ,

$$\|x\|^2 = \|F_N x\|^2 + \sum_{n=0}^{N-1} \|\eta_n F_n x\|^2.$$

The first term is the terminal residue that still continues; the others are the recorded outputs. The theorem theorem 5.4 proves that $F_N^* F_N$ converges strongly to a positive operator T_∞ . One therefore obtains the complete record

$$\mathcal{V} : H_0 \longrightarrow H_0 \oplus \bigoplus_{n \geq 0} H_{n+1}, \quad \mathcal{V}x = (T_\infty^{1/2} x, (\eta_n F_n x)_{n \geq 0}), \quad \mathcal{V}^* \mathcal{V} = I.$$

Indeed, the sum of squared norms is $\|x\|^2$; polarization gives the last identity. Its image $\mathcal{M}$ is closed, and the inverse on it is $\mathcal{V}^*$. The terminal is preserved until its extinction is proved; T_∞ need not be a projector, and convergence of $F_N x$ is not assumed.

Conservation also extends to the operators reading that information. For a bounded operator B on H_0 , the transport $B^\mathcal{V} = \mathcal{V}B\mathcal{V}^*|_{\mathcal{M}}$ preserves products, adjoints, and inner products:

$$(B_1 B_2)^\mathcal{V} = B_1^\mathcal{V} B_2^\mathcal{V}, \quad \langle \mathcal{V}x, B^\mathcal{V} \mathcal{V}y \rangle = \langle x, By \rangle.$$

Inserting $\mathcal{V}^* \mathcal{V} = I$ proves both formulas. In the nonadic reader of theorem 67.6, the same identities are realized with $T_n = (8I + C_n)/9$ and $D_n = \sqrt{8}(C_n - I)/9$, for unitaries C_n on the declared carrier. The composition retains its status as a reader and does not automatically identify every C_n with the generating holonomy Γ_g . Recovery thus preserves the correlations of the record and the transported observables within the precise image on which it acts.

The register K makes this distinction visible in a finite object. Its encoding $\kappa(K)$, in base $B = 1000$ and length twelve, is

$$\kappa(K) = \frac{N(K)}{B^{12} - 1}, \quad N(K) = \sum_{j=1}^{12} K_j B^{12-j}, \quad K \in \{0, \dots, 999\}^{12}.$$

Multiplying by $B^{12} - 1$ and performing twelve Euclidean divisions recovers the register exactly, including its initial zero blocks. The proposition proposition 67.1 further proves that an absolute error less than $1/[2(B^{12} - 1)]$ allows the same integer to be recovered by rounding. Reversibility concerns the register in its chart and with its metadata; it does not identify the register with the entire TPK history. The *holographic arithmetic-geometric coupling constant* denotes, in this article, the object K typed with its readers and relations. Its representation arises from the enriched state and subsequently participates in alpha closure: it is not used as a generating input for π , nor does its name make every scalar reading a universal invariant.

A joint reading of the five principles

The sequence offers a positive criterion for navigating the article. Relational unity fixes what is operated on; prolongation preserves what has been traversed; refinement organizes resolution; double reading keeps value and incidence together; recovery identifies the exact conditions for reconstructing the record and its observables. The proofs are linked because they act on objects of common provenance and respect their transport maps. Wherever a reading forgets information, its domain and its loss fiber remain defined.

The historical extreme and mean ratio shows how a relation between the whole and its parts can determine a proportion. The holographic construction developed here additionally studies relations conserved when resolution, frame, and the distribution of information between continuation and memory change. The self-scaling equation establishes a precise mathematical link with that proportion; the identities of concatenation, refinement, and isometry provide the additional content developed in this article. The historical comparison thus retains its orienting function, while each holographic conclusion rests on its own operations and proofs.

Primary sources for the historical comparison

1. Euclid, *Elements*, Book I, Postulates 1–5. Greek text established by J. L. Heiberg, with a translation by Richard Fitzpatrick, corrected edition of 2008, p. 7; Book VI, Definition 2 in this edition, p. 156. <https://farside.ph.utexas.edu/books/Euclid/Elements.pdf>.
2. Euclid, *Elements*, digital edition by David E. Joyce, Clark University: Book I, Postulates 1–5, <https://mathcs.clarku.edu/~djoyce/elements/bookI/bookI.html>; Book VI, Definition 3, <https://mathcs.clarku.edu/~djoyce/elements/bookVI/defVI3.html>. The difference in Book VI numbering is retained to facilitate comparison between editions.

A

Evolutionary conservation of dimensional unity through double holographic projection

The two arithmetic sheets, tritic orientation and transport with memory constitute the starting point for realizations of the continuum.

One unity, two realizations

The double projection organizes this article from its first construction. One and the same state preserves the data that allow a value to be refined and the data that allow an incidence to be determined. The first realization assembles compatible cylinders until its numerical reading is uniquely determined; the second transports boundary relations. The thesis examines their common origin and the operations that maintain their correspondence throughout refinement. Number and geometrical configuration thereby acquire an explicit genealogy.

Unity is constituted through positions, operations, and transport relations. APP arranges the two nonadic directions on a common support and evaluates its additive and multiplicative sheets. Each evaluation preserves the remainder and its quotient. TRIT distinguishes orientation and the local regime; the TPK composes these data with displacement, carry, boundary, and prolongation. The joint discrete structure of the continuum resides in this enriched construction. Its subsequent realizations receive operations that have already been determined.

The nonadic connection has a constitutive function here. One turn composes the transitions of the complete state. The observable phase returns and the register incorporates the path traversed. This distinction makes it possible to recognize a regularity while the history increases in depth. The continuity of refinement preserves the prefixes that make the two realizations compatible. Each new level of precision belongs to the same history from which the previous levels originate.

The meaning of evolutionary conservation becomes apparent at this point. A normalized quantity may remain unchanged while its distribution, resolution, or representation changes. Recovering the state requires identifying the data that accompany that change. Arithmetic preserves quotients; transport preserves orientation and turns; incidence preserves relations; the operator archive preserves the complements needed to invert the reading. The article develops the maps that articulate these functions.

The holographic extreme and mean ratio expresses this question of unity, proportion, and transformation. Its formulation includes dimensional completion and reversible refinement, and subsequently extends to balances of states and observables. The coupling constant occupies a definite position within this development: the generated register links the arithmetic reading, incidence, and operator recovery. The proofs that follow construct the antecedents of each of these functions.

1 Generation, conservation and realization

The object of Triadic Modular Holography is a state generated by positions and operations that preserves the information of its evolution. Numerical evaluation and boundary incidence are correlated realizations of this antecedent. Contraction of compatible cylinders determines values; prolongation of their incidences preserves relations that distinguish the generating states. This priority of construction also governs Dimensional Mechanics: action, duration, extension and mass arise through readers whose domains preserve orientation, carry and the history of operations.

This article studies a precise structural question: what is conserved when a readout reduces the state, where the complement resides, and how both value and dynamics can be reconstructed from it. A complete answer requires three levels. First, emissions and their refinement are constructed. A transport identity is then proved on the realized state. Finally, we determine what a specific reader observes and what physical consequences follow from its composition with the action and scale readers. The development follows this order in each application.

1.1 Principal thesis and statements

Evolutionary conservation of unity admits an operator formulation. For two transports B, C between spaces equipped with forms b, b' , with

$$B^*b'B = C^*b'C = b,$$

the collective representation of the nine positions yields

$$u(B, C) = \frac{1}{9} \begin{pmatrix} 8B + C & \sqrt{8}(C - B) \\ \sqrt{8}(C - B) & B + 8C \end{pmatrix}, \quad u(B, C)^*(b' \oplus b')u(B, C) = b \oplus b. \quad (1.1)$$

The first column contains the readout and memory generated by a state with no incoming memory. The second column describes the contribution of memory that enters again. The equality preserves a positive form when b is an inner product, and preserves oriented area when b is a symplectic form. These two interpretations use explicitly different structures.

This law yields four connected consequences. The first is stable reconstruction of the complete register, including its terminal component. The second recovers from dodecaphasic memory the direction determining the dimensional projector. The third extends conservation to areas, volumes and higher exterior degrees. The fourth identifies, in a specified quantum realization, the part of memory that changes the purity of a partial readout. Each consequence follows from a specified mathematical operation on the preceding one.

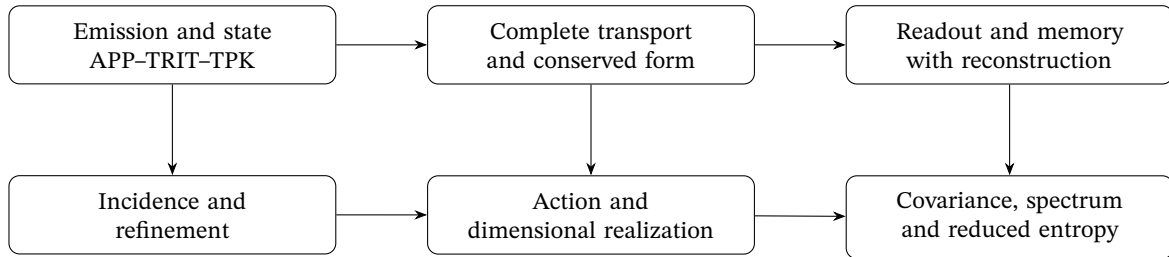


Figure 2: Dependencies between realizations. The arrows represent operations proved in the body of the article; the scalar readout preserves an antecedent richer than its own value.

1.2 Complete memory and partial observation

For a sequence of compressions T_n with complements D_n , write $P_0 = I$, $P_{n+1} = T_n P_n$. The local equality $T_n^* T_n + D_n^* D_n = I$ implies

$$I = A_\infty + \sum_{n \geq 0} P_n^* D_n^* D_n P_n, \quad A_\infty = \text{s-lim}_{N \rightarrow \infty} P_N^* P_N.$$

Reconstruction uses $A_\infty^{1/2} v$ and all the $D_n P_n v$. Strong convergence of the terminal Gram operator suffices for conservation; it is proved by positive monotonicity, independently of convergence of the terminal vector. The register contains amplitudes and correlations, whereas their intensities provide the norm balance. This distinction determines the recoverable information.

In a specific finite register, the incidence of twelve positions in the Witt hexads has Gram matrix $36I + 30\mathbf{11}^*$. Its normalization preserves exactly the centred part of the register. Incorporating the mean recovers its twelve coordinates. The subsequent dimensional map uses this transported information and preserves its dependence on the ordered register K .

1.3 A calculable spectral consequence

Let $J^2 = -I$, $J^\top = -J$, and let H be symplectic in a canonical realization of dimension $2d$. Define

$$D = \frac{\sqrt{8}}{9}(H - I), \quad D_{\text{anti}} = \frac{D + JDJ}{2}, \quad W = \frac{8I + HH^\top}{9}.$$

The article proves

$$\boxed{-(JW)^2 = I + [D, J][D, J]^\top = I + 4D_{\text{anti}}D_{\text{anti}}^\top.} \quad (1.2)$$

For two identical pure Gaussian inputs and transport $\mathcal{U}(I, H)$, the matrix W is the normalized reduced covariance. Hence

$$\text{Tr } \rho_{\text{vis}}^2 = \det(I + 4D_{\text{anti}}D_{\text{anti}}^\top)^{-1/4}.$$

The formula relates a memory operator to the purity functional of the reduced state. Global purity remains equal to one. In particular, mixing depends on the antilinear component relative to the transported complex structure, rather than on an indiscriminate identification of memory with entropy.

The choice of Gaussian representation, input state and observed subsystem forms part of the statement. The genealogy specifies the transports being composed; the protocol specifies the observation whose distribution is calculated. This specification allows reproducible comparison between realizations and separates the theorem from any subsequent experimental attribution.

1.4 Order of exposition

The initial chapters develop the common primitives, prolongation, numerical regions, register and exceptional incidence in full. Action readers and dimensional representations follow. The second part brings together the proofs of conservation, recovery, holonomy and reversible extension. The final part composes these results with physical actions and obtains the spectrum of reduced memory. The bibliography situates the tools of representation after the construction they realize.

Finite symmetry is expressed through actions and intertwiners. Continuous variational symmetry is expressed through an action, a variation and its conserved charge. Noether's theorem applies in this second domain; transport of exceptional incidence preserves its own algebraic content. Their relationship is established through the transported observables and forms, with their differences of type kept explicit.

Preservation is also examined before linear representations are introduced. The emitter's decimal encoding has a check congruence and a complete alphabet of 42 blocks, proved through its signatures and seeds. Incidences between emission halves join the regions while preserving the provenance of every edge. Two records with a common histogram and different correlation establish a concrete

distinction between distribution and order. Archive laws receive that order together with the amplitudes and terminal data stated in their hypotheses.

Normalization of the multiplicative logarithm is distinguished from its inverse by the least-positive-generator criterion. In integer closure, controls of carry direction, terminal endpoint, rotation, and local perturbation keep the remaining data fixed. Boundary selection applies saturation, dual neutrality, and orientation successively. These operations precede recovery readings and retain their domains, so that each inverse acts on the data actually produced by its construction.

The linkage of records is made explicit through the emitter's multigraph and its phase quotient: five minimal closures join the four incidences required for each region. The derivation of boundary occupancy and the parity of readers under oriented reversal complete the description of the data received by bilateral transport and retained by its coupling readings.

2 Formal kernel of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information needed to prolong them. Numerical realizations are formed only afterwards. A constant recognized at the end of this process does not replace the rule that produced its region or the history determining its evaluation.

The name *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not assume that its arithmetic law is the Feynman path integral. The *Topological Phase Kernel*, TPK,³ denotes the composition of selection, transport, memory updating, and prolongation. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks on a line graduated from zero to ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must be distinguished from the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions leads to the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself it does not preserve the integer before reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulation of i repeated j times produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ exchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, rather than a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation preserving both sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^-, q_{ij}^-)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

The term *sheet* here denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the remainders are reduced.

³The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

One support: 81 positions.

Two evaluations:
 $S(i, j) = i + j$, $P(i, j) = ij$.

Each row is generated by accumulation:
 $P(i, j + 1) - P(i, j) = i$.

The chart represents $0 \pmod{9}$ by 9.

The highlighted block is
 $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$,
 with eigenvalues 9 and 5.

Figure 3: APP multiplication table in positive representatives. Non-unit rows and columns are distinguished by gray shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representation edges do not constitute an intrinsic topological boundary of the torus.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The remainders alone do not determine these integer evaluations or their quotients.*

Proof. The first assertion is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same remainder and different quotients. On the two-dimensional support, positions (5, 5) and (8, 2) give the same sum and the same product remainder, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residue projection identifies data that the lifted evaluation distinguishes. $\square$

2.1.1 Cayley graph, toroidal topology, and winding number

In row and column coordinates, the cardinal translations are given by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southwards and the second eastwards. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic grid: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not make every row of APP into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When a path closes, its integer displacement before reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Path reversal changes the sign of this vector. Returning to the same residue position therefore does not require zero winding. This is the first example of a distinction that will reappear in phase return: one coordinate can close while another records nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called a *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of addition of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the schedule into phase and window will use representatives from zero to eight.

The totals of the two evaluations distinguish the visible contribution from the contribution retained in the quotient:

$$\begin{aligned} \sum S = 810, \quad \sum P = 2025, \quad \sum \Sigma = 405, \quad \sum \Pi = 459, \\ \sum q^+ = 45, \quad \sum q^\times = 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Removing it would change the arithmetic being transported, even if the residue table remained identical.

2.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row has this property exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Multiplicative folding therefore already distinguishes units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readouts: its direct reduction modulo three is 0, 0, 0, whereas extraction of the factor three followed by reduction of its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as direct reduction of the original number.

2.1.4 Latin realization of the additive evaluation

The difference between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in every row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a + b$ permutes the nine residues. This gives a realization of a Latin square by rank-one projectors, in the terminology of quantum Latin squares [4, 6]. This construction starts from the additive sheet already defined and makes the map to its operator representation explicit.

The multiplicative evaluation preserves different information: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

Multiplicity is recorded in the operator and characterizes the multiplicative radical. The toroidal support, the Latin property, and the operator resolution of the identity are therefore properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries[5]. This reference provides a precise language for comparing realizations of APP.

2.2 TRIT: orientation and quadratic regimes

Definition 2.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1)$, $(0, 0)$, $(0, +1)$ may have the same zero remainder and different carries.

The signature does not replace the trajectory. A zero remainder does not mean absence of orientation, phase, or quotient information. Nor does it introduce a physical quantity by itself. Its function is to distinguish algebraically the transport classes that will be realized below.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 2.5 (Exponentiation of the three types). *In each algebra (2.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	-I	$\cos t I + \sin t J_\tau$
0	0	$I + t J_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, all terms of degree at least two vanish. The other two cases recover the trigonometric or hyperbolic series. $\square$

The exponential is common to the three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau) \exp(-tJ_\tau) = I$. The same algebraic principle preserves the unit through realizations differing in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction appear jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an order of parametrization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of lifts

Ternary reduction expresses an integer APP evaluation through a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the notation for an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes the compatibility of this rule with sign inversion explicit.

Definition 2.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 belonging to the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the datum needed to reconstruct the preceding one. The quotient is not a perturbation of the remainder: it is its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values +1, -1, 0. The TPK selector defined below uses these classes to determine the operating mode. Reading an evaluation and selecting a mode thus have the same ternary reduction but different domains and functions.

Proposition 2.7 (Arithmetic of balanced pairs). *Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define*

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to pairs through

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product reconstruct as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also lies in the prolongation coordinate, not only in the remainder.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity ensures that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component suppresses this accounting: $1 + 1 = -1 + 3$, whereas the residue readout preserves only -1 . The removed information is determined in this case by $\chi(1, 1) = 1$.

The relation to modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images first fixes r by reduction modulo 3, and then $\bar{c}$. Addition transported through this bijection contains the term $\chi(r, s)$ from (2.7), now reduced modulo 3. The two ternary levels therefore do not compose by independent coordinatewise addition: normalization of the first level modifies the second.

In particular, the classes 3, 6, and 0 of $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving that coordinate prevents the zero-residue sector from being artificially reduced to a single position. Modulus 1000, used later to organize decimal blocks, has a different positional-representation function; its quotient is preserved by the corresponding division. A change of base preserves the integer when it retains both remainder and quotient, whereas agreement of remainders in different bases does not by itself identify their transport states.

2.2.3 Inversion, residue class, and nonvisible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum amplitude 1, these are the three possibilities in the visible fibre over zero; in a general integer lift, this fibre contains every pair $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, remainder, and quotient while differing in orientation, mode sequence, or winding. The decomposition $\mathcal{B}$ is therefore integrated into the transport state rather than replacing it. Arithmetic memory of a division and memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented inversion is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reversed order. In the adopted parametrization, it exchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while preserving the threshold 0. Transport of the remaining trajectory data is specified through the corresponding TPK involution.

This inversion of signatures must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed below, acts within the type already fixed. This distinction prevents algebraic equivalence between different regimes from being attributed to sign inversion.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (2.9) does not replace (2.8); it expresses a different operation in the specified quadratic codomain.

Definition 2.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm is nonzero, the stated formula provides the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm both vanish. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. The algebraic norm here is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

The three types admit faithful realizations through

$$aI + bJ_\tau \mapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. The complex structure, the dual-number structure, and the split real structure are thus represented by a common matrix expression. Each preserves its law and its zero divisors. The form of signature (1, 1) in the last case has a Lorentzian interpretation in dimension $1 + 1$; identifying a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of elements of norm one:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau) \exp(sJ_\tau) = \exp((t + s)J_\tau).$$

Comparing the two components yields the addition laws

$$C_\tau(t + s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.10)$$

$$S_\tau(t + s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.11)$$

Conjugation replaces t with $-t$ and agrees with inversion within this subgroup. It does not change τ : it reverses an evolution in the chosen regime.

In the elliptic type, the conserved form is $a^2 + b^2$, and the exponential trajectory follows the norm-one circle. In the hyperbolic type, the eigen-coordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not state that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

Evolution does not stop because $J_0^2 = 0$. Nilpotency eliminates terms of order at least two but preserves the linear term transporting the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is needed to describe the threshold without identifying it with absence of state or dynamics.

Proposition 2.10 (Composition of affine coordinates). *In the chart where the scalar component is nonzero, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Division by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio between components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a well-defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the complete algebra; a coordinate singularity does not automatically amount to a singularity of the object. This formula will reappear in the elliptic regional readout. Its use requires preserving the denominator and, where appropriate, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. The change of signature organizing a route therefore cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, the domains, and the effective operator acting at each transition.

Within the electron chapter, the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution of square I and two nilpotent directions. The joint realization will be proved on this specific pair of projections; the preceding classification explains why the three quadratic relations can occur in a single matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. This relation equips the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its connection to a real operator square root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of this same relation. Each preserves its characteristic, dimension, and incidence information. The development of the respective operators establishes the connection, rather than equality of their cardinalities.

Norm-one conservation therefore describes the quadratic realizations of evolutions. Conservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties participate in the construction with different functions: one determines the identities of the representations; the other permits composition, reconstruction, and prolongation of the operations from which they originate.

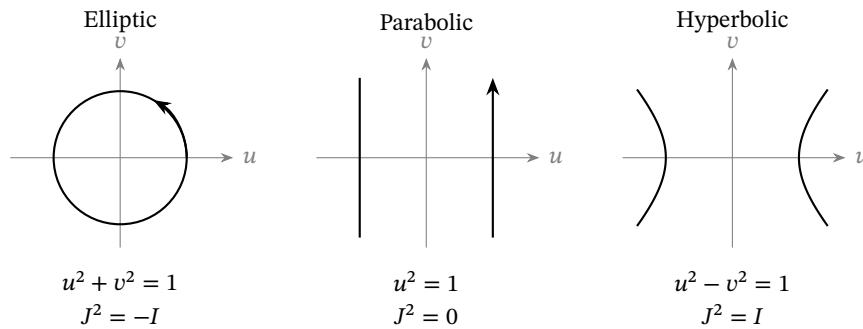


Figure 4: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. Parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: topological phase kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation operates. Cardinal transport moves the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore different functions: a ternary sign is not a displacement on the torus.

We denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of the two cursors, the active mode, the TRIT signature, the phase, the remainders and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object with a reduced version preserving only the visible symbol.

Definition 2.11 (TPK update). Let Sel_t be mode selection through M_{ph} , Tra_t lifted cardinal transport, and Mem_t record updating. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.12)$$

Each map preserves the fields it does not modify and updates the remaining fields through Euclidean division and trajectory concatenation.

The notation (2.12) expresses the order of composition; it does not replace the effective rules. The next section fully specifies its finite two-cursor realization. Subsequent prolongation will be obtained by transporting compatible prefixes and boundaries, rather than by erasing memory and repeating the first emission. In this way, APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamical state, observables, and dependence of operations

The topological phase kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations making prolongations compatible. Equation (2.12) acquires operational content when the coordinates received and modified by each of its factors are specified. Position belongs to the toroidal support; phase belongs to the schedule; orientation determines transport; quotients record the arithmetic evaluation before reduction. These coordinates are related through the transition and preserve different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.13)$$

The subscripts distinguish the additive and multiplicative cursors. The fibres of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their remainders and quotients, and the orientation and boundary records. Precision prolongation adds the prefix and compatible cylinder. Two states with the same projection q can therefore retain different records.

The dependence of operations has three levels. At the local level, the selector activates a sheet, transport modifies its position, and updating calculates the new arithmetic coordinates. At the trajectory level, these transitions compose and preserve the records of preceding steps. At the refinement level, extension relations determine which continuation preserves the prefix, signature, and boundary of its antecedent. The finite schedule describes the phase of these operations; holonomy describes their composition on the history.

2.3.2 Tritic selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the representative of the schedule. The operating phase and dodecaphasic sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.14)$$

The selector already defined can be written as $\varepsilon_9 = M_{\text{ph}}$. Its value vector, in the ordered chart of D_9 , is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_{\times}(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_{\times}(\theta)v_{d_\times} \quad \text{in } X_9. \quad (2.15)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event forms part of the chronology and its record. The cardinal directions determine the vectors v_d , whereas the phase signature determines which of these vectors is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses the positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records preceding crossings, the update is $z' = z + b_d(x, a)$. Thus,

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport datum remains reconstructible after position is reduced on the torus.

Proposition 2.12 (Conservation of lifted displacement). *For a cardinal trajectory of n steps with activation indicators a_t ,*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved when two compatible trajectories are concatenated.

Proof. The one-step identity is (2.16). Summing it over the n steps cancels the intermediate coordinates. Splitting the summation interval into two parts produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the winding number defined previously. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its record simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial crossing memory z_t and the arithmetic quotient q_t have different definitions; their joint preservation retains the geometric and arithmetic provenance of each transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j exchanges N with S and E with O . After (2.15) is applied, both directions are reversed at the end of a block of 27 steps. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)} d_+, x'_\times, j^{h(\theta)} d_\times, \theta + 1). \quad (2.17)$$

The readout forming the emission uses positions before transport, according to the explicit recurrence of Section 4. Observable updating and emission therefore compose in a fixed order.

Proposition 2.13 (Reversibility and period of the observable schedule). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. A block of 54 steps restores positions and directions; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, first recover the preceding phase by subtracting one in Θ_{108} . That phase determines whether the directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector recovers the initial state uniquely.

Each interval of 27 steps contains nine activations of each cursor. The following block preserves the activations and reverses the directions, so the integer displacements cancel over 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any origin of the schedule. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.13). Projection of the history onto the schedule permits this finite check. The trajectory record, its evaluations, and new prefixes are preserved in the dynamical state on which $\mathcal{U}_t$ acts; observable equality does not reinitialize them.

2.3.4 Affine memory transport and composition of states

The observable projection admits fibres of arithmetic and geometric data. Over a configuration q , write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , remainders and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q and S_q denote, respectively, the remainder–quotient data sets, the abelian group of transport memory, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified componentwise in the regional and terminal constructions. This dependence prevents the signature from being treated as a free parameter once the data producing it are fixed.

The elementary remainder–quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives of D_9 congruent modulo 9 are equal. This chart provides local integer reconstruction. The precision towers additionally incorporate truncation maps and the cylinders developed in 4 and 6.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fibre transport is expressed through maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

These maps respect identities and composition, and $C(\gamma)$ is a homomorphism of abelian groups. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.18}$$

The symbol k_{tr} here denotes the transport cocycle; the rational coordinate κ of record K retains its notation in the chapter on dodecaphasic closure.

The transportable coordinates update by

$$\begin{aligned} h' &= H(\gamma)h, \quad c' = C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, \quad \sigma' = \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.19}$$

The second equation is an affine action: the linear term transports the preceding memory and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$ and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.16). This concrete instance links the abstract law to the cardinal operations already defined.

Proposition 2.14 (Compatibility of updating with concatenation). *Updating according to (2.19) by γ_1 , followed by updating by γ_2 , agrees with updating by $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The assertion for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. The states, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves arithmetic transport and boundary prolongability simultaneously. This is the structure in which returns with memory and successive refinements take place.

2.3.5 Dodecaphasic record, phase, and accumulated memory

The 108 transitions are distributed among the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The record preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = \left((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m) \right)_{m=1}^{12},$$

where τ_m is the window subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence record. We adopt the chart developed in 9.2; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal records. Composition of segments and their increments is governed by (2.19). The group C_{12} , by

contrast, describes translation of the window index; Θ_{108} preserves the step schedule. This distinction specifies what transforms and what is observed upon completing a cycle.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, an advance of 9 steps preserves r and increases b by one. Phase addition introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite schedule preserves $[b]_{12}$; a revolution record preserves $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4 proves the nonsplit exact sequence (4.5). Its function is to describe the relation between these coordinates before they are used to construct the signed state.

The signed record receives phase and memory data through the composition developed in Section 9:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between this signed state and K , together with the rational readout of K , is proved in the corresponding chapter. Recording operations precede these transformations. Reconstruction of a record from its cyclic differences verifies its recoverability; its dynamical provenance must be preserved through the events that produced it.

For the coded trace (d_t) preserved by the record, the event reader counts the codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are distinct coordinates of the record. Chapter 9 develops this reader, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data this chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal construction of the channels

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases and let $O_{\text{ph}} = D_9 \setminus \{9\}$ be the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and the schedule origin already fixed; subsequent exceptional realizations will transport these incidences into their own domains.

Proposition 2.15 (Cardinalities and oriented phase incidence). *The sets in (2.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneous translation of the phase origin and the marked return preserves all these cardinalities and their inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Products by 6, 8, 9 give 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A schedule translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve incidence. $\square$

The cardinality 271 here has a realization through phase incidence. The cubic completion of Section 4 has its own construction of the shell 1000 – 729. The correspondence between these realizations must preserve the maps and marks identifying them, as well as cardinality. Similarly, the selector ε_9 , the incidence (90, 120), and the extractor $\varepsilon_{108}^{90,120}$ preserve the domains and operations just distinguished.

2.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The origin of the reversed trajectory is the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and the integer displacement changes sign. These identities follow by reversing step order and replacing each cardinal vector with its opposite. For a loop, this gives $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On records, reversal also requires transport of component order and event orientation. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete record. In particular, the regional involution exchanging the closure and propagation components preserves the self-scaling axis; its formula is presented in Section 6.16. This regional operation acts on a different domain from reversal of a cardinal trajectory.

Extensions over a route $r : q \rightarrow q'$ are expressed through relations between fibres of compatible states. If $\mathcal{F}_q$ denotes the fibre of complete states over q , require

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

A phase revolution composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \dots \circ \mathcal{U}_k. \quad (2.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the complete domain, we retain $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$, with the admissible prolongations developed below. The cyclic base has nine phases, whereas the fibre preserves trajectory, quotients, and boundary. Monodromy is the action of one revolution of this base on the fibres; holonomy $\Gamma_{9,k}$ expresses its realization through the effective TPK transitions.

If g observes phase and M the number of preserved revolutions,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure and the second identifies the new temporal depth. Coinductive prolongation additionally preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section brings together its antecedents without replacing them with the latest observation. This arithmetic of histories and its unit are developed in 4.8–4.10; the regions and their readers are constructed in Section 6.

Phase transfer on already constructed emissions has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the alphabets of additive and multiplicative signatures, of cardinalities 18 and 26. Their product identifies the catalogue $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 4. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for the remaining phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. History transport remains given by (2.22), with its memory and boundary.

The resulting hierarchy establishes the function of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphasic record determines the data entering K and the closure of α . Numerical realization and incidence realization receive this preceding structure. This order permits each consequence to be presented in its chapter while preserving, from the formal kernel, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

Calendar return does not remove increments produced during the journey. To express this distinction in operator form, an accumulated trajectory-event record is constructed first, followed by its generating series. Reduction of memory variables will be performed on this transport, also preserving the source and output reader.

2.4.1 Oriented events and record update

Let (d_t) be the sequence of integer direction codes retained by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification of their alphabets is assumed. We number the windows starting from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

During the first cycle, I_m is window E_{m+1} of the dodecaphase record. Its orientation is $\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1)$. For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign retained in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.23)$$

Each summand is incorporated when its transition occurs. This definition counts trace events; it does not select directions through a value that is to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$ and $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated record in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at a different origin, z_0 retains the preceding record. The frame accompanying the calendar is $y_m = S^{-m} z_m$.

Proposition 2.16 (Update and return with memory). *The record in the moving frame satisfies*

$$y_{m+1} = Ry_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.26)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1}y_m + a_m S^{-1} \mathbf{e}_0$, since $S^{-m} \mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (2.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j} \eta_j = a_j R^{12-j} \mathbf{e}_0 = a_j \mathbf{e}_j$, which proves (2.26). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^\top D_3 + D_4^\top D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^\top B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the record; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.17 (Event-induced balance). *Update (2.24) satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \quad (2.27)$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The coordinate sum is invariant under permutation R , giving the second equality. $\square$

2.4.2 Generating series and uniform memory control

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)). \quad (2.28)$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right). \quad (2.29)$$

The derivative series converges uniformly on every disk $|u| \leq r < 1$.

Proof. Multiplying (2.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} m r^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves its uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with another reader's variable requires preserving that clock and the map relating the two readings.

2.4.3 Memory elimination: operator, source, and reader

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.19 (Schur reduction with source and reader). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.30)$$

Its complete reading satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C) \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting it into the first gives (2.30); applying ℓ_0 and ℓ_1 to the two components gives (2.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2B_1D_uC = \sum_{k \geq 0} u^{k+2}B_1D^kC$ describes departure into memory, k steps within it, and a return. By contrast, $u\ell_1D_uC$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: the source and reader must also be transformed.

The distinction is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.32)$$

Indeed, $R^n\mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example concerns the twelve-window record, not the entire TPK state.

2.4.4 Segment composition and coherence of reduction

Let three composable affine transports be $F_i(y) = R_iy + b_i$, with compatible domains and codomains. The memory produced by the complete journey is

$$F_3F_2F_1(y) = R_3R_2R_1y + b_3 + R_3b_2 + R_3R_2b_1. \quad (2.33)$$

The proof is successive substitution of the three affine formulas. Grouping the first two segments first, or the last two, produces the same expression; the factors transport each increment from its point of origin. In (2.24), $R_i = R$ and b_i is the event of its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2Q_2C_2, & B'_1 &= B_1 + uB_2Q_2D_{21}, \\ C'_1 &= C_1 + uD_{12}Q_2C_2, & D'_{11} &= D_{11} + uD_{12}Q_2D_{21}. \end{aligned} \quad (2.34)$$

Proposition 2.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - uA' - u^2B'_1(I - uD'_{11})^{-1}C'_1 &= I - uA \\ &\quad - u^2(B_1 \ B_2)(I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (2.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting it into the first two produces (2.34). Eliminating the second component then gives the left-hand side of (2.35). Solving the two memory equations together gives the right-hand side, by Proposition 2.19. The formal solution is unique because $I - uT$ has constant term I . Both procedures also transform the source and reader according to (2.30)–(2.31). $\square$

Balance (2.27) describes record increments. The preceding identities allow them to be transported when memory is reused in dodecaphase closure; they do not by themselves determine additional analytic coefficients.

3 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

3.1 Phase selection, transport and enriched state

3.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (3.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t-1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (3.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b-1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

3.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i+1, j+1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (3.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

3.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (3.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

3.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (3.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (3.6)$$

Proposition 3.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = [\lfloor \nu/4 \rfloor]_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (3.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

3.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = \lfloor (n - 1)/9 \rfloor, \quad n = \rho_9(n) + 9q_9(n). \quad (3.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

3.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (3.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

3.3.2 Gauges of the multiplicative observable and generator normalization

The choice of discrete exponent admits a finite comparison in which the APP support, the initial conditions of both cursors, the tritic selector, the reversal calendar, and the emission rule remain fixed. The variation is confined to the observable on the multiplicative sheet. Let ℓ_2 and ℓ_5 be the logarithms of $(\mathbb{Z}/9\mathbb{Z})^\times$ with generators 2 and 5, respectively, extended by zero on $\{3, 6, 9\}$. They are compared with v_3 , evaluated on the positive representatives $1, \dots, 9$, and with the identity reading on those same representatives.

Proposition 3.2 (Normalization of the multiplicative logarithm). *The two isomorphisms from the unit group modulo nine to $\mathbb{Z}/6\mathbb{Z}$, normalized by sending a generator to 1, are ℓ_2 and ℓ_5 . They satisfy*

$$\ell_5(x) \equiv -\ell_2(x) \pmod{6} \quad (x \in (\mathbb{Z}/9\mathbb{Z})^\times).$$

Choosing the least positive generator fixes $\ell_2 = \ell_9$, with the values in (3.8).

Proof. The powers of 2 traverse 1, 2, 4, 8, 7, 5 and close at the sixth step. The generators of a cyclic group of order six correspond to exponents coprime to six: 1 and 5. Their representatives are 2 and $2^5 \equiv 5 \pmod{9}$. Since $5 \equiv 2^{-1} \pmod{9}$, expressing a unit as a power of 5 changes the sign of its exponent in base 2. Normalization by the least positive representative selects 2. $\square$

The homomorphism condition alone also admits maps with proper image. The proposition therefore distinguishes a faithful representation of the unit group from its quotients. Extension by zero also specifies the treatment of the three noninvertible residues; the provenance record retains which residue was observed.

The comparison of the four observables traverses the same 104 976 initial conditions and groups their emissions over six windows. Writing E_g for the emitter with multiplicative observable g , one obtains the following image census.

Both logarithms satisfy the additive exponent law. The valuation v_3 vanishes on every unit but takes the values 1, 1, 2 on 3, 6, 9; this accounts for the difference between its two control columns. The identity retains the positive residue and fails the homomorphism law into $\mathbb{Z}/6\mathbb{Z}$; for example, at the unit 1, the required equality would be $1 \equiv 2 \pmod{6}$. The census of 468 emissions is shared by

Observable	$ \text{im } E_g $	Zero roots	Additive law	Normalization
ℓ_2	468	yes	yes	least generator
ℓ_5	468	yes	yes	inverse generator
v_3	144	no	yes	zero unit image
id	684	no	no	positive residue

Table 1: Comparison of four gauges on the same transport and the same domain of initial conditions.

both logarithms and therefore retains the inversion ambiguity. The least-generator criterion resolves that ambiguity before any regional reading of constants. The cardinalities in the table describe the exhaustive evaluation of these four observables on the fixed domain; their role is to test the emitter's operational normalization.

3.3.3 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 3.3 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_9^2 is crossed.

3.4 Six windows and decimal block emission

3.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 3.4 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (3.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

3.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (3.9)$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \quad (3.10)$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 3.5 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (3.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

3.4.3 Decimal alphabet and check-digit congruence of the emitter

Retain the emission law (3.9), with the APP readings and observable ℓ_9 of (3.8), the cursors of Section 3.1, and the six-window calendar of Section 3.4. The accumulators are S_m, E_m, Z_m , with $Z_m = 3$, and the output digits are denoted $d_{m,1}, d_{m,2}, d_{m,3}$. Proposition 3.5 already recovers both residual signatures. Combining this recovery with the actual accumulator images determines the following alphabet.

Proposition 3.6 (Check-digit congruence for the third digit). *Every emitted block satisfies*

$$d_{m,3} = [5d_{m,2} - 2d_{m,1} + 1]_{10}.$$

Proof. The inverse in Proposition 3.5 gives $[E_m]_{10} = [d_{m,2} - d_{m,1}]_{10}$. Substituting $E_m \equiv d_{m,2} - d_{m,1} \pmod{10}$ and $Z_m = 3$ into the third digit gives $3d_{m,1} + 5(d_{m,2} - d_{m,1}) + 21 \equiv 5d_{m,2} - 2d_{m,1} + 1 \pmod{10}$. $\square$

Proposition 3.7 (Decimal alphabet of forty-two blocks). *The per-window image of the emitter consists exactly of the blocks*

$[E_m]_{10}$	U_m						
0	114	227	443	556	669	885	998
1	129	232	458	561	674	890	903
3	149	252	478	581	694	810	923
5	169	272	498	501	614	830	943
7	189	292	418	521	634	850	963
9	109	212	438	541	654	870	983

Proof. A cardinal trajectory reads three consecutive terms of the additive cycle

$$(2, 3, 4, 5, 6, 7, 8, 9, 1).$$

Their sums are (9, 12, 15, 18, 21, 24, 18, 12, 6); reversing direction preserves the set of sums. Hence

$$\text{Im } S_m = \{6, 9, 12, 15, 18, 21, 24\}, \quad \text{Im}[S_m]_{10} = \{1, 2, 4, 5, 6, 8, 9\}.$$

In the multiplicative channel, a fixed factor divisible by three contributes zero. For invertible factors, the cyclic sums of three readings are

fixed factor	possible sums
1, 8	{1, 3, 7, 9}
2, 7	{3, 5, 9}
4, 5	{1, 5, 7}

Thus $\text{Im } E_m = \{0, 1, 3, 5, 7, 9\}$. Attainability can already be verified in the first window: every cursor in the following two tables has direction E .

S_1	6	9	12	15	18	21	24
(i_+, j_+)	(0, 8)	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)

E_1	0	1	3	5	7	9
(i_x, j_x)	(2, 0)	(0, 0)	(0, 1)	(1, 1)	(0, 2)	(0, 4)

Independence of the two cursors realizes every one of the $7 \cdot 6$ pairs. The first two digits recover the residual pair, so their 42 images are distinct. The check-digit congruence fixes the third digit and yields exactly the stated table. $\square$

The decimal block therefore preserves both residual signatures through an injective encoding and a check digit. This result describes the finite emitter alphabet; continuation of the regional readings additionally retains the phase, orientation and memory conditions of the enriched state.

The per-window image therefore has 42 blocks. Compatible concatenation of six windows preserves the census of 468 emissions in Proposition 3.9, whose ternary reduction yields 243 words within the 729-word ambient space. Each cardinality counts the object of its stage and retains the provenance of the same emitter.

3.4.4 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (3.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 3.8 (Positional recovery of the emissions). *The encoding (3.11) determines the complete word by*

$$U_m = \left[\left\lfloor \frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right\rfloor \right]_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

3.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_\times = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 2: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

The resulting word and its signatures are

$$\begin{aligned} \mathbf{U} &= (501, 614, 498, 169, 272, 272), \\ \mathbf{s} &= (5, 6, 4, 1, 2, 2), \end{aligned} \quad \mathbf{e} = (5, 5, 5, 5, 5, 5).$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

Table 3: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

3.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (3.12)$$

where $G^{(6)}$ applies (3.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

3.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 4: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 5: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 3.9 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 3.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 3.5 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (3.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 3.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

3.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (3.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$$\begin{array}{c|cccccc} |\Psi^{-1}(w)| & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \text{number of words } w & 132 & 66 & 18 & 3 & 6 & 18. \end{array} \quad (3.14)$$

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 3.10 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (3.13).

3.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (3.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (3.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 3.11 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (3.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 3.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 3.12 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 6; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 6: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

3.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $v = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (3.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (3.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (3.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 6.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (3.12); its scope is the same finite domain.

3.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (3.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

3.11 Cyclic correlation and preservation of path order

The record (3.17) preserves the ordered concatenation of samples. The following example isolates order information extracted by two distinct readers from a twelve-position word, retaining the domain of each reading.

Let $\tau = (\tau_1, \dots, \tau_{12}) \in \{-1, 0, 1\}^{12}$ be an ordered word, with indices taken modulo 12. Define

$$\begin{aligned} G_a &= \{a, a + 4, a + 8\} \quad (1 \leq a \leq 4), \\ v_{120}(\tau) &= \sum_{a=1}^4 \text{sgn} \left(\sum_{m \in G_a} \tau_m \right), \\ v_{270}(\tau) &= \sum_{m=1}^{12} \tau_m \tau_{m+3}. \end{aligned}$$

The marginal histogram is $p_j(\tau) = \#\{m : \tau_m = j\}/12$ for $j \in \{-1, 0, 1\}$, and its entropy is $H(\tau) = -\sum_j p_j(\tau) \log p_j(\tau)$, with $0 \log 0 = 0$.

Proposition 3.13 (Order information beyond the marginal distribution). *There exist words with identical histograms and distinct correlations ν_{270} , even when ν_{120} has the same value.*

Proof. Consider

$$\tau^{(a)} = (1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0), \quad \tau^{(b)} = (1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0).$$

Both have $p_1 = 1/6$, $p_0 = 5/6$ and $p_{-1} = 0$, so their marginal entropies coincide. In each case, two of the sets G_a contain a unit and the other two contain only zeros: $\nu_{120} = 2$. In $\tau^{(a)}$, no pair of units has cyclic separation three, whereas in $\tau^{(b)}$ precisely the term $m = 1$ contributes. Thus $\nu_{270}(\tau^{(a)}) = 0$ and $\nu_{270}(\tau^{(b)}) = 1$. $\square$

This separation identifies the information retained by the ordered record: incidences between positions in addition to symbol frequencies. A reduced state retaining only the histogram assigns the same output to both words and loses this correlation. Marginal entropy, uncertainty between bases and entanglement entropy are defined on different objects; relating them requires retaining the distribution, the change of basis, and the state together with its partition, respectively.

Proposition 3.14 (Parity of the readers under oriented reversal). *With cyclic indices in $\mathbb{Z}/12\mathbb{Z}$, let $(C_M\tau)_m = -\tau_{-m}$ and $\text{sgn}(0) = 0$. The two readers satisfy*

$$\nu_{120}(C_M\tau) = -\nu_{120}(\tau), \quad \nu_{270}(C_M\tau) = \nu_{270}(\tau).$$

Proof. The reflection $m \mapsto -m$ permutes the four position classes modulo four. Reversal changes the sign of each class sum and hence, by the oddness of sgn , the sign of ν_{120} . In the second reader, the signs of the two factors cancel:

$$\sum_m (C_M\tau)_m (C_M\tau)_{m+3} = \sum_m \tau_{-m} \tau_{-m-3} = \sum_k \tau_{k+3} \tau_k = \nu_{270}(\tau).$$

$\square$

Orientation thus distinguishes an odd reading from an even correlation. Both depend on the arrangement of samples in the record; their transformation laws specify which information persists when the route is reversed.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

4 Development of the formal kernel: initial conditions, geometry, and refinement

The eighty-one-position support does not coincide with the domain of initial conditions for the dynamics. An initial condition must also specify an orientation, and the additive and multiplicative sheets have independent cursors. This distinction allows exhaustive reconstruction of the emission without selecting a numerical region in advance.

4.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (4.1)$$

The census of residual-emitter signatures in Section 80 enumerates these conditions by identifiers and presents their fibers; the law transforming them is defined here. The rows of an emission table are fibers of this initial-condition domain.

The finite realization uses six windows of nine transitions. At each instant t , the phase signature is $M_{\text{ph}}(1 + ((t - 1) \bmod 9))$. If its value is $+1$, the positive residue $\Sigma(i + 1, j + 1) = \rho_9(i + j + 2)$ of the first cursor is read and that cursor is then moved; if its value is -1 , the positive residue $\Pi(i + 1, j + 1) = \rho_9((i + 1)(j + 1))$ of the second cursor is read and the second cursor is then moved. Zero increments the neutral counter. At the end of transitions twenty-seven and fifty-four, cardinal orientations are reversed. Integer evaluations and their quotients are retained in the record, but the emitter uses the residue readings specified here.

In the unit sector, the multiplicative reader uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For finite emission it is extended by zero to the non-unit sector. This extension is an observable, not an identification of the elements 3, 6, 9 with the multiplicative identity. The sheet and its membership in the radical sector remain in the trajectory record.

Let S_m be the sum of the three positive additive residues Σ in window m , E_m the sum of the three discrete exponents of the multiplicative residues Π , and $Z_m = 3$ the number of neutral events. Thus S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the record. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (4.2)$$

The integer coefficients and calendar of this recurrence are explicit parts of the declared emission law. Values of π, φ, e or α are not received. Distinguishing an emission rule from a constant recognized from its output prevents the genealogy from being hidden in a decimal table.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule is written

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (4.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalog adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (4.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transport of subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with TRIT is performed after (4.4).

Proposition 4.1 (Injective factorization of emission). *The word $U = (U_1, \dots, U_6)$ determines both signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient set of 729 words.*

Proof. The hundreds digit of U_m recovers s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence of (4.3). Hence coupling is injective on the signature product.

Enumerating the 324 conditions for one cursor through the preceding recurrence produces eighteen additive signatures, each with eighteen preimages, and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. Section 80 collects the signatures, their multiplicities, and the exact rule recovering their initial conditions. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\begin{array}{ccc} 432 & 18 & 18 \\ \text{fibers of } 144, & \text{fibers of } 432, & \text{fibers of } 1944. \\ 18 \cdot 24 & & \end{array}$$

The sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (4.4) to the 468 emissions produces exactly 243 words. This last number is the cardinality of the image, not of the space $\mathbb{F}_3^6$. $\square$

This factorization explains the difference between finite exhaustiveness and unlimited depth. Domain (4.1) is complete for the declared support and emitter; it does not contain every prefix of every prolongation in advance. Depth belongs to the system of compatible extensions constructed over those conditions.

4.2 Calendar closure and memory advance

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (4.5)$$

If $t = 9b + r$ with $0 \leq r < 9$, adding two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 4.2 (Nonsplitting of the calendar). *Sequence (4.5) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{lcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring a prolongation prefix. Nonadic holonomy Γ_9 denotes transport through one turn on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (4.6)$$

Phase equality and memory increment are compatible and constitute two observations of the same transport.

4.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by TPK operations on $\hat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \hat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t . The compatibility relation precedes the choice of parameter: the order allows its transitions to be expressed as an evolution. When the term section is used, the projection $p : E \rightarrow \mathcal{T}$ and condition $p \circ \gamma = \text{id}_{\mathcal{T}}$ are also declared. The state space, temporal order, and transported information are thereby specified.

The signature τ selects a quadratic regime and admits a temporal reading relative to this parametrization. Type +1, with $J_{+1}^2 = -I$, represents return and memory; type 0, with $J_0^2 = 0$, represents the transition threshold; type -1, with $J_{-1}^2 = I$, represents opening in two eigendirections. In the model's temporal terminology, these functions correspond, respectively, to past, present, and future. The correspondence concerns the operational position of a transition within a history: retained antecedent, update, and admissible prolongation.

This interpretation acquires content through three distinct operations. Restriction of a trajectory recovers the prefix already constructed. The update $\mathcal{U}_t$ composes selection, transport, and recording in the observed transition. Prolongation considers continuations that preserve the boundary conditions of that prefix. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x , deleting the last step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unlimited continuation is a collection $(\gamma_n)_{n \geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the restrictions each depth imposes on preceding depths. Existence of a compatible collection is established in the corresponding prolongation domain; merely specifying the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the signature sequence, the involution C_{rev} reverses order and changes the sign of each component. The trajectory is also subject to the transformations of position, sheet, and boundary specified by transport. The reversed signature is an observation of the conjugate trajectory, not a replacement for its other coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical record is nonzero.

The relation with exponential geometries is equally precise. In each fixed regime, $\exp(tJ_\tau)$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to that local representation. Phase return of Γ_0 , by contrast, acts on the trajectory with memory, according to (4.6). A trajectory can recover its phase while retaining a new antecedent: historical data have changed although the phase observation is the same. The parabolic threshold retains a linear displacement; hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, whereas the TPK determines their selection and effective concatenation.

The aperiodic suspension in Section 6 will make this distinction explicit through a finite phase factor, an irrational rotation, and a memory degree. The link between TRIT and temporality is thus expressed by operations on trajectories and their quadratic representations. Its content exceeds a classification of signs: it preserves the regime, the direction of transport, and the relation between antecedent, transition, and continuation.

The monodromic representation of this return is developed in Section 6.13. There, the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension in Section 6.15 defines the aperiodic time-crystal model, and Section 6.17 composes this temporality with the three Euler realizations.

4.3 Observable compression and orthogonal conservation

An isometric realization provides a quantitative expression of what a reduced observation loses. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of transport through one turn, $U^*U = I$, and $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (4.7)$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 4.3 (Finite-horizon conservation). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r}D^*DT^r + T^{*n}T^n. \quad (4.8)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r , then summing, produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n}T^n$. $\square$

The final summand preserves the terminal state. Removing it before justifying a limit would alter the equality. This distinction is the operational content of conservation of unity under compression: the norm is distributed among the visible component, memory, and terminal, rather than being replaced by a single residue.

4.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonals modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$, one obtains $B_c = 9P_+ + 5P_-$. The unity of stochastic normalization and conservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \quad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2} \log \frac{9}{5}.$$

For $Q = \text{diag}(1, -1)$, the second matrix satisfies $B_c^T Q B_c = 45Q$. This yields a local realization in $SO^+(1, 1)$; no prior introduction of four-dimensional spacetime is required to prove it.

Uniform averaging over the classes $C_1 = \{1, 4, 7\}$, $C_2 = \{2, 5, 8\}$, and $C_0 = \{3, 6, 9\}$ gives

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_{Π} is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

so its quadratic form has signature $(2, 1)$. The local spectral gap $9 - 5 = 4$, the coefficient 2, the aggregate difference $51 - 45 = 6$, and the integrated defect $9 \cdot 6 = 54$ describe different levels of the same arithmetic comparison. They are not interchangeable indices of a single operator.

Discrete curvature can be described directly before this spectral realization. In residues, let $S(x, y) = x + y$, $P(x, y) = xy$, and let Δ_x, Δ_y be forward differences. Link variables are differences of potentials, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and we set

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}. \quad (4.9)$$

Sum and product, taken separately, induce flat connections; their ordered mixture changes the sign of plaquette curvature. Summing over a rectangle of sides a, b cancels interior edges, and the oriented boundary sum equals $\pm ab$ modulo nine. An exact relation thus appears between composition order and boundary orientation.

4.5 Cubic completion and two notions of boundary

Expanding the cubic support of side nine to that of side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (4.10)$$

The first term corresponds to three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The expansion corona contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (4.10) by 1000 gives a normalized partition of unity. Cube incidence distinguishes six faces, twelve edges, and eight vertices. The subsequent connection to Steiner hexads and octads requires the corresponding combinatorial maps; cube cardinalities do not replace them. Here this completion supplies the relation between bases 729 and 1000, while exceptional prolongation will be developed after numerical generation, the dodecaphasic record, and the constructions of α .

4.5.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description making its operation explicit. Let E_9 be the set of nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, extending E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without changing the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 4.4 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 4.5 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (4.11)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (4.12)$$

The punctured expansion contains 270 elements in addition to the initial cube; the complete expansion contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning one of the nine classes of E_9 to each remaining coordinate. Classification by the number of additional coordinates is exhaustive and its classes are disjoint. The sole element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH in the corpus, here denotes this articulation of the initial body, expansion strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure through restoration of the apex. The added unit therefore has an explicit incidence location.

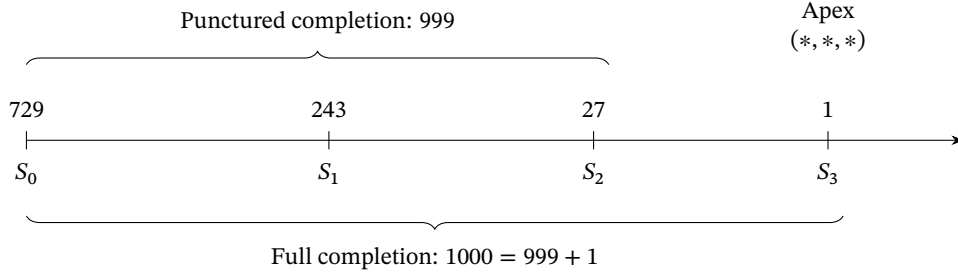


Figure 5: Strata of the cubic completion. The arrangement represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

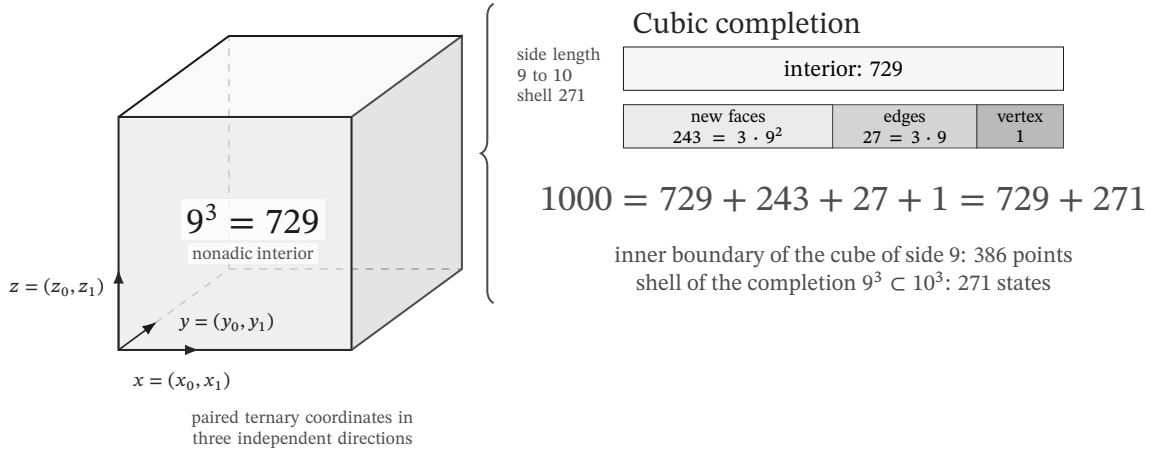


Figure 6: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 5; graphical spacing is schematic.

4.5.2 Conservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\hat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \hat{E}^{d+1} \rightarrow \hat{E}^d$ deletes the last coordinate.

Proposition 4.6 (Projective compatibility of normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\hat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r} / 10^d$, and the masses of all strata sum to one.

Proof. Each point of $\hat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (4.11); the binomial $(9 + 1)^d$ gives total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define a different measure: the distinction is operationally relevant. In particular, conservation of probability mass, conservation of transport records, and thermodynamic dissipation concern different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamic and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\hat{C} \setminus C$ belongs to its expansion. Cubic completion, base-thousand positional expansion, and exceptional-incidence reading share support data; their respective maps determine which information from these data each representation uses.

4.6 Cubic incidence and Lorentzian realization

Cubic completion supplies a relation between interior and boundary. Its geometry can be developed through incidence of the six faces with the eight vertices. Label faces by (i, ε) , $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and vertices by $v \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),v} = \mathbf{1}_{\{v_i=\varepsilon\}}.$$

Let $u = (1, \dots, 1)^\top \in \mathbb{R}^6$, let R_F denote face opposition, and let $P_u = uu^\top/6$, $P_- = (I - R_F)/2$.

Proposition 4.7 (Four-dimensional incidence quotient). *The incidence Gram matrix satisfies*

$$MM^\top = 12P_u + 4P_-.$$

Its eigenvalues 12, 4, 0 have multiplicities 1, 3, 2. Consequently,

$$Q_\square = \mathbb{R}^6 / \ker(MM^\top) \simeq \mathbb{R}u \oplus E_-, \quad E_- = \text{im } P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct non-opposite faces share two vertices. The matrix $12P_u + 4P_-$ has exactly these entries. The uniform subspace has dimension one; differences between opposite faces generate a three-dimensional space orthogonal to it. Its complement has dimension two and belongs to the kernel. $\square$

The decomposition $1 + 3$ comes from incidence. The temporal choice is declared through the bilinear form

$$B_\square([x], [y]) = \langle x, (P_- - P_u)y \rangle.$$

It is well defined on the quotient because both projectors annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \quad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\text{diag}(-1, 1, 1, 1)$. The domain, decomposition, and sign convention producing the Minkowski realization are thus explicit. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_\square = \sqrt{6}P_u + \sqrt{2}P_-, \quad W_\square e_{i,\varepsilon} = t + \varepsilon d_i$$

sends the six face labels to null directions, since $B_\square(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Direction reversal and face opposition thus acquire a concrete geometric realization.

4.6.1 Soldering, reversal, and subdivision

The cellular realization uses an invertible transport $U_m(a)$, preserving the Lorentzian form, between the endpoint fibers of an oriented dual edge a , a face label $\lambda_m(a)$, and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the soldering map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a) \beta_m(a).$$

Identify the source fibers at two consecutive levels through the declared refinement isometry. If the nine subsegments $a_1, \dots, a_9$, at level $m+1$, preserve label and orientation when transported to that common fiber, then

$$\beta_m(a) = \sum_{j=1}^9 U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \quad (4.13)$$

Each transported term equals $\ell_m/9$ times the same null direction, proving the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation brings together direction, scale, and transport, rather than limiting geometric correspondence to a dimension count.

With orientation and the Lorentzian form also fixed, Hodge duality on $\Lambda^2 Q_{\square}^*$ satisfies $\star^2 = -I$. In dimension four, the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, the eigenspaces with eigenvalues $+i$ and $-i$. The electromagnetic realization will use this domain after constructing the electronic section and declaring its coupling.

4.7 Compatibility of modular and positional reductions

The block-decimal base enters through Euclidean division preserving residue and quotient simultaneously:

$$n = 1000q + r, \quad 0 \leq r < 1000.$$

Since $1000 \equiv 1 \pmod{9}$ and $1000 \equiv 1 \pmod{3}$,

$$n \equiv q + r \pmod{9}, \quad n \equiv q + r \pmod{3}.$$

Reduction of a block therefore requires the quotient's contribution to recover the integer's class. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and 3. This is an operational reason for preserving memory through a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated through a single integer remainder. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D 3^L.$$

On adding six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \leq \nu < 729$. If the decimal prefix remains fixed, one obtains

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \quad (4.14)$$

Both identities follow by substitution of the definitions. Comparing endpoints is thus reduced to integer operations retaining each prolongation's contribution. This compatibility makes it possible to refine a region before evaluating its real coordinate.

4.8 Number as a compatible section and value as a reading

Distinguishing state from observation specifies what object is prolonged as resolution increases. At depth n , let X_n be the set of admissible states produced by APP-TRIT-TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. These are not independent coordinates: earlier arithmetic evaluations and transports impose their relations. A prolongation preserves these relations in each antecedent, even while adding new steps and memory data. This is the meaning of the arithmetic of histories developed in (Sections 4 and 6.13).

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined in it.

Definition 4.8 (Compatible numerical section). An HMT number is a compatible history of the state system:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (4.15)$$

A reader is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the reader. The pair consisting of a section and its reader constitutes a read presentation, not a redefinition of the section's identity.

On the Archimedean domain X_∞^{Arch} , the reader associates a closed cylinder with each prefix, preserves nesting, and makes its diameter tend to zero. It thus determines

$$\text{ev}_\mathbb{R} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_\mathbb{R}(x)\} = \bigcap_n I_n(x). \quad (4.16)$$

The concrete construction of these cylinders and the determination of their boundaries are developed in Section 6. Definition (4.15) does not by itself assert that the image of (4.16) is the entire line: a coverage claim requires its own theorem. Nor does equality of values establish equality of histories without an injectivity result. This separation allows precision to be discussed as depth while preserving the distinction between the arithmetic antecedent and its scalar realization.

4.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are its admissible histories. Composition is written $vu = v \circ u$: first traverse u , then v . Elementary steps come from TPK selection, transport, and updating. A local tritic fiber admits the notation $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the transported state; a transition between regimes does not amount to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of this composition. For each object x , a unital associative real algebra A_x , possibly noncommutative, is given. For each $u : x \rightarrow y$, a unital homomorphism $\rho_u : A_x \rightarrow A_y$ is given. Assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (4.17)$$

For each composable pair $u : x \rightarrow y, v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. Require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for every composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (4.18)$$

These are explicit hypotheses on transports and their coefficients, not a consequence of calling the dynamics holonomic. Applying this presentation to a TPK sector requires identifying its ρ_u and $\Omega(v, u)$. Transitions expressed through bimodules require preserving that structure and are not replaced by (4.17).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{I(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (4.19)$$

when the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. Addition gathers histories with coefficients; multiplication concatenates histories while preserving the record multiplier.

Proposition 4.9 (Sectorial associativity with a central multiplier). *Under (4.17) and (4.18), product (4.19) is associative. If the object set is finite, its identity is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to move to the right of that factor; (4.18) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero because of their initial and terminal states. Bilinearity completes the proof. Local identities and normalization of Ω verify the stated identity element. $\square$

Calendar carry provides an effective example. In the category with one object and arrows C_9 , take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a + b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

The two carry sums of a triple are $\lfloor (a + b + c)/9 \rfloor$; hence (4.18) holds. The formal variable z records the turn without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphase record of calendar (4.5); retaining z without that relation distinguishes successive turns. The example realizes this calendar's memory without identifying it with all TPK memory.

4.10 Double variance of refinement and conservation of unity

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (4.20)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ gathers finite-depth observations. This algebra is not identified here with the whole route algebra: prolonging transport operators also requires their compatibility with refinement.

Lemma 4.10 (Identity and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (4.21)$$

and, for every $f \in D_n, y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m . Their indicators are orthogonal and sum to 1_m , proving (4.21). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. $\square$

The global identity is the constant function one on all possibilities; an individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement may increase the number of extensions and the depth of each history while preserving the identity and all evaluations of its antecedents.

4.11 Phase return, Weyl representation, and two readings

Law (4.6) acquires an operator realization in the bilateral suspension constructed in Section 6. There, T is one-step advance, V_9 observes the nonadic phase, W_δ the rotation phase, and D_M the memory degree. Writing $\hat{\Gamma}_9 = T^9$ for the turn in this representation, the Weyl relations (6.35) imply, on finitely supported vectors,

$$\begin{aligned} V_9 \hat{\Gamma}_9 &= \hat{\Gamma}_9 V_9, & W_\delta \hat{\Gamma}_9 &= e^{2\pi i 9\delta} \hat{\Gamma}_9 W_\delta, \\ [D_M, \hat{\Gamma}_9] &= \hat{\Gamma}_9. \end{aligned} \tag{4.22}$$

The first equality follows from $\zeta_9^9 = 1$; the second from nine applications of the covariance of W_δ ; the third expresses the unit memory increment. The finite phase returns while irrational rotation and memory distinguish the next state. Characters and parameter δ are realized subsequently from the numerical construction specified there: they do not select the seeds retrospectively. This representation describes suspension observables and translations; it does not replace all TPK coordinates or operators.

The double reading extends the same distinction between history and observable. On its Archimedean domain, (4.16) applies; on the calibrated incidence domain, the maps Q_n of Section 16.1 preserve words and boundary frames. The equality proved there, $r_{n,m} Q_n = Q_m p_{n,m}$, determines their prolongation Q_∞ . On the common domain of both readings, $\text{ev}_\mathbb{R}(x)$ and $Q_\infty(x)$ can be considered jointly: the former retains the value; the latter, the transported incidence preserved by its codomain. The section precedes both. This articulation makes it possible to pass from the arithmetic of histories to constants and exceptional reading without turning their realizations into new initial data.

5 Correlative construction of the continuum: histories, transport, incidence and terminal

The arithmetic expressed by a normal form preserves a precise readout of the state producing it. To situate that readout within HMT, the histories, sheets and operations acting on them must also be maintained. This chapter brings together that common substrate. Its different constructions are obtained from the same APP-TRIT-TPK execution tree; expository divisions allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented earlier. An admissible emission contains the APP operation, TRIT orientation, sheet change, residue and quotient, together with the effective TPK memory update. When only a residue is expressed, the execution history still belongs to the starting object. Two executions with identical final readouts are not identified on that ground alone.

5.1 Admissible histories and memory by composition

Let X_k be the set of admissible histories of length k . Its elements are constructed by successive emissions from the kernel's initial domain. Write $h\varepsilon$ for prolongation of h by an admissible emission ε , and $\rho_k(h\varepsilon) = h$ for deletion of the last emission. The emission label is preserved, even when different labels produce the same expressed state.

On the two arithmetic sheets, local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \quad 0 \leq r_\diamond < 9, \quad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate likewise admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preservation of the quotient allows these representations to be composed without replacing an emission by its isolated residue.

In a memory chart, an emission's update is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a trajectory $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of these updates. Its linear part and translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \quad \kappa_\alpha = \sum_{j=1}^k C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 5.1 (Affine compatibility of concatenation). *If α precedes β , then*

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β produces $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the trajectory. $\square$

This cocycle identity is the precise content of accumulated memory. The quantity transported by a second trajectory depends on the chart update it performs; it is not, in general, the untransported sum of two registers.

Nonadic return preserves a second distinction. In phase and winding coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its representation is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

During nine steps, exactly one crossing from 8 to 0 occurs. Phase returns and the winding coordinate increases. Enriched holonomy also retains the memory updates accompanying this route.

5.2 Tree refinement and measure of histories

Fix the finite nonempty initial set of admissible seeds, with strictly positive initial masses summing to 1. In this subsection each history has a finite positive number of prolongations. Deletion of emissions defines the refinement maps. Kernel admissibility supplies the prolongations; when a residence uses a neutral emission, that emission explicitly gives a section of ρ_k . Existence of that section concerns the tree including this emission, not every arbitrary restriction of routes.

Let $d(h) \geq 1$ be the number of admissible emissions at h . Equiprobable choice among its successors defines

$$\mu(h\varepsilon) = \frac{\mu(h)}{d(h)}.$$

More generally, a positive local distribution $p(\varepsilon | h)$, summing to 1, defines $\mu(h\varepsilon) = \mu(h)p(\varepsilon | h)$. This second formulation preserves the readout's weights when numbers of prolongations are not uniform.

Proposition 5.2 (Cylindrical compatibility and conditional expectation). *At each level one has*

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

On the spaces $L^2(X_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \quad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^*$, $E_k J_k = I$ and $\|J_k f\| = \|f\|$.

Proof. The first identity is normalization of local probabilities. Applying it to $|f(x)|^2$ gives the isometry. Grouping by predecessors the sum defining $\langle J_k f, g \rangle$ yields the formula for E_k ; on a function constant on each fibre, that formula returns its value. $\square$

Compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(h)$. The measure extends from the cylinder algebra, and uniqueness follows because the cylinders generate the sigma-algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under emission deletion is nonempty. In the cylindrical topology, every nonempty cylinder has positive measure.

This measure distinguishes histories converging to the same representation. The uniform measure on terminal values would be a different readout, since it would remove the multiplicities just preserved by the construction.

5.3 Sheetwise transport and exact accounting of outputs

First consider the space whose orthonormal basis consists of histories at level k . Transport preserving the label of each prolongation is

$$\mathcal{U}_k e_h = \sum_{\varepsilon \text{ admissible at } h} \sqrt{p(\varepsilon | h)} e_{h\varepsilon}.$$

The successor sets of distinct predecessors are disjoint. Normalization of p therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation using history masses is related to this one by the diagonal basis change incorporating $\sqrt{\mu(h)}$.

The sheet label belongs to a history's complete register. Write

$$\hat{h} = ((h_0, s_0), \dots, (h_k, s_k)), \quad s_j \in \{\Sigma, \Pi\},$$

where s_j is the active sheet at step j . Prolongation adds the emission and sheet s_{k+1} determined by the TPK operational selector, preserving the entire prefix, including the initial sheet. In the basis of these lifted histories,

$$\mathcal{U}_k e_{\hat{h}} = \sum_{\varepsilon \text{ admissible at } \hat{h}} \sqrt{p(\varepsilon | \hat{h})} e_{\hat{h}\varepsilon}, \quad P_k e_{\hat{h}} = \mathbf{1}_{s_k=\Sigma} e_{\hat{h}}, \quad Q_k = I - P_k.$$

Thus, P_k reads the current sheet, while the distinction between histories is maintained by their complete prefixes. Two initial histories reaching the same active sheet still have distinct successors. The isometry proof using disjoint collections of prolongations applies verbatim to this basis. Trivial orientation is another register coordinate: the value 0 preserves the threshold readout on each sheet and does not add a third sheet to this decomposition. We now separate sheet preservation from sheet interchange:

$$A_k = P_{k+1} \mathcal{U}_k P_k + Q_{k+1} \mathcal{U}_k Q_k, \quad \eta_k = Q_{k+1} \mathcal{U}_k P_k + P_{k+1} \mathcal{U}_k Q_k.$$

Proposition 5.3 (Sheetwise Gram identity). *For a linear transport U , whether or not it is isometric, the preceding decomposition satisfies*

$$A^* A + \eta^* \eta = P U^* U P + Q U^* U Q.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^* A_k + \eta_k^* \eta_k = I.$$

Proof. In $A^* A$ and $\eta^* \eta$, mixed terms containing $P_{k+1} Q_{k+1}$ vanish. Summing the remaining terms yields $P_k U^* (P_{k+1} + Q_{k+1}) U P_k$ and its analogue with Q_k . The first identity follows from complementarity of the projectors; the second additionally uses $U^* U = I$. $\square$

The first identity is the general formula. Replacing its right-hand side by $U^* U$ requires the cross-blocks of that Gram operator to vanish. Isometric preservation by history transport provides this condition here; it must not be attributed automatically to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$ and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 5.4 (Finite decomposition with terminal residual). *For each horizon N ,*

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing and has a positive strong limit T_∞ . The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 5.3 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. Telescoping summation proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a bounded positive form and, by the representation theorem for bounded forms, an operator T_∞ . Convergence is strong because, for a positive operator $S \leq I$, $\|Sv\|^2 \leq \langle v, Sv \rangle$; apply this to $S = T_N - T_\infty$. $\square$

The terminal residual is preserved until its extinction is proved in the domain considered. This distinction prevents phase return alone from becoming a hypothesis that all histories are exhausted.

5.4 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and the counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)(N_\Sigma(h) - N_\Pi(h)), \quad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a readout expresses the average over prolongations, define $b_c = Eb_f$. Orientation remains in the argument of b_f ; it is not removed before averaging.

Proposition 5.5 (Exact cancellation defect). *The quantity*

$$\kappa_{\text{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\text{can}}, \quad E(b_f^-) = (b_c)^- + \kappa_{\text{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \leq E|b_f|$. Writing the positive and negative parts as $(|b| \pm b)/2$ yields both equalities. $\square$

The information omitted when only the balance is expressed is the common cancellation of opposite contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fibre contains nine equally weighted successors; conditional expectation is the rule that remains valid for general branching and weights.

5.5 Ternary sheet incidence and rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They yield six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus the counts 4, 6, 8 belong to distinct operations on a single sheet module.

On the six pair positions, define

$$(Ba)_{\{L, L'\}} = a_L + a_{L'}, \quad s(q) = \sum_{\{L, L'\}} q_{\{L, L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The readout $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{2}{9}, \quad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fibre contains 3^5 elements. The readout equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These averages describe the uniform ambient space; the distribution induced by a subset of histories must be computed with its own multiplicities, already preserved in X_k .

To fix the ports without losing multiplicities, at depth k retain the labelled readouts $\lambda_\Sigma(h) = (h, \Sigma)$ and $\lambda_\Pi(h) = (h, \Pi)$. Their sets are

$$I_k = \{\lambda_\Sigma(h) : h \in X_k\}, \quad J_k = \{\lambda_\Pi(h) : h \in X_k\}.$$

They are finite and nonempty because X_k is. Values, quotients and residues are expressed on these ports; equality of values does not identify their labels. The product $I_k \times J_k$ is the rectangular-readout domain, within which the support of admissible incidences is recorded.

With this level fixed, write $I = I_k, J = J_k$ and $A_N = \mathbb{Z}/3^N\mathbb{Z}$. Choose reference ports i_0, j_0 . This choice merely provides coordinates for reconstruction formulas. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \quad (\delta w)_{ij} = w_{ij} - w_{i j_0} - w_{i_0 j} + w_{i_0 j_0}$$

define the following sequence, valid over the finite ring A_N :

$$0 \longrightarrow A_N \longrightarrow A_N^I \oplus A_N^J \xrightarrow{\kappa} A_N^{I \times J} \xrightarrow{\delta} A_N^{(I \setminus \{i_0\}) \times (J \setminus \{j_0\})} \longrightarrow 0.$$

Proposition 5.6 (Exactness of rectangular incidence). *The preceding sequence is exact, with first arrow given by the constant diagonal.*

Proof. The equality $u_i - v_j = 0$ for every pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{i j_0}$ and $v_j = w_{i_0 j_0} - w_{i_0 j}$. Then $u_i - v_j = w_{ij}$. Finally, any matrix in the last term extends with zero reference row and column; its image under δ is the initial matrix. The proof uses only addition and subtraction, so it requires no division by a ring unit. $\square$

Reduction from A_{N+1} to A_N , with k, I_k, J_k, i_0, j_0 fixed, commutes with both maps. Reconstruction formulas are the same at each modular depth. This is the arithmetic compatibility proved in this subsection. Prolongation of histories $k \rightarrow k+1$ preserves prefix labels; its maps between port sets are distinct data from the ring change $N \rightarrow N+1$.

These maps are obtained from emission deletion:

$$\rho_k^\Sigma(h\varepsilon, \Sigma) = (h, \Sigma), \quad \rho_k^\Pi(h\varepsilon, \Pi) = (h, \Pi).$$

Fix a reference history with compatible prolongations, whose existence was proved for the tree without terminal leaves. Its two labels define compatible reference ports at every level. Coordinate transport is precomposition:

$$(L_k u)_{i'} = u_{\rho_k^\Sigma(i')}, \quad (L_k v)_{j'} = v_{\rho_k^\Pi(j')}, \quad (L_k w)_{i'j'} = w_{\rho_k^\Sigma(i'), \rho_k^\Pi(j')}.$$

For the sequence's last module, first extend the matrix by zero in the reference row and column, apply this same formula and restrict to the new complement of the reference ports. This rule defines L_k even when a new port has the marked port as its predecessor.

Proposition 5.7 (Naturality of the rectangular complex under prolongation). *The preceding maps satisfy $L_k \kappa_k = \kappa_{k+1} L_k$ and $L_k \delta_k = \delta_{k+1} L_k$. They also commute with coefficient reduction $A_{N+1} \rightarrow A_N$ and compose according to prefix deletion.*

Proof. The first identity follows from $u_{\rho_k^\Sigma(i')} - v_{\rho_k^\Pi(j')}$. In the second, each of the four terms defining δ is transported by the same precomposition. Compatibility of the marked ports identifies their rows and columns; when a predecessor is the marked port, the four terms cancel and agree with extension by zero. All formulas consist of additions and subtractions with integer coefficients, so they commute with modular reduction. Composition of precompositions is precomposition with the composite prefix, proving the last statement. $\square$

5.6 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, at a residence y , defines a free integer complex. Its reduction to A_N uses the same integer register before reduction; successive depths are therefore compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \quad \partial e_y = g_y, \quad \partial b_y = -3c_y g_y, \quad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y through $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a trajectory $\alpha : y \rightarrow y'$ updates memory, its transport in the complex sends the identically named generators to their new representatives, except that

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 5.8 (Naturality of the memory complex). *One has $\partial\Psi_\alpha = \Psi_\alpha\partial$ and $\Psi_{\beta\alpha} = \Psi_\beta\Psi_\alpha$. The maps extend linearly over the chosen coefficient ring.*

Proof. On f_y , the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'}e_{y'} + b_{y'}$. On b_y , the boundary of its image is $-3c_{y'}g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. In a concatenation, memory differences sum telescopically, proving the second identity. $\square$

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v , reduced in the same ring when appropriate, let

$$z^- = r_y, \quad z^+ = r_y + 3c_y v e_y + v b_y, \quad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport also updating v , the correction $K_\alpha = (v_{y'} - v_y)f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α of this local homotopy do not denote the dodecaphasic seal K : they belong to another domain and their role is specified by the preceding identity. That role can also be checked in the equality

$$z_{y'}^+ - \Psi_\alpha z_y^+ = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'}e_{y'} + b_{y'})$.

Proposition 5.9 (Homology and memory of the local complex). *Over $\mathbb{Z}$, or after reduction to A_N , one has*

$$H_0(\Gamma) = H_2(\Gamma) = 0, \quad H_1(\Gamma) \cong A_N[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integer realization.

Proof. Define the degree-1 homotopy by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be projection onto the summand generated by r_y , and ι its inclusion. Checking on each generator gives

$$\partial h + h\partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex homotopy retracts onto the module generated by r_y , concentrated in degree 1, proving the assertion. $\square$

The transport Ψ_α is a determinant-1 shear. Thus this complex explicitly preserves how c_y changes at chain level, but its local homology does not distinguish values of that coordinate. Complete memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinantal readout in addition to this local complex.

5.7 Assembly with terminal paths

For a finite horizon, residences and their prolongations form a path category. To each residence ν , assign the free module $W(\nu) = A_N[\text{Path}(\nu, \text{Term})]$, preserving each terminal continuation as a generator. An initial trajectory acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding subsection acts covariantly through Ψ .

Assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \rightarrow \dots \rightarrow \nu_q} W(\nu_q) \otimes_{A_N} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary deletes an intermediate residence by composing adjacent arrows. At the first endpoint it transports Γ ; at the last it precomposes W . Alternating signs give the path boundary d_{bar} . The total differential is

$$D = d_{\text{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 in pairs. Naturality from Proposition 5.8 implies that d_{bar} commutes with ∂_Γ ; the factor $(-1)^q$ cancels the cross-terms of the square. Thus $D^2 = 0$. Compatibility is proved on path generators and is not introduced through a label asserting membership in a complex.

Terminal evaluation does not require choosing a particular history either. For each continuation h , the effective composition of its updates is applied. The result preserves the pair consisting of h and its readout, with its multiplicity. If $k < \ell < N$, continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N . Proposition 5.1 proves that direct evaluation and evaluation of these two parts produce the same readout. This gives the terminal naturality square.

Path modules, incidences, memory transport and their differentials arise from the same prolongations. Their assembly does not consist of adding five coordinates to a state that would then be preserved by definition: their relations are constructed through sums, boundaries and effective compositions, and verified on their generators.

5.8 Determinants, limit and arithmetic readout

At each residence, the free local complex Γ admits its graded determinant line

$$\text{Det } \Gamma = \bigotimes_r \left(\bigwedge^{\text{rk } \Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The total assembly of the preceding subsection, in turn, has modules $\text{Tot}_d \mathcal{B} = \bigoplus_{q+r=d} \mathcal{B}_{q,r}$. At finite horizon and over the same ring, these are free of finite rank and only finitely many are nonzero. Its line is

$$\text{Det}(\text{Tot } \mathcal{B}) = \bigotimes_{q,r} \left(\bigwedge^{\text{rk } \mathcal{B}_{q,r}} \mathcal{B}_{q,r} \right)^{(-1)^{q+r}}.$$

This formula follows from direct-sum decomposition in each degree. It distinguishes the local complex's line from that of assembly with all paths; their bases are ordered through the preserved labels. Modular reduction commutes with these exterior powers because the modules and their bases arise from the same free integer modules at fixed horizon. The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_N ; a negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, its bases and the relevant homology identifications. To compute Smith factors, the integer matrix produced before reduction is used, not an arbitrary lift of a modular matrix. A square integer block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_N only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences allows stability of the relevant invariant factors to be formulated and checked; these conditions do not follow from existence of the determinant line.

Passage to the limit preserves compatible inverse systems of residues, histories and their multiplicities. The squares proved in the preceding subsections remain commutative because their maps are given by the same formulas at every level. For an Archimedean coordinate, the additional step is to express nonempty cylindrical intervals, compatible by inclusion and with diameter tending to zero. Their nested

compact closures have a one-point intersection: existence follows from compactness and uniqueness from the diameter bound. This is the value of the limit readout. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional expression. The value does not replace the history producing it. The other representation preserves the incidences of the same state. The two destinations share their origin without thereby receiving the same information.

In the following chapters, the coinductive regions, dodecaphasic register and exceptional incidence are expressed from that substrate through their respective maps. The Hilbert-space realization subsequently preserves phase labels and specifies its own unital inclusion; duality and the screens act on the codomains constructed there. The joint operations brought together here transmit histories, sheets, orientation and memory to these representations. Identifying a concrete functional with an external analytic form in turn requires writing its action on generators and verifying its inner products: the conservation identities of this chapter do not replace that composition.

5.9 Provenance, verification and scope

The five operations are recognized by their action on the same histories and by the relations proved in the preceding subsections. The affine memory transport U_α and linear history transport $\mathcal{U}_k$ have the respective domains of Sections 5.1 and 5.3. The sheet grading is $\sigma_k = R_k - Q_k$. For a history $h_j = (\varepsilon_1, \dots, \varepsilon_j)$, accumulation of the balance is made explicit by

$$M_0^\pm = 0, \quad M_k^\pm(h_k) = \sum_{j=1}^k b^\pm(h_j), \quad M_{k+1}^\pm(h_k \varepsilon) = M_k^\pm(h_k) + b^\pm(h_k \varepsilon).$$

Each summand is the signed readout of the corresponding prefix, defined in Section 5.4. Separating the last summand proves the update; the identities $M_k^+ - M_k^- = \sum_{j=1}^k b(h_j)$ and $M_k^+ + M_k^- = \sum_{j=1}^k |b(h_j)|$ follow from the definitions of the positive and negative parts. Accumulation and conditional cancellation are thus distinct operations on the same contributions.

Joint operation	Objects and effective action	Development and proof
Sheet transport and grading	$\mathcal{U}_k : \ell^2(\widehat{X}_k) \rightarrow \ell^2(\widehat{X}_{k+1})$, with $\widehat{X}_k$ consisting of lifted histories; $\sigma_k = R_k - Q_k$, A_k and η_k preserve or interchange the current sheet.	Section 5.3; Proposition 5.3 and Theorem 5.4.
Balance, cancellation and memory	$b : X_k \rightarrow \mathbb{Z}$, $b^\pm : X_k \rightarrow \mathbb{N}_0$; accumulation $M_k^\pm$ over prefixes and cancellation $\kappa_{\text{can}} = (E b_f - Eb_f)/2$ when publishing the conditional average.	Section 5.4; Proposition 5.5.
Ternary sheet incidence	$B : \mathbb{F}_3^4 \rightarrow \mathbb{F}_3^0$, $(Ba)_{\{L,L'\}} = a_L + a_{L'}$; the full ambient space of q is preserved, with $sB = 0$ and $W(q + Ba) = W(q)$.	First part of Section 5.5.
Ports and rectangular complex	$\kappa : A_N^{I_k} \oplus A_N^{J_k} \rightarrow A_N^{I_k \times J_k}$, $\kappa(u, v)_{ij} = u_i - v_j$; the difference δ and precomposition L_k preserve labels and their multiplicities.	Propositions 5.6 and 5.7.
Cellular memory, filling and determinant line	$\Psi_\alpha : \Gamma_y \rightarrow \Gamma_{y'}$, $K_\alpha \in \Gamma_{y',2}$, $\partial h_* = z^- - z^+$; the lines $\text{Det } \Gamma$ and $\text{Det}(\text{Tot } \mathcal{B})$ are formed on the corresponding based complexes.	Sections 5.6–5.8; Propositions 5.8 and 5.9.

The table identifies five actions of one construction: each prefix preserves its sheets, orientation, residues, quotients, memory and terminal prolongations. Port precomposition takes coordinates from one level to its refinement; coefficient reduction takes A_{N+1} to A_N . Their directions and domains are distinct, and their commutative squares are made explicit in Proposition 5.7. The assembly in Section 5.7 simultaneously uses terminal continuations and transport of the memory complex; its composition formulas bring together the table's actions on the same history system. Reading a determinant line as a scalar preserves the conditions on bases, homology and nonsingularity in Section 5.8.

This development brings together the constructions of state, tree and common measure; the correlative intrinsic operations; and terminal assembly with naturality and limit proved in this section. Their

balance, incidence and memory identities are incorporated, with proofs within this chapter. The five continuum constructions are maintained as operations on the same tree. The exposition extends their local proofs and makes explicit the conditions of each specialization: isometric transport, uniform weights, terminal extinction and nonsingularity of a determinantal block.

The accompanying checker compares exact instances of the cocycles, weighted cancellation, rectangular sequence and memory complex. These checks accompany the text's general proofs and verify the operators defined here in their respective domains. The subsequent passage towards exceptional representations and the Hilbert-space realization preserves their actions on generators: Section 18 specifies the phase fibre, Weyl operators and inclusions determining their closures. Documentary preservation and finite checks retain their function of traceability and comparison for these proofs.

$\pi \quad \varphi \quad e$

Coinductive generation

Regional coordinates are constructed through compatible prolongations of arbitrary depth. The state preserves the relations used by each readout.

The depth of a number

Coinductive generation organizes the determination of a number through compatible prolongations. A stage produces a prefix and preserves the conditions for continuing it. Depth can increase while maintaining agreement among all previous readings. The value emerges through the realization of this compatibility; the history also preserves the way in which precision was obtained.

The regions of the circular, golden, and exponential coordinates arise from the same APP-TRIT-TPK development. Their readers perform distinct functions and share the structure that makes it possible to relate them. The triadic extension of Euler makes the realizations of the regimes explicit; the areal-volumetric calendar determines self-scaling; regional emissions and their boundaries preserve the conditions of evaluation. The succession of these steps makes explicit the construction order of the coordinates that subsequently enter into alpha.

The vacancy calendar makes it possible to understand a particular feature of refinement. History advances with every transition, while the capacity of a reading may remain fixed. The word that records these transitions distinguishes acquired capacity from preserved vacancy. Its balance counts both contributions. The regularity of the calendar and the aperiodicity of its itinerary describe related aspects of one and the same evolution.

Monodromy extends this distinction. Returning to a phase means recognizing a position in the cycle; recovering a history means also restoring its turns, orientation, and registers. The regional mirror preserves the axis and the aggregate of the two coordinates it exchanges, while reversing their difference. This datum makes it possible to reconstruct the distribution. The nonadic figure locates the phases and their operations; its marks are positions in the calendar, with a semantics different from that of the real value of a constant.

The fifth boundary, the subsequent regions, and the return are therefore explained through their actions on the state. The regional figures show simultaneously the coincidence of a reading and the differentiation of its antecedents. This differentiation transmits to the exceptional realization the incidence information it uses. Numerical generation thus continues the double projection introduced at the outset: each reading obtains its value from a structure whose organization remains available to the other realizations.

6 Coinductive generation of π, φ, e at arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its function is to produce regions with multiplicity, orientation, and transport rules. Prolongation must preserve these data and determine a compatible collection of prefixes of arbitrary length. The word *generation* denotes this operation; *genealogy* denotes reconstruction of its dependencies. Both notions are involved, but they have different functions.

6.1 Orbits, regions, and characters of closure, propagation, and self-scaling

On a word $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; these permutations therefore realize the dihedral group D_3 . The action preserves the catalogue and multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fibre predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. The catalogue contains exactly one orbit with these properties. To specify the other two selections, write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature preserves this class and the orientation *ES* or *NO*. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the orbits of size six, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, equal to that of closure, selects propagation uniquely; the census $\nu = (2, 2, 2)$ selects self-scaling uniquely. The diagonal support cannot be omitted: without it there are four balanced singleton-fibre orbits of multiplicity 144, rather than one. Finally, the existence of an emission with $d = 6$ and orientation *ES* selects one representative of each orbit and yields

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (6.1)$$

These symbols first denote regions of the catalogue. Scalar identification will be proved through their readers; they are not chosen because they agree with digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transverse	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and constant signature 555555 characterize the transverse region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position of the support would erase the distinction between a numerical region and the electron structure.

The microfibre also has a precise bipartite morphology. Each emission is a product of an additive-cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transverse region has a product 18×24 ; the three positive regions together have another 18×24 ; and the return has a product 18×8 . Decomposition by emissions has five parts, whereas decomposition into connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing form to cardinality.

6.2 Incidences between halves of regional emissions

Let $\mathcal{U}_6$ be the image of six consecutive windows of the emitter of Section 3.4. For $u = (u_1, \dots, u_6)$, define $A(u) = (u_1, u_2, u_3)$ and $B(u) = (u_4, u_5, u_6)$. The half-emission graph has these triples as vertices and an undirected edge between $A(u)$ and $B(u)$ for each emission.

Proposition 6.1 (Cycle of regional junctions). *The vertices*

$$\begin{aligned} A &= (830, 943, 830), & B &= (109, 252, 252), \\ C &= (810, 923, 870), & D &= (169, 272, 272), \\ E &= (850, 963, 890), & F &= (149, 252, 212) \end{aligned}$$

form the cycle $A - B - C - D - E - F - A$.

Proof. Write each seed as (i, j, d) , with residual positions between 0 and 8 and directions $N = (-1, 0)$, $E = (0, 1)$ and $S = (1, 0)$. Under the 54-step recurrence, the following seed pairs produce the six emissions shown:

emission	s_+	$s_\times$
$A \mid B$	$(0, 6, E)$	$(0, 6, N)$
$B \mid C$	$(0, 6, N)$	$(1, 5, E)$
$C \mid D$	$(0, 6, E)$	$(0, 1, S)$
$D \mid E$	$(0, 6, N)$	$(0, 4, S)$
$E \mid F$	$(0, 6, E)$	$(0, 3, E)$
$F \mid A$	$(0, 6, N)$	$(0, 1, S)$

Each row witnesses membership in $\mathcal{U}_6$ by successive evaluation of the two accumulators. Splitting each emission into its two halves yields exactly the six edges of the cycle. $\square$

In the classification (6.1), $A \mid B$, $C \mid D$ and $E \mid F$ respectively produce the words w^{au} , w^{cl} and w^{pr} . The regional readers subsequently identify these outputs with φ , one region of π and e . The other three emissions exhibit their junctions in the graph. The result describes incidences between emitter outputs. Temporal realization of an incidence additionally uses the orientation, phase and memory of the enriched state. Including the APP anchor and all five regions of π imposes further path conditions; the cycle above retains its role as a local junction witness.

The six-edge cycle belongs to this undirected half-emission graph. An enriched temporal path additionally retains the anchoring, orientation and closure data of its construction. Incidence between two halves and transport of a state thus have explicit domains and composition conditions.

6.2.1 Block multigraph and five minimal regional closures

The 468 emissions in $\mathcal{U}_6$, produced by the APP-TRIT-TPK emitter, determine a block multigraph. Identifying its regional anchors makes it possible to solve a joint edge-covering problem: to traverse the APP, propagation, autoscale, and each closure region's incidences within a single closed walk. The local six-junction cycle of Section 6.2 and these covers retain their respective traversal conditions.

Block graph and exact phase quotient. For $U = (U_1, \dots, U_6) \in \mathcal{U}_6$, define

$$A(U) = (U_1, U_2, U_3), \quad B(U) = (U_4, U_5, U_6). \quad (6.2)$$

The undirected multigraph $\tilde{G}$ has as vertices the triples occurring as $A(U)$ or $B(U)$, and one edge labelled by each U , incident at those two triples. All 468 labels are retained: distinct emissions are not identified merely because they share endpoints or because one interchanges the halves of another.

Internal phase acts on each triple by cyclic rotation:

$$(a, b, c) \sim (b, c, a) \sim (c, a, b), \quad \text{can}(a, b, c) = \min_{\text{lex}}\{(a, b, c), (b, c, a), (c, a, b)\}. \quad (6.3)$$

The graph G replaces the endpoints of edge U by $\text{can } A(U)$ and $\text{can } B(U)$, retaining its label. The equivalence acts precisely by the three cyclic rotations of each vertex; every emission retains its identity as an edge.

Enumeration from the same emitter and component searches give

	$ V $	component sizes
$\bar{G}$	96	(28, 28, 28, 4, 4, 4)
G	32	(28, 4)

(6.4)

These data are obtained by collecting distinct endpoints, adding each incidence in both directions, and traversing each unvisited component by breadth-first search. Adjacency may be deduplicated when counting components, but parallel labels are retained when searching for closures with prescribed edges.

APP anchor and regional numbering. The three common edges are

$$\begin{aligned}
 U_{\text{APP}} &= 903|850|850|252|109|252, \\
 U_e &= 850|963|890|149|252|212, \\
 U_\varphi &= 830|943|830|109|252|252.
 \end{aligned}
 \tag{6.5}$$

Fix the endpoint $v_{\text{APP}} = \text{can } A(U_{\text{APP}}) = 850|850|903$. For the five already constructed regions, use the following order, which places the transverse region in the fifth row:

i	$U_\pi^{(i)}$	ρ_i	m_i
1	810 923 870 169 272 272	++	144
2	810 983 810 169 272 272	++	144
3	870 923 810 169 272 272	++	144
4	870 923 810 418 674 521	--	144
5	501 614 498 169 272 272	$\perp$	432

(6.6)

This numbering distinguishes the three ++ regions, the mutated return --, and the transverse region $\perp$. Accordingly, $\rho = (++, ++, ++, --, \perp)$ and $m = (144, 144, 144, 144, 432)$ in this order; their sum remains 1008.

Five complete witnesses with traversal directions. In each table, row j means traversing the edge labelled e_j from v_{j-1} to v_j . Endpoints are canonical phase representatives; since G is undirected, the pair (v_{j-1}, v_j) specifies the traversal direction of the labelled edge. Each table contains exactly eight edges and satisfies $v_0 = v_8 = v_{\text{APP}}$.

The breadth-first search described in the proof yields the following witnesses, one for each set of four target edges. Minimality concerns length; several walks may attain that length.

Witness $i = 1$.

j	v_{j-1}	e_j	v_j
1	850 850 903	109 252 252 850 903 850	109 252 252
2	109 252 252	109 252 252 810 923 870	810 923 870
3	810 923 870	810 923 870 169 272 272	169 272 272
4	169 272 272	169 272 272 850 963 890	850 963 890
5	850 963 890	850 963 890 149 252 212	149 252 212
6	149 252 212	149 252 212 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 2$.

j	v_{j-1}	e_j	v_j
1	850 850 903	169 272 272 850 903 850	169 272 272
2	169 272 272	169 272 272 810 983 810	810 810 983
3	810 810 983	810 983 810 169 272 272	169 272 272
4	169 272 272	169 272 272 850 963 890	850 963 890
5	850 963 890	850 963 890 149 252 212	149 252 212
6	149 252 212	149 252 212 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 3$.

j	v_{j-1}	e_j	v_j
1	850 850 903	169 272 272 850 903 850	169 272 272
2	169 272 272	169 272 272 850 963 890	850 963 890
3	850 963 890	850 963 890 149 252 212	149 252 212
4	149 252 212	149 252 212 870 923 810	810 870 923
5	810 870 923	870 923 810 169 272 272	169 272 272
6	169 272 272	169 272 272 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 4$.

j	v_{j-1}	e_j	v_j
1	850 850 903	169 272 272 850 903 850	169 272 272
2	169 272 272	169 272 272 850 963 890	850 963 890
3	850 963 890	850 963 890 149 252 212	149 252 212
4	149 252 212	149 252 212 870 923 810	810 870 923
5	810 870 923	870 923 810 418 674 521	418 674 521
6	418 674 521	418 674 521 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 5$.

j	v_{j-1}	e_j	v_j
1	850 850 903	109 252 252 850 903 850	109 252 252
2	109 252 252	109 252 252 501 614 498	498 501 614
3	498 501 614	501 614 498 169 272 272	169 272 272
4	169 272 272	169 272 272 850 963 890	850 963 890
5	850 963 890	850 963 890 149 252 212	149 252 212
6	149 252 212	149 252 212 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Theorem 6.2 (Minimality of the five regional edge covers). *For each $i \in \{1, \dots, 5\}$, among closed walks in G traversing every edge of*

$$E_i^* = \{U_{\text{APP}}, U_e, U_\varphi, U_\pi^{(i)}\}$$

at least once, the minimum length, counting traversals with multiplicity, is eight. Vertices and edges may be repeated.

Proof. Every closed walk traversing U_{APP} passes through v_{APP} . Cyclic rotation of its step list allows it to start at that vertex without changing length or coverage. It therefore suffices to solve the problem with a fixed starting vertex.

The search space is the finite set

$$\mathcal{S}_i = V(G) \times \mathcal{P}(E_i^*), \quad |\mathcal{S}_i| \leq 32 \cdot 16 = 512.$$

The initial state is $s_0 = (v_{\text{APP}}, \emptyset)$, and the terminal state is $s_* = (v_{\text{APP}}, E_i^*)$. Traversing an edge e incident at v towards its other endpoint v' updates the state by

$$(v, M) \mapsto (v', M \cup (E_i^* \cap \{e\})). \quad (6.7)$$

The mask records edge labels, not just endpoints; distinct parallel edges must therefore not be collapsed.

Every transition has unit cost. A FIFO queue initialized at s_0 extracts states in increasing distance, marks each state on its first visit, and stores its predecessor and traversed label. Equivalently, with $\mathcal{F}_0 = \{s_0\}$, the new layers are

$$\mathcal{F}_{k+1} = \text{Succ}(\mathcal{F}_k) \setminus \bigcup_{j=0}^k \mathcal{F}_j.$$

Induction shows that $\mathcal{F}_k$ contains exactly the states whose minimum distance from s_0 is k . Finite enumeration over the specified emitter gives, for each of the five regions,

$$s_* \notin \bigcup_{k=0}^7 \mathcal{F}_k, \quad s_* \in \mathcal{F}_8, \quad (\ell_1, \ell_2, \ell_3, \ell_4, \ell_5) = (8, 8, 8, 8, 8).$$

The tables exhibit the five upper bounds, including all four required labels in each case. Exhaustive exploration of the preceding layers proves the lower bound. Together they establish minimality for the declared problem. $\square$

Length eight characterizes the joint coverage of four incidences in each closure region. The quotient retains emission labels and their junctions between phase classes; enriched temporal evolution also preserves orientation and memory. This description links all five regions to the same propagation, autoscale, and APP anchors and complements the local six-junction cycle. Coinductive prolongation retains its operators and cylinders together with these orientation and memory data.

6.3 Finite transport and prolongation boundary

The endomorphisms L_0, L_1 of the ternary word space act before the Archimedean realization. Their determination begins with transitions of the arithmetic state, rather than with two prescribed matrices. If U_i is the composite step of selection, transport, and updating, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl}, \text{pr}, \text{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The source and target matrix pairs obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each six-digit string represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ determine $L_a = X_a^{-1} Y_a$ uniquely. In this row-vector chart, they give

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The schedule (L_0, L_0, L_1, L_1) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace production of those transitions by U_t . The provenance of their record and examination of the sixteen binary schedules are documented in the technical supplement. The stated schedule is the only one of those sixteen that jointly reproduces the three sequences of five blocks.

6.4 The deep coordinate and integer recovery

The six-coordinate chart retains the expressed residue and the integer quotient that determines its prolongation. We fix the integer representative of L_0 displayed in the regional construction:

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad \det A_H = 1. \quad (6.8)$$

Its reduction modulo three equals L_0 . The choice of integer representative is part of the chart: in addition to that reduction, it fixes the action on the deep quotients. For row vectors, define

$$\begin{aligned} \text{Div}_H : \mathbb{Z}^6 &\longrightarrow \{0, 1, 2\}^6 \times \mathbb{Z}^6, \\ x_H &\longmapsto (u_H, x_H^+), \\ y_H = x_H A_H, \quad u_H &= [y_H]_3, \quad x_H^+ = (y_H - u_H)/3. \end{aligned} \quad (6.9)$$

Reduction $[\cdot]_3$ is performed componentwise with representatives 0, 1, 2, including for negative integer components.

Proposition 6.3 (Exact inversion of residue and quotient). *The map Div_H is bijective. Its inverse is*

$$x_H = (u_H + 3x_H^+)A_H^{-1}. \quad (6.10)$$

For a trajectory in this chart, $x_j A_H = u_j + 3x_{j+1}$, reconstruction at depth $T \geq 1$ satisfies

$$x_0 = \sum_{j=0}^{T-1} 3^j u_j A_H^{-(j+1)} + 3^T x_T A_H^{-T}. \quad (6.11)$$

Proof. The determinant in (6.8) is one, so the adjugate formula gives $A_H^{-1} \in M_6(\mathbb{Z})$. Euclidean division of each component of $x_H A_H$ determines a unique pair (u_H, x_H^+) . Conversely, for any $(u, q) \in \{0, 1, 2\}^6 \times \mathbb{Z}^6$, the vector $x = (u + 3q)A_H^{-1}$ is integral and satisfies $x A_H = u + 3q$. Its residue is exactly u , and its quotient is q . This proves both inverse compositions. Successive substitution of $x_j = (u_j + 3x_{j+1})A_H^{-1}$, for $j = 0, \dots, T-1$, gives (6.11); the final term retains the terminal coordinate remaining after the observed prefix. $\square$

The ternary orientation is read subsequently through $0 \mapsto 0$, $1 \mapsto +1$, and $2 \mapsto -1$, retaining the residue u_H and the quotient in (6.9). The other state fields preserve their own records of orientation, carry, route, and boundary. Proposition 79.4 establishes the arithmetic tower for $p = 3$, $d = 6$, and $A = A_H$, together with proofs of compatibility and passage to the limit. Realization of the generated prolongations also uses the APP-TRIT-TPK compatibility conditions specified in Proposition 79.7.

The residual block also retains its index

$$\nu(u_H) = \sum_{j=1}^6 (u_H)_j 3^{6-j} \in \{0, \dots, 728\}.$$

This index enters the prefixes through $N' = 729N + \nu(u_H)$. The mixed chart in Section 7.4 simultaneously retains the ternary and decimal indices and the integer boundary quantities $A(s), B(s)$. Their recurrences and compatibility with truncation are proved in Proposition 7.10 and Theorem 7.11. Integer-coordinate recovery is thereby linked to the exact comparison of the cylinders that express its prefixes.

6.5 Five closure regions and angular compensation

The closure region contains one transverse component and four accompanying components. The symmetric reader acts on the space of its five positions through

$$P_5 = \frac{1}{5} \mathbf{1} \mathbf{1}^T, \quad \mathbf{1} = (1, 1, 1, 1, 1)^T.$$

Invariance $P_5 \lambda = \lambda$ and conservation $\mathbf{1}^T \lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes the unit among reader positions; it does not replace the seed multiplicities 432, 144, 144, 144, 144 with uniform frequencies. The reader's elliptic chart realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. This makes explicit the rule passing from a normalized coordinate to a slope, rather than identifying the two notions without a map. The reader composes the four accompanying positions by the law

$$a \oplus b = \frac{a + b}{1 - ab}. \quad (6.12)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. Slope composition is therefore a realization of the TRIT quadratic structure, rather than ordinary addition of four fractions.

Proposition 6.4 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to slope one determines a unique positive compensator of the form $1/q$, with $q > 1$, and gives $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second gives $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, so $q = 239$. $\square$

The resulting closure character has realization

$$\Pi_{\text{cl}} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (6.13)$$

Machin's angular identity recognizes $\Pi_{\text{cl}} = \pi$. The direction of the construction is precise: the slope and number of compositions are specified by the penta fibre reader; the compensation equation forces 239. Once this character has been produced, its evaluation by alternating series does not reselect the seed orbit.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes in (6.13), they provide rational endpoints with explicit error. The numerical value is thus calculated as a consequence of the exact character, without introducing a target prefix.

6.6 Propagation reader and normalization of the unit

The propagation region is realized through a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not only its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

normalization determines $a_0 = a_1 = 1$. Comparison of the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (6.14)$$

The character at one unit of parameter is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

Normalization of the generator matters. Without this datum, the semigroup law would admit $P_t = e^{ct}$ for different c ; the condition fixing the unit of the parameter in the reader is therefore retained. This condition precedes decimal comparison and belongs to the exposition of the operational rule, rather than to the choice of an experimental value.

Proposition 6.5 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Every subsequent ratio is at most $1/(n+2)$. The geometric majorant of the tail sums to

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

Strict inequality follows from the decrease of successive ratios. □

6.7 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. On the modes Σ, Π , the rule is $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$, $0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Its incidence, in that order of modes, is therefore

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (6.15)$$

The four entries are obtained from the schedule, rather than from an equation that has previously received the value of φ . Repetition of the schedule gives counts $3F_{\text{av}}, 9F_{\text{av}}, 36F_{\text{av}}$ at lengths nine, twenty-seven, and one hundred and eight. These counts are not powers of the matrix.

Theorem 6.6 (Positive self-scaling character). *The incidence (6.15) has a unique strictly positive eigenray. When normalized as $(1, x)^T$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and is $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ produces the polynomial. Only one of its two roots is positive. Since F_{av}^2 has all entries positive, the positive ray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in production of F_{av} . □

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The consecutive ratios F_{n+1}/F_n alternate around φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily narrow.

The measure on all compatible histories does not coincide with the mode frequency of the three-event schedule. The chronological frequency is $(1/3, 2/3)$. The measure of maximal entropy of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2} . The latter enters the areal–volumetric readout; it is not obtained by simply replacing one counting frequency with another.

6.7.1 Trajectory space and invariant measure between sheets

The toroidal topology of APP fixes the support of cardinal translations and their winding classes. The correspondence between sheets arises from another operation on that same support: temporal selection of the additive or multiplicative evaluation. Its geometric typing is

$$\eta_{\text{av}}(\Sigma) = \mathcal{H}_{\text{ar}}, \quad \eta_{\text{av}}(\Pi) = \mathcal{H}_{\text{vol}}.$$

The first sheet corresponds to the additive reading of the boundary; the second, to the multiplicative reading of the interior. This map transports mode labels and preserves the incidence (6.15). The square of the self-scaling ratio is determined through the trajectories of that incidence.

Definition 6.7 (Memory-one symbolic envelope). Let

$$X_{\text{av}} = \{(s_k)_{k \in \mathbb{Z}} \in \{\mathcal{H}_{\text{ar}}, \mathcal{H}_{\text{vol}}\}^{\mathbb{Z}} : (F_{\text{av}})_{s_k, s_{k+1}} = 1 \text{ for every } k\}.$$

Index shift acts on this space. It is the smallest memory-one symbolic system containing the modal sequences and preserving their consecutive pairs: it admits the two cross-transitions and volumetric self-retention. Phase, cardinal orientation, and the trajectory records remain in the TPK state from which this envelope is obtained.

Proposition 6.8 (Weighting trajectories between sheets). *In areal–volumetric order, the invariant measure of maximal entropy of X_{av} has transition matrix and stationary distribution*

$$P_{\text{av}} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \quad (\mu_{\text{ar}}, \mu_{\text{vol}}) = \frac{(1, \varphi^2)}{1 + \varphi^2}. \quad (6.16)$$

In particular, $\mu_{\text{ar}}/\mu_{\text{vol}} = \varphi^{-2}$.

Proof. The already constructed incidence has positive eigenvector $r = (1, \varphi)^T$ and eigenvalue φ . The normalization

$$(P_{\text{av}})_{ij} = \frac{(F_{\text{av}})_{ij} r_j}{\varphi r_i}$$

gives the stated matrix. Its rows sum to one because $\varphi^{-2} + \varphi^{-1} = 1$. The symmetry of F_{av} implies that weights proportional to r_i^2 satisfy detailed balance; therefore, $\mu P_{\text{av}} = \mu$. Irreducibility and volumetric self-retention determine a unique stationary distribution.

To verify maximality, every allowed transition satisfies $\log(P_{\text{av}})_{ij} = \log r_j - \log r_i - \log \varphi$. Under any invariant measure, the endpoint terms cancel on average. If v_i are its vertex weights and $q_i(j) = \nu(s_1 = j \mid s_0 = i)$, for $v_i > 0$, then

$$h_\nu \leq H_\nu(s_1 \mid s_0) = \log \varphi - \sum_{i: v_i > 0} v_i D(q_i \parallel R_i) \leq \log \varphi,$$

where $D(q \parallel p) = \sum_j q_j \log(q_j/p_j)$ is the relative entropy on allowed pairs and $0 \log 0 = 0$ is adopted. The Markov measure in (6.16) attains the bound. Equality in the first inequality requires the Markov property; equality in the second requires $q_i = R_i$. Stationarity and irreducibility then fix μ , establishing uniqueness. This is the Parry measure of the defined envelope. $\square$

The frequency $(1/3, 2/3)$ counts the instants of the primitive calendar; μ weights the trajectories of the envelope. Its quadratic ratio expresses the product of the incoming and outgoing incidence amplitudes. TPK memory and phase restrictions retain their own domain; the entropy of the envelope corresponds to the symbolic system specified in this definition.

6.7.2 Closure operator and limit of marked returns

The closure coordinate π , generated by the regional reader, enters after the construction of modal incidence. Let $e_{\text{ar}} = (1, 0)^\top$, $e_{\text{vol}} = (0, 1)^\top$, and $P_i = e_i e_i^\top$ be the sheet projectors. The reader's conditions are preservation of both sheets, unit volumetric normalization, and an areal mark equal to the closure coordinate. These conditions determine the positive operator

$$D_\pi = \pi P_{\text{ar}} + P_{\text{vol}} = \begin{pmatrix} \pi & 0 \\ 0 & 1 \end{pmatrix}. \quad (6.17)$$

Diagonality expresses preservation of sheets, and the two entries express their normalizations; this makes explicit the uniqueness of the operator for the fixed reader.

Proposition 6.9 (Ratio of boundary returns). *For $n \geq 1$, the marked evaluation*

$$\Theta_n = \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle} = \pi \frac{F_{n-1}}{F_{n+1}} \quad (6.18)$$

satisfies

$$\lim_{n \rightarrow \infty} \Theta_n = \pi \frac{\mu_{\text{ar}}}{\mu_{\text{vol}}} = \frac{\pi}{\varphi^2}. \quad (6.19)$$

Proof. The diagonal entries of F_{av}^n count the paths of length n returning to their initial sheet. The already proved power formula provides F_{n-1} and F_{n+1} . The operator D_π multiplies the first count by π . Finally, F_{n-1}/F_{n+1} is the product of two consecutive ratios converging to φ^{-1} ; its limit is φ^{-2} . The identity with the weight ratio follows from (6.16). $\square$

The closure operator appears only once, in the terminal evaluation of the return. The transfer F_{av} remains determined by the calendar. Information from two readers of the same state is thus composed: modal incidence determines the quadratic weighting, and the closure region contributes its coordinate to the boundary.

The composition also preserves refinement. For positive cylinders $I = [a, b]$ and $J = [c, d]$, the image under the reader $\mathcal{R}(x, y) = x/y^2$ is the interval

$$\mathcal{Q}(I, J) = \left[\frac{a}{d^2}, \frac{b}{c^2} \right].$$

Monotonicity in each argument proves that the nested closure and self-scaling cylinders produce nested intervals for the ratio; positivity of their limits and contraction of their diameters determine π/φ^2 . The scalar expression thus retains a chain of operations compatible with depth, which is applied next in its two orientations.

6.7.3 Multiplicity and recovery of modal itineraries

The incidence F_{av} , constructed by the tritic calendar in the preceding development, acts here on its envelope of modal walks. Write

$$F = F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad 0 = \Sigma, \quad 1 = \Pi.$$

The self-scaling coordinate φ and the positive ray $r = (1, \varphi)$ are the outputs already obtained from this incidence. They determine the multiplicity of the scalar reading and support a reversible representation of each itinerary.

Consider the set of finite directed walks $\gamma = (x_0, \dots, x_n)$ of F . Their endpoints and lengths are distinguished, repetitions are allowed, and trivial walks are included. This domain is the modal envelope; the complete TPK state additionally retains the records of its enriched history. The multiplicative character takes the form

$$\chi(\gamma) = \varphi^n \frac{r_{x_0}}{r_{x_n}} = \varphi^{k(\gamma)}, \quad r = (1, \varphi), \quad k(\gamma) = n + x_0 - x_n. \quad (6.20)$$

The transitions $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$ increment k by 0, 1, and 2, respectively. The first step modifies the path while preserving its character. In particular, both $(1, 1)$ and $(0, 1, 1)$ produce φ under this reading.

Proposition 6.10 (Multiplicity of the self-scaling reading). *Let $\mathcal{F}_k = \{\gamma : k(\gamma) = k\}$ and $N_k = |\mathcal{F}_k|$. Then*

$$N_0 = 3, \quad N_k = 2 \operatorname{tr}(F^k) = 2(\varphi^k + (-\varphi^{-1})^k) \quad (k \geq 1), \quad (6.21)$$

and

$$\lim_{k \rightarrow \infty} N_k^{1/k} = \varphi.$$

Proof. Given endpoints $i, j \in \{0, 1\}$, the length is determined by $n = k - i + j$. The number of walks with those endpoints is $(F^n)_{ij}$. For $k \geq 1$, summing the four sectors gives

$$N_k = \operatorname{tr}(F^k) + (F^{k+1})_{01} + (F^{k-1})_{10}.$$

The identity $F^2 = F + I$ implies by induction that, for $n \geq 1$,

$$F^n = \begin{pmatrix} f_{n-1} & f_n \\ f_n & f_{n+1} \end{pmatrix}, \quad f_0 = 0, \quad f_1 = 1, \quad f_{n+1} = f_n + f_{n-1}.$$

For $k \geq 2$, the two off-diagonal terms are $f_{k+1} + f_{k-1} = \operatorname{tr}(F^k)$; for $k = 1$, their sum is $1 + 0 = 1 = \operatorname{tr}(F)$, using $F^0 = I$. For $k = 0$, the only walks are (0) , (1) , and $(0, 1)$. The eigenvalues of F are φ and $-\varphi^{-1}$, so its trace gives the closed formula. Finally,

$$N_k = 2\varphi^k(1 + (-\varphi^{-2})^k),$$

and the factor in parentheses tends to one; taking k -th roots yields the limit. $\square$

Self-scaling thus coincides with the exponential growth rate of the finite modal genealogies that the scalar character makes indistinguishable.

Refinement by the composition of returns. Let $b(\gamma)$ count the edges $\Pi \rightarrow \Sigma$, and let $c(\gamma)$ count the edges $\Pi \rightarrow \Pi$. The increments in (6.20) give $k = 2b + c$. For formal indeterminates z, t , the graded incidence

$$M(z, t) = \begin{pmatrix} 0 & 1 \\ tz^2 & z \end{pmatrix}$$

records exponent and returns jointly. Its only degree-zero edge is $\Sigma \rightarrow \Pi$, and there is no degree-zero cycle. Each coefficient therefore counts a finite set of walks. If $N_{k,b}$ denotes their multiplicity with values k, b , the sum over all lengths satisfies

$$\sum_{k,b \geq 0} N_{k,b} z^k t^b = \mathbf{1}^\top (I - M(z, t))^{-1} \mathbf{1} = \frac{3 - z + tz^2}{1 - z - tz^2}, \quad \mathbf{1} = (1, 1)^\top. \quad (6.22)$$

Indeed, entry ij of M^n counts walks of length n with those endpoints and their weights. Coefficientwise formal summation yields $(I - M)^{-1}$. The determinant of $I - M$ is $1 - z - tz^2$, and the sum of the four entries of its adjugate is $3 - z + tz^2$.

For $k \geq 1$ and $0 \leq b \leq \lfloor k/2 \rfloor$, this gives

$$N_{k,b} = \frac{2k}{k-b} \binom{k-b}{b}. \quad (6.23)$$

To prove this, the expansion

$$(1 - z - tz^2)^{-1} = \sum_{m \geq 0} (z + tz^2)^m$$

gives coefficient $\binom{k-b}{b}$ at $z^k t^b$. Multiplication by the numerator of (6.22) produces

$$3 \binom{k-b}{b} - \binom{k-b-1}{b} + \binom{k-b-1}{b-1} = 2 \binom{k-b}{b} + 2 \binom{k-b-1}{b-1}.$$

Coefficients outside their range are taken to be zero. For $b \geq 1$, the identity $\binom{k-b-1}{b-1} = \frac{b}{k-b} \binom{k-b}{b}$ yields (6.23); for $b = 0$, both sides equal two. The constant term is treated separately: $N_{0,0} = 3$. For $k = 4$, there are fourteen walks, distributed as 2, 8, and 4 according to whether they contain zero, one, or two returns. The same character therefore admits distinct internal compositions.

Proposition 6.11 (Reversible recovery and transport of codes). *Each walk in the modal envelope admits a code*

$$\mathcal{C}(\gamma) = (k, a), \quad 0 \leq a < N_k,$$

with a decoding map $\mathcal{D}$ such that

$$\mathcal{D}\mathcal{C} = \text{id}, \quad \mathcal{C}\mathcal{D} = \text{id}.$$

If $A_v(\gamma) = \gamma v$ prolongs a walk by an admissible edge and T removes the last state of a positive-length walk, their representations

$$\mathcal{E}_v = \mathcal{C}A_v\mathcal{D}, \quad \hat{T} = \mathcal{C}T\mathcal{D}$$

satisfy, on their respective domains,

$$\hat{T}\mathcal{E}_v = \text{id}, \quad k(A_v\gamma) - k(\gamma) = 1 + x_n - v. \quad (6.24)$$

Proof. For each k , the nonempty sectors are ordered by (n, i, j) and, within each sector, walks are ordered lexicographically by their states. The number of length- m continuations from v to j is $(F^m)_{vj}$. Traversing a walk, one sums the sizes of preceding sectors and, at each position, the sizes of the continuations lexicographically preceding the chosen one. The result is its zero-based index a .

To invert this process, the size of each preceding sector is subtracted from the index until the unique containing sector is identified. At each position, the same procedure is applied to continuation sizes. These sets form a disjoint and exhaustive partition of the starting sector. At length zero, the only walk of a nonempty sector is its vertex. Assuming recovery of continuations of length $m-1$, the index intervals of each first successor uniquely identify that successor and the remaining index of its continuation. Induction on m proves that the procedures are inverse, both on walks and on the indices $0, \dots, N_k - 1$.

Lexicographic order serializes the existing path; the operations generating the path precede this serialization. Since $TA_v = \text{id}$, the two inverse identities give

$$\hat{T}\mathcal{E}_v = \mathcal{C}T\mathcal{D}\mathcal{C}A_v\mathcal{D} = \mathcal{C}TA_v\mathcal{D} = \text{id}.$$

Finally, the exponent definition gives

$$k(A_v\gamma) = (n+1) + x_0 - v, \quad k(\gamma) = n + x_0 - x_n,$$

whose difference is the second identity. □

Distinguishing capacity with a known exponent. With k known, an injective fixed-length binary representation of length ℓ requires $2^\ell \geq N_k$, by the pigeonhole principle. Consequently, $\ell \geq \lceil \log_2 N_k \rceil$. The preceding index a attains this length through its binary representation, padded with leading zeros where necessary. This minimum concerns transmitting the index with k known; transmitting k and delimiting variable-length codes have their own costs. Moreover,

$$\lim_{k \rightarrow \infty} \frac{\log_2 N_k}{k} = \log_2 \varphi.$$

The equality measures the combinatorial capacity to distinguish paths in the modal envelope.

Fiber information under an explicit distribution. For fixed k , let R_k be a walk chosen uniformly from $\mathcal{F}_k$. We use natural logarithms and $H(R_k) = -\sum_{\gamma \in \mathcal{F}_k} p_\gamma \log p_\gamma$, with $p_\gamma = 1/N_k$. All these walks have character φ^k , so

$$H(R_k \mid \chi(R_k)) = \log N_k, \quad H(R_k \mid \mathcal{C}(R_k)) = 0.$$

The first equality follows because the reading is constant on the fiber; the second follows from the unique decoding already proved. In particular, $\log N_k/k \rightarrow \log \varphi$.

For $k = 4$, the character leaves an uncertainty of $\log 14$ nats. The three classes $B = b(R_4)$ have probabilities $2/14$, $8/14$, and $4/14$. Conditional on each class, the distribution remains uniform on its 2, 8, or 4 elements, respectively. By the definition of conditional entropy,

$$H(R_4 \mid B) = \frac{2}{14} \log 2 + \frac{8}{14} \log 8 + \frac{4}{14} \log 4.$$

The index within the class, together with B , uniquely recovers the walk. These values pertain to the declared uniform distribution on the modal fiber. The measure on the complete space of histories, the temporal realization of k , and any thermal reading retain their respective domains and transducers.

After the characters have been generated, the two orientations of the reader

$$R(x, y) = \frac{x}{y^2}$$

provide

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (6.25)$$

The second expression enters the exponent of the electron construction in Section 15.1.1. The first is also obtained as the limit of the marked readout $\pi F_{n-1}/F_{n+1}$. Exchanging the arguments relates the two, but does not turn a reparametrization of the electron equation into a second independent derivation of all its coefficients.

6.8 Iteration of the ratio reader and phase recovery

The reader $R(x, y) = x/y^2$, applied to the positive representations of closure and self-scaling, admits the oriented composition

$$\mathcal{Q}_{\text{rat}}(x, y) = (R(x, y), R(y, x)) = (x/y^2, y/x^2), \quad x, y > 0.$$

This composition receives coordinates already produced by APP-TRIT-TPK. Its study retains both the product $P = xy$ and the oriented ratio $O = x/y$, whose transformations are

$$P \circ \mathcal{Q}_{\text{rat}} = P^{-1}, \quad O \circ \mathcal{Q}_{\text{rat}} = O^3.$$

The notation $\mathcal{Q}_{\text{rat}}$ distinguishes this map from the interval extension of the scalar reader.

Proposition 6.12 (Positive iteration and complex extension). *For $n \geq 0$, let*

$$a_n = \frac{(-1)^n + 3^n}{2}, \quad b_n = \frac{(-1)^n - 3^n}{2}.$$

Then

$$\mathcal{Q}_{\text{rat}}^n(x, y) = (x^{a_n} y^{b_n}, x^{b_n} y^{a_n}). \quad (6.26)$$

The same Laurent expression defines a map on $(\mathbb{C}^\times)^2$. For $n \geq 1$, this map is an unramified covering of degree 3^n , with kernel

$$\ker \mathcal{Q}_{\text{rat}}^n = \{(\zeta, \zeta^{-1}) : \zeta^{3^n} = 1\}. \quad (6.27)$$

Its restriction to the positive quadrant has a unique positive preimage for every positive pair.

Proof. In positive logarithmic coordinates, the matrix is

$$B = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}.$$

The vectors $(1, 1)$ and $(1, -1)$ have eigenvalues -1 and 3 . Decomposition along these two directions gives $B^n = \begin{pmatrix} a_n & b_n \\ b_n & a_n \end{pmatrix}$ and proves (6.26). Since a_n, b_n are integers, the identity extends as a Laurent identity.

Write $m = 3^n$, $\epsilon = (-1)^n$. For an image $(u, v) \in (\mathbb{C}^\times)^2$, the product of a preimage is fixed by $p = xy = (uv)^\epsilon$. The equations are equivalent to

$$x^m = u p^{-b_n}, \quad y = p/x.$$

The first has exactly m distinct roots. The second determines y ; moreover, the product of the two images is $p^\epsilon = uv$, so the second original equation also holds. Every fibre contains exactly m points. For $(u, v) = (1, 1)$, this gives (6.27); the other fibres are its translates in the multiplicative group.

The differential satisfies

$$\det D\mathcal{Q}_{\text{rat}}^n = (-3)^n (xy)^{\epsilon-1} \neq 0.$$

The local description by roots of a nonzero number provides the m local branches and proves the covering assertion. In the positive quadrant only one of these roots is positive. In particular, for one step,

$$x = (uv^2)^{-1/3}, \quad y = (u^2v)^{-1/3},$$

with the positive real roots. □

In the complex extension, the condition $xy = (uv)^{-1}$ couples the two cube roots in the one-step inverse. The coordinates (P, O) also identify (x, y) and $(-x, -y)$; their map has kernel $\{(1, 1), (-1, -1)\}$. The preceding proof retains (x, y) and therefore the distinction between this coordinate identification and the fibres of $\mathcal{Q}_{\text{rat}}^n$.

Proposition 6.13 (Reconstruction from the ternary register). *On the subgroup*

$$(x, y) = (e^{2\pi i \theta}, e^{-2\pi i \theta}),$$

the map induces $\theta \mapsto 3\theta \pmod{1}$. With representatives $\theta_j \in [0, 1)$, let

$$d_j = \lfloor 3\theta_j \rfloor, \quad \theta_{j+1} = 3\theta_j - d_j, \quad C_n = \sum_{j=0}^{n-1} d_j 3^{n-1-j}.$$

Exact recovery is given by

$$\theta_0 = \frac{\theta_n + C_n}{3^n}. \quad (6.28)$$

Proof. Substitution into the Laurent expression gives phase tripling. The recurrence $C_{n+1} = 3C_n + d_n$, with $C_0 = 0$, gives $\vartheta_n = 3^n \vartheta_0 - C_n$ by induction. This proves (6.28). Each d_j belongs to $\{0, 1, 2\}$, so n ternary digits retain the choice among the 3^n preimages of the reduced phase. The branches $B_d(\vartheta) = (\vartheta + d)/3$ satisfy

$$B_a \circ B_b(\vartheta) = \frac{\vartheta + b + 3a}{9}.$$

Thus two steps retain the ordered digit $3d_0 + d_1$. $\square$

The product returns after two iterations, whereas the oriented ratio is raised to the ninth power. The recovery proved here has this reader and its phase register as its domain. Its insertion into a representation of the complete enriched state additionally retains the maps of sheets, orientation, memory and survival belonging to that representation.

6.9 Exponential realizations of the generated coordinates

The regional coordinates already determined allow recognition of concrete realizations of the TRIT quadratic collection. The order is substantive: construction of the regions precedes evaluation of an exponential at parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not participate in selecting the prefixes that produce them.

6.9.1 Ternary cycle, nilpotent transport, and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, consisting of integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. On residue evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. Quotients of the integer lift are retained in the corresponding arithmetic record.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (6.15) gives

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the nilpotent-transport plane, and the modal-incidence plane. Sharing a matrix dimension does not identify these spaces or their coordinates.

Proposition 6.14 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 6.15 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\boxed{\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{\text{av}}^2.} \quad (6.29)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotency removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

Thus the exponential equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the complete turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use different parameters of one common exponential collection.

In this collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI) \exp(tI) = \exp((s+t)I)$. Its role is not restricted to the parabolic type. The coordinate e , previously obtained from its region, is recognized here as the unit evaluation of the exponential; this identity does not replace its coinductive genealogy.

6.9.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

Using Q keeps this operator square distinct from the dodecaphase record K .

Proposition 6.16 (Polynomial regime projectors). *The maps*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I , and recover the three summands of the representation.

Proof. The three polynomials take values $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

One thus obtains

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos tI + \sin t\mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh tI + \sinh t\mathbb{J}). \end{aligned} \quad (6.30)$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The total representation gathers the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation, and memory. The exponential articulation allows recognition of their types without replacing those dynamics by an identification of fibers.

6.10 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A sequence of blocks defines integer prefixes by $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting at $N_0 = m$, where m is the integer part determined by the reader. The preceding brackets place the closure, propagation, and self-scaling characters in (3, 4), (2, 3), and (1, 2), respectively; their initial values are therefore $m = 3, 2, 1$. The subsequent blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \quad (6.31)$$

The half-open convention selects a representative when a value lies on a boundary. Passage to the limit uses the closures of these intervals and identifies the two equivalent boundary expansions. This precision prevents the assumption that every intersection of nested half-open intervals is automatically nonempty.

Theorem 6.17 (Determination of the scalar realization). *A compatible block history determines a unique limit of the left endpoints of (6.31). If an exact character admits rational brackets of arbitrarily small width and is separated from the boundary of every decimal cylinder examined, each decimal prefix of finite length is determined after a finite calculation. The prefixes obtained in this way are mutually compatible.*

Proof. Extension by a block preserves the preceding prefix through Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints form a Cauchy sequence and have a unique limit. For a fixed decimal length, a value separated from the partition boundaries has positive distance from them. A sufficiently narrow bracket lies in a single cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; this decision does not follow from numerical width alone. $\square$

Corollary 6.18 (Finite determination and compatibility of regional prefixes). *For each realization π, φ, e and every finite decimal length, the constructed brackets determine the corresponding prefix by a finite calculation. Prefixes obtained at different depths are compatible under truncation.*

Proof. The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, irrationality of π, e, φ ensures their separation from the rational boundaries of every finite decimal partition. Apply the preceding theorem to each character. $\square$

Arbitrary depth expresses the quantifier in the corollary: for every finite horizon, there is a calculation determining its prefix. The mathematical object is the complete compatible collection; each execution obtains a finite projection of it.

6.11 Mirror involution and synchronization of regions

The nonadic connection transports the three characters jointly. In the ninth-phase chart, the self-scaling axis and the ternary sum of closure and propagation remain fixed under mirror exchange. Helical inversion changes the direction of transport; the mirror of the two channels changes their relative orientation. These are different involutions, although both act on the same system of trajectories.

The agreements of counts illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality of π , e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. In phases eight and nine the relative component vanishes, but the diagonal component changes from $(59, 59, 59)$ to $(43, 43, 43)$. Relative synchronization coexists with a displacement of -16 per channel. The closure leading to α must preserve both kinds of information: relative agreement and evolution of memory.

6.12 Sturmian helical suspension

The relation between bases nine and ten in the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \quad (6.32)$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic readout of refinement has an exact consequence.

Lemma 6.19 (Irrationality of the slope). *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. The factorization of the left-hand side contains the prime three, whereas that of the right-hand side contains only the primes two and five. Equality is impossible. $\square$

The lower mechanical word, with intercept θ , is

$$z_n(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod{1}$ through the interval $[1 - \delta, 1)$. The event sequence has density δ , and the sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, event separations are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word, of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first, rather than being a second free parameter.

6.12.1 Capacity index, vacancies, and induction of the tritic regime

The vacancy sequence is linked to refinement through an increment identity. To exhibit it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \quad \mathcal{C}_t = \lfloor (t+1)\rho \rfloor \quad (t \geq 0).$$

The index $\mathcal{C}_t$ expresses the decimal capacity of refinement in the normalization comparing a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphase record constructed later.

Proposition 6.20 (Complementarity of capacity and vacancy). *For $t \geq 1$,*

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t(0). \quad (6.33)$$

Consequently, if $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t(0) = b - a.$$

Proof. Irrationality of δ gives

$$\mathcal{C}_t = t + 1 - [(t + 1)\delta] = t - [(t + 1)\delta].$$

Subtracting consecutive terms produces (6.33). Summation over the interval telescopes. $\square$

Each transition increases the block counter by one. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity remains unchanged while a transition is incorporated into the history. Here a vacancy is a capacity-retention event, with a determined position and antecedent. The preceding balance preserves length exactly between acquired capacity and recorded vacancies. The observed phase $g_t^{\text{cap}} = [\mathcal{C}_t]_9$ satisfies

$$g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

This capacity phase and chronological phase $[t]_9$ are two different observations. Local capacity retention and a complete holonomy turn have distinct mechanisms, although both allow the phase observation to be preserved while the state changes.

The relation with TRIT can be followed in the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = \mathcal{C}_{p_j}, \quad h_j = p_{j+1} - p_j.$$

The inequality $p_j\delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j . Since $0 < \delta < 1$, it also gives $[(p_j + 1)\delta] = j$, and hence

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$ and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (6.34)$$

An interval of length 22 preserves the class; one of length 21 shifts it by one in the negative direction. The canonical selector is applied to the positive phase $\tau_j = 1 + [v_j]_9$:

$$\tau_j = M_{\text{ph}}(\tau_j).$$

A tritic regime is thus determined by the capacity index retained at each event, in addition to the vacancy's binary encoding.

The positions of the short intervals form the induced mechanical coding of slope $22 - 1/\delta$. Their separations 6 and 7 determine the lengths of plateaus of the class $[v_j]_3$, once the origin and endpoint convention are fixed. The first classes are 2 for 6 events, 1 for 7, and 0 for 7; the first segment of 6 is the initial restriction of a plateau. Their signatures traverse 0, -1 , $+1$. Equations (6.33) and (6.34) make explicit the link between refinement, vacancies, and regime selection. The quadratic reading of TRIT subsequently realizes these types through their respective generators; the temporal event and its geometric representation thus retain an identifiable dependence.

Inverting the monotone index supplies another useful observation:

$$\mathcal{T}(k) = \min\{t : \mathcal{C}_t = k\} = \left\lceil \frac{k}{\rho} \right\rceil - 1 \quad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks needed to gain m units of capacity is

$$\mathcal{T}(k + m) - \mathcal{T}(k) = m + [(k + m)\sigma] - [k\sigma] \in \{m + [m\sigma], m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, whereas return time measured by the other index admits two values. This distinction belongs to refinement dynamics and is not an a posteriori correction of its numerical values.

Theorem 6.21 (Complexity of phase readouts). *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. Preserving the phase $n \bmod 9$ gives complexity $p_9(m) = 9(m + 1)$; preserving only its ternary class gives $p_3(m) = 3(m + 1)$. All three topological entropies vanish.*

Proof. For factors of length m , the predecessors of the two endpoints of the coding interval form $m + 1$ distinct points on the circle: agreement between shifted endpoints removes duplications, and irrationality prevents further agreements. These points determine $m + 1$ distinct intervals of constant itinerary. Every irrational orbit visits each interval, giving $p(m) = m + 1$.

For a fixed phase modulo nine, the positions of that phase follow a rotation of step 9δ , which is also irrational. Density makes every factor occur in each phase, and the first coordinate distinguishes its nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m + 1)) \rightarrow 0$ for $c = 1, 3, 9$. $\square$

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has characters with frequencies

$$\frac{a}{9} + k\delta \pmod{1}, \quad a \in \mathbb{Z}/9\mathbb{Z}, k \in \mathbb{Z}.$$

The spectral module is not $\log 10$: it is the additive module generated by $1/9$ and δ , modulo the integers. The factor $\log 10$ belongs to the normalization in (6.32).

An operator presentation makes memory explicit. On $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n + 1\rangle$, $V_9|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n \delta}|n\rangle$, and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finitely supported vectors,

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = e^{2\pi i \delta} T W_\delta, \quad [D_M, T^9] = T^9. \quad (6.35)$$

The first two relations are Weyl covariances; the third expresses that one nonadic revolution increases memory degree. The base rotation is abelian, whereas the algebra of observables and translations is not. The inversion $|n\rangle \mapsto |-n\rangle$ conjugates forward and backward motion. This realization provides an aperiodic helical suspension with finite phase return, rather than a periodic word of length nine.

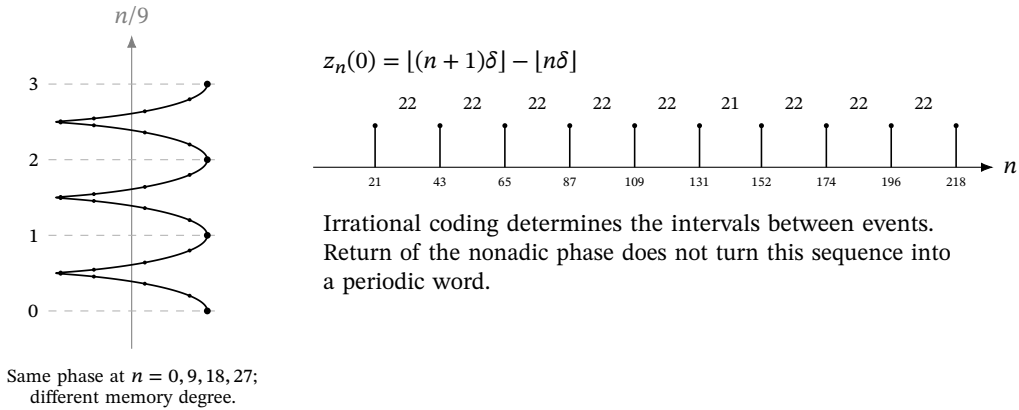


Figure 7: Complementary representations of transport. Left: projection of the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$: advancing nine steps recovers the phase and increases the height. Right: the first ten events of the lower word for $\delta = \log_{10}(10/9)$, with exact separations 21 or 22. The helical representation introduces no additional physical metric.

For this count, fix the intercept $\theta = 0$ and the cumulative endpoints

$$(N_0, \dots, N_4) = (0, 729, 972, 999, 1000),$$

corresponding to the cubic order $(729, 243, 27, 1)$. The lower counts are $\lfloor N_j \delta \rfloor - \lfloor N_{j-1} \delta \rfloor$, giving $(33, 11, 1, 0)$. The upper counts are $\lfloor N_j \delta \rfloor - \lfloor N_{j-1} \delta \rfloor$, with the same initial boundary zero, giving $(34, 11, 1, 0)$. These values depend on the intercept and the ordering of the strata; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a

single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in a common count preceding any unit chart; identifying them with physical exponents additionally requires the corresponding dimensional derivation.

The term *aperiodic crystal* is used here for this symbolic order, with specified complexity, spectrum, and return. Zero topological entropy does not by itself amount to zero physical dissipation: a thermodynamic realization needs defined energy dynamics and an erasure operation. Likewise, this Sturmian readout does not replace all TPK operators or by itself prove periodicity or aperiodicity of the complete output of α . Its function is to make explicit an order structure that simple schedule return does not describe.

6.13 Nonadic connection, holonomy, and monodromic representation

The already constructed regional connection is now studied through three objects: transport of the complete state, the representation of its turns, and the suspension of its itineraries. The APP-TRIT-TPK operations and the initial catalog retain their preceding definitions. The necessary distinction here concerns what each representation retains of the same history (Sections 4 and 6).

At prolongation index k , a state chart preserves

$$\hat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k brings together the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry record, Σ_k the boundary signature, and W_k the incidence information. The phase is $g_k = 1 + [(k-1)]_9$; m_k counts complete turns. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphase position, whereas the integer m_k distinguishes its successive turns.

Definition 6.22 (Transport through one nonadic turn). Let $\mathcal{R}_k \subseteq \hat{X}_k \times \hat{X}_{k+1}$ be the admissible-extension relations of the TPK. The realization of holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation to s turns is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves admissible continuations. On an individualized history, each arrow leads to the state actually prolonged; the isolated visible block may still admit more than one continuation. This difference is precisely the function of memory and boundary conditions in determining the trajectory.

Proposition 6.23 (Iteration of return and accumulated memory). *In every admissible composition of s turns,*

$$g_{k+9s} = g_k, \quad m_{k+9s} = m_k + s, \quad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-turn rule preserves phase and increases the counter by one unit. Applying it to the already transported state repeats exactly those two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. $\square$

The counter thus provides an integer witness to the difference between turns. After 12 turns the pair of finite phases is restored, but the counter increases by 12. The nonsplit extension (4.5) expresses the composition of the calendar through its cocycle; the trajectory additionally retains the integer lift of that datum. Conservation here means transport of previous contributions, not immobility of every coordinate.

6.13.1 Bilateral representation of the return index

The shift T in (6.35) admits an expression in phase and turn coordinates. We number phases from 0 to 8, by shifting the origin relative to g_k , and write $k = 9m + r$, with $0 \leq r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\text{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ expresses that same unitary shift

$$T|r, m\rangle = \begin{cases} |r+1, m\rangle, & r < 8, \\ |0, m+1\rangle, & r = 8. \end{cases}$$

The inverse reduces r by one unit when $r > 0$, and sends $|0, m\rangle$ to $|8, m-1\rangle$. Passage through the phase origin therefore incorporates the exact quotient unit.

Proposition 6.24 (Representation of the monodromy of the cycle). *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m+1\rangle, \quad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\text{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the turn group of the cycle.

Proof. Nine shifts preserve the remainder of division by 9 and increase the quotient. The powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m+s\rangle$ distinguishes every integer s . A permutation of an orthonormal basis is unitary. $\square$

This representation corresponds to the covering of the cycle, whose turn group is $\mathbb{Z}$, and represents phase together with its counter. The TPK state additionally contains the trajectory coordinates already described. The bilateral extension admits negative indices as coordinates of the covering; a trajectory starting from an initial condition uses its admissible half-axis. This provides an operator inverse of the index without turning every extension relation into a bijection of the complete state.

With $D_M|k\rangle = \lfloor k/9 \rfloor |k\rangle$, the identity $[D_M, S] = S$ expresses the memory increment on the dense domain of finitely supported vectors. If $\mathcal{J}|k\rangle = |-k\rangle$ and P_0 projects onto indices divisible by 9, one additionally obtains

$$\mathcal{J}T\mathcal{J} = T^{-1}, \quad \mathcal{J}D_M\mathcal{J} = -D_M - (I - P_0). \quad (6.36)$$

Indeed, for $k = 9m + r$, $\lfloor -k/9 \rfloor = -m$ if $r = 0$ and equals $-m-1$ if $r > 0$. Inversion thus preserves the boundary term of Euclidean division.

6.14 Block clock and capacity clock

The index of an application of refinement and the number of available decimal positions have related but different laws. To express them, we reserve n for the ternary-block index and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho, \quad \mathcal{C}_n = \lfloor (n+1)\rho \rfloor, \quad n \geq 0.$$

The letter $\mathcal{C}_n$ here designates capacity, not the dodecaphase record K . At the canonical phase origin, the preceding mechanical word provides

$$Z_n = \sum_{j=0}^n z_j(0) = \lfloor (n+1)\delta \rfloor.$$

Proposition 6.25 (Exact balance between capacity and vacancies). *For $n \geq 0$ and, in the second identity, $n \geq 1$,*

$$\mathcal{C}_n = n - Z_n, \quad \mathcal{C}_n - \mathcal{C}_{n-1} = 1 - z_n(0).$$

Proof. The irrationality of δ implies $\lfloor (n+1)\delta \rfloor = \lfloor (n+1)\delta \rfloor + 1$. Then

$$\lfloor (n+1)(1-\delta) \rfloor = n+1 - \lfloor (n+1)\delta \rfloor = n - Z_n.$$

Subtracting two consecutive identities completes the proof. $\square$

An event $z_n = 1$ adds a block to the history while maintaining capacity $\mathcal{C}_n$; an event $z_n = 0$ increases both counters. In any interval, capacity increments and vacancies sum to exactly its length. Reducing modulo q , for $q = 9$ or $q = 108$, gives

$$[\mathcal{C}_n]_q = [n]_q - [Z_n]_q.$$

The uniform block phase and the capacity phase are related through the integrated memory of vacancies. In particular, a retention of capacity and a complete turn of the connection have different mechanisms, although both may preserve a phase observation.

Monotone inversion of the clock gives a first-arrival time:

$$T_{\text{cap}}(a) = \min\{n : \mathcal{C}_n = a\} = \left\lceil \frac{a}{\rho} \right\rceil - 1, \quad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time taken to gain b units of capacity is

$$T_{\text{cap}}(a + b) - T_{\text{cap}}(a) = b + [(a + b)\sigma] - [a\sigma].$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. Thus 9 or 10 blocks are obtained to gain 9 units of capacity, and 113 or 114 to gain 108. These are times of the inverse clock, measured from a first arrival; the turn counter is always interpreted in the index to which it belongs.

6.15 Symbolic suspension and aperiodic time crystal

Rotation coding combines phase periodicity, local recurrence, and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0, 1\}^{\mathbb{Z}}$ be the closure, in the product topology, of the translates of the bilateral word $z_n(\theta)$. Denote by σ_{sh} the shift of its sequences. Retaining a nonadic phase, the step is

$$F(r, z) = ([r + 1]_9, \sigma_{\text{sh}} z), \quad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 6.26 (Symbolic aperiodic time crystal). The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = ((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}) / \sim, \quad (y, t + 1) \sim (Fy, t),$$

with flow $\Phi_s[y, t] = [y, t + s]$. This model of temporal order, together with its vacancy coding and representation of the turn counter, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and its return section; aperiodicity characterizes its itineraries. The frequency of ones in every sequence is δ : the uniform discrepancy bound of less than one, obtained by telescoping in any window, persists on taking the closure. A periodic sequence would have rational frequency. Therefore, $\mathcal{X}_\delta$ contains no periodic orbits. The suspension also has no closed orbits: a return of the flow would require an integer time and periodicity of the itinerary.

Finite words do recur. Itineraries of length m arise from a partition of the circle by $m + 1$ points; each interval is visited recurrently by the irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why every word appears in all 9 phases, and explains the already established complexities

$$p(m) = m + 1, \quad p_9(m) = 9(m + 1), \quad p_3(m) = 3(m + 1).$$

Recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

Realization through the observables V_δ, W_δ and shift T expresses the same order through the covariances (6.35). Frequencies of the uniform clock belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[\mathcal{C}_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by the uniform phase when interpreting returns. The term *time crystal* is thus defined through a mathematical dynamical system. Its physical realization requires specification of an energy dynamics, its temporal observables, and its stability, data distinct from the present symbolic construction.

6.16 Specular symmetry of the regions and double realization

Specular exchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$\mathfrak{c}B = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixation refers to the involution within a fiber; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$, and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the distribution that the aggregate preserves without distinguishing it. The inversion $\mathcal{I}$ of the temporal index and the channel involution $\mathfrak{c}$ consequently act on different data.

Phase return has an explicit witness:

$$\begin{aligned} B_1 &= (012222 \mid 121221 \mid 112202), \\ B_{10} &= (101222 \mid 112221 \mid 112002). \end{aligned}$$

Both positions have phase 1; the block, axis, and memory have changed. Likewise, synchronization of the censuses at phases 8 and 9 cancels their relative differences, while the diagonal component passes from $(59, 59, 59)$ to $(43, 43, 43)$. Equality of a relative observable does not exhaust the state determining it.

Double realization preserves that architecture. In Section 16.1, the same prefix determines nested cylinders on the one hand and a sequence of coded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞ . Figure 8 anticipates these two maps. The subsequent incidence development constructs codes, lattices, and the realization of Moonshine in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

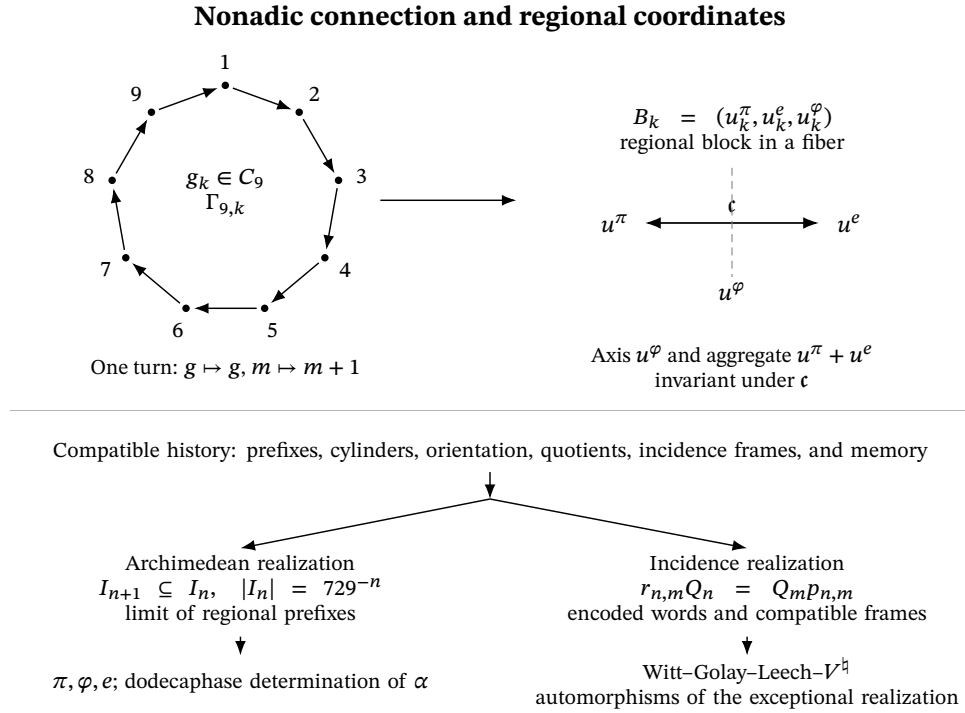


Figure 8: Nine phases of a connection and two realizations of a compatible history. The cycle represents phase; each turn increments the counter. The regional diagram represents the involution c , which interchanges closure and propagation and fixes the self-scaling axis in a fiber. This invariance does not require the axis to remain constant during transport. The two lower maps use the same prefix: one determines its numerical limit, while the other preserves words and incidence frames. The maps of the second realization are specified in Section 16.1; the exceptional construction is developed afterwards.

6.16.1 Rectilinear representation of phases and return

The representation on the segment $[0, 10]$ preserves the positional origin of APP and orders its nine interior marks. In this diagram, the integers $1, \dots, 9$ are phase representatives. Horizontal position indicates the order of the charts; upper labels specify regional operations. Figure 9 thus brings together the initial line, fixation of the self-scaling axis, the specular distribution of the two channels, and return with memory (Section 6.13).

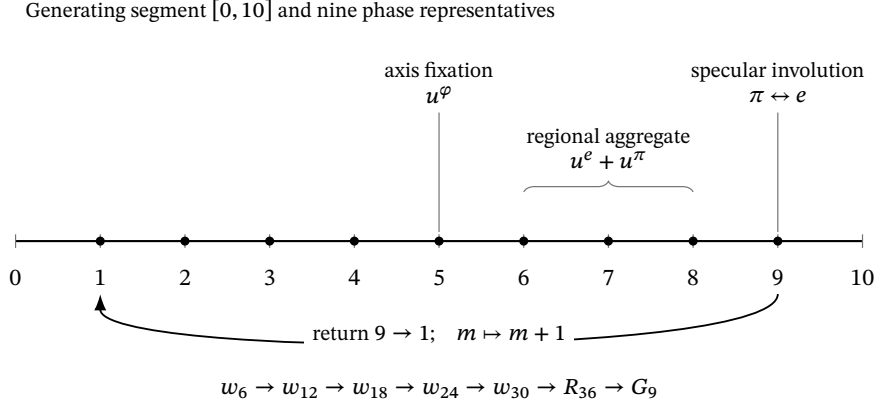


Figure 9: Rectilinear representation of the nonadic connection. The mark 5 locates the first strong saturation and the regional self-scaling axis; the brace 6–8 identifies the ternary aggregate fixed within each fiber; phase 9 selects the closure orientation. Return preserves phase and increases memory. The abscissae number operational positions, while the labels u^π, u^e, u^φ designate blocks of $\mathbb{F}_3^6$.

Fixation of the axis is formulated in the fiber of a given boundary. If $q = u^\pi + u^e + u^\varphi$ and $a = u^\pi + u^e - u^\varphi$, then

$$u^\varphi = 2(q - a), \quad s = u^\pi + u^e = q - u^\varphi \quad \text{in } \mathbb{F}_3^6.$$

The data (q, a) determine the axis and aggregate; the ordered distribution of s retains the specular freedom resolved by prolongation. The fifth phase has a local solution and an extension, with boundary datum $c = 111111$ and residual signature $\rho_W = 000$. This is the meaning of its first strong saturation. The symbol φ placed above the mark 5 identifies the regional axis entering that operation.

The intermediate phases preserve this decomposition within each fiber, while their boundary data evolve. The complete census of local multiplicities and extensions of the distinguished branch is

phase	1	2	3	4	5	6	7	8	9
N_{loc}	3	2	5	4	1	1	3	3	2
N_{ext}	2	5	4	1	1	3	3	2	1

Phase 6 combines local uniqueness and three prolongations; phase 7 resolves a triple specular fiber; phase 8 synchronizes the three regional censuses. Phase 9 preserves two local distributions with the same axis and aggregate, one of which prolongs. These counts express different functions within a single connection and justify the marks on the line.

In particular, for phases 6, 7, 8, the regional sums are respectively 012100, 101001, and 112000. The conservation represented by the brace is an invariance under the specular distribution within each fiber. Subsequent real evaluation applies the readers of compatible cylinders to complete histories; the internal sum represented in the figure belongs to the ternary domain.

Finally, the arc $9 \rightarrow 1$ represents passage from one turn to the next. The formulas of Section 6.13 preserve the phase remainder and increase the turn counter. The rectilinear diagram and the cyclic diagram in Figure 8 therefore express two complementary aspects of the same holonomy: the order of positions and composition of the return on the state with memory.

6.17 Temporal transport and tritic exponential composition

The Euler relations of Section 6.9 describe local realizations of the three regimes. For a fixed signature τ , the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation sends J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau) \exp(tJ_\tau) = \exp((s+t)J_\tau), \quad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is unity. In the elliptic type this is realized through $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, through $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I + tN)(I - tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

The elliptic realization closes a turn; the nilpotent one preserves a linear displacement; the hyperbolic one separates expansion and contraction with unit determinant. At the already generated regional parameters,

$$\exp(\pi J_e) = -I, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.$$

The evaluation $\exp(I) = eI$ recognizes unity in the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transport of a local realization by an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U \exp(tJ) = \exp(tJ')U$. The identity transports the regime and its norm. A transition between different quadratic types belongs, by contrast, to the TPK selection of sheets and regimes; it is not a conjugation changing the value of J^2 . The temporal product therefore preserves the order of transitions and the information at their endpoints.

This articulation distinguishes closure of a local representation from holonomy of a history. The identity $\exp(2\pi J_e) = I$ may hold at the same time as $S|r, m\rangle = |r, m+1\rangle$. The first equality restores the orientation of the regime; the second records another turn. The discrete structure preserves both: local return and the accumulated provenance of the state realizing it.

6.18 Multiplicative irreducibility and return periods

The distinction between structure, evaluation, and phase also appears in the multiplicative sector. APP's local product contains both residue and quotient; iterating it constructs normal forms and factorizations before their decimal periods are examined. This development places the expression *prime mode* in a precise domain and relates irreducibility to the operations already defined, rather than to an isolated coincidence of periods (Section 6.18).

Let

$$\mathcal{W}_9 = \{\emptyset\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \dots, 9\}^\ell.$$

Sequences are ordered from the least significant digit. The evaluation $\text{ev}_9(u) = \sum_j u_j 9^j$, with $\text{ev}_9(\emptyset) = 0$, is bijective onto the nonnegative integers: for a positive integer, division of $n-1$ by 9 uniquely determines the first digit and continuation. This is bijective nonadic numeration; it retains the boundary representative 9.

Addition $\boxplus_\#$ is obtained by applying APP's additive sheet to the digits at each position and propagating their quotients. If there is only one digit, it is the provisional residue; a nonzero incoming quotient is normalized through a second additive operation. The sum of quotients is transported to the next position. To multiply a sequence by a digit d , the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated by

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j > 0. \end{cases}$$

Start with $c_0 = 0$ and incorporate the final quotient when nonzero. During addition, $c_j \leq 2$; during multiplication by a digit, $c_j \leq 9$. Local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_\# u$, with $d \odot_\# \emptyset = \emptyset$.

For $v = (v_0, \dots, v_{m-1})$, the complete product is constructed through the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot_{\#} a_{j+1}) \boxplus_{\#} (v_j \odot_{\#} u), \quad u \boxtimes_{\#} v = a_0.$$

The rule composes both local sheets. Its evaluation satisfies

$$\text{ev}_9(u \boxtimes_{\#} v) = \text{ev}_9(u) \text{ev}_9(v). \quad (6.37)$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the power of 9. The recurrence then gives $\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u)$, whose iteration proves (6.37). Uniqueness of normal form transports associativity and commutativity to the constructed product; its identity is (1).

Definition 6.27 (Reduced multiplicative coproduct). On the finitely supported vector space generated by nonempty normal forms, define

$$\tilde{\Delta}_{\#} e_w = \sum_{\substack{u \boxtimes_{\#} v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A non-unit normal form is irreducible when its image under $\tilde{\Delta}_{\#}$ is zero.

Factorization fibers are finite by (6.37). In the completion $\mathcal{H}_{\#} = \ell^2(\mathcal{W}_9 \setminus \{\emptyset\})$, images of distinct forms have disjoint supports. If $d_{\#}(w)$ counts ordered non-unit factorizations, the domain of the closure is

$$\text{Dom}(\overline{\tilde{\Delta}_{\#}}) = \left\{ x \in \mathcal{H}_{\#} : \sum_w d_{\#}(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is this weighted sum. Hence the operator

$$A_{\#} = (\overline{\tilde{\Delta}_{\#}})^* \overline{\tilde{\Delta}_{\#}}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered non-unit factorizations of w . The projection

$$P_{\text{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|) \mathbf{1}_{\{0\}}(A_{\#})$$

selects exactly the irreducible forms. Through multiplicative evaluation, these correspond to ordinary primes. Selection depends on factorization structure and keeps the unit state separate.

For a value $n \geq 2$ coprime to 10, decimal division induces the residue permutation $r \mapsto 10r \pmod{n}$. Return of residue 1 has period

$$\tau_n = \text{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \text{lcm}_{p^{a_p} \parallel n} \text{ord}_{p^{a_p}}(10).$$

Composite return synchronizes the returns of prime powers. For a prime $p \notin \{2, 3, 5\}$,

$$\tau_p \mid p-1, \quad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \quad M_p \equiv 0 \pmod{9}.$$

The last congruence expresses nonadic closure of decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for every n coprime to 90. In particular, 7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This differentiation specifies the relation with aperiodic refinement. The vacancy sequence determines increments of the capacity index; factorization determines multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, while quotient and factorization retain information that this closure does not distinguish. The connection to the treatise's spectral constructions resides in these operators and their intertwining. Proof of their subsequent analytic results retains its own development; it is not replaced by periodicity of a residue observation.

6.18.1 Vacancy discrepancy and Dirichlet readings

The connection also admits an exact analytic expression. Retain the same coding $z_n = z_n(0)$, for $n \geq 1$, and consider two subsequent observables: the series over all indices and the product over indices selected by irreducibility. For $\text{Re } s > 1$,

$$Z_{\text{vac}}^+(s) = \sum_{n \geq 1} \frac{z_n}{n^s}, \quad Z_{\text{vac}}^\times(s) = \prod_p (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in this half-plane. Their common factor is density δ , but their discrepancies differ.

Proposition 6.28 (Separation of vacancy density). *We have*

$$Z_{\text{vac}}^+(s) = \delta \zeta(s) + H_{\text{vac}}(s), \quad H_{\text{vac}} \in \mathcal{O}(\text{Re } s > 0).$$

In $\text{Re } s > 1$, with logarithms defined by absolutely convergent Euler series,

$$Z_{\text{vac}}^\times(s) = \zeta(s)^\delta G_{\text{vac}}(s), \quad \log G_{\text{vac}}(s) = \sum_p (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centered partial sums satisfy

$$A_N := \sum_{n=1}^N (z_n - \delta) = \lfloor (N+1)\delta \rfloor - N\delta, \quad |A_N| < 1.$$

Abel summation represents the centered term by $s \int_1^\infty A_{\lfloor x \rfloor} x^{-s-1} dx$. The integral converges locally uniformly in $\text{Re } s > 0$, proving holomorphy. The second identity follows by separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. $\square$

The decomposition reveals what the reading on primes adds: it examines the same vacancy sequence on a subset determined by factorization. Controlling discrepancy over all integers gives the first holomorphic continuation; the second observable preserves its own discrepancy on primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation explains their relevance to the arithmetic kernel without transferring the proof of the treatise's terminal results to this article.

K

Holographic arithmetic–geometric coupling constant

Oriented extraction, positional readout and incidence of the generated register.

What the constant couples

The holographic arithmetic-geometric coupling constant is introduced at the point where its register is produced. Its integral representation is denoted by K ; its periodic reading, by κ_{ag} . The phase origin, orientation, and order of the components form part of the definition. The word “constant” names the object determined by this construction and by its readers, which will be presented through explicit operations.

The extraction brings together the oriented channels and a sum datum. Hadamard inversion reconstructs their components in the natural order. Each coefficient of that inversion comes from the declared operator and its integral image. In this way, the register is determined before it is used in the closure of the fine-structure constant. The positional reading then converts the twelve components into a periodic rational expression. Once the base and length are fixed, this conversion admits an exact inverse that restores all the blocks.

The same ordering has a geometrical function. Its readers distinguish a marked tetrad and an incidence configuration; the signature associated with alpha selects a hexad within the corresponding design. The relation to the numerical regions uses the alignment and lifting maps that will be proved in the exceptional part. The periodic reading and the incidence configuration thus receive a common source, through operations of different types.

The subsequent operator development adds a third function. The cyclic shifts of the register form a basis of its carrier. Responses on this basis determine an operator, and the Gram form provides a stability bound for its recovery. When the specified cyclic symmetry is present, one complete vector-valued response makes it possible to reconstruct the others. The isometric archive preserves this capacity for identification.

This is the content of the coupling that structures the exposition: reversible arithmetic encoding, marked incidence, and operator identification. The register is constructed first; what each reader obtains from it is then proved. Recovery theory will be developed once its spaces and transports are available. This position explains both the centrality of the constant and the hierarchy of the proofs that use its different realizations.

7 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

7.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lceil su_c(j) \rceil - 1 = \lfloor sv_c \rfloor. \quad (7.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 7.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lceil su_c(j) \rceil - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (7.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 7.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (7.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (7.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lceil su_c(j) \rceil - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (7.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 7.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (7.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (7.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (7.5) and its residual bound. $\square$

7.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (7.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (7.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 7.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (7.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 7.2. For the second, apply (7.4) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (7.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (7.8) determines its compatibility with every subsequent extension exactly.

Example 7.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (7.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

7.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (7.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 7.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$\begin{aligned} L_0 &= \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, & L_1 &= \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \end{aligned}$$

Both matrices are invertible.

Proof. The matrices

$$\begin{aligned} A_0 &= \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 7.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

7.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (7.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 7.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (7.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (7.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (7.13)$$

Lemma 7.8 (Integer criterion for intersection). *The cylinders in (7.10) have a nonempty intersection if and only if (7.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (7.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (7.13). $\square$

Definition 7.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N+u, D), \\ T_{u,d}(n, k, N, D) = (n+1, k+1, 729N+u, 1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 7.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (7.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (7.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. □

Theorem 7.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)} s_c(n, k) &= s_c(n + 1, k), \\ T_{u_c(n), d_c(1000, k)} s_c(n, k) &= s_c(n + 1, k + 1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 7.2 places v_c simultaneously in both intervals in (7.10); the intersection criterion gives compatibility. The update identities are (7.5) in each base. Recovery follows from (7.4). Thus the defect recurrence and prefix recovery are consequences of the same update. □

Example 7.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

7.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 7.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$\lfloor s\ell_c(j) \rfloor \leq 729P_c(729, n) + u \leq \lceil su_c(j) \rceil - 1. \quad (7.16)$$

Theorem 7.14 (Uniqueness of the prolongation compatible with the reader). *Condition (7.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (7.16) coincide with $P_c(729, n + 1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n + 1)$. By (7.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

7.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (7.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (7.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (7.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (7.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 7.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (7.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (7.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (7.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (7.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (7.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (7.23). $\square$

Example 7.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^{\text{W}} &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 7.17 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (7.24)$$

Proposition 7.18 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (7.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

7.7 Composition theorem for the block, cylinder, and signature

Theorem 7.19 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (7.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (7.14) and (7.15);*
2. *its six trits produce the signature $\mathcal{S}(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (7.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 7.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 7.11 to be applied. The deterministic application of (7.6) produces the bands used by (7.17)–(7.20). The recovery identities are those of Theorem 7.15. Finally, Theorem 7.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

7.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \tag{7.26}$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl, pr, au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \tag{7.27}$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (7.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

8 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

8.1 Regional bands, encoding, and integer signatures

8.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (8.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (8.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (8.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

8.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (8.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (8.4)$$

$$a_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (8.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (8.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 8.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

8.2 Conversion clock and first zero advance

8.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (8.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 8.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (8.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 8.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

8.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (8.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

8.3 Oriented construction of the twenty-four-trit descriptor

8.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative-positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

8.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are $(5, 7, 7, 7, 7, 7)$; this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (8.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase-charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (8.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

8.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\Omega_{24} = (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \quad (8.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 8.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\ 222111\ 211201\ 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (8.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (8.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

8.4 Functional panel and terminal repertoire

8.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (8.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

8.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (8.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its 8! orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (8.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

8.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 7: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 7,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 8.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 8, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 8: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

8.5 Comparative readers, affine reader, and temporal incidence

8.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (8.14)$$

Lemma 8.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (8.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

8.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (8.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

8.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \text{ in } \mathbb{F}_3^6. \quad (8.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

8.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu, o, s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (8.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (8.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 8.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v, u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v, u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (8.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

8.6 Enumeration of cylinders and decimal candidates

8.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \quad (8.19)$$

8.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 8.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lceil \frac{pD}{Q} \right\rceil \leq N < \left\lceil \frac{(p+1)D}{Q} \right\rceil \right\}. \quad (8.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q]$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lceil pD/Q \rceil \leq N$; the second is equivalent to $N < \lceil (p+1)D/Q \rceil$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 8.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 8.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (8.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

8.7 Exceptional incidence and heterotypy condition

8.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (8.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (8.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (8.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

8.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (8.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (8.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (8.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

8.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 &\rightarrow 4, & 1 &\rightarrow 5, & 2 &\rightarrow 6, \\ 3 &\rightarrow 7, & 4 &\rightarrow 8, & 5 &\rightarrow 9, \\ 6 &\rightarrow 10, & 7 &\rightarrow 11, & 8 &\rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

8.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) = & [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ & \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \tag{8.26}$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 8.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_3$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{8.27}$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (8.22); the destination bands T_8 , T_{12} and residues $R_{4,8}$, $R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (8.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (8.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

8.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_2 = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

020022	222111	211201	021101
002001	020111	200110	121012
010122	001001	221110	222110
121020			

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

8.8 Regional composition and continuity of the realizations

8.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{8.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (8.13).

The register is then formed by the twelve Euclidean divisions in (8.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

8.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

8.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

8.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

8.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbitalSelection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (8.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean’s native evaluation; their reports record `Lean.ofReduceBool`, in addition to the library’s customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

9 Dodecaphasic extraction of the arithmetic–geometric coupling constant

Definition 9.1 (Holographic arithmetic–geometric coupling constant). This term denotes the ordered finite register produced by the oriented extraction of the APP–TRIT–TPK state, with its declared phase origin, normalization, and readers, whose positional representation enters the arithmetic closure and whose incidence and geometric realizations act on the same register. The symbol K denotes its canonical integral representation; κ_{ag} denotes its periodic rational readout in the specified chart.

Generation of the regional coordinates does not exhaust the information in the APP–TRIT–TPK state. One Archimedean readout determines a value through compatible cylinders; another preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphasic register belongs to this second readout before entering the closure of alpha. It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A word of twelve catalogue blocks, a list of twelve oriented events, a signed integer vector, and a base-one-thousand dodecaphasic register are distinct objects. Their comparison requires the maps that relate them. Equality of length does not provide these maps, and loss of signs is not merely a notational simplification.

This chapter retains the causal boundary established in the preceding chapters: the nine APP positions on the horizontal and vertical axes produce the arithmetic sheets; TRIT specifies regime and orientation; TPK selects, transports, updates, and prolongs, preserving residue, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite calendar. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ reversed: the last map is a reduced memory readout, not the generator of the signed register.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence between three coordinatizations of a register previously produced by APP–TRIT–TPK. The Hadamard transformation relates the signed register to the integer register; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12}/4} K \xrightleftharpoons[(D_3, D_4, Q)]{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. Once the base 1000, the 12 positions and the phase origin are fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplication by $1000^{12} - 1$ followed by Euclidean divisions recovers the complete register. The proofs in §9.8, §9.9 and §9.11 establish these equivalences in their respective domains.

Two compositions then act on that register. The positional composition combines its components with the closure, propagation and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and obtains

$$K \longrightarrow B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The complete register allows both compositions to be reconstructed. The support B_K specifically preserves membership in the class selected by the threshold. The identity $B_K = H_e \cap P_\pi$ in (16.20) links this integer extraction with the incidence of the regional codes. The negative support of its balance with K completes the flag $B_K \subset H_\alpha^-$ developed in the exceptional extension. Thus, the role of K is defined through an invertible transformation and precise subsequent maps; the additional information on depth and transport remains in the state from which the register arises.

9.1 Cyclic extension and memory grading

The observable calendar contains one hundred and eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \dots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the cyclic translation law. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphasic sector from the internal nonadic position. Neither preserves the total number of turns on its own.

Proposition 9.2 (Calendar carry). *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$, the defect of additivity is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and coincides with the image of ι . Reduction is surjective, and multiplication by nine is injective on the stated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{lcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the stated carry and verifies the section identity. $\square$

The result explains a distinction that will recur for alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts the nonadic turns, the calendar preserves only $\beta = [q_{\text{mem}}]_{12}$. After one hundred and eight steps, β returns, but q_{mem} increases by twelve. The register also preserves the cylinder, the incidence register, and the vertical memory. Closure of a finite chart does not turn this history into a cycle without memory.

9.2 The oriented register and the integral 1 + 5 + 6 chart

For a history produced by TPK, let τ_m be the subroute over E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence entry. The register is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection to the calendar forgets precisely some of the data needed by the signed reader.

An initial chart of the register can be written in terms of oriented events. Oriented extraction specifies the set of event codes $\{2, 4, 6, 8\}$ and the sector orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the direction code at step t , we obtain

$$e_i = \Sigma_{12}(i+1) \# \{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event count separate from the orientation. It does not identify a cardinal direction with a TRIT sign. The vector e , whose components are signed local counts, is not yet the vector U that will appear below.

Opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

we define the integral chart

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The decomposition $1 + 5 + 6$ is not an arbitrary allocation of coordinates: one coordinate preserves charge, five compare the pairs relative to a reference pair, and six preserve the opposition of the half-crowns.

Lemma 9.3 (Inversion of the integral chart). *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, which gives the division by six. Once the p_i have been reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the stated solution. This solution is integral exactly under the parity congruence. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not justify omitting the remaining material. The complete register has a larger domain than this chart of counts. Extraction of the channels feeding U uses that complete register and cannot be deduced from the list (e_i) by identifying names.

9.3 Incidence evaluation preceding the dodecaphasic register

Recovering a register from its cyclic differences and total charge is a different question from evaluating those coordinates on the incidence register. The first problem is solved by the transverse extractor of Proposition 9.13. For the second, a historical presentation of the register provides three integer counts in each window. It is useful to make their composition explicit and determine exactly the information required for their evaluation.

Let $\mathcal{L}$ be a previously generated ledger of visits and oriented traces. Each visit retains the path, APP position, sheet, time, multiplicity, and boundary incidence. Arithmetic evaluation produces the integer and its decomposition

$$n_t = r_t + 9q_t, \quad 1 \leq r_t \leq 9.$$

The integer n_t is computed from the sum or product sheet; its residue is not received as an independent value. The distinction between corona incidence and interior incidence remains explicit for visits with residue 9.

In window E_m , the count-based presentation considers

$$A_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r(v)), \quad (9.1)$$

$$C_m = \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \quad (9.2)$$

$$V_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{9\}}(r(v)) \epsilon_\partial(v). \quad (9.3)$$

Here $w(v)$ is the incidence multiplicity, and $\epsilon_\partial(v) = 1$ on the corona, -1 in the interior. The traces counted by $\nu^\pm$ must be enumerated by the corresponding rule in the article. The integer sum requires finite support or an alternative construction determining its meaning. The number of chronological incidences and the reduced length $120(1 + 3j)$ have different types; relating them requires the reader's unit of length.

Define $[x]_{10} \in \{0, \dots, 9\}$ as the Euclidean representative, retaining the quotients

$$x = [x]_{10} + 10[x/10].$$

The incidence block is

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}. \quad (9.4)$$

Composition yields the map

$$\mathcal{E}_{\text{cnt}}(\mathcal{L}) = \left((z_m - z_{m+3})_{m=1}^{12}, (z_m - z_{m+4})_{m=1}^{12}, \sum_{m=1}^{12} z_m \right). \quad (9.5)$$

Indices are read cyclically modulo 12. In this subsection, $(D_j z)_m = z_m - z_{m+j}$, for $j = 3, 4$, and $Q(z) = \sum_{m=1}^{12} z_m$; hence the preceding composition is $(D_3 z, D_4 z, Q(z))$. This convention is retained in §9.9, where complete integral reconstruction is proved. This formula produces its coordinates from the counts. It does not use K , U , α , or the expected transverse coordinates as arguments. Identifying it with $\mathcal{E}_{108}^{90,120}$ requires composing it with the actual collection and incidences of the terminal state; this does not follow solely from invertibility of the transverse system.

Proposition 9.4 (Necessary and sufficient arithmetic information). *Let $N, N' \in \mathbb{Z}^{12 \times 3}$ be ordered by the counts (A, C, V) . For the map defined by (9.4)–(9.5),*

$$\mathcal{E}_{\text{cnt}}(N) = \mathcal{E}_{\text{cnt}}(N') \iff N - N' \in 10\mathbb{Z}^{36}.$$

In particular, the 36 sectorial residues determine exactly the output of 25 coordinates. Retention of the quotients and provenance constitutes additional information in the register.

Proof. Equality modulo 10 produces the same blocks and hence the same transverse coordinates. Conversely, if the transverse coordinates agree, $D_3(z - z') = D_4(z - z') = 0$. The steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of total charges makes that constant zero. Finally, the decimal representation of an integer between 0 and 999 has three unique digits. This recovers the three residues of each window. The statement is an algebraic identity for every N, N' , not an inference drawn from finite tests. $\square$

9.4 Conjugation of counts and boundary terms

Let $\rho(m) = 13 - m$. Consider an incidence conjugation that reverses the order of the windows, preserves the arithmetic and boundary classes of visits, and exchanges the trace channels. Then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

If $c_m = [C_m]_{10}$, decimal reduction produces the identity

$$z_m^\dagger = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \quad (9.6)$$

Indeed, $[-C]_{10} = 0$ when $[C]_{10} = 0$, and $[-C]_{10} = 10 - [C]_{10}$ otherwise. The formula retains exactly that difference. Negating the channel is not equivalent to negating the entire decimal block.

Applying this identity to a concrete TPK involution requires checking its action on incidences. For an occupancy reader evaluated before the advance, reversal of a path $\gamma = (x_0, \dots, x_n)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0).$$

The endpoint terms are incorporated before reduction modulo 10. They vanish on closed paths; on open windows they may alter the residues. This correction, already present in the bidirectional development of transport, specifies what the connection between the path and the counts must preserve.

Proposition 9.5 (Necessity of the register's nonperiodic information). *Suppose that the observable of a fixed collection of paths satisfies $o(t + 54) = o(t)$, that multiplicity is preserved under this return, and that the same local reader acts on the windows of 9 instants. Then its counts and blocks satisfy*

$$(A, C, V)_{m+6} = (A, C, V)_m, \quad z_{m+6} = z_m.$$

Proof. The translation $t \mapsto t + 54$ maps E_m bijectively onto E_{m+6} , preserving each counted incidence and its multiplicity. The equality passes to the sum and to decimal reduction. $\square$

Thus a presentation with 12 distinct blocks requires the reader to retain some information that does not factor through that observable of period 54: incidence selection, effective orientation, endpoints, multiplicity, or memory. The proposition identifies an inadmissible reduction of the state; it does not assert that the complete dynamics has period 54, nor does it determine its terminal reader by itself.

9.5 Complete census of cylinders and selection of the terminal boundary

The terminal readout jointly uses two kinds of information: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This section constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 6, not values introduced to define the APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent readout of these outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram products, weights, distances, and marginals is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what this profile selects on the specified catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

9.5.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readouts. A length of one thousand is sufficient for the integers that follow, as verified from the two bounds of each interval; no infinite depth is assumed. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (9.7)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphasic register K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 9.6 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1)-1}{h} \right\rfloor. \quad (9.8)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer value that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, which equals the second side because the numerator and denominator are integers. Equality of the endpoints of the range of values proves the assertion. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of the regional transport:

Channel	Thirty-trit prefix	P_χ
π	010211012222010211002111110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the prolongation already constructed; the decimal readout is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (9.9)$$

The catalogue includes all tails for which these two cylinders have a nonempty intersection.

Proposition 9.7 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max \left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r \right), \quad (9.10)$$

$$b_\chi = \min \left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r \right). \quad (9.11)$$

For the specified regional cylinders, no clipping applies and we obtain

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(9.12)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (9.9) intersect exactly when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D / Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1) / Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (9.8) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint can also be obtained directly from $\lfloor T_\chi D / Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 denotes the ternary representation of length six, including its leading zeros. By the table, each catalogue contains fewer than 729 consecutive elements, so this reduction is injective on it. Thus, the three boundary catalogues are completely described, without choosing the final triples:

$$\begin{aligned}\mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}.\end{aligned}\tag{9.13}$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would impose a different condition: it would lose the first cylinder that intersects the boundary and would give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

9.5.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions take integer values; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned}(g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3).\end{aligned}\tag{9.14}$$

The words are ordered as in (9.13). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} that of $[n(u) = 0][w(u) = 3]$.

Theorem 9.8 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (9.14) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331.\tag{9.15}$$

These four counts are, respectively,

$$\begin{aligned}N_g &= \text{tr}(A_{\pi e} A_{e \varphi} A_{\varphi \pi}), \\ N_{gn} &= \text{tr}(D_\pi^n A_{\pi e} D_e^n A_{e \varphi} D_\varphi^n A_{\varphi \pi}), \\ N_{gnw} &= \text{tr}(D_\pi^{nw} A_{\pi e} D_e^{nw} A_{e \varphi} D_\varphi^{nw} A_{\varphi \pi}), \\ N_{gnwh} &= \text{tr}(D_\pi^{nw} A_{\pi e}^h D_e^{nw} A_{e \varphi}^h D_\varphi^{nw} A_{\varphi \pi}^h).\end{aligned}\tag{9.16}$$

All products and traces in this formula are calculated over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace in (9.16) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the cardinality is summed. The final sum runs over $0 \leq i < 196, 0 \leq j < 197$. Substitution of the words in (9.13) and the profile (9.14) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the complete list of 331 triples; its program evaluates these sums with exact binary indicators and checks each survivor by scalar evaluation. No sample or tolerance is involved. $\square$

9.5.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_{\partial}(B) = (1, 2, 2, 1, 2, 0), \quad (9.17)$$

where w_{col} counts the nonzero elements in each column and q_{∂} sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Denote by $\mathcal{F}$ the set of 331 triples from the preceding theorem.

Proposition 9.9 (Reduction of the census to the terminal pair). *The sums of indicators over $\mathcal{F}$, successively imposing rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (9.18)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For each triple in the already enumerated set, the rank is computed by elimination over $\mathbb{F}_3$, the nonzero elements of the six columns are counted, and their entries are summed modulo three. The products of the indicators in (9.17) give (9.18). Substituting the two final indices into (9.13) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing these two matrices as the domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are $(0, 2, 0)$ and $(2, 2, 1)$; only the latter belongs to this line. This recovers the last step of the selector in Section 9.6, now preceded by the construction and enumeration of its complete domain. The stable readout of $[3^{1080}x_{\chi}]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same ranks (129, 197, 148). This subsequent comparison does not define the profile or enter the sums of indicators.

9.5.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 preserves the identity of each candidate here because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled marginals and the charge supply additional effective conditions.

Thus, a complete finite enumeration and its selector have been recovered on output cylinders, with an explicit readout profile. Construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a claim about alpha or about the complete continuum.

9.6 Two readouts of the terminal state

Denote by x_{term} the state produced by the composition of selection, transport, and update up to the terminal cut. In operator notation,

$$x_{\text{term}} = (\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1)(x_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readouts are

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary readout; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The terminal selector operates on the three regional catalogues of sizes 196, 197, and 196, constructed in Section 9.5. The local signatures reduce their $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331, and the column marginal $q_{\partial} = (1, 2, 2, 1, 2, 0)$ leaves two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the latter belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; therefore $B_{\text{term}} = B_1$. This check proves the last step of the selector from the declared census. It does not turn the column marginal into a consequence of the row charge or replace the preceding enumeration with inspection of these two matrices.

The signed reader has the typed composition

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (9.19)$$

The channel register determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}})) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding register, not as parameters fitted using alpha. The reconstruction proved in the following sections starts precisely from this typed output. The documentary trace of the preceding extractor and the scope of its self-contained transcription are preserved in the technical provenance note; the subsequent arithmetic is not presented as a substitute for that trace.

9.7 From the channel register to the signed vector

The harmonic reader Π_H can be written without using either K or alpha. To avoid confusing its orbital sums with the alpha blocks, we denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (9.20)$$

Division by three belongs to the reader of the three orbits. Its integrality is checked on compatible register outputs; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (9.21)$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and hence

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns gives

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (9.22)$$

The presence of negative values and components greater than 999 shows that U is not a word of base-one-thousand digits. Nor is it $U_6 \parallel U_6$, a catalogue concatenation of period six. The distinction concerns the domain and the information preserved, not merely the values.

The constructive order up to this point is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The following inversion produces the dodecaphasic register. Being able subsequently to recover the channels from the dodecaphasic register provides a reversibility check; it does not reverse this causal order.

9.8 Orbitwise Hadamard transform and integer reconstruction

On the cycle of twelve positions, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

Separation into four sectors on each orbit admits, once origin and orientation are fixed, the angular readout of ninety degrees. The transformation does not act on three contiguous quadruples in the natural order.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders according to the three indicated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 9.10 (Hadamard inversion and integrality). *We have*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integer elimination, or expansion of the determinant, gives the negative sign and the value -16 . The integrality condition is exactly divisibility by four of the four coordinates of H_4u . $\square$

Definition 9.11 (Dodecaphasic integral transformation). The sublattice of signed registers and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical register K is the image of the signed register generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines precisely the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves the two inverse compositions. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 9.12 (Canonical integral register). *The inversion $K^{(r)} = H_4U^{(r)}/4$ of the blocks in (9.21) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural order, the dodecaphasic register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (9.23)$$

It is the unique rational preimage of U , and it belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$\begin{aligned} H_4U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4U^{(2)} &= (2172, 2636, 232, 584), \\ H_4U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (9.23). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in terms of lattices. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the minors of order two are even, and at least one has absolute value two; those of order three are multiples of four, and at least one has absolute value four. It follows that

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus, not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by the inversion. The factor 1/4 is forced by the matrix and does not have the status of a free coefficient.

9.9 Cyclic differences and the uniform component

With cyclic indices modulo twelve, let

$$(D_3z)_m = z_m - z_{m+3}, \quad (D_4z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second, by four. The channel names 90 and 120 refer here to these separations in the oriented cycle, not to an identification with a physical operator from another domain.

Proposition 9.13 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphasic register.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. The total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral assertion, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every minor of order twelve must contain the row Q . Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in the row Q . All maximal minors are divisible by twelve, and the tree produces one of absolute value exactly twelve. The last factor is therefore twelve. $\square$

Checking (9.23) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulae (9.20) and (9.21) in turn reconstruct U from these differences. This closes a triangle of exact representations of the same object: transverse register, signed vector, and dodecaphasic register. Equality of results does not merge their codomains into one; it proves that the declared maps recover the relevant information.

9.10 Integral compatibility and determination of the transverse register

The domain of this construction is the representation of the register of an APP-TRIT-TPK history. We distinguish evaluation of the register's events, compatibility of its channels, and inversion of its coordinates. The following result gives a complete integral criterion for the second level and an explicit factorization connecting the first to it. The proof uses the register operators and establishes their relations for every element of the stated integer domains.

9.10.1 Characterization by adjacent increments

Indices belong to $\mathbb{Z}/12\mathbb{Z}$. Let

$$(Sz)_i = z_{i+1}, \quad D_3 = I - S^3, \quad D_4 = I - S^4, \quad Q(z) = \sum_{i=0}^{11} z_i.$$

We retain the transverse extractor

$$\mathcal{D} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}, \quad k \longmapsto (D_3 k, D_4 k, Q(k)).$$

For a triple of integer coordinates (b, c, q) , write

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j \quad (1 \leq i \leq 12), \quad T(w) = \sum_{i=0}^{11} P_i. \quad (9.24)$$

Theorem 9.14 (Integral image of the transverse extractor). *A triple $(b, c, q) \in \mathbb{Z}^{25}$ belongs to $\text{im } \mathcal{D}$ if and only if it satisfies*

$$b_i = w_i + w_{i+1} + w_{i+2} \quad (0 \leq i < 12), \quad (9.25)$$

$$\sum_{i=0}^{11} w_i = 0, \quad (9.26)$$

$$q + T(w) \equiv 0 \pmod{12}. \quad (9.27)$$

Its unique integer preimage is

$$k_i = \frac{q + T(w)}{12} - P_i, \quad 0 \leq i < 12. \quad (9.28)$$

The twelve equations in (9.25) and equation (9.26) are linearly independent over $\mathbb{Q}$.

Proof. For an output of $\mathcal{D}$, we have

$$c_i - b_{i+1} = (k_i - k_{i+4}) - (k_{i+1} - k_{i+4}) = k_i - k_{i+1}.$$

The sum of three increments gives b_i ; the cyclic sum of all of them is zero. Moreover, $P_i = k_0 - k_i$, so $q = 12k_0 - T(w)$. This yields all the conditions and the inverse formula.

Conversely, (9.27) makes formula (9.28) integer-valued; (9.26) allows its cyclic extension. Its adjacent differences are w_i . Therefore, $D_3 k = b$ and

$$(D_4 k)_i = w_i + w_{i+1} + w_{i+2} + w_{i+3} = w_i + b_{i+1} = c_i.$$

Summing the inverse gives $Q(k) = q$. The same formula proves uniqueness. For independence, the invertible integer change of variables $(b, c) \mapsto (b, w = c - Sb)$ converts the first twelve relations into $b = (I + S + S^2)w$. Each solves for a different coordinate of b . The condition $\sum w_i = 0$ is an additional independent relation. $\square$

The rational image has dimension twelve. Its integral structure adds a congruence of order twelve: this is the constructive manifestation of the last Smith factor of $\mathcal{D}$. The eleven increments $w_0, \dots, w_{10}$ and the charge q are sufficient coordinates, with $w_{11} = -\sum_{i=0}^{10} w_i$ and congruence (9.27). The original twenty-five coordinates retain these same twelve degrees of freedom through thirteen relations.

9.10.2 Commutativity with the harmonic readout

Let $\mathcal{H}_{12}$ be the Hadamard transformation of the register, with orbits $(r, r+3, r+6, r+9)$ for $r = 0, 1, 2$. Its block is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4.$$

For compatible (b, c, q) , define

$$B_r = \sum_{j=0}^3 c_{r+3j}, \quad A_0 = \frac{q + 2B_0 + B_1}{3}, \quad A_1 = A_0 - B_0, \quad A_2 = A_1 - B_1.$$

The reader Π_H in (9.21) has blocks

$$(\Pi_H(b, c, q))^{(r)} = (A_r, b_{r+3} - b_{r+9}, b_r + b_{r+6}, b_r - b_{r+6}).$$

Proposition 9.15 (Agreement of the two reconstructions). *On $\text{im } \mathcal{D}$, we have*

$$\Pi_H \mathcal{D} = \mathcal{H}_{12}, \quad \mathcal{D}^{-1} = \frac{1}{4} \mathcal{H}_{12} \Pi_H.$$

The image of Π_H restricted to that domain is exactly

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}.$$

Proof. For $k = \mathcal{D}^{-1}(b, c, q)$, let $a_r = \sum_{j=0}^3 k_{r+3j}$. The shift by four positions takes each orbit to the next, so $B_r = a_r - a_{r+1}$. Together with $q = a_0 + a_1 + a_2$, these identities give $A_r = a_r$. On one orbit, write (x_0, x_1, x_2, x_3) for the coordinates of k and $d_j = x_j - x_{j+1}$. Then

$$d_1 - d_3 = x_0 + x_1 - x_2 - x_3, \quad d_0 + d_2 = x_0 - x_1 + x_2 - x_3, \quad d_0 - d_2 = x_0 - x_1 - x_2 + x_3.$$

These are the three remaining rows of $H_4 k^{(r)}$. Inversion and the integral characterization follow from $\mathcal{H}_{12}^2 = 4I$. $\square$

The condition $\Pi_H(b, c, q) \in \mathcal{L}_H$ controls only the harmonic output. Membership of the twenty-five coordinates in $\text{im } \mathcal{D}$ additionally requires (9.25)–(9.27). For example, adding to c the vector $e_0 - e_3$ preserves the three orbital sums and leaves Π_H unchanged. Starting from $(0, 0, 0)$ thus gives a triple with $\Pi_H = 0$ that violates (9.25). This is an exact negative control for channel compatibility, independent of any terminal evaluation.

9.10.3 Determination and transport on register orbits

Theorem 9.14 determines k once its channels and charge have been produced. As a condition on outputs not yet individualized, the same image is a group isomorphic to $\mathbb{Z}^{12}$. With only a charge q fixed, the solutions form an affine coset of

$$L_0 = \{v \in \mathbb{Z}^{12} : \sum v_i = 0\} = \bigoplus_{i=0}^{10} \mathbb{Z}(e_i - e_{11}).$$

For example, $100\mathbf{1} + e_0$ and $100\mathbf{1} + e_1$ are two distinct registers of charge 1201, with components in $\{0, \dots, 999\}$. Each satisfies all image conditions and all Hadamard congruences through its own channels. These witnesses compare the compatibility equations; they are not assigned the status of terminal HMT histories.

Proposition 9.16 (Cyclic covariance and the register stabilizer). *Let ρ be the window translation on a domain of registers $\mathcal{R}$, and let S be the preceding translation on $\mathbb{Z}^{12}$. On the orbit of $R \in \mathcal{R}$, a covariant reader is determined by a vector*

$$k \in \text{Fix}(S^p),$$

where $p \mid 12$ is the least period of R under ρ . The charge condition $Q(k) = q$ admits integer solutions if and only if $12/p$ divides q ; when they exist, they form an affine coset of rank $p - 1$. In particular, on a free orbit, covariance and charge leave eleven degrees of freedom.

Proof. The rule $F(\rho^j R) = S^j k$ is well defined exactly when $S^p k = k$. Such vectors repeat $12/p$ times a word of length p . Their charge is $\frac{12}{p} \sum_{i=0}^{p-1} k_i$. Divisibility and rank follow from this formula. Channel covariance follows from $D_a S = S D_a$ and $Q S = Q$. On $\mathcal{L}_H$, the corresponding action is $\hat{S}_H = \mathcal{H}_{12} S \mathcal{H}_{12}/4$; by Proposition 9.15, both reconstructions intertwine these actions. $\square$

If the domain carries a marked origin, one must use the action that also transports that mark. The proposition concerns this declared cyclic action; it does not presuppose periodicity of the enriched history. Naturality with respect to truncations beyond the cut at 108 is obtained, for any already defined reader F_{108} , through $F_N = F_{108} \circ \tau_{N,108}$. This naturality preserves the initial evaluation rule; it does not by itself choose among the possible rules on the first one hundred and eight events.

9.10.4 Eventwise evaluation and composition of contributions

The preceding characterization provides a factorization of extraction through two effective readouts of the register:

$$\omega : \mathcal{R} \longrightarrow \{w \in \mathbb{Z}^{12} : \sum w_i = 0\}, \quad q_{\text{reg}} : \mathcal{R} \longrightarrow \mathbb{Z}, \quad q_{\text{reg}}(R) + T(\omega(R)) \equiv 0 \pmod{12}. \quad (9.29)$$

These maps must be evaluated on the events, orientation, incidence, and memory produced by APP-TRIT-TPK. Once their actions on those data are given, the composition

$$R \longmapsto (w, q) \longmapsto ((I + S + S^2)w, (I + S + S^2 + S^3)w, q) \xrightarrow{\Pi_H} u \xrightarrow{\mathcal{H}_{12/4}} k$$

is determined and satisfies all inverse checks. The formula $w = c - Sb$ also connects a producer that already supplies both channels: it is a check on its output, preceding any comparison with the terminal witness.

Proposition 9.17 (Accumulation of contributions and prospective uniqueness). *Suppose that the initial state and each event a_t of the register produce, through rules fixed beforehand, contributions $(\delta w_t, \delta q_t)$ with*

$$\sum_i \delta w_{t,i} = 0, \quad \delta q_t + T(\delta w_t) \equiv 0 \pmod{12}.$$

With compatible initial conditions $(w^{(0)}, q^{(0)})$, the update

$$w^{(t+1)} = w^{(t)} + \delta w_t, \quad q^{(t+1)} = q^{(t)} + \delta q_t$$

produces a unique register at each stage. Its increment is

$$\delta k_{t,i} = \frac{\delta q_t + T(\delta w_t)}{12} - R_i(\delta w_t).$$

The construction respects concatenation of registers, and the readout of a prefix remains unchanged under subsequent prolongations.

Proof. The compatibility conditions are preserved under addition. Formula (9.28) is additive in (w, q) and produces the stated increment. Induction fixes each state successively. The sum over two segments is the concatenated sum, and a prefix receives exactly the same summands before and after the history is prolonged. $\square$

The proposition provides a sufficient criterion, not a requirement of linearity for every TPK reader. If events are transported by nontrivial actions, their contributions can be expressed in the terminal chart through the corresponding affine cocycle. The cocycle equation then preserves the same independence from the partition of the trajectory. In both cases, the event rules and the initial state constitute the constructive data that must come from the article: choosing them to reproduce an expected vector would replace that provenance with a different task.

The result of this development is the exact determination condition (9.29), its proven inverse, and its transport identities. Its scope is the transverse register; the other HMT constructions retain their own domains and proofs.

9.11 Two rational readouts: κ_{36} and κ_{per}

Once K has been constructed, the base-one-thousand chart allows two distinct operations. The first reads a single turn; the second prolongs the dodecaphasic register periodically. Let $B = 1000$, and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (9.30)$$

The leading zeros of a block, such as 058, are part of its three-digit representation. They do not change its integer value, but they allow the twelve blocks to be concatenated unambiguously.

Definition 9.18 (Finite and periodic readouts of the dodecaphasic register). The single-turn readout is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 9.19 (Exact value and period of the prolonged dodecaphasic register). *We have*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12} \kappa_{\text{per}}. \quad (9.31)$$

The infinite word of the dodecaphasic register has least period twelve in base one thousand and least period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series gives $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; the least block period is therefore twelve. The decimal expansion is canonical, since the dodecaphasic register has no tail of blocks 999. If its least decimal period were d , it would divide 36. Grouping digits in threes would produce a block period $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve under this operation. Hence the least decimal period is 36. $\square$

Both readouts are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their proximity does not allow one to replace the other in an exact identity. In particular, the finite closure of alpha uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

9.12 Reversibility of the rational constant and phase change

The rational constant κ_{per} completely determines the register K when the base, length, and readout origin are preserved. Its numerical and structural functions are related through an exact inversion.

Proposition 9.20 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1) \kappa_{\text{per}}.$$

The integer expansion of $I(K)$ into twelve base-1000 blocks recovers all the K_j , including the leading zeros of each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, we have

$$\kappa(\rho K) = 1000 \kappa(K) - K_1. \quad (9.32)$$

Proof. The first identity inverts the definition of the periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. Furthermore,

$$I(\rho K) = 1000 I(K) - K_1 (1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ gives (9.32). $\square$

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the already constructed transformations $D_3 K, D_4 K, Q$. The rational evaluation preserves the signed register and the cyclic differences that depend on it. The total depth and transport operators of a history remain in their own coordinates: the preceding inversion has the integral dodecaphasic register as its domain.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical readout, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

The subsequent incidence also admits an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because this orbit first recovers the integer components. The signed support associated with α additionally uses its regional registers and boundary normalization. The relation between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the function of K : it provides a reversible integral coordinatization of the oriented register, whose periodic evaluation and incidence functions participate in subsequent constructions. The value, transformation, and domain are preserved jointly.

9.13 Periodicity and the cokernel of the transport operator

The periodicity just proved belongs to the dodecaphasic register. It does not imply periodicity of the regional coordinates, the carries of the closure, or alpha. It is useful to make this distinction precise through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written as

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \longmapsto K + \partial_b h, \quad C \longmapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it does not authorize modifying the dodecaphasic register already selected by the register.

Proposition 9.21 (Periodic carry quotient). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module has the presentation $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ amounts to imposing $t = b$, leaving the relation $b^{12} - 1 = 0$. Evaluation of the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $b^{12} - 1$ of κ_{per} and this quotient arise from the same cyclic length, but their objects remain distinct: one is a rational evaluation and the other a quotient of a carry module. The finite closure of alpha will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

The next chapter is thus prepared. The dodecaphasic register has been fixed before solving the alpha recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the register; and its two rational evaluations have separate domains and uses. None of these facts uses alpha as an input, and none yet determines the arithmetic type of the output of the complete closure.

10 Determination of the constant and covariance of its realizations

The holographic arithmetic–geometric coupling constant combines structural determination with a capacity for reconstruction. The former fixes the ordered register compatible with the incidences and chronology of the generated state. The latter establishes how that register is recovered from its representations and how operations acting on it are identified. Their composition gives a precise meaning to its coupling function.

Determination begins with the regions produced by APP–TRIT–TPK. Their six-trit bands retain the position of every symbol, charge, carry and depth. The integer clock compares the ternary and decimal scales; its first zero increment, followed by the nonadic return, determines the phase–charge reading of the descriptor. The terminal repertoire and its incidences retain that provenance. Orderings of this repertoire produce the finite domain on which the joint selection of Section 8 acts.

The exceptional condition restricts that domain to registers whose support face agrees with the face extracted from the regional words. The temporal condition compares, within a single ordering, the comparative and affine readings of the two terminal segments. Thus a coordinate position and the relation between two instants enter the same decision. The exact result is

$$19\,446 \longrightarrow 79 \longrightarrow 1.$$

The unique surviving register determines the integer

$$N_K = 234543140729659824621058914794146601, \quad \kappa_{\text{ag}} = \frac{N_K}{10^{36} - 1}.$$

The rational expression preserves the order of the twelve triads in the specified phase chart. Uniqueness concerns the constructed terminal domain, its two explicit conditions and its orientation data. This fixes what is selected, through which relations, and within which set of alternatives.

The census of the three regional endings serves a different function. Its $196 \cdot 197 \cdot 196$ elements are triples of six-trit words compatible with the declared cylinders. The Gram profile, weights, distances and oriented margins determine the visible ending. The census of 19 446 elements examines ordered twelve-coordinate registers. Both receive information from the same enriched history; their types and filters make it possible to follow their relation while distinguishing a visible triple from the complete register.

Once K has been selected, the Hadamard transformation and its readers establish equivalences between representations of that same register. Selection determines a candidate among alternatives; inversion recovers it from a representation. The bilateral law then provides the conservation identity for the declared isometries, and the archive retains the information needed to invert successive readings. Each result passes to the next an object with specified domain, orientation and normalization.

This hierarchy also specifies the role of K in the conservation law. The bilateral identity is proved for two isometries with a common domain and codomain. In the realizations developed here, the selected register fixes the arithmetic and incidence readings, its centred direction and the reconstruction operations composed with that identity. Generality of the balance and determination of the register are complementary properties: the former describes an operator law; the latter identifies the structure on which the present realizations are calculated.

The closure of the fine-structure constant uses the already-generated regional coordinates and the selected register, preserving the carries of the dodecaphase operation. Its analytic representation develops the same precoordinate through the complete recurrence. Through the other projection, the preserved incidences are transported to the exceptional realizations. Their common provenance relates arithmetic value, incidence and memory while preserving the distinct quantities read by their respective operators.

The subsequent geometric realizations extend this function. The centred direction of K determines the determinant-one diagonal flow, preserves Plücker signs and the covolume of the specified lattices; the archive reconstructs that direction and controls its readout errors. The theta–Mellin transformation preserves the polar structure on the proved domain, while the entire difference records spectral variation. Through the operations already established, temporal selection is thus linked to a structure admitting evaluation, transport and recovery.

The combined contribution is therefore organized into a definite proof sequence: generation of the state, selection of the constant, equivalence of its representations, operator conservation and stable reconstruction. This sequence fixes the position of each theorem and keeps visible both the conditions it receives and the relations it transmits.

α

Fine-structure constant

Regional coordinates and the ordered register enter jointly into the two readouts of the fine-structure constant.

The fine-structure constant within the common generation

The fine-structure constant occupies the point at which the regional coordinates and the ordered register converge in a definite reading. The circular, golden, and exponential coordinates have already been constructed. The signed register originates in the same state and preserves its orientation. The dodecaphasic closure receives these results together with the carries that allow them to be composed.

The arithmetic route provides a precise characterization: its alpha coordinate is the difference between the declared regional aggregate and the periodic reading of the coupling constant. This equality links objects that have been produced previously. Explaining it requires following the blocks, positions, and carries that convert the register into an expansion. The proof will preserve both the value and the data that make the operation reproducible.

The second route uses the vacancy resolvent $E_{21/22}$ and its graded prolongations. Its positive root is studied together with its domain and selection conditions. The comparison of the two routes is formulated through cylinders common to every depth. The shared origin of the constants is expressed by this composition of results, and each evaluation preserves its particular function within it.

Incidence adds another reading of the same closure. The signature distinguishes a hexad on the tetrad determined by the register. As incidence is prolonged towards the binary design, the marked configuration participates in the relation between the regions and their octads. This connection prepares the transition to the exceptional realizations and, subsequently, to the area and information readers.

The importance of alpha in the article follows from this causal position. The coordinate obtained enters into the valuation of action and into the angular readers; these transmit orientation and normalization to the constructions that follow. The exposition advances from the regional origin to these uses, keeping visible the register that accompanies the value. The relation between alpha and the coupling constant thereby acquires an operational definition and an unambiguous place within the double projection.

11 Derivation of the fine-structure constant

In this development, the fine-structure constant receives an arithmetic determination combining the three regional coordinates and the integer dodecaphasic register. The operation combines their blocks with a fixed orientation and transports the quotients of positional normalization. The resulting parameter expresses the compatibility of that oriented difference with the extension of the cylinders. Its value and the incidence data used in its construction will remain related throughout the exceptional development.

The internal order matters. The coordinates π , e and φ , already constructed in the preceding section, are prior coordinates of the same arithmetic state with trajectory memory, and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select alpha. The correlated generation of the four coordinates shares an origin and coinductive compatibility, without requiring all four components to be evaluated in a single step.

The two routes express different mathematical functions of the same parameter. The first performs an integer closure by Euclidean division in base 1000. The second studies the analytic equilibrium of vacancies through the resolvent $E_{21/22}$ and its graded extensions. One route determines blocks and quotients; the other determines a root through an equation and a domain of uniqueness. Their identification is then formulated in the common cylinders at every depth. The exposition distinguishes these three results: positional construction, analytic determination and compatibility of the two.

11.1 First route: positional determination by integer closure

11.1.1 Domain and orientation of the integer closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and extend the dodecaphasic register by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphasic register is subtracted according to the dodecaphasic closure already typed. For $B = 1000$, the local operation distinguishes the signed balance

$$Z_j := P_j + E_j - \Phi_j - K_j$$

from the sum incorporating the incoming quotient. Normalization is

$$s_j = Z_j + c_{j+1}, \quad A_j = s_j - B \left\lfloor \frac{s_j}{B} \right\rfloor, \quad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \quad (11.1)$$

Thus, $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted because the table is printed from left to right.

Definition 11.1 (Finite-depth closure). For a depth N and an integer boundary t , the map $\mathcal{C}_N$ receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (11.1) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division leaves no residual choice of sign: the remainder is fixed in $\{0, \dots, 999\}$ and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of the closure are not anomalies; they are the integer information required to maintain the oriented equality.

11.1.2 Dodecaphasic window and carry conditions

The first twelve triads obtained are

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117). \end{aligned}$$

In the closure channel, the five regions determining the coordinate P retain their distinct antecedents. Sharing this reading does not identify them as regions or as histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. It shows each input, the incoming carry, the total before normalization, the resulting block and the outgoing carry. The table is a reproducible integer calculation, not a list of blocks selected for their resemblance to a target.

j	R_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 9: Initial closure with a zero right boundary. The dependence propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the reading of the window is the rational number

$$\alpha_{12}^{(0)} = \frac{7297352569283800997285105472380663}{10^{36}} = 0.007297352569283800997285105472380663. \quad (11.2)$$

The superscript recalls the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readings becomes visible when the carry from the remote tail is transported.

11.1.3 Telescoping and reversibility in a boundary fiber

The local form of the equality is

$$R_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \quad (11.3)$$

The carry term links adjacent positions. Removing it coordinate by coordinate yields a false equality. Only an evaluation preserving its endpoints allows it to telescope.

Proposition 11.2 (Integer identity at arbitrary depth). *For $\iota_N(z) = \sum_{j=1}^N z_j B^{N-j}$, every execution of $\mathcal{C}_N$ satisfies*

$$\iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(K) - \iota_N(A) = B^N c_1 - c_{N+1}. \quad (11.4)$$

Proof. Multiplying (11.3) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^N (Bc_j - c_{j+1}) B^{N-j} = B^N c_1 - c_{N+1}.$$

Each interior carry appears twice with opposite signs; only the two endpoints survive. $\square$

For Table 9, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphasic register constant appearing here is κ_{36} , not κ_{per} . The domain of this identity contains truncations and a finite boundary; its form does not authorize removing both conditions when passing to the complete section.

The local inversion is also exact:

$$K_j = P_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ , the blocks A and the complete carries, it recovers the dodecaphasic register. Concatenation also recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not be turned into a new causal recipe that starts from alpha and then manufactures the register. The direction of construction remains signed register, dodecaphasic register, closure and output.

11.1.4 Controls of orientation, boundary, and register position

Integer closure distinguishes three operational dependencies: the direction of carry propagation, the boundary condition, and the position of the dodecaphasic register. Their comparison uses the same twelve-block window as Table 9. The regional words P, E, Φ and the already generated register K are retained except for the modification identified in each case. The common reference is $\mathcal{C}_{12}$ with $c_{13} = 0$, whose output is (11.2).

Prospective production of the signed register and its inversion $K^{(r)} = H_4 U^{(r)} / 4$ precede these controls. The normalization of Lemma 9.10 is retained, in particular $H_4^2 = 4I_4$ and $\det H_4 = -16$. Inversion determines the register once U has been produced; its operational selection remains in the construction preceding closure. The following variations of K examine the response of an already defined reader.

Proposition 11.3 (Separation of the dependencies of closure). *For the data in Table 9, the following four modifications have distinct effects:*

1. *Left-to-right quotient propagation, initialized with zero quotient, changes the third block from 352 to 353.*
2. *The condition $c_{13} = 1$, with the original orientation, changes the final block from 663 to 664 and preserves the preceding eleven blocks.*
3. *The rotation $K'_j = K_{j+1}$, with $K_{13} = K_1$, changes the output word while retaining $c_{13} = 0$.*
4. *The perturbation $K'_1 = K_1 + 1 = 235$, with the other eleven coordinates fixed, changes the first block from 007 to 006 and preserves the remaining blocks.*

Proof. For the first case, define the initial quotient $b_0 = 0$ and apply, in the order $j = 1, \dots, 12$,

$$\tilde{s}_j = P_j + E_j - \Phi_j - K_j + b_{j-1}, \quad \tilde{A}_j = [\tilde{s}_j]_{1000}, \quad b_j = \frac{\tilde{s}_j - \tilde{A}_j}{1000}.$$

This orientation produces, in two consecutive halves,

$$\begin{aligned}\tilde{A}_{1:6} &= (007, 297, 353, 570, 284, 800), \\ \tilde{A}_{7:12} &= (995, 285, 106, 471, 379, 663).\end{aligned}$$

The difference already appears at $j = 3$, where the quotient required by the original closure comes from the fourth position.

For the second case, the final sum changes from 663 to 664. Both have zero Euclidean quotient; hence $c_{12} = 0$ in both cases and the recurrence agrees from $j = 11$ to $j = 1$. The resulting word is

$$\begin{aligned}A_{1:6}^{(1)} &= (007, 297, 352, 569, 283, 800), \\ A_{7:12}^{(1)} &= (997, 285, 105, 472, 380, 664).\end{aligned}$$

Identity (11.4) records exactly the unit variation of the right endpoint.

The rotation in the third case, evaluated with the same zero boundary and the original recurrence, gives

$$\begin{aligned}A_{1:6}^{\text{rot}} &= (698, 699, 763, 639, 119, 004), \\ A_{7:12}^{\text{rot}} &= (559, 429, 226, 119, 926, 030).\end{aligned}$$

In this case the left carry is $c_1 = -1$; the right endpoint remains zero. Retaining both endpoints in (11.4) preserves the integer equality as the twelve coefficients change position.

Finally, perturbing K_1 leaves the entire recurrence from $j = 12$ to $j = 2$ unchanged. At $j = 1$, the sum decreases from seven to six, with zero quotient in both cases. This gives

$$\begin{aligned}A_{1:6}^{\text{pert}} &= (006, 297, 352, 569, 283, 800), \\ A_{7:12}^{\text{pert}} &= (997, 285, 105, 472, 380, 663).\end{aligned}$$

□

The four controls separate transport direction, boundary fibre, coefficient position, and local response. Their interpretation retains the finite domain $N = 12$. Tail transport and normalization of the complete section are developed next, with the compatibilities required by passage to arbitrary depth.

11.1.5 Extension and canonical normalization of the complete section

The extension preserves the periodic dodecaphasic register and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} R_j B^{-j}, \quad e_0 = \sum_{j \geq 1} E_j B^{-j}, \quad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2 and 1, so that $p = \pi - 3$, $e_0 = e - 2$ and $f = \varphi - 1$.

Proposition 11.4 (Existence and uniqueness of the complete normalization). *The series*

$$y = p + e_0 - f - \kappa_{\text{per}} = \sum_{j \geq 1} (R_j + E_j - \Phi_j - K_j) B^{-j} \quad (11.5)$$

converges absolutely. In the domain of the blocks obtained, $0 < y < 10^{-2}$. There exists a unique canonical expansion (A_j) of y , with no tail eventually constant at 999, and a unique bounded integer sequence (c_j) satisfying (11.3) with $c_1 = 0$. These carries belong to the invariant set $\mathcal{C} = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B - 1)$, so a geometric series dominates it. If $Z_j = R_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r \geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums

of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; therefore $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y . Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying the equality of the two series by B^{j-1} and separating their first $j - 1$ terms gives

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Moreover, $R_j(A) \in [0, 1)$, hence $-3 < c_j < 2$, which places all carries in $\mathcal{C}$. The shift identities for the tails give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y . The canonical expansion is unique and, once fixed, the tail formula determines all carries. Finally, if a finite execution is applied with $c_{j+1} \in \mathcal{C}$, then $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to $\mathcal{C}$. $\square$

The proposition makes the normalization of the subsequent reader explicit: it does not replace the generation of the three sequences with a conventional series introduced as a datum. Once those sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite procedure preserved in the corpus propagates the five boundaries of $\mathcal{C}$ and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. A boundary is not chosen because it yields favorable digits, nor is compatibility at every depth deduced from observing a long run.

The initial comparison shows why this precaution matters. The stabilized extension gives $c_{13}^\infty = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first turn. The first eleven triads do not change, but the extension begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

An occurrence of the block 999 does not violate the canonical convention: what is excluded is a tail eventually constant at 999. The window with a zero boundary ends in 663; the truncation of the complete section ends in 662. Both statements describe different and exact operations. They must not be merged into a single projection notation.

11.1.6 Operational definition and complete additive equation

Definition 11.5 (Alpha coordinate of the closure). The integer route determines the coordinate

$$\alpha_A = \sum_{j \geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\text{per}}, \quad (11.6)$$

where (A_j, c_j) is the compatible canonical normalization of (11.1). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}.} \quad (11.7)$$

The integer four is not calibrated: it is $3 + 2 - 1$, the combination of the integer parts of the three coordinates. The base one thousand comes from the cubic chart, and the period twelve from the dodecaphasic register. The signs are those of the oriented channel and the closure, and the remote

boundary is the one imposed by the compatible section. These are the effective origins of the coefficients of the equation.

At this level, alpha may be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphasic return, calculated with compatible carry. The term *defect* denotes a determined algebraic difference, not an arbitrary loss or an imperfection of the system. This description does not suppress the genealogy: without the regions, the signed register, the orientation and the carry law, the scalar equality would not be a sufficient exposition of the construction.

Three objects that alpha notation may accompany in subsequent readers must also remain separate: A is a word of blocks; the negative pre-carry support is obtained from the sign of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None of them is interchangeable with the real value α_A .

Signed balance and sum with an incoming quotient. Positional division consists of two consecutive stages. With $K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}$, which denotes the periodic extension also denoted by K_j in (11.1), the balance before quotient transport is

$$Z_j := P_j + E_j - \Phi_j - K_j^{\text{per}}.$$

The variable s_j in (11.1) denotes the sum that already includes the incoming contribution, so that

$$s_j = Z_j + c_{j+1}, \quad A_j = Z_j + c_{j+1} - 1000c_j.$$

The incidence configuration H_α^- uses the signs of the balance Z_j in the initial window; determining the digits also uses the linking quotients c_{j+1} and c_j . This distinction jointly preserves the signed support and the normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and the dodecaphasic register arise from the common APP-TRIT-TPK generation. Once produced, their positional composition has an explicit dependence on the reading depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}, \quad y_N = \sum_{j=1}^N Z_j B^{-j}, \quad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (11.5) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \quad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window with that of its extension, both expressed in the same coordinates.

Proposition 11.6 (Boundary transport and canonical cylinder). *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in \mathcal{C} = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^N A_j^{(N,t)} B^{-j}$. For the registers of the domain under consideration, we have*

$$\alpha_A - a_N^{(t)} = B^{-N} (R_{N+1}(Z) - t), \quad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \quad (11.8)$$

The quotient of the complete extension is $c_{N+1}^\infty = \lfloor R_{N+1}(Z) \rfloor$. With this boundary,

$$0 \leq \alpha_A - a_N^{(c_{N+1}^\infty)} < B^{-N},$$

so the resulting window is the canonical prefix of order N .

Proof. Telescoping (11.3) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, each incoming state belongs to $\mathcal{C}$ and the first dividend lies between 5 and 9, we have $c_1 = 0$. Substituting this equality into the decomposition of the series proves (11.8). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in \mathcal{C}$.

For the complete canonical normalization, $c_{N+1}^\infty = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. The integrality of the quotient implies $c_{N+1}^\infty = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c_{N+1}^\infty)} = B^{-N}R_{N+1}(A)$, proving membership in the canonical cylinder. $\square$

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed at the scale of the next position. The calculation over the five boundaries preserves this information as a finite collection of possibilities until the blocks to the left stabilize. The quantitative contraction arises from depth and tail transport; its object is the normalization of the coordinates already generated.

Graded domain of the analytic resolvent. The analytic composition displayed in §11.2.2 acts on $\text{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation produces exactly $T_{7:9}$; degree 9 is the last component of that domain. The identity $N^3 = 0$ explains why the resolvent is expressed through three summands. To describe an extension to higher degrees, one must also specify the transport between successive graded domains and its action on the initial component of each domain. Local nilpotence and global compatibility thus play distinct mathematical roles.

11.1.7 Constructive determination and role of the dodecaphasic register

The preceding definition clarifies what it means to determine alpha from the structure. The domain of the operation contains four registers of known provenance: the three regional sequences and the integer phase register. Normalization also incorporates a boundary condition compatible with extension. Once these data are fixed, Euclidean division determines each remainder and its quotient. Proposition 11.4 extends that determination to the complete system. Independence from a target physical value is therefore a property of the order of construction: the value compared at the end does not participate in selecting any of the arguments of this map.

This causal independence must be distinguished from algebraic independence between numbers. The former is established by identifying the domain, the operations and the provenance of the coefficients; the latter would be a statement about polynomial relations. The result used here is the former. The construction determines a coordinate through registers produced by APP-TRIT-TPK and allows the relations it satisfies to be checked subsequently. This order makes it possible to use a coordinate already generated in a later operation without turning it into an external parameter.

The register K has a more precise role than that of a numerical correction added to a formula. Its twelve positions arise from the reversible transformation of the signed register. The equation $U = \mathcal{H}_{12}K$ preserves the possibility of returning to that register, while the transverse differences and the total sum allow another reconstruction. In the periodic evaluation, the order of the positions determines the positional weight of each component. The same set of components, arranged in a different order, generally produces a different rational coordinate. Thus, the phase origin and orientation form part of the definition of the number entering (11.7).

The normalization of alpha composes this information with the regions. Its local identity distinguishes two successive operations. First the oriented balance of blocks is formed; then that balance is represented through a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and allows a support to be extracted subsequently. The second allows the prefixes of the real coordinate to be formed. Both readings remain related because the state preserves the data before and after normalization. The signed support and the positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. The interior quotients cancel in pairs when a prefix is evaluated, while their endpoints remain in the equality. This cancellation is a property of the weighted sum; it neither removes the quotients from the state nor allows them to be suppressed before the sum is taken. The right boundary of a window receives the contribution of the continuation. For this reason, evaluating an isolated window and restricting an extended section are

distinct operations. The change of the final block from 663 to 662, proved above, provides an explicit realization of this difference without altering the blocks already stabilized to its left.

In this sense, bidirectionality has a concrete arithmetic formulation. Generation provides compatible extensions; normalization transmits the integer information of their boundary from positions of lower weight towards those of higher weight. The equation preserves the contribution of both directions through its indices and its boundary conditions. Reversibility in the fixed fiber is checked by reconstructing K from the other registers and the complete quotients. This inversion certifies the consistency of the original operation; the genealogical direction remains the one that produces the register first and the alpha coordinate afterwards.

The second determination has complementary content. In it, the immediate object is a balance of vacancies, and the parameter appears as its root in an isolating domain. The positional route provides a canonical normalization of registers; the analytic route studies the vanishing of an operator. Presenting both requires precisely this difference of functions to be preserved. Their relations with the common state must be expressed through the refinement maps and the comparison of their complete realizations, as developed in the corresponding subsection. Agreement of a finite approximation constitutes a localized check; identification of the complete constructions has the domain and scope of its own statement.

Finally, this organization explains the position of alpha in the article. Its equation appears after regional generation and the construction of K , because it uses both results. The electron section then uses that coordinate as a factor in its composition. The exceptional extension receives the registers and their supports, where incidence information is preserved. These two continuations share the same provenance and develop distinct mathematical functions: evaluation of a central section and construction of a boundary configuration. Developing each one allows the structure to be followed through the equation, instead of restricting the comparison to its scalar value.

11.2 Second route: analytic determination through vacancies

11.2.1 Transport invariants and graded resolvent

The analytic route receives invariants of the oriented state, not the output α_A . Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \quad (11.9)$$

and the vacancy condition uses

$$\delta = \log_{10} \frac{10}{9}.$$

The ratio 10/9 compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification reading; it is not a target value of alpha.

The genealogy of the coefficients must be preserved in full, but with its exact logical strength. Recent developments bring together a marked signature of the state. The synchronization matrix and the associated census are

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \quad N_9 = 43.$$

Its determinant yields

$$a = -\frac{\det D_{\text{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \quad k_* = mb - 1 = 20.$$

The vacancy reading in the base-one-thousand chart gives

$$v_- = \lfloor 1000\delta \rfloor = 45, \quad v_+ = \lceil 1000\delta \rceil = 46.$$

These integers are typed antecedents of the evaluation of the truncation. They are not obtained from its root or from a metrological approximation to it.

The lower branch also coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45.$$

The transverse rank of the register in the preceding chapter gives $r_{\text{dod}} = 11$, and the phase half-corona provides $f_{\text{ph}} = 8\binom{6}{2} = 120$. In particular, $45r_{\text{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{J}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{G}_9(\mathcal{J}_9)(x) = & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_-r_{\text{dod}}}. \end{aligned} \quad (11.10)$$

Substitution recovers the coefficients of P_6 and its ninth-order extension. The pair 45/46 crosses the division between the two: the upper branch occupies degree six, and the lower branch reappears in degrees eight and nine.

The factorization in (11.10) is formulated for a state preserving origin, channel order and orientation. Naturality with respect to morphisms preserving those data is not equivalent to uniqueness of every reader after they have been forgotten. The domain of this factorization is determined by the signature (11.10); the extension to all degrees is constructed in Section 11.4, with its recurrence class and its dependence on the state's pre-coordinate explicitly defined.

11.2.2 Nilpotent resolvent in degrees 7, 8 and 9

The first extension admits a particularly explicit finite operator construction. On the space with basis e_7, e_8, e_9 , let

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7.$$

The symbol N here denotes a nilpotent operator, not a truncation depth. Its domain and grading are fixed before the polynomial is evaluated at a number.

Lemma 11.7 (Exact nilpotent extension). *We have $N^3 = 0$ and*

$$(I - N)^{-1}v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

The evaluation $e_n \mapsto x^n$ produces

$$T_{7,9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (11.11)$$

Proof. The operator raises the degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \quad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. $\square$

Here the denominators 120, 45 and 495 are not decimal adjustments: they arise from the half-corona, the APP determinant and its product with the dodecaphasic rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only the symmetry of the cycle and replacing this operator with another symmetric operator does not necessarily preserve the coefficients.

11.2.3 Root of multiplicity one of the ninth-order truncation

Define

$$P_9 = P_6 + T_{7:9}, \quad E_{21/22}^{(9)}(x) = P_9(x) - \delta, \quad I_\alpha = (0, 10^{-2}).$$

This finite operator has its own existence and uniqueness result.

Proposition 11.8 (Root of multiplicity one of the truncation). *The operator $E_{21/22}^{(9)}$ has a unique root ξ_9 in I_α , and that root is simple. The following rational isolation is obtained:*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \quad (11.12)$$

Proof. The bounds on the internal coordinates $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ suffice for existence. At zero, $E_{21/22}^{(9)}(0) = -\delta < 0$. At 10^{-2} , the linear contribution minus the negative terms is greater than 0.0626, so the operator's value is positive.

For $0 \leq x \leq 10^{-2}$, discarding the positive terms of the derivative gives

$$(E_{21/22}^{(9)})'(x) \geq 2\pi - \frac{7}{2} \cdot 10^{-2} - \frac{10}{21} \cdot 10^{-8} - \frac{6}{46} \cdot 10^{-10} - \frac{7}{120} \cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. The narrower interval (11.12) results from a directed rational evaluation. Let x_- and x_+ be its endpoints. To compute $\delta = \log(10/9)/\log 10$, we use, for $z > 1$,

$$L_N(z) = 2 \sum_{j=0}^{N-1} \frac{y^{2j+1}}{2j+1}, \quad y = \frac{z-1}{z+1}, \quad 0 < \log z - L_N(z) < \frac{2y^{2N+1}}{(2N+1)(1-y^2)}.$$

The remainder is bounded by the geometric series of positive terms. This formula is applied to $z = 10/9$ and $z = 3$, with $N = 520$, using $\log 10 = \log(10/9) + 2 \log 3$. Evaluation of π uses the generated thousand-digit regional cylinder, including its width 10^{-1000} . Interval arithmetic in (11.10) gives

$$-\frac{4559}{10^{23}} < E_{21/22}^{(9)}(x_-) < -\frac{4558}{10^{23}}, \quad \frac{1698}{10^{23}} < E_{21/22}^{(9)}(x_+) < \frac{1699}{10^{23}}.$$

These signs and the proved monotonicity establish the isolation. The endpoints are results of evaluating the operator; its coefficients arise from the signature constructed beforehand. $\square$

The result concerns ξ_9 . It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. The root of a truncation may share a prefix with the root of the extended operator and differ at a later position. Simplicity allows the motion of the root under perturbations to be controlled; it does not eliminate those perturbations.

11.3 Compatibility of the two determinations at every depth

Remark 11.9 (Truncation, extension and compatibility). The operations in degrees 6 to 9 determine a finite jet. Positional normalization determines the complete coordinate through its compatible sequence of carries. The next proposition expresses the general criterion for identification through cylinders. Section 11.4 then constructs a correlated analytic chart of all degrees, with a recurrence, uniform convergence, uniqueness of the root and projective compatibility. The construction receives the shared coordinate of the state and the declared class of extensions; it thus preserves the distinction between an analytic representation of that coordinate and an independent selection.

The complete analytic route is organized into a collection of compatible graded readers

$$E_{21/22}^{(6)} \longleftarrow E_{21/22}^{(9)} \longleftarrow E_{21/22}^{(12)} \longleftarrow \dots$$

The symbol

$$E_{21/22} = \varprojlim_m E_{21/22}^{(6+3m)}$$

denotes that complete reader, with its transports and truncations, not a formal series freely chosen to make it pass through α_A . The second route determines its unique positive root in I_α , denoted by α_B , with the uniqueness and compatibility of the complete reader proved in Section 11.4.

The proofs in Section 11.4 establish that, at each depth, both systems determine the same surviving cylinder C_N^α , and that their diameters tend to zero. It is useful to state the mathematical consequence of this compatibility separately.

Proposition 11.10 (Identification of the compatible limits). *If the values of the integer and analytic routes belong to the same cylinders C_N^α for every N , and $\text{diam } C_N^\alpha \rightarrow 0$, then*

$$\alpha_A = \alpha_B = \alpha.$$

Proof. For every N , the distance between the two values is bounded by $\text{diam } C_N^\alpha$. Letting N tend to infinity gives zero distance. The existence of the compatible sections ensures that this is not the empty intersection of a system of approximations. $\square$

This proof identifies the consequence of the compatibility squares. Their construction at arbitrary depth appears in Section 11.4. Finite evaluations check realizations of that construction; the recurrence and its uniform bounds establish the passage to the limit.

The difference can be expressed algebraically. Let $F = \mathbb{Q}(\pi, \delta)$ and let $j_9 : F[[x]] \rightarrow F[x]/(x^{10})$ be the truncation. Then

$$\ker j_9 = x^{10}F[[x]].$$

All the operators

$$E_h(x) = E_{21/22}^{(9)}(x) + x^{10}h(x)$$

have the same ninth-order truncation. However, taking $h = 0$ and $h = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0,$$

so they do not share that root. The observation does not allow a tail foreign to HMT transport to be chosen; it demonstrates precisely why the generated tail and its compatibility must be preserved. Nor does an arbitrary formal series suffice to speak of analytic roots without its domain of convergence or the realization of the cylinder system.

11.4 Compatibility between the dodecaphasic and analytic representations of alpha

Compatibility is established at every depth. The reversible dodecaphasic representation and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal representations with distinct codomains, produced from the same enriched APP–TRIT–TPK state. The jet $E_{21/22}^{(9)}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below is not deduced from the bare jet: it receives the shared coordinate of the state and expresses its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This separation preserves the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \mathcal{X}_\alpha^{\text{enr}} \longrightarrow (P, E, \Phi, U_{12}^{\text{sgn}}, K) \longrightarrow \alpha_{\text{HMT}}.$$

The coordinates P, E, Φ and the register K are earlier outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition, not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\widehat{\mathcal{X}}_r^{\text{enr}, \alpha}$ be the sector of truncated states that extend the internal alpha seed fixed at R_{36} . Write

$$x_r^\alpha = (\tilde{x}_r, \chi_\alpha, \varepsilon_r^\alpha, N_r^\alpha),$$

where χ_α is the internal character of the sector, not the expressed scalar value. Each state preserves remainder, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, register, memory, survival, provenance and depth. If

$$d_r^\alpha = \text{Dig}_{729}(\varepsilon_r^\alpha), \quad \varepsilon_{r+6}^\alpha = 729\varepsilon_r^\alpha - d_r^\alpha, \quad N_{r+6}^\alpha = 729N_r^\alpha + d_r^\alpha,$$

the effective transition is

$$\mathcal{U}_r^\alpha = \text{Upd}_r^\alpha \circ \text{Tra}_r^\alpha \circ \text{Sel}_r^\alpha, \quad \text{Ext}_{r+6,r}^{\alpha,\rightarrow} = \text{graph}(\mathcal{U}_r^\alpha).$$

The single-valuedness of these extensions, with the seed R_{36} fixed, produces a unique compatible section: the phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}_\alpha^{\text{enr}} := \widehat{\Omega}_\alpha := \lim_{\leftarrow r \geq 36} (\widehat{\mathcal{X}}_r^{\text{enr},\alpha}, \text{Ext}_{r+6,r}^{\alpha,\rightarrow}). \quad (11.13)$$

In this section, κ_{per} is the periodic reading κ_{ag} of the register already constructed; both notations denote the same fraction in the fixed base-one-thousand chart.

11.4.1 The dodecaphasic closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \dots, 999\}^N, \quad \mathsf{T}_{\alpha,N}(\mathbf{s}) := (A_1, \dots, A_N), \quad \mathbf{s} \in \mathcal{X}_\alpha^{\text{enr}},$$

where the blocks A_j are obtained through (11.1) with the compatible boundary $c_{N+1}^\infty = [R_{N+1}(Z)]$. If $N' \geq N$, let $\rho_{N',N} : \mathcal{W}_{N'} \rightarrow \mathcal{W}_N$ be the restriction to the first N blocks.

Theorem 11.11 (Complete dodecaphasic output and projective compatibility). *The collection $(\mathsf{T}_{\alpha,N})_{N \geq 1}$ is compatible:*

$$\rho_{N',N} \circ \mathsf{T}_{\alpha,N'} = \mathsf{T}_{\alpha,N} \quad (N' \geq N). \quad (11.14)$$

Its canonical cylinders

$$C_N^A(\mathbf{s}) := \left[\sum_{j=1}^N A_j 1000^{-j}, \sum_{j=1}^N A_j 1000^{-j} + 1000^{-N} \right) \cap I_\alpha \quad (11.15)$$

are nested, have diameter at most 1000^{-N} and satisfy

$$\bigcap_{N \geq 1} C_N^A(\mathbf{s}) = \{\alpha_A(\mathbf{s})\}, \quad \alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}. \quad (11.16)$$

Consequently, the first route expresses a unique infinite section, which we denote by $\alpha_{\text{HMT}} := \alpha_A$.

Proof. Proposition 11.4 provides the unique canonical expansion and the unique bounded sequence of carries. The local identity (11.3) and its telescoping (11.4) preserve the remote boundary; Proposition 11.6 proves that restricting an extended execution recovers exactly the preceding prefix. This gives (11.14). Membership of the full sum in each cylinder follows from the tail identity, and $\text{diam } C_N^A \leq 1000^{-N} \rightarrow 0$; the intersection contains a single point. Finally, (11.7) identifies that point with the expression in (11.16). All its coefficients and signs are fixed before the representation of alpha. $\square$

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^\infty = -1$. In particular,

$$\rho_{36,12}(\alpha_{A,36}) = \alpha_{A,12} \quad (11.17)$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

11.4.2 Analytic representation of the alpha coordinate

The analytic extension is specified through all its coefficients. Let $s \in \mathcal{X}_\alpha^{\text{enr}}$. Before decimal normalization, the same state already expresses the pre-coordinate

$$a(s) := p(s) + e_0(s) - f(s) - \kappa_{\text{per}}(s) \in I_\alpha, \quad (11.18)$$

with $p = \pi_{\text{HMT}} - 3$, $e_0 = e_{\text{HMT}} - 2$ and $f = \varphi_{\text{HMT}} - 1$. Equivalently, $a = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}$. No conventional α is introduced here: p, e_0, f are the three regional characters already generated, and κ_{per} is the rational reading of the terminal register. The integer closure applies to (11.18) the normalization in (11.1); the chart below expresses a compatible analytic representation on the same pre-coordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\text{per}})$ directly: it does not contain an edge $\alpha_A \rightarrow \lambda$, although the equality proved below allows both representation names to be identified subsequently.

Write $q = 1/729$, let F_s be the ordered field generated by the coordinates of s , and set $E_{9,s}(X) = E_{21/22}^{(9)}(s; X)$. We define the linking coefficient

$$\lambda(s) := -\frac{E_{9,s}(a(s))(1 - qa(s)^3)}{a(s)^{10}}. \quad (11.19)$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (11.19) must be read with its exact type: once the first representation of the state has produced $a(s)$, it fixes the unique extension within the geometric class defined below. It is not a second independent selection of $a(s)$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_s$ set

$$E_{s,\mu}(X) := E_{9,s}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(s) < 10^{-2}$ and $qa(s)^3 < 1$, we have

$$E_{s,\mu}(a(s)) = 0 \iff \mu = \lambda(s). \quad (11.20)$$

Thus, the coefficient seed contains no freedom once the state and the class of q -geometric extensions have been fixed. This uniqueness is relative to the declared class; it does not assert that the ninth-order jet by itself selects a tail among all possible analytic series.

The local action on blocks of three coefficients is

$$\mathcal{C}_\alpha : F_s^3 \longrightarrow F_s^3, \quad \mathcal{C}_\alpha(u, v, w) = q(u, v, w), \quad c^{(0)} = (\lambda(s), 0, 0). \quad (11.21)$$

Therefore, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(s), \quad b_{11+3n} = b_{12+3n} = 0. \quad (11.22)$$

This is the rule that computes each coefficient beyond degree nine. No free parameter remains at each extension.

For $M \geq 0$, let

$$E_{s,M}(X) := E_{9,s}(X) + \lambda(s) \sum_{n=0}^M q^n X^{10+3n}. \quad (11.23)$$

The complete function is

$$E_{s,\infty}(X) := E_{9,s}(X) + \lambda(s) \frac{X^{10}}{1 - X^3/729}. \quad (11.24)$$

To formulate the projective system without implicit types, we fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{J}_M := \coprod_{s \in \mathcal{X}_\alpha^{\text{enr}}} \{s\} \times F_s[X]/(X^{d_M+1}). \quad (11.25)$$

If $M' \geq M$, the restriction is

$$\tau_{M',M}(s, [P]_{d_{M'}}) := (s, [P]_{d_M}). \quad (11.26)$$

Proposition 11.12 (Naturality of the analytic recurrence). *Let*

$$\Gamma_M(s) = (s, [E_{s,M}]_{d_M}).$$

If $M' \geq M$, the truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \quad \Gamma_{E21}(s) = \lim_{\leftarrow M} \Gamma_M(s). \quad (11.27)$$

The projection onto the three new degrees is precisely $\mathcal{C}_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (11.22) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ adds only the next block; truncating it recovers the preceding polynomial literally. Compatibility for $M' \geq M$ is obtained by composition. Thus, the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under $\mathcal{C}_\alpha$. $\square$

11.4.3 Uniform convergence, simple root and rational intervals

The regional cylinders and the periodic register provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}\},$$

let P_c be its certified thousand-digit nonadic prefix. The datum used is not the rational number P_c treated as though it were the complete value, but the half-open cylinder and its outer closure

$$\mathcal{C}_c^{(1000)} = [P_c, P_c + \varepsilon), \quad \overline{\mathcal{C}}_c^{(1000)} = [P_c, P_c + \varepsilon], \quad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively with these closures. If P_π, P_e, P_φ denote the three prefixes, we obtain

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\text{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\text{per}}, \\ A_{1000} &:= [a_-, a_+], \quad a(s) \in [a_-, a_+] = A_{1000}, \quad \text{diam } A_{1000} = 3\varepsilon. \end{aligned} \quad (11.28)$$

The natural interval extension of (11.19) then transports A_{1000} , the cylinder of π_{HMT} and the directed rational bound for δ to a closed interval $\Lambda_{1000} \ni \lambda(s)$; none of the three tails is frozen. In particular,

$$\begin{aligned} \frac{7297352569283800997285105472380662}{10^{36}} &< a(s) < \frac{7297352569283800997285105472380663}{10^{36}}, \\ -8.711 \times 10^{-6} &< \lambda(s) < -8.709 \times 10^{-6}. \end{aligned} \quad (11.29)$$

These inequalities use certified prefixes with their rational tail bound; they do not receive a table of alpha values. The package verifier reproduces them from the three nonadic outputs and from the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \leq x \leq 10^{-2}$, the tail beyond M satisfies

$$|E_{s,\infty}(x) - E_{s,M}(x)| \leq 8.711 \times 10^{-26} \frac{r^{M+1}}{1-r}. \quad (11.30)$$

Moreover,

$$\left| \frac{d}{dx} \left(\lambda \frac{x^{10}}{1 - x^3/729} \right) \right| \leq 8.711 \times 10^{-6} \left(\frac{10 \cdot 10^{-18}}{1-r} + \frac{3 \cdot 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \quad (11.31)$$

The tail of the derivatives beyond block M also admits the depth-dependent estimate

$$\sup_{0 \leq x \leq 10^{-2}} |E'_{s,\infty}(x) - E'_{s,M}(x)| \leq 8.711 \times 10^{-24} r^{M+1} \left(\frac{13 + 3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow{M \rightarrow \infty} 0. \quad (11.32)$$

Theorem 11.13 (Limit function and unique simple root of the correlated chart). *The sequence $E_{s,M}$ converges uniformly, together with its derivatives, to $E_{s,\infty}$ on $\bar{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{s,\infty}(a(s)) = 0, \quad \inf_{x \in \bar{I}_\alpha} E'_{s,\infty}(x) > 6.239. \quad (11.33)$$

Consequently, $a(s)$ is the unique simple root of the complete function in I_α .

Proof. The ratio of two consecutive terms of the tail is bounded by $r < 1$, which gives (11.30); term-by-term differentiation and summation of the two geometric series give (11.31) and (11.32). The proof for the jet establishes $E'_{9,s} > 6.24$ on $\bar{I}_\alpha$; hence, the complete perturbation leaves the derivative above 6.239. Finally, substituting (11.19) into (11.24), the two summands evaluated at $a(s)$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with that pre-coordinate. $\square$

To make the identification effective at every depth, directed endpoint evaluations first certify

$$E_{s,\infty}(0) < 0 < E_{s,\infty}(10^{-2}), \quad E'_{s,\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{on } [0, 10^{-2}].$$

By continuity there is a root in the interval and, by the second inequality, it is unique. Only after this existence, monotonicity and the invariant that the bracket contains the root have been established is the mean-value-theorem fallback applied. We start from $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{s,\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}, \delta$ and λ , with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4} (s_j - r_j). \quad (11.34)$$

When $F_j^+ < 0$, the right half is retained; when $F_j^- > 0$, the left half is retained. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{s,\infty} \geq c = 6$ and the mean value theorem imply that any root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4} \right]. \quad (11.35)$$

Indeed, let ξ_j be the root and let $y_j = E_{s,\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (11.34), we have $|y_j| \leq cw_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, proving (11.35). The argument includes the case $y_j = 0$: the root is then at the midpoint, but the algorithm does not need to recognize this equality in order to retain it.

If the available enclosure does not satisfy (11.34), the source cylinders are first refined and the evaluation is repeated; no sign is declared and no insufficient contraction is accepted. In each of the three branches the new interval retains the root and has diameter at most $w_j/2$. For each finite target depth, directed refinement allows the width bound to be met without an uncertified comparison of a real number with zero. Monotonicity proves inductively that

$$a(s) \in D_{j+1} \subseteq D_j, \quad \text{diam } D_{j+1} \leq \frac{1}{2} \text{diam } D_j, \quad \text{diam } D_j \longrightarrow 0, \quad (11.36)$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule to twenty-four base-one-thousand levels and checks that the complete outer closure A_{1000} , not only its lower endpoint, lies inside each expressed cylinder. Its negative tests reject an excessively wide enclosure, an incorrectly ordered enclosure, a point other than the midpoint, a domain without a sign change, and a degenerate bracket. Coinductive extension allows one thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$ the verifier computes

$$j_N^- := \lfloor S_N a_- \rfloor, \quad j_N^+ := \lfloor S_N a_+ \rfloor \quad (11.37)$$

and expresses the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and avoids assigning to the left cell an endpoint lying exactly on its right boundary.

Let $C_N^A(s)$ be the integer cylinder in (11.15). We recursively choose $m(N) > m(N-1)$ so that $\text{diam } D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(s). \quad (11.38)$$

Define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is nonempty because both factors contain $a(s)$. Since the cylinder is half-open, u_N may not belong to $I_{B,N}$. Evaluation at that endpoint is understood through the continuous extension of $E_{s,\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(s), \quad \text{diam } I_{B,N} \longrightarrow 0, \quad E_{s,\infty}(\ell_N) \leq 0 \leq E_{s,\infty}(u_N). \quad (11.39)$$

The bound (11.30) allows a truncation to be chosen that preserves the sign at each endpoint where the complete function does not vanish. If an endpoint coincides with the root, the identity of the complete function and the membership already proved are used; that same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{s,\infty}(x) - E_{s,M}(x) = \lambda(s)x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

In the positive domain under consideration, where $\lambda(s) < 0$, we obtain at the root

$$E_{s,M}(a(s)) = -\lambda(s)a(s)^{10} \frac{(qa(s)^3)^{M+1}}{1 - qa(s)^3} > 0 \quad (M < \infty).$$

Therefore, the certification of the weak signs in (11.39) refers to $E_{s,\infty}$. Its evaluation uses the complete function or the truncation accompanied by the signed remainder. This covers both the included infimum and the supremum that may be excluded from the cylinder.

Proposition 11.14 (Compatibility squares and common cylinders). *The restrictions of words and of analytic representations commute:*

$$\begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\tau_{\alpha,N'}} & \mathcal{W}_{N'} \\ \parallel & & \downarrow \rho_{N',N} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\tau_{\alpha,N}} & \mathcal{W}_N \end{array} \quad \begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{F}_{M'} \\ \parallel & & \downarrow \tau_{M',M} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{F}_M. \end{array}$$

The collection $I_{B,N}$ provides the effective connection between the two squares and satisfies concretely $I_{B,N} \subseteq C_N^A$ for every N .

Proof. The first square is the compatibility of integer normalization; the second is Proposition 11.12. The inclusion follows from the definition (11.38), and nonemptiness and the signs follow from Theorem 11.13. No step identifies the root of a truncation with that of an arbitrary tail. $\square$

We now define, without leaving the prefix reader implicit,

$$\alpha_B(s) := \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x), \quad \text{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \quad \alpha_{B,N} := \text{pref}_N(\alpha_B). \quad (11.40)$$

Theorem 11.15 (Projective equality of the two correlated representations). *For every N , the integer and analytic routes express the same prefix:*

$$\boxed{\alpha_{A,N} = \text{pref}_N(\alpha_A) = \text{pref}_N(\alpha_B) = \alpha_{B,N}.} \quad (11.41)$$

In the limit,

$$\boxed{\alpha_A = \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x) = \alpha_B = \alpha_{\text{HMT}}.} \quad (11.42)$$

Proof. Integer normalization expresses exactly $a(s)$. Theorem 11.13 proves that the analytic chart represents that same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix, and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Therefore, their prefixes coincide for each N ; nesting and diameters tending to zero give equality of the limits. Metrological recognition occurs afterwards. $\square$

The analytic function constitutes a correlated representation of the same state, not a second derivation causally independent of the integer closure. This precision does not weaken the equality proved: it identifies its exact mechanism, prevents attributing to the jet a tail that it does not determine, and avoids promoting a conventional value to a generator.

$h \quad \hbar$

Planck constants and orientation

The determinantal structure fixes the action scale; return coefficients determine the angular coordinates.

From determinantal structure to orientation

The Planck constants are presented after the objects that determine their action sections. The dodecaphasic register contains the local Hadamard block; its integral quotient has a calculable order. The determinantal reader composes this datum with the coordinates already generated and fixes a decimal valuation. The scale and coefficient of a section perform different functions, which the proof keeps explicit.

The return distinguishes the sections before and after the turn. Their coefficients preserve their origin in the character and the regional continuation. The expression of action in physical units is introduced through the corresponding dimensional chart. The construction thus transmits a section, its coefficient, its scale, and the reading rule. The subsequent use of the constant preserves these relations.

Angle and counter-angle are developed next as orientation coordinates. The first reading shifts the blocks of alpha in the completion base; the second composes the dimensionless return coefficient with alpha. The circular chart expresses the result in degrees or radians and preserves the number of turns when the lift is used. This sequence explains why the angular coordinates are important intermediaries between action and the subsequent geometrical readers.

The conjugate channels receive this orientation and, together with marked incidence, evaluate the Barbero–Immirzi functional. The electronic realization uses its trajectory and its return readers, and receives the quotient of the action sections. The narrative preserves the order of its antecedents and distinguishes the dimensionless coefficient from the magnitude of action it helps to represent. The same constant can act at several stages without losing the map through which it was obtained.

This development prepares a concrete physical composition. Action and clock determine an energy; thermal transduction relates this energy to information; the area reader incorporates marked incidence. The longitudinal composition retains the inverse, its memory, and the inscribed incidences, determining a length on the declared radial reference. Its arrival modulus relates energy, conjugate length, and reduced radius; the two normalized identities fix the common coupling. The article preserves the calculations of these compositions to show how the arithmetic and geometrical results participate in a common physical reading.

12 Determinantal structure and oriented action sections

The readouts of π, φ, e, α arise from the APP-TRIT-TPK state of Sections 2–11. The dodecaphasic record of 9 preserves its orientation, its remainders and quotients, and the return memory (4.6). Two successive operations are defined on this record: a local norm determines the valuation of the action section, and a character with regional continuation determines its coordinates before and after the return. Both operations transmit their rules and normalizations to the angular construction.

12.1 Local quotient and determinantal norm

The local step-three block of the dodecaphasic record, identified in 9.8, is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \quad (12.1)$$

The quotient corresponds to this local block. The three blocks of the complete record retain their positions; the following norm operation acts on the sheets of G_H .

Lemma 12.1 (Structure of the local quotient). *The Smith normal form of H_4 is $\text{diag}(1, 2, 2, 4)$. Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even and one has value -2 , so their greatest common divisor is two. The identities $H_4 H_4^\top = 4I$ and $\det H_4 = -16$ show that the cofactors equal ± 4 ; the third determinantal divisor is four. The fourth is 16. The successive quotients of 1, 2, 4, 16 are 1, 2, 2, 4; their product determines the order of the quotient. $\square$

Definition 12.2 (Determinantal action reader). The scalar section α , constant on the sheets of G_H , and one application of the self-scaling factor φ determine

$$\Sigma_H = \varphi N_{G_H}(\alpha), \quad N_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \quad (12.2)$$

Its decimal valuation is

$$\text{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor.$$

The sixteenth power follows from the number of sheets once the multiplicative norm has been fixed. The norm and the single application of φ are part of the stated reader: the group order, considered in isolation, admits other functionals.

Theorem 12.3 (Decimal valuation of the action section). *The preceding reader, applied to the HMT coordinates already generated, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \quad \text{ord}_{10}(\Sigma_H) = -34. \quad (12.3)$$

Proof. We use the rational enclosures of the preceding readouts,

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \quad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}. \quad (12.4)$$

The function $(x, a) \mapsto xa^{16}$ is increasing in both positive variables. The integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \quad 1619 \cdot 7298^{16} < 10^{66}$$

give

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6} \right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6} \right)^{16} < 10^{-33}.$$

The definition of the valuation completes the proof. The enclosures are subsequent checks on the coordinates already produced; the determinantal operation precedes any chart of units. $\square$

12.2 Action character and regional continuation

The central chart retains the matrix

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the step-three orbit has length four, and the census quotient used is $104976/1944 = 54$, with the censuses defined in Section 4. These invariants determine the coefficients assigned to degrees 2, 3, 4, 5 of the character:

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\text{tr}(B_c)/2}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976}. \quad (12.5)$$

The assignment to these degrees and the alternating signs belong to the definition of the analytic character. The operative content combines the support, the orientation, and this assignment rule.

Definition 12.4 (Local character and regional return reader). On the readouts α, φ, π , the fifth-order character, the regional transfer, and its continuation are

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (12.6)$$

$$D_A = \exp\left(-\frac{100\pi\alpha}{9}\right), \quad (12.7)$$

$$c_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad (12.8)$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} c_\pi(\alpha). \quad (12.9)$$

The integer 169 is the regional selector of the readouts constructed in 6: the regional emission $169 | 272 | 272$ is retained, distinguished from $418 | 674 | 521$ in the exceptional construction. The first strict prolongation of its continuation has weight D_A , and each concatenation multiplies the weight by D_A . The initial normalization is unity.

The continuation is formulated before evaluation at $z = \alpha$. With D_A already produced by its reader, write $\mathcal{C}(z) = 169z^6 + \mathcal{R}(z)$, where $\mathcal{R}(z) \in z^7\mathbb{R}[[z]]$. Concatenation increases the degree by one; the initial weight and the transfer rule give

$$\mathcal{R}(z) = D_A z^7 + D_A z \mathcal{R}(z). \quad (12.10)$$

Proposition 12.5 (Uniqueness of the normalized continuation). *Equation (12.10) determines a unique series:*

$$\mathcal{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n. \quad (12.11)$$

Its realization is analytic for $|D_A z| < 1$.

Proof. The constant term of $1 - D_A z$ is one, so this element is invertible in the ring of formal power series. Solving (12.10) gives the rational expression. Equivalently, the coefficient of degree seven is D_A , and each subsequent coefficient is D_A times its predecessor. The geometric series converges absolutely in the stated disk. $\square$

The normalization of the first weight enters the uniqueness statement. The degree-six impulse and the concatenation ratio, without this initial weight, are also compatible with

$$169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}.$$

The unit-normalization rule fixes $\lambda = 1$ before evaluation. For $0 < \alpha < 1$, we have $0 < D_A < 1$ and $D_A \alpha < 1$; therefore,

$$\frac{D_A \alpha^7}{1 - D_A \alpha} = \sum_{n=0}^{\infty} D_A^{n+1} \alpha^{n+7}. \quad (12.12)$$

After the term of index N , the exact remainder is $D_A^{N+2} \alpha^{N+8} / (1 - D_A \alpha)$. The rational expression thus retains the complete continuation at any precision.

12.3 Positive sections and action per revolution

Let $\mathcal{L}_S$ be the oriented action line and $\mathcal{U}_S$ a positive basis. Its sections before and after the return are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (12.13)$$

The return preserves the regional compensation $90\mathcal{C}_\pi(\alpha)/\pi$ and the ratio

$$\delta_{\text{ret}} = H_5 - \eta_{\text{ret}}, \quad R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (12.14)$$

Proposition 12.6 (Positivity and covariance of the action ratio). *On the intervals (12.4), $0 < \eta_{\text{ret}} < H_5$, and $R_{\text{act}} > 1$. The ratio remains fixed under a change of the common positive basis $\mathcal{U}_S$.*

Proof. The terms of $\mathcal{C}_\pi$ are positive. To bound H_5 , it suffices to use $\alpha > 0.007$, $\alpha < 0.008$, and $\varphi < 1.619$. The square-root term in (12.6) is greater than one, and the sum of its negative terms is less than 0.008001; hence $H_5 > 0.99$. With $D_A < 1$ and $\pi > 3$,

$$0 < \frac{90}{\pi} \mathcal{C}_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

Thus, $\eta_{\text{ret}} > 0.989999 > 0$ and $\eta_{\text{ret}} < H_5$. The change of basis divides both action coordinates by the same positive factor, which cancels in their ratio. $\square$

The structural determination of angular action is the pair $(\hbar_{\text{pre}}, \hbar_{\text{ret}})$, with its identification by return. It preserves the valuation of the local norm, the action character, and the regional continuation. In each section $s \in \{\text{pre}, \text{ret}\}$, the action of a complete revolution is

$$h_s = 2\pi \hbar_s. \quad (12.15)$$

The subsequent chart $\mathcal{U}_S \mapsto J$ expresses these sections in units of action. The normalized radius to be constructed remains independent of this common choice of units.

13 Angular readout and channel orientation

The section $\hbar_{\text{ret}}$ belongs to the action line $\mathcal{L}_S$; η_{ret} is its dimensionless coefficient in the basis declared in (12.13). The angular readout then acts on $(\alpha, \eta_{\text{ret}})$. Writing $[t]_{360}$ for the class of t in $\mathbb{R}/(360\mathbb{Z})$, define

$$\mathcal{P}_{\text{ang}}(\alpha, \eta_{\text{ret}}) = ([1000\alpha]_{360}, [2(\eta_{\text{ret}} + \alpha)]_{360}). \quad (13.1)$$

Nonadic completion determines the factor 1000; bidirectional closure determines the factor two. The circular chart with 360 units preserves the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120$. On the positive branch, the lifted representatives are

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha). \quad (13.2)$$

The subscript distinguishes these angular coordinates from the blocks of the dodecaphasic record.

Proposition 13.1 (Shift of the base-thousand readout). *Let $\alpha = \sum_{n \geq 1} b_n 1000^{-n}$, with $b_n \in \{0, \dots, 999\}$, be the compatible expansion already generated. Its lifted coordinate satisfies*

$$A_{\text{deg}} = b_1 + \sum_{n \geq 1} b_{n+1} 1000^{-n}.$$

The readout preserves the blocks from the second onward and places the initial block in the integer part.

Proof. The series converges absolutely. Multiplying each term by 1000 and separating the initial term gives the identity. The operation acts on the constructed expansion, with its blocks and memory; the subsequent circular readout takes its class modulo 360. $\square$

13.1 Circular charts and covariance of action

Definition 13.2 (Charts of the angular coordinate). The degree chart expresses a complete revolution by 360 units; the radian chart expresses it by 2π . The coordinate change and its inverse are

$$\theta_A = \frac{\pi}{180} A_{\text{deg}}, \quad \theta_C = \frac{\pi}{180} C_{\text{deg}}^*, \quad A_{\text{deg}} = \frac{180}{\pi} \theta_A, \quad C_{\text{deg}}^* = \frac{180}{\pi} \theta_C. \quad (13.3)$$

A path with its revolution count retained uses lifted coordinates. The circular quotient gives its class modulo the corresponding revolution.

The record retains orientation, revolutions, and transport memory. The final phase and the lift are different readouts: the latter distinguishes paths that share the former. Each action expression declares which of these coordinates it uses.

Proposition 13.3 (Covariance of the action reader). *For a section $s \in \{\text{pre}, \text{ret}\}$ and a lifted angular coordinate,*

$$\mathcal{S}_s(\theta) = \hbar_s \theta_{\text{rad}} = h_s \frac{\theta_{\text{deg}}}{360}. \quad (13.4)$$

A positive revolution yields h_s .

Proof. Substituting $\theta_{\text{rad}} = \pi \theta_{\text{deg}} / 180$ and $h_s = 2\pi \hbar_s$ gives

$$\hbar_s \frac{\pi \theta_{\text{deg}}}{180} = \frac{h_s \theta_{\text{deg}}}{360}.$$

For one revolution, $\theta_{\text{deg}} = 360$, yielding h_s . $\square$

The periodicity of a representation retains its own condition: if $U(2\pi) = -I$, operator restitution occurs at 4π . The normalization of action per revolution in (13.4) and the return of the represented state thus preserve their respective domains.

13.2 Positive channels and reversal of the odd coordinate

In either chart, the channels are given by

$$q_{\pm} = \exp[-(\theta_A \pm \theta_C)] = \exp\left[-\frac{\pi}{180}(A_{\text{deg}} \pm C_{\text{deg}}^*)\right]. \quad (13.5)$$

Proposition 13.4 (Recovery of the angular coordinates). *In the lifted real chart, the positive channels uniquely determine*

$$A_{\text{deg}} = -\frac{90}{\pi} \log(q_+ q_-), \quad C_{\text{deg}}^* = \frac{90}{\pi} \log \frac{q_-}{q_+}. \quad (13.6)$$

The interchange $(q_+, q_-) \mapsto (q_-, q_+)$ preserves A_{deg} and reverses C_{deg}^ .*

Proof. The sum and difference of the logarithms in (13.5) are $-\pi A_{\text{deg}}/90$ and $-\pi C_{\text{deg}}^*/90$, respectively. The real logarithm is single-valued on the positive channels. Their product is symmetric, and their quotient is inverted upon interchange. $\square$

The antipodal involution $r \mapsto r + 6 \pmod{12}$ of the record forms six pairs. Uniform distribution of the odd coordinate over these pairs assigns $C_{\text{deg}}^*/6$ to each; their sum recovers C_{deg}^* . The unit conversion is applied once to each argument. In particular, $C_{\text{deg}}^*/A_{\text{deg}} = \theta_C/\theta_A$ whenever $A_{\text{deg}} \neq 0$, while the preceding state retains the lifted angles and their memory.

14 Dodecaphasic germ and recovery of unimodal branches

The oriented wheel U_{12}^{sign} , received from the TPK return, determines, in the uniform sector $\eta = 0$, the weights $w_m = 1/12$. Fix the phase functional and its ordered real representatives

$$\phi_m = \pi_{\text{HMT}} \frac{3m-1}{18}, \quad m = 1, \dots, 12. \quad (14.1)$$

The real lift of the phases is part of the reader: the following moments are evaluated on these representatives. The construction proceeds from the wheel, its orientation, weights and functional; a modification of any of these data modifies the resulting collection.

14.1 Normalization and analytic profile

Let $a_m = (3m-1)/18$ and $D_0 = \frac{1}{12} \sum_{m=1}^{12} a_m^2$. The integer sums

$$\sum_{m=1}^{12} (3m-1)^2 = 5394 = 6 \cdot 899, \quad \sum_{m=1}^{12} (3m-1)^4 = 4294614$$

give $D_0 = 899/648$. The quadratic normalization condition $s_0^2 \pi_{\text{HMT}}^2 D_0 = 2$ fixes

$$s_0 = \frac{36}{\pi_{\text{HMT}} \sqrt{899}}, \quad k_m = s_0 \phi_m = \frac{2(3m-1)}{\sqrt{899}}.$$

The angular scale cancels in this normalization before iteration of the profile

$$u_0(x) = 1 - \frac{1}{12} \sum_{m=1}^{12} \cos(k_m x), \quad f_\lambda(x) = 1 - \lambda u_0(x). \quad (14.2)$$

Proposition 14.1 (Exact representation and criticality). *The profile u_0 is an even entire function and satisfies*

$$u_0(x) = x^2 - \frac{715769}{2424603} x^4 + O(x^6).$$

With $y = x/\sqrt{899}$, its closed expression is

$$u_0(x) = 1 - \frac{\cos(37y) \sin(36y)}{12 \sin(3y)} = 1 - \frac{\sin(73y) - \sin y}{24 \sin(3y)}, \quad (14.3)$$

where the quotients extend through their removable singularities. For $0 < \lambda \leq 2/u_0(1)$, f_λ is an even, real-analytic self-map of $[-1, 1]$, with a unique quadratic critical point at zero and negative Schwarzian away from that point.

Proof. The finite cosine sum is entire and even. Its Taylor expansion uses

$$\frac{1}{12} \sum_m k_m^2 = 2, \quad \frac{1}{24 \cdot 12} \sum_m k_m^4 = \frac{715769}{2424603},$$

obtained from the preceding integer sums. For the closed form, the progression of arguments is $(6m-2)y$. Summing the geometric series of exponentials and taking the real part gives

$$\sum_{m=1}^{12} \cos((6m-2)y) = \frac{\sin(36y)}{\sin(3y)} \cos(37y).$$

The identity $2 \cos(37y) \sin(36y) = \sin(73y) - \sin y$ proves (14.3). The finite sum defines its extension at every zero of the denominator.

For $0 < x \leq 1$, all arguments satisfy $0 < k_m x \leq 70/\sqrt{899} < \pi_{\text{HMT}}$. Hence

$$u'_0(x) = \frac{1}{12} \sum_m k_m \sin(k_m x) > 0, \quad u'''_0(x) = -\frac{1}{12} \sum_m k_m^3 \sin(k_m x) < 0.$$

Parity gives the opposite signs on the negative interval and proves uniqueness of the critical point. Moreover, $f''_\lambda(0) = -2\lambda \neq 0$. Since $0 \leq u_0(x) \leq u_0(1)$ on $[-1, 1]$, the stated interval for λ gives $-1 \leq f_\lambda(x) \leq 1$. Finally,

$$S(f_\lambda) = \frac{u'''_0}{u'_0} - \frac{3}{2} \left(\frac{u''_0}{u'_0} \right)^2 < 0 \quad (0 < |x| \leq 1).$$

The first quotient is negative on both sides of the origin, and the remaining term is nonpositive. $\square$

14.2 Branchwise inversion and itinerary memory

Proposition 14.2 (Recovery of the folding). *Let $\lambda > 0$ and $v = (u_0|_{[0,1]})^{-1}$. For $y \in [1 - \lambda u_0(1), 1]$, the preimage on a branch is*

$$x = \sigma v\left(\frac{1-y}{\lambda}\right), \quad \sigma = \text{sgn } x. \quad (14.4)$$

At $y = 1$, the preimage is zero and $\sigma = 0$ is fixed.

Proof. Continuity and strict monotonicity of u_0 on $[0, 1]$ provide the inverse v . The equation $y = 1 - \lambda u_0(x)$, together with parity, determines $|x| = v((1-y)/\lambda)$; the sign determines orientation. At $y = 1$, uniqueness of the minimum of u_0 gives $x = 0$. Away from the critical point, the inverse is differentiable. Its derivative becomes singular at the origin, where $u_0(x) = x^2 + O(x^4)$. $\square$

For a finite distribution on pairs $\pm x_i$, with distinct $x_i > 0$, let $p_i^\pm$ be their probabilities, $p_i = p_i^+ + p_i^-$, $Y = f_\lambda(X)$ and $\Sigma = \text{sgn } X$. A mass at zero is also allowed; its branch-uncertainty contribution is zero. With natural logarithms,

$$H(X | Y, \Sigma) = 0, \quad (14.5)$$

$$H(X | Y) = H(\Sigma | Y) = \sum_{i: p_i > 0} p_i h(p_i^+ / p_i), \quad (14.6)$$

$$h(p) = -p \log p - (1-p) \log(1-p). \quad (14.7)$$

The proof groups the distribution by the fibres of f_λ . Monotonicity distinguishes the moduli x_i , and (14.4) distinguishes the two signs within each fibre. With equal probabilities in each pair and no mass at zero, the uncertainty is $\log 2$.

If $0 < \lambda \leq 2/u_0(1)$, the second iteration is defined on the same interval and satisfies

$$H(X | f_\lambda^2(X)) = H(X | f_\lambda(X)) + H(f_\lambda(X) | f_\lambda^2(X)).$$

Indeed, both intermediate maps are deterministic, and the chain rule for the entropy of $(X, f_\lambda(X))$, conditioned on $f_\lambda^2(X)$, gives the identity. Successive recovery uses the signs at each step. For $\lambda = 0$, the map is constant and additionally requires retention of the input modulus.

14.3 Scope of the rescaled return

For $a = -f(1) \neq 0$, let $h_a(x) = -ax$. The operator

$$\mathcal{R}(f) = h_a^{-1} \circ f^2 \circ h_a$$

is a rescaled return on $[-1, 1]$ when the interval $J = h_a([-1, 1])$ belongs to the domain of f^2 and satisfies $f^2(J) \subseteq J$. The scaling h_a is bijective; the two-step return retains its interpretation with the itinerary and the map f . Reconstruction of an iterative root from $\mathcal{R}(f)$ and a is an additional condition. Proposition 14.1 establishes the germ and its critical class. Existence and uniqueness of a fixed point of the renormalizer, its stable and unstable spectral decomposition, and transversality of this collection are the additional hypotheses required for passage to a universal renormalization limit.

e^-

Algebraic structure of the electron

Central persistence, orientation and action sections in the electronic realization.

Persistence and electronic trajectory

The electronic structure begins at the persistent central position of APP. Cellular inversion distinguishes this position from the conjugate pairs that surround it. Its evaluation preserves the sheets, remainders, and quotients; the trajectory prolongs these data through the operations of the TPK. The exposition of mass subsequently receives the action sections and the readers it uses.

The central vacancy makes visible a decisive difference between residual reading and arithmetic history. Two positions may share the sum and the multiplicative remainder, while their integer products preserve distinct quotients. The change in quotient records the transfer; orientation distinguishes their conjugate positions. The example makes visible the information needed by a reconstruction that restores the operation performed.

The electronic chain adds an explicit chronology. The cellular recurrence, inversions, and reading windows preserve the registers of the persistent section. The calendar records the steps at which capacity remains unchanged; the central defect records the difference in quotient between evaluations with the same remainder. The electronic trajectory preserves this second datum together with its orientation and chronology. Capacity vacancy and quotient defect count distinct operations, with their respective registers.

This continuity makes the distinction between return and recovery pertinent once again. A phase may recur while the register of the section incorporates the path. The subsequent evaluation uses this transported information and keeps the central section identified. Mass, orientation, and the expressions of action are presented together with their antecedents, so that each step can be read as a definite composition.

The electron thus provides a developed case of the article's thesis. Persistence corresponds to a structure; the trajectory describes its evolution; the registers distinguish the changes; the dimensional reader obtains a magnitude. Its relevance to conservation lies in this complete sequence. The unity maintained during transport preserves relations whose content becomes accessible through different readings. The subsequent operator study will abstract the conditions for recovery and show their exact balances.

15 Algebraic structure of the electron and action scale

The electronic construction receives the arithmetic state with trajectory memory produced by APP-TRIT-TPK and its numerical coordinates. At this point in the development, π , φ , e and α designate the coordinates already obtained from the same genealogy, in the constructive order set out above. They are not parameters taken from a physical table. The electronic section is a subsequent realization of that joint discrete structure of the continuum: it preserves the cell of origin, the two arithmetic sheets, orientation, carry, and return memory before producing a scalar.

Three objects should be distinguished from the outset. The first is the central section and its transport history; the second is the dimensionless coordinate obtained from it; the third is its realization on an energy or mass line. Equality between two numerical coordinates does not identify these three levels. In particular, Euler's number e , the electronic coordinate ε_e , and a rest energy E_e will have distinct notations.

The aim of this section is to establish the composition

$$\varepsilon_e^\beta = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4) R_{\text{act}}^{80-54\alpha+6\Delta_4}, \quad (15.1)$$

explaining separately the origin of the prefactor, the two integers of the basal stage, the normalization of Δ_4 , the action ratio, and the return triple. The dimensional chart is introduced only after this composition. The result does not consist in making a dimensionless number acquire units merely through its resemblance to a tabulated quantity.

The composition brings together factors with different functions. The following table identifies their meaning and refers to the corresponding construction; the proofs are developed in the order of their dependencies.

Factor	Meaning in the composition	Construction and proof
$\sqrt{3}/4$	Norm of the commutator of the projectors on the principal plane.	Equation (15.6).
$\exp(\varphi/\pi^2)$	Exponential character of the electronic orientation of the regional quotient.	Equation (15.12); subsections 6.7.1 and 6.7.2.
$-22\alpha^3$	Contribution of the regional complement in the central return.	Equations (15.14) and (15.15).
$1 + 15\Delta_4$	Torsional normalization composed with regional and tritic multiplicity.	Equations (15.14), (15.16) and (15.18).
$R_{\text{act}}^{\kappa_e}$	Quotient of the action sections, raised to the character determined by the two readers of the electronic return.	Equations (15.29), (15.36), (15.37) and (15.38).

Table 10: Factors of the electronic composition and provenance of their rules.

Identity of the section and composition of its observables. The structural identity of the electron is introduced through the central section and its transport operations. Inversion of the APP chart selects the cell (5, 5) by its fixed-point property. The two arithmetic sheets allow the deformations preserving the sum and the product residue to be studied. The result distinguishes the two orientations (8, 2) and (2, 8) and preserves a difference of one quotient unit. This is the first level of information: a cell, an involution, a preserved fibre, and a minimal integer defect. No mass evaluation is required to state or prove it.

The passage to arithmetic projections introduces a second level. The sum and product fibres determine conditional expectations on the same finite space. Their commutator measures dependence on the order of these two projections. The norm $\sqrt{3}/4$ follows from the spectrum of their pairing; the block attaining this norm also preserves the quadratic relations proved below. Thus, the scalar prefactor and the local algebra have a common provenance, but contain different information. The norm retains the magnitude of the commutator, while the matrices preserve its orientation and its composition law.

Areal–volumetric transfer incorporates the regional coordinates at a third level. The mode calendar determines the incidence matrix and its quadratic action. The closure mark produces the ratio π/φ^2 ; interchanging the two arguments provides the orientation φ/π^2 used in the electronic exponential character. The two expressions are related by an explicit involution. This relationship explains why a regional coordinate may appear in the electronic equation with a power or in a position different from that in its first reading: the operation applied is part of the mathematical datum that is preserved.

The cubic alpha term belongs to the regional complement of a marked return. Counting its positions yields $27 - 5 = 22$, while the three sheets of each of the five regions yield $3 \cdot 5 = 15$. The proofs preserve the domain of these counts and the orientation determining their signs. The basal formula therefore composes an incompatibility norm, a transfer character, a position deficit, and a torsional memory. Writing it on a single line expresses the composition of these operations; explaining it requires reconstructing them in that order.

The transition from the basal coordinate to the complete coordinate preserves the same central section. The return factor uses the history records, and the two stated readers recover the triple $(80, 54, 6)$ along that trajectory. Multiplication by $R_{\text{act}}^{80-54\alpha+6\Delta_4}$ modifies the evaluation, but keeps its origin and antecedents identified. The basal value constitutes an intermediate stage of the composition; the complete value also includes this return contribution. The physical comparison must use the stage specified by the formula being evaluated.

The action scale intervenes subsequently with a different function. A ratio of two action sections is dimensionless and cancels their common scale. The absolute realization instead uses an action basis, a temporal reader, and the map to energy or mass. This separation allows the construction of the scale and the identity of the electronic section to be preserved simultaneously. The chronological quantum provides a scale reference; the central section determines the structural factor expressed relative to it. The dimensional equations of this chapter keep that composition explicit.

The electron's relevance to the overall argument arises from this convergence of operations. The same arithmetic that produced the regions now determines a central cell, its deformations, and its projections. The fine-structure constant, constructed earlier, acts in this composition as a closure coordinate. Return information determines the complete evaluation, and dimensional realization allows subsequent comparison. Explaining the electron from its structure consists in preserving this succession of objects, maps, and values at every step of the proof.

15.1 Center, orientation, and vacancy on a preserved fiber

The cellular support under consideration is $\{1, \dots, 9\}^2$, with both APP evaluations, $r + c$ and rc . The central inversion is

$$\rho(r, c) = (10 - r, 10 - c). \quad (15.2)$$

The center is not chosen by a mass: it is characterized by a property of the support that precedes its physical reading.

Proposition 15.1 (Uniqueness of the center). *The inversion ρ has a unique fixed point, $(5, 5)$.*

Proof. The equations $r = 10 - r$ and $c = 10 - c$ are equivalent to $2r = 2c = 10$. Their unique solution in the nonadic chart is $(5, 5)$. $\square$

The preceding property identifies a cell, not a unique deep history. A route adds the initial choice of orientation and its compatible prolongations. In the chart used here, the central oriented section begins at $(5, 5)$ with both orientations positive; it will be denoted by Γ_e . The return of its cellular projection does not erase the transported orientations or the carry record of the section with its trajectory memory.

Vacancy is now considered, after the closure of α , as a local operation on this section. Its arithmetic can be isolated in a finite lemma; this isolation does not make it a substitute for the preceding generator of constants.

Definition 15.2 (Central fiber and quotient defect). The central additive fiber is

$$\mathcal{F}_{10} = \{(5 + t, 5 - t) : t \in \mathbb{Z}, |t| \leq 4\}.$$

In the chart $n = \rho_9(n) + 9q_9(n)$, with $\rho_9(n) \in \{1, \dots, 9\}$, the quotient defect relative to the center is defined as

$$v(t) = q_9(25) - q_9(25 - t^2),$$

when both products have the same nonadic residue.

Lemma 15.3 (Minimal central vacancy). *The only points of $\mathcal{F}_{10}$ that preserve the multiplicative residue of the center are*

$$(5, 5), \quad (8, 2), \quad (2, 8).$$

At the two noncentral points, the quotient defect is exactly one.

Proof. The sum is constantly ten, whereas

$$(5 + t)(5 - t) = 25 - t^2.$$

Preservation of the residue is equivalent to $t^2 \equiv 0 \pmod{9}$, that is, to $3 \mid t$. In the integer interval $[-4, 4]$, the solutions are $t = -3, 0, 3$. For the two nonzero solutions,

$$25 = 7 + 9 \cdot 2, \quad 16 = 7 + 9 \cdot 1.$$

The residue seven is preserved and one unit of quotient is lost. There is no other admissible noncentral displacement in that domain, so the positive defect is minimal. The interchange $t \mapsto -t$ permutes the two outputs and preserves the defect. $\square$

Vacancy here is not a numerical zero: it is a difference between two states with the same residual reading and different quotients. If only the residue were preserved, that difference would disappear. Orientation also distinguishes the two output points. Neither the quotient defect nor that orientation is automatically identified with an electric charge or a mass; their physical realization requires the corresponding readers.

15.1.1 Central coordinates, orientation, and regional records

The central structure relates position, evaluation, and transport. At cell $(5, 5)$, the two APP sheets give

$$\mathcal{A}(5, 5) = ((1, 1), (7, 2)), \quad 5 + 5 = 1 + 9 \cdot 1, \quad 5 \cdot 5 = 7 + 9 \cdot 2. \quad (15.3)$$

The two ones on the additive sheet are, in order, residue and quotient. Their numerical coincidence preserves two distinct arithmetic functions. On the sum-10 fiber, the two oriented vacancies satisfy

$$\mathcal{A}(8, 2) = \mathcal{A}(2, 8) = ((1, 1), (7, 1)). \quad (15.4)$$

The complete additive sheet and the multiplicative residue are preserved; the quotient coordinate of the product decreases by exactly 1. Transport additionally retains which of the two conjugate positions has been reached.

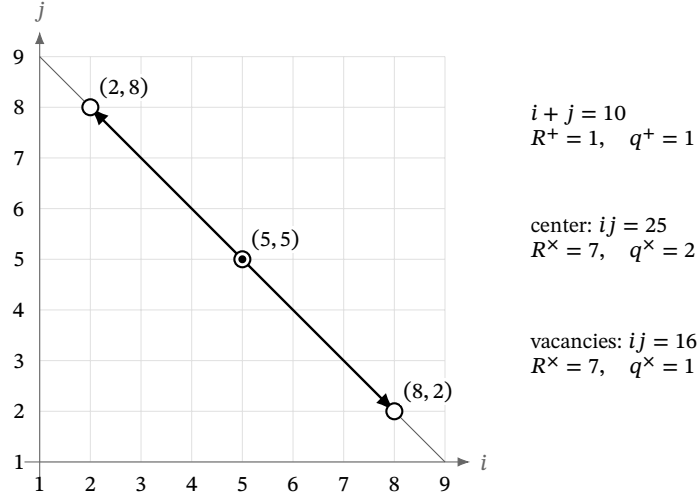


Figure 10: Electronic center and conjugate vacancies in the APP chart. The diagonal is the additive fiber $\mathcal{F}_{10}$; the center is the fixed point of $(i, j) \mapsto (10 - i, 10 - j)$. The two shifts of magnitude 3 preserve the sum and product residue and reduce the multiplicative quotient by one unit. The arrows represent deformations on the fiber.

Central symmetry also characterizes its persistence under changes of chart. If an admissible atlas transformation f intertwines inversion, $f\rho = \rho f$, then

$$\rho f(5, 5) = f\rho(5, 5) = f(5, 5).$$

Uniqueness of the fixed point forces $f(5, 5) = (5, 5)$. This is the internal property making the central location stable: every transformation of that type preserves the cell before evaluating its electronic character (Section 15.1.1).

The central matrix block belongs to evaluation on the two adjacent positions 4, 5. Its decomposition, established in the local development of the kernel, is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-.$$

Here 9 and 5 are eigenvalues of two modes, whereas $(5, 5)$ is a position in the atlas. Concatenations of its entries give $72 + 27 = 77 + 22 = 99$ and $77 - 22 = 55 = 54 + 1$. The first equality preserves the total sum under two arrangements; the second expresses the defect between diagonal and antidiagonal readings. These operations precede electronic composition and preserve their matrix provenance.

The initial orientation is another datum. In the representation of electronic return, $(h_0, v_0) = (1, 1)$ is fixed, that is, positive advance in both coordinates. Simultaneous inversion every 27 steps, read in windows of 9, determines

$$\tau_e = (+1, +1, +1, -1, -1, -1, +1, +1, +1, -1, -1, -1).$$

The segments $(1, 1, 1)$ belong to this orientation sequence: each segment comprises three windows in the same direction. The recurrence in the section on readers proves this record in (15.32), and its two evaluations recover the triple $(80, 54, 6)$ entering (15.1). The positional coordinate, APP quotients, and sign sequence are thus connected through explicit operations.

The signature 555555 belongs to a regional reading in another domain. For a decimal emission $U = (U_1, \dots, U_6)$, we write $U_j = 100d_{1j} + 10d_{2j} + d_{3j}$, with three digits, including leading zeros. The multiplicative-signature reader is

$$\mathcal{E}_\times(U) = ([d_{2j} - d_{1j}]_{10})_{j=1}^6.$$

In the transversal region $U_\pi^\perp = (501, 614, 498, 169, 272, 272)$, the differences are $(-5, -5, 5, 5, 5, 5)$. Consequently,

$$\mathcal{E}_\times(U_\pi^\perp) = (5, 5, 5, 5, 5, 5), \quad \Psi(U_\pi^\perp) = 010211. \quad (15.5)$$

The first map preserves a multiplicative signature of the catalog; the second preserves the ternary word common to the five regions. Figure 12 shows the transversal position of this emission and its multiplicity 432.

Memory required to transport the constant signature. The signature 555555 additionally provides a verifiable example of the trajectory information required by TPK. Let $\mathcal{X}_x = D_9^2 \times \{N, E, S, O\}$, with 324 oriented positions in the product channel. Advance P moves one square in the stored direction, and the half-turn H reverses that direction. In this representation, the reader E_{sig} sums the discrete exponents by windows:

$$\ell_9(1, 2, 4, 8, 7, 5) = (0, 1, 2, 3, 4, 5), \quad \ell_9(3) = \ell_9(6) = \ell_9(9) = 0.$$

During each of the six nine-step windows, $\ell_9(\rho_9(ij))$ is evaluated before every product advance, selected by TRIT at times 2, 5, 8 (mod 9). The window sum is reduced modulo 10. After steps 27 and 54, the direction is reversed. This rule determines the six components of E_{sig} from the initial position and orientation.

Enumeration of this domain produces 26 signatures. The fiber of 555555 contains 24 oriented positions; after one advance, their images are distributed among three signatures, with eight representatives each:

$$555177, \quad 555717, \quad 555771.$$

Three witnesses, in coordinates from 1 to 9, allow the distinction to be checked directly:

x	$E_{\text{sig}}(x)$	$E_{\text{sig}}(Px)$
$(3, 5, N)$	555555	555177
$(3, 4, N)$	555555	555717
$(3, 4, S)$	555555	555771

Thus, the common value of the signature does not determine its next value: the transport class within that fiber forms part of the operational datum. The minimal stable partition is calculated by starting from signature equivalence $\sim_0$ and refining through

$$x \sim_{n+1} y \iff x \sim_n y, \quad Px \sim_n Py, \quad Hx \sim_n Hy.$$

The procedure terminates because the domain is finite. Every stable congruence refining $\sim_0$ refines every $\sim_n$, by induction; its fixed point is therefore the minimal stable congruence required. The census gives $26 \rightarrow 28 \rightarrow 28$, with histogram $27 \cdot 8 + 108 = 324$. The only signature that splits is 555555, into three classes of eight elements. The accompanying certificate reproduces the enumeration, the three classes, and stability under both operators (Section 15.1.1).

The multiplicity 24 here belongs to the product-channel fiber; the multiplicity 432 belongs to the complete regional emission $U_\pi^\perp$. The first proves why the signature needs transport memory; the second places that signature within the joint catalog of the two sheets. Both readings participate in the same architecture, with specified domains and operations.

This distinction of domains makes it possible to bring together the occurrences of unity and the center with their respective meanings:

Record	Object and function
$(5, 5)$	Fixed cell of the central inversion of the APP atlas.
$(R^+, q^+) = (1, 1)$	Residue and quotient of the central additive evaluation.
$(h_0, v_0) = (1, 1)$	Initial orientation of the two shifts.
$(+1, +1, +1)$	Three consecutive windows of τ_e with positive orientation.
$c = 111111$	Datum of the fifth nonadic boundary, Section 6.16.1.
$\mathcal{E}_x = 555555$	Signature of the transversal region of π , with six positions.

Electronic coordination is obtained by composing these levels in their dependency order. APP determines the center and evaluations; TRIT selects regimes and orientations; TPK prolongs the trajectory and preserves its records. The previously generated regions provide the characters π, φ, e ;

the dodecaphase record provides α ; the electronic readers determine the factors of equation (15.1). This composition preserves the identity of the section throughout its different representations, up to dimensional evaluation and subsequent metrological comparison.

15.2 Commutator norm of the arithmetic projections

The electronic prefactor is obtained without using α , torsion, or a dimensional unit. The domain is the third realization of APP, $\mathcal{A}_3 = \{1, \dots, 9\}^3$, equipped with the uniform measure. We define

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \quad \Pi(x) = \rho_9(x_1 x_2 x_3).$$

The conditional expectations R_Σ and R_Π are the orthogonal projections onto functions constant on each additive and multiplicative fiber, respectively. Thus, for a function f ,

$$(R_\Sigma f)(x) = \frac{1}{|\Sigma^{-1}(\Sigma(x))|} \sum_{y: \Sigma(y)=\Sigma(x)} f(y),$$

and analogously for R_Π . The two projections act on the same space; they do not represent two independent arithmetic supports.

Lemma 15.4 (Two projections and principal angles). *If $s \in [0, 1]$ is a singular value of the pairing between orthonormal bases of the ranges of two projections P, Q , its nontrivial block contributes $s\sqrt{1-s^2}$ to $\|[P, Q]\|$.*

Proof. On the corresponding principal plane, one can write

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} s^2 & s\sqrt{1-s^2} \\ s\sqrt{1-s^2} & 1-s^2 \end{pmatrix}.$$

Their commutator is

$$[P, Q] = s\sqrt{1-s^2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The final matrix is orthogonal, so its norm is one. Common or orthogonal blocks have zero commutator. The orthogonal decomposition into principal planes gives the claim and reduces the total norm to the maximum of these contributions. $\square$

Proposition 15.5 (Exact incompatibility norm). *On $\ell^2(\mathcal{A}_3)$,*

$$\|[R_\Sigma, R_\Pi]\| = \frac{\sqrt{3}}{4}. \quad (15.6)$$

Proof. Let $N_{ab} = |\Sigma^{-1}(a) \cap \Pi^{-1}(b)|$. Integer enumeration of the three symbols yields

$$N = 9 \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}.$$

Each row sums to 81; the column sums are

$$(m_1, \dots, m_9) = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

The normalized indicator functions of the fibers have pairing matrix $C_{ab} = N_{ab}/\sqrt{81m_b}$. Consequently,

$$M = CC^T, \quad M_{ac} = \sum_{b=1}^9 \frac{N_{ab}N_{cb}}{81m_b}$$

is a rational matrix whose spectrum consists of the squares of the required singular values.

Exact multiplication shows that $(1, \dots, 1)$, $(1, -1, 0)^3$, and $(1, 1, -2)^3$ are eigenvectors with eigenvalues 1, $1/4$, and $7/132$, respectively. The subspace of vectors supported on $\{3, 6, 9\}$ whose sum is zero has dimension two and eigenvalue $1/18$. The zero-sum vectors supported on $\{1, 4, 7\}$ or $\{2, 5, 8\}$ form a four-dimensional kernel. These subspaces are independent and their dimensions sum to nine. The complete spectrum is therefore established:

$$\text{spec}(M) = \left\{1, \frac{1}{4}, \frac{7}{132}, \frac{1}{18}, \frac{1}{18}, 0, 0, 0, 0\right\}.$$

The maximum of $\lambda(1 - \lambda)$ on this set is $3/16$, attained at $\lambda = 1/4$. The preceding lemma gives $\sqrt{3/16} = \sqrt{3}/4$. $\square$

The mode $(1, -1, 0)^3$ is precisely the distribution of values of the tritic phase selector in the nonadic chart. The scalar norm retains a measure of incompatibility, but does not by itself preserve the orientation of that mode or reconstruct the two projectors. That memory remains in the route state accompanying the scalar reading.

Proposition 15.6 (Algebra of the local electronic block). *On the principal plane attaining (15.6), let*

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad S = 2P - I, \quad J = \frac{4}{\sqrt{3}}[P, Q].$$

Then $S^2 = I$, $J^2 = -I$, $SJ = -JS$, and the real algebra generated by S, J is $M_2(\mathbb{R})$, a realization of $\text{Cl}_{1,1}$. Moreover,

$$n_{\pm} = \frac{S \pm J}{2}, \quad n_{\pm}^2 = 0, \quad n_+ n_- + n_- n_+ = I.$$

Proof. We have $S = \text{diag}(1, -1)$ and $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Multiplication verifies the three relations. The matrices I, S, J, SJ are linearly independent and form a basis of $M_2(\mathbb{R})$, proving the claim about the algebra. Finally, $(S \pm J)^2 = S^2 + J^2 \pm (SJ + JS) = 0$, and the sum of the cross products is $(S^2 - J^2)/2 = I$. $\square$

In this block, an elliptic direction, a hyperbolic direction, and two parabolic null directions coexist. The result describes a local matrix representation of the arithmetic of the two sheets. It does not identify all of APP with $M_2(\mathbb{R})$, or with a quantum Latin square; such claims would require other domains and other relations. Nor does the exponential belong exclusively to the parabolic regime: it can be applied to any of the generators of this block.

15.3 Spinor representation of the electronic block and helicity

The electronic structure has a representation preserving more information than its scalar evaluation. The principal plane of the two three-dimensional APP projectors, selected by the tritic mode $(1, -1, 0)^3$, has already provided the operators S and J . The former is a symmetric involution; the latter is the normalized commutator and preserves sheet orientation. Their composition makes it possible to construct explicitly a rotation representation and separate its two helicity components. This construction acts on the electronic block; the TPK route record and its memory remain as data of the block's provenance.

Let E_e be the real principal plane, equipped with the inner product inherited from the conditional expectations, and let $S_e = E_e \otimes_{\mathbb{R}} \mathbb{C}$ be its complexification. The operators already obtained satisfy

$$S^* = S, \quad J^* = -J, \quad S^2 = I, \quad J^2 = -I, \quad SJ = -JS.$$

Complexification extends the representation field. It preserves the real operators and allows the scalar unit i to be distinguished from the endomorphism J , although both square to -1 in their respective domains.

Proposition 15.7 (Hermitian triple induced by the two sheets). *The operators*

$$\sigma_1 = SJ, \quad \sigma_2 = -iJ, \quad \sigma_3 = S \quad (15.7)$$

are Hermitian and traceless. They satisfy

$$\sigma_a \sigma_b = \delta_{ab} I + i \sum_{c=1}^3 \epsilon_{abc} \sigma_c, \quad \frac{1}{2} \text{tr}(\sigma_a \sigma_b) = \delta_{ab}, \quad (15.8)$$

where $\epsilon_{123} = 1$. They therefore form an orthonormal basis of the real space V_e of traceless Hermitian endomorphisms of $\mathcal{S}_e$, for the inner product $\langle X, Y \rangle = \frac{1}{2} \text{tr}(XY)$.

Proof. The preceding relations give $(SJ)^* = (-J)S = SJ$ and $(-iJ)^* = -iJ$. The squares of all three matrices are I . Furthermore,

$$(SJ)(-iJ) = iS, \quad (-iJ)S = iSJ, \quad S(SJ) = J = i(-iJ).$$

Reversing the order changes the sign. This proves the first identity in (15.8). In the orthonormal basis of the principal plane, the matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Their trace is zero, and taking traces in the product identity proves orthonormality. Every traceless Hermitian endomorphism has the form $\begin{pmatrix} z & x-iy \\ x+iy & -z \end{pmatrix}$, with real x, y, z , and thus belongs to their real span. $\square$

In the usual matrix representation, this triple is called the Pauli matrices. Their use here follows the construction of P, Q, S, J : they are obtained through the products (15.7). The direction space of this representation is the unit sphere of V_e , whose positive form has just been constructed through the trace.

Proposition 15.8 (Rotation generators and directional decomposition). *Define $L_a = \sigma_a/2$. Then*

$$[L_a, L_b] = i \sum_c \epsilon_{abc} L_c, \quad \sum_{a=1}^3 L_a^2 = \frac{3}{4} I. \quad (15.9)$$

For $n = (n_1, n_2, n_3)$ with $\sum_a n_a^2 = 1$, let

$$H_e(n) = \sum_a n_a \sigma_a, \quad h_e(n) = \frac{1}{2} H_e(n), \quad \Pi_{\pm}(n) = \frac{1}{2} (I \pm H_e(n)).$$

The linear formula $H_e(v) = \sum_a v_a \sigma_a$ is also used for v of arbitrary norm. One has

$$\mathcal{S}_e = \text{im } \Pi_+(n) \oplus \text{im } \Pi_-(n), \quad h_e(n) \Pi_{\pm}(n) = \pm \frac{1}{2} \Pi_{\pm}(n). \quad (15.10)$$

The two summands are orthogonal complex lines. Rotation transport around n is

$$U_n(\theta) = \exp\left(-\frac{i\theta}{2} H_e(n)\right) = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} H_e(n), \quad U_n(2\pi) = -I, \quad U_n(4\pi) = I. \quad (15.11)$$

Proof. The product identity of the triple gives the commutators in (15.9); their squares sum to $3I/4$. Antisymmetric terms cancel when multiplying $H_e(n)$ by itself, so that $H_e(n)^2 = I$. It follows that

$$\Pi_{\pm}(n)^2 = \Pi_{\pm}(n), \quad \Pi_+(n) \Pi_-(n) = 0, \quad \Pi_+(n) + \Pi_-(n) = I.$$

They are Hermitian and have trace 1; their rank is therefore 1. Multiplication by $H_e(n)/2$ proves (15.10). The exponential series splits into even and odd powers because $H_e(n)^2 = I$, producing (15.11). It also directly proves unitarity and the law $U_n(\theta + \psi) = U_n(\theta) U_n(\psi)$.

Angular normalization is verified on V_e . For every $v \in \mathbb{R}^3$, the same product identity gives

$$U_n(\theta) H_e(v) U_n(\theta)^* = H_e(v \cos \theta + (n \times v) \sin \theta + n(n \cdot v)(1 - \cos \theta)).$$

The induced action is a rotation of angle θ and axis n . The factor $1/2$ in the spinor generator is therefore the one corresponding to that angular parameterization. A complete rotation restores the direction in V_e and changes the sign of the vector in $\mathcal{S}_e$; two complete rotations restore both. $\square$

The representation is irreducible: a complex subspace invariant under the three matrices would be invariant under all of $M_2(\mathbb{C})$, the algebra they generate. Its only invariant subspaces are 0 and $\mathcal{S}_e$. The relations (15.9) thus identify the spin-1/2 representation. If n is the propagation direction in a kinematic realization, $h_e(n)$ is its dimensionless helicity, with values $\pm 1/2$. The kinematic direction is declared when that realization is made; the two projectors are defined for every direction before a particular propagation state is selected.

Preservation of the electronic structure and types of inversion. Direction inversion has an exact law:

$$H_e(-n) = -H_e(n), \quad \Pi_{\pm}(-n) = \Pi_{\mp}(n).$$

This operation exchanges helicity labels relative to a direction. Charge conjugation, by contrast, acts on the oriented signature of the material route; cellular inversion acts on the atlas coordinates. Their domains and actions remain distinct. Uniqueness of cell (5, 5) is compatible with the two lines in (15.10): an atlas cell is not a vector in the spinor representation it carries.

The electronic evaluation ε_e^β preserves its formula and readers. On $\mathcal{S}_e$, its scalar realization is $\varepsilon_e^\beta I$, which commutes with the generators and with $\Pi_{\pm}(n)$. Helicity decomposition adds rotational information without modifying the evaluation or turning spin into a multiplicative mass coefficient.

Chirality corresponds to the grading of the relativistic representation used: an involution Γ_{ch} determines its projectors $(I \mp \Gamma_{\text{ch}})/2$. The helicity in (15.10) additionally depends on n . For its part, the holonomy of a real Möbius band describes the change of sign of a line when traversing a loop in its base. The law of that line, the spinor return (15.11), and nonadic phase holonomy with memory advance are transport laws on specified objects. Their comparison preserves those objects: the helical form of a trajectory does not replace the representation just constructed or by itself fix chirality or charge.

15.4 Transfer between sheets and coefficients of the basal stage

The primitive tritic calendar visits the modes (Σ, Π, Π) . Counting consecutive pairs, including the closure of the block, yields one transition $\Sigma \rightarrow \Pi$, one $\Pi \rightarrow \Pi$, and one $\Pi \rightarrow \Sigma$. Their primitive incidence is

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

This matrix realization is obtained after the calendar. It is not used to replace the coinductive reader that has already generated φ . Its characteristic polynomial is $X^2 - X - 1$, and recognition of the two roots as φ and $-\varphi^{-1}$ describes the action of that output in modal transfer. The quadratic action has eigenvalues $\varphi^2, -1, \varphi^{-2}$; the last is the positive stable mode. The regional mark π , applied once, gives π/φ^2 .

The trajectory space, its measure, and the closure operator are constructed in subsections 6.7.1 and 6.7.2. Formula (6.18) preserves its provenance as a quotient of returns; the electronic application subsequently composes this reading with the following involution. The regional signature 555555 accompanies the closure region, whose transport preserves the state with memory; the centre (5, 5) fixes the cell of origin of the oriented electronic section. Both kinds of information converge in the composition and retain their respective domains.

Lemma 15.9 (Interchange of the two orientations). *Let $\mathcal{R}(x, y) = x/y^2$ and $\mathcal{S}_{\pi\varphi}(x, y) = (y, x)$, for $x, y > 0$. Then*

$$\begin{aligned} \mathcal{S}_{\pi\varphi}^2 &= I, & \mathcal{R}(\pi, \varphi) &= \frac{\pi}{\varphi^2}, \\ (\mathcal{R} \circ \mathcal{S}_{\pi\varphi})(\pi, \varphi) &= \frac{\varphi}{\pi^2} = \frac{1}{\varphi^3(\pi/\varphi^2)^2}. \end{aligned} \tag{15.12}$$

Proof. Applying the interchange twice gives the identity. Moreover, $\varphi^3(\pi/\varphi^2)^2 = \pi^2/\varphi$; taking its inverse proves the last equality. $\square$

The electronic reader uses the orientation φ/π^2 , followed by the exponential character. The preceding identity explains the interchange, but does not replace the rule that chooses that character or by itself produce the integers of the correction. These have a different provenance.

The phase-compatible central return has minimal length twenty-seven. With the phase and sheet record, its finite space is

$$\mathcal{K}_e = \mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3.$$

The five regions of π are distinct genealogical identities, not five interchangeable copies of a visible expansion. In the regional gauge fixed by the construction, their injection into the return occupies the positions

$$(4, 0), \quad (5, 0), \quad (6, 0), \quad (6, 1), \quad (6, 2). \quad (15.13)$$

The transverse orientation, mutated return, and three positive regions remain separated by that record. Admissible changes of gauge transport the labels jointly; they do not reselect the positions from an electronic value.

Proposition 15.10 (Deficit and retained memory). *Given the central return and the preceding regional injection, the oriented boundary deficit and torsional memory are*

$$n_e = -|\mathcal{K}_e \setminus \iota(\mathcal{R}_\pi)| = -22, \quad m_e^{\text{tor}} = |\mathcal{R}_\pi \times \mathbb{F}_3| = 15. \quad (15.14)$$

Proof. The five positions in (15.13) are distinct. Thus, $|\mathcal{K}_e| = 9 \cdot 3 = 27$, and the complement of the image has cardinality $27 - 5 = 22$. The orientation convention records the deficit removed from the boundary as negative. The retained memory has three sheets per region, so its positive cardinality is $5 \cdot 3 = 15$. $\square$

The same pair has the integer check supplied by the collection's APP chart:

$$396 = 18 + 72 + 108 + 198, \quad \frac{396}{18} = 22, \quad \frac{270}{18} = 15 = 3 \cdot 5 = \binom{6}{2}. \quad (15.15)$$

The weight 18 belongs to the central 2×2 kernel, and 270 to the declared global channel. These equalities check the normalization of the count; the return argument also fixes the signs. The quotient of the alternating rings, $396/(18 + 108) = 22/7$, is rational and is not identified with π . Nor does $\binom{6}{2}$ mean $6/2$: it counts the unordered pairs of a six-element set.

Evaluation of the deficit in the third symmetric realization uses α^3 . This power is the product of three copies of the closure scalar already obtained; it does not assert a universal rule associating any dimension with the same power of α . The specified domain and reader are essential to the composition.

15.5 Global torsion and the basal coordinate

Definition 15.11 (Electronic torsional normalization). The reduced defect d_4 and the complete reader Δ_4 are distinguished:

$$d_4 = \frac{\pi - e \log \pi}{270}, \quad \Delta_4 = d_4 \left(1 + \frac{\pi}{729}\right). \quad (15.16)$$

The denominator 270 corresponds to the census channel, whereas the factor $1 + \pi/729$ retains the coupling with the cubic cell of 729 positions. The electronic reading uses Δ_4 both in its basal stage and in the return exponent.

This definition acts on previously generated π and e . The logarithm is a subsequent operation in their real realization, not a retrospective input of conventional values. Torsion is global: it is not determined independently for each section by an observed mass.

Lemma 15.12 (The two normalizations are not interchangeable). *We have*

$$\Delta_4 - d_4 = \frac{\pi}{729} d_4. \quad (15.17)$$

In particular, $d_4 > 0$ and $\Delta_4 > d_4$ for the coordinates under consideration.

Proof. The identity follows by distributivity. For positivity, the function $f(x) = x - e \log x$, on $x > 0$, has derivative $1 - e/x$ and second derivative $e/x^2 > 0$. Its unique minimum is at $x = e$, where it is zero. Since $\pi \neq e$, we have $f(\pi) > 0$. Both factors in (15.16) are positive and $1 + \pi/729 > 1$. $\square$

Remark 15.13 (Preservation of the normalization). Removing the second factor in (15.16) changes the reader; it is not an algebraic simplification. If d_4 is introduced as an auxiliary coordinate, the same object must be written as $\Delta_4 = (1 + \pi/729)d_4$ in every occurrence. Keeping the integers 15 and 6 and replacing only Δ_4 by d_4 produces a different formula.

Definition 15.14 (Basal electronic coordinate). The composition of incompatibility, modal transfer, deficit, and retained memory is

$$\varepsilon_e^{(0)} = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4). \quad (15.18)$$

The equation combines three operations of different kinds. The norm $\sqrt{3}/4$ quantifies the incompatibility of the sheet projections; its origin is a principal angle of the APP fibers. The exponential acts on the orientation φ/π^2 of the transfer reader. The cubic subtraction records the 22 complementary positions of the marked return, and the torsional factor preserves the 15 components corresponding to three sheets per region. The composition preserves these domains even though its final evaluation is scalar.

The presence of π , φ , and e thus has operational meanings. Euler's number enters through the exponential and the logarithmic normalization of Δ_4 ; φ and π enter through oriented transfer. The constant α acts after its own construction through the cubic closure term. The chronological return must still act on this basal coordinate. Separating the stages makes it possible to explain what changes when an orientation, normalization, or record is modified, while preserving the central structure that identifies the electronic section.

The parentheses in this equation are part of its content. The term $-22\alpha^3$ is subtracted *outside* the exponential; the memory $1 + 15\Delta_4$ multiplies the entire bracket. Placing the deficit in the exponent or applying a different torsion to it would alter the composition.

Proposition 15.15 (Basal positivity). For $0 < \alpha < 10^{-2}$, $\varphi > 0$, and the normalization (15.16), we have $\varepsilon_e^{(0)} > 0$.

Proof. The exponential is greater than one, and $22\alpha^3 < 22 \cdot 10^{-6} < 1$. The bracket is therefore positive. Also, $1 + 15\Delta_4 > 1$ and $\sqrt{3}/4 > 0$. $\square$

The subsequent evaluation of (15.18) begins with

$$\varepsilon_e^{(0)} = 0.5109989224880660725 \dots$$

This number is still a dimensionless coordinate and is not the final stage: the return memory of the route itself remains to be applied.

15.6 Determinantal action scale and decimal valuation

The absolute scale and the relative corrector originate in different objects. The former uses the local block of the step-three dodecaphasic record,

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \quad (15.19)$$

The operation uses a local quotient. The three blocks of the complete record are not replaced by a tensorization that would multiply the number of sheets of this reader again.

Lemma 15.16 (Number of sheets of the local quotient). *The Smith normal form of H_4 is $\text{diag}(1, 2, 2, 4)$. Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. The order-two minors are even, and one has value -2 , so their greatest common divisor is two. Since $H_4 H_4^T = 4I$ and $\det H_4 = -16$, the cofactors are ± 4 ; the third determinantal divisor is four. The fourth is 16. The successive quotients of the divisors 1, 2, 4, 16 are 1, 2, 2, 4. Their product gives the order of the quotient. $\square$

Definition 15.17 (Determinantal action reader). The multiplicative norm of the scalar section α , constant over the sheets of G_H , and one application of self-scaling φ produce

$$\Sigma_H = \varphi N_{G_H}(\alpha), \quad N_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \quad (15.20)$$

The associated decimal chart determines $\text{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor$.

The sixteenth power thus comes from the number of sheets of the quotient, once this local reader is fixed. The factor φ acts once on the basis of the action line. Smith computation alone would not select every possible functional on the quotient: the multiplicative norm and self-scaling form part of the stated operation.

Theorem 15.18 (Decade of the action section). *The reader (15.20), applied to the already generated HMT coordinates, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \quad \text{ord}_{10}(\Sigma_H) = -34. \quad (15.21)$$

Proof. The following rational intervals, much coarser than the available cylinders, suffice:

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \quad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}.$$

The function $(x, a) \mapsto xa^{16}$ is increasing in both positive arguments. The two integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \quad 1619 \cdot 7298^{16} < 10^{66}$$

imply

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6} \right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6} \right)^{16} < 10^{-33}.$$

The definition of decimal order concludes the proof. Neither an SI unit nor a reference mass participates in these inequalities. $\square$

The normalized section begins with $\Sigma_H = 1.0462438381 \dots \times 10^{-34}$. Its mantissa is not identified with the action coordinate before or after return. The determinantal reader fixes the decimal order; the two return coordinates are obtained through the following readers.

15.7 Action sections and dimensionless return quotient

Let $\mathcal{L}_S$ be an oriented action line and $\mathcal{U}_S$ its positive basis. The electronic composition uses two sections of this line and their dimensionless quotient. Its construction requires the determinantal decade already obtained, the fifth-order action character, and the regional contribution of the return.

15.7.1 Action character and provenance of its coefficients

The reduced Planck constant, $\hbar$, is the physical object motivating the realization of the angular action section. Its place in the electronic dependency must be specified at two levels. The determinantal norm Σ_H determines the decade of the scale. The character H_5 and regional return then determine the coordinates of the action sections. This separation allows the result required for the electron to be presented without anticipating the complete developments of other sectors of dimensional mechanics.

The provenance of the fifth-order character can be made explicit through invariants already fixed on the support. In the central chart, consider

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the orbit of step 3 has length 4, and the relevant count quotient is $104976/1944 = 54$. The character coefficients satisfy

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\text{tr}(B_c)/2}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976}. \quad (15.22)$$

The assignment of these invariants to degrees 2, 3, 4, 5, with the alternating signs in (15.23), belongs to the definition of the analytic character used. Making it explicit is essential: the set of cardinalities, considered without the character and the ordering of its degrees, would admit other expressions. The constructive content lies in the composition of the support, orientation, and stated analytic rule.

On this basis, the fifth-order character and regional return term have the expressions

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (15.23)$$

$$D_A = \exp\left(-\frac{100\pi\alpha}{9}\right), \quad (15.24)$$

$$\mathcal{C}_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad (15.25)$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} \mathcal{C}_\pi(\alpha). \quad (15.26)$$

These are subsequent readings of dodecaphasic closure and its regional incidence. The integer 169 is retained as the regional selector already constructed, and the coefficients of H_5 as those of its local character; they are not replaced by an interpolation of the Planck constant. The formulas specify exactly which readings are needed for the electronic stage. No other functionals of mass theory need be introduced here.

For $0 < \alpha < 1$, we have $0 < D_A < 1$, so $D_A\alpha < 1$. The rational term in (15.25) preserves the complete sum of the geometric tail

$$\frac{D_A\alpha^7}{1 - D_A\alpha} = \sum_{n=0}^{\infty} D_A^{n+1} \alpha^{n+7}. \quad (15.27)$$

It is not a truncation that can silently be modified when precision changes. After the term indexed by N , the remaining tail is $D_A^{N+2} \alpha^{N+8} / (1 - D_A\alpha)$, allowing the evaluation to be controlled without recourse to a target quantity.

Definition 15.19 (Action sections and return reader). The sections before and after return are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (15.28)$$

Whenever $\eta_{\text{ret}} > 0$, define

$$\delta_{\text{ret}} = H_5 - \eta_{\text{ret}}, \quad R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (15.29)$$

Proposition 15.20 (Positivity and invariance of the ratio). *On the intervals used in the proof of (15.21), $0 < \eta_{\text{ret}} < H_5$, and therefore $R_{\text{act}} > 1$. The ratio is unchanged when the basis $\mathcal{U}_S$ is replaced by any other common positive basis.*

Proof. The terms of $\mathcal{C}_\pi$ are positive. For an elementary bound, it suffices to use $\alpha < 0.008$, $\varphi < 1.619$, $\alpha > 0.007$, $D_A < 1$, and $\pi > 3$. The square-root term in (15.23) is greater than one, while the sum of the negative terms is less than 0.008001; in particular, $H_5 > 0.99$. Moreover,

$$0 < \frac{90}{\pi} \mathcal{C}_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

Thus $\eta_{\text{ret}} > 0.989999 > 0$ and $\eta_{\text{ret}} < H_5$. Finally, the change of basis divides both action coordinates by the same positive factor, which cancels in their quotient. $\square$

Subsequent evaluation gives

$$\begin{aligned} H_5 &= 1.0545718176461544696 \dots, \\ \eta_{\text{ret}} &= 1.0545718169150390611 \dots, \\ R_{\text{act}} &= 1.0000000006932817630 \dots. \end{aligned}$$

The difference and quotient are two readings of the same sections, one additive and the other multiplicative. They do not represent two independent Planck constants. The passage from angular action to action per full turn is expressed, in each section, by $h = 2\pi\hbar$. The subsequent chart $\mathcal{U}_S \mapsto \mathbf{J}$ assigns a unit but does not act retroactively on the decimal order already proved.

Regional return can also be written in an angular chart. This reparametrization is not an additional indispensable dependency: (15.23)–(15.26) directly provide H_5 , η_{ret} , and R_{act} . If angle and counter-angle are introduced, both must be transported in the same chart; degrees and radians are not mixed within a difference.

15.7.2 Regional continuation and renewal equation

The selector 169 determines the degree-six impulse. Subsequent prolongation is organized by the transfer D_A and a unit normalization of the determinant line. These rules admit a formulation in formal series before substituting $z = \alpha$, with D_A already constructed. Write

$$\mathcal{C}(z) = 169z^6 + R(z), \quad R(z) \in z^7\mathbb{R}[[z]].$$

Concatenating a prolongation increases the degree by one unit and multiplies its weight by D_A . The first strict prolongation has weight D_A . Its renewal equation is therefore

$$R(z) = D_A z^7 + D_A z R(z). \quad (15.30)$$

Proposition 15.21 (Formal uniqueness of the normalized continuation). *Equation (15.30) determines a unique series, and*

$$\mathcal{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n.$$

The realization is analytic in $|D_A z| < 1$.

Proof. The element $1 - D_A z$ has constant term one and is invertible in the ring of formal series. Solving the equation produces the displayed series. Equivalently, the coefficient of degree seven is D_A , and each subsequent coefficient is D_A times the preceding one. The geometric series converges absolutely in the indicated disk. $\square$

Normalization of the first prolongation is an indispensable operational datum. If only the degree-six impulse and concatenation ratio were retained, the series

$$169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

would continue to share both data for every λ . The unit rule fixes $\lambda = 1$ before evaluation. The contribution of each rule to uniqueness of the reader is thus localized, and the rational expression preserves the complete continuation.

15.7.3 Reduced Planck constant and return of the action section

In the dimensional chart of (15.28), the section before return is the model construction intended for comparison with the reduced Planck constant:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S.$$

The subsequent section additionally preserves the regional compensation $90\mathcal{C}(\alpha)/\pi$. Both belong to the same action line, and their ratio R_{act} measures the multiplicative effect of return. Passage to h through 2π uses the relation between angular parameterization and a complete turn.

This construction effectively enters $R_{\text{act}}^{80-54\alpha+6\Delta_4}$. The electron preserves its central coordinate and trajectory; the chronological scale does not replace that structure with an undifferentiated quantum. Nor is a mass already expressed in MeV multiplied again by 10^{-34} : the decade belongs to the action section, cancels in its dimensionless ratio, and reappears only where the corresponding dimensional realization acts.

The numerical comparison of H_5 , the returned section, and $h/(2\pi)$ is presented in Section 62.1. This distinguishes the name of the physical object, the equation the model assigns to it, and the precision with which that equation reproduces it.

15.8 Two readers of the electronic return

Electronic memory is represented by a tritic word of length 108 and a signed twelve-phase word. To make the exponent calculation reproducible, we specify the chronological projection used. Initially, let $(r_0, c_0) = (5, 5)$ and $(h_0, v_0) = (1, 1)$. At each step, the selector repeats $(+1, -1, 0)$. With orientation fixed within each twenty-seven-step block, the visible rule is

mode	integer read	cellular transport
+1	$r + c$	$c \mapsto 1 + (c - 1 + h) \bmod 9$
-1	rc	$r \mapsto 1 + (r - 1 + v) \bmod 9$
0	9	no displacement

The emitted symbol is $\rho_9(\text{integer read}) \bmod 3$. After steps 27, 54, 81, h and v are simultaneously reversed. This rule computes the observable projection; the state update additionally preserves residue, quotient, carry, sheet, boundary, and memory. The record of 108 symbols is not identified with the complete state.

In each complete three-step block, each coordinate advances once; nine such blocks are needed to return to the center in the same phase. The minimal length of this recorded return is therefore 27. If phase is forgotten, the cell is already reached after step 26, before the neutral threshold completing the block; that is not the typed return counted by $\mathcal{K}_e$. The explicit reading of the first two return blocks gives

$$a = 100000220, \quad b = 120200000, \quad \gamma_e = (a^3 b^3)^2. \quad (15.31)$$

Here, powers denote word concatenation. In particular, $\gamma_{e,t+54} = \gamma_{e,t}$. In the record of nine steps per phase, the central orientations give

$$\tau_e = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1). \quad (15.32)$$

The periodicity of these representations does not amount to restarting the history that produces them.

Definition 15.22 (Cyclic discrepancy reader). For a word $\gamma \in \mathbb{F}_3^{108}$, define

$$D_k(\gamma) = \#\{t \in \mathbb{Z}/108\mathbb{Z} : \gamma_t \neq \gamma_{t+k}\}.$$

On the domain of words whose period divides 54, the integer reader is

$$K_D(\gamma) = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right). \quad (15.33)$$

Lemma 15.23 (Integrality and central evaluation). *The D_k on the preceding domain are even. For (15.31),*

k	1	3	4	27	36
D_k	54	56	68	48	32

and $K_D(\gamma_e) = (80, 54, 6)$.

Proof. The map $t \mapsto t + 54$ pairs discrepancy indices without fixed points and preserves their condition; each cardinality is even. For the explicit word a^3b^3 of length 54, the discrepancy count for each displacement is, in the same order, 27, 28, 34, 24, 16. Doubling the word doubles these five cardinalities. Finally,

$$(D_{27} + D_{36}, D_1, (D_4 - D_3)/2) = (48 + 32, 54, (68 - 56)/2) = (80, 54, 6).$$

□

The displacements 27 and 36 are the chronological lifts of the 90- and 120-degree channels in a chart where nine steps correspond to thirty degrees. The angular reading explains that representation, but the integers are computed on the word, without introducing a mass or a target real value.

Definition 15.24 (Dodecaphasic window reader). For $\tau \in \{-1, 0, 1\}^{12}$, with cyclic indices, let

$$\begin{aligned} C_k(\tau) &= \sum_{r=0}^{11} \tau_r \tau_{r+k}, \\ A_\ell(\tau) &= \sum_{r=0}^{11} \left(\sum_{j=0}^{\ell-1} \tau_{r+j} \right)^2, \\ P_6(\tau) &= \sum_{r=0}^5 \tau_r \tau_{r+6}. \end{aligned}$$

Define

$$\Omega(\tau) = 4A_3(\tau) - 3A_4(\tau), \quad K_\Omega(\tau) = (\Omega(\tau), 54, P_6(\tau)). \quad (15.34)$$

The 54 in this reader records the declared chronological half-turn. It is not confused with the local APP jump of value four.

Lemma 15.25 (Primitive contrast and correlation form). *Among integer linear contrasts $uA_3 + vA_4$ that vanish whenever $A_3 = 3a$ and $A_4 = 4a$, the primitive pair, with orientation $90 - 120$, is $(u, v) = (4, -3)$. Moreover,*

$$\Omega = -2C_1 - 4C_2 - 6C_3. \quad (15.35)$$

Proof. Vanishing for every a requires $3u + 4v = 0$. By coprimality, the solutions are $(u, v) = n(4, -3)$, and orientation fixes the sign of the primitive generator. Expanding the squares gives

$$A_\ell = \ell C_0 + 2 \sum_{k=1}^{\ell-1} (\ell - k) C_k.$$

Therefore, $A_3 = 3C_0 + 4C_1 + 2C_2$ and $A_4 = 4C_0 + 6C_1 + 4C_2 + 2C_3$. Their combination $4A_3 - 3A_4$ cancels C_0 and yields (15.35). □

The uniqueness proved belongs to that class of integer linear contrasts; it does not by itself exclude every nonlinear functional of the windows. Specifying the functional class avoids turning an identity of two channels into a claim of universal selection.

Theorem 15.26 (Return triple of the electronic section). *The two readers applied to the records of Γ_e coincide and give*

$$K_D(\gamma_e) = K_\Omega(\tau_e) = (80, 54, 6). \quad (15.36)$$

Their evaluation in the already generated coordinates is

$$\kappa_e = 80 - 54\alpha + 6\Delta_4. \quad (15.37)$$

Proof. The first reader was computed above. For τ_e , the antipodal product is always one, and hence $P_6 = 6$. The cyclic count gives $C_1 = 4$, $C_2 = -4$, $C_3 = -12$; thus

$$\Omega = -2 \cdot 4 - 4(-4) - 6(-12) = 80.$$

Equivalently, $A_3 = 44$, $A_4 = 32$, and $4 \cdot 44 - 3 \cdot 32 = 80$. Both triples are therefore $(80, 54, 6)$. Evaluating the functional $(A, B, C) \mapsto A - \alpha B + \Delta_4 C$ proves the last formula. □

This coincidence concerns the central section. It does not identify the two readers on every route domain: one compares tritic symbols and the other compares windows of a signed record. Nor is 80 obtained from the period of a word assigned to Euler's number. Its origin in this composition is the triple of the electronic route.

15.9 Complete electronic composition and normalization

Definition 15.27 (Multiplicative return). The action of return memory on the basal coordinate is

$$\varepsilon_e^\beta = \varepsilon_e^{(0)} \exp(\kappa_e \log R_{\text{act}}). \quad (15.38)$$

Theorem 15.28 (Positive electronic factorization). *With the declared readers, orientation, and normalization, (15.38) coincides with (15.1). All its factors are dimensionless and the resulting coordinate is positive.*

Proof. The incompatibility norm is (15.6); modal transfer, deficit, and retained memory compose (15.18). The triple (15.36) produces (15.37), and the ratio of the two action sections is (15.29). Substituting those three identities into (15.38) yields exactly (15.1). The basal stage is positive, as is the exponential of a real number. The common factor with the dimension of action has already canceled in forming R_{act} . $\square$

The evaluation of this composition begins with

$$\varepsilon_e^\beta = 0.5109989506900008086082571648 \dots \quad (15.39)$$

The digits are a subsequent evaluation of the formula, not data used to select 22, 15, or (80, 54, 6). The numerical representation of the basal coordinate by a finite prefix introduces only its truncation error; it does not modify the exact formula defining it.

The effect of a different torsional normalization can be located without comparison to a measurement. Write $B = \frac{\sqrt{3}}{4} [\exp(\varphi/\pi^2) - 22\alpha^3]$, $R = R_{\text{act}}$, and

$$\mathcal{E}(d) = B(1 + 15d)R^{80-54\alpha+6d}.$$

For the two defects in (15.16),

$$\frac{\mathcal{E}(\Delta_4)}{\mathcal{E}(d_4)} = \frac{1 + 15\Delta_4}{1 + 15d_4} R^{6(\Delta_4-d_4)} > 1. \quad (15.40)$$

The first torsional inequality and $R > 1$ prove that they are not the same coordinate. If only the defect in the exponent is changed, keeping the complete basal coordinate, the ratio of the two outputs is $R^{6(\Delta_4-d_4)}$, again different from one. A very small difference is still a difference of formula; it cannot be attributed to a mere conversion of units.

15.10 Dimensional realization: energy, mass, and chronological scale

The electronic coordinate finally acts on a dimensional line. Let $\mathcal{L}_E$ be an energy line with positive basis $\mathcal{U}_E$. The realization is

$$E_e^{(0)} = \varepsilon_e^{(0)} \mathcal{U}_E, \quad E_e^\beta = \varepsilon_e^\beta \mathcal{U}_E, \quad m_e^\beta = \frac{E_e^\beta}{c_{\text{int}}^2}. \quad (15.41)$$

The last expression belongs to the mass line when c_{int} has the dimension of velocity in the declared realization. In the comparative chart $\mathcal{U}_E = 1 \text{ MeV}$, (15.39) expresses an energy in MeV; the corresponding mass is expressed in MeV/c^2 . It is common to abbreviate both quantities by taking $c = 1$, but that abbreviation must not conceal the dimensional map.

The action scale also admits an action-clock realization. If t_0 is a positive temporal section, the 108-step clock defines

$$\omega_0 = \frac{2\pi}{108t_0}, \quad E_{\text{clk}} = \hbar_{\text{ret}} \omega_0, \quad m_{\text{clk}} = E_{\text{clk}} c_{\text{int}}^{-2}. \quad (15.42)$$

Action times frequency has the dimension of energy. This chronological quantum is a common dimensional section; it is not, by definition, the electronic section. The route word and its factors are not obtained from the uniformity of the clock.

Proposition 15.29 (Cancellation of the common scale). *The factor $10^{-34}\mathcal{U}_S$ in (15.28) is not part of κ_e . On the same energy line,*

$$\frac{E_e^\beta}{E_e^{(0)}} = R_{\text{act}}^{\kappa_e}. \quad (15.43)$$

The result is independent of the choice of the common dimensional basis.

Proof. In forming $\hbar_{\text{pre}}/\hbar_{\text{ret}}$, the factor $10^{-34}\mathcal{U}_S$ cancels exactly. Dividing the two energy expressions in (15.41) also cancels $\mathcal{U}_E$, and (15.38) gives the stated ratio. $\square$

Thus, a coordinate already expressed in MeV is not multiplied again by 10^{-34} . If E_{clk} is chosen as the basis of $\mathcal{L}_E$, the coordinates must be transported by the change-of-basis factor; retaining the number from the one-MeV chart unchanged would identify two bases without proof. This is the precise point at which the action scale enters the absolute realization and ceases to enter the relative correction.

There is also a linear realization that makes the separation between clock and genealogy visible. On $\ell^2(\mathbb{Z}/108\mathbb{Z}; \mathbb{R})$, the cyclic shift S has fixed space

$$\ker(S - I) = \mathbb{R}n_0, \quad n_0 = \frac{1}{\sqrt{108}} \sum_{t=0}^{107} \delta_t.$$

If g_{Γ_e} is the route symbol in the free linearization of genealogies and $\mathcal{U}_M$ is a basis of the mass line, the rank-one map is defined by

$$\mathfrak{C}_e : \mathbb{R}n_0 \otimes \mathbb{R}g_{\Gamma_e} \longrightarrow \mathcal{L}_M, \quad \mathfrak{C}_e(n_0 \otimes g_{\Gamma_e}) = \varepsilon_e^\beta \mathcal{U}_M. \quad (15.44)$$

The map assembles the preceding factorization once its bases have been declared. The uniform vector n_0 does not select ε_e^β : that coordinate comes from APP, the oriented return, and its readers. An algebraic realization of the null mode is thus distinguished from an identification of the electron with the clock quantum.

The internal conclusion is the positive factorization of a central section, with its operations, memory, and normalizations made explicit. Assignment to a physical electron adds the charge representation, spin, coupling, and corresponding unit chart; it does not follow exclusively from a decimal coincidence. Metrological comparison is presented after the exceptional prolongation and does not alter the inputs of any of the constructions performed here.

15.10.1 Structural dependence of the electronic coordinate

The electronic equation expresses a hierarchical composition. Its first factor, $\sqrt{3}/4$, belongs to the central matrix structure and fixes a geometric normalization. The exponent φ/π^2 uses the already generated regional coordinates in the indicated orientation of the areal–volumetric reader. The term $22\alpha^3$ incorporates the dodecaphasic coupling into the cubic correction. The factor $1 + 15\Delta_4$ preserves the global torsional normalization. Finally, the quotient of the action sections acts with the exponent determined by the return readers. Each operation thus has a different antecedent within the same construction.

The articulation becomes more precise when its dependencies are considered. The central norm can be calculated before the regional coordinates are evaluated. The exponential and cubic correction require those coordinates and the closure of α . Multiplicative memory additionally requires the action sections and the 108-position trajectory. Knowing the basal expression therefore does not amount to having evaluated the complete coordinate. The succession of operations does not add an external mass to the point (5, 5); it develops the readers assigned to its persistent section.

The distinction between scale and structure also provides an interpretation of the physical comparison. Mass is the dimensional realization of the complete coordinate, while the centre, involution,

and vacancy describe the relations that produce it. A common change of unit modifies the dimensional coordinates, but leaves the dimensionless ratios and the proved transport identities unchanged. This is the sense in which the electron preserves its structural character when expressed relative to a chronological scale.

15.11 Lorentzian transport and electromagnetic representation

The electronic section provides a structure and a coordinate before its application to an interaction. The Lorentzian realization of cubic incidence determines, under the stated temporal convention, the space of forms in which that interaction can be expressed. To formulate it, an oriented local realization $(\mathcal{M}, g)$ of dimension four is fixed, with tangent fibres identified with the model $Q_{\square}$. A connection ∇ with $\nabla g = 0$, a potential one-form $A \in \Omega^1(\mathcal{M})$, its curvature $F = dA$, and a charge representation of the section are also specified. These data are the arguments of the physical representation that follows.

Hodge duality satisfies $\star^2 = -I$ on two-forms. The electric–magnetic separation corresponds to a temporal choice in the $1 + 3$ decomposition; F and $\star F$ preserve their relationship within the same space of two-forms. Neutrality of a charge representation means $q = 0$, whereas the section may retain orientation, phase, and internal transport. The absence of electric coupling in that representation has a different content from the absence of phase information.

In this dimensional realization, let $m_e > 0$, let q be the charge, let $\gamma(s)$ be a parametrized trajectory, and let $u = \dot{\gamma}$ be its four-velocity. The Lorentz force is expressed by

$$m_e \nabla_u u^\mu = q F^\mu{}_\nu u^\nu. \quad (15.45)$$

The antisymmetry of F implies $F_{\mu\nu} u^\mu u^\nu = 0$. Consequently,

$$\frac{d}{ds} g(u, u) = 2g(\nabla_u u, u) = 0.$$

The normalization of the trajectory is preserved provided that ∇ is metric and (15.45) holds. The mass used is the dimensional realization of the preceding electronic section; the coupling parameter is identified in the adopted electromagnetic convention. The concrete deduction of the potential and of a field configuration constitutes an additional problem with its own boundary conditions. The result presented here determines the domain, the representation data, and the conservation implied by its equation.

TPK route inversion and the orientation of an electronic line then offer a precise comparison: each has its involution, transport, and representation. The work of Wheeler and Feynman places the question historically; the equivalence of two realizations is decided through their maps and compatibility conditions. This arrangement keeps the electron's arithmetic origin connected to the formulation of its interaction, making the function of each stage explicit.



Incidence and dimensional realization

Exceptional relations, forms and dimensional transports retain their maps and compatibility conditions.

The form preserved by the second projection

The exceptional realization develops the incidence that accompanies numerical generation. Each prefix transmits boundary relations compatible with truncations. The Paley encoding preserves the word and its image; the Witt design organizes its supports; the lift towards Golay introduces binary incidence with its markings. The subsequent development constructs the corresponding lattice and operator realizations.

Incidence stars make it possible to observe this link. The coupling register determines a tetrad. The hexads that contain it distinguish four complementary pairs, and a signature selects the marked hexad. In the figure, the connections represent operations between sets. The lift to octads preserves this configuration and distinguishes a fifth transverse octad. The five numerical regions thus maintain a correspondence with specified incidence relations.

Equality of a ternary reading may bring together regions whose orientation, multiplicity, and position remain distinct. These data participate in the exceptional correspondence. The second projection preserves precisely those relations that the value reading treats differently. Compatibility between the two is expressed on the common enriched state and its refinement maps.

M-theory and T-duality are presented through the dimensional realizations and their compatibility conditions. The Monster action has its carrier in the space constructed for it. The numerical readers, finite configurations, lattices, and representations receive distinct domains within a single genealogy. This precision makes it possible to follow how an incidence structure is transmitted to its subsequent consequences.

The following part preserves the constructions, tables, and proofs that make this development effective. The narrative returns each realization to the guiding question: which relations survive projection, and how are they recovered in the object they reach? The conservation of a Gram form, the covariance of an encoding, and the invariance of an action will have their own proofs. Their articulation supplies the geometrical content of the evolutionary conservation of unity.

16 Exceptional prolongation: from dodecaphasic incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves the two sheets and division with remainder and quotient; the TRIT determines the local regime and its orientation; the TPK composes selection, transport, carry, memory, and return. The Archimedean reading contracts compatible cylinders. The reading now under study instead preserves the incidence of positions and their transport between frames. Both start from the same arithmetic state with trajectory memory. An exceptional geometry is not reconstructed from the final decimals of π , φ , e , or α .

This point clarifies the meaning of the prolongation $\alpha/K \rightarrow$ exceptional incidence. The arrow does not assert that an isolated real number determines a code, a lattice, or a group. It transports the dodecaphasic record and the signed data that entered the determination of α , together with the words and frames arising from APP–TRIT–TPK. In particular, the ordered record K can be recovered from its rational constant κ under the previously fixed base, length, and origin conventions. Projection onto a support, by contrast, preserves only the selected incidences. For its part, K_{ph} denotes a different, reduced-memory reading, subsequent to the holonomy Γ_9 .

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; and subsequently recognizing the resulting objects through the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech, and Monster designate those recognitions and the explicit realizations that make them verifiable. They are not retrospective selectors of the seeds or coefficients of the generation.

The argument begins with compatibility of prefixes and frames, constructs the ternary code, and determines the flag through K and the signed record of α . Marked incidence then selects the lattice origin and the Leech neighbor. The final step realizes the vertex algebra and its automorphisms. At each stage, the structure preserved will be specified: linear combinations, supports, discriminant classes, bilinear form, or graded product. Explanations of the function of K , normalization, and the origin of norm 54 accompany their respective constructions in Sections 16.3 and 16.9.

16.1 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\text{cal}} \rightarrow X_m^{\text{cal}}$ for $m \leq n$. The superscript indicates that the declared initial frame and its transport are preserved, not that a parameter is fitted against an experimental value. We write (w_j, f_j) for the visible six-trit word and the frame at step j . The frames satisfy a causal law

$$f_{j+1} = g_j(x)f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this reading. The return of the phase resets neither the frame nor the memory of the state. Likewise, the census of 468 decimal emissions, the 243 visible six-trit words, and the ambient space $\mathbb{F}_3^6$, with 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a six-trit block b , the Archimedean reading takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \quad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class belongs first to the internal completion $\mathbb{R}_{\text{HMT}}$; its subsequent identification with the usual real line does not enter the choice of blocks.

For the incidence reading we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad w \longmapsto (w, wA). \quad (16.1)$$

Words are written as row vectors. If $A^f = fA f^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{A^f}(w f^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (16.2)$$

The calibrated codomain retains f together with the encoded word. We therefore define

$$Q_n(x) = ((f_j, \gamma_{f_j A_w f_j^{-1}}(w_j)))_{0 \leq j < n}. \quad (16.3)$$

Proposition 16.1 (Compatibility and prolongation of the incidence reading). *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Thus the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves the visible word exactly, since $\text{pr}_1 \gamma_A = \text{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators through index $m - 1$. By induction, they have the same frames at those indices. The first m components of (16.3) agree, proving the square. The universal property of the inverse limit gives Q_∞ . The recovery claim is the projection onto the first six coordinates of each graph. $\square$

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ does not by itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are also transported. The support calculations below are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

16.2 The finite realization and the ternary code

The elliptic operator of the TRIT admits the following finite realization in the six-position chart. These positions coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, with the order $\infty, 0, 1, 2, 3, 4$ declared, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$, and $C_{a,b} = \chi(b - a)$ at the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$, and $\chi(0) = 0$. This is a coordinate realization of the quadratic structure already transported; identifying the six positions with the projective line is chart data, not a new numerical input into the generator.

The quadratic-sum identities give $C^2 = 5I_6$. Reducing modulo three yields

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6. \quad (16.4)$$

The quadratic equality realizes the elliptic regime; it does not identify the field $\mathbb{F}_3$ with an Archimedean realization of the imaginary unit. They are distinct realizations of a common operator relation, with their corresponding domains.

16.2.1 Dual neutrality and orientation of the first joint boundary

The prolongation boundary joins the three regional histories through a relation preserving the axis, lateral aggregate, saturation, and column occupancy. In the established row-vector chart, the final six-position blocks of the words of length thirty are

$$u_4^\pi = 110221, \quad u_4^e = 102011, \quad u_4^\varphi = 010200.$$

The superscripts retain the three regions of (6.1); each word is interpreted as an element of $\mathbb{F}_3^6$, with integer representatives in $\{0, 1, 2\}$. The boundary relation transports the axis by

$$z := u_5^\varphi = u_4^e = 102011, \quad a := u_4^\varphi A_W L_1^3 = 111101. \quad (16.5)$$

Here a is the oriented contrast between the two lateral channels and the axis. The corresponding lateral aggregate is

$$x + y = 210112, \quad q := x + y + z = 012120, \quad a = x + y - z = 111101 \quad \text{in } \mathbb{F}_3^6, \quad (16.6)$$

where $x = u_5^\pi$ and $y = u_5^e$. The sum and the contrast have distinct dual readings: $qA_W = 012000$ and $aA_W = 120210$.

Let $B \in M_{3,6}(\mathbb{F}_3)$ be the matrix with rows x, y, z . Its canonical lift defines, column by column,

$$c_j = \left\lfloor \frac{x_j + y_j + z_j}{3} \right\rfloor, \quad o_j = \mathbf{1}_{x_j \neq 0} + \mathbf{1}_{y_j \neq 0} + \mathbf{1}_{z_j \neq 0}.$$

The law for the boundary at depth six fixes $c_j = 1$. With the axis z and the aggregate q already determined, the integer lateral sums are

$$(x_j + y_j)_{j=1}^6 = (2, 4, 3, 4, 4, 2).$$

The minimum-occupancy rule selects, among decompositions with $x_j, y_j \in \{0, 1, 2\}$, the columns with the fewest nonzero entries, including z_j . For sum two, the extreme pairs $(0, 2), (2, 0)$ occupy one lateral position; for sum three, the pairs $(1, 2), (2, 1)$ occupy two; for sum four, only $(2, 2)$ is possible. Including the axis gives $o = (2, 2, 3, 2, 3, 2)$. At this boundary, with $s = [x + y]_3$, the same rule is expressed by

$$o_j = 2 + \mathbf{1}_{\{z_j \neq 0, s_j \neq 2\}}.$$

At this boundary, saturation and occupancy are

$$c = 111111, \quad o = 223232. \quad (16.7)$$

Thus the integer equality in each column is $x_j + y_j + z_j = q_j + 3$. The carry c records the integer lift of this boundary; its domain consists of the six columns under consideration.

Proposition 16.2 (Eight distributions of a saturated boundary). *Conditions (16.5)–(16.7) leave exactly eight matrices. Their column alternatives are*

j	z_j	$x_j + y_j \text{ in } \mathbb{Z}$	(x_j, y_j)
1	1	2	$(0, 2), (2, 0)$
2	0	4	$(2, 2)$
3	2	3	$(1, 2), (2, 1)$
4	0	4	$(2, 2)$
5	1	4	$(2, 2)$
6	1	2	$(0, 2), (2, 0)$

(16.8)

Proof. Axis transport fixes the third row. The integer equality $x_j + y_j = q_j + 3 - z_j$, together with o_j , gives the alternatives in the table. The first, third, and sixth columns each admit two independent choices; the other three admit one. Their cardinalities have product $2 \cdot 1 \cdot 2 \cdot 1 \cdot 1 \cdot 2 = 8$, and each choice preserves every stated equality. $\square$

Neutrality is evaluated in the dual chart. With $\mathbf{1} = (1, 1, 1, 1, 1, 1)^\top$, the matrix in (16.4) determines

$$\omega = A_W \mathbf{1} = (2, 1, 1, 1, 1, 1)^\top, \quad A_W \omega = -\mathbf{1}. \quad (16.9)$$

The dual closure condition at the boundary is

$$(BA_W)\mathbf{1} = B\omega = 0 \quad \text{in } \mathbb{F}_3^3. \quad (16.10)$$

This condition acts on the collection of saturated boundaries. The quadratic relation for A_W determines transport of the dual unit; neutrality specifies which boundaries satisfy the corresponding closure.

Proposition 16.3 (Dual pair and oriented boundary). *Neutrality (16.10) leaves exactly*

$$B_- = \begin{pmatrix} 0 & 2 & 1 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 & 2 & 0 \\ 1 & 0 & 2 & 0 & 1 & 1 \end{pmatrix}, \quad B_+ = \begin{pmatrix} 2 & 2 & 2 & 2 & 2 & 0 \\ 0 & 2 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \end{pmatrix}. \quad (16.11)$$

The involution interchanging closure and propagation permutes B_- and B_+ . In the half-open chart with the APP orientation fixed for the channels, the canonical boundary is $R_{36} = B_+$.

Proof. Write the three binary choices in the first row as

$$x = (2\varepsilon_1, 2, 1 + \varepsilon_3, 2, 2, 2\varepsilon_6), \quad \varepsilon_1, \varepsilon_3, \varepsilon_6 \in \{0, 1\}.$$

The condition $x\omega = 0$ is equivalent to

$$1 + \varepsilon_1 + \varepsilon_3 - \varepsilon_6 \equiv 0 \pmod{3}.$$

The possibilities are $(0, 0, 1)$ and $(1, 1, 0)$. The axis row satisfies $z\omega = 0$; the lateral aggregate also has zero dual evaluation. Consequently, $y\omega = 0$ in both cases. This gives exactly the two matrices in (16.11).

The final choice applies the APP orientation of the half-open chart to this mirror pair. In that chart, the ordered representations are

$$B_- \mapsto (789, 048, 894), \quad B_+ \mapsto (793, 045, 894).$$

The representative of the fixed orientation is B_+ . This selection preserves the order of the three regions and the boundary convention of their cylinders. The algebraic reduction from eight matrices to two and the choice of oriented representative thus constitute two stages of the same construction. $\square$

The visible charges distinguish these stages precisely:

$$B_- \mathbf{1} = (0, 1, 2)^\top, \quad B_+ \mathbf{1} = (1, 0, 2)^\top, \quad B_\pm \omega = 0. \quad (16.12)$$

Visible charge and dual neutrality are evaluations on distinct vectors. The selected boundary retains three rows of six trits, with ambient indices $(726, 215, 301)$, and supplies the junction state R_{36} for joint prolongation. Its memory includes the transported axis, saturation, column incidence, and the orientation distinguishing the sheets of the pair.

Definition 16.4 (Code of the dodecaphasic chart). The ternary code associated with (16.4) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}.$$

Proposition 16.5 (Self-duality, distance, and the Witt design). *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\text{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (16.13)$$

The set $\mathcal{H}$ has 132 elements and constitutes an $S(5, 6, 12)$ system: every set of five positions is contained in a unique hexad.

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by the symmetry of A_W , satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^T = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (16.13); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two weight-six words have the same support, one of the signs 1, -1 occurs at least three times among the six ratios of their nonzero coordinates. Subtracting the corresponding multiple would produce a nonzero word of weight at most three, which is impossible. Hence a support corresponds to exactly the pair $\{c, -c\}$, and there are $264/2 = 132$ supports. If two distinct supports shared five positions, the same argument would cancel at least three coordinates in a union of size seven, producing weight at most four. Thus a five-element set belongs to at most one hexad. Finally,

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. □

The subsequent recognition is the extended ternary Golay code and the small Witt design [13]. The preceding proof explains which object is recognized. It does not replace the APP-TRIT-TPK genealogy with a search among known codes.

16.3 Dodecaphasic flag associated with K and α

Articulation of regional coordinates and closure incidence. The function of this prolongation is understood through a distinction among three operations: preserving a record, evaluating a coordinate, and extracting an incidence relation. All three act on information produced by APP-TRIT-TPK. Their articulation makes explicit the article's structural thesis: a numerical realization has an operator provenance, and certain data of that provenance additionally admit a combinatorial and lattice realization. The equations make it possible to follow both routes with their domains and the information remaining in each.

The starting point is the terminal state with its compatible prolongations. It preserves the residues and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation, and self-scaling regions are determined in that domain before their readers establish the coordinates π, e, φ . At the same time, the oriented record of the twelve windows determines the signed vector U . The subsequent appearance of K arises from the integral transformation proved in §9.8:

$$U = \mathcal{H}_{12}K, \quad K = \frac{1}{4}\mathcal{H}_{12}U, \quad \mathcal{H}_{12}^2 = 4I_{12}. \quad (16.14)$$

The congruence defining $\mathcal{L}_H$ guarantees integrality of the inverse. Thus, K constitutes a reversible coordinatization of the signed record in the pertinent sublattice. Its positions preserve phase order; its differences D_3K, D_4K preserve transversal relations; the sum $Q(K)$ determines the uniform component. That structure explains why the record enters both numerical composition and selection of a boundary configuration.

The periodic evaluation

$$K \mapsto \kappa_{\text{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable when base, length, and phase origin are fixed. Extracting a support performs a different operation: it preserves the membership of each position in a class defined by a threshold and groups records having the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete record from which they arise.

In the frame inherited from the dodecaphasic record, the data are

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), & (16.15) \\ w_\pi &= 010211, & w_e &= 201101, \\ w_\varphi &= 121200, & w_{\text{APP}} &= 022020. \end{aligned}$$

Applying γ_{A_W} yields the following incidences:

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first is the support of a weight-nine word; the other three are hexads. Through its high-block reading, the record supplies

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi. \quad (16.16)$$

Here the integer 729 marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in (16.4). In this particular record, all integer thresholds between 660 and 729 produce the same support: the support recovers neither the threshold nor the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$Z_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \quad (16.17)$$

Its negative support is

$$H_\alpha^- = \{4, 5, 6, 7, 9, 10\}. \quad (16.18)$$

These signs are not extracted from the decimal expansion of the scalar α ; they are retained in the state preceding its numerical expression.

Normalization of alpha and its signed configuration. The coordinate of α is determined along the positional route by composing regional blocks and the dodecaphase record. Before positional division, the oriented combination has components $Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}$, and the balance with incoming quotient is $s_j = Z_j + c_{j+1}$. Normalization satisfies

$$A_j = s_j - 1000c_j = Z_j + c_{j+1} - 1000c_j, \quad 0 \leq A_j < 1000. \quad (16.19)$$

This identity simultaneously preserves the normalized block and the contribution of the transported quotient. The limit of compatible prefixes determines the scalar closure coordinate. The signed configuration, for its part, preserves which positions in the combination before normalization exhibit negative defect. In the initial window, that configuration is the vector Z_0 of (16.17), and its negative support is H_α^- .

Two contents of the same operation are thus defined. Positional normalization determines a real coordinate through its compatible remainders and quotients; negative support determines a configuration of positions in the twelve-component cycle. The sign refers to the balance of records before normalization. Its incidence reading remains available because the state preserves that balance alongside the normalized result. This articulation corresponds to the proved positional route; comparison with the analytic vacancy operators preserves the precise scope established in the chapter on α .

Proposition 16.6 (Agreement of the two flag selectors). *Within $\mathcal{H}$, the support (16.18) is the unique hexad H satisfying*

$$B_K \subset H \subset P_\pi, \quad (|H \cap H_e|, |H \cap H_\varphi|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Thus the signed reading and the incidence reading select the same flag $B_K \subset H_\alpha^-$.

Proof. The four hexads containing B_K are obtained by adjoining, respectively, the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third pairs are contained in P_π . For the second, the intersection with H_φ has size three, and that with H_{APP} has size three; it does not meet the last condition. For the third, the three intersections have sizes 4, 3, 2. This last pair produces precisely the negative support of (16.17). $\square$

It is useful to retain both the flag and its marked frame:

$$\mathcal{J}_* = (B_K \subset H_\alpha^-), \quad \tilde{\mathcal{J}}_* = (\mathcal{J}_*; P_\pi, H_e, H_\varphi, H_{\text{APP}}).$$

Union, intersection, and difference are equivariant under a simultaneous relabeling of the design. The marked class, not the position numbers taken in isolation, is the transported object.

Lemma 16.7 (Origin selected by the marked incidence). *For the preceding frame,*

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\tilde{\mathcal{J}}_$.*

Proof. The first intersection is $\{1, 3, 4, 9\}$. The subtracted union contains 3, 4, 9 and does not contain 1. Equivariance follows because a bijection commutes with set operations. $\square$

This origin will select a direction of the lattice neighbor once the A_2 metric and the positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

From ternary coding to the marked flag. The regions also provide the codes w_π, w_e, w_φ and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \mapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and adds its transformation by A_W . The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \text{supp}(c)$, preserves the nonzero positions. For weight-six words, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. In this way, the quotient by sign produces the hexads, while the complete code remains available for operations requiring its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \quad (16.20)$$

The left-hand side arises from an integer comparison of the components of K ; the right-hand side, from the intersection of the coded supports of propagation and closure. The threshold belongs to the declared block chart and remains explicit in the reader's definition. The complete record preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes that tetrad to a hexad. The membership condition in support P_π , together with intersection sizes (4, 3, 2) relative to $H_e, H_\varphi, H_{\text{APP}}$, selects a unique hexad. The following proofs

bring together the two determinations: that obtained through the sign of Z_0 and that obtained through incidence. The result is organized as

$$\tilde{J}_* = (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\text{APP}}). \quad (16.21)$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π here recover their internal differentiation. They share w_π as visible ternary code, but retain distinct emissions, signatures, and multiplicities. In the binary realization on a marked dodecad, the transversal region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to lifts of the four hexads of its star. The negative region is distinguished by H_α^- , and chart order individualizes the three positive regions. This correspondence preserves regional roles and the distribution $432 + 4 \cdot 144 = 1008$, in addition to realizing the incidence partition $1 + 4$.

16.4 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p. \quad (16.22)$$

Let $V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}$. The two Euclidean realizations are subsequent languages for the same finite design here.

Proposition 16.8 (Isometry of the dodecaphasic screen). *We have*

$$I^\top I = 36I_{12} + 30J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6}I|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\text{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the automorphisms of the design.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^\top I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I\rho_{12}(g) = \rho_{\mathcal{H}}(g)I$. $\square$

The incidence information also has a concrete spectral reading. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 16.9 (Action of the intersection kernel). *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

Thus G acts by the scalar 30 on E_{inc} .

Proof. The (H, p) entry of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, there are ten four-element sets T containing p , each contained in four hexads, and five not containing it, each with a unique extension of the five-element set $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen four-element subsets of H each give one extension: the total is 15. This yields the identity. The rectangular term vanishes on V_{11} . $\square$

There is indeed a material intertwiner here: its domain, codomain, entries, and normalization are determined. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc} , with $U_W P = Q U_W$. This statement does not identify all the operators acting on these spaces solely by their trace.

16.5 The direction selected by the dodecaphasic register

The register K has a geometric role complementing its rational evaluation and support readout. Once the representation of the twelve positions has been constructed, its components determine a direction in the space of zero-mean variations. The reduction $12 \rightarrow 11$ removes the uniform mode; the next reduction $11 \rightarrow 10$ removes the direction selected by the register itself. The two operations therefore have different antecedents.

To specify the chart used completely, consider the group A_5 of even permutations of five objects and the subgroup $C_5 = \langle (1\ 2\ 3\ 4\ 5) \rangle$. The twelve left cosets $r_p C_5$ are ordered by the following representatives, written as lists of images:

p	r_p	p	r_p
1	(4, 3, 5, 2, 1)	7	(5, 4, 3, 2, 1)
2	(4, 2, 1, 3, 5)	8	(1, 3, 4, 2, 5)
3	(1, 2, 4, 5, 3)	9	(4, 2, 3, 5, 1)
4	(2, 5, 3, 4, 1)	10	(3, 2, 5, 4, 1)
5	(4, 3, 1, 5, 2)	11	(4, 5, 2, 3, 1)
6	(5, 1, 2, 3, 4)	12	(5, 3, 2, 4, 1).

The action by left multiplication defines an orthogonal representation $\rho : A_5 \rightarrow O(\mathbb{R}^{12})$: $\rho(a)e_p = e_q$ when $ar_p C_5 = r_q C_5$. This representation chart acts on the register already obtained; the group characters are not used to choose its twelve values.

Let χ_3 have the values of the three-dimensional character

class	1	2^2	3	5_a	5_b
χ_3	3	-1	0	$(1 + \sqrt{5})/2$	$(1 - \sqrt{5})/2$.

The class 5_a contains the displayed generator of C_5 , and 5_b contains its square. This convention distinguishes the two conjugate three-dimensional sectors of the representation. Define

$$P_3 = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \rho(a), \quad P_{11} = I_{12} - \frac{1}{12} J_{12}. \quad (16.23)$$

Both matrices are determined by finite sums in $\mathbb{Q}(\sqrt{5})$. The appearance of this quadratic field belongs to the realization of the representation, downstream of the regional generation of φ .

Lemma 16.10 (Projection of the register onto the three-dimensional sector). *We have*

$$P_3 = P_3^\top = P_3^2, \quad P_3 \mathbf{1} = 0, \quad \text{tr } P_3 = 3, \quad P_3 P_{11} = P_{11} P_3 = P_3.$$

For the register (16.15), the vector

$$u_K = P_3 P_{11} K \quad (16.24)$$

is nonzero. Its exact norm is

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0. \quad (16.25)$$

Proof. The character sum in (16.23) is the isotypic projector. The real character and the substitution $a \mapsto a^{-1}$ give symmetry. The multiplicity of the sector in the action on A_5/C_5 is the dimension of its subspace fixed by C_5 ,

$$\frac{1}{5} \left(3 + 2 \frac{1 + \sqrt{5}}{2} + 2 \frac{1 - \sqrt{5}}{2} \right) = 1.$$

Hence the rank is three. Orthogonality to the trivial character gives $P_3 \mathbf{1} = 0$ and the two compositions with P_{11} . These identities can also be checked by multiplying the matrices in the finite sum.

Finally, $\sum K_p = 6263$, and substituting the twelve components into that same sum gives

$$\|u_K\|^2 = K^\top P_3 K = \frac{1}{20} \sum_{a \in A_5} \chi_3(a^{-1}) K^\top \rho(a) K = \frac{6638585 + 2275584\sqrt{5}}{20}.$$

This is an exact evaluation of sixty specified terms, reproduced by the accompanying certificate using quadratic-field arithmetic with rational coefficients. The two positive coefficients prove nonvanishing. $\square$

Definition 16.11 (Dodecaphasic direction and transverse complement). The direction selected by K in the declared chart is $\ell_K = \mathbb{R}u_K$. Its transverse complement within $V_{11} = \mathbf{1}^\perp$ is

$$V_{10}(K) = V_{11} \cap u_K^\perp.$$

Proposition 16.12 (Orthogonal reduction determined by K). *The operator*

$$P_{10}(K) = P_{11} - \frac{u_K u_K^\top}{\|u_K\|^2} \quad (16.26)$$

is the rank-ten orthogonal projector onto $V_{10}(K)$. We have

$$V_{11} = \ell_K \oplus^\perp V_{10}(K), \quad P_{10}P_{11} = P_{10}, \quad P_{10}u_K = 0.$$

Proof. Let $E_K = u_K u_K^\top / \|u_K\|^2$. The proven nonvanishing ensures that it is defined. It is symmetric, $E_K^2 = E_K$, has rank one, and, since $u_K \in V_{11}$, $P_{11}E_K = E_K P_{11} = E_K$. Hence

$$(P_{11} - E_K)^2 = P_{11} - 2E_K + E_K = P_{11} - E_K.$$

Its image is $V_{11} \cap u_K^\perp$, and its trace is $11 - 1 = 10$. The decomposition and the remaining identities follow by projection onto the two orthogonal summands. $\square$

This result makes precise a role of K not exhausted by its designation as a register: besides reversibly representing oriented data and selecting the tetrad B_K , it determines a linear polarization. The direction depends on the complete register through P_3K , not merely on the four points of its support. The order of register, representation, direction, and dimensional reduction has thus been retained (Proposition 16.12).

The transformation has a precise covariance law. If the chart is changed simultaneously by an orthogonal permutation matrix S , $K' = SK$ and $P'_3 = SP_3S^{-1}$, then

$$u'_{K'} = Su_K, \quad P'_{10}(K') = SP_{10}(K)S^{-1}.$$

On the other hand, replacing u_K by au_K , with $a \neq 0$, does not change E_K . The direction retains a polarization and does not by itself allow recovery of either the amplitude of u_K or the entire register. Reversibility of $\kappa \leftrightarrow K$ remains within its own domain.

The geometric realization established here is a reduction of inner-product spaces. Its subsequent use as a compact direction requires transporting this line to the corresponding physical realization. Subsection 16.19 presents the algebraic map that keeps this reduction and the lattice screen coordinated.

16.6 The Witt star and the generation of M_{12}

Definition 16.13 (Star on a four-element set). For $B \in \binom{[12]}{4}$, the Witt star is the collection of the four hexads containing B . Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and exchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

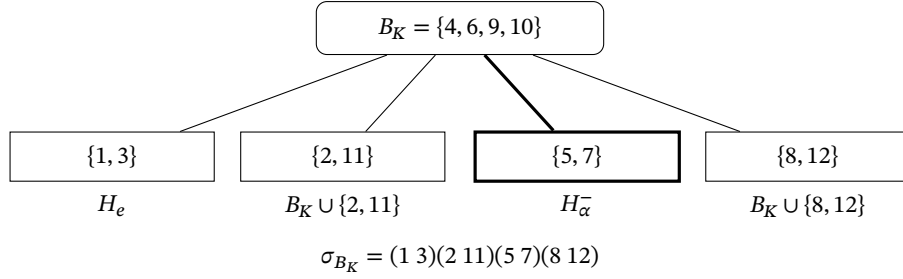


Figure 11: The concrete star of the dodecaphasic flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α . The four pairs exhaust the eight outer positions.

Proposition 16.14 (The collection of stars). *The $\binom{12}{4} = 495$ permutations σ_B preserve $\mathcal{H}$. The group they generate has order 95040 and is the Mathieu group M_{12} in its action on the small Witt design.*

Proof. The preservation claim is a finite check on the 132 hexads of (16.4): for each four-element set the four petals are constructed and one verifies $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$. Closure under composition of these permutations produces 95040 elements. In the lexicographic enumeration, it suffices to add successively six stars not yet contained in the group; the intermediate orders are 2, 6, 18, 360, 7920, 95040, and all 495 stars belong to the final group. The procedure and the six generators are made explicit in the technical provenance note. Finally, one applies the recognition of the automorphism group of $S(5, 6, 12)$ as M_{12} , whose order is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$. $\square$

An individual star generates a cyclic group of order two. It does not by itself generate M_{12} , still less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 16.15 (Equality of traces does not identify operators). On V_{11} , a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (16.22) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

16.7 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. Its weights are 0, 8, 12, 16, 24, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D , the support of a weight-twelve word. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π .

Proposition 16.16 (Residue of a dodecad and lifting of hexads). *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad, } |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D . Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^\perp$ with $O_B^\perp \cap D = B$.

Proof. A five-element set $A \subset D$ belongs to a unique octad O . If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the permitted weights, with $0 \leq j \leq 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \geq 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift by selecting five of its points. Moreover,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \quad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads on B give four distinct octads. The fifth cannot meet D in six points, since it would be another residual hexad; because it contains B , its intersection has size four and equals B . $\square$

A hexad-preserving chart $\ell : [12] \rightarrow D$ identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is the recognition of the design already constructed as the small Witt design; a concrete choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \quad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. The domain of this injectivity is the hexads, not the record K .

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	(501, 614, 498, 169, 272, 272)	555555
$++_1$	144	(810, 923, 870, 169, 272, 272)	339555
$++_2$	144	(810, 983, 810, 169, 272, 272)	393555
$++_3$	144	(870, 923, 810, 169, 272, 272)	933555
$--$	144	(870, 923, 810, 418, 674, 521)	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible tritic word $w_\pi = 010211$, but not the same regional genealogy. The three positive regions preserve the tail 169|272|272; the negative region changes it to 418|674|521; the transverse region has a constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O_{\ell(B_K)}^\perp$, the negative region to $\iota_D(H_\alpha^-)$, and the three positive regions to the other three lifts of the star. This realizes the $4 + 1$ partition of the five octads on $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart order; incidence does not determine it without that datum. The precise claim is therefore a marked correspondence of regions, compatible with their roles and with the flag, not a canonical bijection inferred from the cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also, $1008 = 36 \cdot 28$, and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalog checks, not substitutes for the map ι_D or the regional chart.

16.7.1 Pentadic representation of the microfiber and its incidence

The arrangement of the five regions makes it possible to visualize simultaneously the unity of the ternary quotient and the differentiation preserved by the catalog. The relevant map is the restriction

$$\Psi|_{\mathcal{R}_\pi^{\text{micro}}} : \mathcal{R}_\pi^{\text{micro}} \longrightarrow \{010211\}, \quad \Psi(U) = (-U) \bmod 3.$$

Its five antecedents are the vectors U_6 in the preceding table. Figure 12 recovers their radial organization, preserving chart order, signatures, and multiplicities of the catalog (equation (6.1)).

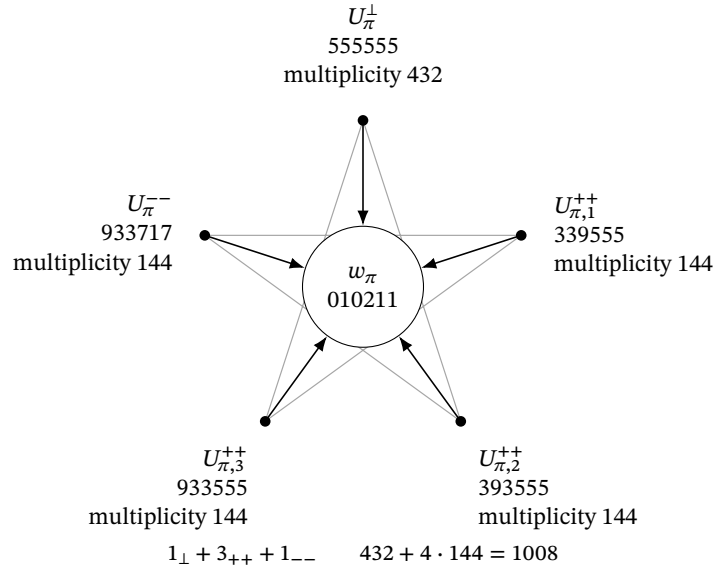


Figure 12: The five regions of π in a pentadic arrangement. Each position preserves its multiplicative signature and multiplicity; the radial arrows represent the common projection $\Psi(U) = 010211$. The star-shaped outline organizes the five positions of the chart. Their realization in the five Witt octads is performed through regional alignment and the $4 + 1$ decomposition in the text.

Projection onto the center identifies the five ternary images; the record of exterior positions retains the regional information making incidence realization possible. The distribution $1_{\perp} + 3_{++} + 1_{--}$ specifies one transversal region, three positive regions, and one negative region. Their normalized weights are $3/7$ and $1/7$ for each of the remaining four: transversal multiplicity is triple, even though all five images under Ψ coincide.

The link with the Witt star is formulated through the flag $B_K \subset H_{\alpha}^{-}$, the alignment ℓ , and the lift ι_D already constructed. The five regional positions are realized in the five octads over $\ell(B_K)$:

$$\begin{aligned} U_{\pi}^{\perp} &\longmapsto O_{\ell(B_K)}^{\perp}, \\ U_{\pi}^{--} &\longmapsto \iota_D(H_{\alpha}^{-}), \\ \{U_{\pi,1}^{++}, U_{\pi,2}^{++}, U_{\pi,3}^{++}\} &\longmapsto \{\iota_D(B_K \cup P) : P \in \mathcal{P}_+\}, \\ \mathcal{P}_+ &= \{\{1, 3\}, \{2, 11\}, \{8, 12\}\}. \end{aligned}$$

Assignment of the three positive positions to the three pairs is performed in an ordered chart. The transversal and negative rows are distinguished by their regional predicates and by the flag.

The two figures are thus articulated: Figure 11 shows the four hexads of the small Witt design over B_K ; Figure 12 shows the five regions that, under alignment, occupy the four residual octads and the transversal octad of the binary design. The law $4 + 1$, proved in Proposition 16.16, specifies the change in incidence. In this composition, K determines the marked tetrad, and the signature of α distinguishes one of its hexads. In addition to their common ternary value, the regions preserve the structure required for that transport toward the exceptional realization.

16.8 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four need not first go through the binary code. It is performed directly from C_W by discriminant gluing. Two compatible prolongations thus remain distinct: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ with Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

Represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with this discriminant. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \quad (16.27)$$

Theorem 16.17 (Ternary gluing). *The object (16.27) is a positive, even, unimodular lattice of rank 24. Its root system is exactly A_2^{12} ; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes stable under addition. Self-orthogonality annihilates the discriminant pairing: the latter is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a glue class is $2 \text{ wt}(c)/3$ modulo $2\mathbb{Z}$. The weights in (16.13) are multiples of three, so the lattice is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(36)^2} = 1.$$

Positivity follows from the orthogonal sum of the twelve A_2 metrics. In a nonzero class of A_2^*/A_2 , the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Hence a vector in a nontrivial glue class has norm at least $6 \cdot 2/3 = 4$. No norm-two roots occur outside A_2^{12} . Within it, the roots are precisely those of its twelve components. $\square$

16.9 Incidence selection of the rootless unimodular neighbor

From the incidence origin to lattice modification. The marked flag determines a position through an equivariant set operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Its function is to distinguish one component among the twelve entering the lattice realization. The complete ternary code acts as a glue code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and self-duality, weights, and distance of the code translate, respectively, into properties of integrality and determinant, parity, and norm bounds for the glue classes. Here the symbolic information of the code performs a function no longer performed by the isolated support.

The A_2 metric and positive radial chamber are fixed in the lattice development. Within that chamber, the congruence required for an even index-three neighbor has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \dots + \rho_{12}, \quad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the declared chamber. The subscript α records the incidence provenance of the selection: its content is the frame that entered closure and now individualizes a lattice direction.

Pairing with v_α determines the index-three sublattice $N_{\alpha,0}$, and adding the class $v_\alpha/3$ produces the neighbor N_α of (16.29). This modification preserves rank and recovers unimodularity and parity; congruence selection eliminates the previous roots, and local bounds exclude roots in the new classes.

Identification with Leech appears after these verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the second receives the ternary code directly as glue data.

The label $o_* = 1$ of the flag is now realized as one of the twelve A_2 components. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \quad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

this congruence is equivalent to $\sum_i b_i \equiv 1 \pmod{3}$. The least norm in the chamber is obtained with a single $b_i = 1$ and all others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \dots + \rho_{12}, \quad v_\alpha^2 = 54. \quad (16.28)$$

The word “least” refers to this positive chamber. It does not assert minimality throughout the residual class: the vector $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not an omissible conclusion.

Let $N = N(C_W)$. Define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3}. \quad (16.29)$$

Theorem 16.18 (Leech neighbor determined by the marked flag). *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank 24, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24} .*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \pmod{3}$ is nonzero: a simple root of a component pairs with $a_i \rho_i$ to give $a_i \equiv 1 \pmod{3}$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$ belonged to N , the integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. It follows that

$$[N_\alpha : N_{\alpha,0}] = 3, \quad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

Hence it is even.

It remains to exclude roots, the decisive step. The roots of N belong to A_2^{12} . Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24} . $\square$

The argument does not recognize Leech from a coincidence of dimension. It constructs the gluing, proves integrality, evenness, and the determinant, and eliminates roots through a local bound. Only then does it apply the classification theorem. The dependence on α is preserved in the flag selecting o_* and in the vector (16.28); the scalar value of the constant is not converted into a length of the lattice.

16.10 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the rank-twenty-four Heisenberg oscillator space; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the central extension of the lattice. The cocycle and the lift of the isometries are part of the construction: a bare vector-space sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ . With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\text{HMT}}^{\natural} = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \quad (16.30)$$

The sign $+$ denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 16.19 (Prolongation to the Moonshine module). *The algebra (16.30), constructed on the neighbor (16.29), is a realization of the Moonshine algebra $V^{\natural}$. It has central charge 24, zero weight-one component, and automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\text{Tr}_{V_{\text{HMT}}^{\natural}} q^{L_0-1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \quad q = e^{2\pi i\tau}. \quad (16.31)$$

Proof. Theorem 16.18 verifies the lattice hypotheses: positivity, evenness, unimodularity, rank twenty-four, and absence of roots. The Frenkel–Lepowsky–Meurman construction and theorem for the Leech lattice then apply. They identify the orbifold with $V^{\natural}$ and establish its character and its automorphism group. Composing the flag, the neighbor, and the VOA construction functor gives the realization stated here. The group is not inferred from the coefficient 196884. $\square$

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof of the FLM construction in this article [11]. Moreover, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It does not introduce π or e as inputs into the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes an alternating sign. Hence

$$\text{Tr}_{V_\Lambda} \theta q^{L_0-1} = t_2(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character of $V^{\natural}$ without insertion. Nor does it by itself identify a Monster conjugacy class: the space, insertion, and lift must be specified before functions are compared.

Grading, unity, and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_α once its properties have been verified. It incorporates the oscillator space, the central extension of the lattice, the vertex product, and the sector twisted by the lift of negation. The result is the algebra $V_{\text{HMT}}^{\natural}$ of (16.30), whose automorphism group is isomorphic to the Monster. Its action domain is thus fixed by an algebraic structure with grading and product. Borchers’s Moonshine theorem subsequently determines the graded traces with insertion of those automorphisms [11, 12].

Unity acquires a precise realization at this level: the weight-zero component is the vacuum line and contributes the coefficient 1 of q^{-1} in the graded character. At the cylinder level, refinement maps preserve the constant function through precomposition. Every conservation statement therefore has its object and map. The development of the proofs makes visible this continuity of construction, from arithmetic operations and memory to incidence, metric, and vertex products.

This articulation places K and the closure of α in the main argument. K reversibly preserves an integral coordinate of the history; normalization of α composes regional records with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. The following proofs develop these maps, distinguishing reversible equivalences, information quotients, and realizations adding metric or algebraic structure. A demonstrable relation between structure, equation, and value is thus expressed, preserving the attribution of each construction and the scope of its theorems.

16.11 State-based construction of the twelve cyclotomic fibres

On the APP marks, multiplication $a : r \mapsto 4r$ in $\mathbb{Z}/9\mathbb{Z}$ satisfies $a^3 = 1$. The sheets satisfy

$$S(ax, ay) = aS(x, y), \quad P(ax, ay) = a^{-1}P(x, y),$$

since $16 \equiv 7 \equiv 4^{-1} \pmod{9}$. The ordered unit orbits are $\mathcal{O}_0 = (1, 4, 7)$, $\mathcal{O}_1 = (2, 8, 5)$. In each, the augmentation module $A(\mathcal{O}) = \ker(\mathbb{Z}[\mathcal{O}] \xrightarrow{\sum} \mathbb{Z})$ inherits the inner product making the e_r orthonormal. In the basis $b_1 = e_r - e_{4r}$, $b_2 = e_{4r} - e_{7r}$, its matrices are

$$G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^\top G_{A_2} C = G_{A_2}.$$

The lattice form therefore originates in the orbits of the marks. For the pairs $(1, 2)$, $(2, 3)$, $(3, 1)$ and sheets $\mu \in \{\Sigma, \Pi\}$, we obtain

$$W_{\text{APP}} = \bigoplus_{(i,j)} \bigoplus_{\mu} \bigoplus_{\varepsilon=0}^1 A(\mathcal{O}_\varepsilon), \quad \ell_{ij}^\Sigma(x) = x_i + x_j, \quad \ell_{ij}^\Pi(x) = x_i x_j \pmod{9}.$$

Each summand preserves pair, sheet, and orbit. The state map is

$$\Psi : (\mathbb{Z}/9\mathbb{Z})^3 \longrightarrow W_{\text{APP}}, \quad \Psi(x)_{ij,\mu,\varepsilon} = \beta_\varepsilon(\ell_{ij}^\mu(x)), \quad \beta_\varepsilon(r) = \begin{cases} e_r - e_{4r}, & r \in \mathcal{O}_\varepsilon, \\ 0, & r \notin \mathcal{O}_\varepsilon. \end{cases}$$

Proposition 16.20 (Equivariance and rank of the state module). *The action of C on the six additive fibres and of C^{-1} on the six multiplicative fibres satisfies $\Psi(ax) = \rho(a)\Psi(x)$. The images span $W_{\text{APP}} \otimes \mathbb{Q}$, of dimension twenty-four.*

Proof. The sheet identities and $\beta(4r) = C\beta(r)$ prove equivariance. For rank, order the coordinates by the three stated pairs, by Σ, Π , by $\mathcal{O}_0, \mathcal{O}_1$, and by b_1, b_2 . In each orbit, β takes successive values $(1, 0)$, $(0, 1)$, $(-1, -1)$. The matrix of the images of the following states, read row by row, has determinant 9:

$$\begin{array}{cccc} (0, 0, 1) & (0, 0, 2) & (0, 0, 4) & (0, 0, 5) \\ (0, 1, 0) & (0, 1, 1) & (0, 1, 2) & (0, 1, 3) \\ (0, 1, 4) & (0, 1, 5) & (0, 1, 6) & (0, 2, 0) \\ (0, 2, 2) & (0, 2, 3) & (0, 4, 0) & (0, 5, 0) \\ (1, 0, 1) & (1, 0, 2) & (1, 0, 4) & (1, 0, 5) \\ (1, 1, 0) & (1, 2, 0) & (1, 4, 0) & (1, 5, 0). \end{array}$$

Successive elimination, reducing each row by its predecessors and normalizing its first nonzero coefficient, gives pivot columns numbered from zero

$$(8, 10, 9, 11, 0, 12, 14, 16, 13, 15, 17, 2, \\ 18, 19, 1, 3, 20, 22, 21, 23, 4, 6, 5, 7)$$

and pivots

$$(1, 1, 1, -1, 1, 1, 1, 1, 1, -1, 3, 1, 1, 3, 1, -1, 1, 1, 1, -1, 1, 1, 1, -1).$$

Their product, multiplied by the sign of the column permutation, is 9. The invertible minor establishes full rank. $\square$

16.12 Dodecaphasic alignment with the Witt positions

The pair index is $i \in \mathbb{Z}/3\mathbb{Z}$, the sheet index is $\mu \in \{0, 1\}$, and the orbit index is $\varepsilon \in \{0, 1\}$. With the dodecaphasic origin fixed, $\lambda(i, \mu, \varepsilon) = 4i + 2\mu + \varepsilon \pmod{12}$ is bijective: the four values for each i form a block, and the three blocks are disjoint. Let $R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Multiplication proves $RCR = C^{-1}$ and $R^\top G_{A_2} R = G_{A_2}$. For the trivializations ι_b and frames P_b transported jointly with Witt incidence, set

$$g_b = \iota_b^{-1} P_b C P_b^{-1} \iota_b, \quad T_{i\mu\varepsilon} = \iota_{\lambda(i, \mu, \varepsilon)}^{-1} P_{\lambda(i, \mu, \varepsilon)} R^\mu.$$

Then

$$g_{\lambda(i, \mu, \varepsilon)} T_{i\mu\varepsilon} = T_{i\mu\varepsilon} C^{(-1)^\mu}.$$

The direct sum is a C_3 -equivariant isomorphism between the twelve state fibres and the twelve lattice positions. It preserves pair, sheet, orbit, and form under simultaneous transport of origin, order, and orientation. These data persist in the lattice gluing and passage to the neighbour constructed below.

16.13 Ternary orientation of the Leech neighbour

The order-two presentation constructed above uses negation of the Leech lattice. The same lattice, obtained from the incidence marked by K and the dodecaphasic support of α , admits a second presentation compatible with the ternary character of the construction. To make this continuity explicit, the gluing is retained, its order-three isometry is constructed, and preservation of the already selected neighbour is verified. This step requires no alteration of K , its rational reading, or the preceding definition of α . The order-three prolongation is developed here, including the lattice operations and the precise conditions of the classical theorems used.

We work in the basis of simple roots of each A_2 factor, with

$$B_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \rho = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (16.32)$$

Multiplication gives

$$C^3 = I_2, \quad C^2 + C + I_2 = 0, \quad C^\top B_2 C = B_2, \quad RCR = C^{-1}, \quad R^\top B_2 R = B_2.$$

These identities fix both the metric and the orientation of the action. In the discriminant $A_2^*/A_2 \cong \mathbb{F}_3$, we take the class of $\varpi = (2, 1)^\top/3$ as generator. We have $C\varpi - \varpi = (-1, 0)^\top$, so C acts trivially on the discriminant; the reflection R reverses its sign.

The Paley–Witt matrix already constructed produces a full-support element by the following operation:

$$\begin{aligned} q_* &= (1, 1, 1, 1, 1, 1), & q_* A_W &= (2, 1, 1, 1, 1, 1), \\ c_* &= (q_*, q_* A_W) = (1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1) \in C_W. \end{aligned} \quad (16.33)$$

Every coordinate of c_* is nonzero. For the twelve positions, define the frames and the isometry

$$P_b(c_*) = \begin{cases} R, & (c_*)_b = 1, \\ I_2, & (c_*)_b = 2, \end{cases} \quad g_c = \bigoplus_{b=1}^{12} P_b(c_*) C P_b(c_*)^{-1}. \quad (16.34)$$

The symbol $P_b(c_*)$ here denotes an invertible frame of an A_2 factor, not the regional projector P_3 .

Proposition 16.21 (Order-three isometry on the marked neighbour). *The operator g_c is an order-three isometry with no nonzero fixed vectors. It preserves the Niemeier lattice $N = N(C_W)$ and the neighbour N_α constructed in (16.29).*

Proof. Each block of g_c is C or C^{-1} ; its minimal polynomial is $X^2 + X + 1$. This proves its order and the absence of fixed vectors. Both blocks act trivially on the discriminant, so they preserve every gluing class and therefore the lattice N .

Write $v = v_\alpha = 4\rho_{o_*} + \sum_{b \neq o_*} \rho_b$, with the frames and incidence origin already fixed. Its radial coefficients are $a_{o_*} = 4$ and $a_b = 1$ for $b \neq o_*$. All satisfy $a_b \equiv 1 \pmod{3}$. In one factor,

$$(C - I_2)\rho = (-2, -1)^\top, \quad (C^{-1} - I_2)\rho = (-1, -2)^\top.$$

After division by three, their discriminant classes are 2 and 1, respectively. Consequently, the vectors

$$y = \frac{g_c v - v}{3}, \quad z = \frac{g_c^{-1} v - v}{3} \quad (16.35)$$

have discriminant words c_* and $-c_*$, both in C_W . Hence $y, z \in N$. The local product is

$$\left\langle \frac{(C^{\pm 1} - I_2)a_b \rho}{3}, a_b \rho \right\rangle = -a_b^2,$$

and summing gives $\langle y, v \rangle = \langle z, v \rangle = -(16 + 11) = -27$. Thus, $y, z \in N_0 = \{x \in N : \langle x, v \rangle \equiv 0 \pmod{3}\}$. For $x \in N_0$,

$$\langle g_c x, v \rangle = \langle x, g_c^{-1} v \rangle = \langle x, v \rangle + 3\langle x, z \rangle \equiv 0 \pmod{3}.$$

The argument applied to g_c^{-1} proves $g_c N_0 = N_0$. Finally, $g_c(v/3) = v/3 + y$ preserves the class generating $N_\alpha = N_0 + \mathbb{Z}(v/3)$. We obtain $g_c N_\alpha = N_\alpha$. $\square$

The proof preserves more than the lattice dimension: it uses the gluing code, its orientation, the flag origin, and the congruence of the radial vector. These data explain why the second presentation remains linked to the dodecaphasic record from which the construction began.

16.14 Ternary trace, twisted weight, and order-three extension

Let $\Lambda = N_\alpha$, and let $\hat{g}_c$ be the standard lift of g_c to the lattice algebra $V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda]$, with the central extension and cocycle already specified. The odd order allows this lift to be chosen of order three. Each complex A_2 factor contributes the eigenvalues ζ_3 and ζ_3^2 , where $\zeta_3^2 + \zeta_3 + 1 = 0$.

Proposition 16.22 (Graded trace and type of the lift). *For $j = 1, 2$,*

$$\mathrm{Tr}_{V_\Lambda}(\hat{g}_c^j q^{L_0-1}) = t_3(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12}. \quad (16.36)$$

The lowest weight of the nontrivial twisted modules is $4/3$, and the lift is of type zero.

Proof. Only the zero vector contributes to the lattice part of the trace, because g_c and g_c^2 have no nonzero fixed vectors. At each oscillator degree, the trace of symmetric powers is the inverse of the determinant $I - g_c^j q^n$. Each A_2 factor contributes $1 + q^n + q^{2n}$, proving (16.36).

The twisted-vacuum weight formula for an order-three lattice automorphism, applied to the two twelve-dimensional eigenspaces, gives

$$\rho_{\hat{g}_c} = \frac{1}{4 \cdot 3^2} (1 \cdot 2 \cdot 12 + 2 \cdot 1 \cdot 12) = \frac{4}{3}.$$

The type is the class of $3^2 \rho_{\hat{g}_c} = 12$ modulo three, which is zero. The weight formula is a result of twisted-module theory; the multiplicities and isometry supplied to it have been constructed here. $\square$

Theorem 16.23 (Second presentation of the Moonshine algebra). *The type-zero cyclic extension*

$$V_{(3)} = \text{Orb}_{\mathbb{Z}/3}(V_{\Lambda}, \hat{g}_c) \quad (16.37)$$

is isomorphic to the vertex operator algebra $V^{\natural}$.

Proof. Theorem 16.18 provides a positive, even, unimodular, rootless lattice of rank twenty-four. Proposition 16.21 constructs an order-three fixed-point-free isometry on that same lattice; Proposition 16.22 verifies the lift and its type. The Chen–Lam–Shimakura theorem on the order-three orbifold of the Leech lattice algebra then applies [17, Theorem 1.1]. This identification is also covered by the uniform treatment of Abe–Lam–Yamada [18, Theorem 4.4 and Sections 2–4]. This application follows construction of the lattice data; equality of characters does not replace the hypotheses of the theorem used. $\square$

To specify the dual symmetry, write $V_{(3)} = W_0 \oplus W_1 \oplus W_2$, where W_i is the integral sector selected in the $\hat{g}_c^i$ -twisted module. The product satisfies $Y(W_i, z)W_j \subset W_{i+j \bmod 3}((z))$. Hence

$$g^{\vee}|_{W_i} = \zeta_3^i I \quad (16.38)$$

defines an order-three automorphism. Indeed, the factor on a product is $\zeta_3^{i+j} = \zeta_3^i \zeta_3^j$, and the vacuum and conformal vector belong to W_0 . The classical identification of this automorphism in the Monster is class 3B, with series $T_{3B} = t_3 + 12$ [19, 12]. This normalization can be checked in the construction. The lattice character is $\text{ch } V_{\Lambda} = J + 24$: its initial pole is q^{-1} , its weight-one space has dimension 24, and the weight-zero modular character is determined by these data. If F is the character of the fixed subspace and H that of each integral twisted sector, projection onto invariants and conjugation of the two sectors give

$$F = \frac{J + 24 + 2t_3}{3}, \quad F + 2H = J, \quad \text{Tr } g^{\vee} q^{L_0-1} = F - H = t_3 + 12.$$

Its domain is the extended algebra, whereas $\hat{g}_c$ acts on the preceding lattice algebra.

16.15 Equivalence of the order-two and order-three presentations

Denote by

$$V_{(2)} = (V_{\Lambda})^+ \oplus (V_{\Lambda}^T)^+$$

the FLM presentation in (16.30). Both presentations preserve the same Leech neighbour as their antecedent. Their extensions employ different automorphisms and vertex products determined by the respective lifting data.

Theorem 16.24 (Graded equivalence and transport of automorphisms). *Given vertex operator algebra isomorphisms*

$$\phi_2 : V_{(2)} \xrightarrow{\cong} V^{\natural}, \quad \phi_3 : V_{(3)} \xrightarrow{\cong} V^{\natural},$$

the composition

$$\Phi_{23} = \phi_2^{-1} \phi_3 : V_{(3)} \xrightarrow{\cong} V_{(2)} \quad (16.39)$$

preserves the vacuum, conformal vector, grading, and vertex product. Conjugation by Φ_{23} identifies the automorphism groups and preserves their graded traces.

Proof. The composition of the two isomorphisms preserves the stated structures. In particular,

$$\Phi_{23} Y_{(3)}(a, z)b = Y_{(2)}(\Phi_{23}a, z)\Phi_{23}b, \quad \Phi_{23} L_0^{(3)} = L_0^{(2)} \Phi_{23}.$$

If h is an automorphism of $V_{(3)}$, then $h' = \Phi_{23} h \Phi_{23}^{-1}$ is an automorphism of $V_{(2)}$. Equality of traces on each finite-dimensional homogeneous component gives

$$\text{Tr}_{V_{(3)}} h q^{L_0-1} = \text{Tr}_{V_{(2)}} h' q^{L_0-1}.$$

$\square$

The choice of ϕ_2 and ϕ_3 forms part of the statement; Φ_{23} is not presented as a canonical identification independent of these choices. Nor does it transform an order-three automorphism into the order-two involution: conjugation preserves order. Its content is that two different extensions realize the same graded structure, with explicit transport of their products, representations, and traces.

16.16 Eta functions, Fricke involution, and vacuum normalization

For $\text{Im } \tau > 0$, the traces of the two lattice automorphisms admit the expressions

$$\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n), \quad t_2 = \left(\frac{\eta(\tau)}{\eta(2\tau)} \right)^{24}, \quad t_3 = \left(\frac{\eta(\tau)}{\eta(3\tau)} \right)^{12}. \quad (16.40)$$

They follow directly from the factorizations $1 - q^{2n} = (1 - q^n)(1 + q^n)$ and $1 - q^{3n} = (1 - q^n)(1 + q^n + q^{2n})$. The order in q of the level-three quotient, before raising it to the twelfth power, is

$$\frac{1}{24} - \frac{3}{24} = -\frac{1}{12}; \quad 12 \left(-\frac{1}{12} \right) = -1.$$

Here $-1/12$ denotes the exact difference between the initial exponents of two eta functions. The operation is specified and does not require interpreting a divergent sum as a convergent one. Agreement with other normalized defects in the corpus preserves their respective realizations; it does not identify their operators through equality of the scalar.

Proposition 16.25 (Fricke involution on the order-two trace). *Let $w_2\tau = -1/(2\tau)$. Then*

$$t_2(w_2\tau) = \frac{2^{12}}{t_2(\tau)}, \quad T_{2A} = t_2 + 24 + \frac{2^{12}}{t_2} \quad (16.41)$$

and $T_{2A}(w_2\tau) = T_{2A}(\tau)$.

Proof. The transformation law $\eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau)$ gives

$$\frac{\eta(-1/(2\tau))}{\eta(-1/\tau)} = \sqrt{2} \frac{\eta(2\tau)}{\eta(\tau)}.$$

The twenty-fourth power yields the first identity. Substitution interchanges the two variable terms of T_{2A} and fixes the constant term. Recognition of this expression as the McKay–Thompson series of class 2A corresponds to the level-two symmetrization of Conway–Norton [19, §§4 and 6, Table 3] and the Moonshine theorem [12]. $\square$

The modular uniformizations of levels two and three, with the preceding normalizations, are

$$j = \frac{(t_2 + 256)^3}{t_2^2}, \quad (16.42)$$

$$J = j - 744 = t_3 + 12 + \frac{10 \cdot 3^9}{t_3} + \frac{4 \cdot 3^{14}}{t_3^2} + \frac{3^{18}}{t_3^3}. \quad (16.43)$$

They are used as classical modular identities on the traces already constructed, not as rules for selecting the code or the neighbour. The second can also be written $j = (t_3 + 27)(t_3 + 243)^3/t_3^3$. The direct product expansion of (16.36) begins with

$$t_3 = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + 1188q^4 + O(q^5).$$

In particular, $[q]J = 54 + 10 \cdot 3^9 = 196884$; this is a consequence of the traces and uniformization. The construction of $V^{\natural}$ retains its lattice and orbifold proof, independently of this check of the first coefficient.

16.17 The Fricke correspondence and differential recovery

The level-two lattice trace and its Fricke completion provide, after the exceptional construction, the coordinates

$$t = t_2, \quad s = \frac{4096}{t}, \quad T = t + s + 24, \quad t \in \mathbb{C}^\times.$$

The involution exchanges t and s . The two ordered modular readings are

$$j_1 = \frac{(t + 256)^3}{t^2}, \quad j_2 = \frac{(s + 256)^3}{s^2}.$$

These formulas use the uniformization already established in (16.42) and retain the choice of the oriented reading.

Proposition 16.26 (Correspondence and discriminant). *The following identities hold:*

$$j_1 - j_2 = (s - t)(T + 23), \quad (16.44)$$

$$j_1 + j_2 = T^2 + T - 7256, \quad (16.45)$$

$$j_1 j_2 = (T + 248)^3. \quad (16.46)$$

The two values are roots of

$$X^2 - (T^2 + T - 7256)X + (T + 248)^3 = 0,$$

whose discriminant is

$$\Delta = (T - 152)(T + 104)(T + 23)^2. \quad (16.47)$$

Proof. Expanding the first reading with $ts = 4096$ gives $j_1 = t + 768 + 48s + s^2$; the second follows by exchanging s, t . Subtraction gives $(s - t)(t + s + 47)$, and addition reduces to $(t + s)^2 + 49(t + s) - 6656$. Substitution of $t + s = T - 24$ yields the first two identities. For the product,

$$j_1 j_2 = \frac{((t + 256)(s + 256))^3}{(ts)^2} = (T + 248)^3.$$

Finally, $(s - t)^2 = (T - 24)^2 - 16384 = (T - 152)(T + 104)$. Multiplication by $(T + 23)^2$ proves (16.47). $\square$

Proposition 16.27 (Recovery of orientation from values and the first jet). *For $T \neq -23$, the ordered pair (T, j_1) recovers*

$$t = \frac{T^2 - j_1 - 3904}{T + 23}. \quad (16.48)$$

At $T = -23$, the two parameters

$$t_{\pm} = \frac{-47 \pm 45i\sqrt{7}}{2}$$

share $(T, j_1) = (-23, -3375)$. The first jet $p = dj_1/dT$, evaluated on the branch parametrized by t , separates them and recovers

$$p_{\pm} = \frac{-45 \mp 45i\sqrt{7}}{2}, \quad s = p - 1, \quad t = -46 - p. \quad (16.49)$$

Proof. The identity $s^2 = (T - 24)s - 4096$ gives

$$j_1 = (T + 23)s + T - 3352 = T^2 - 3904 - (T + 23)t.$$

For $T \neq -23$, this yields (16.48). For $T = -23$, the conditions $t + s = -47$, $ts = 4096$ give $t^2 + 47t + 4096 = 0$, with discriminant $-45^2 \cdot 7$. The same identity gives $j_1 = -3375$ at both roots.

The derivative $dT/dt = 1 - 4096/t^2$ vanishes only at $t = \pm 64$. Neither exceptional root has that value, so T is a local coordinate on both branches. Differentiating the first expression for j_1 ,

$$\frac{dj_1}{dt} = (s+1)\frac{dT}{dt} + (T+23)\frac{ds}{dt}.$$

At $T = -23$, this gives $p = s+1 = -46 - t$, proving (16.49), with opposite signs for the paired indices $t_{\pm}, p_{\pm}$. $\square$

The fixed points of Fricke, $t = \pm 64$, yield respectively $(T, j) = (152, 8000)$ and $(-104, 1728)$. The coincidence at $T = -23$ instead corresponds to two distinct points of the parametrization. The field of definition follows from the equation:

$$\mathbb{Q}(t_{\pm}) = \mathbb{Q}(\sqrt{-7}), \quad \left(\frac{2t+47}{45}\right)^2 = -7.$$

On this fibre, Fricke agrees with quadratic conjugation, with trace -47 and norm 4096 . The jet retains this field, since $(2p+45)^2 = -45^2 \cdot 7$. Reconstruction is exact within this modular correspondence: the differential datum separates the two orientations that share their scalar values.

16.18 Compatibility of the level-two and level-five modular readers

On the modular realization subsequent to the HMT construction, fix $\text{Im } \tau > 0$, $q = e^{2\pi i \tau}$ and $q^{1/5} = e^{2\pi i \tau/5}$. The Rogers–Ramanujan reader is

$$R(q) = \frac{q^{1/5}}{1 + \frac{q}{1 + \frac{q^2}{1 + \dots}}}, \quad x = R(q)^5.$$

The modular identity of this realization, with the normalization of j used in (16.42), is

$$j = -\frac{(x^4 - 228x^3 + 494x^2 + 228x + 1)^3}{x(x^2 + 11x - 1)^5}. \quad (16.50)$$

The identity is the modular-realization premise of the following algebraic transport; the seeds, orientation and self-scaling representations remain those of APP–TRIT–TPK.

Proposition 16.28 (Pentadic quotient and branch selection). *For $x \neq 0$, let $u = x - x^{-1}$. On the domain $u \neq -11$,*

$$j = -\frac{(u^2 - 228u + 496)^3}{(u + 11)^5}. \quad (16.51)$$

The involution $x \mapsto -1/x$ preserves u . Its preimages are

$$x = \frac{u \pm \sqrt{u^2 + 4}}{2}.$$

The quadratic covering ramifies at $u = \pm 2i$; the subsequent rational map $u \mapsto j$ has degree six.

Proof. The exact factorizations are

$$\begin{aligned} x^4 - 228x^3 + 494x^2 + 228x + 1 &= x^2(u^2 - 228u + 496), \\ x^2 + 11x - 1 &= x(u + 11). \end{aligned}$$

Substitution into (16.50) proves (16.51). The fibre equation is $x^2 - ux - 1 = 0$; its discriminant gives the ramification points. The numerator of (16.51) has degree six and the denominator degree five. At $u = -11$, the numerator polynomial has value 3125 , so the two have no common factor. This proves the asserted degree. $\square$

For real u , the two roots have opposite signs. On the complex domain, the branch selected in the parametrization by τ is retained. The reading j alone retains a fibre with generically six values of u .

Proposition 16.29 (Self-scaling limit and compatibility of levels). *In the real limit $q \rightarrow 1^-$,*

$$R(q) \longrightarrow \varphi_{\text{HMT}}^{-1}, \quad x \longrightarrow \varphi_{\text{HMT}}^{-5} = \frac{5\sqrt{5} - 11}{2}, \quad u \longrightarrow -11.$$

When $t = t_2(\tau)$ and u are read on the same modular parameter, they satisfy

$$(t + 256)^3(u + 11)^5 + t^2(u^2 - 228u + 496)^3 = 0. \quad (16.52)$$

Writing $\epsilon = u + 11$, the branches satisfying the stated limits obey

$$\epsilon \rightarrow 0, t \rightarrow 0 : \quad \frac{t^2}{(-\epsilon)^5} \longrightarrow \frac{2^{24}}{5^{15}}, \quad (16.53)$$

$$\epsilon \rightarrow 0, t \rightarrow \infty : \quad t\epsilon^5 \longrightarrow -5^{15}. \quad (16.54)$$

Proof. For $0 < q < 1$, the alternating positive convergents of the continued fraction bracket $R(q)$. As $q \rightarrow 1^-$, each fixed convergent tends to the corresponding convergent of the continued fraction with all numerators equal to one. The two subsequences of these convergents have limit $\varphi_{\text{HMT}}^{-1}$, in its already established quadratic realization. The bracketing proves the first limit. Taking the fifth power and using $\varphi^2 = \varphi + 1$ gives the other two.

Equating (16.42) and (16.51) proves (16.52). Substitution of $u = \epsilon - 11$ gives

$$\frac{(t + 256)^3}{t^2} = -\frac{(3125 - 250\epsilon + \epsilon^2)^3}{\epsilon^5}.$$

On the branch $t \rightarrow 0$, the identity rearranges as

$$\frac{t^2}{(-\epsilon)^5} = \frac{(t + 256)^3}{(3125 - 250\epsilon + \epsilon^2)^3}.$$

Its limit is $256^3/3125^3 = 2^{24}/5^{15}$. On the branch $t \rightarrow \infty$, multiplication by ϵ^5 and expression of the left-hand side as $t\epsilon^5(1 + 256/t)^3$ give the second limit. $\square$

The relation retains the branches of a correspondence over the same j . The choice of τ , or of the corresponding lift, fixes the pair being transported. The pentadic pole and Fricke inversion retain their respective domains and operations.

The three operations occurring in this extension have different domains. The dual symmetry $g^\vee$ acts on sectors of the vertex algebra; Fricke acts on the modular parameter and its function; radius duality interchanges momentum and winding in the compact lattice sector. Their reciprocity formulas can be coordinated through the corresponding maps, but the name “duality” does not replace those maps. This distinction allows the construction to continue towards dimensional screens while preserving incidence, grading, and the origin of each operation.

16.19 Exceptional incidence and coordination of the screens

The direction selected by K and the Leech lattice arise from different representations of the constructed incidence. The former determines the orthogonal reduction $12 \rightarrow 11 \rightarrow 10$; the latter enters the isotropic reduction $26 \rightarrow 24$. Their coordination preserves the preceding space, its relations and the information conveyed by each projection.

Sections 21 and 22 develop this coordination jointly. The former brings together the involution of integer coordinates, unitary conjugation with its domains and the change of residual section retaining the quotient memory. The latter constructs the morphism $\mathcal{R}_M^{\text{alg}}$ of (22.5) and proves the isomorphism of the quotient presentations in Theorem 22.6. The graph preserves the coordinate in V_{11} together with its projection in V_{10} , while the isotropic quotient recovers the Leech lattice. The respective kernels and codomains are thereby preserved.

The action ellipse is constructed before duality is applied and fixes its dimensionless radius together with the reciprocal radius (Section 20). The compact sector then connects exceptional incidence with a determined quadratic form. The fields, action and operators of the subsequent physical realization are presented in their own domain, preserving this algebraic antecedent and its internal proofs.

16.20 Graded Moonshine traces and representation of the Monster

For $g \in \text{Aut}(V_{\text{HMT}}^{\mathfrak{h}})$, the trace with insertion is

$$T_g(\tau) = \text{Tr}_{V_{\text{HMT}}^{\mathfrak{h}}} g q^{L_0 - 1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [12, Theorem 1.1]. The character J , the group $\mathbb{M}$, and the collection $g \mapsto T_g$ are realizations of the algebra already constructed. They do not form a causal sequence in which one first guesses J and then infers the Monster.

The unit has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V_0^{\mathfrak{h}} = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one the dimension is zero, accounting for the absence of the constant term in $J = j - 744$. A morphism of vertex algebras preserves the vacuum vector. This property provides the technical meaning of preservation of the unit at this level; the unit of cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

appears. The first decomposition corresponds to the trivial conformal line and the irreducible component of dimension 196883 in the Monster representation. The other expressions have lattice-theoretic or arithmetic readings. Their equality does not automatically identify their summands with TPK regions, words, or routes. Doing so would require presenting the maps between those spaces and proving that they respect the relevant actions.

In particular, an isometry $N_\alpha \simeq \Lambda_{24}$ induces a realization of (16.30), and a choice of isomorphism with $V^{\mathfrak{h}}$ transports its automorphism group. This does not construct an action of the Monster on the digits of π , on the blocks of K , or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\text{TPK}}(g) = \rho_{V^{\mathfrak{h}}}(g)F,$$

including domain, codomain, and preserved information. The absence of that step from the present claim does not remove the incidence intertwiner U_W , the flag, or the neighbor already constructed; it only delimits a different promotion, namely an action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned} &\text{APP-TRIT-TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, Z_0) \\ &\longrightarrow (C_W, \mathcal{H}; B_K \subset H_\alpha^-, P_\pi, H_e, H_\varphi, H_{\text{APP}}) \\ &\longrightarrow (N(C_W), o_*, \text{metric and chamber}) \longrightarrow N_\alpha \longrightarrow V_{\text{HMT}}^{\mathfrak{h}}. \end{aligned}$$

At the incidence level, the stars generate M_{12} , and the residual chart realizes the regional $4 + 1$ structure in W_{24} . At the lattice and VOA level, the construction gives Leech and the Moonshine module. These are prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber, and lift.

The result of this section is therefore an assembled derivation: the finite and lattice proofs make the passage from dodecaphasic incidence to the hypotheses of the recognition theorems effective. The conclusion comprises the structures and transports specified in the preceding propositions, with their explicit domains and choices. Comparison of constants and experimental validation concern assertions distinct from this algebraic construction.

16.21 Full-support orientations and lattice equivariance

The weight enumerator in (16.13) contains $24y^{12}$. Hence the set

$$C_W^\times := \{u \in C_W : u_b \in \{1, 2\} \text{ for every } b\}$$

has 24 elements. For each $u \in C_W^\times$, define

$$P_b(u) = \begin{cases} R, & u_b = 1, \\ I, & u_b = 2, \end{cases} \quad g_u := \bigoplus_{b=0}^{11} P_b(u) C P_b(u)^{-1}.$$

Proposition 16.30 (Robustness of full-support orientations). *For every $u \in C_W^\times$,*

$$\frac{g_u v - v}{3}, \quad \frac{g_u^{-1} v - v}{3} \in N_0.$$

In particular, each g_u preserves the neighbour constructed with the transported orientation.

Proof. In simple-root coordinates,

$$C\rho - \rho = (-2, -1), \quad C^{-1}\rho - \rho = (-1, -2).$$

The discriminant classes of the two displacements are u and $-u$. For the radial coefficient $a_b \in \{4, 1\}$,

$$\left\langle \frac{(C^{\pm 1} - I)a_b \rho}{3}, a_b \rho \right\rangle = -a_b^2.$$

The sum over the twelve factors again gives -27 . The proof of proposition 16.21 applies component by component. $\square$

To specify the type of the recalibration, let $h \in \text{Aut}(C_W)$ be monomial: a permutation σ of the twelve coordinates followed by multipliers $\epsilon_b \in \{1, 2\}$. Define its lattice lift

$$\tilde{h} := P_\sigma \circ \bigoplus_{b=0}^{11} R^{\nu_b}, \quad \nu_b = \begin{cases} 0, & \epsilon_b = 1, \\ 1, & \epsilon_b = 2. \end{cases} \quad (16.55)$$

Its action on the discriminant is exactly h . If the change simultaneously transports the code, flag, incidence origin, vector v , neighbour, and frames, then

$$g_{hu} = \tilde{h} g_u \tilde{h}^{-1}. \quad (16.56)$$

The ternary matrix h does not act without a lift on A_2^{12} or on the Leech lattice.

17 Finite holonomy, the Schläfli configuration, and E_6 symmetry

The Witt branch does not exhaust the exceptional structures accessible from APP. The parallel branch of this chapter starts from the two mixed polarizations sum-product and product-sum, together with a declared plaquette orientation and central section. The computed curvature determines the alternating form; this defines the central extension. An extension is not inferred from cardinality 27 alone.

17.1 Mixed APP curvatures

On $T_9 = (\mathbb{Z}/9\mathbb{Z})^2$, with $S(x, y) = x + y$ and $P(x, y) = xy$, let

$$\Delta_x f = f(x + 1, y) - f(x, y), \quad \Delta_y f = f(x, y + 1) - f(x, y).$$

The mixed connections are

$$A^{SP} = (\Delta_x S, \Delta_y P) = (1, x), \quad A^{PS} = (\Delta_x P, \Delta_y S) = (y, 1).$$

Their discrete curvature is

$$F_A = \Delta_x A_y - \Delta_y A_x.$$

Proposition 17.1 (Opposite oriented curvatures).

$$F_{ASP} = 1, \quad F_{APS} = -1 \pmod{9}.$$

The circulation around an oriented rectangle with sides r, s is, respectively, rs and $-rs$.

Proof. $\Delta_x x - \Delta_y 1 = 1$ and $\Delta_x 1 - \Delta_y y = -1$. A rectangle contains rs plaquettes; on summing, the interior edges cancel, leaving the boundary circulation. $\square$

For displacements $u = (a, b)$, $v = (c, d)$, the alternating class is

$$\sigma(u, v) = ad - bc \pmod{9}. \quad (17.1)$$

The orientation fixes the sign of σ . The central section used below belongs to the gauge of this realization; changing it by a coboundary produces an isomorphic extension, not a new invariant.

17.2 Central extension and ternary reduction

We define the nonadic Heisenberg group

$$\mathcal{H}_9 = \{(a, b, z) : a, b, z \in \mathbb{Z}/9\mathbb{Z}\}, \quad (a, b, z)(c, d, w) = (a + c, b + d, z + w + bc). \quad (17.2)$$

The commutator of two horizontal displacements is

$$[(a, b, 0), (c, d, 0)] = (0, 0, bc - ad),$$

which records $-\sigma(u, v)$. Reducing each coordinate modulo three yields

$$H_3 = \{(a, b, z) : a, b, z \in \mathbb{F}_3\}, \quad |H_3| = 27, \quad (17.3)$$

with the same law. The kernel of $\mathcal{H}_9 \rightarrow H_3$ is $(3\mathbb{Z}/9\mathbb{Z})^3$, of cardinality 27.

17.3 Weyl–Heisenberg representation and central phase

The nonadic extension admits an explicit unitary realization. Let

$$\zeta_9 := \exp(2\pi i/9), \quad \mathcal{H}_9^{\text{Sch}} := \mathbb{C}^9,$$

with basis $\{|n\rangle : n \in \mathbb{Z}/9\mathbb{Z}\}$. We define

$$X|n\rangle := |n+1\rangle, \quad Z|n\rangle := \zeta_9^n |n\rangle. \quad (17.4)$$

One has

$$ZX = \zeta_9 XZ$$

and, by multiplication,

$$(X^a Z^b)(X^c Z^d) = \zeta_9^{bc} X^{a+c} Z^{b+d}. \quad (17.5)$$

Theorem 17.2 (Faithful Weyl–Heisenberg representation). *The map*

$$\rho_{\text{WH}} : \mathcal{H}_9 \longrightarrow U(\mathcal{H}_9^{\text{Sch}}), \quad \rho_{\text{WH}}(a, b, z) := \zeta_9^z X^a Z^b, \quad (17.6)$$

is a faithful unitary representation. For $u = (a, b)$ and $v = (c, d)$, its commutator satisfies

$$[\rho_{\text{WH}}(u, 0), \rho_{\text{WH}}(v, 0)] = \zeta_9^{bc-ad} I = \zeta_9^{-\sigma(u,v)} I. \quad (17.7)$$

Proof. Identity (17.5) reproduces the term bc of (17.2); hence the homomorphism property. If $\zeta_9^z X^a Z^b = I$, the action on the basis successively forces $a = 0$, $b = 0$, and $z = 0$. This proves faithfulness. Comparing the products in both orders gives ζ_9^{bc-ad} , which is the alternating phase of (17.1). $\square$

The representation distinguishes two conjugate observables. Let

$$Q_{\text{H}} := \text{diag}(0, 1, \dots, 8), \quad F_{jk} := \frac{1}{3} \zeta_9^{jk}, \quad P_{\text{H}} := F^\dagger Q_{\text{H}} F.$$

The eigenbases of Q_{H} and P_{H} are mutually unbiased, since $|F_{jk}|^2 = 1/9$. For every unit vector $\psi \in \mathcal{H}_9^{\text{Sch}}$,

$$\text{Var}_\psi(Q_{\text{H}}) \text{Var}_\psi(P_{\text{H}}) \geq \frac{1}{4} |\langle \psi, [Q_{\text{H}}, P_{\text{H}}] \psi \rangle|^2, \quad (17.8)$$

$$H_{Q_{\text{H}}}(\psi) + H_{P_{\text{H}}}(\psi) \geq \log 9. \quad (17.9)$$

Here $H_{Q_{\text{H}}}$ and $H_{P_{\text{H}}}$ are the Shannon entropies of the Born distributions in the respective eigenbases, with natural logarithm and convention $0 \log 0 = 0$. These operators are dimensionless. A physical units chart is a subsequent realization and does not intervene in the central extension.

17.4 Action of 10 displacements on the Cayley graph

For a linear functional $\ell(a, b) = pa + qb$, set

$$D_\ell(a, b) = \left(a, b, \frac{ab}{2} + \ell(a, b) \right), \quad Z = (0, 0, 1),$$

where $2^{-1} = 2$ in $\mathbb{F}_3$. The inverse-closed set

$$S_\ell = \{D_\ell(v) : v \in \mathbb{F}_3^2 \setminus \{0\}\} \cup \{Z, Z^{-1}\} \quad (17.10)$$

has ten elements and does not contain the identity.

Theorem 17.3 (Independence of the linear gauge). *The nine functionals ℓ produce isomorphic Cayley graphs. The nine sets S_ℓ form a single orbit under $\text{Aut}(H_3)$.*

Proof. The map

$$\phi_\ell(a, b, z) = (a, b, z + \ell(a, b))$$

is a central automorphism and satisfies $\phi_\ell(S_0) = S_\ell$. To write the complete collection in the polarized coordinates of equation (17.2), let $c(v, w) = bc$ for $v = (a, b)$, $w = (c, d)$. Given $M \in \text{GL}(2, 3)$, the defect

$$B_M(v, w) = c(Mv, Mw) - \det(M)c(v, w)$$

is symmetric, because the alternating parts of the two terms coincide. We define the quadratic correction

$$q_M(v) = 2B_M(v, v), \quad q_M(v + w) - q_M(v) - q_M(w) = B_M(v, w), \quad (17.11)$$

where $2 = 2^{-1}$ in $\mathbb{F}_3$. The complete collection of automorphisms is then written

$$(v, t) \mapsto (Mv, \det(M)t + q_M(v) + \ell(v)), \quad M \in \text{GL}(2, 3), \ell \in (\mathbb{F}_3^2)^*. \quad (17.12)$$

The polar identity of equation (17.11) directly proves that the central product is preserved. The collection has $48 \cdot 9 = 432$ elements. Exhaustive enumeration of the $\binom{13}{5} = 1287$ inverse-closed subsets of cardinality ten finds exactly those nine with the parameters derived below. $\square$

Write $\underline{S} = \sum_{s \in S} s$ in $\mathbb{Z}[H_3]$. Counting ordered products gives

$$\underline{S}^2 = 10e + \underline{S} + 5(\underline{H}_3 - e - \underline{S}) = 5\underline{H}_3 + 5e - 4\underline{S}. \quad (17.13)$$

Indeed, the identity receives ten factorizations; an element of S receives one; each of the other sixteen receives five.

Theorem 17.4 (Graph of the twenty-seven lines). *The adjacency matrix A_ℓ of $\Gamma_\ell = \text{Cay}(H_3, S_\ell)$ satisfies*

$$A_\ell^2 = 5I - 4A_\ell + 5J. \quad (17.14)$$

Therefore,

$$\Gamma_\ell = \text{srg}(27, 10, 1, 5), \quad \text{Spec}(A_\ell) = \{10^1, 1^{20}, (-5)^6\}.$$

It is isomorphic to the intersection graph of the twenty-seven lines on a smooth cubic surface.

Proof. The regular representation of equation (17.13) produces equation (17.14). Its entries say that every vertex has ten neighbours, that an adjacent pair has one common neighbour, and a nonadjacent pair has five. On the constant line, A_ℓ acts by 10; on its complement, $J = 0$ and $(A_\ell - I)(A_\ell + 5I) = 0$. Zero trace fixes the multiplicities 20 and 6.

In the model of a cubic obtained by blowing up six points of $\mathbb{P}^2$, the 27 classes are

$$E_i, \quad L_{ij} = H - E_i - E_j, \quad Q_i = 2H - \sum_{j \neq i} E_j.$$

With $H^2 = 1$, $E_i^2 = -1$, $H \cdot E_i = 0$, and $E_i \cdot E_j = 0$ for $i \neq j$, the adjacency matrix defined by intersection between distinct classes has the same parameters. The complete intersection matrix has diagonal -1 . Identification with the classical configuration uses the uniqueness theorem for this realization of the 27 lines [25], with the domain and action specified there. $\square$

The isomorphism statement can be made fully effective through a computed bijection. The labeling is not absolute: another linear section or an automorphism of the surface produces another table in the same class.

Table 11: Explicit bijection between H_3 and the twenty-seven line classes.

abz	line class
000	E_1
001	L_{12}
002	Q_2
010	L_{13}
011	Q_1
012	E_3
020	Q_3
021	E_2
022	L_{23}
100	L_{14}
101	L_{35}
102	L_{26}
110	L_{36}
111	L_{24}
112	L_{15}
120	L_{25}
121	L_{16}
122	L_{34}
200	Q_4
201	L_{46}
202	E_6
210	E_5
211	Q_6
212	L_{56}
220	L_{45}
221	E_4
222	Q_5

As a local check, the ten neighbours of $e = (0, 0, 0)$ are the elements of S_0 . The table transforms them into the ten classes intersecting E_1 : the five L_{1j} and the five Q_j , with $2 \leq j \leq 6$. The complete verification compares all 729 entries of both adjacency matrices.

The complement $B_\ell = J - I - A_\ell$ satisfies

$$B_\ell^2 = 8I + 2B_\ell + 8J, \quad \text{srg}(27, 16, 10, 8), \quad \text{Spec}(B_\ell) = \{16^1, 4^6, (-2)^{20}\}. \quad (17.15)$$

This is the Schläfli graph under the complementary adjacency convention; A_ℓ is the intersection graph with parameters $(27, 10, 1, 5)$.

Each edge belongs to a unique triangle, since $\lambda = 1$. The number of triangles is

$$\frac{27 \cdot 10}{6} = 45.$$

The full automorphism group has order

$$27 \cdot 1920 = 51840$$

and, through its action on the line configuration, is identified with $W(E_6)$, not merely through equality of orders [25].

17.5 The root lattice realizing E_6

To make explicit the object on which that group acts, let

$$\text{Pic}(X) = \mathbb{Z}H \oplus \bigoplus_{i=1}^6 \mathbb{Z}E_i, \quad H^2 = 1, \quad E_i^2 = -1, \quad H \cdot E_i = E_i \cdot E_j = 0 \ (i \neq j), \quad (17.16)$$

be the Picard lattice of the blow-up of six general points of $\mathbb{P}^2$. Its canonical class is

$$K_X = -3H + E_1 + \cdots + E_6.$$

The orthogonal complement

$$L_{E_6} = K_X^\perp \subset \text{Pic}(X), \quad (x, y)_{E_6} := -x \cdot y, \quad (17.17)$$

is positive definite of rank six. A basis of simple roots is

$$\begin{aligned} \alpha_1 &= E_1 - E_2, & \alpha_2 &= E_2 - E_3, & \alpha_3 &= E_3 - E_4, \\ \alpha_4 &= E_4 - E_5, & \alpha_5 &= E_5 - E_6, & \alpha_6 &= H - E_1 - E_2 - E_3. \end{aligned} \quad (17.18)$$

Each α_i is orthogonal to K_X , has norm two, and their products are -1 exactly for the adjacent pairs in the diagram E_6 : the chain $1 - 2 - 3 - 4 - 5$, with node 6 attached to node 3. Therefore, its Gram matrix is the Cartan matrix of E_6 .

Proposition 17.5 (The seventy-two roots and their reflections). *The roots of L_{E_6} are*

$$\begin{aligned} &E_i - E_j && (i \neq j), \\ &\pm (H - E_i - E_j - E_k) && (1 \leq i < j < k \leq 6), \\ &\pm (2H - E_1 - \cdots - E_6), \end{aligned} \quad (17.19)$$

a total of $30 + 40 + 2 = 72$. For each root α ,

$$s_\alpha(x) = x + (x \cdot \alpha)\alpha \quad (17.20)$$

preserves intersection and K_X . The six simple reflections generate $W(E_6)$ and faithfully permute the 27 classes E_i, L_{ij}, Q_i of theorem 17.4.

Proof. Substituting each class from equation (17.19) into equation (17.16) gives $K_X \cdot \alpha = 0$ and $\alpha^2 = -2$. The count of classes is as stated. The reflection formula is the orthogonal reflection for the form $-(\cdot)$; it preserves K_X because $K_X \cdot \alpha = 0$. The simple system in equation (17.18) generates all the roots, and its Coxeter group is $W(E_6)$. Its action on the 27 classes of exceptional curves preserves intersection, so it coincides with the action on the graph already constructed [25]. $\square$

17.6 Stratification of the 729 words

We write a six-trit word as

$$w = (\ell_1, h_1, \ell_2, h_2, \ell_3, h_3),$$

and define two elements of H_3 ,

$$L(w) = (\ell_1, \ell_2, \ell_3), \quad H(w) = (h_1, h_2, h_3),$$

using the central law in their products. According to $L(w)^{-1}H(w)$, the 729 words are divided into

$$\begin{aligned} \mathcal{D}_\ell &= \{w : L(w)^{-1}H(w) = e\}, \\ \mathcal{J}_\ell &= \{w : L(w)^{-1}H(w) \in S_\ell\}, \\ \mathcal{S}_\ell &= \{w : L(w)^{-1}H(w) \notin \{e\} \cup S_\ell\}. \end{aligned}$$

Theorem 17.6 (Hensel–Schläfli partition).

$$729 = 27 + 270 + 432, \quad (17.21)$$

corresponding to the diagonal, incidence, and skewness.

Proof. For each of the 27 values of $L(w)$, there is a unique value of $H(w)$ with identity difference, ten with difference in S_ℓ , and sixteen remaining. Multiply 27 by 1, 10, 16. $\square$

The cardinality 432 of the skew stratum coincides numerically with the multiplicity of the transverse region of π in figure 12. The two objects are not thereby identified: here one counts words classified by $L(w)^{-1}H(w)$, whereas there one counts preimages of a register U_6 . Without an explicit equivariant intertwiner between those domains, $432 = 432$ is only a cardinality check.

17.7 Reversible coordinates for ordered pairs of lines

Table 11 fixes the bijection $\iota : H_3 \rightarrow \mathcal{L}_{27}$, where $\mathcal{L}_{27}$ is the configuration of twenty-seven lines. We retain the central law (17.3):

$$(a, b, z)(c, d, w) = (a + c, b + d, z + w + bc), \quad (a, b, z)^{-1} = (-a, -b, -z + ab).$$

Multiplication in both orders verifies the inverse formula: all three resulting coordinates vanish. All operations in this subsection are performed in $\mathbb{F}_3$.

Proposition 17.7 (Reversible encoding of the 729 words). *For $w = (\ell_1, h_1, \ell_2, h_2, \ell_3, h_3)$, let $L(w) = (\ell_1, \ell_2, \ell_3)$ and $H(w) = (h_1, h_2, h_3)$. The map*

$$\mathcal{E} : \mathbb{F}_3^6 \longrightarrow \mathcal{L}_{27} \times \mathcal{L}_{27}, \quad \mathcal{E}(w) = (\iota L(w), \iota H(w)) \quad (17.22)$$

is bijective. In the relative coordinates $L = (a, b, z)$, $R = L^{-1}H = (u, v, t)$, reconstruction is given by

$$H = (a + u, b + v, z + t + bu), \quad w = (a, a + u, b, b + v, z, z + t + bu). \quad (17.23)$$

Proof. Given an ordered pair (ℓ, h) of lines, the table fixes the unique triples $L = \iota^{-1}(\ell)$ and $H = \iota^{-1}(h)$. Interlacing their coordinates recovers a unique word w , and evaluation of (17.22) returns the original pair. Conversely, separating and then interlacing a word preserves all six coordinates. This establishes both inverse compositions.

The transformation $(L, H) \mapsto (L, L^{-1}H)$ is bijective, with inverse $(L, R) \mapsto (L, LR)$. The central law gives $LR = (a + u, b + v, z + t + bu)$, proving (17.23). In particular, the term bu retains the cocycle in the third coordinate. Omitting it changes the reconstruction by precisely bu ; for example, $L = (0, 1, 0)$ and $R = (1, 0, 0)$ produce $H = (1, 1, 1)$, whereas componentwise addition produces $(1, 1, 0)$. $\square$

The partition in Theorem 17.6 is preserved by this bijection: the relative element R distinguishes the diagonal, incident, and skew strata, of cardinalities 27, 270, and 432. Ordered pairs constitute the full domain of the chart. Adjacency is an additional condition on pairs or their sequences. Prolongation to complete histories is constructed in Theorem 79.8 through the TPK block reader.

17.8 Necessity of the central phase

Enumerating all inverse-closed subsets of cardinality ten in the five groups of order 27 gives

group	abelian	# srg(27, 10, 1, 5)
C_{27}	yes	0
$C_9 \times C_3$	yes	0
C_3^3	yes	0
$C_9 \rtimes C_3$	no	9
H_3	no	9.

Thus, preserving 27 labels but abelianizing composition destroys the configuration. Neither cardinality nor degree suffices.

If $\mathcal{T}$ is the set of 45 triangles, the cubic

$$\mathcal{C}_\Gamma(x_1, \dots, x_{27}) = \sum_{\{i,j,k\} \in \mathcal{T}} x_i x_j x_k$$

recovers adjacency, because each edge lies in a unique triangle. Its permutation stabilizer is exactly

$$\text{Aut}(\Gamma_\ell) \cong W(E_6).$$

For the following diagram we write $\Gamma_{27} = \Gamma_\ell$ and $\mathcal{T}_{45} = \mathcal{T}$.

$$\begin{aligned} \text{APP curvatures } \pm 1 &\longrightarrow \sigma \longrightarrow \mathcal{H}_9 \longrightarrow H_3, \\ H_3 &\longrightarrow \Gamma_{27} \longrightarrow \mathcal{T}_{45} \longrightarrow W(E_6). \end{aligned}$$

Figure 13: Causal chain of the mixed-holonomy branch. It is parallel to $C_W \rightarrow W_{12} \rightarrow M_{12}$, not a link within it.

Criterion 17.8 (Refutation criteria for the E_6 branch). *The branch is refuted if the curvatures are not opposite, if the central cocycle is erased, if the quadratic correction (17.11) does not preserve the product, if (17.13) does not produce the coefficients (10, 1, 5), if an abelian group produces the same configuration, if the partition does not sum to 729, if Table 11 does not intertwine the adjacency matrices, or if $W(E_6)$ is identified solely by its order and not by its action on the twenty-seven classes and the 72 roots.*

18 Hilbert-space realization of the nonadic phase and preservation of the unit

The construction of Triadic Modular Holography first provides the states, their transitions, and the restriction maps. The Hilbert-space realization adds a linear structure and an inner product to their discrete coordinates. This order determines what each operator represents: the basis labels arise from the nonadic construction; linear operations subsequently act on those labels. Genealogical memory remains in the TPK state and accompanies the projections used to represent it.

18.1 Phase fiber, labeled basis, and inner product

For an integer depth $d \geq 1$, let

$$\mathcal{C}_{9,d} = (\mathbb{Z}/9\mathbb{Z})^d, \quad \mathcal{H}_d = \ell^2(\mathcal{C}_{9,d}), \quad \mathcal{A}_d = B(\mathcal{H}_d). \quad (18.1)$$

The inner product is $\langle f, g \rangle = \sum_x \overline{f(x)}g(x)$. The functions δ_x form an orthonormal basis labeled by the phase coordinates. Thus $\dim \mathcal{H}_d = 9^d$ and $\mathcal{A}_d \cong M_{9^d}(\mathbb{C})$. The matrix algebra represents this finite fiber; routes, quotients, and memory omitted by a phase projection still belong to the starting state.

Coordinate factorization induces the unitary map

$$\mathcal{H}_{d+1} \longrightarrow \mathcal{H}_d \otimes \mathbb{C}^9, \quad \delta_{(x,r)} \longmapsto \delta_x \otimes \delta_r. \quad (18.2)$$

Define the normalized trace $\tau_d(a) = 9^{-d} \text{Tr}(a)$. Finite counting and normalization are distinguished: $\text{Tr}(I) = 9^d$, whereas $\tau_d(I) = 1$. Likewise, the group $(\mathbb{Z}/9\mathbb{Z})^3$ and the vector space $\mathbb{F}_3^6$ both have 729 elements, but retain different algebraic laws. Realization of a phase fiber does not replace the ternary encoding of APP emissions.

18.2 Translation and phase: finite Weyl representation

In the complex realization subsequent to the constructions of π and the structure $i^2 = -1$, fix $\omega = \exp(2\pi i/9)$. On $\mathcal{H}_1$, let

$$(Xf)(r) = f(r-1), \quad (Zf)(r) = \omega^r f(r). \quad (18.3)$$

The first operation shifts the phase coordinate; the second evaluates it through a character. Both are unitary and satisfy

$$X^9 = Z^9 = I, \quad ZX = \omega XZ. \quad (18.4)$$

Indeed, $(ZXf)(r) = \omega^r f(r-1)$, whereas $(XZf)(r) = \omega^{r-1} f(r-1)$. The character appears here as a realization of the phase already constructed and does not select the TPK numerical registers.

18.2.1 Oriented area and central extension of translations

The translation support is $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. The mixed APP curvature constructed in Section 4.4 assigns +1 to the oriented plaquette of the sum-product pair and -1 to the pair in the opposite order. Its bilinear extension to two displacements $u = (a, b)$ and $v = (c, d)$ is the alternating form

$$\sigma_{\text{ar}}(u, v) = ad - bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \quad (18.5)$$

Indeed, alternation makes the evaluation vanish on two displacements along the same axis. Values on the oriented basis are $\sigma_{\text{ar}}((1, 0), (0, 1)) = 1$ and $\sigma_{\text{ar}}((0, 1), (1, 0)) = -1$; distributing both arguments gives (18.5). The same formula is the sum of plaquettes of any oriented cellular chain realizing that parallelogram: interior edges cancel. Changing an integer representative by a multiple of nine preserves its representation in $\mathbb{Z}/9\mathbb{Z}$.

The order $X^a Z^b$ of the already defined operators fixes a section of the central extension. Denote its composition factor by

$$c_W(u, v) = bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \quad (18.6)$$

This cocycle preserves the order of the two operations, while σ_{ar} preserves the orientation of the parallelogram.

Proposition 18.1 (Central extension and commutator sign). *The set $\mathfrak{H}_9 = (\mathbb{Z}/9\mathbb{Z})^3$, with product*

$$(a, b, t)(c, d, s) = (a + c, b + d, t + s + bc), \quad (18.7)$$

is a group of order 729. The map

$$\varrho(a, b, t) = \omega^t X^a Z^b$$

is a faithful unitary representation on $\mathcal{H}_1$. With the convention $[g, h] = ghg^{-1}h^{-1}$, one has

$$[\varrho(a, b, 0), \varrho(c, d, 0)] = \omega^{bc-ad} I = \omega^{-\sigma_{\text{ar}}((a,b),(c,d))} I. \quad (18.8)$$

The center and the translation group form the exact sequence

$$0 \longrightarrow \mathbb{Z}/9\mathbb{Z} \xrightarrow{t \mapsto (0,0,t)} \mathfrak{H}_9 \xrightarrow{(a,b,t) \mapsto (a,b)} X_9 \longrightarrow 0.$$

Proof. For $w = (e, f)$, the cocycle identity reads

$$c_W(u, v) + c_W(u + v, w) = bc + (b + d)e = de + b(c + e) = c_W(v, w) + c_W(u, v + w).$$

It proves associativity of (18.7); the identity is $(0, 0, 0)$, and the inverse of (a, b, t) is $(-a, -b, -t + ab)$. The three residues are free, giving order $9^3 = 729$. From $ZX = \omega XZ$ it follows that

$$(X^a Z^b)(X^c Z^d) = \omega^{bc} X^{a+c} Z^{b+d},$$

which proves the representation property. If $\omega^t X^a Z^b = I$, its action on δ_r first forces $a = 0$. Evaluation at $r = 0$ gives $t = 0$, and evaluation at $r = 1$ gives $b = 0$, since ω has order nine. The representation is therefore faithful. Comparing the products in both orders gives the factor $bc - ad$. A central element must satisfy $bc - ad = 0$ for every (c, d) . Taking successively $(c, d) = (1, 0), (0, 1)$ yields $b = a = 0$. This identifies the center and proves exactness. $\square$

The Wilson phase for orientation (18.5) is $\omega^{\sigma_{\text{ar}}(u,v)}$; the commutator of the ordered section $X^a Z^b$ expresses its inverse. In particular, $c_W(u, v) - c_W(v, u) = -\sigma_{\text{ar}}(u, v)$. This identity relates two determined conventions without identifying the polarized cocycle with the alternating form. The extension belongs to the translation group of the APP support. Its representation acts on the phase fiber; the trajectory register and carries retain their residence in the TPK state that produced that fiber. The group order 729 and the cardinality 729 of the ternary ambient space preserve the different types established in (18.1). The center $\mathbb{Z}/9\mathbb{Z}$ of this extension records the commutator phase. The memory C_{12} of the sequence $0 \rightarrow C_{12} \rightarrow C_{108} \rightarrow C_9 \rightarrow 0$, constructed in Proposition 9.2, belongs to the dodecaphasic calendar. Each extension preserves its own composition law and projection.

Proposition 18.2 (Algebra generated by phase and translation). *The 81 operators $X^a Z^b$, with $0 \leq a, b < 9$, form a basis of $\mathcal{A}_1 = M_9(\mathbb{C})$. In particular, the representation (18.3) is irreducible.*

Proof. If $a \neq 0 \pmod{9}$, the matrix $X^a Z^b$ has no diagonal entries. If $a = 0$, its trace is $\sum_{r=0}^8 \omega^{br}$, equal to 9 for $b = 0$ and 0 otherwise. By (18.4), the product $(X^a Z^b)^* X^{a'} Z^{b'}$ is a factor of modulus one times $X^{a'-a} Z^{b'-b}$. Therefore,

$$\tau_1((X^a Z^b)^* X^{a'} Z^{b'}) = \delta_{a,a'} \delta_{b,b'}.$$

These are 81 linearly independent matrices in a space of dimension 81. They generate the whole matrix algebra, whose commutant consists of scalars; irreducibility follows. $\square$

Theorem 18.3 (Uniqueness with fixed primitive central character). *Every irreducible unitary representation of the group presented by $X^9 = Z^9 = c^9 = I$, c central, and $ZX = cXZ$, in which c acts as ωI , is unitarily equivalent to (18.3).*

Proof. Let $v \neq 0$ be an eigenvector of Z , with eigenvalue λ . Since $Z^9 = I$, λ is a power of ω . The vectors $X^k v$, $0 \leq k < 9$, have eigenvalues $\omega^k \lambda$, distinct because the character is primitive. They are orthogonal, and their span is invariant under X , Z , and the center. Irreducibility forces it to be the entire space. Reordering the basis to begin with eigenvalue 1, Z is diagonal with entries $1, \omega, \dots, \omega^8$, and X is cyclic translation. Normalizing the basis produces the asserted unitary equivalence. $\square$

Primitivity of the character and irreducibility are part of the statement. At depth d , operators acting on different factors commute, and tensor products of the preceding bases generate $\mathcal{A}_d$. The phase structure is thus represented at each level without identifying phase return with return of the complete genealogical state.

18.3 Two lifts and two different normalizations

Factorization (18.2) permits preserving the entire new fiber or fixing one of its labels. For $p_j = |\delta_j\rangle\langle\delta_j|$, define

$$\iota_d(a) = a \otimes I_9, \quad J_{d,j}(a) = a \otimes p_j. \quad (18.9)$$

Proposition 18.4 (Preservation of unit and trace). *The inclusion ι_d is unital and preserves normalized trace. The inclusion $J_{d,j}$ is a corner embedding and satisfies*

$$\tau_{d+1}(\iota_d(a)) = \tau_d(a), \quad \iota_d(I) = I, \quad \tau_{d+1}(J_{d,j}(a)) = \frac{\tau_d(a)}{9}, \quad J_{d,j}(I) = I \otimes p_j.$$

Proof. The trace of a tensor product is the product of the traces. Since $\text{Tr}(I_9) = 9$ and $\text{Tr}(p_j) = 1$, dividing by 9^{d+1} produces both formulas. The images of the identity are read directly from (18.9). $\square$

The unital inclusion preserves the set of 9 phases and its global normalization. The corner preserves a marked continuation and its support projector. Both operations are legitimate, but transmit different data to the next depth.

18.4 Inductive closure and tracial realization

Theorem 18.5 (Nonadic uniformly hyperfinite algebra). *The norm closure of the union through the inclusions ι_d is*

$$\mathcal{A}_{9^\infty} = \varinjlim (\mathcal{A}_d, \iota_d) \cong \bigotimes_{n=1}^{\infty} M_9(\mathbb{C}).$$

It is a UHF algebra of type $9^\infty = 3^\infty$, with a unique normalized trace τ , whose restriction to $\mathcal{A}_d$ is τ_d .

Proof. The inclusions are injective and unital. The closure of the increasing union of matrix algebras is precisely the UHF construction. The compatibility in Proposition 18.4 defines τ on the union; positivity and norm one allow its extension to the closure. Every normalized trace on the limit restricts to the unique normalized trace on each matrix algebra. Since the union is dense, those restrictions determine the trace uniquely. $\square$

Let $\mathcal{H}_\tau$ be the completion of $\mathcal{A}_{9^\infty}$ for $\langle a, b \rangle_\tau = \tau(a^*b)$, and let π_τ be its representation by left multiplication. The vector defined by the identity is denoted Ω_τ .

Theorem 18.6 (Tracial factor of complete refinement). *The weak closure $\mathcal{R}_9 = \pi_\tau(\mathcal{A}_{9^\infty})''$ is a hyperfinite factor of type II_1 . The nonadic inclusions preserve its unit and normalized trace.*

Proof. The product trace extends to a faithful finite normal trace in the tracial representation. To see that the center is trivial, let E_d be the tracial expectation onto $\pi_\tau(\mathcal{A}_d)$. On a tensor of length $n > d$, it takes the trace of the last $n - d$ factors. These maps extend continuously to the corresponding orthogonal projections in L^2 . If z is central, $E_d(z)$ commutes with $\pi_\tau(\mathcal{A}_d)$, by bimodularity of E_d . It is therefore scalar and equal to $\tau(z)I$. Density of the matrix union implies $E_d(z) \rightarrow z$ in L^2 , hence $z = \tau(z)I$. The algebra is thus a finite factor. It contains unital matrix algebras of unbounded sizes 9^d , so it is not a finite matrix factor. It is of type II_1 . It is hyperfinite because the same union of matrices is weakly dense. Preservation of unit and trace follows from each finite inclusion and persists upon taking closures. $\square$

18.5 Integer ranks and continuous tracial dimension

In $\mathcal{A}_d$, a projection P has normalized dimension $\tau_d(P) = \text{rank}(P)/9^d$. Ranks remain integers at every finite depth. Their compatible normalizations allow the following passage to the limit.

Proposition 18.7 (Realization of each tracial dimension). *For each $t \in [0, 1]$, there exists a projection $P_t \in \mathcal{R}_9$ with $\tau(P_t) = t$.*

Proof. Let $m_d = \lfloor 9^d t \rfloor$. Then $0 \leq m_{d+1} - 9m_d \leq 8$. After choosing a diagonal projection of rank m_d , its inclusion has rank $9m_d$; add $m_{d+1} - 9m_d$ diagonal positions in its complement. This yields an increasing sequence of projections within $\mathcal{R}_9$. Its strong limit is a projection P_t , and normality of the trace gives

$$\tau(P_t) = \lim_{d \rightarrow \infty} \frac{\lfloor 9^d t \rfloor}{9^d} = t.$$

$\square$

Here t denotes the dimension coordinate represented in the closure, not an input to the APP generator. The assertion describes the scope of the resulting tracial realization. To compare it with a particular observable, it remains necessary to identify the projection and the genealogical information it preserves.

By contrast, the isometries $V_{d,j}\xi = \xi \otimes \delta_j$ realize $J_{d,j}(a) = V_{d,j}aV_{d,j}^*$. In the limit of a marked route, the finite spaces are dense. The corners approximate all finite-rank operators; their norm closure is $\mathcal{K}(\mathcal{H}_\infty)$, and their bicommutant is $B(\mathcal{H}_\infty)$, of type I_∞ . This difference preserves the operational content of the two choices in (18.9): selection of one continuation and unital prolongation of the entire fiber have different limits.

The Hilbert-space realization, its phase operators, and the tracial closure are thus constructed in this section, with their inclusion choices explicit. They constitute the operator-theoretic support to be used subsequently for coordinate interchange and the compact Hamiltonian.

18.6 Polar transport of incidence and preservation of readers

The finite realization of the oriented incidence complex receives the boundary and orientation constructed in the APP-TRIT-TPK state. Once the inner products of this realization are fixed, let $\mathcal{H}_0, \mathcal{H}_1$ be its finite-dimensional spaces and

$$d : \mathcal{H}_0 \longrightarrow \mathcal{H}_1, \quad \Delta_0 = d^*d, \quad \Delta_1 = dd^*.$$

The incidence and return readers are

$$\rho_{\text{int}}(v) = (dv, d^*dv), \quad \rho_{\text{ref}}(a) = (d^*a, dd^*a).$$

The symbols Δ_0, Δ_1 here denote the Laplacians of this Hilbert-space realization. Orientation, metrics and the domain of d are part of its data.

Proposition 18.8 (Unitary equivalence of the positive sectors). *Let*

$$\mathcal{H}_0^+ = (\ker d)^\perp, \quad \mathcal{H}_1^+ = (\ker d^*)^\perp = \text{Ran } d.$$

The map

$$\mathcal{U} = d(\Delta_0|_{\mathcal{H}_0^+})^{-1/2} : \mathcal{H}_0^+ \longrightarrow \mathcal{H}_1^+ \quad (18.10)$$

is unitary and satisfies, on those sectors,

$$\mathcal{U}\Delta_0 = \Delta_1\mathcal{U}, \quad \mathcal{U}f(\Delta_0) = f(\Delta_1)\mathcal{U} \quad (18.11)$$

for every function f defined on the common finite positive spectrum.

Proof. The reduced singular value decomposition is $d = V\Sigma W^*$, where the columns of W and V are orthonormal bases of $\mathcal{H}_0^+$ and $\mathcal{H}_1^+$, and Σ is diagonal with strictly positive entries. Hence

$$\Delta_0|_{\mathcal{H}_0^+} = W\Sigma^2W^*, \quad \Delta_1|_{\mathcal{H}_1^+} = V\Sigma^2V^*, \quad \mathcal{U} = VW^*.$$

It follows that $\mathcal{U}^*\mathcal{U} = I_{\mathcal{H}_0^+}$ and $\mathcal{U}\mathcal{U}^* = I_{\mathcal{H}_1^+}$. Evaluating f on the same diagonal Σ^2 proves (18.11). If $d = 0$, both positive sectors are zero spaces and the identities have their unique interpretation between zero-dimensional spaces. $\square$

For $v \in \mathcal{H}_0^+$ and $h, t \geq 0$, quadratic energy, the resolvent and the semigroup are preserved:

$$\langle \mathcal{U}v, \Delta_1\mathcal{U}v \rangle = \langle v, \Delta_0v \rangle, \quad (18.12)$$

$$\mathcal{U}(I + h\Delta_0)^{-1} = (I + h\Delta_1)^{-1}\mathcal{U}, \quad (18.13)$$

$$\mathcal{U}e^{-t\Delta_0} = e^{-t\Delta_1}\mathcal{U}. \quad (18.14)$$

The first identity follows from unitarity and intertwining. The other two are instances of (18.11), with $f(x) = (1 + hx)^{-1}$ and $f(x) = e^{-tx}$.

18.6.1 Realization on the tritic cycle

In the orthonormal bases of vertices and oriented edges of the three-element cycle,

$$d = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}.$$

Matrix multiplication gives

$$\Delta_0 = \Delta_1 = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = 3I - J_3 = 2I - C - C^{-1},$$

where every entry of J_3 equals one and C is the cyclic permutation. The constant vector spans the kernel; on its orthogonal complement, $J_3 = 0$. Thus the spectrum is $\{0, 3, 3\}$ and $\mathcal{U} = d/\sqrt{3}$ on the positive sector. The two readers share this spectrum because they arise from the same incidence, while retaining their vertex and edge spaces.

18.6.2 Orientation and compatibility of refinements

The replacement $d \mapsto -d$ preserves Δ_0, Δ_1 and changes $\mathcal{U}$ to $-\mathcal{U}$. More generally, for a reversal of edge orientations represented by a diagonal unitary matrix E ,

$$d' = Ed, \quad \Delta'_0 = \Delta_0, \quad \Delta'_1 = E\Delta_1E^*, \quad \mathcal{U}' = E\mathcal{U}.$$

These equalities follow by using $E^*E = I$ in the definitions. Orientation is thus retained by the transporter even when the spectral readings coincide.

In a collection of refinements, the maps $J_m^{(0)} : \mathcal{H}_{0,m}^+ \rightarrow \mathcal{H}_{0,m+1}^+$ and $J_m^{(1)} : \mathcal{H}_{1,m}^+ \rightarrow \mathcal{H}_{1,m+1}^+$ must retain domains and norms and satisfy

$$\mathcal{U}_{m+1} J_m^{(0)} = J_m^{(1)} \mathcal{U}_m$$

in order to transport this identification across levels. The proposition proves finite equivalence of the positive sectors; the preceding square expresses the additional refinement condition. Kernels, nonlinear constraints and passages to the limit retain their own proofs.

19 Genealogical path kernels: finite composition, unitary realization and cylindrical limit

The matrix composition proved in this section establishes exactly that each entry of U^R is the sum of the ordered weights of the routes of length R . The next step preserves the information that a visible route does not express and organizes finite refinements into a compact space of genealogies. In this way, three results are distinguished: the finite combinatorial identity, the cylindrical integral over the path limit, and the realization of that integral as the propagator of a specific physical system.

19.1 Route category and functorial weight

Let P_{TPK} be the free category generated by the admissible TPK transitions. Its objects are enriched boundary states and its morphisms are composable sequences of transitions. Composition is concatenation, and the empty path is the identity. Let P_X be the category of visible routes obtained by applying the reader to the objects and transitions; its compositions are visible concatenations. The map

$$q_{\text{vis}} : P_{\text{TPK}} \longrightarrow P_X$$

expresses the observable route and preserves in its fiber the differences in sheet, orientation, carry, block, boundary and memory.

Definition 19.1 (Genealogical weight). Let $\mathcal{B}$ be a normed real algebra, possibly noncommutative, and let $\mathbf{B}_{\mathcal{B}}$ be the category with a single object $*$ whose endomorphism monoid is $\text{End}(*) = (\mathcal{B}, \cdot, 1)$. A genealogical weight is a functor

$$W : P_{\text{TPK}} \longrightarrow \mathbf{B}_{\mathcal{B}}$$

such that

$$W(\text{id}_x) = 1, \quad W(\gamma_2 \circ \gamma_1) = W(\gamma_2)W(\gamma_1).$$

The product order agrees with the causal order of the route.

For enriched states $\tilde{i}, \tilde{f}$, the kernel at depth R is

$$\mathcal{K}_R(\tilde{f}, \tilde{i}) = \sum_{\substack{\gamma \in P_{\text{TPK}}(\tilde{i}, \tilde{f}) \\ |\gamma| = R}} W(\gamma). \quad (19.1)$$

Two routes with the same visible word remain distinct summands when they belong to different genealogical fibers.

Proposition 19.2 (Composition through an enriched cut). *If every route of length $R + S$ has a unique cut after R transitions, then*

$$\mathcal{K}_{R+S}(\tilde{f}, \tilde{i}) = \sum_{\tilde{x}} \mathcal{K}_S(\tilde{f}, \tilde{x}) \mathcal{K}_R(\tilde{x}, \tilde{i}).$$

The formula preserves the causal order of the factors when $\mathcal{B}$ is noncommutative.

Proof. The set of complete routes is partitioned by the enriched state $\tilde{x}$ reached at the cut. The functorial factorization of the weight turns the sum over each class into the product of the partial kernels. Uniqueness of the cut ensures that each route appears exactly once. $\square$

19.2 Construction of the internal phase through the real action character on genealogical routes

Let (E, J_E) be a real Euclidean space equipped with an orthogonal endomorphism J_E such that $J_E^2 = -I$. If $\mathcal{A}_* : \mathcal{P}_{\text{TPK}} \rightarrow \mathbb{R}$ is additive under concatenation, the phase

$$\Phi_J(\gamma) := \exp[-J_E \mathcal{A}_*(\gamma)] = \cos \mathcal{A}_*(\gamma) I - \sin \mathcal{A}_*(\gamma) J_E \quad (19.2)$$

is multiplicative. Fix a nonzero positive section $\hbar_{\text{int}}$ of the action line $\mathcal{L}_S$ constructed earlier; the section $\hbar_{\text{ret}}$ of the declared profile may be used. For a dimensional action $S_{\text{int}}(\gamma) = \hbar_{\text{int}} \mathcal{A}_*(\gamma)$, this expression internally realizes the phase $\exp[-J_E S_{\text{int}}(\gamma)/\hbar_{\text{int}}]$. The scalar symbol i appears only when a complex chart is chosen for the pair (E, J_E) .

Proposition 19.3 (Phase functor). *The map Φ_J satisfies*

$$\Phi_J(\text{id}_x) = I, \quad \Phi_J(\gamma_2 \circ \gamma_1) = \Phi_J(\gamma_2) \Phi_J(\gamma_1).$$

Proof. The action of the empty path is zero. For composable routes, additivity of $\mathcal{A}_*$ and commutativity of multiples of the same endomorphism J_E allow the exponential of the summed value to be separated. $\square$

19.3 Unitary realization on the chronological register 27×27

Before passing to the cylindrical limit, a finite kernel admits an explicit unitary realization. Let

$$\mathbb{T}_{27}^2 := (\mathbb{Z}/27\mathbb{Z})^2, \quad \mathcal{D} := \{N, S, E, W\},$$

and consider the real space

$$\mathcal{H}_{27}^{\mathbb{R}, \text{reg}} := \ell^2(\mathbb{T}_{27}^2; \mathbb{R}) \otimes \mathbb{R}^{\mathcal{D}} \otimes E.$$

The endomorphism J_E is extended as the identity on the position and direction factors, and the same notation is retained for that extension.

In the ordered basis (N, S, E, W) , the directional mixer is

$$C_4 := \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad C_4^T C_4 = I_4. \quad (19.3)$$

Here C_4 denotes exclusively the Walsh–Hadamard mixer on the directional fiber; when it acts on $\mathcal{H}_{27}^{\mathbb{R}, \text{reg}}$ it is understood as $C_4 \otimes I_E$.

Fix a translational chart in which the action of an elementary edge γ_d depends only on its direction:

$$s_d := S_{\text{int}}(\gamma_d) \in \mathcal{L}_S, \quad a_d := \frac{s_d}{\hbar_{\text{int}}} \in \mathbb{R}.$$

The real diagonal phase operator is

$$D_J := \text{diag}(e^{-J_E a_N}, e^{-J_E a_S}, e^{-J_E a_E}, e^{-J_E a_W}).$$

The conditional shift transports each component in its direction:

$$\begin{aligned} (S\psi)(x, y, N) &= \psi(x, y-1, N), & (S\psi)(x, y, S) &= \psi(x, y+1, S), \\ (S\psi)(x, y, E) &= \psi(x-1, y, E), & (S\psi)(x, y, W) &= \psi(x+1, y, W), \end{aligned}$$

with all coordinates taken modulo 27. The elementary step and its composition of twelve steps are

$$U_{\text{micro}} := S D_J C_4, \quad U_{12}^{\text{reg}} := U_{\text{micro}}^{12}. \quad (19.4)$$

The superscript reg preserves the type: this operator acts on the directional torus 27×27 and is not identified with the other dodecaphasic objects of the construction or with the temporal endomorphism of order 108.

Proposition 19.4 (Unitarity and exact path sum in the register). *The operator U_{micro} is orthogonal on the realification and unitary in the complex chart determined by J_E . In particular, U_{12}^{reg} is unitary. If $\Xi := \mathbb{T}_{27}^2 \times \mathcal{D}$, then, for every $n \geq 0$ and $\xi_i, \xi_f \in \Xi$,*

$$(U_{\text{micro}}^n)_{\xi_f, \xi_i} = \sum_{\substack{\gamma: \xi_i \rightarrow \xi_f \\ |\gamma|=n}} W_{U_{\text{micro}}}(\gamma),$$

where, for $\gamma = (\xi_0, \dots, \xi_n)$,

$$W_{U_{\text{micro}}}(\gamma) := (U_{\text{micro}})_{\xi_n, \xi_{n-1}} \cdots (U_{\text{micro}})_{\xi_1, \xi_0}.$$

The sum ranges over exactly the paths permitted by the nonzero blocks of the operator.

Proof. Since J_E is orthogonal and $J_E^2 = -I$, we have $J_E^\top = -J_E$. Thus, each block $e^{-J_E a_d}$ is an orthogonal rotation and D_J is orthogonal. The operator S permutes an orthonormal basis, and (19.3) proves that C_4 is orthogonal. Consequently,

$$U_{\text{micro}}^\top U_{\text{micro}} = C_4^\top D_J^\top S^\top S D_J C_4 = I.$$

All three factors commute with the extension of J_E ; thus, U_{micro} is complex-linear in the chart $J_E \leftrightarrow i$, and its real orthogonality is equivalent to unitarity. Every power, in particular U_{12}^{reg} , preserves that property.

For $n \geq 1$, block multiplication gives

$$(U_{\text{micro}}^n)_{\xi_f, \xi_i} = \sum_{\xi_1, \dots, \xi_{n-1}} (U_{\text{micro}})_{\xi_f, \xi_{n-1}} \cdots (U_{\text{micro}})_{\xi_1, \xi_i}.$$

Each choice of intermediate states determines a path of length n , and each path determines a unique ordered product. For $n = 0$, the sole term is the empty path and its weight is the identity. $\square$

Translation invariance decomposes the problem into blocks of four directional components, tensored with E . For

$$k_x = \frac{2\pi a}{27}, \quad k_y = \frac{2\pi b}{27}, \quad 0 \leq a, b < 27,$$

we adopt the realified Fourier convention

$$\chi_{a,b}(x, y) := \exp[-J_E(k_x x + k_y y)].$$

Direct substitution into the conditional shift produces

$$S(k_x, k_y) = \text{diag}(e^{J_E k_y}, e^{-J_E k_y}, e^{J_E k_x}, e^{-J_E k_x}),$$

and, since D_J and C_4 do not depend on position,

$$\begin{aligned} U_{\text{micro}}(k_x, k_y) &= S(k_x, k_y) D_J C_4, \\ U_{12}^{\text{reg}}(k_x, k_y) &= U_{\text{micro}}(k_x, k_y)^{12}. \end{aligned} \tag{19.5}$$

Consequently,

$$\text{Spec } U_{12}^{\text{reg}} = \bigcup_{a,b=0}^{26} \text{Spec } U_{12}^{\text{reg}} \left(\frac{2\pi a}{27}, \frac{2\pi b}{27} \right).$$

The matrix power enumerates paths, and the Fourier decomposition expresses the quasienergies of the same finite dynamics.

Fix an elementary duration $\delta t > 0$, and let $\Delta t_{12} := 12 \delta t$. In each unitary block, choose a spectral branch $\mathfrak{b}$ of the logarithm. The operator $\text{Log}_{\mathfrak{b}} U_{12}^{\text{reg}}(k_x, k_y)$ is skew-adjoint and commutes with J_E . Define

$$\begin{aligned} H_{\text{eff}}(k_x, k_y) &:= \frac{\hbar_{\text{int}}}{\Delta t_{12}} J_E \text{Log}_{\mathfrak{b}} U_{12}^{\text{reg}}(k_x, k_y), \\ U_{12}^{\text{reg}}(k_x, k_y) &= \exp \left[-\frac{J_E \Delta t_{12}}{\hbar_{\text{int}}} H_{\text{eff}}(k_x, k_y) \right]. \end{aligned} \tag{19.6}$$

The first line is self-adjoint in the complex chart. The second follows from $-J_E^2 = I$ and proves that the exponential recovers the propagator exactly. The branch $\mathfrak{b}$ fixes the quasienergy zone.

This realization constructs a finite propagator, its exact path sum and its effective generator. Identifying it with a continuous physical Hamiltonian still requires verification of the representation, action and limit hypotheses stated below.

19.4 Cylindrical limit on the space of genealogies

With endpoints fixed, let $\mathcal{P}_n$ be finite sets of paths at resolution n , with surjective restriction maps. The space

$$\Omega_{fi}^{\text{path}} := \varprojlim_n \mathcal{P}_n$$

is compact and totally disconnected. A compatible collection of probability measures μ_n on cylinders determines a Borel measure μ on $\Omega_{fi}^{\text{path}}$.

Let Π_n be the partition into level- n cylinders. For $C \in \Pi_n$, choose a representative γ_C , an approximate action $S_n(C)$ and the weight $\mu_n(C)$. For $F : \Omega_{fi}^{\text{path}} \rightarrow \text{End}(E)$, define

$$\mathcal{I}_n(F) := \sum_{C \in \Pi_n} \mu_n(C) F(\gamma_C) \exp \left[-J_E \frac{S_n(C)}{\hbar_{\text{int}}} \right]. \quad (19.7)$$

Theorem 19.5 (Genealogical path integral). *Suppose that:*

1. *the cylinder measures μ_n are projectively compatible;*
2. *F is continuous and bounded;*
3. *there exists a continuous action $S : \Omega_{fi}^{\text{path}} \rightarrow \mathcal{L}_S$ for which*

$$\max_{C \in \Pi_n} \sup_{\gamma \in C} \left\| \frac{S_n(C) - S(\gamma)}{\hbar_{\text{int}}} \right\| \rightarrow 0;$$

4. *the mesh of Π_n tends to zero.*

Then $\mathcal{I}_n(F)$ converges in operator norm, the limit is independent of the representatives, and it defines the Bochner integral

$$\int_{\Omega_{fi}^{\text{path}}} F(\gamma) \exp \left[-J_E \frac{S(\gamma)}{\hbar_{\text{int}}} \right] d\mu(\gamma). \quad (19.8)$$

Proof. Compatibility of the finite measures constructs μ on the cylinder algebra and, by extension, on the Borel sigma-algebra. The function

$$G(\gamma) = F(\gamma) \exp[-J_E S(\gamma)/\hbar_{\text{int}}]$$

is continuous on a compact space and is therefore uniformly continuous and bounded. Orthogonality of J_E gives the estimate

$$\|e^{-J_E a} - e^{-J_E b}\| \leq |a - b|.$$

Uniform approximation of the action and contraction of the mesh make the cylindrical sums converge to the Bochner integral of G [26]. $\square$

The theorem constructs a rigorous integral on the compact space of genealogies. Its realization as a Feynman oscillatory integral for a specific Hamiltonian is obtained when the finite kernels, action and cylinder measure satisfy these convergence hypotheses. The finite matrix expansion, genealogical integral and physical propagator are thus related by explicit morphisms, without being identified by abbreviation.

19.5 Expressed value and fiber memory

Let $\text{Val}_{\text{path}} : \Omega_{fi}^{\text{path}} \rightarrow \mathcal{Y}_{\text{val}} \subset \mathbb{R}$ be a Borel-measurable path reader, where $\mathcal{Y}_{\text{val}}$ carries its standard Borel structure, and let $\nu = (\text{Val}_{\text{path}})_*\mu$. For every function factoring through the value, $F = \hat{F} \circ \text{Val}_{\text{path}}$, we have

$$\int_{\Omega_{fi}^{\text{path}}} F d\mu = \int_{\mathcal{Y}_{\text{val}}} \hat{F}(x) d\nu(x).$$

Since $\Omega_{fi}^{\text{path}}$ is compact metrizable and μ is a Borel probability, there exists a collection of regular conditional probabilities $(\mu_x)_{x \in \mathcal{Y}_{\text{val}}}$, defined for ν -almost every x . For a general genealogical integrand, disintegration of μ takes the form

$$\int_{\Omega_{fi}^{\text{path}}} G(\gamma) d\mu(\gamma) = \int_{\mathcal{Y}_{\text{val}}} \left[\int_{\text{Val}_{\text{path}}^{-1}(x)} G(\gamma) d\mu_x(\gamma) \right] d\nu(x). \quad (19.9)$$

The inner integral preserves distinct histories publishing the same coordinate. This is the precise link between the sum over all paths, real evaluation and the thesis that the continuum is a projection of a richer genealogical domain.

Domain of the genealogical path integral: cylindrical compactness, projective consistency, kernel convergence and disintegration over fibers The finite expansion, unitary realization on $\mathbb{T}_{27}^2$ and its Fourier reduction are exact on their finite domains. The construction of H_{eff} is exact once the complex structure, elementary translational action, duration and spectral branch have been declared; it does not by itself select a continuous physical Hamiltonian. The enriched route category and internal phase precede the cylindrical assembly. theorem 19.5 is exact under the four declared premises. Realizing a specific continuous quantum model requires verifying those premises for its action and Hamiltonian.

20 Action ellipse and determination of the normalized radius

The angular readout and the action section retain the same APP–TRIT–TPK antecedent. The form constructed below uses the representations (13.2) and a positive section $\hbar = \hbar_s$, with $s \in \{\text{pre}, \text{ret}\}$, from (12.13). The angular ratio will determine the form; the action section will preserve its common scale.

20.1 Domain and labelled semiaxes

For brevity, let $A = A_{\text{deg}}$ and $C^* = C_{\text{deg}}^*$. In the chart $A \neq 0$, $\hbar > 0$, and $|C^*/A| < 1$, define

$$\xi = \frac{C^*}{A}, \quad \eta_{\text{el}} = \text{artanh } \xi, \quad a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}. \quad (20.1)$$

The semiaxes and the balanced coordinates Q, P have units of the square root of action. The axes remain labelled: the sign of η_{el} determines which is larger. The curve and the product of its semiaxes are

$$\gamma(\theta) = (a_S \cos \theta, b_S \sin \theta), \quad \frac{Q^2}{a_S^2} + \frac{P^2}{b_S^2} = 1, \quad a_S b_S = 2\hbar. \quad (20.2)$$

Proposition 20.1 (Membership of the representations in the elliptic chart). *The positive branch of (13.2), with the intervals (12.4), satisfies $A > 0$, $0 < C^* < A$, and $\hbar_s > 0$. Reversal of the odd coordinate preserves $|C^*/A| < 1$.*

Proof. Proposition 12.6 gives $0 < \eta_{\text{ret}} < H_5$ and positivity of both action sections. For an upper bound on H_5 , (12.4) implies $\alpha < 1/125$ and $\varphi > 8/5$. Retaining only the positive terms of (12.6) yields

$$H_5 < \frac{\sqrt{5}}{2} + \frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4}.$$

Since $5 < 81/16$, one has $\sqrt{5}/2 < 9/8$. Moreover,

$$\frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4} = \frac{421882}{11718750000} < \frac{1}{1000}.$$

Thus $H_5 < 9/8 + 1/1000 < 2$. Using $\alpha > 7/1000$ now gives

$$0 < C^* = 2(\eta_{\text{ret}} + \alpha) < 2\left(2 + \frac{1}{125}\right) = \frac{502}{125} < 7 < 1000\alpha = A.$$

The channel interchange in 13.4 replaces C^* by $-C^*$ and preserves A ; it therefore preserves the absolute inequality. $\square$

20.2 Parametric phase and oriented area

Proposition 20.2 (Area law and transport of angular coordinates). *The parametric phase θ of γ and its polar angle ϕ , continuously lifted from zero, satisfy*

$$\phi(\theta) = \arg(a_S \cos \theta + ib_S \sin \theta), \quad \frac{d\phi}{d\theta} = \frac{a_S b_S}{a_S^2 \cos^2 \theta + b_S^2 \sin^2 \theta} > 0. \quad (20.3)$$

In a chart where both tangents are defined, $\tan \phi = (b_S/a_S) \tan \theta$. The oriented area swept from phase zero is

$$\mathcal{A}(\theta) = \frac{1}{2} \int_0^\theta (QP' - PQ') dt = \hbar \theta, \quad \mathcal{A}(1) = \hbar, \quad \mathcal{A}(2\pi) = \hbar_s. \quad (20.4)$$

Proof. The derivative of the argument of a nonzero vector is $(QP' - PQ')/(Q^2 + P^2)$. For the curve (20.2), the numerator equals $a_S b_S = 2\hbar$; this proves the derivative formula and its positivity. The quotient P/Q gives the tangent relation; the continuous lift preserves quadrants and turns. The integral equals $a_S b_S \theta/2 = \hbar \theta$. Finally, $h_s = 2\pi \hbar_s$ gives the identity for the complete turn. $\square$

One radian of parametric phase sweeps an area $\hbar$. The polar angle retains its own derivative in (20.3). In polar coordinates, the equivalent law is

$$d\mathcal{A} = \frac{1}{2} \rho_\partial(\phi)^2 d\phi, \quad \rho_\partial(\phi) = \left(\frac{\cos^2 \phi}{a_S^2} + \frac{\sin^2 \phi}{b_S^2} \right)^{-1/2}.$$

The radial formula follows by substituting $(Q, P) = \rho(\cos \phi, \sin \phi)$ into (20.2). With the orientation of γ , the integral of the canonical one-form retains the sign

$$\oint_\gamma P dQ = -\pi a_S b_S = -h_s. \quad (20.5)$$

Indeed, $P dQ = -a_S b_S \sin^2 \theta d\theta$, and the integral of $\sin^2 \theta$ over one turn is π .

20.3 Shape reciprocity and symplectic orientation

Define the matrices

$$D(\eta) = \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (20.6)$$

Proposition 20.3 (Quadratic form and action under reciprocity). *Both transports satisfy $T^\top D(-\eta)T = D(\eta)$. For $\omega = dQ \wedge dP$, the interchange S_2 is antisymplectic and the rotation J_2 is symplectic:*

$$S_2^* \omega = -\omega, \quad J_2^* \omega = \omega.$$

If $\mathcal{I}(\gamma) = \oint_\gamma P dQ$ on a closed curve, then

$$\mathcal{I}(S_2 \gamma) = -\mathcal{I}(\gamma), \quad \mathcal{I}(J_2 \gamma) = \mathcal{I}(\gamma). \quad (20.7)$$

Proof. Matrix multiplication interchanges the two diagonal entries of D , proving both congruences. The determinants of S_2, J_2 are $-1, 1$; this gives the transformation of the area form. For $\lambda = P dQ$, direct calculation gives

$$S_2^* \lambda = d(PQ) - \lambda, \quad J_2^* \lambda = \lambda - d(PQ).$$

The integral of the exact differential over the closed curve is zero, proving both action identities. $\square$

Orientation distinguishes the two transports even though they express the same quadratic form, including in the circular case. The labelling of the axes and the sign of the action therefore belong to the data preserved in passing to the lattice representation.

20.4 Normalized radius and lattice-readout matrix

Proposition 20.4 (Determination and recoverability of the shape radius). *The labelled ellipse determines a unique positive radius*

$$R_{\star} = \sqrt{\frac{b_S}{a_S}} = e^{-\eta_{\text{el}}/2} = \left(\frac{A - C^*}{A + C^*} \right)^{1/4}, \quad \tau_{\text{el}} = iR_{\star}^2. \quad (20.8)$$

The angular ratio is recovered through

$$\frac{C^*}{A} = \frac{1 - R_{\star}^4}{1 + R_{\star}^4}, \quad R_{\star}^4 = \frac{499\alpha - \eta_{\text{ret}}}{501\alpha + \eta_{\text{ret}}}. \quad (20.9)$$

Interchanging the semiaxes transforms $(\eta_{\text{el}}, R_{\star}, \tau_{\text{el}})$ into $(-\eta_{\text{el}}, R_{\star}^{-1}, -1/\tau_{\text{el}})$ and preserves $a_S b_S = 2\hbar$.

Proof. Dividing the semiaxes gives $b_S/a_S = e^{-\eta_{\text{el}}}$. The map $R \mapsto R^2$ is bijective on the positive reals, so the positive root is unique. For $\xi = C^*/A \in (-1, 1)$,

$$2 \operatorname{artanh} \xi = \log \frac{1 + \xi}{1 - \xi};$$

hence $R_{\star}^4 = (1 - \xi)/(1 + \xi)$. Solving for ξ proves the first identity in (20.9). Substituting $A = 1000\alpha$ and $C^* = 2(\eta_{\text{ret}} + \alpha)$ gives

$$\frac{A - C^*}{A + C^*} = \frac{998\alpha - 2\eta_{\text{ret}}}{1002\alpha + 2\eta_{\text{ret}}} = \frac{499\alpha - \eta_{\text{ret}}}{501\alpha + \eta_{\text{ret}}}.$$

The interchange inverts the semiaxis ratio and changes the sign of η_{el} . The identity $-1/(iR_{\star}^2) = iR_{\star}^{-2}$ gives the modular transport. The product of the semiaxes remains fixed. $\square$

On the positive branch, $0 < R_{\star} < 1$; the reversed shape chart expresses $R_{\star}^{-1} > 1$. The angular ratio is recovered from the radius, while the absolute angles and their history remain in the enriched state. The circular shape corresponds to $C^*/A = 0$, with $R_{\star} = 1$. Changing the positive action section, while retaining the angular pair, modifies the common scale of the semiaxes and preserves their ratio.

Proposition 20.5 (Lattice form determined by the ellipse). *The matrix of normalized squared semiaxes is*

$$D(\eta_{\text{el}}) = \frac{1}{2\hbar} \begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix} = \begin{pmatrix} R_{\star}^{-2} & 0 \\ 0 & R_{\star}^2 \end{pmatrix}. \quad (20.10)$$

Its evaluation on an integer pair produces

$$E_{\star}(m, w) = \begin{pmatrix} m & w \end{pmatrix} D(\eta_{\text{el}}) \begin{pmatrix} m \\ w \end{pmatrix} = \frac{m^2}{R_{\star}^2} + w^2 R_{\star}^2. \quad (20.11)$$

The equation of the ellipse multiplied by $2\hbar$ uses, instead, the inverse matrix $D(-\eta_{\text{el}})$.

Proof. The squares of the semiaxes in (20.1), divided by $2\hbar$, are $e^{\eta_{\text{el}}}$ and $e^{-\eta_{\text{el}}}$. By (20.8), these are $R_{\star}^{-2}$ and $R_{\star}^2$. Multiplication by the integer vector proves (20.11). Finally, $2\hbar/a_S^2 = e^{-\eta_{\text{el}}}$ and $2\hbar/b_S^2 = e^{\eta_{\text{el}}}$, identifying the matrix of the elliptic equation. $\square$

The radius $R_{\star}$ is a dimensionless representation of the labelled shape. The polar radius ρ_{∂} belongs to the square-root-of-action line. Representation by a spatial radius requires its subsequent dimensional chart; this preserves the shape already determined and the independence of its construction from a prescribed length.

20.5 Inductive–capacitive realization of the action form

The action form already constructed admits an electrical realization in a chart equipped with a clock $\omega > 0$ and a reference impedance $Z_* > 0$. These two quantities retain their units and accompany the shape parameter η . Define

$$L_\eta = \frac{Z_*}{\omega} e^{-\eta}, \quad C_\eta = \frac{e^\eta}{Z_* \omega}, \quad Z_\eta = \sqrt{L_\eta / C_\eta}. \quad (20.12)$$

Here C_η denotes capacitance, distinct from the counterangle C^* .

Proposition 20.6 (Conservation of the dynamical product). *The realization (20.12) satisfies*

$$L_\eta C_\eta = \omega^{-2}, \quad \frac{Z_\eta}{Z_*} = e^{-\eta} = \frac{b_S}{a_S}, \quad (20.13)$$

where $a_S = \sqrt{2\hbar} e^{\eta/2}$ and $b_S = \sqrt{2\hbar} e^{-\eta/2}$ are the semiaxes of equation (20.1). With $Q = \sqrt{Z_*} q$ and $P = \Phi / \sqrt{Z_*}$, the energy takes the form

$$H_{LC} = \frac{q^2}{2C_\eta} + \frac{\Phi^2}{2L_\eta} = \frac{\omega}{2} (e^{-\eta} Q^2 + e^\eta P^2).$$

On $(Q, P) = (a_S \cos \theta, b_S \sin \theta)$, the electric and magnetic contributions are $\hbar\omega \cos^2 \theta$ and $\hbar\omega \sin^2 \theta$; the total energy is $\hbar\omega$.

Proof. Multiplying L_η and C_η cancels the factors Z_* and $e^{\pm\eta}$, whereas their ratio is $Z_*^2 e^{-2\eta}$. The positive square root proves the impedance identity. Substitution of $q = Q/\sqrt{Z_*}$ and $\Phi = \sqrt{Z_*} P$ gives the form of H_{LC} . The squared semiaxes cancel the remaining shape factors; the sum uses $\cos^2 \theta + \sin^2 \theta = 1$. $\square$

The Hamiltonian convention $\dot{Q} = \partial_P H$, $\dot{P} = -\partial_Q H$ assigns $\dot{\theta} = -\omega$ to this parametrization. Shape, clock, and dimensional chart remain distinguished. The constitutive vacuum response and the impedance Z_η are related by the channel inversion of proposition 39.4; each retains its definition and normalization.

21 T-duality: integer coordinates, change of section, and operator equivalence

The arithmetic decomposition of APP preserves the residual representative and its quotient. TPK evolution also transports its orientation and the memory of boundary crossings. Interchanging the two coordinates requires preserving this distinction and specifying the domain on which the operation is again defined. An arithmetic involution is first constructed on that domain; its quadratic form subsequently admits a representation by self-adjoint operators.

21.1 Stable domain of the interchange

In the positive residual section,

$$r_9^+(n) = 1 + ((n - 1) \bmod 9), \quad Q_9^+(n) = \frac{n - r_9^+(n)}{9}, \quad n = r_9^+(n) + 9Q_9^+(n). \quad (21.1)$$

An interchange can place an arbitrary quotient in the first coordinate. Its stable domain is therefore $\mathcal{D}_0 = \mathbb{Z}^2 \times \mathbb{R}_{>0}$, not merely the section $1 \leq r \leq 9$. Extending the domain makes it possible to define a total operation; the admissibility of a pair as the readout of a TPK route separately retains its selection rules.

Definition 21.1 (Normalized compact form and interchange). For $d = (m, w, R_0) \in \mathcal{D}_0$, let

$$E_{R_0}(m, w) = \frac{m^2}{R_0^2} + w^2 R_0^2, \quad T_0(m, w, R_0) = (w, m, R_0^{-1}). \quad (21.2)$$

The symbols m, w name the two transported integer coordinates; R_0 is the positive dimensionless scale of the form. Their interpretation through momentum, winding, and physical radius is established by a subsequent realization map.

Theorem 21.2 (Involution and preservation of the quadratic form). *The map T_0 is an involution of $\mathcal{D}_0$, and*

$$E_{R_0^{-1}}(w, m) = E_{R_0}(m, w).$$

Proof. Two interchanges restore (m, w) , and two inversions restore R_0 . Moreover,

$$E_{R_0^{-1}}(w, m) = w^2 R_0^2 + \frac{m^2}{R_0^2} = E_{R_0}(m, w).$$

□

The equivalence holds for each radius and its reciprocal. At $R_0 = 1$, the radius remains fixed and the operation acts on the same form. A specific pair is fixed by the full involution precisely when, in addition, $m = w$.

21.2 Compact Hamiltonian and self-adjoint domain

Let $\mathcal{H}_{\text{lat}} = \ell^2(\mathbb{Z}^2)$, with orthonormal basis $\{|m, w\rangle\}$. Define

$$H(R_0)|m, w\rangle = E_{R_0}(m, w)|m, w\rangle, \quad (21.3)$$

$$D(H(R_0)) = \left\{ \psi : \sum_{m, w \in \mathbb{Z}} E_{R_0}(m, w)^2 |\psi_{m, w}|^2 < \infty \right\}. \quad (21.4)$$

Finitely supported vectors belong to the domain and form a core for the diagonal operator. The associated quadratic form has domain

$$D(H(R_0)^{1/2}) = \left\{ \psi : \sum_{m, w} E_{R_0}(m, w) |\psi_{m, w}|^2 < \infty \right\}.$$

Proposition 21.3 (Self-adjointness and compact resolvent). *The operator $H(R_0)$ is self-adjoint and nonnegative. Its eigenvalues are $E_{R_0}(m, w)$, counted with multiplicity, and $(H(R_0) + I)^{-1}$ is compact.*

Proof. A multiplication operator defined by a real sequence is symmetric on (21.4). If η belongs to the domain of the adjoint, testing the adjoint identity against each vector $|m, w\rangle$ forces its image to have coordinates $E_{R_0}(m, w)\eta_{m,w}$. Membership in ℓ^2 is exactly condition (21.4); hence the adjoint has the same domain and the same action. Nonnegativity follows by summing the nonnegative weights.

With $c_{R_0} = \min(R_0^{-2}, R_0^2) > 0$, we have $E_{R_0}(m, w) \geq c_{R_0}(m^2 + w^2)$. Outside increasing finite sets, the coefficients $(1 + E_{R_0}(m, w))^{-1}$ tend uniformly to zero. Truncating the resolvent to those sets produces finite-rank operators converging in norm, which proves its compactness. The diagonal basis determines the stated spectrum. $\square$

Theorem 21.4 (Unitary conjugation, including domains). *The interchange $U_T|m, w\rangle = |w, m\rangle$ is unitary, $U_T^2 = I$, and satisfies*

$$U_TD(H(R_0)) = D(H(R_0^{-1})), \quad U_TH(R_0)U_T^{-1} = H(R_0^{-1}). \quad (21.5)$$

It also transports the quadratic-form domains.

Proof. The interchange is an involutive permutation of the orthonormal basis. For a sequence ψ , the identity in Theorem 21.2 gives

$$\sum_{m,w} E_{R_0^{-1}}(m, w)^2 |(U_T\psi)_{m,w}|^2 = \sum_{m,w} E_{R_0}(m, w)^2 |\psi_{m,w}|^2.$$

Equality of the domains is established. On each coordinate, $U_TH(R_0)U_T^{-1}$ multiplies by $E_{R_0}(w, m) = E_{R_0^{-1}}(m, w)$. The same sum, with the first power of the weight, proves the result for the quadratic forms. $\square$

The scalar equality and the operator equality are two realizations of the same involution. The second preserves the domains that allow the first to be interpreted as spectral equivalence, not merely as an equality on finitely supported vectors.

21.3 Specialization to the radius of the action ellipse

The radius $R_\star$ determined in Section 20 fixes the realization of this collection that corresponds to the angular representations of the state. The symbol R_0 allows the transport results to be expressed for the entire positive collection; composition with the ellipse evaluates that collection at $R_0 = R_\star$ and also preserves its reciprocal image.

Corollary 21.5 (Elliptic collection stable under duality). *The set*

$$\mathcal{D}_{\text{el}} = \mathbb{Z}^2 \times \{R_\star, R_\star^{-1}\}$$

is stable under T_0 . Its Hamiltonians are self-adjoint, nonnegative, and have compact resolvent. The interchange U_T transports the domain of $H(R_\star)$ onto that of $H(R_\star^{-1})$, including their form domains.

Proof. The positivity of $R_\star$ and its unique determination are established in Section 20. Inversion permutes $R_\star$ and $R_\star^{-1}$, while interchange preserves $\mathbb{Z}^2$. Proposition 21.3 and Theorem 21.4 apply on these two fibres. If $R_\star = 1$, the two fibres coincide; otherwise the operator intertwines two distinct fibres of the same collection. $\square$

The common action scale determines the area of the ellipse; its semiaxis ratio determines $R_\star$. This separation makes it possible to preserve the area and the common action scale while interchanging the semiaxes and obtaining the reciprocal radius. The index interchange on $\ell^2(\mathbb{Z}^2)$ and the transports of the action plane act on different spaces; their respective properties are proved on their own domains.

21.4 Change of representative and quotient memory

The zero section uses $\bar{r} \in \{0, \dots, 8\}$. The change that preserves the integer in (21.1) is

$$C(r, Q) = (\bar{r}, Q + \mathbf{1}_{\{r=9\}}). \quad (21.6)$$

The modification of the quotient is part of the operation. For $n = 9$, the two pairs are $(9, 0)$ and $(0, 1)$. Their uncorrected energies are $81/R_0^2$ and R_0^2 . Representing the same integer therefore does not imply equality of those two insertions into a form that distinguishes the coordinates.

The equivalence relation is preserved by defining

$$L_9 = \mathbb{Z}(9, -1), \quad S(x_1, x_2) = (x_2, x_1), \quad q_{R_0}(x) = \frac{x_1^2}{R_0^2} + x_2^2 R_0^2.$$

For $x \in \mathbb{Z}^2$, let

$$\mathcal{E}_{R_0}^{L_9}([x]) = \min_{k \in \mathbb{Z}} q_{R_0}(x + k(9, -1)). \quad (21.7)$$

This construction operates on classes. The memory distinguishing individual representatives remains in the TPK state preceding the quotient; the function in (21.7) determines the invariant evaluation of the chosen class.

Theorem 21.6 (Covariant duality of sections). *The energy in (21.7) is well defined. The transformation*

$$\mathsf{T}_{\text{cov}}([x]_{L_9}, R_0) = ([Sx]_{SL_9}, R_0^{-1})$$

is a bijection between $(\mathbb{Z}^2/L_9) \times \mathbb{R}_{>0}$ and $(\mathbb{Z}^2/SL_9) \times \mathbb{R}_{>0}$, with inverse given by the second interchange, and

$$\mathcal{E}_{R_0^{-1}}^{SL_9}([Sx]) = \mathcal{E}_{R_0}^{L_9}([x]).$$

Proof. Replacing x by $x + k_0(9, -1)$ reindexes the candidates for the minimum. The minimum exists: the function of k is quadratic and has leading coefficient $81/R_0^2 + R_0^2 > 0$. The change $(9, Q) \mapsto (0, Q+1)$ has difference in L_9 , so it represents the same class. Interchange transports the equivalence relation to SL_9 , and

$$\min_{\ell \in L_9} q_{R_0^{-1}}(Sx + S\ell) = \min_{\ell \in L_9} q_{R_0}(x + \ell).$$

Finally, $S^2 = I$ recovers the initial quotient and radius. $\square$

The minimum can be evaluated exactly using two candidates. If $x = (m, w)$, the real minimum of the quadratic occurs at

$$k_* = \frac{wR_0^4 - 9m}{81 + R_0^4}.$$

The integer minimum is attained at $\lfloor k_* \rfloor$ or $\lceil k_* \rceil$. This formula does not alter the equivalence; it makes its finite evaluation explicit and allows the effect of changing section to be checked without discarding the quotient.

21.5 Radius normalization in the string realization

Let $\alpha' > 0$ be a scale of the string realization. This symbol denotes the string scale and is kept distinct from the fine-structure constant α . Define

$$\mathsf{T}_{\alpha'}(m, w, R) = \left(w, m, \frac{\alpha'}{R}\right), \quad \mathcal{N}_{\alpha'}(m, w, R) = \left(m, w, \frac{R}{\sqrt{\alpha'}}\right). \quad (21.8)$$

Theorem 21.7 (Exact conjugation by the scale). *Normalization satisfies*

$$\mathcal{N}_{\alpha'} \mathsf{T}_{\alpha'} = \mathsf{T}_0 \mathcal{N}_{\alpha'}.$$

If

$$E_{\text{str}}(m, w, R) = \frac{m^2}{R^2} + \frac{w^2 R^2}{\alpha'^2},$$

then $E_{\text{str}} = \alpha'^{-1} E \circ \mathcal{N}_{\alpha'}$. The combinations

$$p_L = \frac{m}{R} + \frac{wR}{\alpha'}, \quad p_R = \frac{m}{R} - \frac{wR}{\alpha'}$$

transform as $p_L \mapsto p_L$ and $p_R \mapsto -p_R$.

Proof. Both compositions in the first equality give $(w, m, \sqrt{\alpha'}/R)$. On the other hand,

$$E_{R/\sqrt{\alpha'}}(m, w) = \frac{\alpha' m^2}{R^2} + \frac{w^2 R^2}{\alpha'} = \alpha' E_{\text{str}}(m, w, R).$$

Substituting $(w, m, \alpha'/R)$ into the expressions for p_L, p_R gives the two asserted transformations, respectively. $\square$

The conjugation fixes what it means to express the same result in a dimensional chart. In the elliptic collection, the radii are

$$R = \sqrt{\alpha'} R_{\star}, \quad R^{\vee} = \sqrt{\alpha'} R_{\star}^{-1}.$$

Their product is α' . The internal determination of the dimensionless semiaxis ratio and the length-squared scale of the realization serve different functions: the former fixes the shape; the latter expresses its radii in a dimensional unit.

The preceding normalization allows the same result to be expressed with normalized or physical radius. Assigning a tension and a unit to α' belongs to the realization map and has its own origin; it is not a datum entering the production of APP-TRIT-TPK registers.

21.6 Relationship with the modular transformation

Radius inversion has a precise representation on the imaginary axis of the upper half-plane. Let $\iota(R_0) = iR_0^2$ and $S(\tau) = -1/\tau$. Then

$$S \circ \iota(R_0) = \iota(R_0^{-1}). \quad (21.9)$$

The equality follows from $-1/(iR_0^2) = iR_0^{-2}$. For the modular function $J = j - 744$, whose invariance is preserved in the Moonshine development, it follows that $J(iR_0^2) = J(iR_0^{-2})$.

The map ι expresses a semiconjugacy of parameters. The interchange U_T acts on the coordinates of $\ell^2(\mathbb{Z}^2)$; S acts on the upper half-plane; the dual automorphism of an orbifold acts on its graded sectors. Writing out their domains makes it possible to compose specific relationships between them while preserving the information of each representation. The results assembled here establish coordinate interchange with explicit operator domains and proofs.

22 Dimensional screens and coordinated state reduction

The ranks appearing in the exceptional prolongation and in compactification belong to different objects. Relating them requires retaining their presentation: generators, relations, inner product, and selected direction. Dimension is then a consequence of the channel module and its relations. Its transport is formulated through maps acting on that module, as well as on the state determining it.

22.1 Channels, relations, and effective dimension

Let $\mathcal{S}_n$ be the set of surviving states of depth n , with compatible restrictions $r_{m,n} : \mathcal{S}_m \rightarrow \mathcal{S}_n, m \geq n$. The boundary datum $\beta_n(x)$ preserves the information required to decide compatibility of a prolongation: sheet, orientation, quotient, memory, and incidence class according to the readout considered. The restrictions satisfy

$$r_{\ell,n} = r_{m,n} r_{\ell,m}, \quad \beta_n r_{m,n} = b_{m,n} \beta_m.$$

These equalities are the naturality conditions of the presentation and retain the same data under restriction in stages or directly.

Definition 22.1 (Channel screen). Fix a field $\mathbb{K}_n$, and let $M_n(x)$ be the space generated by the prolongation channels and $\mathcal{R}_n(x)$ the subspace of relations. The screen is

$$\Sigma_n(x) = (\mathbb{K}_n, M_n(x), \beta_n(x), \mathcal{R}_n(x)),$$

and its effective dimension is

$$d_{\Sigma_n}(x) = \dim_{\mathbb{K}_n}(M_n(x)/\mathcal{R}_n(x)). \quad (22.1)$$

If $M_n(x) \cong \mathbb{K}_n^{g_n}$ and the columns of $R_n(x)$ generate the relations, then $d_{\Sigma_n}(x) = g_n - \text{rank } R_n(x)$.

Proposition 22.2 (Invariance of presentation and transport). *Dimension (22.1) is preserved under invertible changes of basis. For two screens over the same field $\mathbb{K}_n = \mathbb{K}_m = \mathbb{K}$, if the linear map $F_\gamma : M_n(x) \rightarrow M_m(T_\gamma x)$ sends $\mathcal{R}_n(x)$ into $\mathcal{R}_m(T_\gamma x)$, it induces a unique linear map between the screen quotients.*

Proof. A change of generators and relations transforms R into URV , with U, V invertible, preserving its rank. For descent, define $\bar{F}_\gamma([v]) = [F_\gamma(v)]$. If two representatives differ by a relation, linearity implies that their images differ by a relation of the terminal screen; the map is well defined. The quotient projection is surjective and guarantees uniqueness. $\square$

The dimensional change during transport is determined by

$$\Delta d(\gamma; x) = d_{\Sigma_m}(T_\gamma x) - d_{\Sigma_n}(x).$$

This expression retains the state and its channel presentation. An isolated rank permits reconstruction of neither the relations nor the map that produced it.

22.2 Ternary puncturing and syndrome screen

The code $C_W = [12, 6, 6]_3$, constructed from the exceptional incidence, produces another screen by puncturing one coordinate. Denote that map by $p : C_W \rightarrow \mathbb{F}_3^{11}$ and its image by $G_{11} = p(C_W)$.

Proposition 22.3 (Punctured code and perfect partition). *The map p is injective. The code G_{11} has parameters $[11, 6, 5]_3$, and its Hamming balls of radius 2 partition $\mathbb{F}_3^{11}$.*

Proof. A vector in the kernel of p would have support contained in the sole removed coordinate. The minimum distance 6 of C_W excludes every nonzero vector of that form. Thus the dimension remains 6. Puncturing decreases weight by at most one, so the minimum distance of the image is at least 5. The balls of radius 2 are therefore disjoint. Each contains

$$1 + 2\binom{11}{1} + 2^2\binom{11}{2} = 1 + 22 + 220 = 243$$

vectors. Since $|G_{11}| = 3^6$ and $3^6 \cdot 243 = 3^{11}$, the balls cover the ambient space. To verify that the distance is exactly 5, take a vector of weight 3. Its decoding ball cannot have centre zero and has a centre of weight at most $3 + 2 = 5$; that centre must have weight at least 5. A word of weight 5 is thereby obtained. $\square$

The integer $243 = 3^{11-6}$ is here the number of classes of the syndrome quotient $\mathbb{F}_3^{11}/G_{11}$. Its function is fixed by the puncturing and distance of the code; equality in cardinality with another fibre does not replace a map between them. This combinatorial screen and the following real screen retain their own operations.

22.3 The dodecaphase direction and rank reduction

The construction in 16.5 starts from the record K , already obtained by APP-TRIT-TPK, and the representation chart specified on $\mathbb{R}^{12}$. In that chart,

$$V_{11} = \mathbf{1}^\perp, \quad P_{11} = I_{12} - \frac{1}{12}J_{12}, \quad u_K = P_3 P_{11} K. \quad (22.2)$$

Here J_{12} is the all-ones matrix and P_3 is the isotypic projector specified by the action of A_5 . The finite sum defining P_3 , its representatives, and the generation of K are retained in the cited section of this same article. In particular,

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

The positive norm defines $\ell_K = \mathbb{R}u_K$, and the projector

$$P_{10} = P_{11} - \frac{u_K u_K^\top}{\|u_K\|^2} \quad (22.3)$$

has image $V_{10} = V_{11} \cap u_K^\perp$.

Proposition 22.4 (Two orthogonal reductions). *One has*

$$\mathbb{R}^{12} = \mathbb{R}\mathbf{1} \oplus^\perp V_{11}, \quad V_{11} = \ell_K \oplus^\perp V_{10}, \quad \dim V_{10} = 10.$$

Moreover, $P_{10}P_{11} = P_{10}$, $P_{10}u_K = 0$, and P_{10} is an orthogonal projector.

Proof. Proposition 16.12, proved before the screens are constructed, establishes idempotence, symmetry, rank, and the two operator identities for this same u_K and this same P_{10} . The first decomposition follows from $P_{11} = I - J_{12}/12$; the second is the orthogonal decomposition of that proposition. The single proof of the reduction is thus retained, and its two stages are made explicit here. $\square$

The first reduction removes only the uniform mode. The second removes the direction determined by K in the fixed representation. If the chart is simultaneously changed by an orthogonal matrix S , the vector transforms as $u_K \mapsto Su_K$ and the projector as $P_{10} \mapsto SP_{10}S^{-1}$. This covariance preserves the geometric operation although its coordinates change.

22.4 Lorentzian lattice screen

Let $\Lambda = N_\alpha$ be the Leech lattice constructed in 16.9. Consider the integral hyperbolic plane $U = \mathbb{Z}e_+ \oplus \mathbb{Z}e_-$, with

$$\langle e_+, e_+ \rangle = \langle e_-, e_- \rangle = 0, \quad \langle e_+, e_- \rangle = 1.$$

The lattice $L_{26} = \Lambda \oplus U$ is even and unimodular, of signature $(25, 1)$. Each vector is written $y = \lambda + ae_+ + be_-$. Since $\langle y, e_+ \rangle = b$,

$$e_+^\perp = \Lambda \oplus \mathbb{Z}e_+, \quad e_+^\perp / \mathbb{Z}e_+ \cong \Lambda. \quad (22.4)$$

The quotient map is $q_{24}(\lambda + ae_+) = \lambda$; it is surjective and has kernel $\mathbb{Z}e_+$. The rank decreases by two through an orthogonality restriction followed by an isotropic quotient. This is therefore an operation of a different type from (22.3).

22.5 Joint screen morphism

With the domain $\mathcal{D}_0$ of 21.1, let

$$\begin{aligned} \mathcal{G}_K &= \{(v_{11}, P_{10}v_{11}) : v_{11} \in V_{11}\}, \\ \mathcal{S}_{\text{HMT}} &= \mathbb{R}^{12} \times \mathcal{D}_0 \times e_+^\perp, \quad \mathcal{S}_M^{\text{alg}} = \mathcal{G}_K \times \mathcal{D}_0 \times \Lambda. \end{aligned}$$

Definition 22.5 (Algebraic map of coordinated screens). Define

$$\mathcal{R}_M^{\text{alg}}(v, d, y) = ((P_{11}v, P_{10}v), d, q_{24}(y)). \quad (22.5)$$

Theorem 22.6 (Quotient isomorphism and intertwining of duality). *The map (22.5) induces an isomorphism*

$$\overline{\mathcal{R}}_M^{\text{alg}} : (\mathbb{R}^{12}/\mathbb{R}\mathbf{1}) \times \mathcal{D}_0 \times (e_+^\perp / \mathbb{Z}e_+) \longrightarrow \mathcal{S}_M^{\text{alg}}.$$

It intertwines the transformations acting by T_0 on the factor $\mathcal{D}_0$ and by the identity on the other two factors. The form $E_{R_0}(m, w)$ is preserved under that duality.

Proof. The map $[v] \mapsto P_{11}v$ is an isomorphism from $\mathbb{R}^{12}/\mathbb{R}\mathbf{1}$ onto V_{11} . Since $P_{10}P_{11} = P_{10}$, the map $[v] \mapsto (P_{11}v, P_{10}v)$ identifies that quotient with $\mathcal{G}_K$; its inverse takes the first component and its class. By (22.4), q_{24} induces the lattice isomorphism of the third factor. The second is transported by the identity. The product of the three maps proves the isomorphism.

The projectors and q_{24} do not modify d , while duality acts only on d . The two compositions are therefore identical. Preservation of the form is Theorem 21.2. $\square$

The graph $\mathcal{G}_K$ retains the preceding coordinate v_{11} together with its projection. The theorem proves equivalence of the quotient presentations; the isolated reduction $V_{11} \rightarrow V_{10}$ retains its kernel ℓ_K . This precision establishes what information the isomorphism retains and what information is removed by each projection composing it.

Corollary 22.7 (Screens over the elliptic collection). *The same morphism restricts to an isomorphism*

$$(\mathbb{R}^{12}/\mathbb{R}\mathbf{1}) \times \mathcal{D}_{\text{el}} \times (e_+^\perp / \mathbb{Z}e_+) \cong \mathcal{G}_K \times \mathcal{D}_{\text{el}} \times \Lambda,$$

and preserves the intertwining of duality.

Proof. The morphism and its inverse preserve the second factor exactly. Corollary 21.5 proves that duality preserves $\mathcal{D}_{\text{el}}$. Thus both the bijection and the already proved intertwining square restrict to it. $\square$

22.6 Coordination with the exceptional realization

On the domain $\mathcal{X}_{\text{HMT}}$, with phase, orientation, quotient, boundary, incidence, grading, and memory, the readouts ε_{exc} and ε_{scr} extract the exceptional and screen data. The corridor constructs the graded algebra $\mathfrak{B}(x) \cong V^{\natural}$, with its vertex product and automorphisms; $\mathfrak{S}_M(x)$ assembles the preceding screens, quotients, and operators. The joint assignment $x \mapsto (\mathfrak{B}(x), \mathfrak{S}_M(x))$ therefore has an algebra as its first component and a collection of spaces and maps as its second. The Monster action belongs to the first; the reduction directed by K and compact duality belong to the second. The two readouts establish their common provenance, and each construction retains its type. The isomorphism in this section establishes the screen correspondence with its factors and proof made explicit.

23 Action on trajectories and eleven-dimensional dynamical realization

Extending the structure to dimensional dynamics requires joint transport of the state, channels and boundary data. This section begins with a functional defined on TPK trajectories and its exact composition law. The worldsheet and eleven-dimensional codomain are introduced afterwards, with the maps that must preserve that composition and the operators already constructed.

23.1 Descent of an evolution and sufficiency of the observed datum

Let $q : \mathcal{X}_{\text{HMT}} \rightarrow Y$ be a surjective readout of states with memory, and let $U : \mathcal{X}_{\text{HMT}} \rightarrow \mathcal{X}_{\text{HMT}}$ be a complete evolution. An evolution on observed values exists precisely when the datum preserved by q suffices to determine the next observed value.

Theorem 23.1 (Exact criterion for descent along fibres). *There exists a unique map $\bar{U} : Y \rightarrow Y$ satisfying $qU = \bar{U}q$ if and only if*

$$q(x) = q(y) \implies q(Ux) = q(Uy). \quad (23.1)$$

Proof. If $\bar{U}$ exists, applying the same function to $q(x) = q(y)$ proves the condition. Conversely, define $\bar{U}(q(x)) = q(Ux)$. Condition (23.1) guarantees independence of the representative. Surjectivity of q defines the map on all of Y and proves uniqueness. $\square$

The theorem determines what memory a readout must preserve. When two representatives have different observed images, the retained datum may be enlarged, the domain restricted, or a multivalued correspondence used. These are distinct mathematical choices. In particular, return of a phase coordinate does not authorize identification of the evolutions of two histories retaining different memories.

23.2 Cocycle of directional and arithmetic changes

An admissible TPK trajectory carries, on each edge e_j , a direction d_j and an arithmetic type t_j . To count changes in a concatenation, the preceding datum is also preserved. The augmented edge is $\tilde{e}_j = (e_j, d_{j-1}, t_{j-1})$; the pair (d_0, t_0) belongs to the initial boundary. Define

$$w_{\text{giro}}(\tilde{e}_j) = \mathbf{1}_{\{d_j \neq d_{j-1}\}}, \quad w_{\text{tipo}}(\tilde{e}_j) = \mathbf{1}_{\{t_j \neq t_{j-1}\}}.$$

For $\tilde{\gamma} = (\tilde{e}_1, \dots, \tilde{e}_R)$, let

$$\mathbf{I}(\tilde{\gamma}) = \left(\sum_{j=1}^R w_{\text{giro}}(\tilde{e}_j), \sum_{j=1}^R w_{\text{tipo}}(\tilde{e}_j) \right) = (N_{\text{giro}}, N_{\text{tipo}}). \quad (23.2)$$

Two trajectories compose when their endpoints agree and the preceding datum of the second agrees with the terminal datum of the first.

Proposition 23.2 (Additivity with preserved boundary). *For every composable pair,*

$$\mathbf{I}(\tilde{\gamma}\tilde{\eta}) = \mathbf{I}(\tilde{\gamma}) + \mathbf{I}(\tilde{\eta}). \quad (23.3)$$

Thus, $\mathbf{I}$ is an additive cocycle with values in $\mathbb{N}^2$ on the category of augmented trajectories. For every $\lambda > 0$, the functional

$$I_\lambda(\tilde{\gamma}) = N_{\text{giro}}(\tilde{\gamma}) + \lambda N_{\text{tipo}}(\tilde{\gamma})$$

is also additive.

Proof. Concatenation does not change the preceding datum of any edge: for the first edge of $\tilde{\eta}$, the composition condition makes it agree with the terminal datum of $\tilde{\gamma}$. The sum over all edges splits exactly into the two sums in (23.3). Applying the linear functional $(a, b) \mapsto a + \lambda b$ proves the last statement. $\square$

Boundary information is an effective part of the proof. If two one-edge segments are evaluated separately while ignoring the connection between them, both may have zero turn count even though their directions differ. The concatenated path then contains a turn. Preserving (d_{j-1}, t_{j-1}) records that term on the first edge of the second segment and restores additivity.

For fixed endpoints and boundary data, the corpus's minimum-interaction principle selects minimizers of I_λ among admissible routes. The principle is a selection rule, whereas the following result establishes a finite-existence property.

Proposition 23.3 (Finite existence of minimizers). *If the set $\Gamma(a, b)$ of admissible trajectories with the fixed boundary is finite and nonempty, there exists $\gamma_* \in \Gamma(a, b)$ such that $I_\lambda(\gamma_*) \leq I_\lambda(\gamma)$ for all $\gamma \in \Gamma(a, b)$.*

Proof. The image of $\Gamma(a, b)$ under I_λ is a nonempty finite set of real numbers, which has a minimum element and at least one preimage. $\square$

The proposition does not select the value of λ , nor does it confuse existence with uniqueness. The value and selection rule used in a particular realization must be part of its declared data. The additive functional recovered here quantifies changes of the path itself; its transport to a dimensional action is a later operation.

23.3 Extended persistence and worldsheet

Realization of a route as an extended object distributes its genealogy along an internal coordinate. For a string, the history has two local coordinates σ^a : one internal spatial coordinate and one temporal coordinate. A realization is described by a map $X : \Sigma \rightarrow \mathcal{M}$, a metric h_{ab} on the worldsheet and background fields in the target space. Conditions on $\partial\Sigma$ belong to the variational domain. For a brane of spatial dimension p , the history is a worldvolume of dimension $p + 1$.

The string variational codomain used by the corpus is the Polyakov action

$$S_P[X, h] = -\frac{T_s}{2} \int_\Sigma d^2\sigma \sqrt{-h} h^{ab} g_{MN}(X) \partial_a X^M \partial_b X^N, \quad (23.4)$$

where T_s is tension. The notation T_s distinguishes it from duality T_0 , cardinal transport T_d and complete evolution U .

Composition of augmented trajectories already has the property proved in 23.2. A continuous realization of that composition must specify the map from paths to configurations, its transport of boundary conditions and the normalization comparing I_λ with the continuous action. Integral (23.4) specifies the codomain and its fields; its parameters do not retrospectively select TPK emissions or registers.

23.4 Eleven-dimensional realization and graded operators

The reduction directed by K provides $V_{11} = \ell_K \oplus V_{10}$, and Theorem 22.6 coordinates this reduction with the lattice screen and duality. The physical realization is typed through

$$\Xi_M : \mathcal{S}_M^{\text{alg}} \longrightarrow \mathcal{X}_{11}^{\text{phys}}.$$

The codomain carries a metric g_{MN} , a three-form C_{MNP} , a gravitino ψ_M , an action, gauge constraints and supersymmetry transformations. An operator realization includes

$$\mathfrak{S}_M = (\mathcal{H}_{\text{phys}}, H, Q), \quad \mathcal{H}_{\text{phys}} = \mathcal{H}_+ \oplus \mathcal{H}_-.$$

The grading is expressed through an involution Γ , equal to $+I$ on $\mathcal{H}_+$ and $-I$ on $\mathcal{H}_-$. The supercharge is odd, $\Gamma Q = -Q\Gamma$, and the Hamiltonian is nonnegative. The relation $Q^2 = H$ is understood as an equality of operators, including their domains.

Let $\mathcal{H}_{\text{HMT}}$ be a Hilbert-space realization of the relevant genealogical states, with operators $\hat{H}, \hat{Q}$. An isometry $W : \mathcal{H}_{\text{HMT}} \rightarrow \mathcal{H}_{\text{phys}}$ must transport domains, grading and observables. The operator conditions are written

$$W\hat{H} = HW, \quad W\hat{Q} = QW, \quad WU_T^{\text{HMT}} = U_T^{\text{phys}}W. \quad (23.5)$$

Theorem 23.4 (Compatible transport of the supercharge). *Suppose $\hat{Q}^2 = \hat{H}$, $WD(\hat{Q}) \subset D(Q)$, and the intertwining relations of (23.5) hold on their domains. Then, for $\psi \in D(\hat{H}) = D(\hat{Q}^2)$,*

$$W\psi \in D(Q^2), \quad Q^2W\psi = HW\psi.$$

If physical transport of the radii also respects (21.5), the transported compact sector preserves its spectrum under reciprocal radii.

Proof. Since $\psi \in D(\hat{Q}^2)$, both ψ and $\hat{Q}\psi$ belong to $D(\hat{Q})$. Thus, $W\psi \in D(Q)$ and $QW\psi = W\hat{Q}\psi \in D(Q)$. One may apply Q again, and

$$Q^2W\psi = W\hat{Q}^2\psi = W\hat{H}\psi = HW\psi.$$

Spectral equality of the compact sector is transported by the isometry and the unitary T-intertwiner; on the closed image of W , this is a unitary equivalence of the corresponding realizations. $\square$

The theorem explicitly preserves the operators and the domain on which the identity is proved. The grading organizes two sectors; construction of fermionic fields and their occupation relations has its own structure. Thermal distributions and the species catalogue are not used in the proof of the screens or in the compact duality of this article.

23.5 Compatible compactification and conclusion of the development

A compactification of the state is described through a prolongable section x_∞ , a map κ_{comp} to the chosen moduli space and a transport $\mathcal{U}$ of the metric, orientation, boundary and memory data. The subscript distinguishes this map from the rational readout κ of the dodecaphasic register. Its compatibility requires four concrete properties:

1. the finite restrictions of x_∞ belong to admissible TPK prolongations;
2. the compactification map intertwines the dualities, $\kappa_{\text{comp}}\mathsf{T} = \mathsf{T}_M\kappa_{\text{comp}}$;
3. observables and the spectrum descend to the chosen quotients according to criterion 23.1;
4. the realization transports the direction ℓ_K , boundary conditions and operator domains, and preserves the declared action and supersymmetry equations.

The preceding proofs provide the phase Hilbert-space realization, tracial closure, screens with their quotient maps, radius involution, its unitary equivalence and the additive trajectory cocycle. The eleven-dimensional realization specifies the fields and compatibilities it uses when receiving that structure. Its complete physical identification also retains the conditions on tension, oscillators, amplitudes, anomalies and the low-energy limit enumerated by the corpus. These conditions do not alter the proved algebraic identities: they delimit the subsequent claim obtained by transporting them.

The exceptional and dimensional prolongations remain linked by their common origin. The Monster acts on the graded construction $V^{\mathfrak{h}}$; the compact Hamiltonian and screens receive data from other maps of the same state. This formulation preserves the positive content of the dual realization, with its operations, proofs and the information transmitted at each stage (Theorem 23.4).

U

Information conservation and recovery

Refinement, recovery, holonomy and symmetries of an evolution with a complete register.

What a transformation preserves

Operator theory gives a calculable form to the conservation of the state and its registers. A partial reading obtains one component; the complement collects the contribution that allows the complete balance to be restored. The sum of the squared norms of the component read and its complement reproduces the squared norm of the initial state. The recovery formulas specify which data and which operators make it possible to invert the reading.

As transport is iterated, each stage incorporates a complement and prolongs a terminal component. The finite balance explicitly preserves this terminal component. The passage to arbitrary depth studies convergence and determines when the archive accounts for the entire norm. This order makes it possible to interpret memory as an operation of conservation and recovery, with verifiable conditions.

The exterior extension shows how areas and volumes of the linear realization are transported. Mixed contributions participate in the exterior products and their balances. The change of representation preserves the relations measured by these objects. Recovery maintains the link with the orientation and forms used by each proof.

Monodromy and loops between geometrical regimes add a definite dynamics. The article composes elliptic, parabolic, and hyperbolic transformations, preserves their memory, and calculates their response. The symmetries of the transport action produce the variational charges of that domain. The invariance of an observable, in turn, is formulated through its relation to transport. These two forms of conservation are linked while preserving their particular definitions.

The physical realizations subsequently recover the action sections, area readers, and scales generated previously. The Gaussian example makes it possible to calculate the purity and spectrum of a reduced reading from a specified joint preparation. Its correlations preserve the information of the complete state. This result shows an operational consequence of the argument: transport determines both what an observation receives and the information needed to understand its relation to the whole.

24 Refinement of histories and conservation of unity

24.1 Generation of emissions and realization domain

The construction starts from APP emissions and their enriched prolongations. An emission preserves the two sheets and their divisions

$$\delta^\diamond = \text{res}^\diamond + 9 \text{ quo}^\diamond, \quad \diamond \in \{\Sigma, \Pi\}.$$

TRIT produces orientation and carry through

$$\Delta(\epsilon) = \text{or}(\epsilon) + 3\kappa(\epsilon), \quad \text{or}(\epsilon) \in \{-1, 0, +1\}.$$

TPK transports the complete pair of sheets and updates memory through

$$U_{\epsilon_2 \epsilon_1} = U_{\epsilon_2} U_{\epsilon_1}, \quad U_\epsilon(m) = C_\epsilon m + \kappa(\epsilon), \quad \kappa(\epsilon_2 \epsilon_1) = \kappa(\epsilon_2) + C_{\epsilon_2} \kappa(\epsilon_1).$$

The cocycle makes affine composition associative. The path boundary telescopes, but its ordered record is an ordered word, not the sum of its entries. The odometer increments vertical memory when the phase returns; return does not identify the complete states. The reduced catalogue transfer does not replace the action of emissions on the tree; the construction in section 5.1 and proposition 5.1 preserves this distinction.

The enriched state preserves the intermediate cells, both sheets, remainders, quotients, orientation, carry, clock, memory, boundary, path, and ordered record. An admissible emission jointly updates these fields. Truncation erases exactly its final contribution. A neutral prolongation gives a section of truncation but records a step: it is not an absence of history. This yields finiteness, nonemptiness, and surjectivity at every level, followed by total survival.

This development stops at the analytic realization of this system and its memory. No target numerical values or conventional constants enter as inputs. The following probabilistic coefficients arise solely from the number of effective outgoing emissions.

24.2 Refinement on the complete multisection

Fix an actual enriched prefix x of depth k . Write

$$H_n = \text{Term}_{k,k+n}(x), \quad \rho_n : H_{n+1} \longrightarrow H_n, \quad \Omega_x = \varprojlim_n (H_n, \rho_n).$$

Each element of H_n is a complete prolongation to that horizon, with its emission witnesses. For $h \in H_n$, the successors are $h\epsilon$, for all emissions $\epsilon \in \text{Out}(h)$. Operation multiplicities are preserved even when reduced terminal data coincide. By section 5.2,

$$p_h(\epsilon) = \frac{1}{d(h)} > 0, \quad d(h) = \# \text{Out}(h), \quad \mu_{n+1}(h\epsilon) = \mu_n(h) p_h(\epsilon).$$

Therefore,

$$\sum_{\epsilon \in \text{Out}(h)} \mu_{n+1}(h\epsilon) = \mu_n(h), \quad (\rho_n)_* \mu_{n+1} = \mu_n. \quad (24.1)$$

Proposition 5.2 constructs the unique probability measure μ_x on Ω_x with these cylinder masses; every nonempty cylinder has positive mass. The measure neither selects a history nor imposes a filter subsequent to the existence of the five constructions.

Proposition 24.1 (Refinement of observables is isometric and unital). *Let $\mathcal{L}_n = L^2(H_n, \mu_n)$ and*

$$R_n : \mathcal{L}_n \longrightarrow \mathcal{L}_{n+1}, \quad (R_n f)(h\epsilon) = f(h).$$

Then $R_n^ R_n = I$, $R_n 1 = 1$, and its adjoint is*

$$(E_n g)(h) = \sum_{\epsilon \in \text{Out}(h)} p_h(\epsilon) g(h\epsilon). \quad (24.2)$$

In particular, $E_n R_n = I$, $E_n 1 = 1$, and $R_n E_n$ is the orthogonal projection onto the observables depending only on the prefix.

Proof. Applying (24.1),

$$\|R_n f\|_{n+1}^2 = \sum_h \mu_n(h) |f(h)|^2 \sum_{\epsilon} p_h(\epsilon) = \|f\|_n^2.$$

The same grouping in the inner product gives (24.2). The statements concerning unity hold pointwise; the projection statement follows from $E_n = R_n^*$ and $E_n R_n = I$. Moreover, $R_n(fg) = R_n f R_n g$ and $R_n \bar{f} = \overline{R_n f}$: on finite observables, it is also a unital algebra homomorphism. $\square$

Upon iteration, $R_{m,n} = R_{m-1} \cdots R_n$ coincides with pullback along the composed truncation. The cylindrical pullbacks $R_{\infty,n}$ realize the inductive limit of these Hilbert spaces in $L^2(\Omega_x, \mu_x)$. Their union is dense: cylinders form an algebra generating the sigma-algebra, and simple functions from that algebra are dense in L^2 . Constructing the limit requires no choice of a trajectory.

24.2.1 The same isometry in the history basis

The unitary identification $W_n f(h) = \sqrt{\mu_n(h)} f(h)$ maps $\mathcal{L}_n$ to $\ell^2(H_n)$. Conjugating R_n gives

$$\hat{R}_n \delta_h = \sum_{\epsilon \in \text{Out}(h)} \sqrt{p_h(\epsilon)} \delta_{h\epsilon}. \quad (24.3)$$

The vectors on the right have disjoint supports for distinct predecessors, because $\rho_n(h\epsilon) = h$. Each column has norm one by normalization. This directly proves isometry on the enriched histories already generated, without imposing that a numerical projection preserve all memory. The unit vector transforms as

$$\Omega_n = \sum_{h \in H_n} \sqrt{\mu_n(h)} \delta_h, \quad \hat{R}_n \Omega_n = \Omega_{n+1}.$$

No artificial copy of h is appended to a result in order to recover it: $h\epsilon$ is precisely the prolongation already produced by the APP-TRIT-TPK domain.

24.3 Correlative prolongation of the five operations

The construction in section 5.9 defines a single record ϵ_ϵ for each emission. Its spectral, solenoidal, gauge, cohomological, and determinantal operations depend on this same emission, with the same source, target, orientation, carry, boundary, path, and ordered record. The cellular action satisfies

$$\partial \text{Cell}(\epsilon) = \text{Cell}(\epsilon) \partial, \quad \text{Cell}(\epsilon_2 \epsilon_1) = \text{Cell}(\epsilon_2) \text{Cell}(\epsilon_1).$$

The complex and its determinant line belong to the same object; the constructed filling satisfies $\partial h_* = z_- - z_+$. Totality of the constructor does not require a specialized trivialization to be acyclic or invertible. Joint existence is thus preserved before any specialized reader.

The naturality in propositions 5.7 and 5.8 is expressed jointly as

$$u_{(\ell, N'), (k, N)} \mathfrak{D}_{\ell, N'}(\hat{x}_\ell) = \mathfrak{D}_{k, N}(\tau_{\ell, k} \hat{x}_\ell). \quad (24.4)$$

Its proof does not identify an isolated fine edge with a fictitious coarse edge: it restricts the emission history and preserves the actions of its entries. Transport composes over the same cylinder probabilities; balances use conditional expectation and its corrector; incidence and ports use their cellular maps; homology and determinant use the same map of complexes. These are not five limits of separately selected histories.

Two directions must be distinguished. State and coefficient maps run from fine to coarse. The pullbacks R_n of Proposition 24.1 run from coarse observables to fine observables. Expectation E_n returns in the reduction direction. This distinction makes it possible to realize (24.4) analytically without reversing its domains.

The terminal construction in theorem 5.4 and sections 5.7 and 5.8 prolongs the complete terminals and simultaneously preserves polar telescoping, Gram and losses, incidence with measure, the return cone, and Smith–Bockstein data. Promoting one of these actions to a bounded operator requires the corresponding norm and bound (see the passage to the limit proved at the end of this section). A compatible finite collection does not become a bounded operator on the limit merely by being named as such.

24.4 Memory fibres: Gram criterion and conservation of unity

For a nonscalar realization, assign to each prefix h a Hilbert fibre $\mathcal{M}_h$, and to its emission a typed linear transport $I_\epsilon : \mathcal{M}_h \rightarrow \mathcal{M}_{h\epsilon}$. This linear operator is not identified with the affine update U_ϵ without specifying the representation realizing it.

On $\mathcal{H}_n = \bigoplus_{h \in H_n} \mathcal{M}_h$, define

$$(V_n \xi)_{h\epsilon} = \sqrt{p_h(\epsilon)} I_\epsilon \xi_h.$$

Disjointness of the successors proves the exact identity of proposition 5.3:

$$V_n^* V_n|_{\mathcal{M}_h} = \sum_{\epsilon \in \text{Out}(h)} p_h(\epsilon) I_\epsilon^* I_\epsilon. \quad (24.5)$$

The necessary and sufficient criterion for V_n to be isometric is that the sum in (24.5) equal I at every predecessor. Each I_ϵ being isometric is sufficient. It is also necessary if all I_ϵ are already contractions: each $I - I_\epsilon^* I_\epsilon$ is then positive, and a sum with strictly positive weights can vanish only when every summand vanishes. The Gram criterion characterizes isometry of the total map; isometry of each transport is a sufficient condition, which is also necessary within the class of contractions.

For the scalar histories of (24.3), $I_\epsilon = 1$, so the condition has been proved. In additional analytic fibres, (24.5) is retained and the precise assumption allowing passage to isometry is identified.

If the fibres have distinguished unit vectors u_h and $I_\epsilon u_h = u_{h\epsilon}$, then

$$V_n \left(\bigoplus_h \sqrt{\mu_n(h)} u_h \right) = \bigoplus_{h\epsilon} \sqrt{\mu_{n+1}(h\epsilon)} u_{h\epsilon}. \quad (24.6)$$

Isometry alone does not assert (24.6): preserving the norm and preserving a distinguished unit vector are distinct properties.

24.5 Symmetries and readers on the common domain

A coherent reversible symmetry of the tree transports all emissions and their multiplicities. Invariance of multiplicities in section 5.2 proves that it preserves degrees and cylinder probabilities. It may map Ω_x to Ω_{gx} ; only when it fixes the prefix does it act within the same prolongation fibre.

Therefore,

$$(K_g f)(g\omega) = f(\omega)$$

defines a unitary map $L^2(\Omega_x, \mu_x) \rightarrow L^2(\Omega_{gx}, \mu_{gx})$. Naturality of truncation implies $K_{g,n+1}R_n = R'_n K_{g,n}$, and the corresponding identity for expectations follows by taking adjoints. In memory fibres, one additionally requires the square

$$S_{g,h\epsilon} I_\epsilon = I_{g\epsilon} S_{g,h}$$

with $S_{g,h}$ unitary. This is the intertwiner that turns a tree symmetry into a symmetry of this memory representation; it is not attributed merely by name to every subsequent exceptional group.

For the double readout, take the initial gauge and its prefix transport specified in section 16.1 and proposition 16.1. Work on a common multisection Ω_x where the readers are defined; for the Archimedean reader, this includes the nested compact intervals and diameter contraction specified in (6.31). This is a premise concerning the reader realization, rather than a new existence criterion for the joint constructor. The finite identities are

$$r_{n,m} Q_n = Q_m \rho_{n,m}, \quad a_{n,m} R_n^{\text{arq}} = R_m^{\text{arq}} \rho_{n,m}. \quad (24.7)$$

They determine the joint readout at the limit. If $L = (\text{ev}_{\text{arq}}, Q_\infty)$ and $\nu = L_* \mu_x$, then

$$C_L : L^2(Y, \nu) \longrightarrow L^2(\Omega_x, \mu_x), \quad C_L f = f \circ L,$$

is an isometry, since equality of norms is the definition of the pushforward measure. It preserves unity, and its ranges consist of the observables measurable with respect to the information read out by L . For compatible finite readers, (24.7) also implies consistency of the pushforward measures.

This isometry does not imply that the reader distinguishes every state. The construction in proposition 16.1 distinguishes the fact that the same visible stream with different hidden memory may have the same incidence. The adjoint of C_L is expectation with respect to the reader; $C_L C_L^*$ preserves exactly this observable part. If an enlarged realization (L, M) is injective and Borel, with a measurable inverse on its image—for example, a continuous injection from the compact history space into an appropriate Hausdorff space—its pullback does identify the whole of L^2 . The required memory is the memory already preserved in the state, not a new retrospective selection.

24.6 Passage-to-the-limit theorem for the observable/memory charge

This section specifies the types in the connection with a memory conservation identity. Canonical refinement of observables supplies the required isometric maps. An additional realization in memory fibres uses the Gram criterion of (24.5).

Let $(\mathcal{H}_n, R_n)$ and $(\mathcal{K}_n, S_n)$ be two inductive systems of Hilbert spaces with isometric inclusions. For each n , let

$$J_n : \mathcal{H}_n \rightarrow \mathcal{K}_n \text{ be isometric, } P_n = J_n J_n^*, \quad U_n \text{ be unitary on } \mathcal{K}_n,$$

and let $\tilde{A}_n$ be bounded and self-adjoint, with $\sup_n \|\tilde{A}_n\| < \infty$. Define

$$A_n = J_n^* \tilde{A}_n J_n, \quad T_n = J_n^* U_n J_n, \quad \eta_n = (I - P_n) U_n J_n.$$

Assume conservation and separation at every level,

$$U_n^* \tilde{A}_n U_n = \tilde{A}_n, \quad [\tilde{A}_n, P_n] = 0, \quad (24.8)$$

and the following refinement squares:

$$J_{n+1} R_n = S_n J_n, \quad P_{n+1} S_n = S_n P_n, \quad (24.9)$$

$$U_{n+1} S_n = S_n U_n, \quad \tilde{A}_{n+1} S_n = S_n \tilde{A}_n. \quad (24.10)$$

Theorem 24.2. *These collections induce bounded operators on the inductive limits, with J_∞ isometric and U_∞ unitary, and preserve*

$$A_\infty = T_\infty^* A_\infty T_\infty + \eta_\infty^* \tilde{A}_\infty \eta_\infty. \quad (24.11)$$

The same identity holds at every finite depth. If $\tilde{A}_n \geq 0$, the retained charge is positive. Without positivity, a self-adjoint form is preserved, rather than necessarily a positive energy.

Proof. Decompose $U_n J_n = J_n T_n + \eta_n$. The two terms belong to $P_n \mathcal{K}_n$ and $(I - P_n) \mathcal{K}_n$. Commutation in (24.8) makes the cross terms of the form $\tilde{A}_n$ vanish. Applying the other identity in (24.8),

$$A_n = J_n^* U_n^* \tilde{A}_n U_n J_n = T_n^* A_n T_n + \eta_n^* \tilde{A}_n \eta_n.$$

The second equality in (24.9), together with the first, implies $J_{n+1}^* S_n = R_n J_n^*$: write $P_{n+1} S_n = S_n J_n J_n^* = J_{n+1} R_n J_n^*$ and multiply by J_{n+1}^* . By (24.9)–(24.10),

$$T_{n+1} R_n = R_n T_n, \quad \eta_{n+1} R_n = S_n \eta_n, \quad A_{n+1} R_n = R_n A_n.$$

The operators are thus defined on the algebraic unions of the systems. Isometries and the uniform bound allow extension by continuity to the completions. Compatibility of U_n^* also follows from compatibility of U_n , because both levels are unitary; the limit is therefore unitary, not merely isometric. The projections P_n induce a projection whose range is the closure of the union of the ranges of J_n ; it is $J_\infty J_\infty^*$. All identities persist first on finite-level vectors and then by density. This proves (24.11). $\square$

Compatibility of J_n alone does not replace the second equality in (24.9). A refinement preserving the observable inclusion while mixing its complement with it may alter the memory defect.

24.6.1 Explicit compatible nonadic realization

Suppose C_n is unitary on $\mathcal{H}_n$, natural with respect to R_n , and A_n is bounded and self-adjoint, uniformly bounded, natural, and satisfies $[A_n, C_n] = 0$. On $\mathcal{K}_n = \mathcal{H}_n^{\oplus 9}$, define

$$J_n v = \frac{1}{3}(v, \dots, v), \quad S_n = R_n^{\oplus 9}, \quad U_n(v_1, \dots, v_9) = (C_n v_9, v_1, \dots, v_8), \quad \tilde{A}_n = I_9 \otimes A_n.$$

The factor $1/3$ normalizes the nine components. The formulas directly verify (24.8)–(24.10), including the projection. The compression is $T_n = (8I + C_n)/9$, and

$$\eta_n^* \tilde{A}_n \eta_n = A_n - T_n^* A_n T_n = \frac{8}{81}(I - C_n)^* A_n (I - C_n). \quad (24.12)$$

The final equality uses $C_n^* A_n C_n = A_n$. By the theorem, (24.12) also holds at the limit. Nonadic refinement or complete history refinement transports the charge and its memory when C_n and A_n satisfy the indicated squares.

For $A = I$, finite telescoping gives

$$\|v\|^2 = \|T^N v\|^2 + \sum_{j=0}^{N-1} \|\eta T^j v\|^2.$$

Thus, $v \mapsto (\eta v, \eta T v, \dots, \eta T^{N-1} v, T^N v)$ is an isometry. The corresponding charge formula replaces the norms by the forms of A and $\tilde{A}$. The terminal term is not removed without proving convergence to zero; depth compatibility alone does not prove this temporal stability.

If $D_{N,n}$ denotes this isometry at depth n and temporal horizon N , the preceding squares also prove

$$(S_n^{\oplus N} \oplus R_n) D_{N,n} = D_{N,n+1} R_n.$$

Indeed, each component satisfies $\eta_{n+1} T_{n+1}^j R_n = S_n \eta_n T_n^j$, and the terminal component satisfies $T_{n+1}^N R_n = R_n T_n^N$. Finite-horizon memory accumulation and resolution refinement commute. The same calculation proves equivariance for a symmetry represented by unitaries intertwining $J, U, P, \tilde{A}$; it does not extend the action to symmetries for which these squares have not been constructed.

25 Conservation of observables and the complete record

25.1 Balance of a conserved observable

Let $\mathcal{H}, \mathcal{K}$ be realizations of a readout space and its enriched extension. Let $J : \mathcal{H} \rightarrow \mathcal{K}$ be an isometry, $P = JJ^*$, $U : \mathcal{K} \rightarrow \mathcal{K}$ a bounded transport, and $\tilde{A}$ a bounded self-adjoint observable. Suppose

$$U^* \tilde{A} U = \tilde{A}, \quad [\tilde{A}, P] = 0.$$

Define

$$A = J^* \tilde{A} J, \quad T = J^* U J, \quad \eta = (I - P) U J.$$

Proposition 25.1. *The charge decomposes exactly between readout and complement:*

$$A = T^* A T + \eta^* \tilde{A} \eta. \quad (25.1)$$

Proof. The identity $UJ = JT + \eta$ is an orthogonal decomposition. Expanding $J^* U^* \tilde{A} U J$, the cross terms vanish because $\tilde{A}$ reduces the subspaces $P\mathcal{K}$ and $(I - P)\mathcal{K}$. The left-hand term is A ; the two diagonal terms are those in (25.1). This also proves the identity of sesquilinear forms for two distinct states. If $\tilde{A} \geq 0$, both summands are positive. For $\tilde{A} = I$, the conservation hypothesis is $U^* U = I$. $\square$

If the readout mixes sectors of the observable, the balance additionally retains

$$T^* J^* \tilde{A} \eta + \eta^* \tilde{A} J T.$$

These terms represent interference between the two spaces carrying the charge. The reducing condition stated above ensures their vanishing for the entire decomposition. In particular realizations, they may also vanish for other reasons, for example when $\eta = 0$.

25.2 Specialization to the nonadic connection

Let C be the unitary realization of memory transport on $\mathcal{H}$. On $\mathcal{H}^9$, transport of the nine phases and their uniform inclusion are

$$S_C(v_0, \dots, v_8) = (Cv_8, v_0, \dots, v_7), \quad Jv = \frac{1}{3}(v, \dots, v).$$

The factor $1/3$ follows from normalization of nine copies. The compression and its complement are

$$T = \frac{8I + C}{9}, \quad \eta v = \frac{1}{27}(8(C - I)v, -(C - I)v, \dots, -(C - I)v).$$

If $A = A^*$ is bounded and $[A, C] = 0$, the observable $\tilde{A} = I_9 \otimes A$ satisfies the hypotheses of Proposition 25.1. Therefore,

$$A = T^* A T + \frac{8}{81}(I - C)^* A (I - C). \quad (25.2)$$

The coefficients arise from the distribution among the nine components: $8^2 + 8 = 72$, divided by 27^2 , gives $8/81$. The identity is the sesquilinear balance already documented in the spectral article, expressed here for a general observable.

Telescopic iteration gives, for every $N \geq 1$,

$$A = T^{*N} A T^N + \sum_{j=0}^{N-1} T^{*j} \eta^* \tilde{A} \eta T^j. \quad (25.3)$$

For $A = I$, the map

$$V_N v = (T^N v, \eta v, \eta T v, \dots, \eta T^{N-1} v)$$

is isometric. The step $N \rightarrow N + 1$ decomposes only the terminal component $T^N v$ into its new readout and complement, retaining every preceding record. Conservation is thus verified at any depth.

Distinction of operations. One has $S_C^9 = I_9 \otimes C$. Iteration T^N describes successive compressions with storage of their complements. Enriched transport without intermediate compressions is S_C^N . Both operations have defined domains and different functions; the proof preserves this distinction.

25.3 Terminal sector and the unlimited-depth record

Let P_{fix} be the orthogonal projection onto $\ker(I - C)$.

Theorem 25.2. *For every unitary transport C , the nonadic compression satisfies*

$$\text{s-lim}_{N \rightarrow \infty} T^N = P_{\text{fix}}, \quad \boxed{I = P_{\text{fix}} + \sum_{j=0}^{\infty} T^{*j} \eta^* \eta T^j.} \quad (25.4)$$

The series converges in the strong topology. For each bounded self-adjoint observable A commuting with C , one also obtains

$$\boxed{A = P_{\text{fix}} A P_{\text{fix}} + \sum_{j=0}^{\infty} T^{*j} \eta^* (I_9 \otimes A) \eta T^j.} \quad (25.5)$$

Proof. In the spectral representation of the unitary operator C , compression corresponds to $t(z) = (8 + z)/9$. On the unit circle,

$$|t(z)|^2 = 1 - \frac{8}{81} |1 - z|^2.$$

Hence $t(z)^N \rightarrow 0$ for $z \neq 1$, while $t(1)^N = 1$. Dominated convergence with respect to each vector spectral measure proves $T^N \rightarrow P_{\text{fix}}$ strongly. The same argument holds for T^{*N} . Since A is bounded, $T^{*N} A T^N \rightarrow P_{\text{fix}} A P_{\text{fix}}$ strongly. Taking the limit in (25.3) gives (25.5), and specializing to $A = I$ gives (25.4). $\square$

In particular,

$$V_{\infty} v = (P_{\text{fix}} v, \eta v, \eta T v, \eta T^2 v, \dots)$$

is an isometry. It preserves the complete inner product as well as the norm.

Corollary 25.3 (Bilateral memory). *For the shift $Ce_n = e_{n+1}$ on $\ell^2(\mathbb{Z})$, $P_{\text{fix}} = 0$: a fixed sequence would be constant, and its only square-summable realization is zero. Consequently, the entire initial norm is distributed through the complement record. This specifies the unlimited behaviour of the bilateral memory realization present in the corpus.*

Corollary 25.4 (Finite calendar). *On a cycle of d positions, the uniform sequence determines a one-dimensional fixed subspace. This sector remains as the terminal component, while the complement accumulates the remainder. The finite calendar and the bilateral record therefore have different terminal components.*

In the bilateral case, strong convergence may coexist with $\|T^N\| = 1$ for every N . The proof assumes no uniform contraction rate. It also distinguishes this compression from the mode $(5/9)C$, whose geometric contraction is uniform and is studied separately in the sources.

25.4 Symmetry conservation and chart transport

Let R be unitary and define $C' = RCR^*$, $A' = RAR^*$. The explicit compression formula gives

$$T(C')R = RT(C), \quad \eta(C')R = R^{\oplus 9}\eta(C). \quad (25.6)$$

Proof. The first equality follows by expanding $(8I + RCR^*)R/9$. The second follows from $(C' - I)R = R(C - I)$ in each of the nine components of η . $\square$

Iteration of (25.6) intertwines the finite records. Moreover, $P_{\text{fix}}(C') = RP_{\text{fix}}(C)R^*$, by transport of the kernel of $I - C$, so the unlimited record is also intertwined. The observable balance is thus covariant when the chart, operator, and observable are transported jointly. Symmetries commuting with C also act within a single fixed chart. This distinction allows incorporation of exceptional transformations with their frames, flags, and orientations effectively transported.

The composition in lemma 9.10, section 16.4, and proposition 16.8 uses the record's Hadamard matrix, $H_{12}^2 = 4I$, and the incidence J of the twelve positions in the 132 hexads:

$$J_H = H_{12}/2, \quad J^*J = 36I + 30\mathbf{1}\mathbf{1}^*, \quad U_W = J/6|_{\mathbf{1}^\perp}.$$

Both normalized operators are isometric on the indicated domains. The composition $B = U_W J_H$ on $W = J_H(\mathbf{1}^\perp)$ preserves inner products and intertwines the transported representation of M_{12} with the action on the hexads. The integer record K additionally preserves the integral inversion $H_{12}/4$ on its congruence lattice; this inversion and isometric normalization have different functions. The proof, the APP–Witt–Leech prolongation, and their domains are developed here.

In this composition, incidence conservation has concrete operator content. Its extension through other transports uses (25.6) and the effective intertwiners from the sources, respecting their respective groups and domains.

25.4.1 Complete transport of the twelve coordinates of K

Separation of the centred subspace can be completed by also retaining its mean. Let $P_0 = I - \mathbf{1}\mathbf{1}^*/12$, $\mu(k) = \mathbf{1}^*k/12$, and

$$F(k) = \left(\mu(k), \frac{1}{6}JP_0k \right) \in \mathbb{R} \oplus E_{\text{inc}}.$$

Equip the codomain with $\|(\mu, y)\|^2 = 12|\mu|^2 + \|y\|^2$. The incidence Gram operator proves

$$\|F(k)\|^2 = \|k\|^2, \quad \boxed{F^{-1}(\mu, y) = \mu\mathbf{1} + \frac{1}{6}P_0J^*y.} \quad (25.7)$$

Indeed, $J^*JP_0 = 36P_0$, so the inverse formula recovers P_0k , while the first term recovers its mean. The action of M_{12} is trivial on the mean and permutes the hexads; hence F is equivariant. Composing FJ_H transports the normalized signed coordinates. All twelve record coordinates are thereby preserved, rather than only the centred part. The domain of this reconstruction is the dodecaphasic record; the complete history memory retains its other coordinates and its prolongation.

25.4.2 Specialization to the exceptional operator of order three

The exceptional construction produces the orthogonal operator g_c on the realization of the Leech neighbour, through the twelve APP–Witt fibres of type A_2 , and proves $g_c^3 = I$ and $\ker(I - g_c) = 0$. Its construction, intertwining, and lattice preservation are located in sections 16.11 and 27.3.

Now construct, as an operator composition of this output with the nine-phase chart,

$$T_3 = \frac{8I + g_c}{9}, \quad \eta_3 = (I - JJ^*)S_{g_c}J.$$

Corollary 25.5. *This composition satisfies*

$$\boxed{T_3^* T_3 = \frac{19}{27} I, \quad \eta_3^* \eta_3 = \frac{8}{27} I, \quad \|T_3^N v\|^2 = \left(\frac{19}{27}\right)^N \|v\|^2.} \quad (25.8)$$

Proof. The vector $(I + g_c + g_c^2)v$ is fixed by g_c ; the absence of fixed vectors implies $I + g_c + g_c^2 = 0$. Since $g_c^* = g_c^2$, one has $g_c + g_c^* = -I$. Substitution gives

$$T_3^* T_3 = \frac{65I + 8(g_c + g_c^*)}{81} = \frac{57}{81} I = \frac{19}{27} I.$$

Balance (25.2) with $A = I$ gives the second coefficient. Repeated application of the norm identity gives the power for every N . $\square$

The unlimited record is isometric, and its terminal component tends to zero at the explicit rate in (25.8). Here, the exceptional structure determines the two complementary coefficients; both arise from the polynomial relation of g_c and the nine phases. This is a mathematical composition developed in this research. Its definition neither identifies g_c with Γ_9 nor asserts that S_{g_c} is the canonical temporal operator of TPK: it preserves the distinction between exceptional prolongation and temporal holonomy while composing them explicitly.

25.5 History refinement and compatibility of balances

For enriched histories of depth n , let $p_h(\epsilon)$ be the probability of a surviving extension, normalized by $\sum_\epsilon p_h(\epsilon) = 1$. Canonical refinement on the history bases is

$$R_n \delta_h = \sum_{\epsilon \in \text{Out}(h)} \sqrt{p_h(\epsilon)} \delta_{h\epsilon}.$$

Extensions of different histories have different prefixes and disjoint supports. Hence $R_n^* R_n = I$. This is the isometric realization of projective consistency of the measure: the coefficients arise from the survival weights already constructed.

To transport (25.1) between depths, inclusions, projections, transports, and observables are preserved jointly. With refinement isometries $R_n, \tilde{R}_n$, the required squares are

$$J_{n+1} R_n = \tilde{R}_n J_n, \quad P_{n+1} \tilde{R}_n = \tilde{R}_n P_n, \quad U_{n+1} \tilde{R}_n = \tilde{R}_n U_n, \quad \tilde{A}_{n+1} \tilde{R}_n = \tilde{R}_n \tilde{A}_n.$$

These squares imply $T_{n+1} R_n = R_n T_n$ and $\eta_{n+1} R_n = \tilde{R}_n \eta_n$. With uniform bounds, the identities pass to the inductive limit of the realizations. Histories, by contrast, form the projective system of prefixes; the opposite directions of the two types of maps remain distinguished. The measure, Gram, and passage-to-the-limit proofs are included.

25.6 Variational specialization: action and orientation

The elliptic action in sections 20 and 30 supplies a physical specialization of the principle, with parameters constructed at their HMT stages. In its canonical coordinates,

$$H_\eta(Q, P) = \frac{\omega}{2} (e^{-\eta} Q^2 + e^\eta P^2), \quad I_\eta = H_\eta / \omega.$$

The canonical transformation $Q = e^{\eta/2} q, P = e^{-\eta/2} p$ turns I_η into $(q^2 + p^2)/2$. Rotations of (q, p) , transported to (Q, P) , form a continuous symmetry. For the Lagrangian $L = P\dot{Q} - \omega I_\eta$, its Noether charge is I_η . On the source contour $I_\eta = \hbar$, the oriented action is $\oint P dQ = 2\pi\hbar = h$, with the orientation of the Hamiltonian orbit. The proof distinguishes the symplectic and antisymplectic transformations: they can preserve the same energy while acting with different signs on oriented action. The full variational proof is presented in section 30.

25.6.1 Updating the form and the connection term

The form matrix $M_\eta = \text{diag}(e^{\eta/2}, e^{-\eta/2})$ allows development of transport between ellipses with different parameters. Fix $J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $R(\theta) = \exp(-\theta J_2)$, the Hamiltonian rotation in the (q, p) -plane. The transport

$$z' = M_{\eta'} R(\delta\theta) M_\eta^{-1} z, \quad z = (Q, P)^T,$$

is symplectic and satisfies exactly $I_{\eta'}(z') = I_\eta(z)$. The proof consists of moving both states to their normalized coordinates and using preservation of $q^2 + p^2$ under rotation. It preserves action while changing the form.

For a differentiable realization $\eta(t)$, with angular advance $\dot{\theta} = \omega$, differentiation of this transport gives

$$\dot{Q} = \omega e^\eta P + \frac{1}{2} \dot{\eta} Q, \quad \dot{P} = -\omega e^{-\eta} Q - \frac{1}{2} \dot{\eta} P.$$

Its Hamiltonian generator, obtained from these two equations, is

$$H_{\text{trans}} = \omega I_\eta + \frac{\dot{\eta}}{2} QP. \quad (25.9)$$

The second summand is the connection term for the varying form. With the convention $\{f, g\} = \partial_Q f \partial_P g - \partial_P f \partial_Q g$,

$$\partial_t I_\eta = \frac{\dot{\eta}}{2} (-e^{-\eta} Q^2 + e^\eta P^2), \quad \left\{ I_\eta, \frac{\dot{\eta}}{2} QP \right\} = \frac{\dot{\eta}}{2} (e^{-\eta} Q^2 - e^\eta P^2).$$

The two contributions cancel, and $dI_\eta/dt = 0$. For the prescribed transport, (25.9) is unique up to a function of time. The differentiable realization is a declared specialization of transport between forms; the preceding discrete result applies directly between two generated parameters. The specific TPK update retains its own rule for selecting these parameters and their times.

This gives a precise connection: an observable conserved by the complete dynamics is distributed between readout and memory through (25.1). When this observable is the generator of a continuous symmetry of the constructed action, the distributed quantity is a Noether charge. The norm, the symmetry charge, and the energy functional are related through their specific operators.

An effective bounded specialization is obtained from the 108-state calendar. Its realized Hamiltonian is $H_{\text{clk}} = \hbar \omega_0 N_{108}$, with $N_{108} = \sum_{k=0}^{107} k P_k$ and P_k its Fourier projectors. The clock action has charge $\hbar \langle \psi, N_{108} \psi \rangle = E_{\text{clk}}/\omega_0$. Transports and inclusions satisfying the hypotheses of (25.1) distribute this charge between readout and complement. All operators here are finite and bounded. The elliptic action of (25.9) has been proved in its classical symplectic realization; a quantization with unbounded observables additionally retains its own domain conditions.

26 Recovery of the state and its dynamics

26.1 Recoverable representation of transport

Let C be the unitary realization of memory transport on $\mathcal{H}$. The nine-phase chart, its uniform inclusion, and its compression are

$$S_C(v_0, \dots, v_8) = (Cv_8, v_0, \dots, v_7), \quad Jv = (v, \dots, v)/3, \quad T = (8I + C)/9.$$

Let $\eta = (I - JJ^*)S_CJ$. Its action is

$$\eta v = \frac{1}{27}(8(C - I)v, -(C - I)v, \dots, -(C - I)v).$$

Normalization of nine copies determines $1/3$; the distribution of the seam determines $8/81$ in

$$I - T^*T = \eta^*\eta = \frac{8}{81}(I - C)^*(I - C). \quad (26.1)$$

Write $P = \text{proj ker}(I - C)$. The preceding theorem proves strong convergence $T^N \rightarrow P$. Therefore,

$$\mathcal{V}v = (Pv, \eta v, \eta T v, \eta T^2 v, \dots) \quad \text{in} \quad \mathcal{R} = P\mathcal{H} \oplus \ell^2(\mathbb{N}_0; \mathcal{H}^9) \quad (26.2)$$

is an isometry. Denote its closed image by $\mathcal{M}$.

On the record space, define the block shift

$$\mathcal{B}(a, m_0, m_1, m_2, \dots) = (a, m_1, m_2, \dots). \quad (26.3)$$

The first component preserves the fixed sector; the remaining components are the complete vector records, including their correlations.

Theorem 26.1. *The original transport has the exact representation*

$$\boxed{\mathcal{V}C = (9\mathcal{B} - 8I)\mathcal{V}}. \quad (26.4)$$

The restriction $\mathcal{C} = (9\mathcal{B} - 8I)|_{\mathcal{M}}$ is unitary on $\mathcal{M}$, and

$$C = \mathcal{V}^*\mathcal{C}\mathcal{V}. \quad (26.5)$$

Proof. Since $PT = P$, definition (26.2) gives, componentwise,

$$\mathcal{V}T v = (Pv, \eta T v, \eta T^2 v, \dots) = \mathcal{B}\mathcal{V}v.$$

The algebraic identity $C = 9T - 8I$ implies (26.4). The isometry $\mathcal{V} : \mathcal{H} \rightarrow \mathcal{M}$ is surjective by definition of its image. Equality (26.4) identifies the restriction with $\mathcal{V}C\mathcal{V}^{-1}$, which is unitary because C is unitary. This also proves $\mathcal{C}\mathcal{M} = \mathcal{M}$ and (26.5). $\square$

Restriction to the subspace of compatible records is essential: $9\mathcal{B} - 8I$ on the whole of $\mathcal{R}$ is not unitary. The theorem characterizes transport on records effectively arising from the same linear trajectory of the documented chart.

Corollary 26.2. *For $v, w \in \mathcal{H}$ and $k \in \mathbb{Z}$,*

$$\langle C^k v, w \rangle = \langle \mathcal{C}^k \mathcal{V}v, \mathcal{V}w \rangle. \quad (26.6)$$

All cross moments of the realized dynamics are thus preserved. Negative powers belong to the inverse of $\mathcal{C}$ on $\mathcal{M}$, rather than to an inversion of the unilateral shift on the ambient space.

Corollary 26.3. *For every bounded observable A , its realization on the image is $\hat{A} = \mathcal{V}A\mathcal{V}^*|_{\mathcal{M}}$. Products, adjoints, and commutators are preserved. In particular,*

$$[A, C] = 0 \iff [\hat{A}, \mathcal{C}] = 0. \quad (26.7)$$

The proof uses $\mathcal{V}^*\mathcal{V} = I$ and the fact that $\mathcal{V}\mathcal{V}^*$ is the identity on $\mathcal{M}$. Symmetry conservation is expressed through complete transport of its operators, in addition to an equality of norms.

26.2 Explicit reconstruction and record depth

Let $m_j = \eta T^j v$. The telescopic identity gives, for each N ,

$$v = T^{*N} T^N v + \sum_{j=0}^{N-1} T^{*j} \eta^* m_j. \quad (26.8)$$

Passing to the strong limit,

$$v = Pv + \sum_{j=0}^{\infty} T^{*j} \eta^* m_j. \quad (26.9)$$

Partial reconstruction with fixed terminal component, $v_N = Pv + \sum_{j < N} T^{*j} \eta^* m_j$, has the exact error

$$v - v_N = T^{*N} T^N (I - P)v. \quad (26.10)$$

The series converges for every state. If the complete record has an error δr measured in the norm of $\mathcal{R}$, its reconstruction through $\mathcal{V}^*$ changes by at most $\|\delta r\|$ in norm, because $\|\mathcal{V}^*\| = 1$. This is uniform stability of the complete readout.

26.2.1 Difference between finite and unlimited depth

For the bilateral shift $Ce_k = e_{k+1}$ on $\ell^2(\mathbb{Z})$, one has $P = 0$. Consider

$$v_L = L^{-1/2} \sum_{k=1}^L e_k.$$

Then $\|(I - C)v_L\|^2 = 2/L$. Commutation of T with $I - C$, contractivity of T , and (26.1) imply

$$\sum_{j=0}^{N-1} \|\eta T^j v_L\|^2 \leq \frac{16N}{81L}. \quad (26.11)$$

For any fixed depth N , the right-hand side tends to zero as $L \rightarrow \infty$, although $\|v_L\| = 1$. The fixed-depth memory record, separated from its terminal component, has no uniform lower bound. By contrast, the unlimited record has norm exactly one for each v_L .

The difference corresponds to two distinct quantifiers: every state is preserved upon completion of its record; a prescribed depth does not uniformly recover every state without its terminal component. Conservation at arbitrary depth here has an exact functional expression.

26.2.2 Vector information and intensity

The theorem preserves m_j , not merely $\|m_j\|^2$. For example, in the bilateral realization, $v = e_0$ and $w = e_1$ have identical intensities at every depth because the shift commutes with T and transports η unitarily. Their vector records differ, and (26.9) recovers the two distinct states. Intensities serve the balance; records with their correlations additionally serve reconstruction.

26.3 Naturality of recovery under refinement

Let $\iota_n : \mathcal{H}_n \rightarrow \mathcal{H}_{n+1}$ be the refinement isometries obtained from the common emissions, and let C_n be unitary realizations with

$$C_{n+1}\iota_n = \iota_n C_n. \quad (26.12)$$

The relation also holds for adjoints: multiplication by C_{n+1}^* and C_n^* gives $C_{n+1}^*\iota_n = \iota_n C_n^*$. The images are therefore reducing. The polynomial formulas for T_n , η_n , and the strong limit of T_n^N give

$$T_{n+1}\iota_n = \iota_n T_n, \quad \eta_{n+1}\iota_n = \iota_n^{\oplus 9} \eta_n, \quad P_{n+1}\iota_n = \iota_n P_n. \quad (26.13)$$

Let $\mathfrak{I}_n$ be the refinement applying ι_n to the terminal component and $\iota_n^{\oplus 9}$ to every record. Then

$$\boxed{\mathcal{V}_{n+1}\iota_n = \mathfrak{I}_n \mathcal{V}_n, \quad \mathcal{B}_{n+1}\mathfrak{I}_n = \mathfrak{I}_n \mathcal{B}_n.} \quad (26.14)$$

Proof. Each component of the first equality is an identity from (26.13), iterated to the corresponding power of T_n . The second equality states that refining every block commutes with shifting the block index, as verified directly. Both maps are bounded, so the equalities pass from finitely supported records to their completion. $\square$

By (26.4) and (26.14), reconstruction of the dynamics also commutes with refinement. The genealogical-depth index n and the successive-compression index j remain distinct. The equality expresses their compatibility when they act through the documented squares, rather than identifying them by name.

26.4 Two exact regimes of memory distribution

For a unit state v with $Pv = 0$, define

$$a_N = \|T^N v\|^2, \quad b_N = \|\eta T^N v\|^2 = a_N - a_{N+1}. \quad (26.15)$$

One has $a_0 = 1$, $a_N \rightarrow 0$, $b_N \geq 0$, and $\sum b_N = 1$. The weights b_N describe the distribution of the memory record across depths. This index counts compressions; identifying it with a physical duration would require application of the corresponding temporal reader.

26.4.1 Exceptional transport of order three

The fixed-point-free orthogonal operator g_c satisfies $g_c^2 + g_c + I = 0$. For the explicit composition $C = g_c$, the preceding development obtained

$$T^*T = \frac{19}{27}I, \quad a_N = \left(\frac{19}{27}\right)^N, \quad b_N = \frac{8}{27} \left(\frac{19}{27}\right)^N. \quad (26.16)$$

The mean depth, counting the first record as depth one, is

$$\sum_{N \geq 0} (N+1)b_N = \frac{27}{8}. \quad (26.17)$$

This is a result of this composition of the nonadic chart with the already constructed exceptional transport; it preserves the domain and does not identify g_c with complete temporal holonomy merely from its order.

26.4.2 Bilateral record of revolutions

For $Ce_k = e_{k+1}$ and $v = e_0$, the binomial formula determines every term:

$$T^N e_0 = 9^{-N} \sum_{k=0}^N \binom{N}{k} 8^{N-k} e_k, \quad a_N = 81^{-N} \sum_{k=0}^N \binom{N}{k}^2 64^{N-k}. \quad (26.18)$$

This formula is exact at any depth. The subsequent circular representation gives

$$a_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{65 + 16 \cos \theta}{81} \right)^N d\theta. \quad (26.19)$$

Theorem 26.4. *In this bilateral realization,*

$$\boxed{a_N \sim \frac{9}{4\sqrt{2\pi}} N^{-1/2}, \quad b_N \sim \frac{9}{8\sqrt{2\pi}} N^{-3/2}.} \quad (26.20)$$

Proof. Write $r(\theta) = (65 + 16 \cos \theta)/81$ and $q = 8/81$. Near zero,

$$r(\theta) = 1 - q\theta^2 + O(\theta^4), \quad \log r(\theta) = -q\theta^2 + O(\theta^4).$$

For a sufficiently small fixed neighbourhood, positive constants c_1, c_2 bound $-\log r(\theta)$ between $c_1\theta^2$ and $c_2\theta^2$. Outside this neighbourhood, $r \leq \rho < 1$. The substitution $x = \sqrt{N}\theta$, the Gaussian bound, and dominated convergence give

$$\lim_{N \rightarrow \infty} \sqrt{N} a_N = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-qx^2} dx = \frac{1}{2\sqrt{\pi q}} = \frac{9}{4\sqrt{2\pi}}.$$

For the second limit, apply the same procedure directly to

$$b_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} r(\theta)^N (1 - r(\theta)) d\theta.$$

In the variable x , $N(1 - r(x/\sqrt{N})) \rightarrow qx^2$, locally dominated by a constant times x^2 ; outside the neighbourhood, the contribution is exponentially small even after multiplication by $N^{3/2}$. Therefore,

$$\lim_{N \rightarrow \infty} N^{3/2} b_N = \frac{q}{2\pi} \int_{\mathbb{R}} x^2 e^{-qx^2} dx = \frac{1}{4\sqrt{\pi q}} = \frac{9}{8\sqrt{2\pi}}.$$

This proof of the second limit uses its own integral, without differentiating an uncontrolled asymptotic equivalence. $\square$

By Tonelli's theorem and the tail-sum identity for a positive sequence,

$$\sum_{N \geq 0} (N+1) b_N = \sum_{N \geq 0} a_N = +\infty. \quad (26.21)$$

The entire norm is preserved in memory, but its mean depth diverges for this bilateral state. The distribution law depends on the effective transport representation: the exceptional sector of order three has a geometric rate, whereas the bilateral record considered here has an algebraic rate.

26.5 Composition with Euler's identity and incidence

The extended Euler identity of Section 6.14 realizes the three regimes through the Coxeter operator of A_2 , a nilpotent transport, and F_{av}^2 , with their stated antecedents. Their normalized generators satisfy $J_\tau^2 = -\tau I$ and have zero trace. In their two-dimensional charts,

$$C_\tau(t) = \exp(tJ_\tau), \quad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad C_\tau(t)^\top \Omega C_\tau(t) = \Omega.$$

Zero trace gives $\det C_\tau(t) = 1$, and the two-dimensional identity $M^\top \Omega M = (\det M)\Omega$ proves the final equality. For

$$T_\tau = \frac{8I + C_\tau}{9}, \quad D_\tau = \frac{\sqrt{8}}{9}(C_\tau - I),$$

one obtains a common oriented conservation law:

$$\boxed{\Omega = T_\tau^\top \Omega T_\tau + D_\tau^\top \Omega D_\tau.} \quad (26.22)$$

Proof. Expanding the two terms, the cross contributions of C_τ have opposite coefficients. The remaining terms are $72\Omega/81 + 9C_\tau^\top \Omega C_\tau/81 = \Omega$. $\square$

This alternating form corresponds to oriented area in the canonical plane. Its preservation permits comparison of the three regimes while respecting their differences. For the concrete representatives given by the Coxeter operator C , $I + N$, and F_{av}^2 , respectively,

$$(\det T, \det D) = \left(\frac{19}{27}, \frac{8}{27}\right), \quad (1, 0), \quad \left(\frac{89}{81}, -\frac{8}{81}\right). \quad (26.23)$$

The sum is one in all three cases. In the parabolic case, D has rank one and may preserve information although its area is zero. In the hyperbolic case, the complement has the opposite orientation; the balance concerns signed area. Theorem 26.1 uses a unitary, positive realization: the hypotheses of each result are expressly retained.

The action and varying-form development proves the version between different charts, its integration over closed curves, and its compatibility with refinement. The incidence development composes reconstruction with $F(k) = (\mu(k), \mathcal{P}_0 k/6)$, retaining all twelve coordinates of K ; it also transports operators and their local frames. The same conservation identity thus admits typed realizations of norm, incidence, and oriented action.

27 Exceptional incidence and transport of forms

27.1 Generation of the record and conservation of incidence

27.1.1 Dodecaphasic record of the enriched history

Section 9.2 defines

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

It preserves the window, subpath, orientation, carry, and incidence ledger. Section 9.6 fixes

$$x_{\text{term}} = (\mathcal{U}_{108} \cdots \mathcal{U}_1)x_{\text{seed}}, \quad \mathcal{U}_t = \text{Upd}_t \text{Tra}_t \text{Sel}_t,$$

$$R_{\text{sgn}} = \Pi_H \mathcal{E}_{108}^{90,120} \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \rightarrow \mathbb{Z}^{12}.$$

The following algebraic reconstruction begins with this typed output. It does not replace evaluation of the preceding events or obtain their coordinates by imposing K or alpha as a target.

27.1.2 Integral normalization and isometric normalizations

For the step-three orbits (1, 4, 7, 10), (2, 5, 8, 11), (3, 6, 9, 12), Lemma 9.10 fixes the symmetric matrix $H = H_{12}$ and proves $H^2 = 4I$.

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : Hu \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}Hu, \quad \mathcal{K}^{-1}(k) = Hk.$$

This is an integral isomorphism between the sublattice of signed records and $\mathbb{Z}^{12}$. The factor 1/4 implements inversion, not a Euclidean isometry.

As an immediate consequence of the same identity, $J_H = H/2$ satisfies $J_H^* J_H = I$: this is the real isometric transport. If $u = Hk$, then $u/2 = J_H k$ and $\|u\|^2 = 4\|k\|^2$. Confusing $H/4$ with $H/2$ would change the balance by a factor of four.

In section 16.4 and proposition 16.8, the constructed design determines

$$I : \mathbb{R}^{12} \rightarrow \mathbb{R}^{132}, \quad (Iv)(H') = \sum_{p \in H'} v_p, \quad I^* I = 36I_{12} + 30J_{12}.$$

The coefficients arise from the incidences: each point belongs to 66 hexads, and each pair to 30. On $V = \mathbf{1}^\perp$, the term J_{12} vanishes, so $U_W = I/6 : V \rightarrow E_{\text{inc}}$ is an isometry. The factor 1/6 normalizes this screen; it is not the inverse map for K .

27.1.3 Prefix refinement and event partitions

Section 16.1 constructs causal frames $f_{j+1} = g_j(x)f_j$ and prefix encodings Q_n . It proves

$$r_{n,m}Q_n = Q_m p_{n,m}, \quad \gamma_{A'}(wf^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}).$$

The proof uses the fact that a prolongation preserves preceding words, transports, and frames. The inverse limit produces Q_∞ . Recovery of the visible word follows from the first component of γ ; the text expressly distinguishes this scope from recovery of all deep memory and passage to supports.

The additive and affine composition of the record, described in section 9.2 and proposition 5.1, proves a second conservation property: with compatible prospective contributions $(\delta w_t, \delta q_t)$, record inversion is additive. Concatenating segments or subdividing the same list of events gives the same final record. The version with nontrivial transport uses the affine cocycle. Preserving the total under a change of partition does not assert that the record stops growing when new events are added.

27.2 K-Witt transport of norms, operators, and balances

A formalization of the recovered results is assembled here, together with the algebraic proof below. This assembly attributes no new priority to its starting identities.

Let $P_0 = I - J_{12}/12$, $V = P_0 \mathbb{R}^{12}$, $W_H = J_H V$, and

$$B := U_W J_H|_{W_H} : W_H \longrightarrow E_{\text{inc}}.$$

Since $J_H^2 = I$, this is an isometric isomorphism between eleven-dimensional spaces. For a record k , its normalized signed coordinate is $J_H k$. If $\hat{P}_0 = J_H P_0 J_H$,

$$B \hat{P}_0(J_H k) = U_W P_0 k,$$

$$\|\hat{P}_0(J_H k)\|^2 = \|U_W P_0 k\|^2 = \sum_{p=1}^{12} k_p^2 - \frac{(\sum_p k_p)^2}{12}.$$

No constant value has been introduced to obtain this equality: it follows from H^2 and the incidence Gramian. The removed mean must be retained separately to recover the whole of k .

For $g \in \text{Aut}(\mathcal{H})$, let $\rho_{12}(g)$ be its point permutation and $\rho_{\text{hex}}(g)$ its hexad permutation. The action on signed coordinates is

$$\hat{\rho}_H(g) = J_H \rho_{12}(g) J_H = \frac{1}{4} H \rho_{12}(g) H.$$

It is orthogonal, preserves W_H , and also preserves the sublattice $\mathcal{L}_H$: if $u = Hk$, its image is $H\rho_{12}(g)k$. The explicit intertwining in section 16.4 gives

$$B \hat{\rho}_H(g) = \rho_{\text{hex}}(g) B.$$

Proof. Substitute the definitions, cancel J_H^2 , and apply $U_W \rho_{12}(g) = \rho_{\text{hex}}(g) U_W$. The integral record, centred inner product, and incidence action are thus preserved, each on its own domain. $\square$

For an operator A already constructed on W_H , one obtains $A_{\text{inc}} = BAB^*$. By substitution,

$$BA = A_{\text{inc}} B, \quad \langle z, Az \rangle = \langle Bz, A_{\text{inc}} Bz \rangle.$$

Consequently, if a transport C on W_H preserves A , so that $C^* A C = A$, its transport $C_{\text{inc}} = B C B^*$ preserves A_{inc} . The condition that a particular C be a readout of TPK dynamics remains part of the causal composition producing it. The isometry alone does not turn an incidence permutation into the nonadic cycle.

Commutation need not be imposed to compare transported charts. If R is unitary, $C' = R C R^*$, and $T(C) = (8I + C)/9$, then $T(C')R = R T(C)$ by substitution. Together with $A' = R A R^*$, the balance in one chart is conjugated to that in the other. Commutation $[R, C] = 0$ concerns only the additional problem of keeping that same seam C fixed. This distinction agrees with the lattice naturality of Section 27.3.

One available concrete finite action is the star selected by the flag:

$$\sigma_{B_K} = (1\ 3)(2\ 11)(5\ 7)(8\ 12).$$

Section 16.6 proves its construction and distinguishes it from the group generated by the 495 stars. The individual star gives C_2 ; the collection gives M_{12} . Transporting the mark and origin gives covariance between frames. For a symmetry of a fixed frame, the appropriate domain is its stabilizer, rather than automatically the whole of M_{12} . If it must additionally commute with C , one uses the subgroup satisfying the concrete equation $[\hat{\rho}_H(g), C] = 0$.

27.3 Order-three action on the lattice fibres

The construction in sections 16.11 and 16.12 produces twelve A_2 fibres labelled by $(i, \mu, \varepsilon) \in \mathbb{Z}/3\mathbb{Z} \times \{0, 1\} \times \{0, 1\}$. The alignment $\lambda(i, \mu, \varepsilon) = 4i + 2\mu + \varepsilon \pmod{12}$ is bijective. If C_{A_2} is the Coxeter operator and R the reflection satisfying $RC_{A_2}R = C_{A_2}^{-1}$, the trivializations and transported frames fix

$$g_b = \iota_b^{-1} P_b C_{A_2} P_b^{-1} \iota_b, \quad T_{i\mu\varepsilon} = \iota_{\lambda(i, \mu, \varepsilon)}^{-1} P_{\lambda(i, \mu, \varepsilon)} R^\mu,$$

$$g_\lambda T_{i\mu\varepsilon} = T_{i\mu\varepsilon} C_{A_2}^{(-1)^\mu}.$$

The direct sum $T_\lambda : W_{\text{APP}} \otimes \mathbb{C} \rightarrow \bigoplus_{b=1}^{12} (A_{2,b} \otimes \mathbb{C})$ is a C_3 -equivariant isomorphism. The proof consists of conjugation and the bijection of the twelve labels: it preserves pair, sheet, and orbit. It is not a coincidence of dimensions.

The constructions in equation (16.33) and proposition 16.21 use $q^\star = (1, 1, 1, 1, 1)$ and its full-support word $c^\star = (q^\star, q^\star A_W)$ to orient the twelve blocks. The resulting g_c has order three, has no nonzero fixed vectors, and preserves the Niemeier gluing. For the incidence vector v of squared norm 54,

$$N_0 = \ker(\langle \cdot, v \rangle \bmod 3), \quad N_v = N_0 + \mathbb{Z} \frac{v}{3},$$

$$y_* = (g_c v - v)/3 \in N_0, \quad z_* = (g_c^{-1} v - v)/3 \in N_0.$$

The proof identifies their discriminant classes as $c^\star$ and $-c^\star$, and their pairings with v as -27 . Hence $g_c N_0 = N_0$ and $g_c(v/3) = v/3 + y_*$, giving $g_c N_v = N_v$. The preserved structure includes the gluing, congruence kernel, and class generating the Leech neighbour.

Prolongation to the 24 full-support orientations and the lattice lift of a monomial code automorphism h are developed in Section 16.21. With all marks transported,

$$g_{hu} = \tilde{h} g_u \tilde{h}^{-1}.$$

This C_3 is an admissible group genuinely intertwined with the APP fibres and exceptional structure. It is not identified here with C_9 or its quotient by a mere relation of orders. Nor is the lattice lift replaced by the ternary matrix without its action on the A_2 fibres.

27.4 Refinement of energy and its differential

The energy operator of (50.1) has the form $D_{9,m}^* W_m D_{9,m}$, with U_a transporting orientation and carry. The domains in section 50.1 are

$$\mathcal{H}_m = \bigoplus_v H_v, \quad \mathcal{K}_m = \bigoplus_{a \in \mathcal{E}_m^+} H_{t(a)}, \quad (D_m f)_a = f_{t(a)} - U_a f_{s(a)}, \quad E_m = \langle D_m f, W_m D_m f \rangle.$$

The fixed maps J_m and K_m satisfy, for the C^1 collection under consideration,

$$D_{m+1}(t)J_m = K_m D_m(t), \quad K_m^* W_{m+1}(t)K_m = W_m(t).$$

The result in theorem 50.5 proves $E_{m+1}(J_m f) = E_m(f)$ by substitution, and equality of differentials follows by differentiating both identities. For parameter-dependent refinement, the term $D_{m+1} \dot{J}_m f$ is also retained.

With $v_a = U_a f_s$ and $q = W_m D_m f$, the response to $\dot{U}_a = A_a U_a$ is

$$\mathbf{j}_m(A) = -2 \operatorname{Re} \sum_a \langle q_a, A_a v_a \rangle, \quad \mathcal{J}_a = v_a q_a^* - q_a v_a^*.$$

The transport product $U_\gamma = U_N \cdots U_1$ satisfies

$$\dot{U}_\gamma U_\gamma^{-1} = \sum_{k=1}^N B_k A_k B_k^{-1}, \quad B_k = U_N \cdots U_{k+1}.$$

This formula retains the order of each incidence and transports its covector contragrediently. Corollary 51.5 composes this differential with connection variations and proves, for compatible refinements of the action,

$$s_m = P_m^\top s_{m+1}, \quad M_m \sigma_m = P_m^\top M_{m+1} \sigma_{m+1}.$$

Here, P_m carries connection variations, and M_m is the cellular pairing; they are neither the mean-zero projector nor the Hadamard matrix.

27.4.1 Assembly with incidence

In a realization of fibres W_H whose frame actions are $\hat{\rho}_H(g_a)$, conjugation by B carries every transport, field, and weight to E_{inc} . By the intertwining proved in Section 27.2, the difference operators satisfy $D_{\text{inc}} B_{\text{vertices}} = B_{\text{edges}} D_H$. Transporting J_m , K_m , and W_m as well, the two preceding refinement squares remain the same conjugated squares. By substitution, they preserve energy; for fixed B , they also preserve the differential and covectors.

The balance law and its covectors are transported to the exceptional representation while retaining compatibility with refinement. Each map uses the transports of the specified dynamics and the subspaces they preserve. The variation domain is defined in section 50.1.

28 Incidence reconstruction and covariance

28.1 The complete isometry and its domain

Let $H = \mathbb{R}^{12}$, with the Euclidean inner product; let $P_0 = I - \mathbf{1}\mathbf{1}^*/12$, $\mu(k) = \mathbf{1}^*k/12$, and let $\mathcal{J} : H \rightarrow \mathbb{R}^{132}$ be the incidence matrix of section 16.4 and proposition 16.8. Write $E_{\text{inc}} = \mathcal{J}(\mathbf{1}^\perp)$ and

$$Y = \mathbb{R} \oplus E_{\text{inc}}, \quad \langle (\mu, y), (\nu, z) \rangle_Y = 12\mu\nu + \langle y, z \rangle.$$

The complete isometry constructed in Section 25 is

$$F : H \longrightarrow Y, \quad Fk = (\mu(k), \mathcal{J}P_0k/6),$$

$$F^{-1}(\mu, y) = \mu\mathbf{1} + P_0\mathcal{J}^*y/6.$$

The Gram identity $\mathcal{J}^*\mathcal{J} = 36I + 30\mathbf{1}\mathbf{1}^*$ proves both inverse compositions and $F^*F = I_H$, $FF^* = I_Y$, with the indicated weighted inner product. Hence F is orthogonal between H and Y . It can be complexified without changing the proofs.

The codomain is the twelve-dimensional space Y , not the whole of $\mathbb{R} \oplus \mathbb{R}^{132}$. Extending an operator from Y to its complement in that ambient space requires an additional choice; this is not part of the uniquely transported dynamics. In normalized signed coordinates, one uses FJ_H , with $J_H = H_{12}/2$. Integral inversion remains $H_{12}/4$ on its sublattice, rather than an alternative metric normalizer.

28.2 Covariant lifting theorem on the image

Theorem 28.1. *For a bounded operator D on H , there exists a unique operator on Y satisfying $D_Y F = F D$. It is*

$$D_Y = F D F^{-1}.$$

The correspondence preserves sums, products, adjoints, norm, positivity, and identity. In particular, it maps unitaries to unitaries and self-adjoint observables to self-adjoint observables. For a unitary C and an observable A ,

$$C^*AC = A \iff C_Y^*A_YC_Y = A_Y.$$

If R is a unitary chart change, $R_Y = F R F^{-1}$, then

$$F(R D R^{-1})F^{-1} = R_Y D_Y R_Y^{-1}.$$

Proof. The first formula follows by multiplying the intertwining relation by F^{-1} . Uniqueness follows from surjectivity onto Y . The product and adjoint identities follow from $F^{-1} = F^*$, and equality of norms follows from the surjective isometry. The final equality is an associative composition of these maps. No commutation hypothesis between R and D enters. $\square$

For $g \in \text{Aut}(\mathcal{H})$, Proposition 16.8 supplies

$$F\rho_{12}(g) = (1 \oplus \rho_{\text{hex}}(g))F.$$

Thus, the chart change has a concrete combinatorial action on the hexads. For a general unitary D , however, D_Y is an operator of the incidence representation, not necessarily a permutation of the design. Preserving the inner product of that representation does not make all its unitaries combinatorial automorphisms or elements of the Monster.

28.2.1 Covariance under a noncommuting chart change

Let C be the cyclic advance $Ce_p = e_{p+1}$ on the twelve positions, namely the reverse direction of the convention S in section 9. This is an operation of the window chart, not complete temporal holonomy. Let R be the already constructed star

$$R = (1\ 3)(2\ 11)(5\ 7)(8\ 12).$$

Then $RCE_1 = e_{11}$, whereas $CRE_1 = e_4$. They do not commute. Nevertheless, for $C' = RCR^{-1}$, the identities $T(C')R = RT(C)$ and $\eta(C')R = R^{\oplus 9}\eta(C)$ hold exactly, as do their versions on Y through F . Chart covariance therefore works in an example where commutation in a fixed chart fails.

28.3 Explicit recovery from the incidence record of complements

Let C be unitary, $T = (8I + C)/9$, $\eta = (I - JJ^*)S_C J : H \rightarrow H^9$, and let P be the projection onto $\text{Fix}(C)$, as in Section 25. Write $F_9 = F^{\oplus 9}$. The incidence version of the finite record is

$$\mathcal{W}_N k = (FT^N k, F_9 \eta k, F_9 \eta T k, \dots, F_9 \eta T^{N-1} k).$$

Theorem 28.2. *This map is isometric and has the explicit left inverse*

$$\mathcal{W}_N^*(a, b_0, \dots, b_{N-1}) = T^{*N} F^{-1} a + \sum_{j=0}^{N-1} T^{*j} \eta^* F_9^{-1} b_j.$$

In particular, applying this formula to $\mathcal{W}_N k$ recovers k exactly. The infinite record

$$\mathcal{W}_\infty k = (FPk, (F_9 \eta T^j k)_{j \geq 0})$$

is also isometric, and its left inverse is

$$k = PF^{-1} a + \sum_{j \geq 0} T^{*j} \eta^* F_9^{-1} b_j \quad \text{for } (a, (b_j)) = \mathcal{W}_\infty k.$$

The series converges in norm. For arbitrary square-summable data, the same formula defines the bounded adjoint, although not every datum in the codomain belongs to the image of $\mathcal{W}_\infty$.

Proof. Applying F and F_9 to each component of V_N or V_∞ preserves the inner product. Taking the adjoint of this composition gives the formulas. The telescopic identity in Section 25 proves $\mathcal{W}_N^* \mathcal{W}_N = I$, and its strong limit proves $\mathcal{W}_\infty^* \mathcal{W}_\infty = I$. Convergence of the series follows by applying a bounded operator to truncations of a sequence in a Hilbert direct sum. $\square$

This reconstruction does not preserve k as an intact first coordinate: the first component is the compressed terminal component or its fixed sector; the rest arises from successive complements. When $P = 0$, the incidence complements alone recover k . The contribution recovered from the first N blocks is

$$k_N = \sum_{j < N} T^{*j} \eta^* F_9^{-1} b_j = (I - T^{*N} T^N) k.$$

The remainder is $T^{*N} T^N k$, which tends to zero if $P = 0$. With a nonzero fixed sector, FPk must be retained; complement memory alone does not recover that sector. Reconstruction rates and dynamics on the block index are developed in Section 26.

If A is self-adjoint and $[A, C] = 0$, the observable form is also preserved:

$$\langle k, Ak \rangle = \langle FPk, A_Y FPk \rangle_Y + \sum_{j \geq 0} \langle F_9 \eta T^j k, (I_9 \otimes A_Y) F_9 \eta T^j k \rangle.$$

The scalar series converges absolutely because A is bounded and the sum of squared component norms is finite. The hypothesis expresses preservation of A by C ; it does not require C to commute with exceptional chart changes.

28.4 Natural lifting of refinement with local frames

The construction in sections 5.1 and 5.2 supplies sets Z_n of enriched cylindrical states, truncations, and projective measures with full support. For a descendant z' of z ,

$$w_n(z'|z) = \sqrt{\mu_{n+1}(z')/\mu_n(z)}, \quad \sum_{\tau z'=z} w_n(z'|z)^2 = 1.$$

The label z' preserves the ordered record, orientation, carry, boundary, memory, path, and terminal component. Refinement of a state does not identify distinct descendants.

Let $q(z)$ be the integer memory record, C a unitary on H , and R_z orthogonal changes of the fibre frame. For concrete incidence frames, one may use $R_z = \rho_{12}(g_z)$, with $g_z \in \text{Aut}(\mathcal{H})$, transporting their marks. Define

$$G_z = R_z C^{q(z)}, \quad U_{z',z} = G_{z'} G_z^* = R_{z'} C^{q(z')-q(z)} R_z^*.$$

This construction prolongs the explicit gauge equivalence of section 16.1. It does not require $[R_z, C] = 0$ and keeps the order of operations explicit.

On $\mathcal{H}_n = \ell^2(Z_n) \otimes H$,

$$\mathcal{U}_n(e_z \otimes v) = \sum_{\tau z'=z} w_n(z'|z) e_{z'} \otimes U_{z',z} v.$$

Define $\mathcal{F}_n = I_{\ell^2(Z_n)} \otimes F$, with codomain $\mathcal{Y}_n = \ell^2(Z_n) \otimes Y$.

Theorem 28.3. *The refinement is isometric and satisfies*

$$\mathcal{U}_n^Y \mathcal{F}_n = \mathcal{F}_{n+1} \mathcal{U}_n, \quad \mathcal{U}_n^Y = \mathcal{F}_{n+1} \mathcal{U}_n \mathcal{F}_n^{-1}.$$

The formula for each term is the preceding one with $U_{z',z}^Y = F U_{z',z} F^{-1}$. Systems at several depths preserve composition and admit a surjective isometry between their inductive limits. All fibre coefficients are preserved exactly with their labels z .

Proof. Descendants of distinct predecessors are orthogonal; each $U_{z',z}$ is unitary; and the squared weights sum to one. This proves isometry. Along a trajectory, the intermediate factors $G_z^* G_z$ cancel and the measure ratios telescope. Composition does not depend on the partition of the trajectory. The square with F is verified on $e_z \otimes v$, term by term. The maps and their inverses are compatible isometries, so they extend as mutual inverses to the inductive completions. $\square$

28.4.1 Compatible observables and dynamics

If A_0 is a bounded fibre observable, let

$$(\mathcal{A}_n \psi)_z = G_z A_0 G_z^* \psi_z.$$

Then $\mathcal{A}_{n+1}\mathcal{U}_n = \mathcal{U}_n\mathcal{A}_n$, whence $\mathcal{U}_n^*\mathcal{A}_{n+1}\mathcal{U}_n = \mathcal{A}_n$. This follows by substituting $U_{z',z} = G_{z'}G_z^*$. It is conservation of an observable field transported by refinement; it does not assert that A_0 commutes with C in a fixed chart.

The unitary dynamics at each level,

$$(\mathcal{C}_n\psi)_z = G_z C G_z^* \psi_z$$

satisfies $\mathcal{C}_{n+1}\mathcal{U}_n = \mathcal{U}_n\mathcal{C}_n$. Their adjoints are also intertwined: since both $\mathcal{C}$ are unitary, multiply the identity by their inverses. Thus, T , its nonadic complements, and its fixed projections are intertwined. The finite and infinite records of Section 28.3 are natural with respect to this refinement. Upon applying $\mathcal{F}_n$, the same squares hold in the incidence fibres. For $\mathcal{A}_n$ also to be conserved by these dynamics at a fixed level, the precise condition is $[A_0, C] = 0$; covariance between frames still does not require $[R_z, C] = 0$.

28.4.2 Bilateral-memory specialization with varying frames

On the reversible orbit of section 26, a coordinate readout may be represented by $\psi = (\psi_m)_{m \in \mathbb{Z}} \in \ell^2(\mathbb{Z}; H)$. Its advance is $(C_0\psi)_m = \psi_{m-1}$. The map F acts pointwise. For frames $R_m = \rho_{12}(g_m)$, the unitary change $(\mathcal{R}\psi)_m = R_m\psi_m$ produces

$$(C'\psi)_m = R_m R_{m-1}^* \psi_{m-1}.$$

The incidence operator has the same formula with $1 \oplus \rho_{\text{hex}}(g_m g_{m-1}^{-1})$. Neither constant frames nor commutation with the shift is imposed. Since C' is conjugate to C_0 , it has no nonzero fixed vectors in this square-summable space. Section 28.3 recovers the complete linear readout from its incidence complements. This concerns the represented orbit and coordinates, rather than an identification of all TPK histories with $\mathbb{Z}$.

28.5 Direct extension to the twelve A_2 fibres

The exceptional construction preserves twelve A_2 fibres and the order-three operator g_c developed in Section 27.3. Let $V_{A_2} = A_2 \otimes_{\mathbb{Z}} \mathbb{R}$ be their two-dimensional Euclidean realization. The same positional isometry can be tensored over $\mathbb{R}$ with this space:

$$F_{A_2} = F \otimes I_{V_{A_2}} : \mathbb{R}^{12} \otimes_{\mathbb{R}} V_{A_2} \longrightarrow Y \otimes_{\mathbb{R}} V_{A_2}.$$

Its first component is the vector mean, and its second consists of the centred incidence sums, now valued in V_{A_2} . The incidence Gram operator, tensored with the identity, again proves the complete isometry and inverse. Hence $g_c^Y = F_{A_2} g_c F_{A_2}^{-1}$ is orthogonal, of order three, and fixed-point-free. The nonadic composition of Section 25 can be read out and recovered on this V_{A_2} -valued screen through Sections 28.2 and 28.3. The twelve scalars of K are not confused with the twenty-four coordinates of the A_2 fibres, nor is the action turned into a hexad permutation when the lift also acts within the fibres.

28.6 Exact scope with respect to histories

Let $k : X \rightarrow H$ be the record reader on a history domain, and let $\ell = F \circ k$. Then

$$\ell(x) = \ell(y) \iff k(x) = k(y).$$

The complete screen adds zero loss relative to K , but it does not change the fibres of this reader. A map $d : X \rightarrow X$ descends to dynamics on the values of K if and only if

$$k(x) = k(y) \implies k(dx) = k(dy).$$

The same condition is necessary and sufficient for descent to the incidence image. The proof consists of defining $D(k(x)) = k(dx)$; the condition is equivalent to independence of the representative. If this D is linear and unitary in the stated realization, Section 28.2 applies. If it is nonlinear, F still conjugates it as a map but does not make it unitary.

The criterion distinguishes descent to K from the enriched dynamics. Section 9.6 establishes that the reader uses subpath, orientation, carry, and incidences; Section 16.1 distinguishes recovery of the visible word, support, and deep memory. On $\ell^2(Z_n) \otimes H$, Section 28.4 preserves enriched labels and transports the fibre representation. The equivalence acts on the previously generated history domain.

For an already constructed cut, naturality of the record in proposition 9.2 fixes $K_N = K_{108} \circ \tau_{N,108}$: F preserves this naturality exactly. A sequence of new events may update K ; the additive or affine composition in section 9 preserves independence of partition, not constancy of the value when events are added.

29 Affine holonomy and memory symmetries

29.1 Effective composition of memory

For an admissible emission ϵ , the construction in proposition 5.1 defines

$$U_\epsilon(m) = C_\epsilon m + \kappa_\epsilon, \quad \kappa_{\epsilon_2 \epsilon_1} = \kappa_{\epsilon_2} + C_{\epsilon_2} \kappa_{\epsilon_1}. \quad (29.1)$$

The domain is the memory module at the source; the codomain is that of the next emission. The product preserves order. On a path γ of length r , one obtains

$$C_\gamma = C_r \cdots C_1, \quad \kappa_\gamma = \sum_{j=1}^r C_r \cdots C_{j+1} \kappa_j, \quad m_\gamma = C_\gamma m_0 + \kappa_\gamma. \quad (29.2)$$

The empty product in the sum is the identity. The proof is induction using (29.1): appending the next emission multiplies the preceding record by its linear transport and adds its increment. The positional boundary may cancel while the transported sum in (29.2) remains.

Proposition 29.1 (Memory class of a return). *Let M be an abelian module, and consider a path closed in the base whose memory transport is $H(m) = Cm + \kappa$, with C an automorphism of M . Its return class is*

$$\mathfrak{m}(H) = [\kappa] \in M/(I - C)M. \quad (29.3)$$

A memory-coordinate change $m' = Rm + b$, with R invertible, transforms the return into

$$C' = RCR^{-1}, \quad \kappa' = R\kappa + (I - C')b. \quad (29.4)$$

The map induced by R takes $[\kappa]$ to $[\kappa']$. The transport H has a fixed point exactly when its class (29.3) is zero.

Proof. Substituting $m = R^{-1}(m' - b)$ into H gives (29.4). Since $R(I - C) = (I - C')R$, the map R induces an isomorphism between the quotients. The term $(I - C')b$ represents zero in the target quotient. Finally, the fixed-point equation $Cm + \kappa = m$ is equivalent to $(I - C)m = \kappa$. This proves the final assertion. $\square$

Formula (29.3) determines whether a change of origin removes the increment or merely redistributes it among coordinates. In particular, when $C = I$, the class is the vector κ itself. A nonzero translation remains nonzero under every invertible affine change.

Proposition 29.2 (Return symmetry versus a change of representation). *An affine transformation $A(m) = Rm + b$ commutes with H exactly when*

$$RC = CR, \quad (I - C)b = (I - R)\kappa. \quad (29.5)$$

Proof. The linear parts of AH and HA are RC and CR ; their constant parts are $R\kappa + b$ and $Cb + \kappa$. Equating them gives (29.5). If H is merely transported through (29.4), commutation is not required: the same law is obtained in another representation. By contrast, a symmetry of the fixed return requires both equalities in (29.5). $\square$

This precisely distinguishes covariance of the law from the stabilizer of a specific memory. The distinction is relevant to studying variations in state symmetry without modifying the composition laws.

29.2 Application to the effective record of the observable revolution

Section 47 defines two ordered counters: g counts orientation reversals, and s counts effective sheet changes. Their canonical count per 27-step block is

$$c_{27} = (1, 18).$$

The complete observable phase returns after four such blocks. In the path groupoid, the record module is $M = \mathbb{Z}^2$, and

$$c_{108} = (4, 72) = 4v, \quad v = (1, 18), \quad H(g, s) = (g + 4, s + 72). \quad (29.6)$$

Identity (29.6) is a corpus result used here; the verifier checks its consequences rather than recounting the original emissions. The symbol c_{108} denotes this two-counter increment. It differs from the dodecaphasic seal K and from the central cocycle of the Heisenberg representation.

Theorem 29.3 (Complete record invariants under full revolutions). *The orbits of H and its inverse on $\mathbb{Z}^2$ are classified exactly by*

$$\boxed{\mathcal{I}(g, s) = s - 18g, \quad \nu(g, s) = [g]_4.} \quad (29.7)$$

Consequently,

$$\mathbb{Z}^2 / \mathbb{Z}(4, 72) \cong \mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}. \quad (29.8)$$

Proof. Adding $(4, 72)$ preserves both expressions in (29.7). If two records have the same invariants, then $g' - g = 4N$ for an integer N , and $s' - s = 18(g' - g) = 72N$; thus they differ by exactly $N(4, 72)$. The map in (29.7) is surjective: given an integer j and r modulo four, choose a representative g of r and set $s = j + 18g$. This proves (29.8). $\square$

The description covers every orbit of this memory reader. Orbits of the complete enriched state additionally preserve their remaining coordinates. For a forward-directed prolongation, a later record is obtained when the preceding integer N is nonnegative; the bilateral equivalence in (29.8) belongs to the groupoid.

The quantity $\mathcal{I}$ is the unique primitive integer linear charge, up to sign, invariant under these translations. Indeed, for $L(g, s) = ag + bs$, invariance requires $4a + 72b = 0$, so $(a, b) = b(-18, 1)$. A revolution accumulates longitudinal memory and preserves a transverse relation determined by the increment.

The factor four has a precise origin here: it is $\gcd(4, 72)$ and, in the preceding count, the number of 27-step blocks completing the observable return. The class ν preserves the information lost when the increment is divided by four and only the primitive direction v is retained.

Corollary 29.4 (Joint block and phase record). *Write the phase of these four blocks as $a \in \mathbb{Z}/4\mathbb{Z}$. The update*

$$(a, g, s) \longmapsto (a + 1, g + 1, s + 18)$$

preserves $(s - 18g, [g - a]_4)$. Both data classify its bilateral orbits: the difference $N = g' - g$ simultaneously satisfies $a' - a = [N]_4$ and $s' - s = 18N$. This brings together memory advance and phase return without replacing either by the other.

29.3 Parabolic generator determined by memory

The ordered pair of counters equips the module M with the unimodular alternating form

$$\Omega_M((g, s), (g', s')) = gs' - sg'.$$

Its domain is the record module. This form is not identified with a dimensional physical area or with the form on the APP torus, whose domain differs. The choice of order (g, s) fixes its orientation.

Theorem 29.5 (Unimodular stabilizer of the increment). *Define, directly from the primitive vector v in (29.6),*

$$N_v m = v \Omega_M(v, m) = v \mathcal{J}(m), \quad N_v = \begin{pmatrix} -18 & 1 \\ -324 & 18 \end{pmatrix}. \quad (29.9)$$

Then $N_v^2 = 0$, $N_v \neq 0$, and

$$\text{Stab}_{SL_2(\mathbb{Z})}(c_{108}) = \{I + nN_v : n \in \mathbb{Z}\}. \quad (29.10)$$

Every transformation in (29.10) preserves Ω_M , the increment c_{108} , and the charge $\mathcal{J}$. For nonzero n , it is a nonidentity unipotent matrix of trace two, of parabolic type.

Proof. The identity $\Omega_M(v, v) = 0$ implies $N_v^2 m = v \Omega_M(v, v) \Omega_M(v, m) = 0$. Its upper-right entry is one, so it is nonzero. The matrix

$$B = \begin{pmatrix} 1 & 0 \\ 18 & 1 \end{pmatrix}$$

has determinant one, maps the first basis vector to v , and satisfies

$$N_v = BNB^{-1}, \quad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (29.11)$$

An integer automorphism fixes $4v$ exactly when it fixes v , because M is torsion-free. A determinant-one matrix fixing the first basis vector necessarily has the form $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$. Conjugation by B proves (29.10). The rest follows from this expression or directly from (29.9). $\square$

Formula (29.9) makes the construction independent of the second vector chosen to complete v to an oriented unimodular basis: any other complement differs by an integer multiple of v and produces the same N_v . Under an oriented transformation $R \in SL_2(\mathbb{Z})$, one has $N_{Rv} = RN_v R^{-1}$.

The action can be written particularly clearly as

$$(I + nN_v)m = m + n\mathcal{J}(m)v. \quad (29.12)$$

The transverse charge determines how far this symmetry shifts along the memory direction. Summing revolutions produces the displacement $4Nv$; stabilizing symmetries preserve this advance law.

On the quotient (29.8), the transformation $I + nN_v$ acts as $(\mathcal{J}, [g]_4) \mapsto (\mathcal{J}, [g + n\mathcal{J}]_4)$. It therefore preserves the orbit collection and may permute its residual classes; v is an invariant of the return H , not of every stabilizing transformation. This distinction follows directly from (29.12).

Conjugation (29.11) supplies an explicit algebraic map from the parabolic TRIT model, defined in proposition 6.14, to the record module extended to real scalars. One also obtains $\exp(tN_v) = I + tN_v$ and $B \exp(tN) = \exp(tN_v)B$. This intertwining concerns operators: the particular motion selected by TPK also uses its emissions and calendar. The complete arithmetic group in (29.10) fixes this reader; the

admissible symmetry group of the total state is obtained by intersecting it with the actions preserving the remaining data and relations.

In the translated system (29.6), affine changes $m \mapsto Rm + b$, with R in (29.10) and arbitrary b , commute with H . Those additionally preserving the fixed initial record must satisfy $Rm_0 + b = m_0$. Distinguishing the two conditions separates symmetry of the law from symmetry of a marked trajectory.

Theorem 29.6 (Variational realization of the charge and stabilizer). *The scalar extension of the record, $M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}$, preserves the generated coordinates (g, s) and the form $\Omega_M = dg \wedge ds$. It is a subsequent realization of the integer module. With the convention $\iota_{X_H} \Omega_M = dH$, one has*

$$X_J = v, \quad X_{\frac{1}{2}J^2} = Jv = N_v m. \quad (29.13)$$

The constructed translation and parabolic collection thus admit explicit Hamiltonian generators. The flow of J is $m \mapsto m + uv$; at u equal to four, it reproduces (29.6). The flow of $J^2/2$ is $m \mapsto (I + uN_v)m$. The parameter u belongs to this continuous realization; the discrete return step and intermediate emissions retain their own TPK calendar.

Proof. For a vector $X = (X_g, X_s)$, $\iota_X(dg \wedge ds) = X_g ds - X_s dg$. Since $dJ = ds - 18 dg$, the first field is $(1, 18) = v$. The chain rule gives the second. The quantity J remains constant along both fields because $dJ(v) = 0$. The second flow therefore integrates $dm/du = J(m_0)v$, equivalent to $\exp(uN_v) = I + uN_v$. Nondegeneracy of Ω_M implies that the Hamiltonian generating N_v is $J^2/2$ plus an additive constant: its differential is fixed by $\iota_{N_v m} \Omega_M$. $\square$

The mathematical action

$$\mathcal{A}[m] = \int_{u_0}^{u_1} \left[s \frac{dg}{du} - \frac{1}{2}(s - 18g)^2 \right] du \quad (29.14)$$

yields, by variation, the equations $dg/du = J$, $ds/du = 18J$. The translation $g \mapsto g + \varepsilon$, $s \mapsto s + 18\varepsilon$ preserves the Hamiltonian and changes the Lagrangian by the total derivative $d(18\varepsilon g)/du$, for constant ε . Its Noether charge is

$$\underbrace{s \delta g}_s - \underbrace{18g}_{\text{boundary term}} = s - 18g = J. \quad (29.15)$$

The variation is interpreted per unit of ε . The equations directly verify $dJ/du = 18J - 18J = 0$. The discrete translation (29.6) itself is also realized by the Hamiltonian $H_{\text{ret}} = 4J$, whose flow at parameter one is H ; the two Hamiltonians Poisson-commute. The stabilizer flow $I + uN_v$ and the return $m + 4v$ are distinct commuting actions: on the line $J = 0$, the former fixes every point, while the latter continues to accumulate memory.

More generally, every C^1 Hamiltonian on this plane invariant under translations $m \mapsto m + uv$ has the form $F(J)$. To prove this, the global coordinates (g, J) express the symmetry as translation in g ; invariance removes dependence on g from H . Its field is $F'(J)v$. This classification makes explicit both the common conservation law and the additional choice of specific dynamics.

Action (29.14) is dimensionless and belongs to the realized record plane. Its proof supplies an exact variational relation between memory, charge, and stabilizer. Conversion to physical action and dimensional time is performed through the corresponding action and clock readers; these units are not assigned to the counter u .

29.4 APP curvature and conservation of the central phase

Section 4.4 works on $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. The sheets are $S(x, y) = x + y$ and $P(x, y) = xy$. The mixed connections use differences of these potentials, not the potentials themselves as links:

$$F(S, P) = \Delta_x \Delta_y P - \Delta_y \Delta_x S = 1, \quad F(P, S) = -1 \pmod{9}. \quad (29.16)$$

The oriented sum of links around a rectangle is ab or $-ab$, depending on order. The alternating area is $\sigma(u, v) = ad - bc$ for $u = (a, b)$, $v = (c, d)$. Section 18.2.1 represents these transports through $D(u) = X^a Z^b$, $ZX = \omega XZ$, with ω the ninth root in the subsequent representation. Then

$$D(u)D(v) = \omega^{bc} D(u+v), \quad D(u)D(v)D(u)^{-1}D(v)^{-1} = \omega^{-\sigma(u,v)} I. \quad (29.17)$$

The position resulting from the commutator returns. The central phase retains oriented area with the sign fixed in (29.17). This is retention modulo nine: an area divisible by nine has trivial central phase, although a lift with quotient and record may distinguish its paths. The phase does not replace that lift.

Proposition 29.7 (Exact covariance of central transport). *Since two is invertible modulo nine, define*

$$W(a, b) = \omega^{ab/2} D(a, b), \quad W(u)W(v) = \omega^{-\sigma(u,v)/2} W(u+v). \quad (29.18)$$

For $R \in SL_2(\mathbb{Z}/9\mathbb{Z})$, the map

$$\omega^z W(u) \longmapsto \omega^z W(Ru) \quad (29.19)$$

is an automorphism of the central extension. It preserves its products and commutation phases.

Proof. The phase in the product of the two W is $\frac{ab}{2} + \frac{cd}{2} + bc - \frac{(a+c)(b+d)}{2} = \frac{bc-ad}{2}$. This proves (29.18). The identity $\sigma(Ru, Rv) = \det(R)\sigma(u, v) = \sigma(u, v)$ shows that (29.19) preserves the law; invertibility of R supplies the inverse automorphism. If $\det R = -1$, the corresponding automorphism also changes $z \mapsto -z$; the area sign and centre are reversed jointly. $\square$

The complete record and its central phase have different types. Covariance preserves the area law while the presentation changes; the affine class in (29.3) preserves memory content while its origin changes. Both properties act on typed path readers, and their maps must remain separate when composed with the exceptional readout of the complete state.

Proposition 29.8 (Persistence under local frames and subdivisions). *Let γ be a loop based at x , with invertible links U_e . The local change $U'_e = G_t U_e G_s^{-1}$ transforms its holonomy into $H'_\gamma = G_x H_\gamma G_x^{-1}$: frames at interior vertices cancel successively. If $H_\gamma = \omega^k I$, the central phase remains exactly the same. For refined links whose product reproduces the preceding link, subdivision likewise preserves H_γ , by associativity. On APP plaquettes, cancellation of interior edges realizes this property through finite Stokes.*

An elementary rectangle has additive Wilson value one and nontrivial central phase. This mixed transport therefore distinguishes actual holonomy from the identity that links of the form $G_t G_s^{-1}$ would produce. Conservation under subdivision concerns compatible loops and links: adding another area changes the transport.

The integer lift of holonomy distinguishes oriented area from its nonadic remainder: for a rectangle with sides four and seven, $28 = 1 + 9 \cdot 3$, whereas $-28 = 8 + 9(-4)$ corresponds to the reverse orientation. The residual phase coincides for areas one and twenty-eight, while their quotients distinguish the accumulations. This example exhibits the difference between finite phase and complete record; its area quotient and the temporal counter q_{mem} retain distinct domains.

29.5 Composition between the three regimes and continuity of the balance

The realizations in propositions 6.14 and 6.15 construct generators $J_k^2 = -\tau_k I$ on their own fibres, of real dimension two. Take effectively specified links $B_k : V_k \rightarrow V_{k+1}$ preserving the alternating forms, $B_k^* \Omega_{k+1} B_k = \Omega_k$. The symbol $*$ in this section means transpose/bilinear pullback. The operators $E_k = \exp(t_k J_k)$ preserve Ω_k : in dimension two, zero trace of the generator implies $J_k^* \Omega_k + \Omega_k J_k = 0$. The symplectic identity is the hypothesis that would be retained in extending the lemma to higher dimensions.

For the nonadic compression and its recorded complement,

$$T_k = B_k(8I + E_k)/9, \quad D_k = B_k \sqrt{8(E_k - I)}/9,$$

algebraic expansion gives

$$\Omega_k = T_k^* \Omega_{k+1} T_k + D_k^* \Omega_{k+1} D_k. \quad (29.20)$$

Indeed, the cross terms cancel, and the numerator is $72\Omega_k + 9E_k^* \Omega_k E_k = 81\Omega_k$. For $P_0 = I$, $P_{k+1} = T_k P_k$, successive substitution in path order gives

$$\Omega_0 = P_N^* \Omega_N P_N + \sum_{k=0}^{N-1} (D_k P_k)^* \Omega_{k+1} (D_k P_k). \quad (29.21)$$

The terminal component and every complement remain present. The identity is oriented and bilinear; positivity results apply in the positive profiles proved in the preceding sections. The conserved form does not require a link between distinct regimes to intertwine their generators: if $BJ_\tau = J_{\tau'} B$, squaring with invertible B would impose $\tau = \tau'$. Conservation of a common structure allows the three regimes to remain distinct.

The complete transport $U_N = (B_{N-1} E_{N-1}) \cdots (B_0 E_0)$ maps V_0 to V_N . Matrix monodromy requires a typed closure $K : V_N \rightarrow V_0$; only then is KU_N an endomorphism whose trace can be computed. Compatible basis changes preserve this trace by conjugation. Phase return, fibre closure, and return of the vector remain different conditions.

If the links have exclusively the form $B_{yx} = G_y G_x^{-1}$, their product along a path is $G_{\text{final}} G_{\text{initial}}^{-1}$. Closing with the same frame gives the identity. The holonomy in (29.17) and the increment in (29.6) exhibit concrete mechanisms that this mere substitution of frames cannot represent by itself.

29.6 Compatibility with refinement and arbitrary extent

The memory action is compatible with a restriction homomorphism of modules $p : M' \rightarrow M$ when

$$pC' = Cp, \quad p\kappa' = \kappa. \quad (29.22)$$

Then $pH' = Hp$, and p induces a quotient map

$$M'/(I - C')M' \longrightarrow M/(I - C)M, \quad [\kappa'] \longmapsto [\kappa]. \quad (29.23)$$

Proof. The first equality in (29.22) maps $(I - C')M'$ into $(I - C)M$; the second fixes the image of the increment. Compatibility of H follows by substitution. Concatenations and refinements compose the same squares by virtue of law (29.1). $\square$

A nonzero return class appearing at a preserved level remains nonzero in every compatible lift: a zero class would project to zero. This statement applies to the system of memories and restrictions satisfying (29.22); the joint constructor and the maps of its inverse system are preserved in section 5.

In the record of (29.6), every integer N satisfies $H^N(m) = m + N(4, 72)$. The proof is algebraic for arbitrary N , rather than an inference from finitely many iterations. In forward prolongation, every prefix preserves its finite record; the complete history is described by the compatible collection of these records, without replacing its counters by an “infinite” integer quantity.

If complete dynamics has the reader q_{mem} with $q_{\text{mem}}(\Gamma_9 x) = q_{\text{mem}}(x) + 1$, it likewise cannot have a periodic state of positive period N on that domain: applying the reader would yield $q_{\text{mem}}(x) = q_{\text{mem}}(x) + N$. This corollary uses the complete integer degree; its residual quotients may have finite periods, as appropriate to the observable calendar in the corpus.

29.7 Foundational consequence and scope of the composition

The specific connection obtained is

$$\text{return increment } 4v \longrightarrow \mathcal{J}(m) = \Omega_M(v, m) \longrightarrow N_v m = v\mathcal{J}(m) \longrightarrow I + nN_v. \quad (29.24)$$

The accumulated record determines a conserved charge and a parabolic collection stabilizing its direction. Evolution and conservation belong to the same action: g and s change, their relation $s - 18g$ remains, and the increment retains an explicit stabilizer. This specifies part of the connection between memory and symmetry that motivated the research.

Equality (29.21) connects this research with the preceding balances: an evolution may redistribute the form between readout and complements while preserving the total. The APP identities (29.16)–(29.19) show, in another concrete reader of the same origin, how transport order produces a central datum even when position returns.

The charge (29.7) is an algebraic invariant of transport and, through the explicit action (29.14), a Noether charge of the record realization. Theorem 29.6 and its variational development construct this relation, its symmetry, and its boundary term. The preceding physical action and current developments retain their domains and units; transporting them to this realization requires preservation of their dimensional maps.

Composition with dimensional readers must transport the pair (v, Ω_M) and the charge $\mathcal{J}$, preserving their units and the variational current. Every subsequent relation with π, ϕ, e, α and K must arise from this composition; the present formulas specify the information to be preserved.

30 Variational conservation and oriented action

30.1 Symmetry of the elliptic action and Noether charge

Fix a positive action section $\hbar$, the angular pair A, C^* in a common unit, $A \neq 0$, $|C^*/A| < 1$, and

$$\eta = \operatorname{artanh}(C^*/A).$$

The realized plane $\mathcal{P}_\eta$ has coordinates (Q, P) , both with the dimension of the square root of action, with

$$\lambda = P dQ, \quad \Omega = dQ \wedge dP = -d\lambda.$$

The construction in section 20 gives

$$\mathcal{J}_\eta(Q, P) = \frac{1}{2}(e^{-\eta}Q^2 + e^\eta P^2), \quad H_\eta = \omega \mathcal{J}_\eta, \quad \omega > 0. \quad (30.1)$$

The HMT ellipse is $\mathcal{J}_\eta = \hbar$. Its area is $2\pi\hbar$; $\mathcal{J}_\eta$ has the dimension of action and H_η that of energy. Define the already realized matrices

$$M_\eta = \operatorname{diag}(e^{\eta/2}, e^{-\eta/2}), \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_\eta(\varepsilon) = M_\eta e^{-\varepsilon J_2} M_\eta^{-1}. \quad (30.2)$$

Proposition 30.1. *The group R_η is a canonical symmetry of the action*

$$\mathcal{S}_\eta[z] = \int_{t_a}^{t_b} (P\dot{Q} - \omega \mathcal{J}_\eta(Q, P)) dt \quad (30.3)$$

and its Noether charge is $\mathcal{J}_\eta$.

Proof. The determinant of M_η is one, and $M_\eta^\top J_2 M_\eta = J_2$. Moreover, $\mathcal{J}_\eta(M_\eta y) = \|y\|^2/2$, which is invariant under $e^{-\varepsilon J_2}$. The generator

$$X_\eta = e^\eta P \partial_Q - e^{-\eta} Q \partial_P$$

satisfies $\iota_{X_\eta} \Omega = d\mathcal{J}_\eta$. By Cartan's formula,

$$\mathcal{L}_{X_\eta} \lambda = dF_\eta, \quad F_\eta = \lambda(X_\eta) - \mathcal{J}_\eta = e^\eta P^2 - \mathcal{J}_\eta.$$

Since $X_\eta H_\eta = 0$, the variation of the action density is dF_η/dt . Its charge is

$$\frac{\partial L}{\partial \dot{Q}} X_\eta Q - F_\eta = P e^\eta P - F_\eta = \mathcal{J}_\eta.$$

The equations $\dot{Q} = \omega e^\eta P$, $\dot{P} = -\omega e^{-\eta} Q$ directly verify $d\mathcal{J}_\eta/dt = 0$. □

The positive parametrization of the ellipse is $Q = a_S \cos \theta$, $P = b_S \sin \theta$. These equations assign it $\dot{\theta} = -\omega$. Thus, over a turn traversed according to the dynamical flow, $\oint P dQ = 2\pi \mathcal{J}_\eta$; with increasing θ , the integral is $-2\pi \mathcal{J}_\eta$. This is the orientation distinction established in section 20.2. On the radional contour, the magnitude is $h = 2\pi\hbar$.

30.2 Change of form: determined transport and connection

For two already expressed forms η, η' and a phase advance ϑ , define

$$\mathcal{T}_{\eta', \eta}^{\vartheta} : \mathcal{P}_{\eta} \rightarrow \mathcal{P}_{\eta'}, \quad \mathcal{T}_{\eta', \eta}^{\vartheta} = M_{\eta'} e^{-\vartheta J_2} M_{\eta}^{-1}. \quad (30.4)$$

Proposition 30.2. *This transport is symplectic and satisfies*

$$\mathcal{J}_{\eta'}(\mathcal{T}_{\eta', \eta}^{\vartheta} z) = \mathcal{J}_{\eta}(z), \quad \mathcal{T}_{\eta'', \eta'}^{\vartheta_2} \mathcal{T}_{\eta', \eta}^{\vartheta_1} = \mathcal{T}_{\eta'', \eta}^{\vartheta_1 + \vartheta_2}. \quad (30.5)$$

Proof. All factors are symplectic. The first equality reduces to invariance of $\|y\|^2/2$ under rotation. In the second, the adjacent matrices $M_{\eta'}^{-1} M_{\eta'}$ cancel and the phases add. $\square$

The formula receives the forms and advances from the reader; it does not select their values. The complete list of transitions can be retained alongside the composite transport: two lists yielding the same composite map are not identified as histories.

Proposition 30.3. *A differentiable realization $\eta(t), \vartheta(t) \in C^1$, with $\dot{\vartheta} = \omega(t)$, of the transport (30.4) has generator*

$$H_{\text{tr}}(Q, P, t) = \omega(t) \mathcal{J}_{\eta(t)}(Q, P) + \frac{\dot{\eta}(t)}{2} QP. \quad (30.6)$$

This generator preserves $\mathcal{J}_{\eta(t)}(z(t))$.

Proof. Differentiating $z(t) = M_{\eta(t)} e^{-\vartheta(t) J_2} y_0$ gives

$$\dot{Q} = \omega e^{\eta} P + \frac{\dot{\eta}}{2} Q, \quad \dot{P} = -\omega e^{-\eta} Q - \frac{\dot{\eta}}{2} P.$$

These are Hamilton's equations for (30.6). The connection term realizes exactly $\dot{M}_{\eta} M_{\eta}^{-1} z$, and its Hamiltonian is determined up to a function of t alone. Indeed,

$$\begin{aligned} \frac{d\mathcal{J}_{\eta}}{dt} &= \frac{\dot{\eta}}{2} (-e^{-\eta} Q^2 + e^{\eta} P^2) + \left\{ \mathcal{J}_{\eta}, \frac{\dot{\eta}}{2} QP \right\} \\ &= \frac{\dot{\eta}}{2} (-e^{-\eta} Q^2 + e^{\eta} P^2) + \frac{\dot{\eta}}{2} (e^{-\eta} Q^2 - e^{\eta} P^2) = 0. \end{aligned} \quad (30.7)$$

Equivalently, $Q = e^{\eta/2} q, P = e^{-\eta/2} p$ give

$$P\dot{Q} - H_{\text{tr}} = p\dot{q} - \frac{\omega(t)}{2} (q^2 + p^2). \quad (30.8)$$

The rotational symmetry and its charge are transported exactly. $\square$

Here the connection term has been derived from transport, not from an external force. The fixed-form LC Hamiltonian preserves $\mathcal{J}_{\eta}$. During a deformation, omitting the second term of (30.6) leaves the first term of (30.7) uncanceled. Variational conservation also requires transporting the canonical form.

The proposition determines the generator of the declared interpolation; it does not assign a unique physical interpolation to every TPK history. The discrete identity (30.4)–(30.5) works without interpolation.

30.3 Orientation and boundary of the action

The transformations in section 20.3 distinguish $S(Q, P) = (P, Q)$ and $J_2(Q, P) = (-P, Q)$. Both exchange the form $\eta \leftrightarrow -\eta$, but

$$S^* \Omega = -\Omega, \quad J_2^* \Omega = \Omega, \quad S^* \lambda = -\lambda + d(PQ), \quad J_2^* \lambda = \lambda - d(PQ). \quad (30.9)$$

Likewise,

$$SR_\eta(\varepsilon)S^{-1} = R_{-\eta}(-\varepsilon), \quad J_2R_\eta(\varepsilon)J_2^{-1} = R_{-\eta}(\varepsilon). \quad (30.10)$$

These are verified using $SM_\eta = M_{-\eta}S$, $J_2M_\eta = M_{-\eta}J_2$, $SJ_2S^{-1} = -J_2$. For an oriented open curve $z : [a, b] \rightarrow \mathcal{P}_\eta$,

$$\int_{J_2z} \lambda = \int_z \lambda - [PQ]_a^b, \quad \int_{Sz} \lambda = - \int_z \lambda + [PQ]_a^b. \quad (30.11)$$

The boundary terms vanish on a cycle and cancel when compatible endpoints are concatenated. Quadratic energy alone does not determine these signs or these terms.

Nonadic monodromy records phase return and $q_{\text{mem}} \mapsto q_{\text{mem}} + 1$. Constancy of a charge does not require the counter to remain fixed. In the elliptic realization, a turn preserves $\mathcal{I}_\eta$ while increasing the accumulated oriented integral by $2\pi\mathcal{I}_\eta$. Charge, integrated action, phase, and register have distinct types. An angular turn is not identified here with an arbitrary TPK cycle without specifying its reader.

30.4 Variational clock charge and balance between reading and memory

30.4.1 An explicit action charge of the clock

The calendar is realized on $\mathcal{H}_{\text{clk}} = \mathbb{C}^{108}$, with Fourier projectors $R_k = |\tilde{k}\rangle\langle\tilde{k}|$, and determines

$$N_{108} = \sum_{k=0}^{107} kR_k, \quad H_{\text{clk}} = \hbar\omega_0 N_{108}, \quad \omega_0 = \frac{2\pi}{108t_0}. \quad (30.12)$$

Its propagator over t_0 is the shift by one cell. The clock action is

$$\mathcal{S}[\psi] = \int \left[\frac{i\hbar}{2} (\langle\psi, \dot{\psi}\rangle - \langle\dot{\psi}, \psi\rangle) - \langle\psi, H_{\text{clk}}\psi\rangle \right] dt. \quad (30.13)$$

The inner product is antilinear in the first variable. The equation is $i\hbar\dot{\psi} = H_{\text{clk}}\psi$.

Proposition 30.4. *For $B = B^*$ with $[B, H_{\text{clk}}] = 0$, the symmetry $\psi \mapsto e^{-i\varepsilon B}\psi$ has charge*

$$\mathcal{Q}_B(\psi) = \hbar\langle\psi, B\psi\rangle. \quad (30.14)$$

Proof. Localizing the parameter, $\delta\psi = -i\varepsilon(t)B\psi$, gives $\delta L = \hbar\dot{\varepsilon}\langle\psi, B\psi\rangle$. Integration by parts in the variational equations gives $d\mathcal{Q}_B/dt = 0$. Directly,

$$\frac{d}{dt}\langle\psi, B\psi\rangle = \frac{i}{\hbar}\langle\psi, [H_{\text{clk}}, B]\psi\rangle = 0.$$

With $B = I$, norm is conserved; with $B = N_{108}$, the charge is the action E_{clk}/ω_0 . $\square$

The symplectic form of the underlying real space is $\Omega_{\text{clk}}(u, v) = 2\hbar \text{Im}\langle u, v \rangle$; its one-form is $\Theta_\psi(v) = -\hbar \text{Im}\langle \psi, v \rangle$, with $\Omega_{\text{clk}} = -d\Theta$.

30.4.2 Exact charge balance

Let $\mathcal{H}, \mathcal{K}$ be finite-dimensional Hermitian spaces, $J : \mathcal{H} \rightarrow \mathcal{K}$ an isometry, U unitary on $\mathcal{K}$, and

$$\Pi = JJ^*, \quad T = J^*UJ, \quad \eta_{\text{mem}} = (I - \Pi)UJ. \quad (30.15)$$

The operator η_{mem} is distinct from the scalar parameter η .

Proposition 30.5. *If*

$$\tilde{B} = \tilde{B}^*, \quad U^*\tilde{B}U = \tilde{B}, \quad [\tilde{B}, \Pi] = 0, \quad B = J^*\tilde{B}J, \quad (30.16)$$

then

$$B = T^*BT + \eta_{\text{mem}}^*\tilde{B}\eta_{\text{mem}}, \quad \mathcal{Q}_B(\psi) = \mathcal{Q}_B(T\psi) + \mathcal{Q}_{\tilde{B}}(\eta_{\text{mem}}\psi). \quad (30.17)$$

Proof. Expand $UJ\psi = JT\psi + \eta_{\text{mem}}\psi$ in $J^*U^*\tilde{B}UJ = J^*\tilde{B}J$. Commutation with Π cancels the cross terms. Evaluating the operator identity and multiplying by $\hbar$ gives the action balance. $\square$

By iteration,

$$\mathcal{Q}_B(\psi) = \mathcal{Q}_B(T^n\psi) + \sum_{j=0}^{n-1} \mathcal{Q}_{\tilde{B}}(\eta_{\text{mem}}T^j\psi). \quad (30.18)$$

This sum records successive complements of the compressed dynamics. The complete dynamics has another exact formula, with $z_n = U^nJ\psi$:

$$\mathcal{Q}_B(\psi) = \mathcal{Q}_B(J^*z_n) + \mathcal{Q}_{\tilde{B}}((I - \Pi)z_n). \quad (30.19)$$

Without an additional property, T^n is not identified with J^*U^nJ : the complete state can reabsorb memory. If $B, \tilde{B} \geq 0$, all summands in (30.18) are nonnegative; for signed charges, the equality remains, without this positivity interpretation.

If U is the propagator of an action such as (30.13), commutation of $\tilde{B}$ with its Hamiltonian supplies Noether conservation. The reduction condition involving Π specifies the readings that separate charge without interference terms.

Without imposing (30.16), the general identity preserves the additional terms:

$$\begin{aligned} & \mathcal{Q}_B(T\psi) + \mathcal{Q}_{\tilde{B}}(\eta_{\text{mem}}\psi) + 2\hbar \text{Re}\langle JT\psi, \tilde{B}\eta_{\text{mem}}\psi \rangle \\ &= \mathcal{Q}_B(\psi) + \hbar\langle \psi, J^*(U^*\tilde{B}U - \tilde{B})J\psi \rangle. \end{aligned} \quad (30.20)$$

The physical charge uses its reader, for example N_{108} ; a norm identity corresponds to $B = I$. The sections $\hbar_{\text{pre}}$ and $\hbar_{\text{ret}}$ preserve the return reader of section 12.3 and equation (12.14); their difference and the memory complement have their respective domains of definition.

30.5 Covariant balance of the total tensor

The variational elimination of contorsion and the Noether identity of the covariant action, developed in Proposition 53.6, provide

$$\mathcal{T}_{\mu\nu} = T_{\mu\nu}^{\text{phys,LC}} + U_{\mu\nu}^{\text{phys,spin}}, \quad \mathring{\nabla}^\mu \mathcal{T}_{\mu\nu} = 0. \quad (30.21)$$

Its domain is the realized HMT–MD chart, with the couplings held constant during variation. For a Killing field ξ ,

$$j_\xi^\mu = \mathcal{T}^{\mu\nu} \xi_\nu, \quad \mathring{\nabla}_\mu j_\xi^\mu = 0. \quad (30.22)$$

The proof uses (30.21), the symmetry of $\mathcal{T}$, and the antisymmetry of $\mathring{\nabla}_\mu \xi_\nu$. Upon integration, the difference between the charges of the hypersurfaces is the flux through the lateral boundary.

Conservation belongs to the total tensor; each component may exchange charge. Theorem 58.3 explicitly determines $\mathcal{J}_\nu^{\text{tr}} = \partial_\nu \text{tr}_g(T^{\text{res}})/4$ and the opposite balances of the trace and traceless part. This is recovered as a covariant instance of the joint-balance criterion, not as a new gravitational equation.

31 Conservation of action forms and oriented area in the tritic regimes

31.1 Balance between two action forms

The angular outputs A, C^* , in a common unit, determine the form $\eta = \text{artanh}(C^*/A)$ on the domain $A \neq 0, |C^*/A| < 1$. Given two successive forms η_n, η_{n+1} , define

$$M_n = \text{diag}(e^{\eta_n/2}, e^{-\eta_n/2}), \quad G_n = M_n^{-T} M_n^{-1}, \quad I_n(z) = \frac{1}{2} z^T G_n z.$$

Here $z = (Q, P)^T$, with both coordinates having the dimension of the square root of action. The scale $\hbar$, when it fixes the contour $I_n = \hbar$, is the previously constructed action output; it does not serve as a fitting target.

Let R_n be the canonical rotation of angular advance θ_n . The transport without angular advance and the transport with that advance are, respectively,

$$B_n = M_{n+1} M_n^{-1}, \quad C_n = M_{n+1} R_n M_n^{-1}. \quad (31.1)$$

Both go from chart n to chart $n+1$. Define the compressed reading and its normalized complementary register:

$$T_n = \frac{8B_n + C_n}{9}, \quad D_n = \frac{\sqrt{8}}{9}(C_n - B_n). \quad (31.2)$$

The operator D_n is an isometrically normalized realization of the unique pattern of differences among the nine phases. The nine-component register is $(8z, -z, \dots, -z)/27$, with $z = (C_n - B_n)v$; the sum of its inner products gives the same factor $8/81$. The nine-phase register is retained when the individual labels enter into other readers.

Theorem 31.1. *Action is conserved between reading and memory:*

$$\boxed{G_n = T_n^T G_{n+1} T_n + D_n^T G_{n+1} D_n.} \quad (31.3)$$

Proof. The normalization of the charts gives $B_n^T G_{n+1} B_n = G_n = C_n^T G_{n+1} C_n$. Upon expanding (31.2), the cross contributions have coefficients $+8$ and -8 . The remaining diagonal coefficients are 72 for B_n and 9 for C_n , divided by 81 . Their sum is G_n . $\square$

With $z_{n+1} = T_n z_n$, $m_n = D_n z_n$, the scalar form is

$$\boxed{I_n(z_n) = I_{n+1}(z_{n+1}) + I_{n+1}(m_n).} \quad (31.4)$$

The telescoping sum, retaining each memory in the chart where it was emitted, gives

$$I_0(z_0) = I_N(z_N) + \sum_{n=0}^{N-1} I_{n+1}(m_n). \quad (31.5)$$

The transport B_n is an effective part of this composition: replacing it by the identity when the forms differ changes the domains and may invalidate (31.3).

31.2 Exact accumulation law with variable phase

In normalized coordinates $y_n = M_n^{-1} z_n$, the reading is $y_{n+1} = (8I + R_n)y_n/9$. For a planar rotation,

$$\left(\frac{8I + R_n}{9}\right)^T \left(\frac{8I + R_n}{9}\right) = r_n I, \quad r_n = 1 - \frac{32}{81} \sin^2(\theta_n/2). \quad (31.6)$$

Therefore,

$$I_N(z_N) = I_0(z_0) \prod_{n=0}^{N-1} r_n. \quad (31.7)$$

Theorem 31.2. *For $I_0(z_0) > 0$, the terminal component tends to zero exactly when*

$$\sum_{n \geq 0} \sin^2(\theta_n/2) = +\infty. \quad (31.8)$$

Proof. Let $c = 32/81$ and $u_n = \sin^2(\theta_n/2) \in [0, 1]$. We have $cu_n \leq -\log(1 - cu_n) \leq cu_n/(1 - c)$, by integrating $1/(1 - s)$ between zero and cu_n . The sum of these logarithms diverges exactly when $\sum u_n$ diverges. Applying the logarithm to the product in (31.7) gives the equivalence. $\square$

Equation (31.5) is preserved in both cases. A sequence of angular advances may leave a positive terminal component or transfer all action to the register, depending on the sum in (31.8). Change of form alone preserves action; angular dispersion determines its allocation in this realization. The phases θ_n must come from the specific reader of the trajectory: the proof does not retrospectively select a sequence to impose a result.

31.3 Extended Euler and common oriented area

The generators of the three quadratic regimes are

$$J_{+1} = (2C_{A_2} + I)/\sqrt{3}, \quad J_0 = N, \quad J_{-1} = (2F_{av}^2 - 3I)/\sqrt{5}.$$

In their two-dimensional charts, they have zero trace and satisfy $J_\tau^2 = -\tau I$. The Coxeter operator comes from the APP antecedent, the nilpotent from threshold transport, and F_{av} from the descent of TPK modes. The exponential functions constitute the subsequent analytic realization:

$$E_\tau(t) = \exp(tJ_\tau) = c_\tau(t)I + s_\tau(t)J_\tau,$$

where (c_τ, s_τ) are $(\cos, \sin)$, $(1, t)$, and $(\cosh, \sinh)$, respectively. Zero trace gives determinant one. With

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

we obtain $E_\tau^\top \Omega E_\tau = \Omega$. In variable charts, take $C_n = M_{n+1} E_{\tau_n}(t_n) M_n^{-1}$ and the same B_n as in (31.1).

Theorem 31.3. *The three regimes satisfy*

$$\boxed{\Omega = T_n^\top \Omega T_n + D_n^\top \Omega D_n}, \quad (31.9)$$

with T_n, D_n from (31.2).

Proof. In dimension two, $L^\top \Omega L = (\det L)\Omega$. The determinants of M_n , B_n , and C_n are one. The expansion used in (31.3), applied to the alternating form, proves (31.9). $\square$

In a fixed chart, the two determinants are

$$\det T_\tau = \frac{65 + 16c_\tau(t)}{81}, \quad \det D_\tau = \frac{16(1 - c_\tau(t))}{81}. \quad (31.10)$$

This follows from $\det(aI + C) = a^2 + a \operatorname{tr} C + \det C$ and $\operatorname{tr} E_\tau = 2c_\tau$. Their sum is one. For the representatives C_{A_2} , $I + N$, F_{av}^2 , the pairs are $(19/27, 8/27)$, $(1, 0)$, $(89/81, -8/81)$.

The parabolic regime may produce a nonzero rank-one memory with zero area. The hyperbolic regime preserves oriented area through a complement of opposite sign. The elliptic regime additionally permits a positive action balance between the forms of the two fibres. The common alternating form and the metric forms of each regime retain their respective types.

31.3.1 Integrated action and orientation

Let γ be a closed piecewise C^1 curve in the canonical plane. For any real matrix L of order two,

$$\oint_{L\gamma} P dQ = (\det L) \oint_{\gamma} P dQ.$$

Indeed, writing $Q' = aQ + bP$, $P' = cQ + dP$, the integrals of $Q dQ$ and $P dP$ are zero, and $\oint Q dP = -\oint P dQ$. The coefficient $ad - bc$ remains. Applying (31.10),

$$\oint_{\gamma} P dQ = \oint_{T_n\gamma} P dQ + \oint_{D_n\gamma} P dQ. \quad (31.11)$$

The signs are those of the transported curves. The identity is preserved for the three regimes and for changes of form with determinant one.

31.4 The common factor of the exponential reader

The radional reader in section 20 and proposition 6.15 also contains $e^{-x} \exp(-yJ_{\tau})$. This factor is retained when it acts in the reader. Let

$$C_n = e^{-x_n} M_{n+1} E_{\tau_n}(t_n) M_n^{-1}.$$

Now $C_n^T \Omega C_n = e^{-2x_n} \Omega$. The same expansion gives

$$T_n^T \Omega T_n + D_n^T \Omega D_n = \frac{8 + e^{-2x_n}}{9} \Omega. \quad (31.12)$$

For $x_n \geq 0$, define a complementary scale register

$$D_{\text{esc},n} = \frac{\sqrt{1 - e^{-2x_n}}}{3} B_n.$$

The complete identity is then

$$\Omega = T_n^T \Omega T_n + D_n^T \Omega D_n + D_{\text{esc},n}^T \Omega D_{\text{esc},n}. \quad (31.13)$$

This is a composition defined here to represent explicitly the reader's scale defect. Its coefficient comes from (31.12). The name of this register does not identify it with another physical sector of the corpus; the map is retained so that this comparison can be made with their respective operators and domains.

31.5 Action refinement over the complete emissions

For each enriched prefix h , let M_h be its chart and $G_h = M_h^{-T} M_h^{-1}$. For each surviving emission ϵ , let R_{ϵ} be a realized rotation and

$$U_{\epsilon} = M_{h\epsilon} R_{\epsilon} M_h^{-1}, \quad (\mathcal{U}z)_{h\epsilon} = \sqrt{p_h(\epsilon)} U_{\epsilon} z_h.$$

The weights come from the emission measure of the common construction; $\sum_{\epsilon} p_h(\epsilon) = 1$, with multiplicities preserved. From $U_{\epsilon}^{\top} G_{h\epsilon} U_{\epsilon} = G_h$ it follows that

$$\sum_{h,\epsilon} I_{h\epsilon}((\mathcal{U}z)_{h\epsilon}) = \sum_h I_h(z_h). \quad (31.14)$$

The proof first sums the weights of the descendants of one predecessor. Along a trajectory, the intermediate factors $M_h^{-1} M_h$ cancel, and the rotations compose in the recorded order. This preserves labels and memory without mixing successors of different predecessors. The direct-sum alternating form satisfies the same transport identity for the three symplectic regimes, replacing R_{ϵ} by $E_{\tau_{\epsilon}}(t_{\epsilon})$.

The metric completions are taken with $\sum_h z_h^{\top} G_h z_h$, so these maps are isometric. Further identifying this norm with a fixed Euclidean direct sum requires uniform bounds on M_h and M_h^{-1} ; that additional identification is not used in (31.14).

32 Explicit reconstruction from complete returns

32.1 State domain and bilateral extension

Consider a chart of the surviving domain generated by APP-TRIT-TPK. A state in this chart preserves the operator sheets, the tritic orientation, the remainder, the quotient, the carry, the route, the boundary, the memory record, and the schedule origin. The update acts on this complete state.

Let X_0 be a nonempty compact Hausdorff space and $F : X_0 \rightarrow X_0$ a continuous surjective update. These are the hypotheses of the result in the chart under consideration. The bilateral extension of the update is

$$\mathcal{X}_F = \{(x_k)_{k \in \mathbb{Z}} \in X_0^{\mathbb{Z}} : F(x_k) = x_{k+1} \text{ for every } k \in \mathbb{Z}\}, \quad (32.1)$$

with the topology induced by the product.

Lemma 32.1 (Existence and bilateral transport). *The space $\mathcal{X}_F$ is compact, Hausdorff, and nonempty. Every coordinate projection $\mathcal{X}_F \rightarrow X_0$ is surjective. The shift*

$$\sigma((x_k)_{k \in \mathbb{Z}}) = (x_{k+1})_{k \in \mathbb{Z}}$$

is a homeomorphism of $\mathcal{X}_F$.

Proof. Each condition $F(x_k) = x_{k+1}$ defines a closed subset of the compact Hausdorff product $X_0^{\mathbb{Z}}$. For a fixed index k_0 and state $z \in X_0$, any finite collection of these conditions, together with $x_{k_0} = z$, admits a solution: complete forwards using F and backwards using preimages, which exist by surjectivity. The conditions therefore have the finite intersection property. Compactness produces a bilateral history with $x_{k_0} = z$.

The shifts σ and $\sigma^{-1}((x_k)) = (x_{k-1})$ preserve the relations in (32.1); both are continuous by their coordinate action. $\square$

32.2 Tritic sampling and reconstruction of every step

With the schedule origin fixed, the typed return times are

$$\mathcal{T}_+ = \{k \in \mathbb{Z} : 1 + (k \bmod 9) \in \{1, 4, 7\}\} = 3\mathbb{Z}.$$

The complete state x_k is preserved at each such time. Define the space of return records by

$$\mathcal{Y}_+ = \{(y_j)_{j \in \mathbb{Z}} \in X_0^{\mathbb{Z}} : F^3(y_j) = y_{j+1} \text{ for every } j \in \mathbb{Z}\}. \quad (32.2)$$

Theorem 32.2 (Recovery from complete returns). *The map*

$$r = \text{Ret}_+ : \mathcal{X}_F \longrightarrow \mathcal{Y}_+, \quad r(x)_j = x_{3j},$$

is a homeomorphism. Its inverse is given explicitly by

$$\boxed{(r^{-1}(y))_{3j+s} = F^s(y_j), \quad j \in \mathbb{Z}, \quad s \in \{0, 1, 2\}. \quad (32.3)}$$

Proof. For $x \in \mathcal{X}_F$, $F^3(x_{3j}) = x_{3j+3}$, so $r(x) \in \mathcal{Y}_+$. Conversely, every integer k has a unique expression $k = 3j + s$, with $j \in \mathbb{Z}$ and $0 \leq s < 3$, including when $k < 0$. Formula (32.3) therefore defines every coordinate of a sequence x .

If $s = 0$ or $s = 1$, then $F(x_{3j+s}) = F^{s+1}(y_j) = x_{3j+s+1}$. At the boundary between two blocks,

$$F(x_{3j+2}) = F^3(y_j) = y_{j+1} = x_{3j+3}.$$

Thus $x \in \mathcal{X}_F$. Reading its coordinates indexed by $3j$ returns y_j . In the reverse direction, the relations in (32.1) give $x_{3j+s} = F^s(x_{3j})$, so reconstruction from $r(x)$ returns x in full.

The map r is continuous because it selects coordinates. Each coordinate of its inverse is the composition of a coordinate projection with one of the continuous maps I, F, F^2 . The inverse is therefore continuous. $\square$

An additional discrete coordinate, such as the index $m \in \mathbb{Z}$ of a realization chart, is preserved through

$$\tilde{r} : \mathbb{Z} \times \mathcal{X}_F \longrightarrow \mathbb{Z} \times \mathcal{Y}_+, \quad \tilde{r}(m, x) = (m, r(x)).$$

Its inverse is $(m, y) \mapsto (m, r^{-1}(y))$. The same result holds for a fixed schedule origin a : replace x_{3j} and x_{3j+s} with x_{a+3j} and x_{a+3j+s} .

32.3 Conjugation of transport and preservation of readouts

Let $\sigma_+(y)_j = y_{j+1}$ be the shift of records. The preceding reconstruction gives

$$r\sigma^3 = \sigma_+r, \quad (r\sigma r^{-1}(y))_j = F(y_j). \quad (32.4)$$

The map in the second equality is reversible on $\mathcal{Y}_+$; its inverse is

$$y \longmapsto (F^2(y_{j-1}))_{j \in \mathbb{Z}}.$$

Indeed, both compositions reduce to $F^3(y_{j-1}) = y_j$. Reversibility of the bilateral history is thus preserved even when the update F on an isolated state is not injective. If the nonadic revolution is realized as $\Gamma_9 = \sigma^9$, then

$$r\Gamma_9 = \sigma_+^3 r. \quad (32.5)$$

Corollary 32.3 (Joint readouts and recovered transport). *Let $a : \mathcal{X}_F \rightarrow \mathcal{R}$ and $q : \mathcal{X}_F \rightarrow \mathcal{E}$ be two continuous readers of the same state, and let $d : \mathcal{X}_F \rightarrow \mathcal{X}_F$ be a homeomorphism of the calibrated domain. Then*

$$a_+ = a \circ r^{-1}, \quad q_+ = q \circ r^{-1}, \quad d_+ = r \circ d \circ r^{-1}$$

are continuous, d_+ is a homeomorphism, and

$$(a, q) = (a_+, q_+) \circ r, \quad rd = d_+r. \quad (32.6)$$

For every integer $N > 0$,

$$d^N x = x \iff d_+^N r(x) = r(x).$$

Proof. Compose the maps with the homeomorphism of theorem 32.2. The inverse of d_+ is $rd^{-1}r^{-1}$. The identity $d_+^N r = rd^N$, together with injectivity of r , proves the last equivalence. $\square$

Application to the double readout takes $\mathcal{R} = R_{\text{HMT}}$ and the incidence codomain constructed in section 16.1 and proposition 16.1. The result establishes a recoverable presentation of the same histories, their readers, and their transport. Its object is this dynamical equivalence of presentations.

33 Observable return and persistence of the register

33.1 Joint recovery from complete returns

Let X be the domain of calibrated complete histories in equation (32.1), including the calendar origin and, when used, the discrete chart index. The states preserve the transitions and the memory register. Write $r = \text{Ret}_+ : X \rightarrow Y_+$ for sampling at complete returns, with $Y_+ = r(X)$.

The result in theorem 32.2 provides its explicit inverse: $x_{3j+s} = F^s(y_j)$, for $s = 0, 1, 2$. Reconstruction uses the complete state preserved at each return and the effective update; (32.4) also preserves transport.

Let $a : X \rightarrow R_{\text{HMT}}$ be numerical evaluation by cylinders and $q : X \rightarrow E_\infty$ incidence with its gauge, according to section 6, section 16.1, and proposition 16.1. Let $d : X \rightarrow X$ be a reversible continuous transport, with the gauge transported. On Y_+ , define

$$a_+ = a \circ r^{-1}, \quad q_+ = q \circ r^{-1}, \quad d_+ = r \circ d \circ r^{-1}.$$

Then

$$(a, q) = (a_+, q_+) \circ r, \quad rd = d_+r. \quad (33.1)$$

Proof. This is corollary 32.3: the compositions with r and r^{-1} cancel, and the inverse of d_+ is $rd^{-1}r^{-1}$. Both readers are transported from the same complete state. $\square$

In particular, for $N \in \mathbb{Z}_{>0}$, $d^N x = x$ if and only if $d_+^N r(x) = r(x)$. Conjugation preserves periodicity or aperiodicity of the complete state. The dynamics of a partial reader is determined through its own transport factor.

33.2 Explicit subsequent representation of transport

On the already generated set of histories, consider the subsequent realization $H = \ell^2(X)$, with orthonormal basis $(\delta_x)_{x \in X}$ and counting measure. Every vector has at most countable support. Reversible transport determines

$$C\delta_x = \delta_{dx}, \quad C^{-1}\delta_x = \delta_{d^{-1}x}.$$

The bijection d proves that C is unitary. This representation preserves the complete labels of the histories. The space X and its transport precede this linear realization.

The nonadic chart in section 25 determines

$$T = \frac{8I + C}{9}, \quad \eta v = \frac{1}{27}(8(C - I)v, -(C - I)v, \dots, -(C - I)v),$$

$$I - T^*T = \eta^*\eta = \frac{8}{81}(I - C)^*(I - C).$$

Let P be the orthogonal projection onto $\ker(I - C)$. The spectral proof in sections 25 and 26 gives $T^N \rightarrow P$ strongly. Telescoping provides the isometry

$$\mathcal{V}v = (Pv, \eta v, \eta T v, \eta T^2 v, \dots), \quad \mathcal{M} = \text{Im } \mathcal{V}.$$

The register shift B preserves the first component and removes the first memory block. On the compatible image,

$$\widehat{C} = \mathcal{V}C\mathcal{V}^{-1} = (9B - 8I)|_{\mathcal{M}}. \quad (33.2)$$

The inverse of the representation is given by

$$v = Pv + \sum_{j \geq 0} T^{*j} \eta^* (\eta T^j v). \quad (33.3)$$

Proof. The defect identity telescopes up to N ; its limit preserves the norm exactly. The identities $\mathcal{V}T = B\mathcal{V}$ and $C = 9T - 8I$ give (33.2). The isometry $\mathcal{V}$ is surjective onto $\mathcal{M}$; taking its adjoint gives (33.3), with norm convergence of the series. Unitarity of the recovered transport is established on $\mathcal{M}$. $\square$

For $A \in \mathcal{B}(H)$, let $\hat{A} = \mathcal{V}A\mathcal{V}^{-1}$ on $\mathcal{M}$. This correspondence preserves norm, products, adjoints, and commutators, since $\mathcal{V}^{-1} = \mathcal{V}^*|_{\mathcal{M}}$. It jointly represents the numerical observables, the incidence observables, and their relations with C .

33.3 Numerical convergence and correlations with incidence

The cylinders in (6.31) supply six trits per block in the chart of integer index $m(x)$. Write

$$\xi_n(x) = \sum_{k=1}^{6n} d_k(x) 3^{-k}, \quad \xi(x) = a(x) - m(x), \quad 0 \leq \xi(x) - \xi_n(x) \leq 3^{-6n}.$$

Here $d_k \in \{0, 1, 2\}$ is the subsequent ternary reading of the trit; its balanced orientation preserves its own typing. The chart retains m separately, including the endpoint convention, and reconstructs $a = m + \xi$. At endpoints with dual representation, ξ is interpreted in that chart.

Let M_f denote multiplication by f on $\ell^2(X)$. The normalization $0 \leq \xi_n \leq \xi \leq 1$ makes M_ξ and M_{ξ_n} contractions. For a set B in the finite incidence codomain of q_j , define $E_{j,B} = M_{\mathbf{1}_{q_j^{-1}(B)}}$. This is the projector onto the histories satisfying that incidence.

Theorem 33.1 (Simultaneous conservation of precision and transport). *For every $t \in \mathbb{Z}$,*

$$\left\| \hat{C}^{-t} \hat{M}_\xi \hat{C}^t - \hat{C}^{-t} \hat{M}_{\xi_n} \hat{C}^t \right\| = \|M_\xi - M_{\xi_n}\| \leq 729^{-n}. \quad (33.4)$$

The relations among the incidence projectors, their products, and their transport are preserved exactly. For an ordered product of contractions containing r numerical observables M_ξ , evaluated at arbitrary integer times, and any finite number of exact incidence or unitary factors, replacing those r factors by M_{ξ_n} changes the product in norm by at most

$$r 729^{-n}. \quad (33.5)$$

The same bound holds for any matrix element between unit vectors. The factors may be noncommutative; their order is preserved.

Proof. The norm of a multiplication operator on $\ell^2(X)$ is $\sup_{x \in X} |f(x)|$; the ternary tail proves the first bound. Conjugation by C^t and by $\mathcal{V}$ preserves the norm. For products $A_1 \cdots A_s$ and $B_1 \cdots B_s$, use

$$\prod_{i=1}^s A_i - \prod_{i=1}^s B_i = \sum_{j=1}^s B_1 \cdots B_{j-1} (A_j - B_j) A_{j+1} \cdots A_s.$$

The exact factors give zero difference; each of the remaining r contributes norm at most 729^{-n} . Submultiplicativity and the bound one for the other factors prove (33.5). Cauchy–Schwarz gives the assertion about matrix elements. $\square$

This uniformity preserves precision when the complete approximate observable is transported. The operator C and incidence remain exact in (33.4). The index n measures cylinder depth; t measures iterations of the transport d ; j in (33.3) measures compression iterations.

33.4 Covariance of the memory operator under return

Restrict X and its atomic representation to the graded reversible sector $X_{\text{rev,grad}}$ of the clock in (6.35). The integer memory degree q_{mem} is preserved in each history. For $d = \Gamma_9$, its update is $q_{\text{mem}}(dx) = q_{\text{mem}}(x) + 1$. The memory operator D is multiplication by q_{mem} , with domain

$$\text{Dom } D = \{v : \sum_{x \in X} |q_{\text{mem}}(x)|^2 |v_x|^2 < \infty\}.$$

The index change $x \mapsto dx$ and the inequalities $(q \pm 1)^2 \leq 2q^2 + 2$ prove that C and C^{-1} preserve this domain. On it,

$$C^*DC = D + I, \quad \widehat{C}^*\widehat{D}\widehat{C} = \widehat{D} + I_{\mathcal{M}}, \quad \text{Dom } \widehat{D} = \mathcal{V} \text{Dom } D. \quad (33.6)$$

Proof. On δ_x , the left-hand side equals $q_{\text{mem}}(dx)\delta_x = (q_{\text{mem}}(x) + 1)\delta_x$. The identity extends to the indicated domain by truncating the support in the graph norm of D . The second equality is obtained by unitary conjugation to $\mathcal{M}$. $\square$

By induction, $q_{\text{mem}}(d^N x) = q_{\text{mem}}(x) + N$. Every integer $N > 0$ changes the complete state. Every orbit of d is infinite; a square-summable function constant on each of these orbits is zero. In this bilateral realization, $P = 0$. Formula (33.3) then reconstructs the complete linear state from its memory complements. This conclusion uses the effective grading of the clock.

Phase return coexists with the memory increment in (33.6). Sturmian aperiodicity and its complexity belong to the symbolic factor in section 6; the present result preserves the graded dynamics of the complete state.

33.5 Incidence, refinement, and three regimes

The register K also has the complete realization in section 28, $\mathcal{F}k = (\mu(k), IR_0 k/6)$. Its inverse preserves the twelve coordinates, including the mean. Applying $\mathcal{F}$ to the corresponding fibres and retaining the history labels transports this representation. The map $\mathcal{F} \circ K$ preserves exactly the fibres of the reader K ; recovery of complete histories comes from r or from their effective labels.

The isometric refinement systems in sections 24, 26 and 28 preserve $\mathcal{V}$ and C through their intertwining squares. The proof uses the levelwise realizations constructed there. The atomic representation $\ell^2(X)$ and the projective-measure realizations L^2 retain their corresponding domains and measures.

The connection with folding and unfolding is developed in propositions 6.14 and 6.15: Coxeter, nilpotent transport, and the action of F_{av}^2 . For each two-dimensional realization E preserving the alternating form Ω , the nonadic chart satisfies

$$\Omega = T_E^* \Omega T_E + D_E^* \Omega D_E, \quad T_E = (8I + E)/9, \quad D_E = \sqrt{8(E - I)}/9. \quad (33.7)$$

The algebraic expansion cancels the cross terms and uses $E^* \Omega E = \Omega$. In the real realization, $*$ means transpose. The oriented law brings together the three regimes through the forms and transports of sections 30 and 31. Its realization as action also uses the normalizations and connections specified there.

The relation to connection dynamics is formulated through the constructions in sections 29 to 31. The identities (33.2)–(33.7) preserve their representation, the precision of the observables, and the transported forms.

34 Reversible coupling and nonstationary transport

34.1 Reversible extension of the balance between two forms

Let $B, C : V \rightarrow V'$ be isomorphisms preserving a nondegenerate form b between two realizations:

$$B^*b'B = C^*b'C = b.$$

The form may be a positive metric or an alternating form. On real spaces, $*$ denotes transposition with respect to the chosen coordinates; the equality expresses transport of the form between the specified realizations. On Hilbert spaces, the operators and their inverses are bounded.

With $s = \sqrt{8}$ and $Q_V = \frac{1}{3} \begin{pmatrix} sI & -I \\ I & sI \end{pmatrix}$, define

$$\mathcal{U}(B, C) = Q_V^* \text{diag}(B, C) Q_V = \frac{1}{9} \begin{pmatrix} 8B + C & \sqrt{8}(C - B) \\ \sqrt{8}(C - B) & B + 8C \end{pmatrix}. \quad (34.1)$$

Write $T = (8B + C)/9$, $D = \sqrt{8}(C - B)/9$, and $R = (B + 8C)/9$. The complete update is

$$\begin{pmatrix} x' \\ m' \end{pmatrix} = \begin{pmatrix} T & D \\ D & R \end{pmatrix} \begin{pmatrix} x \\ m \end{pmatrix}. \quad (34.2)$$

Theorem 34.1. *The map (34.1) is reversible and preserves the form $b \oplus b$:*

$$\mathcal{U}(B, C)^*(b' \oplus b')\mathcal{U}(B, C) = b \oplus b, \quad \mathcal{U}(B, C)^{-1} = \mathcal{U}(B^{-1}, C^{-1}). \quad (34.3)$$

Proof. We have $Q^*Q = I$ because $8 + 1 = 9$. Its scalar blocks make Q preserve $b \oplus b$ for any form b . The diagonal map preserves the forms by hypothesis. Composition proves (34.3), and inversion of the three factors gives the declared inverse. $\square$

For incoming memory $m = 0$, the balance $b = T^*b'T + D^*b'D$ is recovered. The second column determines the response to arbitrary incoming memory. Conservation of positive norms and conservation of oriented area are the respective specializations of (34.3).

34.1.1 Collective realization in the nine copies

On V^9 , let $W_E = \text{diag}(I, \dots, I, E)$, with E in the ninth position. Let $A(a, b) = (a/\sqrt{8}, \dots, a/\sqrt{8}, b)$ and $L = AQ$. Then $L(x, 0) = Jx$ and

$$W_E L = L \mathcal{U}(I, E). \quad (34.4)$$

Proof. $W_E A = A \text{diag}(I, E)$; multiplying by Q and using $AQ = L$ gives (34.4). Orthogonality of the mean of the first eight positions to their seven differences also shows that the complement remains fixed under W_E . $\square$

The original cyclic step is $S_E = P_{\text{cyc}} W_E$. Hence $S_E L = (P_{\text{cyc}} L) \mathcal{U}(I, E)$, with distinct input and output charts. The collective reduction preserves the corresponding pattern: $S_E^9 = I_9 \otimes E$, whereas $W_E^9 = \text{diag}(I_8, E^9)$. When a reader uses the nine individual labels, they are retained in their original representation.

34.2 Exact composition and memory re-entry

For $B_1, C_1 : V_0 \rightarrow V_1$ and $B_2, C_2 : V_1 \rightarrow V_2$,

$$\boxed{\mathcal{U}(B_2, C_2)\mathcal{U}(B_1, C_1) = \mathcal{U}(B_2B_1, C_2C_1).} \quad (34.5)$$

Proof. In (34.1), the factors $Q_{V_1} Q_{V_1}^*$ cancel, and the two diagonal blocks multiply in the indicated order. Induction extends the formula to any finite number of transports, preserving their order. $\square$

The reading block contains the identity

$$\boxed{T_2T_1 + D_2D_1 = \frac{8B_2B_1 + C_2C_1}{9}.} \quad (34.6)$$

Starting from $(x, 0)$, the first stage produces (T_1x, D_1x) . The second reading is $T_2T_1x + D_2D_1x$. The second term is exactly the contribution of the reintroduced memory. Storing D_1x outside the next coupling and continuing only with T_1x gives the separate-register procedure, whose reading is T_2T_1x . Equality (34.6) makes the difference between the two procedures precise.

The realization U preserves the coupling state and the ordered products $B_N \cdots B_1, C_N \cdots C_1$. Conservation of all prefixes additionally includes the HMT emission register, which retains each factor together with its order.

34.2.1 Covariant increment equation

For the separate register $x_{n+1} = T_nx_n, m_n = D_nx_n$, we have

$$x_{n+1} - B_nx_n = m_n/\sqrt{8}. \quad (34.7)$$

With $G_0 = I$ and $G_{n+1} = B_nG_n$, telescoping gives

$$x_0 = G_N^{-1}x_N - \frac{1}{\sqrt{8}} \sum_{n < N} G_{n+1}^{-1}m_n. \quad (34.8)$$

Memory is the increment relative to the transport between forms B_n , with the declared normalization. Equality (34.8) is a finite algebraic reconstruction; its passage to an infinite series requires convergence. The following theorem constructs the stable inverse through the limit of positive operators.

34.3 Memory of a nonstationary sequence

In a Hilbert realization H , let C_n be unitary, with their composition order declared. Write

$$T_n = (8I + C_n)/9, \quad D_n = \sqrt{8}(C_n - I)/9, \quad P_0 = I, \quad P_{n+1} = T_nP_n, \quad A_N = P_N^*P_N.$$

The index N counts compressions with separate storage, unlike complete transport with re-entry. For variable metrics with B_n isometric isomorphisms, conjugation by G_n takes the transports to a fixed Hilbert space and produces this same case, with $\tilde{C}_n = G_{n+1}^{-1}C_nG_n$. The assertion uses the fibre metrics and the declared isometries between them.

Theorem 34.2 (Positive terminal component and nonstationary recovery). *There exists a positive operator $0 \leq A_\infty \leq I$, the strong limit of A_N , and*

$$\boxed{I = A_\infty + \sum_{n \geq 0} P_n^* D_n^* D_n P_n.} \quad (34.9)$$

The map

$$\boxed{\mathcal{V}v = (A_\infty^{1/2}v, (D_nP_nv)_{n \geq 0})} \quad (34.10)$$

is isometric and has a stable left inverse

$$\mathcal{V}^*(a, (m_n)) = A_\infty^{1/2}a + \sum_{n \geq 0} P_n^* D_n^* m_n, \quad \|\mathcal{V}^*\| = 1. \quad (34.11)$$

Proof. The local identity $T_n^* T_n + D_n^* D_n = I$ gives $A_N - A_{N+1} = P_N^* D_N^* D_N P_N \geq 0$. The quadratic forms of the decreasing positive sequence converge to that of a positive operator A_∞ . For $B_N = A_N - A_\infty$, the inequality $0 \leq B_N^2 \leq B_N$ makes this convergence strong. Finite telescoping and the limit prove (34.9). Evaluation on v proves (34.10). Its adjoint is (34.11); the series converges in norm because truncations of a square-summable register converge and the adjoint is bounded. $\square$

Reconstruction with the limiting terminal component and N registers has exact error $(A_N - A_\infty)v$. The vector $A_\infty^{1/2}v$ is the canonical positive realization of the norm terminal component. Its existence follows from the Gram limit, independently of vector convergence of $P_N v$.

34.4 Three distinct properties: exhaustion, distinguishability, and stability

Let $Mv = (D_n P_n v)_n$. By (34.9), $M^* M = I - A_\infty$. Therefore:

1. The entire norm of v passes to the register exactly when $A_\infty v = 0$.
2. The register distinguishes every state exactly when $\ker(I - A_\infty) = \{0\}$.
3. Recovery from memory alone has a bounded inverse exactly when $I - A_\infty \geq \varepsilon I$ for some $\varepsilon > 0$.
In that case, a left inverse is $(I - A_\infty)^{-1} M^*$.

Moreover,

$$\ker M = \bigcap_{n \geq 0} \text{Fix}(C_n). \quad (34.12)$$

Proof. If $Mv = 0$, then $D_0 v = 0$, whence $C_0 v = v$ and $P_1 v = v$. Inductively, $D_n P_n v = 0$ gives $C_n v = v$ and $P_{n+1} v = v$. The converse is immediate. The three equivalences follow from the Gram matrix $M^* M$ and the characterization of operators bounded below. $\square$

34.4.1 An exact example separating the three properties

In the algebraic example $C_0 = -I$ and $C_n = I$ for $n \geq 1$,

$$A_\infty = \frac{49}{81}I, \quad M^* M = \frac{32}{81}I, \quad m_0 = -\frac{2\sqrt{8}}{9}v, \quad v = -\frac{9}{2\sqrt{8}}m_0. \quad (34.13)$$

Memory contains 32/81 of the squared norm and makes it possible to recover the complete vector. The terminal component retains 49/81 and is a positive operator distinct from a projector. Recoverability depends on the kernel and conditioning of the reader. Reading and memory constitute correlated components of an isometry.

This example separates the three logical consequences of the theorem. Application to a physical trajectory additionally preserves the selection and effective origin of its C_n through the TPK.

34.5 Quantitative criterion for complete transfer

For $P_{j,n} = T_{j-1} \cdots T_n$ and $P_{n,n} = I$, a window of length L has Gram matrix

$$G_{n,L} = \sum_{j=n}^{n+L-1} P_{j,n}^* D_j^* D_j P_{j,n} = I - P_{n+L,n}^* P_{n+L,n}.$$

If $G_{n,L} \geq \kappa I$, with $\kappa > 0$, in each consecutive block of length L , then

$$\|P_N\|^2 \leq (1 - \kappa)^{\lfloor N/L \rfloor}, \quad A_\infty = 0. \quad (34.14)$$

Each block contracts the squared norm by $1 - \kappa$; composition and contractivity of the remaining steps prove (34.14).

A sufficient criterion preceding the products is

$$\sum_{j=n}^{n+L-1} D_j^* D_j \geq \alpha I \implies G_{n,L} \geq \frac{\alpha}{[1 + 2(L-1)/9]^2} I. \quad (34.15)$$

Proof. For $y_j = P_{j,n} v$, we have $y_{j+1} - y_j = D_j y_j / \sqrt{8}$ and $\|D_j\| \leq 2\sqrt{8}/9$. Hence

$$\|D_j v\| \leq \|D_j y_j\| + \frac{2}{9} \sum_{i=n}^{j-1} \|D_i y_i\|.$$

The triangular matrix of partial sums has norm at most $L - 1$. Applying the window's ℓ^2 norm and the Gram hypothesis proves (34.15). $\square$

The absence of a common fixed vector gives distinguishability, whereas (34.15) adds a quantitative uniform separation. Neither condition is presumed merely by mentioning symmetries: each applies to the effective transports of the sequence.

34.6 Complete incidence realization and changes of regime

Section 28.1 constructs, for the twelve-coordinate register, $Fk = (\mu(k), \mathcal{I}P_0 k/6)$, with an inverse and weighted inner product on $Y = \mathbb{R} \oplus \mathcal{I}(\mathbf{1}^\perp)$. The transformation $F \oplus F$ takes the complete coupling to

$$\mathcal{U}_Y = (F' \oplus F') \mathcal{U}(B, C) (F^{-1} \oplus F^{-1}) = \mathcal{U}(F' B F^{-1}, F' C F^{-1}). \quad (34.16)$$

This identity preserves the second column, re-entry, and composition. For variable forms, the form is also transported through F . In the real realization of the twelve A_2 fibres, use $F \otimes \text{id}_{V_{A_2}}$, with $V_{A_2} = A_2 \otimes_{\mathbb{Z}} \mathbb{R}$. Its domain is the real space of fibres; preservation of the integer lattice corresponds to the lattice transport already proved. The map F recovers K and retains the remaining genealogical labels in the complete state.

The positive version of (34.9)–(34.15) uses unitary transports in the declared metrics. The alternating-form version of (34.1)–(34.8) admits the three Euler regimes. With $E = F_{\text{av}}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, we obtain $\det T = 89/81$, $\det D = -8/81$. The finite oriented sums telescope, and the area terminal components grow as $(89/81)^N$. The positive Hilbert limit uses the specific unitary hypotheses of Theorem 34.2.

The balance brings together forms and domains, reversible evolution with re-entry, and unlimited recovery in the positive realizations. The subsequent physical applications preserve the readers, connections, and scales of this operator composition.

35 Dimensional reconstruction and exterior conservation

35.1 Two memories recover the centred register

35.1.1 Register-reading operators

In the ordered chart of the register K from section 9 and theorem 9.14, let $V = \mathbb{R}^{12}$ be its subsequent linear realization and let S be the permutation $(Sx)_i = x_{i+1}$, with indices modulo 12. The readers $I - S^3$ and $I - S^4$ are those of the transverse extractor in theorem 9.14. The nonadic compression in section 34 provides, for $j = 3, 4$,

$$T_j = \frac{8I + S^j}{9}, \quad D_j = \frac{\sqrt{8}}{9}(S^j - I), \quad T_j^* T_j + D_j^* D_j = I. \quad (35.1)$$

The partition $8 + 1$ and normalization 9 come from the nine positions and their collective contrast. Consider the following explicit composition of readers, applied to the generated register:

$$m_0 = D_3 K, \quad m_1 = D_4 T_3 K, \quad MK = (m_0, m_1). \quad (35.2)$$

The second memory acts on the reading produced by the first stage. The complements are archived separately. The complete evolution with re-entry corresponds to the reversible coupling in section 34.

35.1.2 Exact inverse

Since $(S^3)^4 = I$,

$$T_3^{-1} = \frac{512I - 64S^3 + 8S^6 - S^9}{455}. \quad (35.3)$$

Indeed, $(8I + S^3)(512I - 64S^3 + 8S^6 - S^9) = 4095I$ and $4095 = 9 \cdot 455$. Hence the memories provide

$$b = -\frac{9}{\sqrt{8}}m_0 = (I - S^3)K, \quad c = -\frac{9}{\sqrt{8}}T_3^{-1}m_1 = (I - S^4)K. \quad (35.4)$$

Commutation of T_3 with S^4 justifies the second equality. Define

$$w_i = c_i - b_{i+1}, \quad B_0 = 0, \quad B_i = \sum_{j < i} w_j.$$

The identity $w_i = K_i - K_{i+1}$ proves that $B_i = K_0 - K_i$. If $q = \sum_{i=0}^{11} K_i$, we recover exactly

$$K_i = \frac{q + \sum_{j=0}^{11} B_j}{12} - B_i. \quad (35.5)$$

This is the inverse (9.28), now composed with compression and memory. With $q = 0$, the formula returns $R_{11}K$, where

$$R_{11} = I - \frac{1}{12}\mathbf{11}^*.$$

The two memories recover the eleven zero-mean components. The uniform charge is recovered through q . The kernel of M is precisely the uniform line: vanishing of both differences implies fixation by S^3 and S^4 ; since $S = S^4(S^3)^{-1}$, it implies fixation by S .

35.1.3 Optimal stability bound

The Gram matrix of the procedure is

$$G = M^*M = D_3^*D_3 + T_3^*D_4^*D_4T_3 = I - (T_4T_3)^*(T_4T_3). \quad (35.6)$$

In mode k of S , the eigenvalue of $T_j^*T_j$ is

$$\frac{65 + 16 \cos(2\pi jk/12)}{81}.$$

Substituting it into the preceding Gram matrix gives, with multiplicities,

$$\text{Spec } G = \{0^{[1]}, (16/81)^{[2]}, (8/27)^{[2]}, (32/81)^{[1]}, (952/2187)^{[4]}, (1256/2187)^{[2]}\}. \quad (35.7)$$

The zero mode is the mean. Let L be the least-squares inverse of M on $\mathbf{1}^\perp$, extended to the register space by

$$L = (G|_{\mathbf{1}^\perp})^{-1}M^*.$$

Then

$$LM = P_{11}, \quad \|L\| = \frac{9}{4}. \quad (35.8)$$

The bound is optimal because modes $k = 3, 9$ attain $16/81$. For perturbed data and reconstruction with L ,

$$\|\delta K\|^2 \leq \frac{|\delta q|^2}{12} + \frac{81}{16}(\|\delta m_0\|^2 + \|\delta m_1\|^2). \quad (35.9)$$

The uniform and centred components are orthogonal. Formula (35.5) is an exact inverse on compatible data; (35.8) specifies the stable extension used for noisy data, which need not satisfy the compatibility conditions of (35.5).

35.1.4 Example using the expressed register

The register in (9.23) is

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601). \quad (35.10)$$

It is an output of the preceding generator, used as an application case. The coefficients and readers are fixed before this register is evaluated. To store the memories using rational arithmetic, write $z_i = m_i/\sqrt{8}$. The two effective registers are

$$9z_0 = (495, 116, 684, -108, -601, 90, 173, 88, -313, -560, 397, -461),$$

$$81z_1 = (2729, 2503, 3818, -5843, 2583, -920, -4051, 4338, -5312, -1583, 233, 1505). \quad (35.11)$$

Substituting these registers into the reconstruction formulas, with $q = 6263$, reproduces the twelve components of K . With $q = 0$, it reproduces $K - (6263/12)\mathbf{1}$. The supplementary certificate first reconstructs the register from the memories and then compares it with (9.23).

Adding a third memory $D_3T_4T_3K$ raises the smallest positive eigenvalue of the Gram matrix to $8/27$. The optimal amplification decreases from $9/4$ to $\sqrt{27/8}$. A third memory $D_4T_4T_3K$ retains $16/81$. The choice between repeating step 3 or step 4 has a stability effect calculable before evaluating the register.

35.2 Reconstruction of the dimensional projector from two memories

35.2.1 Composition with the incidence already constructed

The exceptional chart in section 16.5 defines the orthogonal representation of A_5 on the twelve classes of A_5/C_5 , with explicit representatives and character. Its three-dimensional isotypic projector is (16.23):

$$P_3 = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \rho(a), \quad P_3 P_{11} = P_3, \quad P_3 \mathbf{1} = 0. \quad (35.12)$$

The generated register precedes this representation. The result in lemma 16.10, together with (16.25) and (16.26), establishes

$$u_K = P_3 P_{11} K, \quad \|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0, \quad (35.13)$$

$$P_{10}(K) = P_{11} - \frac{u_K u_K^*}{\|u_K\|^2}.$$

The new composition of (35.8), (35.12), and (35.13) is

$$u_K = P_3 L(m_0, m_1), \quad P_{10}(K) = P_{11} - \frac{P_3 L(m) [P_3 L(m)]^*}{\|P_3 L(m)\|^2}. \quad (35.14)$$

Proof. The identities $LM = P_{11}$ and $P_3 P_{11} = P_3$ give $P_3 LMK = P_3 K = u_K$. Nonvanishing is evaluated in (16.25). The rank-one quotient therefore coincides with that of (16.26). The composition recovers the complete rank-ten orthogonal projector from the two memories, independently of the charge q and terminal reading $T_4 T_3 K$. $\square$

The reduction $12 \rightarrow 11$ removes the uniform direction, and the reduction $11 \rightarrow 10$ uses the vector selected by K . The composition recovers the full projector matrix on the linear incidence carrier. Its dimensional realizations retain the maps of proposition 16.12.

35.2.2 Projector error

Let ε be the joint norm of the error in the two memories, and let $\hat{u} = P_3 L(m + \delta m)$. Then $\|\hat{u} - u_K\| \leq 9\varepsilon/4$. If $9\varepsilon/4 < \|u_K\|$, the vector $\hat{u}$ is nonzero and

$$\|\hat{P}_{10} - P_{10}\| \leq \frac{9\varepsilon}{4\|u_K\|}. \quad (35.15)$$

Proof. The operator-norm distance between two orthogonal projectors onto lines is the sine of the angle between them. The distance from u_K to the line of $\hat{u}$ is at most $\|u_K - \hat{u}\|$. Division by $\|u_K\|$ proves the bound. Here $\|u_K\| \approx 765,733$, and the coefficient is approximately 0,00293836. $\square$

For registers close to $\ker P_3$, stability of the direction depends on $\|u_K\|$. The bound $9/4$ for the centred register remains valid.

35.2.3 A scalar descriptor of K , with explicit invariances

The register K , its scalar readings, and its invariants are distinguished. For $K \notin \mathbb{R}\mathbf{1}$, define

$$\eta_K = \frac{\|P_3 P_{11} K\|^2}{\|P_{11} K\|^2}. \quad (35.16)$$

For the register (9.23), $\sum_i K_i^2 = 4251257$ and $\|P_{11} K\|^2 = 11789915/12$. Thus,

$$\eta_K = \frac{3(6638585 + 2275584\sqrt{5})}{5 \cdot 11789915} = \frac{13089003 + 13653504\varphi_{\text{HMT}}}{58949575} \approx 0,5967954228259091. \quad (35.17)$$

The second expression uses the subsequent quadratic recognition of φ_{HMT} . We have $0 \leq \eta_K \leq 1$. The ratio is invariant under $K \mapsto aK + b\mathbf{1}$, with $a \neq 0$, and under simultaneous orthogonal changes of K, P_{11}, P_3 . In the fixed chart, it is invariant under $\rho(A_5)$, since both projectors commute with that representation. It is recoverable from the memories because both norms defining it are recoverable. It measures the quadratic fraction of the centred register lying in the chosen isotypic sector. The complete register also preserves the information identified by this scalar descriptor. The provenance of this additional formalization is documented in the supplement; the affine invariances proved here belong to the linear realization.

35.3 Application to an effective vector of the exceptional construction

The construction in section 16.13 and proposition 16.21 uses twelve A_2 fibres, their gluing, and the lattice neighbour. The tritic word is $c^\star = (1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1)$. Its blocks act by C^{-1} when the component is 1 and by C when it is 2, in the charts specified by equations (16.33) and (16.34). The resulting operator g satisfies $g^3 = I$, $g^2 + g + I = 0$, and is orthogonal for the sum of the A_2 metrics. Descent to the gluing and the neighbour is proved in proposition 16.21. The following calculations act on its real carrier of dimension 24.

In each fibre,

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

On the sum of the twelve fibres, define

$$T = \frac{8I + g}{9}, \quad D = \frac{\sqrt{8}}{9}(g - I).$$

With the adjoint relative to that metric, $g^* = g^{-1} = -g - I$, and we obtain

$$T^*T = \frac{19}{27}I, \quad D^*D = \frac{8}{27}I, \quad D^{-1} = -\frac{3}{\sqrt{8}}(g + 2I). \quad (35.18)$$

The inverse comes from $(g - I)(g + 2I) = -3I$. The neighbour construction fixes the vector in (16.28), $v = 4\rho_1 + \rho_2 + \dots + \rho_{12}$, with a root ρ_b of squared norm 2 in each fibre. Therefore,

$$\|v\|^2 = 54, \quad \|Tv\|^2 = 38, \quad \|Dv\|^2 = 16, \quad v = -\frac{3}{\sqrt{8}}(g + 2I)Dv. \quad (35.19)$$

A single complement recovers this complete vector. The reading preserves 38 units of squared norm and the register contains 16; their sum is 54. The result uses the vector in (16.28) and its lattice fibres.

35.4 Conservation law in every exterior degree

35.4.1 Statement and proof

Let $P_0 = I$, $P_n = T_{n-1} \cdots T_0$, and

$$\mathcal{J}_N v = (P_N v, D_0 P_0 v, \dots, D_{N-1} P_{N-1} v).$$

The law in sections 31 and 34 preserves the form G_0 through the orthogonal sum of the corresponding forms: $\mathcal{J}_N^* G_{\text{sal}} \mathcal{J}_N = G_0$. For each admissible finite exterior degree p , it follows that

$$(\Lambda^p \mathcal{J}_N)^* (\Lambda^p G_{\text{sal}}) (\Lambda^p \mathcal{J}_N) = \Lambda^p G_0. \quad (35.20)$$

The induced form is defined by

$$\langle x_1 \wedge \cdots \wedge x_p, y_1 \wedge \cdots \wedge y_p \rangle_{\Lambda^p G} = \det(\langle x_i, y_j \rangle_G)_{i,j=1}^p.$$

Proof. Substitute the degree-one balance into each entry of this determinant. The equality holds on decomposable vectors and extends bilinearly because these span the exterior power. $\square$

The relevant decomposition is

$$\Lambda^p \left(\bigoplus_a H_a \right) \simeq \bigoplus_{\sum n_a = p} \bigotimes_a \Lambda^{n_a} H_a.$$

In the positive case, if $\mathcal{J}_{p,\mathbf{n}}$ denotes the components of that map,

$$\|\xi\|^2 = \sum_{\sum n_a = p} \|\mathcal{J}_{p,\mathbf{n}} \xi\|^2. \quad (35.21)$$

The purely visible, purely registered, and mixed components are preserved. For $p = 2$ and $p = 3$, the squared area and volume norms, respectively, are preserved. For an alternating or indefinite form, the exterior identity retains its bilinear interpretation.

35.4.2 Exact distribution among the exterior sectors

In the preceding Coxeter realization, $a = 19/27$ and $b = 8/27$. Normalizing T and D gives two isometries, so the component with k directions in memory has squared norm

$$\|\mathcal{J}_{p,k} \xi\|^2 = \binom{p}{k} a^{p-k} b^k \|\xi\|^2. \quad (35.22)$$

This is proved by exterior expansion of $x \mapsto (\sqrt{a}x, \sqrt{b}x)$. Components with distinct memory positions are orthogonal; the normalized isometries T and D preserve their norms.

For a unit-norm bivector,

Sector	Squared norm
Two directions in the reading	361/729
One direction in the reading and one in memory	304/729
Two directions in memory	64/729

The sum is one. The mixed sector contains exactly 304/729, approximately 41.7%, of the squared-area identity. In degree three, the four components are (6859, 8664, 3648, 512)/19683; the mixed components sum to 12312/19683.

The law applies to $0 \leq p \leq 24$ on the carrier of the twelve A_2 fibres, and functorially to other generated carriers. Exterior degree, refinement resolution, the scale of a metric, and physical dimension have distinct typings. The change $G \mapsto s^2 G$, with $s > 0$, multiplies the squared norm of degree p by s^{2p} ; both sides of the exterior identity transform in the same way. The realization of dimensions 10 and 11 preserves the previously specified dimensional maps.

35.4.3 Naturality and arbitrary depth

The generated refinement of histories preserves measure through

$$R\delta_h = \sum_{\epsilon \in \text{Out}(h)} \sqrt{p_h(\epsilon)} \delta_{h\epsilon}.$$

The probabilities come from the multiplicities and normalization of the APP–TRIT–TPK emissions in section 5.2 and section 24. The register identifies the preceding state of each emission. If the transport square is $\mathcal{J}'_N R = (\bigoplus_a R_a) \mathcal{J}_N$, then

$$(\Lambda^p \mathcal{J}'_N)(\Lambda^p R) = \Lambda^p(\bigoplus_a R_a)(\Lambda^p \mathcal{J}_N). \quad (35.23)$$

Apply $\Lambda^p(AB) = \Lambda^p(A)\Lambda^p(B)$. The squares of the five correlative operations and their common limit are those of propositions 5.7 and 5.8 and section 5.9; all act within the same construction of the continuum.

In the nonstationary positive case, section 34 proves $P_N^* P_N \downarrow A_\infty$ strongly, and

$$\mathcal{J}v = (A_\infty^{1/2}v, (D_n P_n v)_{n \geq 0}), \quad \mathcal{J}^* \mathcal{J} = I.$$

By density, $\Lambda^p \mathcal{J}$ is isometric for each finite degree p . The exterior sum extends to finitely supported multi-indices over the countable Hilbert direct sum; Parseval justifies passage to the limit. The terminal component $A_\infty^{1/2}$ and the memories jointly preserve the initial form at arbitrary depth.

35.4.4 Precision of area and volume reconstruction

For the reader M , order the eleven positive eigenvalues of G , with multiplicity, as $\lambda_1 \leq \dots \leq \lambda_{11}$. In degrees $1 \leq p \leq 11$ of $\mathbf{1}^\perp$, the singular-value decomposition induces

$$\|(\Lambda^p M)^\dagger\| = (\lambda_1 \dots \lambda_p)^{-1/2}. \quad (35.24)$$

The exterior modes use p distinct directions. The computed spectrum gives the optimal constants 9/4, 81/16, 243 $\sqrt{6}$ /64, and 2187/128 for $p = 1, 2, 3, 4$. In degree twelve, $\Lambda^{12} M = 0$; recovery of the complete volume also incorporates the uniform charge. The tensor product of p independent readers on the zero-mean subspace has bound $(9/4)^p$. Tensoring the reader with an identity preserves the bound 9/4. The complete register $\mathcal{J}$ has an inverse of norm one on its image in every degree.

35.5 Transport memory and relative holonomy

The reversible coupling in section 34 is

$$U(B, C) = \frac{1}{9} \begin{pmatrix} 8B + C & \sqrt{8}(C - B) \\ \sqrt{8}(C - B) & B + 8C \end{pmatrix} = Q^* \text{diag}(B, C)Q,$$

with Q constant and orthogonal, fixed by the partition $8 + 1$. Hence $U(B_2, C_2)U(B_1, C_1) = U(B_2B_1, C_2C_1)$, preserving the order of factors. On a loop γ , let H_B, H_C be the ordered products of the two transports. The memory block of the complete evolution is

$$D_\gamma = \frac{\sqrt{8}}{9}(H_C - H_B).$$

We reconstruct exactly

$$\boxed{H_B^{-1}H_C = I + \frac{9}{\sqrt{8}}H_B^{-1}D_\gamma.} \quad (35.25)$$

The equality is an operator equality: recording the block's response on a basis recovers the complete relative holonomy. Its evaluation on a vector recovers the corresponding action. Under a chart change at the base point, both sides are conjugated. If $H_B = I$, the complement determines $H_C - I$. This relates memory and transport defect on a circuit; the differential realization additionally uses its connection and the limit of circuits.

35.5.1 An elliptic–parabolic loop produces hyperbolic self-scaling

The construction in proposition 6.14 contains the Coxeter operator C and the nilpotent

$$N_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

on their respective carriers. Consider their realizations on a common oriented plane, using the indicated bases and the form

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The Coxeter operator C and transport $I + N_0$ have determinant one and preserve Ω . This chart allows them to be composed while preserving the distinction between the generators of their regimes and their own positive metrics.

Let

$$P = I + N_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Since $N_0^2 = 0$, we have $P^n = I + nN_0$ for every $n \in \mathbb{Z}$. The ordered loop of four transports gives

$$H_n = C^{-1}P^{-n}CP^n = \begin{pmatrix} 1+n & n^2 \\ n & n^2 - n + 1 \end{pmatrix}. \quad (35.26)$$

For $n \neq 0$, its normalized difference is

$$F_n = \frac{H_n - I}{n} = \begin{pmatrix} 1 & n \\ 1 & n - 1 \end{pmatrix}.$$

Multiplying these matrices gives, without introducing spectral values,

$$\boxed{F_n^2 = H_n, \quad F_n^2 - nF_n - I = 0.} \quad (35.27)$$

With $H_B = I$, memory is $D_{\gamma,n} = (\sqrt{8}/9)nF_n$. Consequently,

$$\boxed{H_n = \left(\frac{9}{n\sqrt{8}}D_{\gamma,n} \right)^2.} \quad (35.28)$$

The matrix-valued memory response reconstructs hyperbolic self-scaling. For $n > 0$, subsequent spectral recognition identifies

$$\rho_n = \frac{n + \sqrt{n^2 + 4}}{2}, \quad \text{Spec } F_n = \{\rho_n, -\rho_n^{-1}\}, \quad \text{Spec } H_n = \{\rho_n^2, \rho_n^{-2}\}. \quad (35.29)$$

The case $n = 1$ produces the golden polynomial. The relation to the operator $F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ generated by the calendar in (6.15) preserves the chart change:

$$F_1 = PF_{\text{av}}P^{-1}, \quad H_1 = PF_{\text{av}}^2P^{-1}. \quad (35.30)$$

The transport P , of determinant one, intertwines the two matrices and preserves Ω . The case $n = 2$ produces the spectral readings $1 + \sqrt{2}$ and $3 + 2\sqrt{2}$. The composition relates these realizations while preserving their respective genealogies; the selection of a four-factor word remains specified as part of the reader.

In the same example $n = 1$, the oriented balance is

$$D_\gamma^\top \Omega D_\gamma = -\frac{8}{81} \Omega, \quad T_\gamma^\top \Omega T_\gamma = \frac{89}{81} \Omega. \quad (35.31)$$

The sum of the two contributions is Ω . The memory term has opposite orientation: the visible contribution $89/81$ is compensated by $-8/81$. This alternating balance and the positive allocation $19/27 + 8/27 = 1$ preserve their respective forms.

The expression for H_n holds for every $n \in \mathbb{Z}$. The definition of F_n and reconstruction through its square use $n \neq 0$; the preceding spectral parametrization uses $n > 0$. In general,

$$D_{\gamma,n}^\top \Omega D_{\gamma,n} = -\frac{8n^2}{81} \Omega, \quad T_{\gamma,n}^\top \Omega T_{\gamma,n} = \left(1 + \frac{8n^2}{81}\right) \Omega.$$

The identities $N_0^2 = 0$ and $C^2 + C + I = 0$ prove the complete collection; the supplementary program checks its exact specializations. The provenance of this additional formalization is documented in the supplement.

35.6 Symmetry, action, and conserved charge

Let $(E_n)_{n \in \mathbb{Z}}$ be finite-dimensional real vector spaces and $U_n : E_n \rightarrow E_{n+1}$ the invertible transports of the complete evolution. The cotangent lift acts on $q_n \in E_n$ and $p_n \in E_n^*$. In dual bases, define the auxiliary discrete Lagrangian

$$L_n(q_n, q_{n+1}, p_{n+1}) = p_{n+1}^\top (q_{n+1} - U_n q_n).$$

The action is used in a local sense: for each finitely supported variation, sum L_n over a finite interval containing all affected terms. The first variation is independent of any enlargement of that interval. Specifically,

$$\delta S = \sum_{n \in \mathbb{Z}} [\delta p_n^\top (q_n - U_{n-1} q_{n-1}) + \delta q_n^\top (p_n - U_n^\top p_{n+1})], \quad (35.32)$$

where the sum is finite for every admitted variation. Stationarity with respect to all these variations is equivalent to

$$q_{n+1} = U_n q_n, \quad p_n = U_n^\top p_{n+1} \quad (n \in \mathbb{Z}).$$

The second equation is $p_{n+1} = U_n^{-\top} p_n$, that is, the cotangent lift of transport.

Proposition 35.1 (Variational charge of intertwined transport). *Let $A_n \in \text{End}(E_n)$ be a collection satisfying*

$$A_{n+1}U_n = U_nA_n.$$

On the stationary solutions of (35.32), the charge

$$\boxed{J_n = p_n^\top A_n q_n = J_{n+1}} \quad (35.33)$$

is constant.

Proof. The evolution equations and intertwining give

$$J_{n+1} = p_{n+1}^\top A_{n+1} U_n q_n = p_{n+1}^\top U_n A_n q_n = p_n^\top A_n q_n.$$

Conservation also has an explicit variational derivation. For constant parameter ε , the transformations

$$q_n \mapsto e^{\varepsilon A_n} q_n, \quad p_n \mapsto e^{-\varepsilon A_n^\top} p_n$$

preserve every L_n , because $e^{\varepsilon A_{n+1}} U_n = U_n e^{\varepsilon A_n}$. If the parameter is replaced by a finitely supported sequence (ε_n) , the first variation is

$$\delta L_n = (\varepsilon_{n+1} - \varepsilon_n) p_{n+1}^\top U_n A_n q_n.$$

On a solution, the last factor is J_n . Finite summation by parts gives

$$\delta S = \sum_{n \in \mathbb{Z}} \varepsilon_n (J_{n-1} - J_n).$$

Stationarity for every finitely supported sequence produces (35.33). □

For the preceding cyclic readers, take

$$A_0 = S - S^{-1}, \quad \mathcal{A} = \text{diag}(A_0, A_0)$$

on the complete carrier $V \oplus V$. The matrix $\mathcal{A}$ commutes with $U(I, S^3)$ and $U(I, S^4)$, since its blocks are polynomials in S . Consequently, $p_n^\top \mathcal{A} q_n$ preserves an oriented correlation between neighbouring positions.

The construction specifies a Lagrangian, its symmetry, and the corresponding conserved charge, in the variational sense developed in section 30. Its domain is the cotangent realization of the readers. The physical action and its units retain the operators and normalizations of the corresponding constructions. The auxiliary Lagrangian is obtained after the transport U_n .

36 Canonical realization of covariance

36.1 Canonical covariance and reciprocity of the action ellipse

The preceding construction determines two complementary invariants: the product of the semiaxes fixes the action area, and their quotient preserves the anisotropy of the angular pair. We now specify the canonical realization in which those semiaxes are two standard deviations. The starting structure is the ellipse already obtained from the angular and action representations of APP–TRIT–TPK; the operator realization is established after that construction.

In this subsection we write $\eta = \eta_{\text{el}} = \text{artanh}(C^*/A)$, with $|C^*/A| < 1$ and $\hbar > 0$. The angular quotient uses the same unit for its two components. In the balanced coordinates (Q, P) , both having the dimension of the square root of action, define

$$V_\eta := \frac{1}{4} \begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}. \quad (36.1)$$

This positive matrix has entries of action dimension. Its interpretation as quantum covariance is specified by the following realization.

Proposition 36.1 (Gaussian realization of action covariance). *Consider the canonical representation on $L^2(\mathbb{R}, dQ)$, with $\hat{Q}f(Q) = Qf(Q)$ and $\hat{P}f(Q) = -i\hbar f'(Q)$, initially on the Schwartz space $\mathcal{S}(\mathbb{R})$. The normalized state*

$$\psi_\eta(Q) = (\pi\hbar e^\eta)^{-1/4} \exp\left(-\frac{Q^2}{2\hbar e^\eta}\right) \quad (36.2)$$

has zero means and covariance matrix V_η . In particular,

$$(\Delta Q)^2 = \frac{\hbar e^\eta}{2}, \quad (\Delta P)^2 = \frac{\hbar e^{-\eta}}{2}, \quad \text{Cov}(Q, P) = 0, \quad \det V_\eta = \frac{\hbar^2}{4}. \quad (36.3)$$

This realization saturates the Robertson–Schrödinger inequality.

Proof. The preceding operators are essentially self-adjoint on $\mathcal{S}(\mathbb{R})$, which is a common invariant core; on this space, $[\hat{Q}, \hat{P}] = i\hbar I$. For $\hat{Y} = (\hat{Q}, \hat{P})$, the symmetric covariance is

$$V_{jk} = \frac{1}{2} \langle (\hat{Y}_j - \langle \hat{Y}_j \rangle)(\hat{Y}_k - \langle \hat{Y}_k \rangle) + (\hat{Y}_k - \langle \hat{Y}_k \rangle)(\hat{Y}_j - \langle \hat{Y}_j \rangle) \rangle.$$

With $s = \hbar e^\eta > 0$, let $I_s = \int_{\mathbb{R}} e^{-Q^2/s} dQ$. The product of two equal integrals, evaluated in polar coordinates, gives $I_s^2 = 2\pi \int_0^\infty r e^{-r^2/s} dr = \pi s$. Furthermore, the integral of $\frac{d}{dQ}(Qe^{-Q^2/s})$ is zero, by decay at both ends; hence $\int_{\mathbb{R}} Q^2 e^{-Q^2/s} dQ = sI_s/2$. These identities and parity prove normalization, zero mean, and $\langle \hat{Q}^2 \rangle = s/2$. Moreover, $\hat{P}\psi_\eta = ie^{-\eta}Q\psi_\eta$. Thus $\langle \hat{P} \rangle = 0$, $\|\hat{P}\psi_\eta\|^2 = \hbar e^{-\eta}/2$, and $\text{Re}\langle \hat{Q}\psi_\eta, \hat{P}\psi_\eta \rangle = 0$.

For any normalized state in that domain, Cauchy–Schwarz applied to $(\hat{Q} - \langle \hat{Q} \rangle)\psi$ and $(\hat{P} - \langle \hat{P} \rangle)\psi$ gives

$$(\Delta Q)^2(\Delta P)^2 \geq \text{Cov}(Q, P)^2 + \frac{1}{4} |[\hat{Q}, \hat{P}]|^2 = \text{Cov}(Q, P)^2 + \frac{\hbar^2}{4}.$$

The computed variances attain equality. Equivalently, the operator

$$a_\eta = \frac{e^{-\eta/2}\hat{Q} + ie^{\eta/2}\hat{P}}{\sqrt{2\hbar}}$$

satisfies $[a_\eta, a_\eta^\dagger] = I$ and $a_\eta\psi_\eta = 0$, by direct substitution. $\square$

Proposition 36.2 (Area, dispersion, and reciprocal transport). *The action ellipse is exactly the covariance sublevel set*

$$\mathcal{E}_\eta = \{y = (Q, P)^\top : y^\top V_\eta^{-1} y \leq 4\} = \left\{ \frac{Q^2}{a_S^2} + \frac{P^2}{b_S^2} \leq 1 \right\}. \quad (36.4)$$

Its area is $h = 2\pi\hbar$, its semiaxes are $a_S = 2\Delta Q$, $b_S = 2\Delta P$, and

$$\frac{\Delta P}{\Delta Q} = e^{-\eta} = \frac{b_S}{a_S} = R_{\text{el}}^2. \quad (36.5)$$

The canonical rotation J_2 of Section 20 transports V_η to $V_{-\eta}$ and preserves the symplectic form.

Proof. Inverting the diagonal matrix (36.1) gives (36.4). Its semiaxes are the positive roots of four times its variances, and

$$\text{Area}(\mathcal{E}_\eta) = 4\pi\sqrt{\det V_\eta} = 2\pi\hbar = h,$$

in agreement with the areal law (20.4). The ratio of the positive roots proves (36.5). If $M_\eta = \text{diag}(e^{\eta/2}, e^{-\eta/2})$, then

$$V_\eta = M_\eta V_0 M_\eta^\top, \quad M_\eta^\top J_2 M_\eta = J_2, \quad J_2 V_\eta J_2^\top = V_{-\eta}.$$

These identities follow by multiplication. In operator terms, $(\hat{Q}', \hat{P}') = (-\hat{P}, \hat{Q})$ preserves $[\hat{Q}', \hat{P}'] = i\hbar I$. The permutation $S_2 : (Q, P) \mapsto (P, Q)$ also exchanges the variances but reverses the sign of the commutator and of the symplectic form. Thus covariance transport respects the distinction between canonical reciprocity and orientation reversal already established in Section 20. $\square$

Position, momentum, and dimensional transduction. Let $\lambda_{xp} > 0$ be a factor of the dimensional chart, with dimension momentum per unit length. The transformation

$$Q = \sqrt{\lambda_{xp}} x, \quad P = \frac{p_x}{\sqrt{\lambda_{xp}}}$$

realizes the balanced pair as a position x and its conjugate momentum p_x ; it preserves $dQ \wedge dP = dx \wedge dp_x$ and $[\hat{x}, \hat{p}_x] = i\hbar I$. By transformation of covariance,

$$V_\eta^{(x, p_x)} = \frac{\hbar}{2} \begin{pmatrix} e^\eta/\lambda_{xp} & 0 \\ 0 & \lambda_{xp}e^{-\eta} \end{pmatrix}, \quad \Delta x \Delta p_x = \frac{\hbar}{2}, \quad \frac{\Delta p_x}{\Delta x} = \lambda_{xp}e^{-\eta}. \quad (36.6)$$

The dispersion quotient here acquires the dimension momentum per length. By contrast, (36.5) belongs to the balanced chart. For the inductive–capacitive realization already defined, (20.13) supplies the dimensionless identity $Z_\eta/Z_* = e^{-\eta} = \Delta P/\Delta Q$. Each reading uses its declared transduction and preserves the same action determinant.

Contour energy and mean energy of the state. The inductive–capacitive realization has Hamiltonian

$$H_\eta(Q, P) = \frac{\omega}{2} (e^{-\eta} Q^2 + e^\eta P^2), \quad \omega > 0.$$

On $\partial\mathcal{E}_\eta$, its value is $H_\eta = \hbar\omega$, by the semiaxis equations. In the quantum realization of Proposition 36.1,

$$\langle \hat{H}_\eta \rangle_{\psi_\eta} = \frac{\omega}{2} \left(e^{-\eta} \frac{\hbar e^\eta}{2} + e^\eta \frac{\hbar e^{-\eta}}{2} \right) = \frac{\hbar\omega}{2}. \quad (36.7)$$

The two-standard-deviation contour therefore has twice the mean energy of this Gaussian state. The areal law fixes action normalization; the angular pair fixes its distribution between the two coordinates; the canonical representation and explicit state realize that distribution as covariance. Saturation of uncertainty is thus established in the realization specified by its operators and its state.

37 Sheet return, areal–volumetric incidence and constitutive orientation

The sheet structure enters before electromagnetic quantities are evaluated. Its function is to distinguish closure of a projection, restoration of an orientation and evolution of the transported register. These operations act on related objects of different types: a projected trajectory may have closed while its lift remains on the opposite sheet; orientation may have been restored while memory preserves the two windings performed. The constitutive construction must receive this information together with the structure determining it.

37.1 Projected return and restoration of the oriented cover

Let ℓ be a return loop of a surviving extension, and let $\varepsilon : \langle \ell \rangle \rightarrow \{+1, -1\}$ be the character of its lift, with $\varepsilon(\ell) = -1$. The associated interval bundle has the presentation

$$\mathcal{M} = ([0, 1] \times [-1, 1]) / (0, u) \sim (1, -u). \quad (37.1)$$

The decisive datum is the sign of transport around the specified loop. The identification of endpoints geometrically realizes that character.

Proposition 37.1 (Two closure orders). *One winding of the loop closes the base and reverses the fibre orientation. Two windings restore orientation. In the angular parametrization of the boundary readout, these returns correspond, respectively, to the areal closure 2π and the volumetric closure 4π of the oriented cover.*

Proof. The equivalence relation of (37.1) identifies the terminal fibre with the initial one by $u \mapsto -u$. Composing two transports gives $u \mapsto -(-u) = u$. Parametrizing one winding of the base by 2π assigns 4π to its square. The statement concerns this lift and this readout. $\square$

TPK memory also preserves the trajectory traversed and the register updates. The square of the sign character is the identity, but it does not impose vanishing of these updates. The spinorial representation, when used, must transport this datum through its own map.

37.2 Descent of incidence from the tritic calendar

In the primitive block $(+1, -1, 0)$, the TPK updates the arithmetic mode through $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$ and $0 : m \mapsto m$. Its cyclic orbit is (Σ, Π, Π) . The sheet readout assigns Σ to the areal sheet and Π to the volumetric sheet. In the basis order $(e_{\text{ar}}, e_{\text{vol}})$, edge incidence is represented by

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (37.2)$$

The three consecutive pairs are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$ and $\Pi \rightarrow \Sigma$, each with multiplicity one. The calendars of lengths 9, 27 and 108 contain 3, 9 and 36 copies of the primitive block. Their incidence matrices are therefore these multiples of F_{av} .

This construction preserves the distinction between chronological frequency and the measure of the continuation space. The former assigns weights $(1/3, 2/3)$ to the mode cycle. The latter is defined on the minimal memory-one symbolic envelope admitting exactly the preceding pairs. Its adjacency matrix is F_{av} ; trajectory counts belong to that envelope.

Proposition 37.2 (Return weights of the two sheets). *Let $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$. For $n \geq 1$,*

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}. \quad (37.3)$$

The Parry measure of this envelope assigns the sheets weights proportional to $(1, \varphi^2)$, and the ratio of diagonal counts converges to φ^{-2} .

Proof. The power formula follows by induction using the recurrence. The square of F_{av} is strictly positive. Its positive Perron vector is $(1, \varphi)$, with eigenvalue φ , since $\varphi^2 = \varphi + 1$. As the matrix is symmetric, the Parry weights are proportional to the squares of that vector's components. The other root of the characteristic polynomial is $-\varphi^{-1}$; its modulus is strictly less than φ , giving the limit of the diagonal ratios. $\square$

Spectral identification of this incidence follows its construction from the calendar. The regional coordinate φ retains its earlier coinductive genealogy: recognition of the eigenvalue does not select the calendar.

37.3 Boundary marking and areal–volumetric ratio

Let $\Pi_{\text{HMT}}(X_\infty)$ be the value of the regional closure readout already constructed. With the diagonal sheet projectors, define

$$D_\pi = \Pi_{\text{HMT}}(X_\infty)P_{\text{ar}} + P_{\text{vol}}, \quad \Theta_n = \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle}. \quad (37.4)$$

Volumetric normalization is unity; regional closure marks the areal return only once. Applying (37.3),

$$\Theta_n = \Pi_{\text{HMT}}(X_\infty) \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi}{\varphi^2}. \quad (37.5)$$

The return ratio is determined by incidence, measure and marking. The quotient $2\pi/4\pi$, by contrast, describes the cover's two closure orders. Both relations participate in the sheet readout without replacing each other.

37.4 Transport of orientation to the constitutive channels

In a two-channel representation, let $\mathcal{R}$ be a self-adjoint involution and $\mathcal{S}$ a unitary interchange operator satisfying $\mathcal{S}\mathcal{R}\mathcal{S}^{-1} = -\mathcal{R}$. Write the angular coordinates already constructed as $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$, with $x > |y|$. Then

$$B = xI + y\mathcal{R}, \quad T = e^{-B}, \quad \mathcal{S}T(y)\mathcal{S}^{-1} = T(-y). \quad (37.6)$$

Indeed, conjugation transforms $B(y)$ into $B(-y)$ and commutes with its convergent exponential series. The eigenvalues $q_\pm = e^{-x \mp y}$ are interchanged. To fix the response order, the chamber $x > y > 0$ is used, where $0 < q_+ < q_- < 1$. The map $(x, y) \mapsto (x, -y)$ transports it to the opposite chamber $x > -y > 0$, where $0 < q_- < q_+ < 1$. Both belong to the contractive domain $x > |y|$. On the boundary $y = 0$ the two channels coincide. Conjugation interchanges the chambers; it does not preserve the strict ordering of their labels. It acts on the projectors through $\mathcal{S}P_\pm\mathcal{S}^{-1} = P_\mp$, with $P_\pm = (I \pm \mathcal{R})/2$. This concrete map transmits orientation to the constitutive operator developed next.

38 Discrete polarization and the tritic mode of arithmetic projections

Refinement and comparison between the APP sheets provide two complementary observations. The first follows the accumulated discrepancy between the number of transitions and the capacity acquired; the second preserves the mode of the tritic selector in the pairing of the additive and multiplicative fibres. Discrete polarization and the electronic prefactor are obtained from these respective operations. Their articulation with the vacuum requires preserving both the temporal information and the spaces on which the operators act.

38.1 Refinement discrepancy and endpoint convention

The cardinalities of the ternary block and its decimal representation fix

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho = \log_{10}(10/9).$$

We have $0 < \delta < 1$. Moreover, δ is irrational: an equality $\delta = p/q$, with $q > 0$, would imply $10^{q-p} = 9^q$, which is incompatible with prime factorization. This mismatch already enters the encoding of refinement. Here its centred readout is made explicit, preserving the phase origin and the endpoint convention.

Definition 38.1 (Lower and upper encodings). For $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$, define

$$z_n^\downarrow(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor, \quad (38.1)$$

$$z_n^\uparrow(\theta) = \lceil (n+1)\delta + \theta \rceil - \lceil n\delta + \theta \rceil. \quad (38.2)$$

Both take values in $\{0, 1\}$. The arrow distinguishes only the endpoint convention for the intervals; tritic orientation retains its earlier definition.

The lower encoding coincides with the one used in the development of refinement. The upper encoding recovers the formulation of polarization using the ceiling function. The two expressions are related by an exactly determined boundary term.

Proposition 38.2 (Change of convention and boundary term). Let $b_n(\theta) = \mathbf{1}_{\mathbb{Z}}(n\delta + \theta)$. Then

$$z_n^\uparrow(\theta) - z_n^\downarrow(\theta) = b_n(\theta) - b_{n+1}(\theta). \quad (38.3)$$

In particular, for $N \geq 0$,

$$\sum_{n=0}^{N-1} (z_n^\uparrow - z_n^\downarrow) = b_0 - b_N.$$

Proof. For every real x , $\lceil x \rceil = \lfloor x \rfloor + 1 - \mathbf{1}_{\mathbb{Z}}(x)$. Applying this identity to both endpoints of each increment gives (38.3); its sum telescopes. The irrationality of δ implies that, for fixed θ , at most one integer index satisfies $n\delta + \theta \in \mathbb{Z}$. $\square$

With $\theta = 0$, the first upper symbol is 1 and the lower one is 0; for $n \geq 1$ they coincide. The initial term belongs to the register and is preserved when the convention is changed. The densities and discrepancy bounds are transported with that term, without modifying the TPK update or restarting the history at each phase return.

38.2 Bounded polarization and exact temporal balance

Definition 38.3 (Centred discrete polarization). The polarization of the upper encoding, relative to the density δ , is

$$P_N^\uparrow(\theta) = \sum_{n=0}^{N-1} (z_n^\uparrow(\theta) - \delta), \quad P_0^\uparrow = 0. \quad (38.4)$$

Its temporal increment is

$$J_N^\uparrow(\theta) = P_{N+1}^\uparrow(\theta) - P_N^\uparrow(\theta) = z_N^\uparrow(\theta) - \delta. \quad (38.5)$$

Centring separates uniform growth of density δ from phase information. Thus, $P_N^\uparrow$ measures an accumulated imbalance, not the total number of vacancies. The increment $J_N^\uparrow$ preserves the local pulse and the background relative to which it is evaluated.

Theorem 38.4 (Uniform bound and discrete temporal continuity). *For $N \geq 0$ and every phase θ ,*

$$P_N^\uparrow(\theta) = [N\delta + \theta] - [\theta] - N\delta, \quad |P_N^\uparrow(\theta)| < 1. \quad (38.6)$$

For any integers $0 \leq a < b$,

$$\sum_{n=a}^{b-1} J_n^\uparrow(\theta) = P_b^\uparrow(\theta) - P_a^\uparrow(\theta), \quad \left| \sum_{n=a}^{b-1} J_n^\uparrow(\theta) \right| < 1. \quad (38.7)$$

The density of the vacancy sequence is δ .

Proof. Summing the ceiling increments in (38.2) yields the first identity. Writing $r(x) = [x] - x \in [0, 1)$ gives $P_N^\uparrow = r(N\delta + \theta) - r(\theta)$; its absolute value is less than one. On the interval $[a, b)$, one similarly obtains $r(b\delta + \theta) - r(a\delta + \theta)$, proving the second bound, which is sharper than the sum of two individual bounds. Finally, $N^{-1} \sum_{n=0}^{N-1} z_n^\uparrow = \delta + N^{-1} P_N^\uparrow \rightarrow \delta$. $\square$

The lower encoding admits the same proofs with $r(x)$ replaced by $[x] - x \in (-1, 0]$. In particular, $P_N^\uparrow - P_N^\downarrow = b_0 - b_N$. The bounded discrepancy previously used in the Dirichlet readouts and the present polarization therefore preserve an explicit relationship of origin and endpoints. Nor does aperiodicity introduce secular drift in this imbalance: the history can extend indefinitely while the centred discrepancy remains uniformly bounded.

38.3 Realization on the charge and time lines

To realize the balance dimensionally, specify a charge line $\mathcal{L}_Q$ with a nonzero section u_Q and an increasing sequence of times t_N on an oriented time line. With $\Delta t_N = t_{N+1} - t_N > 0$, the centred charge and the mean current of each interval are defined by

$$Q_N = P_N^\uparrow u_Q, \quad I_N = \frac{Q_{N+1} - Q_N}{\Delta t_N} = \frac{u_Q}{\Delta t_N} (z_N^\uparrow - \delta). \quad (38.8)$$

Thus, $Q_N \in \mathcal{L}_Q$ and $I_N \in \mathcal{L}_Q \otimes \mathcal{L}_T^{-1}$. The section u_Q and the time intervals are data of this realization; the preceding arithmetic construction has already determined $P_N^\uparrow$ and $J_N^\uparrow$.

Proposition 38.5 (Preservation of balance under dimensional realization). *For $a < b$,*

$$Q_b - Q_a = \sum_{n=a}^{b-1} \Delta t_n I_n, \quad \left| \frac{Q_b - Q_a}{u_Q} \right| < 1. \quad (38.9)$$

If all intervals have length t_0 , the two possible currents are $-\delta u_Q/t_0$ and $(1 - \delta)u_Q/t_0$.

Proof. Multiplying each identity in (38.8) by Δt_n and summing telescopes the charges. The bound is (38.7); the two local values follow from $z_n^\uparrow \in [0, 1]$. $\square$

The continuity proved is temporal and refers to the centred charge. A spatial realization also incorporates the localization of charges, incidence between cells, and identification of boundary fluxes. In a constitutive description by channels, the coupling is specified through an excitation map $\iota : \mathcal{L}_Q \rightarrow \mathcal{H}_{\text{ch}} \otimes \mathcal{L}_Q$ and an observation map $\ell : \mathcal{H}_{\text{ch}} \otimes \mathcal{L}_Q \rightarrow \mathcal{L}_Q$, where $\mathcal{H}_{\text{ch}}$ is a real realization of the channel space. If both are linear and independent of the time index, the response of a fixed operator $\mathcal{V}$ is transported by

$$Q_N^{\text{resp}} = \ell(\mathcal{V} \otimes I_{\mathcal{L}_Q}) \iota(Q_N), \quad I_N^{\text{resp}} = \frac{Q_{N+1}^{\text{resp}} - Q_N^{\text{resp}}}{\Delta t_N}.$$

Linearity preserves the telescoping balance. This composition makes explicit the data of a constitutive identification: the scalar sequence determines the pulse, while ι , $\mathcal{V}$, and ℓ determine its excitation, transport, and observation through channels. Identities (38.4)–(38.9) establish the first construction independently of that choice of realization. The vacuum operator is constructed in Section 39 on its own channels and projectors.

38.4 Pairing the additive and multiplicative fibres

The second observation comes from the three-dimensional realization of APP, $\mathcal{A}_3 = D_9^3$, with uniform measure and $D_9 = \{1, \dots, 9\}$. The observables preserved are

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \quad \Pi(x) = \rho_9(x_1 x_2 x_3),$$

where $\rho_9(n) = 1 + [(n-1) \bmod 9]$ for $n > 0$. In $\ell^2(\mathcal{A}_3)$, let $R_{\Sigma,3}$ and $R_{\Pi,3}$ be the conditional expectations on their fibres. For example,

$$(R_{\Sigma,3}f)(x) = \frac{1}{|\Sigma^{-1}(\Sigma(x))|} \sum_{y: \Sigma(y)=\Sigma(x)} f(y).$$

They are orthogonal projections defined on the same support. The incidence matrix $N_{ab} = |\Sigma^{-1}(a) \cap \Pi^{-1}(b)|$ results from enumerating the 729 triples, adding and multiplying their coordinates before applying ρ_9 :

$$N = 9 \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}. \quad (38.10)$$

The rows sum to 81, and the column sums are

$$(m_1, \dots, m_9) = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

Therefore, the pairing of the normalized indicator functions of the fibres is $C_{ab} = N_{ab}/\sqrt{81m_b}$. Normalization incorporates the multiplicities produced by APP, not a choice of singular values.

Proposition 38.6 (Singular mode of the tritic selector). *In the canonical chart, the selector vector is $m_{\text{ph}} = (1, -1, 0)^3$. We have*

$$Cm_{\text{ph}} = C^T m_{\text{ph}} = -\frac{1}{2}m_{\text{ph}}. \quad (38.11)$$

Thus, the corresponding singular value is $s = 1/2$.

Proof. In the six columns where m_{ph} is nonzero, $m_b = 36$ and $C_{ab} = N_{ab}/54$. Multiplying the rows of (38.10) by $(1, -1, 0)^3$ yields $-m_{\text{ph}}/2$. For the transposed product, the signed sums of columns 3, 6, 9 are zero; the other six give the same values $-m_{\text{ph},b}/2$. This proves both identities and, by composition, $CC^T m_{\text{ph}} = m_{\text{ph}}/4$. $\square$

38.5 Commutator norm and electronic prefactor

Theorem 38.7 (Exact norm of arithmetic incompatibility). *The two projections satisfy*

$$\|[R_{\Sigma,3}, R_{\Pi,3}]\| = \frac{\sqrt{3}}{4}. \quad (38.12)$$

The maximum is attained in the principal plane corresponding to the tritic mode of (38.11).

Proof. The rational matrix $M = CC^\top$ has entries $M_{ac} = \sum_b N_{ab}N_{cb}/(81m_b)$. Its complete spectrum is verified through the following independent subspaces: the vectors $(1, \dots, 1)$, $(1, -1, 0)^3$, and $(1, 1, -2)^3$ have eigenvalues 1, $1/4$, and $7/132$, respectively; the subspace of vectors supported on $\{3, 6, 9\}$ whose sum is zero has dimension 2 and eigenvalue $1/18$; the zero-sum vectors supported on $\{1, 4, 7\}$ or $\{2, 5, 8\}$ form a kernel of dimension 4. Substitution into the matrix verifies these actions, and the dimensions sum to 9. Consequently,

$$\text{spec } M = \{1, 1/4, 7/132, 1/18, 1/18, 0, 0, 0, 0\}.$$

For two orthogonal projections, a singular value $s \in (0, 1)$ of the pairing determines a principal plane in which their matrices are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} s^2 & s\sqrt{1-s^2} \\ s\sqrt{1-s^2} & 1-s^2 \end{pmatrix}.$$

This form is obtained by taking a unit vector u in the first image with $PQu = s^2u$ and completing it with $(Qu - s^2u)/(s\sqrt{1-s^2})$. Their commutator is $s\sqrt{1-s^2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The sectors with singular value 0 or 1 contribute zero. Therefore, the total norm is the maximum of $\sqrt{\lambda(1-\lambda)}$ over the spectrum of M . That maximum is $\sqrt{3/16}$, attained at $\lambda = 1/4$. $\square$

The norm is the prefactor entering the electronic composition. Its derivation preserves the three-dimensional support, the fibres, and the selector mode; scalar evaluation does not replace that structure. In particular, knowing $\sqrt{3}/4$ gives the maximum magnitude of the incompatibility, while the projectors preserve the directions in which it acts.

38.6 Quadratic defect and response of the 90/120 sectors

The tritic mode simultaneously determines

$$1 - s^2 = \frac{3}{4} = \frac{90}{120}, \quad s = \frac{1}{2}. \quad (38.13)$$

The first expression is the quadratic defect of the APP principal plane. The last preserves the lengths of the two calendar sectors. The constitutive response uses those sectors on another object: a contractive channel $q \in (0, 1)$. Its evaluation is

$$r(q) = \frac{1 - q^{90}}{1 - q^{120}}.$$

The relationship between the sector ratio and this response can be made precise without identifying their operators.

Proposition 38.8 (Limit of the response ratio). *For $0 < q < 1$,*

$$\frac{3}{4} < r(q) < 1, \quad \lim_{q \uparrow 1} r(q) = \frac{3}{4} = 1 - s^2. \quad (38.14)$$

Proof. With $u = q^{30} \in (0, 1)$, factoring $1 - u^3$ and $1 - u^4$ gives $r(q) = (1 + u + u^2)/(1 + u + u^2 + u^3)$. The limit at $u = 1$ is $3/4$. Moreover, $4(1 + u + u^2) - 3(1 + u + u^2 + u^3) = (1 - u)(3u^2 + 2u + 1) > 0$, and the denominator exceeds the numerator by $u^3 > 0$. This yields both inequalities. $\square$

The exact coincidence concerns the defect of the singular mode and the limit of the sector response. In the contractive interior, the response depends on q and is strictly greater than $3/4$. The constitutive development preserves its two channels $q_{\pm}$ and its projectors $P_{\pm}$; the projectors $R_{\Sigma,3}$ and $R_{\Pi,3}$, for their part, preserve the arithmetic fibres that determined the prefactor. An operator equivalence between the two realizations requires a map between their spaces that intertwines the corresponding actions. Equations (38.11), (38.12), and (38.14) specify the shared information and keep the data of each realization visible.

39 Constitutive operator and reconstruction of the vacuum channels

The constitutive quantities are obtained by an operation on angular transport. The starting information comprises the channels, their projectors, and the orientation distinguishing them. This organization makes it possible to follow separately the construction of the operator, the evaluation of its spectral functions, and its subsequent dimensional realization.

39.1 Positive response on two sheets

In the oriented chamber $x > y > 0$ of Section 37, let $P_{\pm} = (I \pm \mathcal{R})/2$ and $T = q_+ P_+ + q_- P_-$, with $0 < q_+ < q_- < 1$. The incidences constructed by the calendar fix the sectors 90 and 120. The constitutive response is

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}. \quad (39.1)$$

This definition receives the transport arising from the TPK and preserves the sheet assignments. The response rule and its domain are an explicit part of the construction.

The quotient arises from comparing the two calendar incidences on the same transport, not from fitting four quantities separately. If $S = T^{30}$ and $\Sigma_j(S) = I + S + \dots + S^{j-1}$, factoring $I - S^j$ gives

$$\mathcal{V} \Sigma_4(S) = \Sigma_3(S), \quad \mathcal{V} = \Sigma_3(S) \Sigma_4(S)^{-1}. \quad (39.2)$$

The factors Σ_3 and Σ_4 express, respectively, the sectors $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$. Since the spectrum of S is contained in $(0, 1)$, $\Sigma_4(S)$ is invertible: the completion equation determines a unique response for the fixed transport. This uniqueness concerns the declared constitutive rule and its incidences. Electromagnetic identification adds the oriented assignment of sheets and the dimensional lines specified below.

Proposition 39.1 (Decomposition and positivity). *The response is well-defined and*

$$\mathcal{V} = r_+ P_+ + r_- P_-, \quad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}. \quad (39.3)$$

Its eigenvalues satisfy $3/4 < r_- < r_+ < 1$ in this chamber.

Proof. Each coefficient $1 - q_{\pm}^{120}$ is positive. The inverse of $I - T^{120}$ is the sum of the projectors divided by those coefficients, proving (39.3). With $s = q^{30}$, one obtains

$$f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \quad f'(s) = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0. \quad (39.4)$$

The limits at 0 and 1 are 1 and 3/4, respectively. Monotonicity and the channel ordering complete the proof. $\square$

39.2 Four coordinates of a common response

Define the normalized evaluations

$$\hat{\varepsilon} = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{Z} = \frac{r_-}{r_+}, \quad \hat{c} = \frac{1}{r_+ r_-}. \quad (39.5)$$

The first pair preserves the quadratic responses assigned to each sheet. The product captures the common mode; the quotient distinguishes their relative orientation. The following hold exactly:

$$\hat{\mu} \hat{\varepsilon} = \hat{c}^{-2}, \quad \frac{\hat{\mu}}{\hat{\varepsilon}} = \hat{Z}^2, \quad r_+ = \frac{1}{\sqrt{\hat{Z} \hat{c}}}, \quad r_- = \sqrt{\frac{\hat{Z}}{\hat{c}}}. \quad (39.6)$$

The positive roots fix the inversion. Under sheet conjugation, read relative to the fixed markings $P_{\pm}$, $\hat{\varepsilon}$ and $\hat{\mu}$ are interchanged, $\hat{Z}$ is inverted, and $\hat{c}$ is preserved. The response is then transported to the chamber $x > -y > 0$: the formulas remain valid, with the orderings $q_- < q_+$ and $r_+ < r_-$ reversed. Consequently, common mode and orientation are data recoverable through complementary observables.

Proposition 39.2 (Determination of the quadratic evaluations). *Let $W = r_+P_+ + r_-P_-$ be a positive response on the two sheets. With the product and quotient readouts fixed,*

$$\hat{c}^{-1} = \det W = r_+r_-, \quad \hat{Z} = r_-/r_+,$$

there exists a unique positive pair $(\hat{\varepsilon}, \hat{\mu})$ satisfying $\hat{\mu}\hat{\varepsilon} = \hat{c}^{-2}$ and $\hat{\mu}/\hat{\varepsilon} = \hat{Z}^2$. It is precisely (r_+^2, r_-^2) .

Proof. Positivity permits taking roots unambiguously:

$$\hat{\mu} = \frac{\hat{Z}}{\hat{c}} = \frac{r_-}{r_+}r_+r_- = r_-^2, \quad \hat{\varepsilon} = \frac{1}{\hat{Z}\hat{c}} = \frac{r_+}{r_-}r_+r_- = r_+^2.$$

These expressions satisfy both relations and are their only positive roots. $\square$

Thus the squares are not two independent choices added to the channels: they are determined by the multiplicative and oriented readouts once the constitutive relations are fixed. This proposition establishes uniqueness within that system of rules. Interpreting the rules as an electromagnetic response retains its additional physical-representation content.

The determinantal readout of transport and the constitutive response must also preserve their order. For $T = e^{-xI-yR}$, one has $\det T = e^{-2x}$, whereas

$$\det \mathcal{V} = f(q_+^{30})f(q_-^{30}) = \hat{c}^{-1}.$$

Both arise from the same two-sheet operator but evaluate different functions of its spectrum. In general, applying the determinant first and then a rational function is not equivalent to taking the determinant of that function of the operator. Keeping both eigenvalues until the response is completed preserves orientation information that an isolated determinant no longer distinguishes.

39.3 Cubic inversion and angular recovery

Theorem 39.3 (Constitutive inversion in the contractive domain). *For each $r \in (3/4, 1)$, there exists a unique $s \in (0, 1)$ such that $f(s) = r$. This s is the unique root in that interval of*

$$rs^3 + (r-1)(1+s+s^2) = 0. \quad (39.7)$$

The labeled response reconstructs $s_{\pm}$, $q_{\pm} = s_{\pm}^{1/30}$, and

$$x = -\frac{1}{2} \log(q_+q_-), \quad y = \frac{1}{2} \log \frac{q_-}{q_+}. \quad (39.8)$$

Proof. The derivative and limits in (39.4) prove that f is a decreasing bijection between the stated intervals. Multiplying $f(s) = r$ by its denominator produces (39.7). The positive root of order 30 recovers the channel. Finally, adding and subtracting $\log q_{\pm} = -x \mp y$ gives (39.8). $\square$

Reconstruction preserves the sheet labels. The complete operator $\mathcal{V}$ also preserves the projectors, and the inverse function is applied by spectral calculus. Recovering two angular coordinates is an operation defined on this representation; the complete historical TPK trace preserves additional data.

The interval $r \in (3/4, 1)$ and the inverse function of this theorem belong to the internal normalization of (39.5). If coordinates expressed in another unit chart are available, first recover this normalization through the transformation in Equation (39.11); then apply cubic inversion.

39.4 Logarithmic coordinates and refinement

Let $\mathcal{E} = \log(r_+ r_-)$ and $\mathcal{B} = \log(r_-/r_+)$. Then

$$\log \mathcal{V} = \frac{1}{2} \mathcal{E} I - \frac{1}{2} \mathcal{B} \mathcal{R}. \quad (39.9)$$

This decomposition displays the invariant and oriented components. In the amplification $\mathcal{V}_m = \mathcal{V} \otimes I_{g_m}$, normalized partial expectations satisfy $E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m$. For every polynomial p , $p(\mathcal{V}_m) = p(\mathcal{V}) \otimes I_{g_m}$; hence $E_m(p(\mathcal{V}_{m+1})) = p(\mathcal{V}_m)$. Uniform continuity extends the identity to continuous functions on the spectrum. Thus projectors, powers, and the logarithm are preserved in the stated tower.

39.5 Dimensional types of the four evaluations

With the action, charge, and velocity lines, equipped with sections u_S, u_Q, u_C , the dimensional realization reads

$$Z_{\text{vac}} = \hat{Z} u_S u_Q^{-2}, \quad c_{\text{vac}} = \hat{c} u_C, \quad (39.10)$$

$$\mu_{\text{vac}} = \hat{\mu} u_S u_C^{-1} u_Q^{-2}, \quad \epsilon_{\text{vac}} = \hat{\epsilon} u_S^{-1} u_C^{-1} u_Q^2. \quad (39.11)$$

Multiplying and dividing these expressions preserves $\mu_{\text{vac}} \epsilon_{\text{vac}} = c_{\text{vac}}^{-2}$ and $\mu_{\text{vac}}/\epsilon_{\text{vac}} = Z_{\text{vac}}^2$. The SI chart subsequently evaluates the sections. Physical comparison requires presenting their construction and normalization alongside evaluation; the four normalized coordinates in (39.5) retain their dimensionless type.

39.6 Reconstruction of the action shape from the constitutive response

The constitutive response and the inductive–capacitive realization receive the same angular pair constructed from APP–TRIT–TPK. The former evaluates rational functions of the two channels; the latter uses the ratio of their exponents to determine the shape of the action ellipse. Cubic inversion allows the two readouts to be composed without identifying their numerical ratios.

Let $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$, with $x > |y|$, be the lifted coordinates already constructed, and preserve their sheet labels. Write

$$q_{\pm} = e^{-(x \pm y)}, \quad s_{\pm} = q_{\pm}^{30}, \quad r_{\pm} = f(s_{\pm}), \quad f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}. \quad (39.12)$$

The contractive domain gives $s_{\pm} \in (0, 1)$ and $r_{\pm} \in (3/4, 1)$. In the chamber $x > y > 0$, we have $s_+ < s_-$ and $r_+ > r_-$; sheet interchange transports this chamber to the opposite orientation.

The image of the two normalized constitutive coordinates is the domain

$$\mathcal{U} = \{(z, v) \in (0, \infty)^2 : (zv)^{-1/2} \in (3/4, 1), \quad (z/v)^{1/2} \in (3/4, 1)\}. \quad (39.13)$$

Here $(z, v) = (\hat{Z}, \hat{c})$. The restrictions express membership of the reconstructed channels in the domain of the inversion theorem; they do not constitute a subsequent choice of their values.

Proposition 39.4 (Constitutive composition of the shape and LC impedance). *For $(\hat{Z}, \hat{c}) \in \mathcal{U}$, let $s_{\pm} \in (0, 1)$ be the unique roots of*

$$r_{\pm} s_{\pm}^3 + (r_{\pm} - 1)(1 + s_{\pm} + s_{\pm}^2) = 0, \quad r_+ = \frac{1}{\sqrt{\hat{Z}\hat{c}}}, \quad r_- = \sqrt{\frac{\hat{Z}}{\hat{c}}}. \quad (39.14)$$

The angular ratio, the anisotropy of the action ellipse, and the relative LC impedance are reconstructed exactly by

$$\frac{y}{x} = \frac{\log s_+ - \log s_-}{\log s_+ + \log s_-}, \quad (39.15)$$

$$\eta_{\text{el}} = \frac{1}{2} \log \frac{\log s_+}{\log s_-}, \quad (39.16)$$

$$\frac{Z_{\text{LC}}}{Z_*} = e^{-\eta_{\text{el}}} = \sqrt{\frac{\log s_-}{\log s_+}}. \quad (39.17)$$

In the last identity, $Z_{\text{LC}} = Z_{\eta_{\text{el}}}$ is the impedance of the LC chart in (20.12).

Proof. The cubic inversion in (39.7) uniquely determines $s_{\pm}$, because f is a decreasing bijection from $(0, 1)$ onto $(3/4, 1)$. Their positive roots of order 30 recover $q_{\pm}$. The identities

$$\log s_+ = -30(x + y), \quad \log s_- = -30(x - y)$$

prove (39.15) by difference and sum. Both logarithms are negative; their ratio is positive and their sum is nonzero. Therefore,

$$\frac{1}{2} \log \frac{\log s_+}{\log s_-} = \frac{1}{2} \log \frac{x + y}{x - y} = \text{artanh}(y/x) = \eta_{\text{el}}.$$

The relation $Z_{\eta_{\text{el}}}/Z_* = e^{-\eta_{\text{el}}}$, proved in (20.13), gives (39.17). $\square$

The composition preserves the information needed for each reconstruction. With $\hbar$, one recovers $a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}$ and $b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}$; with the chart (ω, Z_*) , one recovers the inductance and capacitance already defined. Action, clock, and dimensional reference accompany the shape as typed data. Inverting two dimensionless coordinates does not reconstruct their units or the entire memory of the state.

Normalization enters before the cubic equations are solved. If Z_{vac} and c_{vac} are received in a dimensional chart, relations (39.11) first determine

$$\hat{Z} = \frac{Z_{\text{vac}}}{u_S u_Q^{-2}}, \quad \hat{c} = \frac{c_{\text{vac}}}{u_C}.$$

These are the coordinates to which (39.14) applies. The reference Z_* belongs to the LC realization and is explicitly preserved in (39.17).

The two ratios are related through the channels, but evaluate different functions:

$$\hat{Z} = \frac{f(s_-)}{f(s_+)}, \quad \frac{Z_{\text{LC}}}{Z_*} = \sqrt{\frac{\log s_-}{\log s_+}}. \quad (39.18)$$

The preceding reconstruction provides the map between the two readouts. In particular, interchanging the sheets inverts both ratios, changes η_{el} to $-\eta_{\text{el}}$, and preserves $\hat{c}$. In the symmetric limit $y = 0$, the channels coincide and both ratios equal 1. Outside that case, a common origin does not impose equality of the two functions. The action shape is recoverable from the constitutive response while preserving orientation and normalization.

39.7 Constitutive speed and kinematic realization of the clock

The constitutive response and the kinematic readout realize the same speed section. Their identification preserves the order of construction: the APP sheets, their TRIT orientation, and the angular transport expressed by TPK precede the speed readouts and the choice of units. In the two-sheet representation of Section 37, with $x > |y|$ and rank-one projectors $P_{\pm}$, the transport is

$$T = e^{-xI-y\mathcal{R}} = q_+P_+ + q_-P_-, \quad q_{\pm} = e^{-x \mp y}, \quad 0 < q_{\pm} < 1.$$

Its temporal blocks of lengths 90 and 120 produce the response $\mathcal{V}$ in (39.1). The coordinates x, y and the channels are outputs of the preceding angular development; this composition applies its readouts to that already constructed transport.

The symmetric logarithmic readout of temporal transport is

$$\mathcal{E} = \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})]. \quad (39.19)$$

On the dimensional speed line, with positive section u_C , the kinematic readout expresses $c_{\text{int}} = \exp(-\mathcal{E})u_C$. The constitutive response expresses $c_{\text{vac}} = \hat{c}u_C$ through (39.5).

Proposition 39.5 (Constitutive and kinematic identity). *In the same dimensional chart,*

$$c_{\text{int}} = c_{\text{vac}} = \hat{c}u_C, \quad \hat{c} = (r_+r_-)^{-1} = \exp(-\mathcal{E}). \quad (39.20)$$

The identification preserves sheet interchange.

Proof. All factors in (39.19) are positive. The logarithm law and (39.3) give

$$\mathcal{E} = \log(r_+r_-) = \log \det \mathcal{V}, \quad e^{-\mathcal{E}} = (\det \mathcal{V})^{-1}.$$

The determinant corresponds to the realization with two rank-one sheets. Both definitions therefore express the same section. Sheet interchange permutes r_+ and r_- and preserves their product. The determinant $\det T = e^{-2x}$ corresponds to elementary transport, whereas $\det \mathcal{V} = r_+r_-$ corresponds to its temporal response; (39.20) uses the latter. $\square$

39.7.1 Path length and elementary duration

The positive dimensional frame (u_S, u_C, u_T, u_Q) induces the length basis $u_L = u_C u_T$. For an enriched path γ with $n = |\gamma|$ transitions and constant channel, the additive readouts are

$$T_{\text{int}}(\gamma) = nu_T, \quad L_{\text{int}}(\gamma) = n\hat{c}u_L. \quad (39.21)$$

Concatenation adds the numbers of transitions and their realizations. Enriched routes also preserve their orientation and memory.

Corollary 39.6 (Normalization of the elementary length). *For $n > 0$, $L_{\text{int}}(\gamma)/T_{\text{int}}(\gamma) = c_{\text{vac}}$. In the normalization $t_0 = u_T$,*

$$\ell_0 = c_{\text{int}}t_0 = \hat{c}u_L. \quad (39.22)$$

In a temporal chart $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, one obtains $\ell_0 = \tau_0 \hat{c}u_L$.

Proof. Dividing the two expressions in (39.21) cancels n and uses $u_L/u_T = u_C$. Multiplying the speed section by t_0 proves both elementary formulas. $\square$

When the channel depends on the edge, the readout retains the additive form $L_{\text{int}}(\gamma) = \sum_{e \in \gamma} \hat{c}(e)u_L$. Its quotient by nu_T is the mean of the speeds of those edges. The constant-channel expression is the homogeneous specialization of this readout.

39.7.2 Change of chart and adapted normalization

Proposition 39.7 (Covariance of the identification). *Let $u'_C = du_C$ and $u'_T = \eta u_T$, with $d, \eta > 0$. The coordinates of the same sections satisfy*

$$\hat{c}' = d^{-1}\hat{c}, \quad \tau'_0 = \eta^{-1}\tau_0, \quad u'_L = d\eta u_L. \quad (39.23)$$

If another representation writes $c' = \hat{c}'u'_C$ with $u'_C = du_C$, equality of sections is exactly equivalent to $d\hat{c}' = \hat{c}$.

Proof. Equality of each section in the two bases gives $\hat{c}'u'_C = \hat{c}u_C$ and $\tau'_0u'_T = \tau_0u_T$. In particular,

$$\tau'_0\hat{c}'u'_L = \frac{\tau_0}{\eta} \frac{\hat{c}}{d} d\eta u_L = \tau_0\hat{c}u_L = \ell_0.$$

The final equivalence follows from the same substitution for speed. □

The adapted choice $d = \hat{c}$ gives $u'_C = c_{\text{vac}}$ and $\hat{c}' = 1$: it expresses the constructed section in an adapted basis. The coefficient in the internal chart remains $(r_+r_-)^{-1}$, and that is the normalization used by cubic inversion. Keeping the action and charge bases, (39.11) yields

$$\hat{\mu}' = d\hat{\mu}, \quad \hat{\varepsilon}' = d\hat{\varepsilon}, \quad \hat{\mu}'\hat{\varepsilon}' = (\hat{c}')^{-2}.$$

In the adapted chart, $\hat{\mu}' = r_-/r_+$ and $\hat{\varepsilon}' = r_+/r_-$, by substitution of $\hat{\mu} = r_-^2$ and $\hat{\varepsilon} = r_+^2$.

39.7.3 Preservation of the readout under amplification

For the refinement $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$, the normalized trace $\tau_m = \text{Tr}/(2 \cdot 9^m)$ satisfies

$$\exp[-2\tau_m(\log \mathcal{V}_m)] = \exp[-\log r_+ - \log r_-] = \hat{c}. \quad (39.24)$$

Indeed, each eigenvalue appears 9^m times; that multiplicity cancels against the denominator of the normalized trace. The ordinary amplified determinant is $(r_+r_-)^{9^m}$. The two-sheet normalization, equivalently expressed by (39.24), preserves the constitutive readout during amplification. This realization preserves the enriched transport from which it originates.

40 Charge section and electrical quantization relations

The constitutive response establishes a relation between the two sheets and a relative propagation scale. The charge section is obtained by composing this response with the fine-structure constant and the action, both constructed in the preceding chapters. This order has a precise consequence: charge does not retrospectively determine the dodecaphase record. It is the record, together with the action and the vacuum response, that determines the charge section on the corresponding line.

40.1 Positive determination of the charge section

Consider the positive sections $\hbar_a$, with $a \in \{\text{pre}, \text{ret}\}$, and $h_a = 2\pi\hbar_a$. The impedance Z_{vac} has already been constructed through (39.11). The quotient h_a/Z_{vac} belongs to the quadratic charge line. This dimensional compatibility permits the following definition without yet choosing an SI chart.

Definition 40.1 (Positive charge section). The charge section corresponding to action orientation a is

$$e_a = \left(\frac{2\alpha h_a}{Z_{\text{vac}}} \right)^{1/2}. \quad (40.1)$$

The root is taken on the positive charge half-line. Its conjugate section is $-e_a$; the sign of the latter is not identified with the action return index.

Proposition 40.2 (Existence, uniqueness, and compatibility of the coupling). *The equation $Z_{\text{vac}} e_a^2 = 2\alpha h_a$ determines a unique positive section. In the constitutive realization, one also has*

$$\alpha = \frac{Z_{\text{vac}} e_a^2}{2h_a} = \frac{e_a^2}{4\pi\epsilon_{\text{vac}} \hbar_a c_{\text{vac}}}. \quad (40.2)$$

Proof. Positivity of α , h_a , and Z_{vac} guarantees a unique positive root. The constitutive identities give $\epsilon_{\text{vac}} c_{\text{vac}} = Z_{\text{vac}}^{-1}$. Substituting this equality and $h_a = 2\pi\hbar_a$ into the first expression of (40.2) yields the second. $\square$

The final expression is the customary form of electromagnetic coupling in a physical chart. Here it appears as a subsequent identity of the constructed sections. The proof does not recalculate α from a measured charge: it establishes the algebraic coherence of the preceding result with the section just defined. The physical identification additionally requires retaining the unit chart and coupling regime used in the comparison. An identity between quantities and an experimental comparison serve different functions in that recognition.

40.2 Resistance, conductance, flux, and frequency

The following four quantities use the same section e_a . Their dependencies can be fully resolved before any decimal is evaluated.

Theorem 40.3 (Composition of the electrical quanta). *The sections*

$$R_K = \frac{h_a}{e_a^2} = \frac{Z_{\text{vac}}}{2\alpha}, \quad G_0 = \frac{2e_a^2}{h_a} = \frac{4\alpha}{Z_{\text{vac}}}, \quad (40.3)$$

$$\Phi_{0,a} = \frac{h_a}{2e_a} = \sqrt{\frac{h_a Z_{\text{vac}}}{8\alpha}}, \quad K_{J,a} = \frac{2e_a}{h_a} = \Phi_{0,a}^{-1} \quad (40.4)$$

satisfy exactly

$$R_K G_0 = 2, \quad G_0 Z_{\text{vac}} = 4\alpha, \quad K_{J,a} \Phi_{0,a} = 1, \quad K_{J,a}^2 R_K = \frac{4}{h_a}. \quad (40.5)$$

Proof. Substituting $e_a^2 = 2\alpha h_a/Z_{\text{vac}}$ into the first two definitions cancels h_a and gives (40.3). Squaring the positive section $h_a/(2e_a)$ gives $h_a Z_{\text{vac}}/(8\alpha)$; positivity fixes the root in (40.4). Its inverse is $2e_a/h_a$. The first three products in (40.5) reduce, respectively, to 2, 4α , and 1. Finally, $(4e_a^2/h_a^2)(h_a/e_a^2) = 4/h_a$. $\square$

The physical realization recognizes in these sections the von Klitzing resistance, the conductance quantum, the flux quantum, and the Josephson constant. The factor 2 in the conductance belongs to the normalization of the quantity defined here; identifying it with a multiplicity of channels in a material system requires specifying that representation. The four equations do not constitute four independent generators: they share α , the action, and the vacuum response. Precisely this dependency produces the cross-identities and makes it possible to check an evaluation without altering the operations from which it arose.

40.3 Action return and transport of the sections

The quotient $R_{\text{act}} = h_{\text{pre}}/h_{\text{ret}}$, obtained before the angular pair, preserves the distinction between the two action sections. It does not change the sheet assignments of the constitutive response. These are two distinct operations: action return and channel interchange.

Proposition 40.4 (Covariance under return). *With common α and Z_{vac} , one has*

$$\frac{e_{\text{pre}}}{e_{\text{ret}}} = R_{\text{act}}^{1/2}, \quad R_K^{\text{pre}} = R_K^{\text{ret}}, \quad G_0^{\text{pre}} = G_0^{\text{ret}}, \quad (40.6)$$

$$\frac{\Phi_{0,\text{pre}}}{\Phi_{0,\text{ret}}} = R_{\text{act}}^{1/2}, \quad \frac{K_{J,\text{pre}}}{K_{J,\text{ret}}} = R_{\text{act}}^{-1/2}. \quad (40.7)$$

Proof. The first identity follows by dividing the two versions of (40.1) and taking the positive root. In (40.3), the action has already cancelled. The other two identities follow from the powers $h_a^{1/2}$ and $h_a^{-1/2}$ in (40.4). $\square$

Resistance and conductance preserve the constitutive quotient, whereas flux and frequency also retain the action orientation. This separation gives a structural reading of their dependencies: some quantities are invariants of the sheet quotient, while others additionally measure the action section used. The metrological comparison must specify which section it evaluates; the numerical proximity of the two actions does not authorize their identification.

γ

Barbero–Immirzi parameter

Hexadic–octadic incidence, action, area and information.

Action, area, and information

The Barbero–Immirzi parameter is presented on the basis of the antecedents of action, orientation, and incidence already constructed. The angular coordinates and conjugate channels act together with the marked inclusion between hexadic and octadic configurations. The functional receives these data and determines its evaluation. The relation between arithmetic and geometry remains visible in each of its arguments.

The incidence structure organizes which elements are distinguished and how they are prolonged. A hexad is a support of six positions in the design; its lift uses a declared dodecad and alignment. The resulting octad belongs to the binary realization. The area readers preserve the marking that makes this correspondence meaningful. The figures help to follow the transport of the sets and their intersections.

Action and clock subsequently make it possible to express the energy associated with a reading. The Boltzmann constant enters into thermal transduction; the Barbero parameter and the areal unit enter into the geometrical increment. The gravitational constant enters the realization that relates action and Planck area. When these expressions are composed, the same action section can cancel in the quotient between energy and area increment. The cancellation exhibits a concrete structural dependence.

The law of state change also preserves the relative-entropy term and the reference with respect to which the comparison is made. The information of a reduced observation depends on the correlations maintained by the joint state. This distinction extends the argument of the archive: the content of the whole and that of a partial reader are related through explicit operations.

Conservation theory thus acquires a definite physical application. The state, its orientation, the area assigned to it, and the thermal reading form a composition with identified antecedents. The development preserves these antecedents and their proofs. The subsequent synthesis will bring this composition together with the operator balances and with the calculable example of entanglement.

41 Hexadic–octadic incidence and the Barbero–Immirzi functional

Let $H \hookrightarrow O$ be the marked inclusion of a hexad in an octad, $|H| = 6$, $|O| = 8$, constructed through the binary residue of Section 16.16. The hexad, dodecad, and regional chart determining this inclusion are preserved. The incidence objects

$$\mathcal{F}_6 = H \times \binom{H}{2}, \quad \mathcal{F}_8 = O \times \binom{H}{2}$$

have cardinalities 90 and 120. Their normalized difference gives

$$\frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24\binom{6}{2}} = -\frac{1}{12}.$$

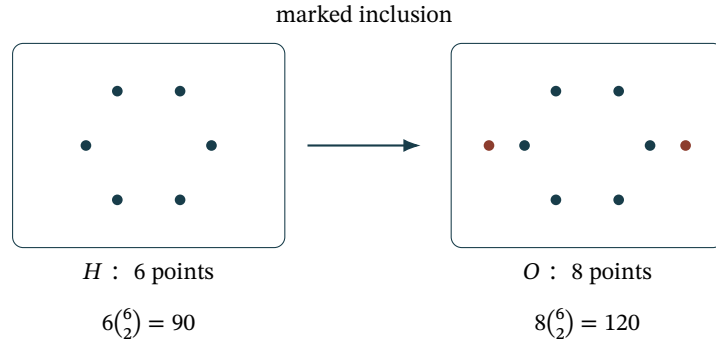


Figure 14: Marked incidence from six to eight points. The diagram represents the sets and the inclusion; it does not purport to turn the faces or vertices of a cube into Steiner blocks without their incidence map.

41.1 The weights precede the pole coefficient

In the free module of oriented events, fix the fundamental chain of the revolution

$$c_{12}^+ = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\text{ret}}, \mathcal{F}_8].$$

The multiplicity twelve corresponds to the sectors, and the multiplicity one to the oriented seam. The involution reverses the orientation of the entire chain. Let

$$S_m(q) = \frac{q^m}{1 - q^{3m}}, \quad 0 < q < 1.$$

The linear reader assigns S_{90} to each sector and S_{120} to the seam; consequently,

$$\mathcal{R}(c_{12}^+) = 12S_{90} - S_{120}.$$

Definition 41.1 (Oriented evaluation of hexadic–octadic incidence). The parameter γ_{HMT} is the conjugate evaluation of the chain c_{12}^+ , completed by its angular component. On the domain $0 < q_{\pm} < 1$, it is defined by

$$\gamma_{\text{HMT}} = \frac{\pi AC^*}{180} + [\mathcal{R}(c_{12}^+)](q_-) - [\mathcal{R}(c_{12}^+)](q_+).$$

The marked incidence determines the event spaces; the oriented chain fixes the multiplicities, and the channels of Section 13.4 determine the evaluation.

Theorem 41.2 (Pole coefficient and conjugate evaluation). *The image of the fundamental chain has coefficient $1/24$ with respect to $(1 - q)^{-1}$ and residue $-1/24$ with respect to $(q - 1)^{-1}$. Its evaluation in the previously generated channels determines*

$$\gamma_{\text{HMT}} = \frac{\pi AC^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)]. \quad (41.1)$$

Proof. The factorization $1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$ implies $\lim_{q \uparrow 1} (1 - q)S_m(q) = 1/(3m)$. By linearity,

$$\frac{12}{270} - \frac{1}{360} = \frac{1}{24}.$$

The coordinate $q - 1$ reverses the sign of the residue. The conjugate evaluation and the angular component of Definition 41.1 yield (41.1). $\square$

The Diophantine certificate $4a - 3b = 45$ admits the pairs $(12 + 3t, 1 + 4t)$, $t \geq 0$; it certifies the minimality of $(12, 1)$, but does not replace the origin of the multiplicities in the chain. The evaluation of (41.1) is

$$\gamma_{\text{HMT}} = 0.27400767269445927474725526077398041940 \dots$$

This numerical value is an output of the oriented functional, rather than an external value used to select A , C^* , or their coefficients.

41.2 Parity and structural sensitivity

Reversing C^* interchanges q_+ and q_- and reverses the sign of the angular component. Hence

$$\gamma_{\text{HMT}}(A, -C^*) = -\gamma_{\text{HMT}}(A, C^*).$$

The functional preserves orientation, whereas other readers, such as q_+q_- , are even. This distinction precedes their physical interpretation.

42 Longitudinal return, radial reciprocity, and the memory archive

The coupling constant retains its ordered register and invertible transformations. The following realization composes those operators with the incidences already constructed, the local action norm, and the constitutive response. The APP–TRIT–TPK state and the joint discrete structure of the continuum precede their two readouts. The coordinates π, φ, e, α inherit the genealogy established in the preceding sections; the dimensional radius is subsequently constructed on an explicit length section.

42.1 Constitutive restitution and oriented moment

The response in (39.3) and the functional in (41.1) share the labelled channels. For $s = q^{30}$, they retain the function

$$f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \quad 0 < s < 1.$$

The oriented depth-two moment and its differential operator are

$$\mathcal{C}_2(s) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}, \quad \mathcal{D} = s \frac{d}{ds}.$$

Local uniform convergence gives $\mathcal{D}^2 \mathcal{C}_2 = s/(1 + s^2)$, hence

$$f(s) = \frac{1 + s + s^2}{s(1 + s)} \mathcal{D}^2 \mathcal{C}_2(s), \quad r_{\pm} = f(q_{\pm}^{30}). \quad (42.1)$$

The condition $\mathcal{C}_2(s) \rightarrow 0$ as $s \downarrow 0$ fixes the restitution

$$\mathcal{C}_2(s) = \int_0^s \log(s/t) \frac{1+t}{1+t+t^2} f(t) dt.$$

Indeed, integrating twice with respect to $\log s$ inverts $\mathcal{D}^2$; the difference of two solutions is $a \log s + b$, and the zero limit forces $a = b = 0$. The integral reduces to $\int_0^s \log(s/t)/(1+t^2) dt$. The original series converges absolutely at $s = 1$ and gives Catalan's constant; its second derivative at the endpoint is interpreted through the Abel limit. The complete parametric function is thus retained alongside its scalar evaluation.

The analytic chart retains the canonical representatives of the fundamental chamber: $x = \pi A_{\text{deg}}/180$, $y = \pi C_{\text{deg}}^*/180$, and $q_{\pm} = e^{-x \mp y}$. These lifted representatives fix the real exponentials. With $\mathcal{D}_2(z) = \sum_{n \geq 1} z^n/n^2$, the common potential is

$$\mathcal{F}(x, y) = \sum_{\sigma = \pm} \left[\frac{\mathcal{D}_2(q_{\sigma}^{120})}{120} - \frac{\mathcal{D}_2(q_{\sigma}^{90})}{90} \right], \quad x > |y|.$$

The identity $z \mathcal{D}'_2(z) = -\log(1-z)$ proves $\partial_x \mathcal{F} = \log \hat{c}$ and $\partial_y \mathcal{F} = \log \hat{Z}$. Its symmetric Hessian retains $\partial_y \log \hat{c} = \partial_x \log \hat{Z}$; the condition $\mathcal{F} \rightarrow 0$ as $q_+, q_- \rightarrow 0$ fixes the additive constant.

Theorem 42.1 (Joint constitutive coordinates). *The map $(r_+, r_-) \mapsto (\hat{c}, \gamma)$ is a diffeomorphism from $(3/4, 1)^2$ onto $(1, 16/9) \times \mathbb{R}$. It preserves the assignment of sheets.*

Proof. The derivative of f in (39.4) is strictly negative; let $\psi = f^{-1}$. Define

$$\mathcal{H}_B(s) = \frac{12s^3}{1-s^9} - \frac{s^4}{1-s^{12}} - \frac{(\log s)^2}{20\pi}, \quad F_B = \mathcal{H}_B \circ \psi.$$

One verifies $\gamma = F_B(r_-) - F_B(r_+)$. Moreover,

$$\mathcal{H}'_B(s) = \frac{36s^2(1+2s^9)}{(1-s^9)^2} - \frac{4s^3(1+2s^{12})}{(1-s^{12})^2} - \frac{\log s}{10\pi s} > 0.$$

The ratio of the first two positive terms, before subtraction, exceeds 9; the last summand is positive. Thus F_B decreases from $+\infty$ to $-\infty$. For fixed $v = \hat{c}$, write $r_{\pm} = v^{-1/2}e^{\mp z/2}$. The admissible interval is $|z| < \min\{\log v, \log(16/(9v))\}$. The derivative of $F_B(r_-) - F_B(r_+)$ is $[r_-F'_B(r_-) + r_+F'_B(r_+)]/2 < 0$, and its limits are $+\infty$ and $-\infty$. A unique solution exists for every γ , with a smooth inverse because that derivative is nonzero. $\square$

On the generated image, $s_{\pm} = \psi(r_{\pm})$ recovers the sexagesimal coordinates $A_{\deg} = -3 \log(s_+s_-)/\pi$, $C_{\deg}^* = 3 \log(s_-/s_+)/\pi$, followed by $\alpha = A_{\deg}/1000$ and $\eta_{\text{ret}} = C_{\deg}^*/2 - \alpha$. The last operation inverts the action section already constructed. This recovery fixes the channels; the markings, units, and longitudinal reader retain their own roles.

42.2 Marked inscription and linear return

The hexadic–octadic incidence and the dodecaphasic orbits define the support

$$\Omega = \{(j, k, q, u, v) : j, k \in \mathbb{Z}/12\mathbb{Z}, \quad k - j \in \{0, 3, 6, 9\}, \quad 1 \leq q \leq 8, \quad u < v, \quad u, v \in \{1, 2, 4, 5, 7, 8\}\}.$$

Its cardinality is $N = 12 \cdot 4 \cdot 8 \binom{6}{2} = 5760$. The difference $k - j$, computed in $\mathbb{Z}/12\mathbb{Z}$, the octad index, and the hexad pair provide the 480 auxiliary labels per sector. In this ordering, the integral inverse from Section 9.8 extends as

$$B_{\Omega} = (\mathcal{H}_{12}/4) \otimes I_{480}, \quad B_{\Omega}^* B_{\Omega} = B_{\Omega} B_{\Omega}^* = \frac{1}{4}I.$$

The identities $\mathcal{H}_{12}^* = \mathcal{H}_{12}$ and $\mathcal{H}_{12}^2 = 4I$, obtained from the three order-four Hadamard blocks and their joint permutation, prove the equality and preserve the orientation of the three orbits. Set $u = N^{-1/2}\mathbf{1}$, $P = uu^*$, $C_0 = (I - P)B_{\Omega}$, $M_0 = PB_{\Omega}$, and $U = 2B_{\Omega}$. Then

$$C_0^* C_0 + M_0^* M_0 = \frac{1}{4}I, \quad C_0 C_0^* = \frac{1}{4}(I - P), \quad U^* U = U U^* = I.$$

The archive $(2C_0v, 2M_0v)$ is isometric, and its sum recovers Uv . The transport U acts on this finite register; the TPK evolution U_t additionally retains the depth and history of the section.

Proposition 42.2 (Response of the marked inscription). *For the atomic projections $P_i = e_i e_i^*$, the unique expectation onto their diagonal algebra that fixes each P_i and is bimodular is*

$$\Delta_{\Omega}(T) = \sum_i P_i T P_i, \quad \Delta_{\Omega}(C_0 C_0^*) = r_N I, \quad r_N = \frac{N-1}{4N} = \frac{5759}{23040}.$$

Proof. On each matrix unit E_{ij} , bimodularity imposes $\Delta(E_{ij}) = P_i \Delta(E_{ij}) P_j$. Diagonality makes this expression zero when $i \neq j$; fixing P_i determines the remaining case. Linearity fixes the whole map. Since $P_{ii} = 1/N$, the diagonal of $(I - P)/4$ is $r_N I$. The common component contributes $I/(4N)$; the two add to $I/4$. $\square$

The isometry $W e_i = e_i \otimes e_i$ realizes $W^*(T \otimes I)W = \Delta_{\Omega}(T)$ while retaining the extended register. Uniformity here concerns the common mode; every state on the marked algebra evaluates the same scalar r_N . An auxiliary memory is transported through $\Delta_{\Omega} \otimes \text{id}$.

Definition 42.3 (Longitudinal composition of the return). On a reference $L_* > 0$ in the length line, the local norm α^{16} , constructed through the Hadamard cokernel in the action section, acts linearly on the inscribed return:

$$D = L_* \alpha^{16} \Delta_{\Omega}(C_0 C_0^*) U = L U, \quad L = L_* \alpha^{16} r_N. \quad (42.2)$$

This composition assembles the inverse, its complementary archive, and the longitudinal section into an explicit rule. On an oriented cochain β , it multiplies by $\alpha^{16} r_N$, preserving addition, orientation reversal, and subdivision into nine edges. The norm of $C_0^* a e_i$ is $|a| \sqrt{r_N}$, whereas the inscribed linear response is $a r_N$: these are different operations. The coefficient in (42.2) follows from the specified expectation as well as its linear type.

42.3 Arrival modulus and common coupling

Let $X = X^* > 0$ be the full character on the nonzero finite fibre of the register and let $Y = UXU^*$. The radial readout of this realization is the arrival modulus

$$R_X = (DX^2D^*)^{1/2} = LY.$$

The radion retains the action law $S(\theta) = \hbar_a \theta$. In the circular realization of the same L , $\omega L = c$, hence

$$Q_a = \frac{\hbar_a c}{L}, \quad H_X = Q_a Y, \quad M_X = H_X/c^2, \quad \Lambda_X = LY^{-1}. \quad (42.3)$$

Here Q_a denotes an energy; the preceding projections and registers retain their own notation. The physical identification of R_X is the reduced gravitational radius $R_X = (G_a/c^2)M_X$. This identification and the longitudinal composition define the realization whose coefficients are determined below.

Theorem 42.4 (Radial rigidity and normalized reciprocity). *For fixed D and X , R_X is the unique positive operator satisfying $\|R_X v\|^2 = \|XD^*v\|^2$ for every arrival vector. The preceding readouts satisfy*

$$R_X \Lambda_X = L^2 I, \quad H_X \Lambda_X = \hbar_a c I, \quad \boxed{G_a = \frac{c^3 L^2}{\hbar_a}}. \quad (42.4)$$

The same G_a acts on every admissible positive character of this realization.

Proof. Polarization of the quadratic-form identity gives $R_X^2 = DX^2D^* = L^2 UX^2 U^*$. Its unique positive square root is $LUXU^*$. The products in (42.3) yield the two identities. Right multiplication by Λ_X^{-1} proves $R_X = L^2 H_X / (\hbar_a c)$; this deduction retains every cross term. Substituting $H_X = c^2 M_X$ and the radial identification gives (42.4). The invertibility of M_X fixes the coefficient. $\square$

Operator uniqueness holds for each fixed X , whereas the coefficient is common to the whole fibre. The null sector retains its separate treatment outside the inverse operation. The native law $L_{-p}(z^{-1}) = -L_p(z)$, with $L_p(z) = (z^p - 1)/p$ and $L_0(z) = \log z$, expresses oriented inversion before this gravitational realization; its proof is the direct substitution $(z^{-1})^{-p} = z^p$.

42.4 Clock, mass, and dimensional sections

The compatible chronological chart satisfies

$$108\ell_0 = 2\pi L, \quad \ell_0 = ct_0 = \hat{c} u_L, \quad t_0 = \frac{\pi L}{54c}, \quad \frac{L_*}{u_L} = \frac{54\hat{c}}{\pi\alpha^{16}r_N}. \quad (42.5)$$

This distinguishes the radial reference L_* , the base $u_L = u_C u_T$, and the advance ℓ_0 . The period $108t_0$ is $2\pi L/c$. The mode $X = I$ determines the clock quantum; the electronic coordinate additionally retains its normalization $b_e = E_e^{(0)}/Q_{\text{ret}}$ and its factor $R_{\text{act}}^{\kappa_e}$. When comparing $L' = sL$ with fixed $\hbar, c, E_e^{(0)}$, one has $Q' = Q/s$, $b'_e = sb_e$, and $Q'b'_e = E_e^{(0)}$. The fixed mass retains its value while its coordinate is transported to the new clock.

The two action sections retain $G_{\text{pre}}/G_{\text{ret}} = R_{\text{act}}^{-1}$ and $\ell_P^2 = \hbar_a G_a/c^3 = L^2$. In the chart $L_* = 1 \text{ m}$, $\mathcal{U}_S = 1 \text{ J s}$, with $c = 299792458 \text{ m s}^{-1}$, the return composition gives

$$\begin{aligned} L &= 1.616254982625126330168262501334561962 \dots 10^{-35} \text{ m}, \\ t_0 &= 3.136499962536549073434219319400884017 \dots 10^{-45} \text{ s}, \\ G_{\text{ret}} &= 6.674299659414022706367776189182194954 \dots 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}. \end{aligned}$$

The constitutive section retains $u_C = c/\hat{c}$. These digits evaluate the composition in the declared sections; the normalized identities and their domains constitute its proof.

The positive scales recognized in this realization are $t_P = L/c$, $m_{P,a} = \hbar_a/(cL)$, and $E_{P,a} = \hbar_a c/L = Q_a$. Substituting (42.4) into $\sqrt{\hbar_a G_a/c^5}$, $\sqrt{\hbar_a c/G_a}$, and $\sqrt{\hbar_a c^5/G_a}$ proves the three identities, respectively. For a mode $y > 0$, $m(y) = m_{P,a} y$, $r_g(y) = Ly$, and $\bar{\lambda}(y) = L/y$. Their product is L^2 , whereas equality of the lengths characterizes $y = 1$. One has $t_0 = (\pi/54)t_P$; a horizon $R_h = \zeta_h L$ retains its geometric variable $\zeta_h > 0$.

42.5 Reduction with a residual and transport of symmetries

The memory energy in Section 50 retains the Hermitian product antilinear in its first argument, the unitary transports U_a , and the positive global weight W_m , including its cross terms. Its full variation (50.4) combines the field, transport $\dot{U}_a = A_a U_a$, $A_a^* = -A_a$, and weight $\dot{W}_m$. The first-order expansion proof and Theorem 50.5 remain within the article; the latter transports the response through fixed inclusions J_m, K_m . Continuous variations belong to the declared reader; a TPK route additionally retains its typed realization. The following reduction applies to a fixed positive fibre of that energy and explicitly retains its residual. For a common split between boundary and memory, write

$$Y = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix} > 0, \quad S = A - BC^{-1}B^*, \quad r = y + C^{-1}B^*x.$$

Completing the square and factoring the matrix proves

$$(x, y)^* Y(x, y) = x^* S x + r^* C r, \quad y = r - C^{-1}B^*x, \quad (Y^{-1})_{bb} = S^{-1}.$$

By homogeneity of the Schur complement, the reduced responses are

$$R_{\text{eff}} = LS, \quad H_{\text{eff}} = Q_a S, \quad \Lambda_b = LS^{-1}, \quad R_{\text{eff}} \Lambda_b = L^2 I, \quad H_{\text{eff}} \Lambda_b = \hbar_a c I.$$

The dual length compresses the inverse; the archive (x, r) retains the states that share a minimum. An invertible boundary coordinate T transports R, H by $T^{-*}(\cdot)T^{-1}$ and Λ by $T(\cdot)T^*$, preserving both products. The metric is transported jointly.

Under unitary changes T at the source and S_0 at the target, one jointly transports $B'_\Omega = S_0 B_\Omega T^*$, $P'_i = S_0 P_i S_0^*$, $P' = S_0 P S_0^*$, $X' = T X T^*$, and $\Delta'(Z) = S_0 \Delta(S_0^* Z S_0) S_0^*$. Substitution gives $D' = S_0 D T^*$, $R' = S_0 R S_0^*$, $H' = S_0 H S_0^*$, and $\Lambda' = S_0 \Lambda S_0^*$. The scalar coupling remains fixed. A representation ρ already constructed on the target fibre is transported by $\rho'(g) = S_0 \rho(g) S_0^*$, preserving all its relations by cancellation of $S_0^* S_0$. The preceding exceptional constructions retain their own carriers and maps.

Refinement retains an additional phase constraint. If $\theta \neq 0$ and $e^{-ia\theta/9^n} = e^{-i\theta/9^n}$ for every n , then $(a-1)\theta/9^n \in 2\pi\mathbb{Z}$ tends to zero and is eventually zero. Thus $a = 1$. The lifted action and its refined phases retain the full normalization.

43 Action, gravity, and thermal information

43.1 Dodecaphasic period and circular realization of action

The longitudinal composition in (42.2) determines $L = L_* \alpha^{16} 5759/23040$, and its compatible clock satisfies $t_0 = \pi L / (54 c_{\text{int}})$. The calendar already constructed determines $T_{108} = 108 t_0$. This section transports that realization to the circular reader, retaining $R_{108} = L$. The constitutive velocity, duration, and path length are obtained from the identities in Proposition 39.5 and equations (39.21) and (39.22). In its circular realization,

$$\omega_{108} = \frac{2\pi}{108 t_0}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_{108}} = \frac{54}{\pi} \ell_0, \quad \ell_0 = c_{\text{int}} t_0.$$

The unit ℓ_0 and the time and velocity charts remain explicit; the SI chart does not select the cycle. For each oriented action section $a \in \{\text{pre}, \text{ret}\}$,

$$E_{108,a} = \hbar_a \omega_{108}, \quad m_{108,a} = \frac{E_{108,a}}{c_{\text{int}}^2}.$$

The reduced and full lengths are, respectively,

$$\frac{\hbar_a}{m_{108,a} c_{\text{int}}} = R_{108}, \quad \frac{h_a}{m_{108,a} c_{\text{int}}} = 2\pi R_{108}.$$

The same action and duration enter both readouts.

Proposition 43.1 (Radius-coincidence selector). *In the dimensional collection*

$$G_a(\vartheta) = \vartheta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a}, \quad \vartheta > 0,$$

the circular-realization condition $G_a(\vartheta) m_{108,a} / c_{\text{int}}^2 = R_{108}$ has the unique solution

$$\vartheta_{108}^\circ = (54/\pi)^2.$$

Proof. The first radius is $r_g(\vartheta) = \vartheta(\pi/54)\ell_0$. Comparing it with $(54/\pi)\ell_0$, with $\ell_0 > 0$, reduces the equality to a strictly increasing linear equation. $\square$

Circular coincidence is the restriction to the unit mode of the positive radial reader in Theorem 42.4. Its solution retains the longitudinal and metric compositions declared there. The reduced radius is Gm/c^2 ; the Schwarzschild radius is $2Gm/c^2$ and carries its corresponding factor of two.

Proposition 43.2 (Reciprocal action and gravity, conserved area). *In the two preceding charts,*

$$\frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}, \quad \hbar_a G_a = \vartheta_{108}^\circ c_{\text{int}}^5 t_0^2, \quad \ell_P^2 = \frac{\hbar_a G_a}{c_{\text{int}}^3} = \vartheta_{108}^\circ \ell_0^2.$$

Proof. The definition of G_a contains $\hbar_a^{-1}$, while all remaining factors are common to the two charts. Taking the ratio and product yields the three equalities. $\square$

This reciprocity explains why orientation information can vary while the area remains unchanged. It thus provides a precise relation between the reciprocal action sections, the Planck scale, and the area operator of Section 44.

44 Area operator and reconstruction of sector information

Regrouping six trits constructs a bijection between $(\mathbb{Z}/9\mathbb{Z})^3$ and $(\mathbb{Z}/27\mathbb{Z})^2$, both of cardinality 729. The bijection is not presented as an additive isomorphism. On a 9×9 boundary block, the source decomposes 36 links into two sets of 18, labelled by the channels 90 and 120.

44.1 Tritic regrouping and boundary incidence

The correspondence preserves an ordered word as well as its cardinality. Write each coordinate $x_j \in \mathbb{Z}/9\mathbb{Z}$, for $j = 1, 2, 3$, using its unique positional representatives $x_j = r_{2j-1} + 3r_{2j}$, with $r_k \in \{0, 1, 2\}$. The map is

$$\beta_{729}(x_1, x_2, x_3) = (r_1 + 3r_2 + 9r_3, r_4 + 3r_5 + 9r_6) \in (\mathbb{Z}/27\mathbb{Z})^2. \quad (44.1)$$

Proposition 44.1 (Positional correspondence between interior and boundary). *The map β_{729} is bijective and preserves the order of the six trits. Its inverse extracts the two triples of digits and regroups them into three pairs.*

Proof. The positional expansions of the representatives $0, \dots, 8$ and $0, \dots, 26$ are unique. Extracting the six digits, regrouping them, and extracting them again returns the same word. The inverse operation thus restores each of the three initial coordinates. $\square$

The carry retains its position in the enriched word. The additive operations of the two groupings have different boundaries: for example, $\beta_{729}(0, 2, 0) = (18, 0)$ and $\beta_{729}(0, 1, 0) = (9, 0)$, whose sum in the torus of order twenty-seven is $(0, 0)$, whereas $\beta_{729}(0, 3, 0) = (0, 1)$. This distinction permits reuse of the positional correspondence while keeping the operations typed.

In the torus $(\mathbb{Z}/27\mathbb{Z})^2$, the block $B_\partial = \{0, \dots, 8\}^2$ has the oriented-link sets

$$\partial B_{90} = \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \leq j \leq 8\}, \quad (44.2)$$

$$\partial B_{120} = \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \leq i \leq 8\}. \quad (44.3)$$

All coordinates are read modulo twenty-seven. Each collection contains eighteen links, and the two collections are disjoint. Their labels preserve the incidence channels; the number of links is $18 + 18$.

Theorem 44.2 (Minimum capacity of the cubic cut). *In the cubic graph with vertices $\{0, \dots, N-1\}^3$, where $N \geq 2$, links between neighbours, and unit capacity per link, the minimum cut between two opposite faces has capacity N^2 . For $N = 9$, a maximally mixed product state on the trits of a minimum layer has information*

$$\text{cap}_{\min} = 81, \quad s_{\text{corte}} = 81 \log 3. \quad (44.4)$$

Proof. The N^2 lines parallel to the normal of the faces form edge-disjoint paths. Every separating cut intersects each path and therefore contains at least N^2 links. The links crossing a transverse layer constitute a cut of that cardinality. Each uniform trit contributes $\log 3$ nats; entropy is additive for the product state of the eighty-one trits in the layer. $\square$

The capacity of the cubic cut and the state of the thirty-six links in (44.3) belong to observables defined on different cuts. The genealogy preserves both domains and their measures.

44.2 Transport of collective incidence modes

The spaces $\mathcal{F}_6$ and $\mathcal{F}_8$ of Section 41 supply the uniform vectors

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \quad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f)$$

in $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$. In $\ell^2(\partial B_{90}) \oplus \ell^2(\partial B_{120})$, construct

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{90}} (\delta_e, 0), \quad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{120}} (0, \delta_e).$$

Both pairs are orthonormal. Denote their respective spans by $\mathcal{H}_{\text{inc}}^{\text{coll}}$ and $\mathcal{H}_{\partial}^{\text{coll}}$.

Theorem 44.3 (Collective intertwining of sector labels). *There exists a unique surjective linear isometry between these planes such that*

$$\mathcal{J}_{6 \rightarrow 8} u_m = v_m \quad (m = 90, 120). \quad (44.5)$$

For the operators

$$\begin{aligned} D_{\text{inc}} &= 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \\ D_{\partial} &= 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|, \end{aligned}$$

one has

$$\mathcal{J}_{6 \rightarrow 8} D_{\text{inc}} = D_{\partial} \mathcal{J}_{6 \rightarrow 8}. \quad (44.6)$$

The sector projectors and their oriented combination with coefficients 12 : −1 are also transported.

Proof. The image of a basis fixes a unique linear map. Norms and inner products are preserved because both bases are orthonormal. Identity (44.6) holds on every vector u_m , since both sides equal mv_m . The projectors are expressed as $(120I - D)/30$ and $(D - 90I)/30$; their linear functional calculus and every combination of the two preserve the intertwining relation. $\square$

This transport acts on the two uniform modes and their spectral labels. Individual incidences and fluctuations within each sector remain in their complete spaces. The connection preserves exactly the collective datum that the area weighting will use.

44.3 Angular character and area operator

Let $N_e = \text{diag}(0, 1, 2)$ on each link, and let N_{90}, N_{120} be the sums over each sector. With the scale

$$\theta_{\text{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \quad a_0 = \frac{1 + 2 \cos(10^\circ)}{3} \theta_{\text{clk}},$$

fixed, the angular factor is the normalized real character of the triplet

$$\chi_{10} = \frac{1}{3} \text{Tr} \text{diag}(1, e^{i\pi/18}, e^{-i\pi/18}) = \frac{1 + 2 \cos(10^\circ)}{3}, \quad a_0 = \chi_{10} \theta_{\text{clk}}. \quad (44.7)$$

Summing the conjugate phases proves the equality. On the positive branch under consideration, $\theta_{\text{clk}} > 0$, $\chi_{10} > 0$, and $\gamma_{\text{HMT}} > 0$; hence a_0 and the area scale are positive. The angular triplet and its normalization precede the entropic distribution on the links. With these quantities, the normalized area operator is

$$\hat{\mathcal{A}}/\ell_P^2 = \gamma_{\text{HMT}} a_0 (N_{90} + N_{120}).$$

The reciprocity in (42.4) identifies $\ell_P^2 = L^2$ here in both action sections. The area readout therefore gives $N = \hat{\mathcal{A}}/(L^2 \gamma_{\text{HMT}} a_0)$; the modular Hamiltonian below recovers the second coordinate. The spectrum of $\hat{\mathcal{A}}/L^2$ has values $n\gamma_{\text{HMT}} a_0$, $0 \leq n \leq 72$, with multiplicities given by $(1 + z + z^2)^{36}$. The proof consists of summing the eigenvalues 0, 1, 2 of the 36 tensor factors.

Theorem 44.4 (Recovery of the boundary sectors). *Let*

$$N = N_{90} + N_{120}, \quad D = 90N_{90} + 120N_{120}.$$

Then

$$N_{90} = 4N - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3N. \quad (44.8)$$

Thus, the total and the weighted reader jointly recover the two occupations.

Proof. The matrix $\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}$ has determinant 30. Its inverse gives (44.8). On the domain of integer occupations, the image satisfies $D \in 30\mathbb{Z}$ and $90N \leq D \leq 120N$. $\square$

The modular Hamiltonian in the source is $K_A = -\log \rho_A = cI + \Delta_{\text{mod}} D$. For $\Delta_{\text{mod}} \neq 0$ and known normalization, $D = (K_A - cI)/\Delta_{\text{mod}}$ reads out the second observable. This operator describes the spectral structure of the reduced state; notation alone does not identify it with the generator of temporal evolution. Area preserves the total occupation, and the pair adds provenance. For example, $(N_{90}, N_{120}) = (1, 0)$ and $(0, 1)$ share $N = 1$, but have $D = 90$ and $D = 120$.

44.4 State normalization and the meaning of the modular Hamiltonian

The second observable is made explicit through the state defining it. The preceding tensor decomposition is retained: eighteen qutrits per sector, each with number operator $\text{diag}(0, 1, 2)$; N_m is the sum of these operators, and the sectors act on distinct factors. Let

$$z_m = 1 + e^{-m\Delta_{\text{mod}}} + e^{-2m\Delta_{\text{mod}}}, \quad m \in \{90, 120\}.$$

The state collection in the source has the product realization

$$\rho_A = Z^{-1} e^{-\Delta_{\text{mod}}(90N_{90} + 120N_{120})}, \quad Z = z_{90}^{18} z_{120}^{18}.$$

Commutativity of the number operators and tensor factorization prove the expression for Z . The state has full rank for real, finite Δ_{mod} , and its spectral logarithm gives

$$K_A = -\log \rho_A = (\log Z)I + \Delta_{\text{mod}} D.$$

For $\Delta_{\text{mod}} \neq 0$, K_A weights the excitations of the two sectors differently. The joint readout with N recovers both occupations by equation (44.8).

The need for the joint readout is apparent in two complementary examples. The occupation states $(1, 0)$ and $(0, 1)$ have the same total $N = 1$ and different modular weights. Conversely, $(4, 0)$ and $(0, 3)$ have the same $D = 360$ and different totals. The pair (N, D) separates both cases. This recovery concerns sector occupations, rather than each microscopic vector within a degeneracy of those occupations.

When $\Delta_{\text{mod}} = 0$, the state is proportional to the identity and K_A is scalar: this readout ceases to distinguish the sectors. For $\Delta_{\text{mod}} \neq 0$, the known normalization permits recovery of D and its composition with area. This defines the role of the modular Hamiltonian: it characterizes the reduced state and its information distribution. Its inclusion in the article is motivated by this mathematical function, whereas the evolution Hamiltonian belongs to the dynamical development of action.

44.5 Reduced state, Schmidt decomposition, and entanglement entropy

If $\rho_A = \sum_{\nu} p_{\nu} |\nu\rangle\langle\nu|$, the purification

$$|\Psi\rangle = \sum_{\nu} \sqrt{p_{\nu}} |\nu\rangle_A |\nu\rangle_B$$

satisfies $\text{Tr}_B |\Psi\rangle\langle\Psi| = \rho_A$. Thus, $-\text{Tr}(\rho_A \log \rho_A)$ is the entanglement entropy of this specific pure bipartition. The enlarged state specifies which system supports that interpretation.

For a two-sheet involution with one eigenvector of each sign,

$$\rho_y = \frac{e^{-yR}}{2 \cosh y}, \quad R^2 = I,$$

diagonalization yields

$$S(\rho_y) = \log(2 \cosh y) - y \tanh y, \quad D(\rho_y \| \rho_{-y}) = 2y \tanh y.$$

The entropy is even in y ; the divergence explicitly compares the two orientations. This two-level entropy is not identified with the capacity $81 \log 3$ of the triadic cut.

45 Genealogical memory, informational reduction, and thermal transduction

The thermal construction uses two readouts of the same APP–TRIT–TPK development. The first preserves the states, admissible paths, and record information; the second supplies the oriented action and the return duration. Their composition determines an energy per tritic decision. Temperature and energy coordinates are introduced subsequently through explicit dimensional maps. This sequence identifies which information is preserved, which information is read out by a reader, and which quantity converts that information into dimensional entropy.

45.1 Reader, memory, and state recovery

Fix a finite level of the development and let Ω denote the set of admissible enriched states. The record preserves the sheet, orientation, remainder, quotient, carry, path, phase, and memory on the domains where each coordinate enters. Let

$$q : \Omega \longrightarrow Y, \quad M : \Omega \longrightarrow \mathcal{M}$$

be the visible readout and the retained memory, respectively. For a distribution p on Ω , write $H(p) = -\sum_x p_x \log p_x$, with $0 \log 0 = 0$. The logarithm is applied to dimensionless probabilities.

Definition 45.1 (Genealogical recovery of a readout). The pair $\hat{q} = (q, M)$ is a recoverable reader on the support Ω_p when there exists

$$d : \hat{q}(\Omega_p) \longrightarrow \Omega_p, \quad d \circ \hat{q} = \text{id}_{\Omega_p}.$$

The fibre information of q is $H_{\text{fib}}(q; p) = H(X \mid q(X))$, where X has law p .

Proposition 45.2 (Recovery identity and cost of reduction). *For a recoverable reader,*

$$H(X \mid q(X), M(X)) = 0, \tag{45.1}$$

$$H(X) = H(q(X)) + H(X \mid q(X)), \tag{45.2}$$

$$H(X \mid q(X)) = H(M(X) \mid q(X)). \tag{45.3}$$

Proof. The map d recovers X from the pair $(q(X), M(X))$, so the first conditional entropy vanishes. Since q is a function of X , the chain rule gives the second equality. For the third, expand $H(X, M(X) \mid q(X))$ in two ways. One yields $H(X \mid q(X))$, because M is a function of X ; the other yields $H(M(X) \mid q(X)) + H(X \mid q(X), M(X))$. $\square$

Euclidean division on the APP sheets exhibits the elementary arithmetic mechanism. For $i, j \in \{1, \dots, 9\}$, one preserves

$$i + j = 9q_{ij}^+ + R_{ij}^+, \quad ij = 9q_{ij}^\times + R_{ij}^\times, \quad 1 \leq R_{ij}^\sigma \leq 9 \quad (\sigma \in \{+, \times\}).$$

The remainder–quotient pair recovers the integer value of the operation. Summing the differences over the eighty-one positions gives

$$2025 - 810 = 54 + 9 \cdot 129.$$

The quotient memory restores the contribution not read out by the residual reduction. Recoverability of the complete state additionally requires preservation of the relevant position and path coordinates; the remainder–quotient pair here identifies the result of the operation.

45.2 Reversible updating, tritic decision, and erasure

Let $U : \Omega \rightarrow \Omega$ be the TPK update on a domain where the enriched transport has an inverse. The transported law is $p'_y = p_{U^{-1}y}$. The bijection preserves the multiset of probabilities and therefore

$$H(U(X)) = H(X), \quad H(X | U(X)) = 0. \quad (45.4)$$

If the observed output is $q(U(X))$, the information not read out is quantified by $H(X | q(U(X)))$. The complete state, its readout, and the resetting of a record thus define three distinct maps.

For the three oriented states of the TRIT, let $p = (p_{-1}, p_0, p_{+1})$ be a normalized distribution. Its information is

$$I_{\text{TRIT}}(p) = - \sum_{a=-1}^1 p_a \log p_a, \quad 0 \leq I_{\text{TRIT}}(p) \leq \log 3. \quad (45.5)$$

The upper endpoint corresponds to $p_a = 1/3$. Indeed,

$$D(p \| (1/3, 1/3, 1/3)) = \sum_a p_a \log(3p_a) = \log 3 - I_{\text{TRIT}}(p) \geq 0;$$

the inequality follows from the convexity of $t \log t$, and equality characterizes the uniform distribution. The lower endpoint corresponds to a distribution concentrated on one state.

The ideal Landauer realization uses a memory with energetically degenerate logical states, an isothermal environment at temperature Θ , and a balance that preserves the output of the logical map and accounts for the auxiliary records. In this model, the decrease in logical entropy is $k_B[H(X) - H(F(X))] = k_B H(X | F(X))$ for a deterministic map F . The entropy balance requires at least this amount to be transferred to the environment; its energetic expression is $Q \geq k_B \Theta H(X | F(X))$. The distinction between reversible logical evolution and physical erasure corresponds to the Landauer-Bennett realization [20]. The invertible update (45.4) has zero logical contribution; resetting a uniform triple requires the transfer of $\log 3$ nats to the environment and satisfies $Q \geq k_B \Theta \log 3$. The control energy and dissipation of a specific physical realization belong to its additional thermodynamic balance.

45.3 Action, frequency, and decision energy

Let $\mathcal{L}_{\text{act}}$ and $\mathcal{L}_T$ be the dimensional lines of action and duration; the energy line is $\mathcal{L}_E = \mathcal{L}_{\text{act}} \otimes \mathcal{L}_T^*$. The preceding development supplies the positive sections $\hbar_a \in \mathcal{L}_{\text{act}}$, with $a \in \{\text{pre}, \text{ret}\}$, and the elementary duration $t_0 \in \mathcal{L}_T$. The complete calendar supplies

$$T_{108} = 108t_0, \quad \omega_{108} = \frac{2\pi}{T_{108}}, \quad h_a = 2\pi\hbar_a.$$

The dimensional contraction between action and frequency defines

$$\epsilon_a := \hbar_a \omega_{108} = \frac{h_a}{108t_0} = \frac{\hbar_a}{t_P}, \quad t_P = \frac{54}{\pi} t_0. \quad (45.6)$$

The final expression uses the same reduced circular period as the gravitational section. The three expressions correspond to a single energy section, with the orientation a retained in every factor. The normalized information of a uniform decision yields the energy section per nat

$$\epsilon_a := \frac{\epsilon_a}{\log 3} \in \mathcal{L}_E^+. \quad (45.7)$$

The previously constructed action determines the energy scale once the duration is fixed; the cardinality of the TRIT alternatives determines its informational normalization. The value $\log 3$ enters through the counting and measure specified in (45.5).

45.4 Dimensional definition of Boltzmann's constant and coordinate selection

Let $\mathcal{L}_\Theta$ be the oriented temperature line. The dimensional entropy space is $\mathcal{L}_{\text{ent}} = \mathcal{L}_E \otimes \mathcal{L}_\Theta^*$. A positive linear map $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ is equivalently a positive element of $\mathcal{L}_{\text{ent}}$. The same object can therefore act on a temperature or multiply dimensionless information.

Definition 45.3 (Thermal normalization of the tritic decision). A thermal pair compatible with the oriented section a consists of a positive transducer k_B and a positive section $\Theta_{\text{clk},a} \in \mathcal{L}_\Theta$ such that

$$k_B(\Theta_{\text{clk},a}) = \epsilon_a, \quad k_B \Theta_{\text{clk},a} \log 3 = \epsilon_a. \quad (45.8)$$

Boltzmann's constant serves as the energy-temperature transducer of tritic information in this realization.

Theorem 45.4 (Uniqueness relative to the thermal section and dimensional covariance). *For a fixed positive section $\Theta_{\text{clk},a}$, there exists a unique positive linear map satisfying (45.8). For positive bases u_E, u_Θ , if*

$$\epsilon_a = \epsilon_a^{\text{num}} u_E, \quad \Theta_{\text{clk},a} = \theta_a u_\Theta, \quad k_B(u_\Theta) = b u_E,$$

its coordinate is

$$b = \frac{\epsilon_a^{\text{num}}}{\theta_a \log 3}. \quad (45.9)$$

Under $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$, with $s_E, s_\Theta > 0$, one has

$$b' = \frac{s_\Theta}{s_E} b, \quad \theta'_a = \frac{\theta_a}{s_\Theta}, \quad (\epsilon_a^{\text{num}})' = \frac{\epsilon_a^{\text{num}}}{s_E}.$$

Proof. A nonzero section spans a line. Specifying its image determines a unique linear map; positivity of both sections proves positivity of the map. Substitution of the coordinates into (45.8) gives $b \theta_a \log 3 = \epsilon_a^{\text{num}}$. The three transformation laws follow by expressing the same sections and map in the new bases. $\square$

The same Boltzmann constant must act in both orientations. Since t_P is common, simultaneous compatibility is equivalent to

$$\frac{\Theta_{\text{clk,pre}}}{\Theta_{\text{clk,ret}}} = \frac{\epsilon_{\text{pre}}}{\epsilon_{\text{ret}}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}. \quad (45.10)$$

Indeed, an isomorphism of lines preserves ratios of sections. Conversely, if the preceding equality holds, the isomorphism determined in one orientation also satisfies the normalization in the other. Thus, the return changes the thermal section of the cycle in the same proportion as the energy, while the transducer remains common to both readouts.

The preceding equation makes explicit the choice required to read out an individual value. If the thermal section has not yet been selected by an independent reader, then, for every $\lambda > 0$, the pair

$$(k_B, \Theta_{\text{clk},a}) \longmapsto (\lambda k_B, \lambda^{-1} \Theta_{\text{clk},a})$$

preserves ϵ_a . Consequently, presenting an internally selected individual coordinate requires specification of the reader for $\Theta_{\text{clk},a}$, the bases, and their dimensional transport. The action and information equations remain determined before this specification. The subsequent identification of the bases with the joule and kelvin belongs to the metrological readout of the result.

45.5 Dimensional entropy and information of the reduced state

For a distribution p of admissible paths, or a reduced state with density matrix ρ , let

$$s(p) = - \sum_x p_x \log p_x, \quad s(\rho) = - \text{Tr}(\rho \log \rho).$$

Their dimensional readouts are

$$S(p) = k_B s(p), \quad S(\rho) = k_B s(\rho) \quad \text{in } \mathcal{L}_{\text{ent}}. \quad (45.11)$$

Evaluation at Θ yields an energy $\Theta S = k_B \Theta s$. For the uniform triple at the cycle temperature, (45.8) gives exactly

$$S_{\text{TRIT}} = k_B \log 3, \quad \Theta_{\text{clk},a} S_{\text{TRIT}} = \epsilon_a.$$

If ρ_A is the reduction of the bipartite purification constructed in the area section, $s(\rho_A)$ measures the entanglement of that bipartition. Unit transport preserves its eigenvalues, orientation, and sector occupations. The area scale $\gamma a_0 \ell_P^2$ converts occupations into area; the transducer k_B converts information into dimensional entropy. Both readouts use the same record through distinct operators.

45.6 Spectral normalization of thermal histories

Let $\mathcal{G}$ be an already constructed finite graph of admissible extensions, with adjacency matrix A . Each edge e preserves its path coordinates and receives a dimensionless action increment $a_*(e) > 0$ from the corresponding reader. Once the energy section ϵ_* has been chosen, set $t = \beta \epsilon_*$, where $\beta = (k_B \Theta)^{-1}$. The thermal operator is

$$(L_t f)(y) = \sum_{e: x \rightarrow y} \exp[-t a_*(e)] f(x). \quad (45.12)$$

All its exponents are dimensionless. The condition $\rho(L_t) = 1$ normalizes the weighted growth of histories.

Proposition 45.5 (Existence and uniqueness of normalized pressure). *Suppose that A is irreducible, $\rho(A) > 1$, and $0 < m \leq a_*(e) \leq M$ for every edge. There exists a unique $t_* > 0$ such that $\rho(L_{t_*}) = 1$, and*

$$\frac{\log \rho(A)}{M} \leq t_* \leq \frac{\log \rho(A)}{m}. \quad (45.13)$$

Proof. Entrywise comparison yields $e^{-tM} A \leq L_t \leq e^{-tm} A$ for $t \geq 0$. Monotonicity of the spectral radius of nonnegative matrices implies the corresponding bounds for $\rho(L_t)$. The spectral radius is continuous, equals $\rho(A) > 1$ at zero, and tends to zero at infinity. Moreover, for $s > 0$, $L_{t+s} \leq e^{-sm} L_t$, so that $\rho(L_{t+s}) \leq e^{-sm} \rho(L_t) < \rho(L_t)$. Continuity and strict decrease prove existence and uniqueness. Substituting the value one into the initial bounds proves (45.13). $\square$

The local realization of three equally weighted alternatives, represented by a vertex with three loops and $a_*(e) = 1$, satisfies $\rho(L_t) = 3e^{-t}$. Its normalization gives $t_* = \log 3$. With $\epsilon_* = \epsilon_a$, one obtains $\beta_{\text{clk}} = \log 3 / \epsilon_a$, exactly the relation (45.8). This calculation identifies the local tritic coincidence. For a history graph with additional constraints, coincidence with the same clock requires verification of

$$\rho(L_{(\log 3 / \epsilon_a) \epsilon_*}) = 1$$

with its effective increments. Local normalization and the pressure of the complete system thus retain their respective domains.

The resulting relation brings together structure, value, and meaning. The record determines recoverable information; the probabilities on the continuations determine its entropic readout; action and period determine ϵ_a ; the energy–temperature transducer reads out dimensional entropy. Area and entanglement are related through the reduced state and its boundary operators, while keeping explicit which datum produces each quantity.

46 Oriented incidence and the gravitational tensor representation

The exceptional representation of the state preserves the incidence accompanying its arithmetic readout. The ternary matrix A_W in (16.4) provides a precise material antecedent here. Its balanced lift $C = \text{bal}(A_W)$, with representatives 0, 1, -1 , satisfies

$$C = C^T, \quad C^2 = 5I_6, \quad \text{Tr } C = 0.$$

The six positions have the order $(\infty, 0, 1, 2, 3, 4)$. This incidence first produces a three-dimensional space; its quadratic readout then produces the five-component space on which the gravitational coupling acts.

46.1 Signed covering and calendar transport

Define

$$E_C = \frac{1}{2}(I_6 + C/\sqrt{5}), \quad \mathcal{H}_C = \text{im } E_C, \quad v_{u,\sigma} = \sigma\sqrt{2}E_C e_u, \quad \sigma \in \{1, -1\}.$$

Proposition 46.1 (Twelve oriented positions). *The vectors $v_{u,\sigma}$ are twelve distinct unit vectors in a three-dimensional space. Their incidence is determined by $\langle v_{u,\sigma}, v_{v,\tau} \rangle = 1/\sqrt{5}$. Orientation reversal acts by the antipodal map.*

Proof. The quadratic identity for C implies $E_C^2 = E_C = E_C^*$; its trace is three. Since the diagonal of C is zero,

$$\langle v_{u,\sigma}, v_{v,\tau} \rangle = \begin{cases} \sigma\tau, & u = v, \\ \sigma\tau C_{uv}/\sqrt{5}, & u \neq v. \end{cases}$$

This proves normalization, distinctness of the twelve vectors, and incidence. Changing σ to $-\sigma$ changes the sign of the vector. $\square$

An explicit Euclidean expression is obtained by setting $\nu = \sqrt{1 + \varphi^2}$ and

$$B = \frac{1}{\nu} \begin{pmatrix} 0 & -1 & -\varphi & 0 & \varphi & 1 \\ -1 & -\varphi & 0 & 1 & 0 & -\varphi \\ -\varphi & 0 & -1 & -\varphi & -1 & 0 \end{pmatrix}.$$

The relation $\varphi^2 = \varphi + 1$, for the already generated coordinate, gives $B^*B = 2E_C$. Consequently, $B/\sqrt{2}|_{\mathcal{H}_C}$ is an isometry and maps $v_{u,\sigma}$ to $\sigma B e_u$. The image is

$$\mathcal{V} = \frac{1}{\nu}(\{0\} \times \{\pm 1\} \times \{\pm \varphi\} \cup \{\pm 1\} \times \{\pm \varphi\} \times \{0\} \cup \{\pm \varphi\} \times \{0\} \times \{\pm 1\}).$$

This relates the internal coordinate, the incidence, and its twelve geometric representatives.

The calendar coordinate is written as $(b, r) \in \mathbb{Z}/12\mathbb{Z} \times \{0, \dots, 8\}$. A route of length n induces

$$k(r, n) = \left\lfloor \frac{r+n}{9} \right\rfloor, \quad (b, r) \mapsto (b + k(r, n) \bmod 12, (r + n) \bmod 9).$$

The explicit alignment of its positions $p = b + 1$ with the signed chart is

p	1	2	3	4	5	6	7	8	9	10	11	12
u	∞	0	0	∞	1	3	3	1	2	4	4	2
σ	+	-	+	-	-	+	-	+	-	+	-	+

This table fixes a trivialization of the carrier. Each change of position jointly transports the vectors and their incidence matrix.

Proposition 46.2 (Frame covariance under return). *If $Se_p = e_{p+1 \bmod 12}$, the matrix representation of a carrier operator P is transported as $P_k = S^k P S^{-k}$. This transport preserves inner products, incidences, and route composition. After twelve returns the chart returns, while the memory accumulated in the source state is preserved.*

Proof. S is orthogonal and $S^{12} = I$; the metric properties and chart periodicity follow. For composition, with $r' = (r + n) \bmod 9$, Euclidean division gives $k(r, n + m) = k(r, n) + k(r', m)$. The exponents of S add exactly with the carry. The operation acts on the calendar coordinate; the route's memory updates remain present in its enriched domain. $\square$

46.2 Quadratic readout and the five-component space

Let $\mathbb{R}^{\mathcal{V}}$ be the space of twelve coordinates, $Ae_v = e_{-v}$ its antipodal involution, and $J = \mathbf{1}\mathbf{1}^T$. Define

$$P_5 = \frac{1}{2}(I + A) - \frac{1}{12}J, \quad Q_v = vv^T - \frac{1}{3}I_3, \quad \Gamma_0 x = \sum_{v \in \mathcal{V}} x_v Q_v.$$

Theorem 46.3 (Five-component projector and tensor isometry). *P_5 is the rank-five orthogonal projector onto the space of coordinates even under the antipodal map and having zero sum. Moreover,*

$$\Gamma_0^* \Gamma_0 = \frac{8}{5}P_5, \quad \Gamma_0 \Gamma_0^* = \frac{8}{5}I_{\text{Sym}_0^2(\mathbb{R}^3)}.$$

Thus

$$\Gamma_5 = \sqrt{\frac{5}{8}} \Gamma_0|_{\text{im } P_5}$$

is an isometry onto the space of symmetric trace-free tensors.

Proof. The even space has one coordinate for each antipodal pair, so its dimension is six. The constant vector belongs to it; subtracting its rank-one projector gives P_5 , of rank five.

Since $\langle Q_v, Q_w \rangle_F = (v^T w)^2 - 1/3$, the entries of $\Gamma_0^* \Gamma_0$ equal $2/3$ for $w = \pm v$ and $-2/15$ in the other ten positions of each row. These are exactly the entries of $(8/5)P_5$.

The identity can also be checked in tensor space without appealing to a classification of representations. The vertex list gives

$$\sum_v v_i^4 = \frac{12}{5}, \quad \sum_v v_i^2 v_j^2 = \frac{4}{5} \quad (i \neq j),$$

and moments containing any odd power vanish under sign changes. Indeed, $(1 + \varphi^2)^2 = 5\varphi^2$ and $1 + \varphi^4 = 3\varphi^2$. For a symmetric tensor Q , these sums give

$$\sum_v (v^T Q v) vv^T = \frac{8}{5}Q + \frac{4}{5}(\text{tr } Q)I.$$

With $\text{tr } Q = 0$, and using $\sum_v v^T Q v = 4 \text{tr } Q = 0$, we obtain $\Gamma_0 \Gamma_0^* Q = (8/5)Q$. The two identities prove the normalized isometry and its surjectivity. $\square$

The gravitational section in 43 determines the operator $G_a P_5$. Its spectrum has G_a with multiplicity five and zero with multiplicity seven. Orientation is preserved in the carrier and its transport; G_a fixes the common coupling.

46.3 Transverse projection and helicities

For $n \in S^2$, set $\Pi = I - nn^\top$ and

$$\Lambda_n Q = \Pi Q \Pi - \frac{1}{2} \text{tr}(\Pi Q \Pi) \Pi.$$

Proposition 46.4 (Rank-two transverse reduction). Λ_n is the orthogonal projector onto $\{Q \in \text{Sym}_0^2(\mathbb{R}^3) : Qn = 0\}$, of dimension two.

Proof. $\Pi n = 0$ gives transversality, and $\text{tr} \Pi = 2$ gives zero trace. A transverse trace-free tensor remains fixed. The formula is self-adjoint for the Frobenius inner product. In a frame (e_1, e_2, n) , the image has orthonormal basis

$$E_+ = \frac{e_1 e_1^\top - e_2 e_2^\top}{\sqrt{2}}, \quad E_\times = \frac{e_1 e_2^\top + e_2 e_1^\top}{\sqrt{2}}.$$

□

The other three vectors in an orthonormal basis of the carrier are

$$E_1 = \frac{e_1 n^\top + n e_1^\top}{\sqrt{2}}, \quad E_2 = \frac{e_2 n^\top + n e_2^\top}{\sqrt{2}}, \quad E_0 = \frac{2nn^\top - e_1 e_1^\top - e_2 e_2^\top}{\sqrt{6}}.$$

Under a rotation of the transverse frame through an angle θ , the pairs $(E_+, E_\times)$ and (E_1, E_2) rotate through 2θ and θ , respectively, while E_0 remains fixed. The complexification therefore has weights $2, -2, 1, -1, 0$. Λ_n preserves the first two and annihilates the rest. This establishes the chain

$$\mathbb{R}^{12} \xrightarrow{P_5} \text{im } P_5 \xrightarrow{\Gamma_5} \text{Sym}_0^2(\mathbb{R}^3) \xrightarrow{\Lambda_n} \mathcal{H}_{\text{TT}}(n), \quad 12 \longrightarrow 5 \longrightarrow 5 \longrightarrow 2.$$

The five-component representation and its reduction are algebraic results preceding the choice of field equations. In the massless gravitational realization, the gauge constraints select the transverse image. This distinction preserves both the five structural coordinates and the two radiative helicities.

47 Phase return and representation of accumulated memory

The chronology of enriched routes distinguishes position return, orientation return, and phase return. The observable representation identifies certain arrival states, while the cumulative register preserves the sequence traversed. The geometric representation must transport both data together.

47.1 Reversal cocycle and change of sheet

Let F_{obs} be the chronological representation of TPK transport. Its canonical blocks have the hierarchy

27	position return with orientation reversal,
54	position and orientation return,
108	complete return of the observable phase.

The register counts orientation reversals $g(a)$ and effective changes of sheet $s(a)$. For a route,

$$c(\gamma) = \sum_{a \in \gamma} (g(a), s(a)), \quad c(\gamma_2 \gamma_1) = c(\gamma_2) + c(\gamma_1). \quad (47.1)$$

The canonical computation for the block of length 27 expresses $c_{27} = (1, 18)$. Composition preserves these increments in the four blocks completing the observable period. On forward-oriented routes the register is cumulative; its extension to the route groupoid uses $\mathbb{Z}^2$ and $c(\bar{\gamma}) = -c(\gamma)$.

Proposition 47.1 (Memory of the complete observable turn). *Writing $\mathcal{K}_{27} = F_{\text{obs}}^{27}$, the lift to the register, for $x \in \mathcal{O}_{108}$ and $z \in \mathbb{Z}^2$, satisfies*

$$\tilde{\mathcal{K}}_{27}(x, z) = (\mathcal{K}_{27}x, z + (1, 18)), \quad \tilde{\mathcal{K}}_{27}^4(x, z) = (x, z + (4, 72)).$$

After N complete turns, the increment is $N(4, 72)$.

Proof. The observable projection of $\mathcal{K}_{27}$ has order four. The preceding canonical computation fixes $c_{27} = (1, 18)$ constantly on that observable orbit. Additivity in (47.1) sums the four increments and gives $c_{108} = 4(1, 18) = (4, 72)$. Repetition applies the same concatenation law and produces Nc_{108} . Return of the first coordinate preserves the increment of the second. $\square$

47.2 Affine representation of the cocycle

The discrete realization uses an element R_4 of order four in $\text{Spin}^+(1, 3)$, a unit timelike vector u fixed by its vector action, and a unit of length $\ell_0 > 0$. The scale ℓ_0 retains its provenance in the dimensional section. Define

$$\rho_C^{\text{disc}}(\gamma) = (R_4^{c_s(\gamma)}, \ell_0 c_s(\gamma)u) \in \text{Spin}^+(1, 3) \ltimes \mathbb{R}^{1,3}. \quad (47.2)$$

The power acts on the vector through the associated vector representation of the spin group.

Theorem 47.2 (Composition and translational amplitude of return). *The map ρ_C^{disc} preserves the identity, concatenation, and inversion. On complete returns,*

$$\rho_C^{\text{disc}}(\gamma_{108}) = (I, 72\ell_0 u), \quad \rho_C^{\text{disc}}(\gamma_{108}^N) = (I, 72N\ell_0 u). \quad (47.3)$$

Inversion changes the sign of the translation.

Proof. The semidirect product is $(R, a)(S, b) = (RS, a + Rb)$. Since $R_4 u = u$,

$$\begin{aligned} \rho_C^{\text{disc}}(\gamma_2)\rho_C^{\text{disc}}(\gamma_1) &= (R_4^{c_s(\gamma_2)+c_s(\gamma_1)}, \ell_0(c_s(\gamma_2) + c_s(\gamma_1))u) \\ &= \rho_C^{\text{disc}}(\gamma_2\gamma_1). \end{aligned}$$

The cocycle of the identity is zero, and that of the inverse is its opposite. For the complete turn, substitute $c_{108} = (4, 72)$ and $R_4^4 = I$. Concatenation of N turns keeps the rotational part equal to the identity and sums their translations. $\square$

The amplitude $72\ell_0$ is a length associated with a finite route. A realization through a co-tetrad and connection transports it to its affine holonomy. The local contribution to the gravitational equations is obtained subsequently through the curvature of that connection, the variation of the matter action, and the normalization of its density. The representation thus preserves the exact cumulative content that must enter that realization.

48 Boundary soldering and energy response

The cubic realization of transport preserves six oriented incidences. Its geometric quotient, Lorentzian form, and energy response are obtained from the same incidence matrix. Transport originates in the enriched APP–TRIT–TPK route: face, sign, carry, and memory accompany each edge. This section develops the linear operation acting on that boundary representation.

48.1 Cubic incidence and the rank-four quotient

Let $H_F = \mathbb{R}^F$, with $F = \{(i, \epsilon) : 1 \leq i \leq 3, \epsilon \in \{-1, 1\}\}$, its Euclidean inner product, and its basis $e_{i,\epsilon}$. Face opposition is the orthogonal involution $Re_{i,\epsilon} = e_{i,-\epsilon}$. Define

$$u = \sum_{i,\epsilon} e_{i,\epsilon}, \quad P_u = \frac{uu^\top}{6}, \quad P_- = \frac{I - R}{2}, \quad P_0 = \frac{I + R}{2} - P_u, \quad P_\square = P_u + P_-.$$

The vertices are $V = \{-1, 1\}^3$. The incidence matrix $M : \mathbb{R}^V \longrightarrow H_F$ is determined by

$$M_{(i,\epsilon),\nu} = \mathbf{1}_{\{\nu_i=\epsilon\}}.$$

Proposition 48.1 (Incidence decomposition). *We have*

$$MM^\top = 12P_u + 4P_-, \quad \ker M^\top = P_0H_F.$$

Consequently, the quotient $Q = H_F/P_0H_F$, orthogonally identified with $P_\square H_F$, has dimension four.

Proof. A face contains four vertices. Two opposite faces have no common vertices, and two faces on different axes have two common vertices. Writing $\mathbf{J} = uu^\top$, we obtain

$$MM^\top = 4I + 2(\mathbf{J} - I - R) = 2(I - R) + 2\mathbf{J} = 4P_- + 12P_u.$$

The three projectors P_u, P_-, P_0 are orthogonal and sum to I ; their ranks are respectively one, three, and two. The identity $\|M^\top x\|^2 = \langle x, MM^\top x \rangle$ identifies the stated kernel. The complementary rank is four. $\square$

On Q , the form

$$B = P_- - P_u$$

has signature $(3, 1)$. The time orientation is fixed by u . With

$$t = \frac{u}{\sqrt{6}}, \quad d_i = \frac{e_{i,+} - e_{i,-}}{\sqrt{2}}, \quad W = \sqrt{6}P_u + \sqrt{2}P_-,$$

we have $We_{i,\epsilon} = t + \epsilon d_i$. These six vectors are null for B , whereas

$$3t + \nu_1 d_1 + \nu_2 d_2 + \nu_3 d_3$$

has quadratic norm -6 . The same incidence satisfies $MM^\top = 2W^*W$ on Q . Proper rotations of the cube commute with the projectors, preserve both forms, and fix t . These identities determine the metric of the cubic representation.

48.2 Cellular soldering and memory transport

For each oriented edge a , representation of the route provides the face $\lambda(\underline{a})$, the sign $\sigma(a)$, the source and target fibres, and the transport $U(a) : Q_{s(a)} \longrightarrow Q_{t(a)}$. Inversion satisfies $U(\bar{a}) = U(a)^{-1}$. Boundary half-edges retain their formal inversion and their incidence on the boundary. With $\ell_m = \ell_0 9^{-m} > 0$, the soldering is

$$\beta_m(a) = \ell_m \sigma_m(a) n_{\lambda(\underline{a})}^{s(a)}, \quad n_{i,\epsilon} = t + \epsilon d_i. \quad (48.1)$$

The face and sign are transported together with the route memory. The inversion convention requires

$$\beta_m(\bar{a}) = -U_m(a)\beta_m(a). \quad (48.2)$$

Applying this relation twice recovers $\beta_m(a)$; the carry register separately preserves the ordered composition.

Proposition 48.2 (Naturality of soldering under refinement). *Let $a = a_1 \cdots a_9$ be a compatible refinement of an edge. Suppose that the nine successor incidences, transported to the predecessor fibre, preserve face and sign, and that $\ell_{m+1} = \ell_m/9$. Let $\mathcal{U}_j : Q_{m,s(a)} \rightarrow Q_{m+1,s(a_j)}$ be the accumulated transport including the refinement identification and the path to the source of successor j . Then*

$$\beta_m(a) = \sum_{j=1}^9 \mathcal{U}_j^{-1} \beta_{m+1}(a_j).$$

The equality is compatible with inversion and with concatenation of the memory register.

Proof. Each summand, expressed in the fibre at the predecessor's source, is $(\ell_m/9)\sigma_m(a)n_{\lambda(\underline{a})}$. The sum of nine terms reproduces the predecessor's soldering. Inversion reverses the order of the transports and replaces each by its inverse; relation (48.2) supplies the overall sign. Memory is concatenated in that same order, so refinement preserves both the vector representation and its enriched register. $\square$

48.3 Positive response, uniqueness, and minimal extension

The linear screen soldering is $\bar{\beta}_m = \ell_m W|_Q$. It is invertible, with

$$\bar{\beta}_m^{-1} = \ell_m^{-1} \left(\frac{1}{\sqrt{6}} P_u + \frac{1}{\sqrt{2}} P_- \right).$$

The eigenvalue $5/9$ of the antisymmetric sector of the normalized block $B_c/9$, constructed in Section 4.4, determines the compression coefficient $q_{\text{vis}} = 5/9$. We reserve this notation to distinguish it from the Euclidean quotient $q_9(n)$ of the arithmetic kernel. Its realization on the screen uses an isometry $V_9 : Q \rightarrow \mathcal{K}$. The visible contribution $C_9 = q_{\text{vis}} V_9$ and its complement $D_9 = \sqrt{1 - q_{\text{vis}}^2} V_9$ satisfy

$$C_9^* C_9 + D_9^* D_9 = I_Q, \quad D_9^* D_9 = \frac{56}{81} I_Q.$$

The coefficient 56 is $9^2 - 5^2$; it belongs to the quadratic defect of this compression reader.

Theorem 48.3 (Energy response determined by soldering). *There exists a unique operator $\Lambda_m : Q \rightarrow Q$ such that $\bar{\beta}_m^* \Lambda_m \bar{\beta}_m = D_9^* D_9$. It is positive definite, and*

$$\Lambda_m = \frac{28}{243 \ell_m^2} (P_u + 3P_-) = \frac{112}{81 \ell_m^2} (MM^T|_Q)^{-1}. \quad (48.3)$$

On H_F , the corresponding identity has right-hand side $(56/81)P_{\square}$.

Proof. Congruence by the invertible soldering necessarily determines

$$\Lambda_m = \bar{\beta}_m^{-*} \frac{56}{81} I_Q \bar{\beta}_m^{-1}.$$

Substituting the inverse, the coefficient of P_u is $56/(486 \ell_m^2) = 28/(243 \ell_m^2)$; that of P_- is $28/(81 \ell_m^2)$. Both are positive. By Proposition 48.1, $(MM^T|_Q)^{-1} = P_u/12 + P_-/4$, which gives the second expression. Extension to six faces annihilates $P_0 H_F$; the identity therefore extends through $P_{\square}$. $\square$

Proposition 48.4 (Characterization by minimal extension). *For every $b \in Q$,*

$$\langle b, \Lambda_m b \rangle = \frac{112}{81\ell_m^2} \min_{Mx=b} \|x\|^2.$$

The minimizing extension is unique: $x_{\min} = M^\top(MM^\top|_Q)^{-1}b$.

Proof. The stated vector satisfies $Mx_{\min} = b$ and belongs to $\text{im } M^\top = (\ker M)^\perp$. Every other solution is $x_{\min} + z$, with $z \in \ker M$; by orthogonality, its squared norm is $\|x_{\min}\|^2 + \|z\|^2$. Moreover,

$$\|x_{\min}\|^2 = \langle b, (MM^\top|_Q)^{-1}b \rangle.$$

The formula for Λ_m completes the proof. $\square$

The operator $MM^\top|_Q$ realizes the Dirichlet-to-Neumann response of this incidence formulation; its inverse is the Neumann-to-Dirichlet response. The preceding energy corresponds to the latter, with the normalization of soldering and of the compression defect. The Schur complement of $\begin{pmatrix} I & M^\top \\ M & 0 \end{pmatrix}$ is $-MM^\top$, providing the same variational elimination.

48.4 Conservation of density under refinement

Consider the sector of cubic transports, orthogonal for the energy inner product and compatible with the form B . Fix a common label space Q^{lab} , identified isometrically with the predecessor's screen. The solderings have types $\tilde{\beta}_m : Q^{\text{lab}} \rightarrow Q_m$ and $\tilde{\beta}_{m+1,j} : Q^{\text{lab}} \rightarrow Q_{m+1,j}$; the operators Λ act in their respective target fibres. If $U_j : Q_j \rightarrow Q$ transports the successor fibre to the predecessor fibre, then

$$\tilde{\beta}_{m+1,j} = \frac{1}{9}U_j^{-1}\tilde{\beta}_m, \quad \Lambda_{m+1,j} = 81U_j^{-1}\Lambda_m U_j.$$

The factor 81 comes from $\ell_{m+1}^{-2} = 81\ell_m^{-2}$.

Theorem 48.5 (Invariance of the normalized density). *Each successor preserves the predecessor's quadratic response:*

$$\tilde{\beta}_{m+1,j}^* \Lambda_{m+1,j} \tilde{\beta}_{m+1,j} = \frac{56}{81} I_{Q^{\text{lab}}}.$$

For r refinements, the joint response, expressed in the predecessor fibres, is

$$E_{m+r} = \frac{56}{81} I_{Q^{\text{lab}}} \otimes I_{9^r}.$$

The expectation with normalized trace $\tau_{9^r}(A) = 9^{-r} \text{Tr}(A)$ satisfies

$$(\text{id} \otimes \tau_{9^r})(E_{m+r}) = \frac{56}{81} I_{Q^{\text{lab}}}$$

at every finite depth.

Proof. A successor's congruence contains the factors $1/9, 81, 1/9$; their product is one. Orthogonality cancels the intermediate transports. The direct sum of nine equal responses identifies the first refinement with $(56/81)I_Q \otimes I_9$. Iteration produces the tensor with I_{9^r} . Finally, $\tau_{9^r}(I_{9^r}) = 1$. Expectations compose through the identity $\tau_{9^{r+s}} = \tau_{9^r} \otimes \tau_{9^s}$. $\square$

The unital inclusions $A \mapsto A \otimes I_9$ are compatible with this collection of traces. In the inductive limit of the matrix algebras, the response is the central element $(56/81)I_Q \otimes \mathbf{1}$. The unnormalized extensive sum instead gives $9^r(56/81)I_Q$. This distinction specifies the quantity preserved: the quadratic screen density.

48.5 Variational realization of the response

The matter realization uses an isometry $J_{\text{scr}} : Q \longrightarrow \mathcal{H}_{\partial}$ compatible with the spin action and refinement. The boundary co-tetrad is $e = J_{\text{scr}} \tilde{\beta}_m$. Let Λ_{CE} be the boundary response obtained by second variation of the matter and geometric action, with its boundary conditions and elimination of the interior variables. The compatibility defect is

$$\mathcal{R}_{\text{grav}} = \tilde{\beta}_m^* (J_{\text{scr}}^* \Lambda_{\text{CE}} J_{\text{scr}} - \Lambda_m) \tilde{\beta}_m = e^* \Lambda_{\text{CE}} e - D_9^* D_9. \quad (48.4)$$

By invertibility of $\tilde{\beta}_m$,

$$\mathcal{R}_{\text{grav}} = 0 \quad \Longleftrightarrow \quad J_{\text{scr}}^* \Lambda_{\text{CE}} J_{\text{scr}} = \Lambda_m.$$

The isometry J_{scr} transports the boundary states; the variational spin current is a bivector-valued three-form. Each enters the realization of the response with its own domain.

The construction determines a positive energy form, its minimizing interior extension, and the density conserved under refinement. The amplitude of a specific matter contribution is obtained by evaluating the current of that realization in its variational quadratic form. This distinguishes the operator fixing the response, the current on which it acts, and the gravitational observable expressed by the evaluation.

49 Geometric realization of transport and soldering

The discrete representation of return and screen soldering determine two aspects of transport: memory accumulation along routes and the comparison of their fibres. Their geometric realization retains both operations. This section specifies the target space, the dimensional normalization of the connection, and the identities used in its variation.

49.1 Screen, co-tetrad, and connection

The screen quotient $Q_\square$, its Lorentzian form, and the soldering β_m come from 48. In each cell, a screen realization $J_{\text{scr},m}$ transports the form and soldering to the corresponding geometric fibre. Its cellular image is

$$e_m = J_{\text{scr},m} \beta_m.$$

The scale ℓ_m remains within the soldering. Changes of frame act jointly on $J_{\text{scr},m}$, the connection, and the transport of the cells.

A smooth realization of these data on an oriented four-dimensional manifold $\mathcal{U}$ consists of a nondegenerate co-tetrad $e = (e^I)$, a Lorentz connection $\omega = (\omega^I_J)$, and a route realization $\mathfrak{R}_C$ satisfying

$$\text{Hol}_{\mathcal{A}_{\ell_0}}(\mathfrak{R}_C(\gamma)) = \rho_C(\text{Hol}_{\text{TPK}}(\gamma)). \quad (49.1)$$

In this matrix equality, the representation is normalized by

$$N_{\ell_0}(R, a) = (\Lambda(R), a/\ell_0), \quad \rho_C = N_{\ell_0} \circ \rho_C^{\text{disc}},$$

where $\Lambda : \text{Spin}^+(3, 1) \rightarrow \text{SO}^+(3, 1)$ is the vector representation. The map N_{ℓ_0} preserves the semidirect product: dividing the translation $a + \Lambda(R)b$ by ℓ_0 is equivalent to composing the two normalized translations. The full cycle therefore expresses $72u$ in the normalized connection and $72\ell_0 u$ in the dimensional chart. The spin lift separately retains $R_4^{c_g(\gamma)}$; the vector representation has central kernel $\{1, -1\}$, so this lift is retained when spinor fields are used. Identity, concatenation, inversion, and subdivision are maintained in the corresponding realization. The screen pairing and the holonomy condition specify the data of the realization used in the field equations.

With $\eta = \text{diag}(-1, 1, 1, 1)$, define $g = \eta_{IJ} e^I \otimes e^J$. For a positive elementary scale ℓ_0 , constant in the chart, the affine connection is

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1} e \\ 0 & 0 \end{pmatrix}. \quad (49.2)$$

The co-tetrad has the dimension of length; the factor ℓ_0^{-1} makes its translational component homogeneous.

Proposition 49.1 (Curvature and torsion of the realized connection). *The curvature of (49.2) is*

$$\mathcal{F}_{\ell_0} = d\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_\omega & \ell_0^{-1} T \\ 0 & 0 \end{pmatrix}, \quad F_\omega = d\omega + \omega \wedge \omega, \quad T = de + \omega \wedge e. \quad (49.3)$$

Proof. The product of matrices of forms produces $\omega \wedge \omega$ in the diagonal block and $\ell_0^{-1} \omega \wedge e$ in the translational block. The exterior derivative contributes $d\omega$ and $\ell_0^{-1} de$, since ℓ_0 is constant. Their sum proves the equality. $\square$

Curvature describes the defect in frame transport, and torsion the defect of the co-tetrad. Both belong to the same realized connection; their components retain distinct tensorial domains.

Proposition 49.2 (Covariance and Bianchi identities). *Under a change of frame $g_L : \mathcal{U} \rightarrow SO^+(3, 1)$,*

$$e' = g_L e, \quad \omega' = g_L \omega g_L^{-1} - dg_L g_L^{-1},$$

one has $T' = g_L T$ and $F_{\omega'} = g_L F_{\omega} g_L^{-1}$. Furthermore,

$$D_{\omega} T = F_{\omega} \wedge e, \quad D_{\omega} F_{\omega} = 0. \quad (49.4)$$

Proof. In $d(g_L e) + \omega' \wedge g_L e$, the terms containing $dg_L \wedge e$ cancel. The same substitution into the curvature gives its transformation by conjugation. For the first Bianchi identity, expand

$$D_{\omega}(de + \omega \wedge e) = (d\omega + \omega \wedge \omega) \wedge e.$$

In $dF_{\omega} + \omega \wedge F_{\omega} - F_{\omega} \wedge \omega$, the terms containing $d\omega$ and the cubic products cancel in pairs. This proves the second identity. $\square$

49.2 The three quadratic realizations of curvature

The TRIT sheet $\tau \in \{+1, 0, -1\}$ is represented in the Cartan algebra by generators $J_{ab} = -J_{ba}$ and $P_a^{(\tau)}$, with relations

$$\begin{aligned} [J_{ab}, J_{cd}] &= \eta_{bc} J_{ad} - \eta_{ac} J_{bd} - \eta_{bd} J_{ac} + \eta_{ad} J_{bc}, \\ [J_{ab}, P_c^{(\tau)}] &= \eta_{bc} P_a^{(\tau)} - \eta_{ac} P_b^{(\tau)}, \\ [P_a^{(\tau)}, P_b^{(\tau)}] &= -\tau J_{ab}. \end{aligned}$$

For constant $\ell > 0$, write

$$\mathcal{A}_{\tau}^{\text{Cart}} = \frac{1}{2} \omega^{ab} J_{ab} + \ell^{-1} e^a P_a^{(\tau)}.$$

Proposition 49.3 (Curvature of the tritified realization). *With the preceding relations,*

$$\mathcal{F}_{\tau}^{\text{Cart}} = \frac{1}{2} (F_{\omega}^{ab} - \tau \ell^{-2} e^a \wedge e^b) J_{ab} + \ell^{-1} T^a P_a^{(\tau)}. \quad (49.5)$$

Proof. Expand $d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}]$. The bracket of the Lorentz sector produces F_{ω} , the mixed bracket produces $D_{\omega} e$, and the translational bracket contributes $-\frac{1}{2} \tau \ell^{-2} e^a \wedge e^b J_{ab}$. $\square$

The central sheet recovers the affine connection; the extreme sheets retain the two signs of the constant-curvature term. The representation thus maintains the tritic regime of the route before the variational equations are applied. The holonomy equality (49.1) corresponds to the central sheet. Each extreme sheet uses the representation $\rho_{C, \tau}$ in its Cartan group and its respective holonomy condition. The current and metric tensor use the fields and representation of the same realization.

50 Variational response of memory energy

The transport of a route identifies its endpoint fibres and preserves the orientation and memory entering that identification. The associated energy compares the final field with the transported initial field. Its differential quantifies how that comparison responds to a variation of transport. This section obtains the response directly from the discrete functional and develops its composition, covariance, and inversion.

50.1 Discrete functional and space of variations

At a finite level m , let $\mathcal{V}_m$ be the vertices of the route representation and $\mathcal{E}_m^+$ a choice of orientation for each edge. Each vertex carries a finite-dimensional Hermitian fibre H_v . Define

$$\mathcal{H}_m = \bigoplus_{v \in \mathcal{V}_m} H_v, \quad \mathcal{K}_m = \bigoplus_{a \in \mathcal{E}_m^+} H_{t(a)}.$$

The Hermitian inner product is antilinear in the first variable. For unitary $U_a : H_{s(a)} \rightarrow H_{t(a)}$, the difference operator is

$$(D_m f)_a = f_{t(a)} - U_a f_{s(a)}, \quad E_m(f, U) = \langle D_m f, W_m D_m f \rangle, \quad W_m = W_m^* > 0. \quad (50.1)$$

This is the energy of the term $D_m^* W_m D_m$ in the discrete generator. The weight W_m may couple different edges. The soldering operator W of Section 48 and this energy weight have different domains and roles.

The transport U is a representation of the enriched state. The field f and weight W_m remain explicit as arguments of the functional; each matter application preserves the rules determining them. We consider differentiable collections of unitary realizations of transport,

$$\dot{U}_a = A_a U_a, \quad A_a^* = -A_a.$$

The physically used variations form the subspace preserving the conditions of the corresponding realization. The differential is first calculated in the unitary tangent space and then restricted to that subspace.

Proposition 50.1 (Transport-response covector). *Write*

$$v_a = U_a f_{s(a)}, \quad r = D_m f, \quad q = W_m r.$$

At fixed field and weight, the energy differential is

$$\dot{\mathbf{j}}_m(A) = -2 \operatorname{Re} \sum_{a \in \mathcal{E}_m^+} \langle q_a, A_a v_a \rangle. \quad (50.2)$$

Its representative for the real Hilbert–Schmidt inner product is

$$\mathcal{J}_a = v_a q_a^* - q_a v_a^*, \quad \mathcal{J}_a^* = -\mathcal{J}_a, \quad \dot{\mathbf{j}}_m(A) = \sum_a \operatorname{Re} \operatorname{Tr}(\mathcal{J}_a^* A_a). \quad (50.3)$$

For simultaneous variations of field, transport, and weight, we have

$$\dot{E}_m = 2 \operatorname{Re} \sum_a \langle q_a, \dot{f}_{t(a)} - U_a \dot{f}_{s(a)} - A_a v_a \rangle + \langle r, \dot{W}_m r \rangle. \quad (50.4)$$

Proof. The first-order update, written in the algebra of dual numbers with $\epsilon^2 = 0$, gives

$$r + \epsilon \dot{r}, \quad \dot{r}_a = \dot{f}_{t(a)} - U_a \dot{f}_{s(a)} - A_a v_a.$$

The coefficient of ϵ in the energy is

$$\langle \dot{r}, W_m r \rangle + \langle r, W_m \dot{r} \rangle + \langle r, \dot{W}_m r \rangle.$$

Self-adjointness of the weight yields (50.4). Fixing field and weight leaves (50.2). Moreover,

$$\operatorname{Tr}(\mathcal{J}_a^* A_a) = v_a^* A_a q_a - q_a^* A_a v_a.$$

Anti-Hermiticity of A_a implies $v_a^* A_a q_a = -\overline{q_a^* A_a v_a}$, giving the stated representative. $\square$

The response is thus evaluated: on each edge the field is transported, its difference from the final field is calculated, and the energy weight is applied. The antisymmetrized product of these two vectors determines $\mathcal{J}_a \in \mathfrak{u}(H_{t(a)})$. Through the real Hilbert–Schmidt inner product, this operator represents the response covector on unitary variations of the target fibre.

50.2 Composition and transport of the response

Proposition 50.2 (Composition rule along a route). *For a route formed by N transports, let $U_\gamma = U_N \cdots U_1$ and $B_k = U_N \cdots U_{k+1}$, with $B_N = I$. If $\dot{U}_k = A_k U_k$, then*

$$\dot{U}_\gamma U_\gamma^{-1} = \sum_{k=1}^N B_k A_k B_k^{-1}. \quad (50.5)$$

Thus a representative $\mathcal{J}_\gamma$ in the final fibre induces in factor k the contribution $B_k^ \mathcal{J}_\gamma B_k$. When a factor belongs to several routes in the functional, its contributions are summed with the incidences used by that functional.*

Proof. The product rule gives

$$\dot{U}_\gamma = \sum_k U_N \cdots U_{k+1} A_k U_k \cdots U_1.$$

Multiplication by U_γ^{-1} eliminates the factors to the right of A_k and yields (50.5). By unitarity and cyclicity of the trace,

$$\operatorname{Re} \operatorname{Tr}(\mathcal{J}_\gamma^* B_k A_k B_k^{-1}) = \operatorname{Re} \operatorname{Tr}((B_k^* \mathcal{J}_\gamma B_k)^* A_k).$$

Linearity of the differential proves the sum of contributions. $\square$

This formula preserves the position of each operation within the route. Transport following an incidence determines how its contribution is represented in the final fibre; the local response is recovered by transporting it to the fibre of that incidence.

Proposition 50.3 (Covariance under unitary frame changes). *Let g_v be unitary frame changes independent of the variation parameter, and let $G_\mathcal{E} = \bigoplus_a g_{t(a)}$. Under the transformations*

$$f'_v = g_v f_v, \quad U'_a = g_{t(a)} U_a g_{s(a)}^{-1}, \quad W'_m = G_\mathcal{E} W_m G_\mathcal{E}^{-1}, \quad A'_a = g_{t(a)} A_a g_{t(a)}^{-1},$$

we have

$$\mathcal{J}'_a = g_{t(a)} \mathcal{J}_a g_{t(a)}^{-1}, \quad \dot{\mathbf{j}}'_m(A') = \dot{\mathbf{j}}_m(A). \quad (50.6)$$

Proof. We obtain $r' = G_\mathcal{E} r$, $q' = G_\mathcal{E} q$, and $v'_a = g_{t(a)} v_a$. Substitution into (50.3) proves the first identity. Invariance of the Hermitian inner product, or cyclicity of the trace, proves the second. $\square$

When frame changes depend on the parameter, their derivatives also act on the field and weight. The corresponding covariant expression is then the complete differential (50.4).

50.3 Oriented inversion and transport of the weight

Inversion can be seen explicitly for an edge-local energy, $W_m = \bigoplus_a W_a$. For $a : s \rightarrow t$, its inverse carries

$$U_{\bar{a}} = U_a^{-1}, \quad r_{\bar{a}} = -U_a^{-1} r_a, \quad W_{\bar{a}} = U_a^{-1} W_a U_a.$$

These rules preserve the edge energy exactly.

Proposition 50.4 (Differential compatibility with inversion). *If W_a and the fields at both endpoints remain fixed, then*

$$A_{\bar{a}} = -U_a^{-1}A_aU_a, \quad \dot{W}_{\bar{a}} = U_a^{-1}[W_a, A_a]U_a.$$

The complete variation of the inverted edge energy agrees with the original. In particular, if $[W_a, A_a] = 0$, the partial transport covector satisfies

$$\dot{\mathbf{j}}_{\bar{a}}(U_a^{-1}A_aU_a) = -\dot{\mathbf{j}}_a(A_a). \quad (50.7)$$

Proof. The derivative of U_a^{-1} is $-U_a^{-1}A_a$, giving the two formulae. Writing $v_a = U_a f_s$ and $f_t = r_a + v_a$, the part due solely to the inverted transport is

$$\dot{\mathbf{j}}_{\bar{a}}(A_{\bar{a}}) = -2 \operatorname{Re}\langle r_a, W_a A_a f_t \rangle = \dot{\mathbf{j}}_a(A_a) - 2 \operatorname{Re}\langle r_a, W_a A_a r_a \rangle.$$

Variation of the weight contributes

$$\langle r_{\bar{a}}, \dot{W}_{\bar{a}} r_{\bar{a}} \rangle = \langle r_a, [W_a, A_a] r_a \rangle = 2 \operatorname{Re}\langle r_a, W_a A_a r_a \rangle.$$

The two terms cancel. If the commutator vanishes, this contribution disappears; linearity in A_a gives (50.7). $\square$

Energy is summed over one orientation per edge. A sum over both orientations carries the factor $1/2$. This convention preserves the same functional and hence the same response.

50.4 Compatibility with refinement

Theorem 50.5 (Transport of the differential under refinement). *Let $J_m : \mathcal{H}_m \rightarrow \mathcal{H}_{m+1}$ and $K_m : \mathcal{K}_m \rightarrow \mathcal{K}_{m+1}$ be fixed maps. For a C^1 collection, suppose the refinement identities*

$$D_{m+1}(t)J_m = K_m D_m(t), \quad K_m^* W_{m+1}(t) K_m = W_m(t). \quad (50.8)$$

Then $E_{m+1}(J_m f) = E_m(f)$. The transport covectors agree on variations linked by the first identity. When the weights vary, their contributions to the differential also agree.

Proof. By substitution,

$$\langle D_{m+1} J_m f, W_{m+1} D_{m+1} J_m f \rangle = \langle D_m f, K_m^* W_{m+1} K_m D_m f \rangle = E_m(f).$$

Differentiating the first identity in (50.8) yields $\dot{D}_{m+1} J_m = K_m \dot{D}_m$. Consequently,

$$2 \operatorname{Re}\langle D_{m+1} J_m f, W_{m+1} \dot{D}_{m+1} J_m f \rangle = 2 \operatorname{Re}\langle D_m f, W_m \dot{D}_m f \rangle.$$

The derivative of the second identity is $K_m^* \dot{W}_{m+1} K_m = \dot{W}_m$, proving agreement of the weight-variation terms. $\square$

The statement uses compatibility of the transport collection. If J_m depends on the parameter, the chain rule also retains the term

$$2 \operatorname{Re}\langle D_{m+1} J_m f, W_{m+1} D_{m+1} \dot{J}_m f \rangle.$$

The resulting naturality jointly transports the functional, its incidences, and its differential.

50.5 Matter realization of the response

The result of this section is the explicit covector $\mathbf{j}_m$, with its representative $\mathcal{J}_a$. The matter realization composes this covector with the transport variation induced by the connection and with the action normalization. If $\mathcal{L}_m$ denotes this differential map, the covector on connection variations is $\mathcal{L}_m^* \mathbf{j}_m$. The composition rule (50.5) evaluates its part corresponding to finite products of transports.

In Section 53, the current σ_{IJ} is a three-form defined by variation of the matter action. Identification between the two realizations preserves the field, representation, weight, dimensional normalization, and limiting process used by that action. These data allow the contraction entering the torsional density to be evaluated. The present development supplies the discrete differential composed in that calculation.

51 Realization of the current in connection coordinates

Memory energy assigns each route a response that preserves its incidences and the order of its transports. Inserting it into a geometric action requires its components with respect to connection variations. This step is obtained by differentiating the transport that already realizes the route: the connection current is the composition of that differential with the energy covector of the preceding section.

The elliptic, parabolic, and hyperbolic regimes of the TRIT require retaining the full differential before restricting it to a particular representation. Unitary rotations constitute one domain of that construction. The following extension also retains the dilation and transport directions admitted by the geometric realization being used.

51.1 Differential on invertible transports

The fibres, fields, and weight of (50.1) are retained. Now consider invertible transports and differentiable collections with $\dot{U}_a = A_a U_a$, where A_a belongs to the tangent space of the chosen representation. Write

$$v_a = U_a f_{s(a)}, \quad r_a = f_{t(a)} - v_a, \quad q = W_m r, \quad E_m = \langle r, W_m r \rangle.$$

Proposition 51.1 (Full representative of the differential). *At fixed field and weight,*

$$d_U E_m(A) = -2 \operatorname{Re} \sum_a \langle q_a, A_a v_a \rangle = \sum_a \operatorname{Re} \operatorname{Tr}(\mathcal{G}_a^* A_a), \quad \mathcal{G}_a = -2 q_a v_a^*. \quad (51.1)$$

The anti-Hermitian and Hermitian projections of this representative are

$$\frac{\mathcal{G}_a - \mathcal{G}_a^*}{2} = v_a q_a^* - q_a v_a^* = \mathcal{J}_a, \quad \frac{\mathcal{G}_a + \mathcal{G}_a^*}{2} = -(q_a v_a^* + v_a q_a^*). \quad (51.2)$$

Restriction to anti-Hermitian generators recovers (50.3) exactly.

Proof. The identity $\dot{r}_a = -A_a v_a$ and self-adjointness of the weight give $\dot{E}_m = 2 \operatorname{Re} \langle q, \dot{r} \rangle$. Furthermore,

$$\operatorname{Tr}(\mathcal{G}_a^* A_a) = -2 \operatorname{Tr}(v_a q_a^* A_a) = -2 q_a^* A_a v_a.$$

The two projections follow by taking the adjoint. For the real Hilbert–Schmidt inner product, the Hermitian and anti-Hermitian subspaces are orthogonal; hence the Hermitian part has zero contraction with an anti-Hermitian generator. $\square$

The formula allows weights that couple edges: their action is computed in $q = W_m r$ before separating the contributions of each incidence. If the field or weight varies, the full expression (50.4) is retained; it also holds for the invertible transports considered here.

51.2 Quadratic representations of the TRIT

In a two-dimensional matrix realization of $\mathbb{A}_\tau$, one may take

$$J_+ = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J_- = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$J_+^2 = -I, \quad J_-^2 = I, \quad J_0^2 = 0.$$

Their exponentials realize the elliptic, hyperbolic, and parabolic regimes, respectively. The first is orthogonal for the fixed positive inner product. The second preserves the form $B = \operatorname{diag}(-1, 1)$, since $J_-^T B + B J_- = 0$. The third represents the algebra of dual numbers. These representations specify examples of the quadratic types; their use on a screen of another dimension retains the corresponding representation map.

A calculation distinguishes the components of the differential. On an edge, take $U = I$, $f_s = (1, 1)^T$, $f_t = 2f_s$, and $W = I$. Then $q = v = f_s$, $\mathcal{J} = 0$, whereas

$$dE(J_-) = -4, \quad dE(J_0) = -2. \quad (51.3)$$

The full representative retains both responses. This distinction identifies the space of variations before the matter contraction is performed.

51.3 Composition of routes and oriented inversion

Proposition 51.2 (Contragredient transport of the differential). *For $U_\gamma = U_N \cdots U_1$, let $B_k = U_N \cdots U_{k+1}$ and $B_N = I$. Then*

$$\dot{U}_\gamma U_\gamma^{-1} = \sum_k B_k A_k B_k^{-1}.$$

The representative $\mathcal{G}_\gamma$ in the final fibre induces, at incidence k ,

$$\mathcal{G}_k = B_k^* \mathcal{G}_\gamma (B_k^{-1})^*. \quad (51.4)$$

Proof. The product rule, followed by multiplication by U_γ^{-1} , proves the first equality. Cyclicity of the trace gives

$$\text{Re Tr}(\mathcal{G}_\gamma^* B_k A_k B_k^{-1}) = \text{Re Tr}((B_k^* \mathcal{G}_\gamma (B_k^{-1})^*)^* A_k).$$

For unitary B_k , the conjugation of 50.2 is recovered. $\square$

Proposition 51.3 (Inversion by energy congruence). *For an edge-local energy, inversion is represented by*

$$U_{\bar{a}} = U_a^{-1}, \quad r_{\bar{a}} = -U_a^{-1} r_a, \quad W_{\bar{a}} = U_a^* W_a U_a. \quad (51.5)$$

It preserves energy. If W_a remains fixed,

$$A_{\bar{a}} = -U_a^{-1} A_a U_a, \quad \dot{W}_{\bar{a}} = U_a^* (A_a^* W_a + W_a A_a) U_a.$$

The full variation of the inverted energy agrees with the original.

Proof. Substitution into $r_{\bar{a}}^* W_{\bar{a}} r_{\bar{a}}$ yields $r_a^* W_a r_a$. The derivative of the inverse is $\dot{U}_a^{-1} = -U_a^{-1} A_a$; that of the weight follows from the product rule. Equality of energies holds throughout the collection, so their derivatives agree, including the term $\langle r_{\bar{a}}, \dot{W}_{\bar{a}} r_{\bar{a}} \rangle$. $\square$

Congruence preserves positivity for every invertible transport. In the unitary representation, it reduces to the inversion rule of 50.4. The sum uses one orientation per edge, or half the sum over both orientations.

51.4 Evaluation of current components

Let $\theta = (\theta_b)$ be a real variation of the connection coordinates of the realization being used. The effective differential of transport defines the operators

$$A_a(\theta) = \dot{U}_a U_a^{-1} = \sum_b B_{ab} \theta_b. \quad (51.6)$$

For a finite composition, their contributions are obtained from (51.4). The operators B_{ab} are the derivatives of the connection representation that realizes the route.

Theorem 51.4 (Connection current of the memory action). *For a matter term $S_{\text{mem}} = \mathbf{a}_m E_m$, where $\mathbf{a}_m$ is its normalization in the action line, at fixed field, weight, and normalization one has*

$$\delta S_{\text{mem}} = -\frac{1}{2} \sum_b \theta_b s_b, \quad s_b = 4\mathbf{a}_m \text{Re} \sum_a \langle q_a, B_{ab} v_a \rangle. \quad (51.7)$$

If the cellular pairing between the connection and current bases is an invertible matrix M , defined by $\delta S_{\text{mem}} = -\frac{1}{2} \theta^\top M \sigma$, the coordinates of the current are

$$\sigma = M^{-1} s. \quad (51.8)$$

Proof. Substitute (51.6) into (51.1), multiply by $\mathbf{a}_m$, and collect the coefficients of θ_b . The variational convention $-1/2$ fixes the factor 4. Comparing both expressions for every variation θ gives $M\sigma = s$; invertibility of the pairing proves existence and uniqueness of these coordinates. $\square$

The formula explicitly realizes the composition mentioned in 50.5. To evaluate each component, the field is transported, the residual is computed, the weight is applied, and contraction is taken with the connection variation of the same realization. The pairing M preserves the orientation and volume of the cells. A diagonal pairing gives the corresponding division by their volumes; other bases retain the full matrix.

If the connection also enters f , W_m , or $\mathbf{a}_m$, (50.4) is applied and $E_m \delta \mathbf{a}_m$ is added. Contributions from all matter terms are summed before composing the Cartan response.

Corollary 51.5 (Compatibility of the current with refinement). *For the compatible collections of 50.5, let P_m be the fixed map taking connection variations from level m to level $m+1$. If the memory actions agree on these refined fields and transports, their covectors satisfy*

$$s_m = P_m^\top s_{m+1}, \quad M_m \sigma_m = P_m^\top M_{m+1} \sigma_{m+1}.$$

Proof. Differentiating the equality of actions in the direction $P_m \theta$ gives $-\frac{1}{2} \theta^\top s_m = -\frac{1}{2} \theta^\top P_m^\top s_{m+1}$. The arbitrariness of θ and the cellular representations of both covectors prove the two identities. Composition of these maps iterates the equality through every finite number of refinements. $\square$

51.5 Insertion into the Cartan equation

Section 53 normalizes the matter three-form by $\delta_\omega S_m = -\frac{1}{2} \int \delta \omega^{IJ} \wedge \sigma_{IJ}$. The preceding theorem provides its cellular components for the declared memory action and pairing. The Cartan response uses this current together with the co-tetrad, the Barbero–Immirzi parameter, and the gravitational coupling of the same realization.

The dependence of matter on contorsion determines how this equation is solved. For affine matter, the current is independent of contorsion, and the linear inversion and quadratic elimination proved in the next section apply. For a holonomy action, the differential is evaluated at the current connection, and that dependence is retained when solving the equation. For example, an edge with $f_s = f_t = 1$, unit weight, and $U(t) = e^{it}$ has

$$E(t) = 2 - 2 \cos t, \quad E'(t) = 2 \sin t.$$

The current here depends on t . The full formula retains this dependence instead of replacing it by its value at a particular connection.

Finally, the gravitational density is obtained from the metric variation of the effective action, with its field and normalization. The representation of the current, the contorsion it produces, and the resulting density are the successive operations of this composition. The subsequent dilution law transports the amplitude thus evaluated in the corresponding matter section.

52 Differential of the minimum boundary extension

Cubic incidence simultaneously determines the minimum-extension field and the weight of its response. For a boundary datum, both are computed through the inverse incidence Gramian. We now develop the differential of that same operation, retaining variations of the boundary, transport, and normalization scale. In this way, the field entering the response is selected by the screen variational problem.

52.1 Minimizing field and response weight

Let H and Q be finite-dimensional Hermitian spaces, with fixed inner products antilinear in the first variable. The real case is obtained by replacing adjoints with transposes. On an open set of real parameters Ω , consider C^1 collections

$$C(\theta) : H \longrightarrow Q, \quad b(\theta) \in Q, \quad \chi(\theta) > 0, \quad \text{Ran } C(\theta) = Q.$$

Surjectivity is equivalent to full row rank in orthonormal bases. Define the Gramian, the conjugate potential, and the extension

$$L = CC^*, \quad y = L^{-1}b, \quad f_{\min} = C^*y, \quad \Lambda = \chi L^{-1}. \quad (52.1)$$

The incidence C and the cellular pairing M of (51.8) are different operators.

Proposition 52.1 (Variational selection of the extension). *The extension $f_{\min}$ uniquely minimizes the norm among fields $f \in H$ satisfying $Cf = b$. Its energy is*

$$E(\theta) = \chi \min_{Cf=b} \|f\|^2 = \chi \langle b, L^{-1}b \rangle = \langle b, \Lambda b \rangle. \quad (52.2)$$

The field $f_{\min}$, potential y , and energy E depend C^1 -smoothly on the parameters.

Proof. Surjectivity of C makes C^* injective, so $\langle z, Lz \rangle = \|C^*z\|^2 > 0$ for $z \neq 0$. Thus L is invertible and $Cf_{\min} = Ly = b$. Every solution is $f_{\min} + z$, with $z \in \ker C$. Since $f_{\min} \in \text{Ran } C^* = (\ker C)^\perp$,

$$\|f_{\min} + z\|^2 = \|f_{\min}\|^2 + \|z\|^2, \quad \|f_{\min}\|^2 = \langle y, Ly \rangle = \langle b, L^{-1}b \rangle.$$

These identities prove the minimum, its uniqueness, and its value. Inversion is a differentiable operation on the open set of invertible matrices; the expressions in (52.1) give the asserted regularity. $\square$

The cubic realization of 48.1 corresponds to $H = \mathbb{R}^{\{-1,1\}^3}$, the screen quotient $Q = P_\square \mathbb{R}^6$, and the incidence $C = M : H \rightarrow Q$. Its Gramian and normalization are

$$L_\square = 12P_u + 4P_-, \quad \chi_m = \frac{112}{81\ell_m^2}, \quad \Lambda_m = \frac{28}{243\ell_m^2}(P_u + 3P_-). \quad (52.3)$$

The factor 112/81 comes from soldering and the quadratic compression defect of (48.3). If b has the dimension of length, so do y and $f_{\min}$; incidence is dimensionless and E is dimensionless. Conversion to action uses the normalization in the action line specified below.

52.2 Full differential of the minimum

Theorem 52.2 (Response to variations of incidence and boundary). *For every real tangent direction $\delta\theta$,*

$$\delta E = \delta\chi \langle b, y \rangle + 2\chi \operatorname{Re} \langle y, \delta b - (\delta C)f_{\min} \rangle. \quad (52.4)$$

If $\chi = 112/(81\ell_m^2)$, with variable $\ell_m > 0$,

$$\delta E = -2E \frac{\delta\ell_m}{\ell_m} + 2\chi_m \operatorname{Re} \langle y, \delta b - (\delta C)f_{\min} \rangle. \quad (52.5)$$

Proof. Differentiating $LL^{-1} = I_Q$ gives

$$\delta L^{-1} = -L^{-1}(\delta L)L^{-1}, \quad \delta L = (\delta C)C^* + C(\delta C)^*.$$

Applying the product rule to (52.2) and using self-adjointness of L^{-1} gives

$$\delta E = \delta\chi \langle b, y \rangle + 2\chi \operatorname{Re} \langle y, \delta b \rangle - \chi \langle y, (\delta L)y \rangle.$$

The two terms in the last inner product are conjugates of one another, and $C^*y = f_{\min}$; hence

$$\langle y, (\delta L)y \rangle = 2 \operatorname{Re} \langle y, (\delta C)f_{\min} \rangle.$$

This yields (52.4). Finally, $\delta\chi_m = -2\chi_m \delta\ell_m/\ell_m$ gives (52.5). $\square$

The formula incorporates variation of the minimizing field through the equation determining it. This can also be checked directly: $C\delta f_{\min} = \delta b - (\delta C)f_{\min}$ and $f_{\min} = C^*y$ imply

$$\delta \|f_{\min}\|^2 = 2 \operatorname{Re} \langle f_{\min}, \delta f_{\min} \rangle = 2 \operatorname{Re} \langle y, \delta b - (\delta C)f_{\min} \rangle.$$

For example, if C is fixed and $b = \ell_m b_0$, with fixed b_0 , the two terms of (52.5) cancel and the energy remains constant. This check jointly preserves the dimension of the boundary datum and its normalization.

52.3 Fixed quotient and changes of frame

The preceding derivatives are taken in fixed spaces H, Q with fixed Hermitian inner products. In the six-face presentation, the inverse always acts on $Q = P_{\square}H_F$. For a collection of quotients $Q(\theta)$, first fix an isometric trivialization $V_Q(\theta) : Q \rightarrow Q(\theta)$, and similarly for the domain. The derivatives of these maps are included when differentiating the expressions transported to the fixed spaces. The formulas apply on each open set where rank is preserved. Changes in the energy inner product require differentiating that inner product as well; a Lorentzian metric and the positive inner product used in minimization retain their distinct types here.

Proposition 52.3 (Covariance of the minimum). *Let g_H, g_Q be unitary. Under*

$$C' = g_Q C g_H^*, \quad b' = g_Q b, \quad \chi' = \chi,$$

one has $y' = g_Q y$, $f'_{\min} = g_H f_{\min}$, and $E' = E$. When changes of frame depend on the parameter, the full differential preserves this equality. In particular, a pure frame variation has $\delta E = 0$.

Proof. The transformed Gramian is $L' = g_Q L g_Q^*$, which gives the transformations of $y, f_{\min}$ and equality of energies. For a pure frame variation, at a point with identity frames, let $A_Q^* = -A_Q$ and $A_H^* = -A_H$ be their generators. Then

$$\delta C = A_Q C - C A_H, \quad \delta b = A_Q b, \quad \delta b - (\delta C)f_{\min} = C A_H f_{\min}.$$

Its contribution to (52.4) is $2\chi \operatorname{Re} \langle f_{\min}, A_H f_{\min} \rangle = 0$. The proof thus includes the terms arising from moving frames. $\square$

52.4 Transported incidence and connection components

Connection dependence is specified before contraction is performed. An explicit block realization of incidence is obtained as follows. Let $\mathcal{K}$ be a common Hermitian fibre, $F = \{(i, \epsilon) : 1 \leq i \leq 3, \epsilon = \pm 1\}$, and $V = \{-1, 1\}^3$. For each incidence $v_i = \epsilon$, the realization being used declares a route $\gamma_{i,\epsilon,v}$ from the vertex fibre to the face fibre, with transport $U_{i,\epsilon,v}$. The trivializations identify these fibres with $\mathcal{K}$. Define

$$\begin{aligned} (B_U f)_{i,\epsilon} &= \sum_{v_i=\epsilon} U_{i,\epsilon,v} f_v, \\ C_U &= (q_{\square} \otimes I_{\mathcal{K}}) B_U : \bigoplus_{v \in V} \mathcal{K} \longrightarrow Q \otimes \mathcal{K}, \quad q_{\square} = P_{\square} : H_F \rightarrow Q. \end{aligned} \quad (52.6)$$

This is an explicit variational continuation of cubic incidence in the declared route representation. For identity transports, $C_U = M \otimes I_{\mathcal{K}}$, and (52.3), tensored with the fibre, is recovered. The Gramian remains positive in a neighbourhood of that realization, since positive definiteness is an open condition. On each full-rank domain, the weight and field are fixed by (52.1). The routes realizing the incidences, their orientations, and their memories are data of this realization of APP-TRIT-TPK transport and accompany the construction.

If $\omega = (\omega_j)$ are real connection coordinates, the derivative of C_U is computed by substituting in each block

$$\partial_j U_{\gamma} = \sum_{k=1}^N U_N \cdots U_{k+1} (\partial_j U_k) U_{k-1} \cdots U_1, \quad U_{\gamma} = U_N \cdots U_1. \quad (52.7)$$

Empty products are identities. The formula follows from the product rule and preserves the position of each operation along the route. Thus $C_j := \partial_j C_U$ is obtained from a finite sum of matrices known in the chosen realization, followed by the projection $q_{\square} \otimes I$. If a moving quotient is used, its derivative is incorporated as in 52.3. The fixed integer cubic matrix corresponds to the case $C_j = 0$; the connection response comes from the transport dependence actually declared in (52.6).

Theorem 52.4 (Variational components of the minimum extension). *For the screen action $S_{\min} = \mathbf{a}_m E$, where $\mathbf{a}_m$ is its normalization in the action line, write*

$$b_j = \partial_j b, \quad \chi_j = \partial_j \chi, \quad a_j = \partial_j \mathbf{a}_m.$$

With the convention $\delta S_{\min} = -\frac{1}{2} \sum_j s_j \delta \omega_j$, the components are

$$\begin{aligned} s_j &= 4\mathbf{a}_m \chi \operatorname{Re} \langle y, C_j f_{\min} - b_j \rangle \\ &\quad - 2\mathbf{a}_m \chi_j \langle b, y \rangle - 2a_j E. \end{aligned} \quad (52.8)$$

For the cubic normalization, the second term is equivalent to $4\mathbf{a}_m E \partial_j \ell_m / \ell_m$. If the cellular pairing of this same action has invertible matrix M_{cel} , its current coordinates are $\sigma = M_{\text{cel}}^{-1} s$.

Proof. The product rule gives $\partial_j S_{\min} = a_j E + \mathbf{a}_m \partial_j E$. Substituting (52.4) and multiplying by -2 proves (52.8). The derivative of χ_m gives the scale expression. Finally, comparing $\delta S_{\min} = -\frac{1}{2} \delta \omega^T s$ with $-\frac{1}{2} \delta \omega^T M_{\text{cel}} \sigma$ for every variation gives $M_{\text{cel}} \sigma = s$, and its inverse determines the coordinates. $\square$

The construction recovers a field and a weight jointly selected by incidence and evaluates their response to variations of the same realization. The boundary datum b , the route configuration, the scale ℓ_m , the normalization $\mathbf{a}_m$, and the cellular pairing retain their representation rules. Interpretation as a matter current uses these objects and the full action. The Cartan response and its subsequent metric density are composed with it in the corresponding matter domain, retaining any dependence of the current on the connection.

53 Spin current and algebraic Cartan–Holst response

The realization of enriched transport in Section 49 provides the co-tetrad and connection; the dimensional sections and the incidence parameter in Sections 41 and 43 fix their couplings. Their composition determines a first-order variational problem. The matter current occupies a precise position within it: it is the derivative of the action with respect to the connection and determines the torsional response with the same normalization.

53.1 Geometric domain and current normalization

We work on a smooth oriented four-dimensional manifold M , with internal Lorentzian bundle E , metric $\eta = \text{diag}(-1, 1, 1, 1)$, and nondegenerate co-tetrad $e \in \Omega^1(M; E)$. In a local frame, write e^I , with $I = 0, 1, 2, 3$. The metric connection $\omega^{IJ} = -\omega^{JI}$ acts through D_ω . When the fields Ψ are spinorial, a spin structure and the corresponding matter representation are retained. Define

$$\Sigma^{IJ} = e^I \wedge e^J, \quad F^{IJ} = d\omega^{IJ} + \omega_K^I \wedge \omega^{KJ}, \quad T^I = D_\omega e^I. \quad (53.1)$$

The operator $\star$ acts on internal bivector indices,

$$(\star X)^{IJ} = \frac{1}{2} \epsilon^{IJ}_{KL} X^{KL}, \quad \star^2 = -I, \quad P_\gamma = \star + \gamma^{-1}I. \quad (53.2)$$

This duality is thus distinguished from the exterior duality of forms on M . Internal indices are raised and contracted with η .

Fix a chart with $G_{\text{int}} > 0$, $c_{\text{int}} > 0$, $\gamma = \gamma_{\text{HMT}} \in \mathbb{R} \setminus \{0\}$, and Λ_{int} constant on it. Abbreviate $\kappa = 8\pi G_{\text{int}}/c_{\text{int}}^4$ and $c = c_{\text{int}}$. These sections remain fixed during variation. The action is

$$S[e, \omega, \Psi] = \frac{1}{2\kappa c} \int_M \Sigma_{IJ} \wedge (P_\gamma F)^{IJ} - \frac{\Lambda_{\text{int}}}{\kappa c} \int_M \text{vol}_e + S_m[e, \omega, \Psi]. \quad (53.3)$$

The volume and orientation convention is the one giving $\Sigma_{IJ} \wedge (\star F)^{IJ} = R \text{vol}_e$ in the Levi–Civita sector. The preceding dimensional type makes each summand an action section.

Connection variations have compact support in the interior, or satisfy $\delta\omega|_{\partial M} = 0$. A boundary completion whose variational problem cancels the same boundary term may also be used. At fixed co-tetrad and matter fields, the spin current is the three-form with values in dual internal bivectors determined by

$$\delta_\omega S_m = -\frac{1}{2} \int_M \delta\omega^{IJ} \wedge \sigma_{IJ}. \quad (53.4)$$

Pairing with an arbitrary compactly supported variation uniquely determines σ . This definition retains the action unit of the current and the normalization factor entering its geometric response.

53.2 Connection variation and Holst inversion

Theorem 53.1 (Variational Cartan–Holst equation). *On the preceding domain, stationarity with respect to the connection is equivalent to*

$$D_\omega(P_\gamma \Sigma)^{IJ} = \kappa c \sigma^{IJ}. \quad (53.5)$$

Proof. The curvature variation is $\delta F = D_\omega \delta\omega$. Internal duality is parallel and self-adjoint for bivector contraction; since γ is constant, $D_\omega P_\gamma = P_\gamma D_\omega$. Setting $B_{IJ} = (P_\gamma \Sigma)_{IJ}$ gives

$$B_{IJ} \wedge D_\omega \delta\omega^{IJ} = d(B_{IJ} \wedge \delta\omega^{IJ}) - D_\omega B_{IJ} \wedge \delta\omega^{IJ}.$$

The last form changes sign when its factors of degrees three and one are interchanged. The boundary conditions eliminate the integral of the exact term. The cosmological term is independent of ω . Therefore,

$$\delta_\omega S = \frac{1}{2\kappa c} \int_M \delta\omega^{IJ} \wedge [D_\omega B_{IJ} - \kappa c \sigma_{IJ}].$$

The arbitrariness of the variation proves the equation, including its sign and coefficient. $\square$

Lemma 53.2 (Real inverse of the Holst operator). *For $\gamma \in \mathbb{R} \setminus \{0\}$,*

$$P_\gamma^{-1} = \frac{\gamma^2}{1 + \gamma^2}(-\star + \gamma^{-1}I). \quad (53.6)$$

Proof. Both factors are polynomials in $\star$ and commute. Their product before normalization is

$$(\star + \gamma^{-1}I)(-\star + \gamma^{-1}I) = -\star^2 + \gamma^{-2}I = (1 + \gamma^{-2})I.$$

The denominator $1 + \gamma^2$ is positive on the fixed real domain. Multiplication by $\gamma^2/(1 + \gamma^2)$ gives the identity. $\square$

The current thus determines the bivector-valued three-form

$$Y^{IJ} := \kappa c \frac{\gamma^2}{1 + \gamma^2}(-\star + \gamma^{-1}I)\sigma^{IJ}, \quad D_\omega \Sigma^{IJ} = Y^{IJ}. \quad (53.7)$$

53.3 Pointwise reconstruction of torsion and contorsion

The Leibniz rule gives

$$D_\omega \Sigma^{IJ} = T^I \wedge e^J - e^I \wedge T^J =: L_e(T)^{IJ}. \quad (53.8)$$

At each point, L_e acts from $\Lambda^2 T^*M \otimes E$ to $\Lambda^3 T^*M \otimes \Lambda^2 E$. Both spaces have dimension twenty-four. Let E_I be the frame dual to the co-tetrad, so that $t_{E_I} e^J = \delta_I^J$.

Theorem 53.3 (Algebraic inversion of the torsional equation). *For a nondegenerate co-tetrad, L_e is an isomorphism. Its inverse on the form Y is written*

$$B^I = \sum_J t_{E_J} Y^{IJ}, \quad b = \frac{1}{4} \sum_I t_{E_I} B^I, \quad T^I = B^I - e^I \wedge b. \quad (53.9)$$

Equation (53.5) therefore determines torsion pointwise from e and the current appearing in it.

Proof. First let $Y = L_e(T)$ and set $t = \sum_J t_{E_J} T^J$. Contracting (53.8), using the fact that T^I has degree two and that the co-tetrad has four elements, gives

$$\sum_J t_{E_J} (T^I \wedge e^J) = 2T^I, \quad \sum_J t_{E_J} (e^I \wedge T^J) = T^I - e^I \wedge t.$$

Thus $B^I = T^I + e^I \wedge t$. A second contraction yields

$$\sum_I t_{E_I} B^I = t + 3t = 4t.$$

This recovers $t = b$ and then $T^I = B^I - e^I \wedge b$. In particular, the kernel of L_e is zero. Equality of dimensions proves surjectivity and makes the formula valid for every form Y in the codomain. $\square$

The connection decomposes uniquely as

$$\omega^{IJ} = \hat{\omega}^{IJ}(e) + K^{IJ}, \quad K^{IJ} = -K^{JI}, \quad T^I = K^I_J \wedge e^J, \quad (53.10)$$

where $\hat{\omega}(e)$ is the Levi-Civita connection and K is contorsion. To make the inverse explicit, write $K_{IJ} = K_{IJK} e^K$ and $T_I = \frac{1}{2} T_{IJK} e^J \wedge e^K$. Then

$$T_{IJK} = K_{IKJ} - K_{IJK}, \quad K_{IJK} = \frac{1}{2} (T_{KIJ} - T_{IJK} - T_{JKI}). \quad (53.11)$$

The second equality follows by cyclically combining the first and using $K_{IJK} = -K_{JIK}$; substitution recovers the components of T . Torsion and contorsion are consequently two equivalent presentations of the algebraic response at fixed co-tetrad.

Corollary 53.4 (Zero-current sector and minimal linear response). *If $\sigma = 0$, then $T = 0$ and $\omega = \hat{\omega}(e)$. For a matter coupling whose current $\sigma[e, \Psi]$ is independent of K , equations (53.7), (53.9), and (53.11) give a unique response $K_*(e, \Psi)$, linear in σ .*

Proof. The three reconstruction maps are linear at fixed e , γ , and κc . Zero current produces $Y = 0$, then $T = 0$, and finally $K = 0$. $\square$

In the minimal action, the connection equation contains this pointwise response. Its evolution follows by evolving e and Ψ : $K_* = K_*(e, \Psi)$ changes with them. If a matter action retains additional algebraic dependence on K , the Cartan equation retains its variational form and must be solved with that current; the preceding linear response belongs to the specified minimal domain.

53.4 Elimination of contorsion and quadratic contribution

Now consider the minimal coupling affine in K , expressed by

$$S_m[e, \hat{\omega} + K, \Psi] = S_m[e, \hat{\omega}, \Psi] - \frac{1}{2} \int_M K^{IJ} \wedge \sigma_{IJ}[e, \Psi]. \quad (53.12)$$

This condition identifies the matter action to which quadratic elimination applies; the current remains the one defined by (53.4).

The curvature expansion is

$$F(\hat{\omega} + K) = F(\hat{\omega}) + D_{\hat{\omega}}K + K \wedge K, \quad (K \wedge K)^{IJ} = K^I_L \wedge K^{LJ}. \quad (53.13)$$

Since $D_{\hat{\omega}}\Sigma = 0$, the linear term is the exterior derivative of $\Sigma_{IJ} \wedge (P_\gamma K)^{IJ}$. The corresponding boundary contribution remains explicit:

$$S_\partial[e, K] = \frac{1}{2\kappa c} \int_{\partial M} \Sigma_{IJ} \wedge (P_\gamma K)^{IJ}. \quad (53.14)$$

On an interior domain, or with the compatible boundary completion, the action density depending on contorsion is

$$\mathcal{Q}_{e,\gamma}(K) - \frac{1}{2} K^{IJ} \wedge \sigma_{IJ}, \quad \mathcal{Q}_{e,\gamma}(K) = \frac{1}{2\kappa c} \Sigma_{IJ} \wedge P_\gamma(K \wedge K)^{IJ}. \quad (53.15)$$

The density $\mathcal{Q}_{e,\gamma}$ is a differential four-form, homogeneous of degree two in K .

Theorem 53.5 (Effective response quadratic in the current). *Under (53.12), substitution of the algebraic solution K_* produces the effective spin density*

$$\mathcal{L}_{\text{spin}}^{\text{eff}} = -\frac{1}{4} K_*^{IJ} \wedge \sigma_{IJ} = -\mathcal{Q}_{e,\gamma}(K_*). \quad (53.16)$$

At fixed co-tetrad and couplings, it is homogeneous of degree two in the current. The effective action also retains $S_\partial[e, K_]$, or its declared boundary completion.*

Proof. By Corollary 53.4, K_* is linear in σ . The connection equation is precisely the stationarity of (53.15). Since this is algebraic, the radial variation $\delta K = K_*$ applies pointwise; it may also be multiplied by a compactly supported function. Euler's identity for a quadratic form gives

$$2\mathcal{Q}_{e,\gamma}(K_*) - \frac{1}{2} K_*^{IJ} \wedge \sigma_{IJ} = 0.$$

Substitution into the action density proves (53.16). If $\sigma \mapsto \lambda\sigma$, then $K_* \mapsto \lambda K_*$ and the effective density is multiplied by λ^2 . The expansion (53.13) separately identifies the boundary term accompanying this elimination. $\square$

The specific current, its Lorentzian contraction, duality, and the matter-action conventions determine the sign and coefficient of this contribution. The quadratic identity provides the composition rule that evaluates them once these data are specified.

53.5 Effective metric source and covariant balance

The normalization of the metric source is fixed in the same action line. After eliminating K , define

$$\delta_g S_m[e, \dot{\omega}, \Psi] = -\frac{1}{2} \int_M \text{vol}_e \mathcal{T}_{\mu\nu}^{S, \text{LC}} \delta g^{\mu\nu}, \quad (53.17)$$

$$\delta_g S_{\text{spin}}^{\text{eff}} = -\frac{1}{2} \int_M \text{vol}_e \mathcal{U}_{\mu\nu}^{S, \text{spin}} \delta g^{\mu\nu}. \quad (53.18)$$

Here interior variations are considered, with boundary terms treated according to the fixed variational problem. For affine coupling, the second source is the metric variation of (53.16). The variation includes the dependence of the matter current on e and Ψ .

Proposition 53.6 (Metric equation and conservation of the total source). *With the preceding conventions, the metric equations are*

$$G_{\mu\nu}[g] + \Lambda_{\text{int}} g_{\mu\nu} = \kappa c (\mathcal{T}_{\mu\nu}^{S, \text{LC}} + \mathcal{U}_{\mu\nu}^{S, \text{spin}}) = \kappa (T_{\mu\nu}^{\text{phys, LC}} + U_{\mu\nu}^{\text{phys, spin}}), \quad (53.19)$$

where $T^{\text{phys, LC}} = c \mathcal{T}^{S, \text{LC}}$ and $U^{\text{phys, spin}} = c \mathcal{U}^{S, \text{spin}}$ when the co-tetrad takes values in the length line. Every solution satisfies

$$\overset{\circ}{\nabla}^\mu (T_{\mu\nu}^{\text{phys, LC}} + U_{\mu\nu}^{\text{phys, spin}}) = 0. \quad (53.20)$$

Proof. Eliminating K preserves the equations for the remaining variables: in the chain rule, the term multiplying δK_* vanishes by the connection equation. In the Levi–Civita sector, the first Bianchi identity makes the term $\Sigma_{IJ} \wedge F^{IJ}$ vanish, and the gravitational part is written $(2\kappa c)^{-1} \int_M (R - 2\Lambda_{\text{int}}) \text{vol}_e$. Its interior variation is

$$\frac{1}{2\kappa c} \int_M \text{vol}_e (G_{\mu\nu} + \Lambda_{\text{int}} g_{\mu\nu}) \delta g^{\mu\nu}.$$

Adding (53.18) gives (53.19). The contracted Bianchi identity for $\overset{\circ}{\nabla}$, metric compatibility, and constancy of κ and Λ_{int} in the chart yield (53.20). $\square$

Before eliminating the connection, the geometric identities retain the form

$$D_\omega T^I = F^I_J \wedge e^J, \quad D_\omega F^{IJ} = 0. \quad (53.21)$$

The first is $D_\omega^2 e = F \wedge e$; the second follows by expanding $D_\omega(d\omega + \omega \wedge \omega)$ and cancelling terms. In a covariant matter action, total metric balance agrees with the Noether identity on the matter equations. Geometry and matter thus retain a joint source; the division between its Levi–Civita contribution and its spin contribution is determined by the specialized action.

54 Torsional elimination and metric response of the minimum extension

The minimum extension provides a field, a response weight, and a current through a single variational operation. Its transported incidence depends on the connection through the route products in (52.7). This dependence is retained when composing the current with the Cartan–Holst equation. The connection is therefore eliminated from the full action of this extension. The affine case of 53.5 is contained as an exact specialization of the following construction.

54.1 Variational composition at a cellular level

Fix a finite level, its routes, the transport representation, and the cellular pairing of 52.4. Let p be real coordinates of the co-tetrad and matter fields in a declared local trivialization, and $k \in \mathbb{R}^d$ the coordinates of contorsion K . The collection $p \mapsto e(p)$ retains nondegeneracy of the co-tetrad. The connection acting along each route is

$$\omega(p, k) = \hat{\omega}(e(p)) + K(k), \quad F(p, k) = S_{\min}[e(p), \omega(p, k), \Psi(p)] = \mathbf{a}(p, k)E(p, k). \quad (54.1)$$

The spaces and bases in this expression are transported to fixed spaces before differentiation. We work on an open set where incidence $C(p, k)$ is surjective and the collections are C^2 . This additional regularity allows examination of the action Hessian. The couplings $G_{\text{int}}, c_{\text{int}}, \gamma_{\text{HMT}}$ retain the chart and variation convention of 53.1.

Let $Q(p, k)$ be the integral, or the declared cellular realization, of the quadratic form (53.15). In real bases,

$$Q(p, k) = \frac{1}{2}k^\top A(p)k, \quad A(p)^\top = A(p). \quad (54.2)$$

The operator A comes from the gravitational action and its pairing, whereas the Gramian CC^* comes from the extension problem. Positivity of the latter guarantees existence of the minimizing field; the gravitational form retains its own signature. Boundary terms are retained according to (53.14); the following interior formulas apply with admissible variations cancelling their contribution, or with the corresponding boundary completion.

Define the response covector by

$$d_k F(p, k)[h] = -\frac{1}{2}h^\top s(p, k), \quad s(p, k) = M_{\text{cel}}(p, k)\sigma(p, k). \quad (54.3)$$

Here s and σ retain the distinct types of (52.8): the first contains the coefficients of the differential, and the second the coordinates of the current in the matter pairing. A change of basis in k also transports this pairing. Differentiating $s = M_{\text{cel}}\sigma$ includes derivatives of both factors when the pairing depends on the parameter.

Proposition 54.1 (Composed equation of extension and contorsion). *Interior stationarity of the action with respect to contorsion is*

$$\mathcal{R}(p, k) := A(p)k - \frac{1}{2}s(p, k) = 0. \quad (54.4)$$

Each component of s is computed from the same extension:

$$\begin{aligned} s_j = & 4\mathbf{a}\chi \operatorname{Re}\langle y, (\partial_{k_j} C)f_{\min} - \partial_{k_j} b \rangle \\ & - 2\mathbf{a}(\partial_{k_j} \chi)\langle b, y \rangle - 2(\partial_{k_j} \mathbf{a})E. \end{aligned} \quad (54.5)$$

When the cellular pairing realizes the connection equation of 53.1, this equation is equivalent to composing $\sigma(p, k)$ with the Holst inverse, the inverse of L_e , and contorsion reconstruction:

$$K = C_e^{\text{tor}} L_e^{-1}(\kappa c P_\gamma^{-1} \sigma[e, \hat{\omega}(e) + K, \Psi]). \quad (54.6)$$

The symbol C_e^{tor} denotes exclusively the map $T \mapsto K$ of (53.11), at fixed co-tetrad.

Proof. The differential of $Q + F$ in an arbitrary direction h is

$$d_k(Q + F)[h] = h^\top (Ak - \frac{1}{2}s).$$

Its vanishing in every direction proves (54.4). Theorem 52.4, applied to ∂_{k_j} , gives (54.5). At fixed p , $\delta\omega = \delta K$. The Cartan equation is transformed successively through (53.6), (53.9), and (53.11); each operation is invertible on its domain. This proves equivalence with (54.6) when both expressions use the same matter and cellular realization. $\square$

The composition retains the connection on both sides. Routes passing through several cells may couple their current components. Pointwise reconstruction of T from a given current and joint solution of (54.4) are then successive operations. This finite formulation keeps explicit the route representation entering the functional.

54.2 Local existence of a stationary branch

Theorem 54.2 (Local continuation of the torsional solution). *Suppose (p_0, k_0) belongs to the preceding open set, $\mathcal{R}(p_0, k_0) = 0$, and the total Hessian*

$$H_0 = A(p_0) + \partial_k^2 F(p_0, k_0) = A(p_0) - \frac{1}{2}\partial_k s(p_0, k_0) \quad (54.7)$$

is invertible. There are neighbourhoods of p_0 and k_0 in which a unique C^1 branch, $k = k_(p)$, satisfies (54.4) and passes through k_0 . On this branch,*

$$\partial_{p_i} k_* = -[A + \partial_k^2 F]^{-1} [(\partial_{p_i} A)k_* + \partial_{p_i} \partial_k F]. \quad (54.8)$$

Proof. Gramian inversion is C^2 on the full-rank open set. Therefore F is C^2 and $\mathcal{R}$ is C^1 . Its derivative with respect to k at (p_0, k_0) is H_0 . The implicit function theorem provides the branch, its regularity, and its uniqueness in a neighbourhood of the point. Existence can also be seen through the map

$$\mathcal{T}_p(k) = k - H_0^{-1} \mathcal{R}(p, k).$$

Its derivative with respect to k vanishes at (p_0, k_0) . On a sufficiently small ball it has norm at most $q < 1$. Shrinking the neighbourhood of p_0 ensures $\|\mathcal{T}_p(k_0) - k_0\| < (1 - q)r$, where r is the radius of the ball. Thus $\mathcal{T}_p$ maps the closed ball into itself and is contractive. Its fixed point is the local solution. Differentiating $\mathcal{R}(p, k_*(p)) = 0$ and solving for $\partial_{p_i} k_*$ gives (54.8). $\square$

The hypothesis is checked on the composed action: Gramian rank ensures the screen minimum, and the total Hessian controls the torsional branch. The asserted uniqueness is local around the specified solution; other branches may coexist on the same full-rank domain.

54.3 Effective action and non-affine contribution

Theorem 54.3 (Exact elimination identity). *Let $k_*(p)$ be a stationary branch. Suppose also that the segment $t \mapsto (p, tk_*)$, $0 \leq t \leq 1$, remains within the open set where F is defined. The action correction relative to the Levi-Civita realization is*

$$\begin{aligned} \Delta S(p) &:= Q(p, k_*) + F(p, k_*) - F(p, 0) \\ &= \frac{1}{4} k_*^\top s(p, k_*) - \frac{1}{2} \int_0^1 k_*^\top s(p, tk_*) dt. \end{aligned} \quad (54.9)$$

Equivalently,

$$\Delta S = -\frac{1}{4} k_*^\top s(p, k_*) + \frac{1}{2} \int_0^1 k_*^\top [s(p, k_*) - s(p, tk_*)] dt. \quad (54.10)$$

These identities refer to the integrated or cellular action. In a realization with a local density, they apply to the density with its corresponding pairing of forms. The effective action separately retains the declared boundary term.

Proof. By the chain rule and (54.3),

$$\frac{d}{dt}F(p, tk_*) = -\frac{1}{2}k_*^\top s(p, tk_*).$$

Integration between 0 and 1 gives the difference of matter actions. The stationarity equation, evaluated in the direction k_* , and quadratic homogeneity of Q give

$$2Q(p, k_*) = \frac{1}{2}k_*^\top s(p, k_*).$$

Adding the two identities proves (54.9); adding and subtracting $\frac{1}{2}k_*^\top s(p, k_*)$ gives (54.10). $\square$

Corollary 54.4 (Affine specialization). *If $s(p, k) = s(p)$, then*

$$\Delta S = -\frac{1}{4}k_*^\top s(p) = -Q(p, k_*).$$

This is the integrated realization of (53.16).

Proof. The difference within the integral in (54.10) vanishes. The second equality follows from the radial stationarity identity. $\square$

54.4 Metric differential of the eliminated action

Theorem 54.5 (Response of the geometric variables). *On the branch of Theorem 54.2,*

$$\partial_{p_i}\Delta S = \frac{1}{2}k_*^\top (\partial_{p_i}A)k_* + (\partial_{p_i}F)(p, k_*) - (\partial_{p_i}F)(p, 0), \quad (54.11)$$

where partial derivatives keep the coordinates k fixed. In fixed Hermitian inner products, each matter derivative is computed by

$$\begin{aligned} \partial_{p_i}F &= (\partial_{p_i}a)E + a(\partial_{p_i}\chi)(b, y) \\ &\quad + 2a\chi \operatorname{Re}\langle y, \partial_{p_i}b - (\partial_{p_i}C)f_{\min} \rangle. \end{aligned} \quad (54.12)$$

If p_i varies the co-tetrad, $\partial_{p_i}C$ includes variation of $\hat{\omega}(e(p)) + K(k)$, of the realized routes, and of the trivializations used, according to their declared dependencies.

Proof. The total derivative of $Q(p, k_*) + F(p, k_*)$ contains, in addition to its partial derivatives, the term

$$[\partial_k Q(p, k_*) + \partial_k F(p, k_*)]\partial_{p_i}k_*.$$

The bracket is zero by the connection equation. Subtracting the derivative of $F(p, 0)$ proves (54.11). The product rule and (52.4) prove (54.12). The composition (54.1) requires differentiating the Levi-Civita connection when e varies at fixed k . $\square$

When the positive extension inner product also varies, its contribution is retained explicitly. In coordinates, let $h(p, k) > 0$ be its Hermitian matrix, and consider the minimum of f^*hf subject to $Cf = b$. Then

$$L = Ch^{-1}C^*, \quad y = L^{-1}b, \quad f_{\min} = h^{-1}C^*y, \quad E = \chi b^*y. \quad (54.13)$$

Here y represents the constraint covector in fixed bases. The positive matrix h retains its energy role, distinct from that of the Lorentzian metric.

Lemma 54.6 (Variation of the extension inner product). *With the preceding conventions,*

$$\delta E = (\delta\chi)b^*y + 2\chi \operatorname{Re}[y^*(\delta b - \delta C f_{\min})] + \chi f_{\min}^*(\delta h)f_{\min}. \quad (54.14)$$

Proof. The formula $\delta h^{-1} = -h^{-1}(\delta h)h^{-1}$ gives

$$\delta L = (\delta C)h^{-1}C^* + Ch^{-1}(\delta C)^* - Ch^{-1}(\delta h)h^{-1}C^*.$$

On differentiating $E = \chi b^* L^{-1} b$, the term $-\chi y^*(\delta L)y$ produces the two terms with δC and the positive term with δh . This proves the full identity. $\square$

In a covariant matter realization whose variation admits the metric pairing of (53.18), the correction determines

$$\delta_g \Delta S = -\frac{1}{2} \int_M \text{vol}_e \mathcal{U}_{\mu\nu}^{S,\min} \delta g^{\mu\nu}, \quad U_{\mu\nu}^{\text{phys},\min} = c \mathcal{U}_{\mu\nu}^{S,\min}. \quad (54.15)$$

Equations (54.11) and (54.12), supplemented by (54.14) where appropriate, evaluate its coefficients in the declared metric variations. The screen minimum, torsional solution, and metric derivative are three consecutive operations on the same action. The reduced action retains any intercell couplings present in the original routes.

The homogeneous specialization of 56 also uses the realization of the tensor obtained: its spatial symmetry, the sign of its contraction with the observer, and the scaling law. When this evaluation gives $\rho_{\text{corr}}(a) = -s_0 a^{-6}$, the amplitude is read in the reference state as $s_0 = -a_{\text{ref}}^6 \rho_{\text{corr}}(a_{\text{ref}})$. This readout retains the matter state and its normalization; the elimination theorem supplies the action to be contracted.

54.5 Exact non-affine checking model

The following finite model checks the coefficients and non-affine dependence of the identities. Its parameters belong to the algebraic example and keep the matter-realization problem of the routes separate.

Proposition 54.7 (Exact elimination of a rank-one extension). *Let $H = \mathbb{R}^2$, $Q = \mathbb{R}$, $C(k) = (1 \ k)$, $b = 1$, $\chi = 1$, $a = \lambda > 0$, and $Q_{\text{ex}}(k) = k^2/2$. Then*

$$f_{\min} = \frac{(1, k)^T}{1 + k^2}, \quad F(\lambda, k) = \frac{\lambda}{1 + k^2}, \quad s(\lambda, k) = \frac{4\lambda k}{(1 + k^2)^2}, \quad \mathcal{R}(\lambda, k) = k \left(1 - \frac{2\lambda}{(1 + k^2)^2} \right). \quad (54.16)$$

For $\lambda = 25/2$, the stationary solutions are 0, 2, -2. At $k_* = 2$,

$$H_* = \frac{16}{5}, \quad s_* = 4, \quad Q_{\text{ex}}(k_*) = 2, \quad F(\lambda, k_*) = \frac{5}{2}, \quad \Delta S = -8. \quad (54.17)$$

The affine expression $-k_* s_*/4 = -2$ differs from the exact correction. On the positive branch,

$$\partial_\lambda \Delta S = -\frac{4}{5}, \quad \partial_\lambda k_* = \frac{1}{20} \quad \text{at } \lambda = 25/2. \quad (54.18)$$

Proof. The Gramian is $1 + k^2 > 0$, so the minimum has the indicated expressions. Differentiating F gives $s = -2\partial_k F$ and then $\mathcal{R} = k - s/2$. For $\lambda = 25/2$, the equation is $k[(1 + k^2)^2 - 25] = 0$; since $1 + k^2 > 0$, its roots are precisely 0, ± 2 . The total Hessian is

$$1 - \frac{2\lambda(1 - 3k^2)}{(1 + k^2)^3},$$

which equals 16/5 at both nonzero roots. The correction is $2 + 5/2 - 25/2 = -8$. In the radial identity,

$$\int_0^1 k_* s(\lambda, tk_*) dt = \int_0^1 \frac{200t}{(1 + 4t^2)^2} dt = 20,$$

since an antiderivative is $-25/(1 + 4t^2)$. Thus $\Delta S = 8/4 - 20/2 = -8$, including the non-affine term. For $\lambda > 1/2$, the positive branch satisfies $k_*^2 = \sqrt{2\lambda} - 1$ and $\Delta S = \sqrt{2\lambda} - 1/2 - \lambda$. Their derivatives give (54.18). The envelope formula gives the same result: $(\partial_\lambda F)(\lambda, k_*) - (\partial_\lambda F)(\lambda, 0) = 1/5 - 1 = -4/5$. $\square$

55 Spinor current and torsional amplitude of the matter state

The preceding development and the spinor realization below articulate two variational problems with common geometric antecedents. The minimum-extension action $S_{\min} = \mathbf{a}_m E$, constructed in Section 52, incorporates the dependence of transported incidence and its minimum on the connection; its torsional elimination retains this full dependence in Section 54. The spinor action S_D of (55.3) couples the same connection to the fields of its representation and is affine in contorsion at fixed co-tetrad and fields. Each current is obtained by varying its own action. A minimizing preparation may determine the initial matter state of the spinor realization; substitution of connection-dependent fields also retains the terms of the composed differential. This distinction places the amplitude calculation below within its specific action and variational domain.

The spinor representation of transport and the algebraic Cartan–Holst response allow torsional intensity to be calculated as a functional of the matter state. Its evaluation retains three data: the field representation, the second moment of its current, and the volume relative to which the density is defined. The coupling coefficient results from the action and its inversion; the initial state determines the matter amplitude.

55.1 Spinor representation and affine matter action

We retain the co-tetrad, orientation, and couplings of Section 53. The spinor realization uses the four-dimensional complex module obtained by complexifying the real matrices

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (55.1)$$

$$g^0 = J \otimes I_2, \quad g^1 = R \otimes I_2, \quad g^2 = Z \otimes R, \quad g^3 = Z \otimes Z.$$

Multiplication gives $\{g^I, g^J\} = 2\eta^{IJ}I_4$, with $\eta = \text{diag}(-1, 1, 1, 1)$. The matrices are distinguished from the scalar parameter γ of the Holst operator. Define

$$g_I = \eta_{IJ}g^J, \quad \mathbb{B}_{IJ} = \frac{1}{4}[g_I, g_J], \quad A_D = -ig^0, \quad \bar{\psi} = \psi^\dagger A_D, \quad \Omega = \frac{1}{2}\omega^{IJ}\mathbb{B}_{IJ}. \quad (55.2)$$

The Clifford product satisfies

$$[\mathbb{B}_{IJ}, g^K] = \delta_J^K g_I - \delta_I^K g_J.$$

Thus the same metric connection acts on the fields through $D\psi = d\psi + \Omega\psi$ and $D\bar{\psi} = d\bar{\psi} - \bar{\psi}\Omega$. Transport along a route is obtained by integrating this connection in its spinor representation; it retains the connection and route of the preceding geometric realization.

Let $\eta_I = \iota_{E_I} \text{vol}_e$, so that $e^J \wedge \eta_I = \delta_I^J \text{vol}_e$. For a field of dimensional type $[\psi] = L^{-3/2}$, take the minimal matter action

$$S_D = \int_M \left\{ \frac{\hbar}{2} [\bar{\psi} g^I D\psi - (D\bar{\psi}) g^I \psi] \wedge \eta_I - mc \bar{\psi} \psi \text{vol}_e \right\}. \quad (55.3)$$

Here $\hbar = \hbar_{\text{int}}$, $c = c_{\text{int}}$, and the mass m is the readout of the matter species in the chosen chart. These values are outputs preceding this realization. The matrix A_D is Hermitian and $A_D g^I$ is anti-Hermitian. The symmetric kinetic term therefore has the real convention corresponding to (55.1). In this signature, the torsion-free equation has principal term $\hbar g^I D_I$; the equivalent notation $i\hbar \Gamma^I D_I$, with $\Gamma^I = -ig^I$, uses the Clifford relation with opposite signature. This specification jointly fixes the adjoint, signature, and imaginary factor of the action.

The matter fields and co-tetrad remain fixed during connection variation. A preparation from the minimum extension in Section 52 may determine the initial datum of the field or state. Each action retains its own variation: substituting a connection-dependent minimizing field into an action also requires its composed differential, or the envelope theorem under its stationarity hypotheses. Action (55.3) here realizes a coupling affine in contorsion.

Proposition 55.1 (Spinor current with variational normalization). *Let $B^{IJK} = \bar{\psi} g^{[I} g^J g^{K]} \psi$, where brackets denote normalized antisymmetrization. The current defined in (53.4) by action (55.3) is*

$$\sigma_{JK} = -\frac{\hbar}{2} \bar{\psi} \{g^I, \mathbb{B}_{JK}\} \psi \eta_I = -\frac{\hbar}{2} B^I_{JK} \eta_I. \quad (55.4)$$

It is independent of contorsion when e and ψ are fixed. The inversion of Section 53.3 therefore applies to this current.

Proof. The variation of Ω is $\delta\Omega = \frac{1}{2} \delta\omega^{JK} \mathbb{B}_{JK}$. The two kinetic summands give the product with g^I on the left and right, respectively. Consequently,

$$\delta_\omega S_D = \frac{\hbar}{4} \int_M \delta\omega^{JK} \wedge \eta_I \bar{\psi} \{g^I, \mathbb{B}_{JK}\} \psi.$$

Comparison with (53.4) fixes the factor and sign in (55.4). Expanding the anticommutator and using the Clifford relation cancels the degree-one contractions, leaving $\{g^I, \mathbb{B}_{JK}\} = g^{[I} g_J g_K]$. The action is linear in Ω , proving the final assertion. $\square$

55.2 Effective contraction and second-moment coefficient

The Lorentzian quadratic form of the trivector is written

$$q(B) = \frac{1}{3!} B^{IJK} B_{IJK} = -(B^{012})^2 - (B^{013})^2 - (B^{023})^2 + (B^{123})^2. \quad (55.5)$$

This contraction retains the signature of the spinor module. Its second moment depends on the state and the realization of field products.

Theorem 55.2 (Effective interaction of the spinor current). *With matter action (55.3), algebraic elimination of contorsion gives*

$$\mathcal{L}_{\text{spin}}^{\text{eff}} = C_\gamma q(B) \text{vol}_e, \quad C_\gamma = \frac{3\kappa c \hbar^2}{16} \frac{\gamma^2}{1 + \gamma^2}. \quad (55.6)$$

The coefficient is obtained from the current and the Holst inverse with the normalizations of the preceding action.

Proof. By (53.16), it suffices to evaluate $-\frac{1}{4} K_*^{IJ} \wedge \sigma_{IJ}$, with

$$Y^{IJ} = \kappa c \frac{\gamma^2}{1 + \gamma^2} (-\star + \gamma^{-1} I) \sigma^{IJ}, \quad K_* = C_e(Y).$$

Here C_e is the explicit composition of (53.9) with (53.11). Linearity of these two inverses reduces the calculation to the $-\star$ and I parts. To give the full contraction, order the trivector components as (012, 013, 023, 123), factor out $\kappa c \hbar^2 \gamma^2 / (1 + \gamma^2)$, and substitute $\sigma_{JK} = -\frac{1}{2} B^I_{JK} \eta_I$. The two matrices of the remaining quadratic form are

$$B_{-\star} = \frac{3}{16} \text{diag}(-1, -1, -1, 1), \quad B_I = 0_{4 \times 4}. \quad (55.7)$$

These matrices are obtained using only $e^I \wedge \eta_J = \delta^I_J \text{vol}_e$: for each component B^{abc} , form σ , compute Y , then $T^I = \sum_J \iota_{E_J} Y^{IJ} - \frac{1}{4} e^I \wedge \sum_{L,J} \iota_{E_L} \iota_{E_J} Y^{LJ}$, and finally $K_{IJK} = (T_{KIJ} - T_{IJK} - T_{JKI})/2$. Contraction gives the four diagonal entries shown; the six polarizations between distinct components are zero for both operators. This calculates every coefficient of both quadratic forms. The $\gamma^{-1} I$ part vanishes, and the $-\star$ part produces exactly (55.5), with the coefficient in (55.6). $\square$

55.3 Finite fermionic realization and second moment of the state

Fix a periodic comoving cell, an orthonormal frame adapted to its observer, and the spatially homogeneous mode of the four spinor components. This finite realization has state space $\mathcal{F} = \bigoplus_{n=0}^4 \Lambda^n \mathbb{C}^4$. Let $a_r, a_r^\dagger, r = 1, \dots, 4$, be its canonical operators, with $\{a_r, a_s^\dagger\} = \delta_{rs}$ and $\{a_r, a_s\} = 0$. The occupation vacuum fixes normal ordering. For a matrix M , write

$$d\Gamma(M) = \sum_{r,s} a_r^\dagger M_{rs} a_s, \quad \hat{N} = d\Gamma(I_4).$$

The letter Γ in this notation denotes second quantization; nonadic holonomy retains its own notation.

The four matrices of the bilinears are computed directly:

$$\begin{array}{c|cccc} abc & 012 & 013 & 023 & 123 \\ \hline M_{abc} = A_D g^a g^b g^c & iJ \otimes R & iJ \otimes Z & iI_2 \otimes J & iZ \otimes J \end{array} \quad (55.8)$$

All are Hermitian, have zero trace, and satisfy $M_{abc}^2 = I_4$. The quartic observable corresponding to (55.5), with this declared normal ordering, is

$$\begin{aligned} \hat{Q}_{\text{sp}} &= \sum_{a < b < c} \eta_{aa} \eta_{bb} \eta_{cc} : d\Gamma(M_{abc})^2 : \\ &= \sum_{a < b < c} \eta_{aa} \eta_{bb} \eta_{cc} d\Gamma(M_{abc})^2 + 2\hat{N}. \end{aligned} \quad (55.9)$$

Normal ordering is part of the specified finite realization; its definition makes subtraction of single-occupation contractions explicit.

Proposition 55.3 (Evaluation by the canonical relations). *The operator $\hat{Q}_{\text{sp}}$ is self-adjoint, preserves every occupation sector, and has restrictions*

$$\hat{Q}_{\text{sp}}|_{\Lambda^0 \oplus \Lambda^1} = 0, \quad \hat{Q}_{\text{sp}}|_{\Lambda^3} = 4I_4, \quad \hat{Q}_{\text{sp}}|_{\Lambda^4} = 8I_1. \quad (55.10)$$

In the basis of $\Lambda^2 \mathbb{C}^4$ whose occupation subsets are (12, 13, 23, 14, 24, 34), its matrix is

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 2 & 0 & 0 & 6 & 0 \\ 0 & 0 & 2 & -6 & 0 & 0 \\ 0 & 0 & -6 & 2 & 0 & 0 \\ 0 & 6 & 0 & 0 & 2 & 0 \\ -4 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (55.11)$$

Proof. The relation $a_s a_t^\dagger = \delta_{st} - a_t^\dagger a_s$ applied to $d\Gamma(M)^2$ separates its single-occupation contraction and gives

$$: d\Gamma(M)^2 : = d\Gamma(M)^2 - d\Gamma(M^2).$$

Since the first three signs in (55.8) are negative and the last positive, the sum of contractions is $-2\hat{N}$. This proves (55.9). Self-adjointness and occupation conservation follow from these same expressions.

On Λ^0 , all summands vanish. On Λ^1 , $d\Gamma(M) = M$, and the squares sum to $-2I_4$, which cancels with $2\hat{N}$. On Λ^4 , $d\Gamma(M) = \text{tr } M = 0$, leaving eight. Identification by complements between $\Lambda^3 \mathbb{C}^4$ and the dual of $\mathbb{C}^4$ takes $d\Gamma(M)$ to $(\text{tr } M)I - M^T = -M^T$. Their squares again sum to $-2I_4$, whereas $2\hat{N} = 6I_4$; the restriction is therefore $4I_4$. Finally, applying the four explicit matrices to the six exterior products in the indicated basis gives (55.11). This evaluates the operator on all sixteen dimensions of $\mathcal{F}$. $\square$

For a state operator $\varrho \geq 0$, $\text{Tr } \varrho = 1$, define $q_\varrho = \text{Tr}(\varrho \hat{Q}_{\text{sp}})$. Its dependence on second moments is explicit:

$$q_\varrho = \sum_{a < b < c} \eta_{aa} \eta_{bb} \eta_{cc} \{ \langle \hat{B}_{abc} \rangle_\varrho^2 + \text{Var}_\varrho(\hat{B}_{abc}) - \langle \hat{N} \rangle_\varrho \}, \quad \hat{B}_{abc} = d\Gamma(M_{abc}). \quad (55.12)$$

Covariance and normal subtraction are retained even when mean currents vanish. In particular, if P_3 is the projection onto $\Lambda^3\mathbb{C}^4$,

$$\mathfrak{g}_3 = \frac{1}{4}P_3, \quad \langle \hat{B}_{abc} \rangle_{\mathfrak{g}_3} = 0, \quad q_{\mathfrak{g}_3} = 4 > 0. \quad (55.13)$$

This is a positive state of trace one and an explicit witness of the positive domain of the functional. The two-occupation sector also contains negative values; for example, the vector $(|12\rangle + |34\rangle)/\sqrt{2}$ has value -4 . The sign of the intensity thus comes from the contraction and the state, separately from the selection of cosmological conditions.

55.4 Density, pressure, and conservation of comoving amplitude

In the spatially homogeneous realization, fix

$$e^0 = N(t)c \, dt, \quad e^j = a(t) \, dx^j, \quad \mathcal{V}(t) = a(t)^3 V_c,$$

with lapse $N(t) > 0$, dimensionless scale factor $a(t) > 0$, and comoving volume $V_c > 0$. The frame and periodic mode of Subsection 55.3 remain declared. The homogeneous-field normalization is $\hat{\psi} = \mathcal{V}^{-1/2}(a_1, a_2, a_3, a_4)^\top$. Each bilinear carries a factor $\mathcal{V}^{-1}$, and its normally ordered product carries $\mathcal{V}^{-2}$. The symmetric temporal kinetic form retains this canonical normalization: terms arising from differentiating $\mathcal{V}^{-1/2}$ cancel between its two summands.

Theorem 55.4 (Torsional amplitude as a state functional). *Holding the canonical state and product realization fixed during metric variations, the effective spin term has reduced action*

$$S_{\text{spin}}^{\text{hom}} = \int dt \, N(t) \frac{cC_\gamma}{a(t)^3 V_c} q_{\mathfrak{g}}. \quad (55.14)$$

Its energy density and isotropic pressure are

$$\rho_{\text{spin}} = p_{\text{spin}} = -\frac{cC_\gamma}{V_c^2} q_{\mathfrak{g}} a^{-6}. \quad (55.15)$$

In any evolution preserving $q_{\mathfrak{g}}$, the corresponding constant amplitude is

$$s_0[\mathfrak{g}, V_c] = \frac{3\kappa c^2 \hbar^2}{16} \frac{\gamma^2}{1 + \gamma^2} \frac{\text{Tr}(\mathfrak{g} \hat{Q}_{\text{sp}})}{V_c^2}. \quad (55.16)$$

Proof. Spatial integration of (55.6) multiplies the factor $\mathcal{V}^{-2}$ by $Nc\mathcal{V} \, dt$, giving (55.14). In canonical variables, lapse variation defines energy through $\delta_N S_m = -\int E \, \delta N \, dt$. Therefore,

$$E_{\text{spin}} = -\frac{cC_\gamma q_{\mathfrak{g}}}{a^3 V_c}, \quad \rho_{\text{spin}} = \frac{E_{\text{spin}}}{a^3 V_c}.$$

Isotropic spatial variation satisfies $\delta_a S_m = \int 3Na^2 V_c p \, \delta a \, dt$. Applied to (55.14), it gives

$$3Na^2 V_c p_{\text{spin}} = -\frac{3NcC_\gamma q_{\mathfrak{g}}}{a^4 V_c}.$$

This simultaneously proves the sign, pressure, and density in (55.15). When $q_{\mathfrak{g}}$ is conserved, the coefficient of $-a^{-6}$ is constant, yielding (55.16). Dimensions are consistent: $[cC_\gamma] = \text{energy } L^3$, so s_0 has the dimension of energy density. $\square$

Corollary 55.5 (Conserved positive-amplitude domain). *Let a unitary evolution of $\mathcal{F}$ preserve $\Lambda^3\mathbb{C}^4$. For every initial trace-one state supported on this sector, $q_{\mathfrak{g}(t)} = 4$, and*

$$s_0 = \frac{3\kappa c^2 \hbar^2}{4V_c^2} \frac{\gamma^2}{1 + \gamma^2} > 0. \quad (55.17)$$

Proof. Invariance of the sector preserves the support of the state; there, (55.10) gives $\hat{Q}_{\text{sp}} = 4I$. Trace conservation then fixes its expectation at four, independently of the internal evolution of its coherences. Positivity of the couplings, $V_c > 0$, and $\gamma \neq 0$ proves the formula and its sign. $\square$

For general states, conservation of $\langle \hat{N} \rangle$ determines dilution of the occupation density, $n(t) = \langle \hat{N} \rangle / (a^3 V_c)$. Conservation of the quartic moment is checked separately. If $i\hbar \dot{\varrho} = [H(t), \varrho]$, then

$$\frac{dq_{\varrho}}{dt} = \frac{i}{\hbar} \text{Tr}(\varrho[H(t), \hat{Q}_{\text{sp}}]). \quad (55.18)$$

Commutation, invariance of a sector on which the observable is scalar, or conservation proved for the specific state provides the relevant conditions. If the moment evolves, (55.15) retains its instantaneous evaluation; its balance includes

$$\dot{\rho}_{\text{spin}} + 6H\rho_{\text{spin}} = -\frac{cC_{\gamma}}{V_c^2} a^{-6} \dot{q}_{\varrho}, \quad H = \dot{a}/a$$

in proper time. This term records exchange with the remaining matter degrees of freedom.

The composition is determined: spinor representation of transport, matter action, variational current, Cartan–Holst inversion, second moment, and normalized density. The domain $s_0 > 0$ of Section 56 thus has an evaluable functional and explicit witness states. Its use in cosmological dynamics also retains the matter, geometric, and conservation conditions stated in that section.

56 Memory dilution and persistence of the bounce

Memory evolution and spatial expansion are two readouts of the same sequence of states. The refinement index can be eliminated between them; this elimination determines the dilution exponent that the cosmological realization transmits to its torsional contribution. The initial amplitude depends on evaluation of the specific material current in its variational response. The scaling law and bounce theorem below preserve this dependence.

Theorem 55.4 determines this amplitude as a functional of the material state and comoving volume. Corollary 55.5 provides an explicit domain of states with positive, conserved amplitude. In this domain, evaluation of s_0 precedes the cosmological integration developed below. The ordinary density $\rho_0 > 0$ and its barotropic law are specified additionally in Section 56.2: the amplitude-positivity witness does not by itself evaluate this material density.

56.1 The a^{-6} dilution law

Proposition 56.1 (Elimination of the memory index). *Let $a_{\text{ref}} > 0$, $\varphi > 1$, and let the route readers be*

$$\frac{a_n}{a_{\text{ref}}} = \varphi^{n/3}, \quad M_n = \varphi^{-2n}.$$

Then

$$M_n = \left(\frac{a_n}{a_{\text{ref}}} \right)^{-6}$$

for every index in the sequence. If the quadratic-density realization is $s_n^2 = s_^2 M_n$, with $s_*^2 > 0$, then*

$$s_n^2 = s_*^2 \left(\frac{a_n}{a_{\text{ref}}} \right)^{-6}.$$

Proof. Raising the first equality to the power -6 gives φ^{-2n} , which is exactly the second reader. Multiplication by the initial amplitude yields the second statement. The exponent is determined by the ratio of the exponents of the two readers; the amplitude is the evaluation of the reference state. $\square$

The continuous homogeneous realization preserves this constitutive law: the torsional density is written $s_0 a^{-6}$, where the normalization of the factor a , the material coefficient and the reference amplitude have been absorbed into s_0 . The scaling law and evaluation of s_0 are successive operations of the construction.

56.2 Homogeneous threshold and transversality

Consider the spatially flat homogeneous chart, in units $c = 1$, with constant $G > 0$, positive densities ρ_0, s_0 , and constant barotropic parameter $w < 1$. The specified realization satisfies $\Lambda_{\text{eff}} = 0$ and the equations

$$\rho_m(a) = \rho_0 a^{-3(1+w)}, \quad \rho_s(a) = s_0 a^{-6}, \quad (56.1)$$

$$H^2 = \frac{8\pi G}{3}(\rho_m - \rho_s) =: F_0(a), \quad \dot{H} = -4\pi G((1+w)\rho_m - 2\rho_s). \quad (56.2)$$

The couplings belong to the dimensional section already constructed. The expression for ρ_s incorporates the sign of its contribution in this gravitational realization.

Theorem 56.2 (Existence and uniqueness of the transverse threshold). *There is a unique $a_b > 0$ with $F_0(a_b) = 0$:*

$$a_b^{3(1-w)} = \frac{s_0}{\rho_0}.$$

The region permitted by $H^2 \geq 0$ is $a \geq a_b$. At the threshold, with $\rho_b = \rho_m(a_b) = \rho_s(a_b) > 0$,

$$\dot{H}_b = 4\pi G(1-w)\rho_b > 0.$$

The solution crossing $H = 0$ has a strict minimum of a , with local expansion

$$a(t) = a_b \left[1 + 2\pi G(1-w)\rho_b(t-t_b)^2 + O((t-t_b)^3) \right].$$

Proof. The factorization

$$F_0(a) = \frac{8\pi G}{3} a^{-6} (\rho_0 a^{3(1-w)} - s_0)$$

reduces its zeros to that of a strictly increasing function from $(0, \infty)$ onto $(-s_0, \infty)$. This proves existence, uniqueness and the sign of F_0 on either side of the threshold. The second equation in (56.2), evaluated at equality of the densities, gives the formula for $\dot{H}_b$. Since $\dot{a} = aH$, at the threshold $\ddot{a}_b = a_b \dot{H}_b > 0$. The equations are smooth for $a > 0$, so their Taylor expansion gives the stated expression. The change of sign of H is transverse: it passes from contraction to expansion. $\square$

For $w = 1$, both terms have the same exponent, and equality of amplitudes determines a degenerate case. For $w > 1$, the sign of $\dot{H}$ at a coincidence of densities is negative. The theorem's domain $w < 1$ expresses precisely the order between the two dilution exponents.

56.3 Local persistence under controlled perturbations

Persistence of the threshold uses two independent estimates: control of the sign at the endpoints and of the derivative on the interval. Let $I = [a_-, a_+] \subset (0, \infty)$, with $a_- < a_b < a_+$, be chosen so that

$$F_0(a_-) < 0 < F_0(a_+), \quad m_I := \inf_{a \in I} F'_0(a) > 0.$$

Such an interval exists because

$$F'_0(a_b) = \frac{8\pi G}{a_b} (1-w)\rho_b > 0.$$

Define $\delta_I = \min\{-F_0(a_-), F_0(a_+)\} > 0$.

Theorem 56.3 (Local persistence of the transverse bounce). *Let $R \in C^1(I)$ be a perturbation satisfying*

$$\|R\|_{\infty, I} < \delta_I, \quad \|R'\|_{\infty, I} < m_I.$$

Then $F_R = F_0 + R$ has a unique zero $a_R \in (a_-, a_+)$, and $F'_R(a_R) > 0$. The realization $H^2 = F_R(a)$ admits a local transverse bounce, with

$$\dot{H}_R = \frac{a_R}{2} F'_R(a_R) > 0$$

at the threshold. If $R(a, \eta)$ is jointly C^1 , the persistent zeros depend locally in a C^1 manner on the parameter η .

Proof. The first bound preserves $F_R(a_-) < 0 < F_R(a_+)$; the intermediate value theorem provides an interior zero. The second gives $F'_R(a) \geq m_I - \|R'\|_{\infty, I} > 0$ on I , so the zero is unique and simple.

For $a > a_R$ close to the zero, $F_R(a) = F'_R(a_R)(a - a_R) + o(a - a_R)$. The integral

$$t - t_b = \int_{a_R}^a \frac{d\xi}{\xi \sqrt{F_R(\xi)}}$$

is finite and strictly increasing. Its inverse defines the expanding branch; time reflection defines the contracting branch. They join with $\dot{a}(t_b) = 0$ and

$$a(t) - a_R = \frac{a_R^2 F'_R(a_R)}{4} (t - t_b)^2 + o((t - t_b)^2).$$

Away from the threshold, differentiating $H^2 = F_R(a)$ gives $\dot{H} = aF'_R(a)/2$; its continuous limit gives the positive expression at a_R . The C^1 regularity of F_R ensures continuity of the acceleration at this junction. Finally, $\partial_a F_R(a_R) > 0$ permits application of the implicit function theorem when the perturbation depends jointly on (a, η) . $\square$

The first bound controls existence; the second ensures local uniqueness and transversality. The stability proved here concerns this threshold in the selected interval. Torsional amplitude enters its location, whereas the relation between exponents determines the temporal orientation of the crossing.

57 Explicit homogeneous solution and causal completeness

Determining the amplitude permits a complete realization of the bounce to be developed in addition to the local argument. The spatially flat chart of (56.2) is retained, in units $c = 1$, with $\rho_0, s_0, G > 0$, and the material sector is specialized to dust, $w = 0$, with $\Lambda_{\text{eff}} = 0$. The coefficient s_0 is obtained from the material state in the corresponding realization; the following integration uses this coefficient and preserves its domain.

Theorem 57.1 (Exact integration of the dust bounce). *Set*

$$a_b^3 = \frac{s_0}{\rho_0}, \quad \tau^2 = \frac{s_0}{6\pi G \rho_0^2}, \quad u = t - t_b.$$

The solution of the homogeneous system with $a(t_b) = a_b$ and $H(t_b) = 0$ is, for every $t \in \mathbb{R}$,

$$a(t) = a_b \left(1 + \frac{u^2}{\tau^2}\right)^{1/3}, \quad H(t) = \frac{2u}{3(\tau^2 + u^2)}, \quad \dot{H}(t) = \frac{2(\tau^2 - u^2)}{3(\tau^2 + u^2)^2}. \quad (57.1)$$

It has a unique minimum at t_b ; it is contracting before and expanding afterwards. The evolution is smooth and satisfies $a(t) \geq a_b > 0$.

Proof. Let $y = a^3$. The Friedmann equation gives

$$\dot{y}^2 = 24\pi G(\rho_0 y - s_0).$$

The function $y = s_0/\rho_0 + 6\pi G \rho_0 u^2$ satisfies this identity and the initial conditions. Its derivatives give the three expressions in (57.1). To verify the evolution equation as well, write

$$\rho_b = \frac{\rho_0^2}{s_0}, \quad x = \frac{u}{\tau}.$$

Then

$$\rho_m = \frac{\rho_b}{1 + x^2}, \quad \rho_s = \frac{\rho_b}{(1 + x^2)^2}, \quad 4\pi G \rho_b = \frac{2}{3\tau^2}.$$

Substitution into $\dot{H} = -4\pi G(\rho_m - 2\rho_s)$ gives exactly the third formula. Thus the complete system $(\dot{a}, \dot{H})$ is satisfied, and its vector field is smooth for $a > 0$. Local uniqueness, prolonged along the explicit solution, determines the evolution from these initial conditions. The denominator is positive for every t , and the statements about the minimum and sign follow from H . $\square$

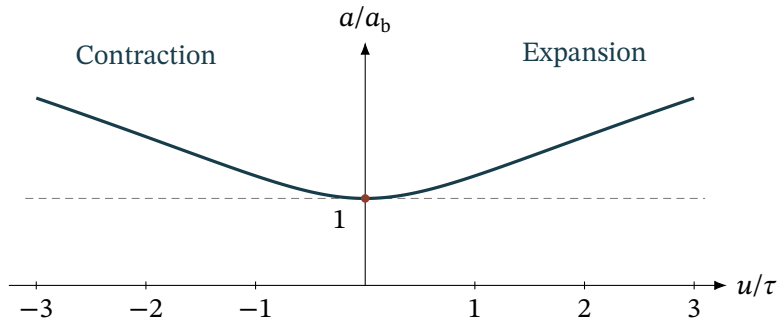


Figure 15: Dimensionless profile of the proved solution: $a/a_b = (1 + (u/\tau)^2)^{1/3}$. The parameters a_b and τ are calculated from ρ_0, s_0, G ; the diagram represents the entire rescaled collection.

57.1 Finite curvature and extension of geodesics

The metric realized in this chart is $g = -dt^2 + a(t)^2 dx^2$ on $\mathbb{R} \times \mathbb{R}^3$. With the convention in which the flat-model scalar curvature is $R = 6(\dot{H} + 2H^2)$, the solution gives

$$R(t) = \frac{4\tau^2 + \frac{4}{3}u^2}{(\tau^2 + u^2)^2}.$$

In particular, $R(t_b) = 4/\tau^2$ is finite. In an orthonormal frame, the Riemann components depend on H^2 and $\dot{H} + H^2$, which are bounded functions on $\mathbb{R}$. The same holds for polynomial contractions of these components; for example,

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 12[(\dot{H} + H^2)^2 + H^4].$$

Proposition 57.2 (Completeness of causal geodesics). *The preceding metric realization is geodesically complete for the timelike and null geodesics of its Levi-Civita connection, in both directions.*

Proof. Spatial translations provide the constants $p_i = a^2 dx^i/d\lambda$. Causal normalization gives

$$\left| \frac{dt}{d\lambda} \right| = \sqrt{\epsilon + \frac{|\mathbf{p}|^2}{a(t)^2}}, \quad \epsilon = 1 \text{ in the timelike case, } \epsilon = 0 \text{ in the null case.}$$

A nonconstant null geodesic has $|\mathbf{p}| > 0$. Since $a \geq a_b$, in the timelike case

$$\left| \frac{d\lambda}{dt} \right| \geq \left(1 + \frac{|\mathbf{p}|^2}{a_b^2} \right)^{-1/2} > 0,$$

and in the null case $|d\lambda/dt| = a/|\mathbf{p}| \geq a_b/|\mathbf{p}| > 0$. Thus reaching $t = \pm\infty$ requires infinite affine length. On a compact time interval, a and its derivatives are smooth and a is bounded away from zero. The explicit equations above and $dx^i/d\lambda = p_i/a^2$ allow the geodesic to be continued through its finite endpoints. Both possibilities for prolongation are covered. $\square$

The completeness result belongs to this homogeneous dust realization, with its conditions on space, matter and amplitude. Local persistence under threshold perturbations preserves the bounds of Theorem 56.3; this local property and global completeness of the explicit solution are distinct conclusions. The minimum scale follows from the relation between material density and torsional response. Memory thus enters a verifiable geometric evolution, with a regular realization of contraction and expansion.

58 Residual tensor, trace decomposition, and covariant balance

The geometric realization of the enriched state provides a metric, while the matter transducer expresses the source to which the readout refers. The residual tensor retains the difference between both representations in a common tensor space. Its decomposition distinguishes the sector proportional to the metric from the traceless sector and determines the exchange current corresponding to that division.

58.1 Chart data and construction of the residual tensor

Fix an HMT state x and a connected chart $\mathcal{U}$ of its geometric realization. On $\mathcal{U}$, the nondegenerate co-tetrad of Section 49 determines a smooth metric $g = g_x$ of signature $(-+++)$. Denote its Levi-Civita connection by $\mathring{\nabla}$, and write

$$G_{\mu\nu}[g] := \text{Ric}_{\mu\nu}[g] - \frac{1}{2}R[g]g_{\mu\nu}.$$

This distinguishes the Einstein tensor $G[g]$ from the gravitational coupling G_{int} . The derivative used in the balances of this section is $\mathring{\nabla}$, consistently with the effective metric equation of Section 53.5.

We work in the dimensional chart $c_{\text{int}} = 1$. Assume $G_{\text{int}}(x) > 0$ and $\Lambda_{\text{ref}}(x)$ constant on $\mathcal{U}$, and set $\kappa_x = 8\pi G_{\text{int}}(x)$. The symbol x identifies the state supplying these readers; covariant differentiation acts on the points of $\mathcal{U}$. The genealogy of the couplings remains in Section 43.

Let T_b be the smooth symmetric tensor expressed by the chart's matter transducer; in the baryonic readout, it is its baryonic source. Define

$$\mathcal{E}_{\mu\nu}^{\text{ref}} := \kappa_x^{-1}(G_{\mu\nu}[g] + \Lambda_{\text{ref}}g_{\mu\nu}), \quad T_{\mu\nu}^{\text{res}} := \mathcal{E}_{\mu\nu}^{\text{ref}} - T_{b,\mu\nu}. \quad (58.1)$$

The tensor T^{res} is the effective residual tensor, also denoted T_{osc} when interpreting the geometric sector remaining to be expressed relative to T_b . The construction retains g_x , G_{int} , Λ_{ref} and T_b as its effective arguments.

Theorem 58.1 (Uniqueness of the residual and balance with the matter source). *For the preceding data, there is a unique symmetric tensor T^{res} satisfying $\mathcal{E}^{\text{ref}} = T_b + T^{\text{res}}$. Its divergence is*

$$\mathring{\nabla}^\mu T_{\mu\nu}^{\text{res}} = -\mathring{\nabla}^\mu T_{b,\mu\nu}. \quad (58.2)$$

In particular, if the matter source is separately conserved, $\mathring{\nabla}^\mu T_{b,\mu\nu} = 0$, then T^{res} is also conserved.

Proof. The difference of two tensors satisfying the stated equation is zero; expression (58.1) supplies existence. The contracted Bianchi identity and Levi-Civita metric compatibility give

$$\mathring{\nabla}^\mu G_{\mu\nu} = 0, \quad \mathring{\nabla} g = 0.$$

Since κ_x and Λ_{ref} are constant on the chart, $\mathring{\nabla}^\mu \mathcal{E}_{\mu\nu}^{\text{ref}} = 0$. Differentiating the difference defining T^{res} proves (58.2), including the assertion under separate matter conservation. $\square$

Spatial and temporal constancy of the couplings on the chart fixes the domain of this balance. A collection of states may express different couplings in different charts; each preceding identity applies to the chart and state specifying it.

58.2 Canonical decomposition and exchange current

Define the trace function and the two tensors

$$\theta := g^{\mu\nu}T_{\mu\nu}^{\text{res}}, \quad T_{\mu\nu}^\Lambda := \frac{1}{4}\theta g_{\mu\nu}, \quad T_{\mu\nu}^0 := T_{\mu\nu}^{\text{res}} - T_{\mu\nu}^\Lambda. \quad (58.3)$$

The superscript Λ identifies the component proportional to the metric; Λ_{ref} is the reference parameter of (58.1). Their functions are distinct.

Proposition 58.2 (Trace projectors and uniqueness of the decomposition). *On symmetric covariant tensors of order two, the maps*

$$P_{\text{tr}}(S) = \frac{1}{4} \text{tr}_g(S)g, \quad P_0(S) = S - P_{\text{tr}}(S)$$

satisfy

$$P_{\text{tr}}^2 = P_{\text{tr}}, \quad P_0^2 = P_0, \quad P_{\text{tr}}P_0 = P_0P_{\text{tr}} = 0, \quad P_{\text{tr}} + P_0 = I. \quad (58.4)$$

In particular, $T^{\text{res}} = T^\Lambda + T^0$, $\text{tr}_g T^0 = 0$, and this decomposition is unique.

Proof. In four dimensions, $g^{\mu\nu}g_{\mu\nu} = 4$. Therefore, $\text{tr}_g(P_{\text{tr}}S) = \text{tr}_g S$, which proves idempotence of P_{tr} , vanishing of the trace of P_0S , and, by expansion, the other identities. If $S = fg + Q$ with $\text{tr}_g Q = 0$, taking the trace gives $f = \text{tr}_g S/4$. This also determines $Q = S - fg$, establishing uniqueness. $\square$

Theorem 58.3 (Covariant exchange between the two sectors). *On the domain of (58.1), let $b_\nu := \mathring{\nabla}^\mu T_{b,\mu\nu}$. The trace exchange current is the one-form*

$$\mathcal{J}_\nu^{\text{tr}} := \frac{1}{4} \partial_\nu \theta = \mathring{\nabla}^\mu T_{\mu\nu}^\Lambda. \quad (58.5)$$

The full balances are

$$\mathring{\nabla}^\mu T_{\mu\nu}^0 = -b_\nu - \mathcal{J}_\nu^{\text{tr}}, \quad \mathring{\nabla}^\mu (T^\Lambda + T^0)_{\mu\nu} = -b_\nu. \quad (58.6)$$

If $b = 0$, the two sectors exchange exactly opposite currents. In that case, they are separately conserved if and only if θ is constant on the connected chart.

Proof. Metric compatibility allows the calculation

$$\mathring{\nabla}^\mu \left(\frac{1}{4} \theta g_{\mu\nu} \right) = \frac{1}{4} g^{\mu\alpha} (\partial_\alpha \theta) g_{\mu\nu} = \frac{1}{4} \partial_\nu \theta.$$

Subtracting this equality from (58.2) gives (58.6). With $b = 0$, separate conservation is equivalent to $d\theta = 0$; a smooth function with zero differential is constant on every connected component. $\square$

The one-form $\mathcal{J}^{\text{tr}}$ measures exchange in the tensorial division. It retains a type distinct from the transport covector of Section 50 and the spin current $\sigma_{IJ} \in \Omega^3(\mathcal{U}; \Lambda^2 E^*)$ of (53.4).

58.3 Link to the effective Cartan–Holst source

On a solution of (53.19), in the same units $c = 1$, the residual has expression

$$T^{\text{res}} = T^{\text{phys,LC}} + U^{\text{phys,spin}} - T_b + \frac{\Lambda_{\text{ref}} - \Lambda_{\text{int}}}{\kappa_x} g. \quad (58.7)$$

Indeed, substituting $G + \Lambda_{\text{int}}g = \kappa_x(T^{\text{phys,LC}} + U^{\text{phys,spin}})$ into (58.1) gives equality term by term. When the matter transducer expresses precisely $T_b = T^{\text{phys,LC}}$ and $\Lambda_{\text{ref}} = \Lambda_{\text{int}}$, one obtains $T^{\text{res}} = U^{\text{phys,spin}}$. The condition of separate conservation of T_b in Theorem 58.1 retains its own function: (53.20) establishes balance of the total source, and the division among its components depends on the matter action being used.

58.4 Trace sector and acceleration of the selected charts

The TRIT reader of state x expresses $s = \tau(x) \in \{+1, 0, -1\}$, and its positive clock expresses

$$H_* = \frac{2\pi}{108t_0(x)} > 0. \quad (58.8)$$

Let h_s be a three-dimensional metric of constant sectional curvature s . The three charts of the selected realization are written

$$g_s = -dt^2 + a_s(t)^2 h_s, \quad a_+(t) = H_*^{-1} \cosh(H_* t), \quad a_0(t) = \exp(H_* t), \quad a_-(t) = H_*^{-1} \sinh(H_* t). \quad (58.9)$$

The closed and flat charts are considered for $t \in \mathbb{R}$; the open chart for $t > 0$. The sign s and the clock H_* retain their antecedents in the state selecting the realization.

Lemma 58.4 (Curvature of the three charts of the same clock). *For the functions in (58.9),*

$$\frac{\ddot{a}_s}{a_s} = H_*^2, \quad \left(\frac{\dot{a}_s}{a_s}\right)^2 + \frac{s}{a_s^2} = H_*^2, \quad \text{Ric}[g_s] = 3H_*^2 g_s. \quad (58.10)$$

Consequently, $R[g_s] = 12H_*^2$ and $G[g_s] = -3H_*^2 g_s$.

Proof. The first two equalities follow by differentiating the three functions: the closed and open charts use $\cosh^2 z - \sinh^2 z = 1$, and the flat chart uses $\dot{a}_0 = H_* a_0$. To verify the tensorial identification, the connection coefficients involving the temporal coordinate are

$$\Gamma^0_{ij} = a_s \dot{a}_s (h_s)_{ij}, \quad \Gamma^i_{0j} = (\dot{a}_s/a_s) \delta_j^i,$$

and the purely spatial ones agree with those of h_s . Contracting the curvature gives

$$\text{Ric}_{00} = -3\ddot{a}_s/a_s, \quad \text{Ric}_{0i} = 0, \quad \text{Ric}_{ij} = (a_s \ddot{a}_s + 2\dot{a}_s^2 + 2s)(h_s)_{ij}.$$

Substituting the first two identities of (58.10) gives $\text{Ric}_{00} = -3H_*^2$ and $\text{Ric}_{ij} = 3H_*^2 a_s^2 (h_s)_{ij}$. Taking the trace and forming the Einstein tensor proves the other assertions, with the curvature convention of (53.19). $\square$

Theorem 58.5 (Density, pressure, and sign of the selected acceleration). *In the preceding charts, for $\Lambda_{\text{ref}} = 0$ and $T_b = 0$, the residual tensor is*

$$T^{\text{res}} = -\rho_* g_s, \quad \rho_* := \frac{3H_*^2}{8\pi G_{\text{int}}} > 0, \quad T^\Lambda = T^{\text{res}}, \quad T^0 = 0. \quad (58.11)$$

For a unit comoving observer $u = \partial_t$, the density and isotropic pressure are ρ_* and $p_* = -\rho_*$. The acceleration equation evaluates to

$$\frac{\ddot{a}_s}{a_s} = -\frac{4\pi G_{\text{int}}}{3}(\rho_* + 3p_*) = H_*^2 > 0. \quad (58.12)$$

Proof. The preceding lemma and (58.1) give $T^{\text{res}} = -(3H_*^2/\kappa_x)g_s$. Since $g_s(u, u) = -1$, $T_{\mu\nu}^{\text{res}} u^\mu u^\nu = \rho_*$. The spatial projector $q^{\mu\nu} = g_s^{\mu\nu} + u^\mu u^\nu$ satisfies $q^{\mu\nu}(g_s)_{\mu\nu} = 3$, and hence

$$p_* := \frac{1}{3} q^{\mu\nu} T_{\mu\nu}^{\text{res}} = -\rho_*.$$

The trace is $-4\rho_*$, so (58.3) produces the two sectors in (58.11). Finally, $\rho_* + 3p_* = -2\rho_*$, and substitution into the right-hand side of (58.12) gives $\kappa_x \rho_*/3 = H_*^2$, equal to the geometric value already calculated. Positivity uses exactly $G_{\text{int}} > 0$ and $H_* > 0$. $\square$

In this realization, the trace sector expresses a constant density and an opposite pressure, and the internal clock fixes the positive acceleration. In a general chart, T^Λ describes the pointwise isotropic component, while T^0 retains the traceless residual, and both obey the exchange in (58.6). Identifying these sectors with observational dark sources requires a single realization to reproduce expansion, lensing, rotation, structure growth, and the cosmic background jointly. The present construction fixes the tensor, its balances, and the acceleration of the selected charts entering that comparison.

59 Pure bipartite states, reduced entropy, and horizon area laws

The enriched state retains paths and their registers during evolution; a subsystem readout determines which part of that information remains accessible. The construction in this section uses the angular channels of Section 13, the incidence and area operator of Section 44, and the units of action, area, and temperature of Sections 43 and 45. These are earlier representations of the same APP-TRIT-TPK construction. Each transmits its domain, orientation, normalization, and realization conditions here.

Three spaces are distinguished: the binary sheet register, the tritic factors of sectorial incidence, and the geometric realization of the exterior regions of a horizon. Purification, transport of observables, and dimensional conversion specify the relations between them. This distinction preserves the structure accompanying each entropy value.

59.1 Sheet register and bipartite purification

Write $a = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$ for the angular coordinates already constructed. The channels and their probabilities are

$$q_{\pm} = e^{-a\mp y}, \quad p_{\pm} = \frac{q_{\pm}}{q_{+} + q_{-}} = \frac{e^{\mp y}}{2 \cosh y}. \quad (59.1)$$

The signs index these two channels. On $\mathcal{H}_h = \text{span}\{|+\rangle, |-\rangle\}$ define

$$\rho_h(y) = p_{+}|+\rangle\langle+| + p_{-}|-\rangle\langle-|.$$

The auxiliary space $\mathcal{H}_{\text{ar}} = \text{span}\{|\text{adv}\rangle, |\text{ret}\rangle\}$ carries two orthonormal labels. With a phase χ_0 fixed, the joint state is

$$|\Psi_{\text{ar}}\rangle = \sqrt{p_{+}}|+\rangle \otimes |\text{adv}\rangle + e^{i\chi_0}\sqrt{p_{-}}|-\rangle \otimes |\text{ret}\rangle. \quad (59.2)$$

Proposition 59.1 (Accessible information in the sheet register). *The vector (59.2) is normalized, and its partial trace over $\mathcal{H}_{\text{ar}}$ is $\rho_h(y)$. Its entanglement entropy, expressed in nats, is*

$$s_h(y) = \log(2 \cosh y) - y \tanh y. \quad (59.3)$$

Exchanging the labels + and – conjugates $\rho_h(y)$ to $\rho_h(-y)$ and preserves s_h .

Proof. The squared norm is $p_{+} + p_{-} = 1$. Orthogonality of the auxiliary labels eliminates the cross terms in the partial trace. The squared Schmidt coefficients are precisely $p_{\pm}$. By (59.1), $p_{+} - p_{-} = -\tanh y$; substituting $\log p_{\pm} = \mp y - \log(2 \cosh y)$ into $-\sum_{\pm} p_{\pm} \log p_{\pm}$ proves the formula. Exchanging the signs permutes the two eigenvalues and preserves entropy. $\square$

The common factor e^{-a} cancels when the binary register is normalized. The relative phase χ_0 remains in the joint state even though this diagonal reduction does not express it. Logical reversibility concerns transport of the complete register, with its labels and correlations; phase return preserves the memory advance described in Section 47. A thermal implementation also has its own energy and dissipation law. In particular, s_h measures this binary subsystem and its specific bipartition.

59.2 Purification of the incidence sectors

The boundary incidence constructed earlier distinguishes two sets of eighteen links, with sectorial labels $m = 90, 120$. Each link carries $\mathbb{C}^3$, with basis $|0\rangle, |1\rangle, |2\rangle$ and occupation operator $\text{diag}(0, 1, 2)$. Let N_m be their sectorial sums, and

$$N = N_{90} + N_{120}, \quad D_{\partial} = 90N_{90} + 120N_{120}. \quad (59.4)$$

On the positive branch, γ_{BI} denotes the Barbero–Immirzi parameter already obtained, and $a_0 > 0$ its angular area factor. This notation distinguishes the parameter from the connection and its holonomies. Area is expressed as

$$\hat{\mathcal{A}}_m = \ell_P^2 \gamma_{\text{BI}} a_0 N_m, \quad \hat{\mathcal{A}} = \hat{\mathcal{A}}_{90} + \hat{\mathcal{A}}_{120}. \quad (59.5)$$

For the finite parameter $\Delta = \Delta_{\text{mod}} \in \mathbb{R}$ of the reduced state, put

$$x_m = e^{-m\Delta}, \quad z_m = 1 + x_m + x_m^2, \quad Z(\Delta) = z_{90}^{18} z_{120}^{18}, \quad (59.6)$$

$$\rho_{\partial}(\Delta) = Z(\Delta)^{-1} e^{-\Delta D_{\partial}}. \quad (59.7)$$

The trace in this section is the ordinary matrix trace, and the states have trace one. The parameter Δ retains the selection and normalization of the antecedent state.

Proposition 59.2 (Tensor-product purification of the boundary). *For a finite real weight w , let*

$$|\text{TFD}(w)\rangle = \frac{|0\rangle|0\rangle + e^{-w/2}|1\rangle|1\rangle + e^{-w}|2\rangle|2\rangle}{\sqrt{1 + e^{-w} + e^{-2w}}}. \quad (59.8)$$

The product

$$|\Psi_{\Delta}\rangle = |\text{TFD}(90\Delta)\rangle^{\otimes 18} \otimes |\text{TFD}(120\Delta)\rangle^{\otimes 18} \quad (59.9)$$

is a normalized purification of $\rho_{\partial}(\Delta)$.

Proof. At each link, the sum of the squares of the three coefficients is one. The partial trace eliminates terms with distinct indices and returns $\text{diag}(1, e^{-w}, e^{-2w})/(1 + e^{-w} + e^{-2w})$. The partial trace commutes with the tensor product. Multiplying the thirty-six reductions gives the exponential and the partition function in (59.7). $\square$

59.3 Information equation and area fluctuations

The occupation means and variances per link are

$$f_m = \frac{x_m + 2x_m^2}{z_m}, \quad v_m = \frac{x_m + 4x_m^2}{z_m} - f_m^2. \quad (59.10)$$

Write $s(\rho) = -\text{Tr}(\rho \log \rho)$, with $0 \log 0 = 0$. In the preceding purification, $s(\rho_{\partial})$ is its entanglement entropy.

Proposition 59.3 (Mean area, entropy, and sectorial response). *For real, finite Δ , we have*

$$\langle N \rangle_{\Delta} = 18(f_{90} + f_{120}), \quad (59.11)$$

$$\langle D_{\partial} \rangle_{\Delta} = 18(90f_{90} + 120f_{120}), \quad (59.12)$$

$$\langle \hat{\mathcal{A}} \rangle_{\Delta} = 18\ell_P^2 \gamma_{\text{BI}} a_0 (f_{90} + f_{120}), \quad (59.13)$$

$$s(\Delta) = \log Z(\Delta) + \Delta \langle D_{\partial} \rangle_{\Delta}, \quad (59.14)$$

$$\frac{d\langle D_{\partial} \rangle_{\Delta}}{d\Delta} = -\text{Var}_{\Delta}(D_{\partial}) = -18(90^2 v_{90} + 120^2 v_{120}), \quad (59.15)$$

$$\frac{ds}{d\Delta} = -\Delta \text{Var}_{\Delta}(D_{\partial}), \quad (59.16)$$

$$\frac{d\langle \hat{\mathcal{A}} \rangle_{\Delta}}{d\Delta} = -18\ell_P^2 \gamma_{\text{BI}} a_0 (90v_{90} + 120v_{120}). \quad (59.17)$$

Proof. The local weights are $(1, x_m, x_m^2)/z_m$, whose first two moments give f_m and v_m . Additivity of the occupation operators and factorization of the state give the first three formulas. Taking the trace of $-\rho_{\partial} \log \rho_{\partial} = \rho_{\partial}(\log Z + \Delta D_{\partial})$ gives entropy. Differentiation of the finite sums yields $\partial_{\Delta} \log Z = -\langle D_{\partial} \rangle_{\Delta}$ and $\partial_{\Delta}^2 \log Z = \text{Var}_{\Delta}(D_{\partial})$. The independent factors add their variances. In the derivative of (59.14), the two first-moment terms cancel. Finally, $\partial_{\Delta} f_m = -mv_m$ gives the area response. $\square$

Loss of uniformity is quantified in dimension $d_{\partial} = 3^6$:

$$I_{\partial}(\Delta) = 36 \log 3 - s(\Delta) = \mathcal{D}(\rho_{\partial}(\Delta) \| I/d_{\partial}) \geq 0. \quad (59.18)$$

Here $\mathcal{D}$ denotes relative entropy, defined and justified below. Substituting $\log(I/d_{\partial}) = -36 \log 3$ proves the identity. Convexity of $t \log t$ gives positivity; equality corresponds to the uniform distribution, which in this collection occurs exactly at $\Delta = 0$. The deficit measures dimensionless information; the scale $\gamma_{\text{BI}} a_0 \ell_P^2$ measures area per occupation.

Theorem 59.4 (Occupation reversal and complementary area). *Let $\mathcal{R}_\partial$ be the involution exchanging $|0\rangle$ and $|2\rangle$ and fixing $|1\rangle$ in each factor. Then*

$$\mathcal{R}_\partial \mathbf{N} \mathcal{R}_\partial^* = 72I - \mathbf{N}, \quad (59.19)$$

$$\mathcal{R}_\partial \mathbf{D}_\partial \mathcal{R}_\partial^* = 7560I - \mathbf{D}_\partial, \quad (59.20)$$

$$\rho_\partial(-\Delta) = \mathcal{R}_\partial \rho_\partial(\Delta) \mathcal{R}_\partial^*, \quad (59.21)$$

$$s(-\Delta) = s(\Delta), \quad (59.22)$$

$$\langle \hat{\mathcal{A}} \rangle_{-\Delta} + \langle \hat{\mathcal{A}} \rangle_\Delta = 72\gamma_{\text{BI}} a_0 \ell_P^2, \quad (59.23)$$

$$\mathcal{D}(\rho_\partial(\Delta) \| \rho_\partial(-\Delta)) = \Delta(7560 - 2\langle \mathbf{D}_\partial \rangle_\Delta). \quad (59.24)$$

Proof. Each occupation transforms by $n \mapsto 2 - n$. In each sector, $\mathbf{N}_m \mapsto 36I - \mathbf{N}_m$, so the constant term of the weighted reader is $36(90 + 120) = 7560$. Moreover, $z_m(-\Delta) = e^{2m\Delta} z_m(\Delta)$ and $Z(-\Delta) = e^{7560\Delta} Z(\Delta)$. Substitution into the normalized state proves conjugacy. Its spectrum preserves entropy; the transformation of $\mathbf{N}$ gives the complementary area. Subtracting the logarithms of the two states and taking the expectation in $\rho_\partial(\Delta)$ gives the last equality. $\square$

At $\Delta = 0$, $s = 36 \log 3$. As $\Delta \rightarrow +\infty$, the state concentrates on zero occupation and entropy tends to zero; reversal takes this limit to maximum occupation. For $\Delta > 0$, entropy decreases strictly because the weighted reader has more than one eigenvalue and the state has full rank. The complete configuration, area, and entropy thus retain distinct roles.

59.4 Modular law and finite area variations

Fix $\rho_0 = \rho_\partial(\Delta_0)$, with real, finite Δ_0 , and hold $\Delta_0, \gamma_{\text{BI}}, a_0, \ell_P$ constant. The normalized areas $\mathcal{A}_m^n = \gamma_{\text{BI}} a_0 \mathbf{N}_m$ express the modular operator as

$$\mathbf{K}_0 = -\log \rho_0 = c_0 I + \sum_{m=90,120} \beta_m \mathcal{A}_m^n, \quad \beta_m = \frac{m\Delta_0}{\gamma_{\text{BI}} a_0}, \quad c_0 = \log Z(\Delta_0). \quad (59.25)$$

We have $\beta_{120} = \frac{4}{3}\beta_{90}$, including when both are zero. The ratio is $4/3$ when $\Delta_0 \neq 0$. $\mathbf{K}_0$ is the logarithm of this reference state; its type distinguishes it from temporal transport and the gravitational connection.

Theorem 59.5 (First modular law and finite correction). *For any normalized state σ on the same space, let $\Delta_\sigma \langle B \rangle = \text{Tr}[(\sigma - \rho_0)B]$. Then*

$$s(\sigma) - s(\rho_0) = \sum_{m=90,120} \beta_m \Delta_\sigma \langle \mathcal{A}_m^n \rangle - \mathcal{D}(\sigma \| \rho_0), \quad (59.26)$$

where $\mathcal{D}(\sigma \| \rho_0) = \text{Tr}[\sigma(\log \sigma - \log \rho_0)]$ is finite and nonnegative. If $\sigma(\varepsilon)$ is differentiable, normalized, and $\sigma(0) = \rho_0$, we obtain

$$\left. \frac{ds(\sigma(\varepsilon))}{d\varepsilon} \right|_0 = \sum_{m=90,120} \beta_m \left. \frac{d\langle \mathcal{A}_m^n \rangle_{\sigma(\varepsilon)}}{d\varepsilon} \right|_0. \quad (59.27)$$

Proof. By definition, $\mathcal{D}(\sigma \| \rho_0) = -s(\sigma) + \text{Tr}(\sigma \mathbf{K}_0)$, whereas $s(\rho_0) = \text{Tr}(\rho_0 \mathbf{K}_0)$. Subtraction and (59.25) prove the finite identity, since normalization cancels the term $c_0 I$.

For positivity, diagonalize σ with eigenvalues p_i and vectors u_i , and ρ_0 with eigenvalues $q_j > 0$ and vectors v_j . If $r_i = \langle u_i, \rho_0 u_i \rangle > 0$, concavity of the logarithm gives

$$\sum_j |\langle u_i, v_j \rangle|^2 \log q_j \leq \log r_i.$$

Consequently, $\mathcal{D}(\sigma \| \rho_0) \geq \sum_i p_i \log(p_i/r_i) \geq 0$. The last inequality is convexity of $t \log t$ with weights r_i , whose sum is one. Faithfulness of ρ_0 keeps its logarithms finite; zero eigenvalues of σ are interpreted through $0 \log 0 = 0$.

For the infinitesimal formula, near the faithful state ρ_0 , the derivative of the trace of $-\sigma \log \sigma$ is $-\text{Tr}[\dot{\sigma}(\log \rho_0 + I)]$. This identity follows from the spectral derivative of the trace. The condition $\text{Tr} \dot{\sigma} = 0$ leaves $\text{Tr}(\dot{\sigma} \mathbf{K}_0)$, and substituting (59.25) completes the proof. $\square$

For physical areas, the coefficients are β_m/ℓ_P^2 . Applying the common transducer k_B gives $S = k_B S$ and retains the full term $k_B \mathcal{D}$. The derivative in (59.17) varies Δ ; law (59.27) holds Δ_0 fixed and varies the state around the reference. These are different operations.

59.5 Action, decision energy, and the area unit

Fix a common oriented action section and its gravitational and thermal sections, on the positive branch. The antecedents provide

$$\begin{aligned} \ell_P^2 &= \frac{\hbar G}{c^3}, & t_P &= \frac{\ell_P}{c}, \\ T_{108} &= 2\pi t_P, & h &= 2\pi\hbar, \\ E_P &= \frac{\hbar}{t_P} = k_B \Theta_{\text{clk}} \log 3. \end{aligned} \quad (59.28)$$

The thermal equality retains the pair and normalization previously constructed; when their dimensional basis changes, k_B and Θ_{clk} are transported together. The elementary area increment is $\delta\mathcal{A} = \gamma_{\text{BI}} a_0 \ell_P^2$.

Proposition 59.6 (Conversion coefficient between decision and area). *The ratio of decision energy to the area increment satisfies*

$$\begin{aligned} \mathcal{T}_{I/A} &:= \frac{E_P}{\delta\mathcal{A}} = \frac{k_B \Theta_{\text{clk}} \log 3}{\gamma_{\text{BI}} a_0 \ell_P^2} \\ &= \frac{c^3}{\gamma_{\text{BI}} a_0 G t_P} = \frac{c^4}{\gamma_{\text{BI}} a_0 G \ell_P}. \end{aligned} \quad (59.29)$$

Its dimension is energy per area.

Proof. Substitution of $E_P = \hbar/t_P$ and $\ell_P^2 = \hbar G/c^3$ cancels the common action section. The identity $\ell_P = ct_P$ gives the final expression. Positivity of the factors ensures that every quotient is defined. $\square$

The capacity 81 of the cubic cut and the bipartition of the thirty-six sectorial links correspond to different domains. In the cubic graph $\{0, \dots, N-1\}^3$, the N^2 parallel lines between two opposite faces are edge-disjoint: every cut intersects all N^2 , and a transverse layer attains this bound. For $N = 9$, the uniform product state of the trits in that layer has $s_{\text{corte}} = 81 \log 3$. Assigning one decision of the same cycle to each link expresses

$$S_{\text{corte}}^{\text{fis}} = 81 k_B \log 3, \quad E_{\text{corte}}^{(1)} = 81 E_P = \Theta_{\text{clk}} S_{\text{corte}}^{\text{fis}}. \quad (59.30)$$

The proof combines the minimum cut, additivity of the product-state entropy, and (59.28). For the sectorial state, the relevant finite equation remains (59.26), with its relative entropy.

59.6 Cosmological normalization of the horizon

For an areal radius $R > 0$, the cosmological realization considered uses the triad

$$E_\Lambda(R) = \frac{c^4 R}{2G}, \quad S_H(R) = \frac{k_B \pi R^2}{\ell_P^2}, \quad T_{\text{cos}}(R) = \frac{\hbar c}{2\pi k_B R}. \quad (59.31)$$

The factor 2π belongs to this temperature normalization. The area $\mathcal{A}_H = 4\pi R^2$ expresses the same entropy as $S_H/k_B = \mathcal{A}_H/(4\ell_P^2)$.

Theorem 59.7 (Gravitational cell of half action). *In (59.31),*

$$T_{\text{cos}}(R) S_H(R) = E_\Lambda(R), \quad (59.32)$$

$$T_{\text{cos}}(\ell_P) S_H(\ell_P) = \frac{E_P}{2} = \frac{k_B \Theta_{\text{clk}} \log 3}{2}, \quad (59.33)$$

$$E_\Lambda(\ell_P) t_P = \frac{\hbar}{2}, \quad E_\Lambda(\ell_P) T_{108} = \frac{h}{2}. \quad (59.34)$$

Holding the chart constants fixed and varying R ,

$$T_{\cos} dS_H = 2 dE_\Lambda, \quad S_H dT_{\cos} = -dE_\Lambda. \quad (59.35)$$

Proof. Multiplying temperature and entropy gives $\hbar c R / (2\ell_P^2) = c^4 R / (2G)$. At $R = \ell_P$, the result is $\hbar / (2t_P) = E_P / 2$. The two durations in (59.28) give the action relations. For the differentials, $dE_\Lambda = c^4 dR / (2G)$, $dS_H = 2\pi k_B R dR / \ell_P^2$, and $dT_{\cos} = -T_{\cos} dR / R$. Substitution proves both identities. $\square$

The product $E_\Lambda = T_{\cos} S_H$ expresses a state identity. Its radial variations retain both terms of (59.35); a complete thermodynamic interpretation also specifies the corresponding temporal generator, work, and material realization. In this chart, action and gravitation fix the unit ℓ_P^2 , while Boltzmann converts information into dimensional entropy. Transport of sections with fixed $G\hbar$ preserves that unit when the section c is held fixed, and preserves S_H / k_B at fixed radius.

In the cell $R = \ell_P$, $S_H / k_B = \pi$, so the effective multiplicity is $\Omega_{\text{eff}} = e^\pi$. For the tritic hyperbolic involution H_{trit} , with spectrum $\{1, -1\}$, diagonalization gives $\text{spec}(e^{\pi H_{\text{trit}}}) = \{e^\pi, e^{-\pi}\}$. This spectral relation specifies the expanding value; a microscopic identification of both spaces additionally retains their realization map and state.

59.7 Kruskal chart and exchange of exterior regions

Fix a expressed state x , its radius $R = R(x) > 0$, and the positive sections c, G . In the following chart, x, R, c, G are fixed parameters; t, r are the spacetime coordinates. The exterior realization of the two sheets is specified by

$$f(r) = 1 - \frac{R}{r}, \quad ds^2 = -f(r)c^2 dt^2 + f(r)^{-1} dr^2 + r^2 d\Omega_2^2, \quad r > R. \quad (59.36)$$

For the sheet reader $\sigma \in \{+1, -1\}$, define

$$r_* = r + R \log\left(\frac{r}{R} - 1\right), \quad (59.37)$$

$$\mathcal{U} = -\sigma \exp\left(-\frac{ct - r_*}{2R}\right), \quad \mathcal{V} = \sigma \exp\left(\frac{ct + r_*}{2R}\right). \quad (59.38)$$

The symbols $\mathcal{U}, \mathcal{V}$ are coordinates and are distinguished from path-transport operators.

Theorem 59.8 (Geometric realization and sheet involution). *The chart (59.38) realizes the two sheets as the two exterior regions $\mathcal{U}\mathcal{V} < 0$ of the Kruskal extension. We have*

$$-\mathcal{U}\mathcal{V} = \left(\frac{r}{R} - 1\right) e^{r/R}, \quad ds^2 = -\frac{4R^3}{r} e^{-r/R} d\mathcal{U} d\mathcal{V} + r^2 d\Omega_2^2. \quad (59.39)$$

The involution

$$\mathcal{C}_H : (\sigma, \mathcal{U}, \mathcal{V}) \mapsto (-\sigma, -\mathcal{U}, -\mathcal{V}) \quad (59.40)$$

is isometric, fixes the areal radius, exchanges the exterior regions, and preserves the set of horizons $\mathcal{U}\mathcal{V} = 0$. The metric is regular at $r = R$ and satisfies the vacuum equations on the regular domain $r > 0$.

Proof. Multiplying the coordinates gives the first identity. Since $dr_*/dr = f^{-1}$,

$$d\mathcal{U} = \frac{\mathcal{U}}{2R} (-c dt + f^{-1} dr), \quad d\mathcal{V} = \frac{\mathcal{V}}{2R} (c dt + f^{-1} dr).$$

Their product reproduces the radial and temporal part of the metric. The function $z \mapsto (z - 1)e^z$ has derivative $ze^z > 0$ for $z > 0$, which determines r from the product. In the exterior one recovers $\sigma = \text{sign } \mathcal{V}$ and $t = (R/c) \log(-\mathcal{V}/\mathcal{U})$. At $r = R$, the radial coefficient of the extended metric is $-4R^2/e$, finite and nonzero. Reversal preserves the product, radius, and $d\mathcal{U}d\mathcal{V}$, and exchanges the two sign choices.

To check the vacuum equations in the exterior chart, the independent mixed components of the Einstein tensor reduce to

$$G^t_t = G^r_r = \frac{rf' + f - 1}{r^2}, \quad G^\theta_\theta = G^\phi_\phi = \frac{f'}{r} + \frac{f''}{2}.$$

With $f' = R/r^2$ and $f'' = -2R/r^3$, all vanish. The regular expression (59.39) extends the result across the horizons on the domain $r > 0$. $\square$

This realization preserves $E_H = c^4 R/(2G)$ and $S_H = k_B \pi R^2/\ell_P^2$, which depend on the areal radius. The temperature T_{cos} in (59.31) belongs to the cosmological realization declared there. The exterior chart (59.36) retains its own temporal normalization. The geometric map establishes the exchange of exterior regions; comparison with collapse formation, radiation spectra, or quasinormal modes also uses their respective physical conditions.

59.8 Transport of states, observables, and refinements

Let F_x denote the chart of the two exterior regions just constructed. In each regular angular chart, its temporal and radial Jacobian is

$$\left| \frac{\partial(\mathcal{U}, \mathcal{V})}{\partial(t, r)} \right| = \frac{cr}{2R^3} e^{r/R} > 0. \quad (59.41)$$

The identity follows from the two differentials in the preceding proof. The state's remaining registers are transported as fibre coordinates; the chart preserves their identity and memory.

Proposition 59.9 (Unitary equivalence of sheet and exterior readouts). *Let μ be a sheet measure and $\nu = (F_x)_* \mu$ its transported measure. The change of variables defines a unitary*

$$W_H : L^2(\mu) \longrightarrow L^2(\nu), \quad (W_H \psi)(z) = \psi(F_x^{-1} z). \quad (59.42)$$

With densities relative to fixed coordinate measures, the same map incorporates the corresponding Jacobian factor. The conjugation

$$\mathfrak{S}_H(B) = W_H B W_H^* \quad (59.43)$$

is a unital completely positive $$ -isomorphism. It transports the state, its expectations, and its modular functional calculus. For the symmetric measure on the two sheets, it also intertwines their involutions.*

Proof. The definition of pushforward measure gives $\int |\psi \circ F_x^{-1}|^2 d\nu = \int |\psi|^2 d\mu$. The chart is invertible on the exterior regions, so the isometry is surjective. Unitary conjugation preserves products, adjoints, and the identity; at each matrix amplification it conjugates by $I_n \otimes W_H$, proving complete positivity. For a state ρ , cyclicity of the trace gives $\text{Tr}(W_H \rho W_H^* \mathfrak{S}_H(B)) = \text{Tr}(\rho B)$. Functional calculus implies $g(W_H \rho W_H^*) = W_H g(\rho) W_H^*$ on its domain, in particular for the logarithm and modular flow of a state faithful on its support. The equality $F_x \circ \mathcal{C}_{\text{hoja}} = \mathcal{C}_H \circ F_x$ proves the intertwining of involutions; the symmetric measure realizes them unitarily. $\square$

If refinements preserve (σ, t, r, R) and use the same isometry on the fibre registers, the inclusions j_m^{hoja} and j_m^{ext} satisfy

$$W_{H,m+1} j_m^{\text{hoja}} = j_m^{\text{ext}} W_{H,m}. \quad (59.44)$$

Indeed, the change of variables acts on the preserved geometric coordinates and the inclusion on the same fibre coordinates; evaluating both compositions gives the same function and the same transported density. The equality extends the map to the refined system without resetting the registers.

For a bipartition $\mathcal{H}_A \otimes \mathcal{H}_B$, a factorized transport $U_A \otimes U_B$ satisfies

$$\rho'_A = U_A \rho_A U_A^*, \quad s(\rho'_A) = s(\rho_A).$$

The first identity follows from the partial trace and the second from the spectrum. For a general unitary, the equivalent formulation also transports the subsystem algebra: $\mathcal{B}(\mathcal{H}_A) \otimes I \mapsto U(\mathcal{B}(\mathcal{H}_A) \otimes I)U^*$. This preserves the same operational question about accessible information. The entropy equalities apply to states with the relevant finite entropies.

The relation between information and gravitation is expressed through distinct, composable operations: the state and bipartition fix accessible information; incidence and Barbero–Immirzi fix area resolution; action and gravitation construct the unit ℓ_P^2 ; Boltzmann realizes entropy dimensionally; the horizon chart transports geometry and its observables. The finite laws retain relative entropy, and the geometric realizations retain their measures, temporal normalization, and domains.

60 Observer, memory, and relational response

The reading operation acts on the transport expressed by the enriched APP–TRIT–TPK state. Its domain retains route, sheet, orientation, carry, and memory; the result accessible to the observer is a map from that domain. This distinction makes it possible to describe three operations jointly: the choice of an oriented representative of the transport, the inscription of its register in an apparatus, and the local response induced by the global boundary. Each operation conveys different information. The following construction specifies their maps and the conditions under which they can be composed.

60.1 Boundary transport and intrinsic readout

The finite representation of the transport provides an oriented cycle C_n , with $n \geq 1$, vertices $0, \dots, n-1$, and edges $e_j : j \rightarrow j+1 \pmod{n}$. We fix the lattice realization L of the register, its vector space $W = L \otimes_{\mathbb{Z}} \mathbb{R}$, a positive inner product, and the nonzero primitive representative $v \in L$ of the distinguished residual class, in the declared chamber. The sum and product sheets of APP provide the polarizations of the transport; the orientation of the TRIT and the TPK update precede this linear representation. The cycle, the lattice realization, and the residual class are the data that this section receives from the transport constructed above.

A 0-cochain is a function on the vertices with values in W ; a 1-cochain assigns a vector to each oriented edge. The coboundary acts by

$$(d\Phi)(e_j) = \Phi(j+1) - \Phi(j).$$

We denote by $\mathbf{1}_E$ the scalar cochain that is constant on the edges. For a transport $\omega \in C^1(C_n; W)$, its accumulated representation Γ_ω is obtained by partial sums and affine interpolation; its closure defect is

$$B(\omega) = \Gamma_\omega(1) - \Gamma_\omega(0) = \sum_{j=0}^{n-1} \omega(e_j). \quad (60.1)$$

Theorem 60.1 (Exact decomposition and harmonic mode of the cycle). *Every transport has a unique decomposition*

$$\omega = d\Phi + u\mathbf{1}_E, \quad \Phi(0) = 0, \quad u = \frac{1}{n} \sum_j \omega(e_j).$$

In particular, $B(d\Phi) = 0$ and $B(\omega) = nu$.

Proof. Let $y_j = \omega(e_j) - u$. The sum of the y_j is zero. Define $\Phi(k) = \sum_{j=0}^{k-1} y_j$; the value for $k = n$ coincides with $\Phi(0) = 0$, so the function is well defined on the cycle, including its final edge. We have $d\Phi(e_j) = y_j$. If $d\Phi + u\mathbf{1}_E = d\Psi + u\mathbf{1}_E$, summing over edges gives $nu = nw$. Then $d(\Phi - \Psi) = 0$; the connectivity of the cycle and the normalization at the initial vertex imply $\Phi = \Psi$. The identity $\sum_j (\Phi(j+1) - \Phi(j)) = 0$ directly proves the telescopic cancellation. $\square$

Definition 60.2 (Admissible transport and normalized readout). The transport is admissible when its harmonic mode belongs to $\mathbb{R}v$. On the space $\mathcal{T}_{\text{adm}}$ of such transports we define the global readout and its density per phase:

$$\mathbf{a}_v(\omega) = \frac{\langle B(\omega), v \rangle}{\langle v, v \rangle}, \quad \mu_v(\omega) = \frac{\mathbf{a}_v(\omega)}{n}. \quad (60.2)$$

We have $B(\omega) = \mathbf{a}_v(\omega)v$.

Theorem 60.3 (Determination of the intrinsic readout). *The quotient $\mathcal{T}_{\text{adm}}/dC^0(C_n; W)$ is one-dimensional, generated by the class of $v\mathbf{1}_E$. The map μ_v is the unique linear functional on $\mathcal{T}_{\text{adm}}$ satisfying*

$$F(\omega + d\Phi) = F(\omega), \quad F(v\mathbf{1}_E) = 1.$$

Proof. The preceding decomposition writes each admissible transport as $d\Phi + \lambda v\mathbf{1}_E$. Its class modulo exact cochains is determined by λ . The class of $v\mathbf{1}_E$ is nonzero because its sum is $nv \neq 0$. By linearity and invariance, every functional with the stated properties has value $F(\omega) = \lambda F(v\mathbf{1}_E) = \lambda$. Formula (60.2) gives precisely $\mu_v = \lambda$. The global readout is $\mathfrak{a}_v = n\mu_v$. $\square$

Definition 60.4 (Boundary observer). With the polarization of the expressed transport fixed, a boundary observer is a pair $\mathcal{O} = (\varepsilon, \Psi)$, where $\varepsilon \in \{+1, -1\}$ and $\Psi \in C^0(C_n; W)$. Its reading action is

$$\omega^{\mathcal{O}} = \varepsilon\omega + d\Psi. \quad (60.3)$$

The sign fixes the orientation and the cochain fixes the trivialization. Two choices of polarization are compared through their expressed transport; relation (60.3) describes exactly those representatives that differ by orientation and coboundary.

Proposition 60.5 (Covariance of the readout). *The action preserves admissibility and satisfies*

$$B(\omega^{\mathcal{O}}) = \varepsilon B(\omega), \quad \mathfrak{a}_v(\omega^{\mathcal{O}}) = \varepsilon \mathfrak{a}_v(\omega), \quad \mu_v(\omega^{\mathcal{O}}) = \varepsilon \mu_v(\omega).$$

Trivializations with the same orientation express the same closure.

Proof. The sum of the coboundary $d\Psi$ is zero. Therefore, the harmonic mode is transformed into εu , which still belongs to $\mathbb{R}v$, and the defect is transformed into εB . The two scalar equalities follow from the normalized projection. $\square$

The unit of this readout is the residual representative v , with its internal normalization. The freedom of trivialization acts on the cochain representative; the closure information resides in its class. The description makes it possible to compare observers while keeping the genealogical transport, its linear representation, and the scalar that evaluates the residual class separate.

60.2 Boundary identity and cellular interior

Let K be a finite oriented two-dimensional cell complex, equipped with a fundamental 2-chain c_K such that $\partial c_K = [C_n]$. This condition expresses that the interior incidences cancel with their multiplicities and orientations. Let $\Omega \in C^1(K; W)$ be an extension of ω , and $F_{\Omega} = d\Omega$ its additive discrete curvature.

Theorem 60.6 (Cellular holographic identity of the readout). *With the preceding data,*

$$B(\omega) = \langle \Omega, \partial c_K \rangle = \langle d\Omega, c_K \rangle, \quad \mathfrak{a}_v(\omega) = \frac{\langle \langle d\Omega, c_K \rangle, v \rangle}{\langle v, v \rangle}. \quad (60.4)$$

In particular, $F_{\Omega} = 0$ implies $\mathfrak{a}_v(\omega) = 0$. If the closure is nonzero, every extension of the transport on that complex has nonzero discrete curvature.

Proof. The pairing between chains and cochains satisfies $\langle d\Omega, c_K \rangle = \langle \Omega, \partial c_K \rangle$. On expansion, each interior edge appears with the sum of the incidence numbers of the adjacent cells; the condition on ∂c_K leaves exactly the oriented boundary cycle. The restriction of Ω to that cycle is ω , whose sum is $B(\omega)$. Projection onto v gives the second formula and the flatness implication. $\square$

The equality compares two evaluations of the same extended cochain: the accumulated boundary defect and the coboundary integrated over the interior. Its curvature takes values in W and uses the additive cellular coboundary. Realization by a geometric connection additionally preserves its own coefficient space and transport law.

60.3 Isometric inscription of the register in the apparatus

Let $\mathfrak{M}_n$ be a nonempty finite set of registers expressed at level n , with result, route, carry, sheet, and memory. Its register space is $\mathcal{H}_n = \ell^2(\mathfrak{M}_n)$, with basis $|m\rangle$. The visible update F_n has a register lift

$$\widehat{F}_n(m) = (F_n(m), m).$$

As registers of the next level we take a set containing these pairs. The second coordinate retains the antecedent and makes the lift injective. This construction specifies the memory retained by the inscription, including when the visible update merges values.

In a finite apparatus space $\mathcal{K}_{A,n}$, we fix an orthonormal collection $(|a_m\rangle)_{m \in \mathfrak{M}_n}$ and define

$$\iota_n : \mathcal{H}_n \longrightarrow \mathcal{K}_{A,n}, \quad \iota_n|m\rangle = |a_m\rangle, \quad V_n : \mathcal{H}_n \longrightarrow \mathcal{H}_{n+1}, \quad V_n|m\rangle = |\widehat{F}_n(m)\rangle.$$

Let $\mathcal{P}_n = \iota_n \mathcal{H}_n$ be the pointer subspace.

Theorem 60.7 (Compatibility between updating and inscription). *The maps ι_n and V_n are isometries. There is a unique isometry $W_n^0 : \mathcal{P}_n \longrightarrow \mathcal{K}_{A,n+1}$ satisfying*

$$W_n^0 \iota_n = \iota_{n+1} V_n. \quad (60.5)$$

If $\dim \mathcal{K}_{A,n+1} \geq \dim \mathcal{K}_{A,n}$, it extends to an isometry on all of $\mathcal{K}_{A,n}$. Using auxiliary complements that equalize the dimensions gives a unitary extension between the enlarged spaces.

Proof. The images of the bases under ι_n are orthonormal; the images under V_n are also orthonormal by the injectivity of $\widehat{F}_n$. For $z = \iota_n \psi$ define $W_n^0 z = \iota_{n+1} V_n \psi$. The representation of z is unique, and the three maps preserve its norm. The formula determines the map on a basis of $\mathcal{P}_n$ and proves the square. If the target dimension is sufficient, the orthogonal complement of $W_n^0 \mathcal{P}_n$ contains an orthonormal collection of the size of $\mathcal{P}_n^\perp$; sending a basis of the latter to that collection constructs the extension. Finally, both spaces are enlarged to the same dimension and the two prescribed collections are completed to orthonormal bases. The map between those bases is unitary. $\square$

When the register includes labels of a finite instrument, this same inscription realizes its environment: if the operators $K_{y,a} : \mathcal{H}_S \rightarrow \mathcal{H}'_S$ satisfy $\sum_{y,a} K_{y,a}^* K_{y,a} = I$, the map

$$V_{\text{inst}} \psi = \sum_{y,a} K_{y,a} \psi \otimes |y, a\rangle$$

is isometric, since its squared norm is $\sum_{y,a} \langle \psi, K_{y,a}^* K_{y,a} \psi \rangle = \|\psi\|^2$. The inscription ι transports each label to its corresponding pointer. The model thus fixes the premeasurement register; a material device additionally requires its interaction and calibration.

60.4 Outcome map and record conditioning

The preceding inscription composes with the apparatus readout on the complete genealogical state. Coupling retains the operation that produces the correlation; the outcome map expresses the result; conditioning restricts the extensions compatible with that record.

Let (Ω_S, Σ_S) and (Ω_M, Σ_M) be the measurable spaces of genealogical states of the system and apparatus, obtained from their compatible cylinders and records. A measurement context is written $\mathfrak{M}_a = (U_a, q_a, \mathcal{C}_a, m_0)$: m_0 is the apparatus preparation, $\mathcal{C}_a$ specifies compatibilities and the extension horizon, and

$$U_a : \Omega_S \times \Omega_M \longrightarrow \Omega_S \times \Omega_M, \quad q_a : \Omega_M \longrightarrow Y_a$$

are, respectively, the measurable coupling and the measurable readout into the outcome space (Y_a, Σ_{Y_a}) . In a reversible regime, U_a is bijective and bimeasurable; the corresponding unitarity is also required in the Hilbert-space realization. The individual outcome is the composition

$$\text{Out}_a(x) = q_a(\text{pr}_M U_a(x, m_0)). \quad (60.6)$$

It is determined by the complete state, apparatus preparation, and context. The statistical description of a preparation is another readout of the set of compatible states.

Proposition 60.8 (Outcome measure and restriction to a fibre). *Let $\Omega_a \in \Sigma_S$ be the set of states compatible with the preparation and context, endowed with a probability measure μ_a on $(\Omega_a, \Sigma_S|_{\Omega_a})$. The map $\text{Out}_a|_{\Omega_a}$ is measurable and induces the outcome probability measure*

$$\nu_a(B) = \mu_a(\text{Out}_a^{-1}(B) \cap \Omega_a), \quad B \in \Sigma_{Y_a}. \quad (60.7)$$

For an outcome y whose singleton is measurable, the fibre $\Omega_{a,y} = \{x \in \Omega_a : \text{Out}_a(x) = y\}$ is measurable. Whenever $\mu_a(\Omega_{a,y}) > 0$,

$$\mu_{a,y}(E) = \frac{\mu_a(E \cap \Omega_{a,y})}{\mu_a(\Omega_{a,y})} \quad (60.8)$$

for $E \in \Sigma_S|_{\Omega_a}$ defines a probability measure concentrated on that fibre.

Proof. The inclusion $x \mapsto (x, m_0)$, coupling, apparatus projection, and readout are measurable; their composition is therefore measurable. Preimages preserve disjoint unions and the whole space, so (60.7) is countably additive with total mass one. The fibre is the preimage of the singleton $\{y\}$ within Ω_a . Restricting μ_a to that fibre gives a positive, countably additive measure; division by its positive mass normalizes it. $\square$

For a prefix s of depth n , the extensions following the record form

$$\text{Fut}_{a,y}(s) = \{x \in \Omega_{a,y} : x|_n = s\}.$$

Restriction preserves the prefix already traversed and the coupling record, while changing the collection of admissible extensions. Outcome (60.6) acts on complete states; the outcome law (60.7) is obtained by pushing the preparation measure forward through that map. A density-matrix representation expresses the statistical level of the quantum realization and retains its conditions of correspondence with genealogical fibres.

60.5 Accessible algebra and pointer complement

Define $P_m = |a_m\rangle\langle a_m|$, $P = \sum_m P_m$, and $P_\perp = I - P$. The algebra of the pointer subspace is $P\mathcal{B}(\mathcal{K}_{A,n})P$, whose unit is P . On it, the diagonal reader is

$$\mathcal{D}_P(A) = \sum_m P_m A P_m.$$

Proposition 60.9 (Register expectation and unital extension). *The map $\mathcal{D}_P$ is a completely positive, trace-preserving, idempotent conditional expectation on the pointer algebra, with $\mathcal{D}_P(P) = P$. Its image is $\bigoplus_m \mathbb{C}P_m$. On the algebra of the full apparatus, the map*

$$\mathcal{E}_P(A) = \sum_m P_m A P_m + P_\perp A P_\perp \quad (60.9)$$

is a unital, completely positive, trace-preserving conditional expectation. Its image is

$$\left(\bigoplus_m \mathbb{C}P_m \right) \oplus \mathcal{B}(\mathcal{P}_n^\perp).$$

Both maps are orthogonal projections for the Hilbert–Schmidt inner product. Their spectra are contained in $\{0, 1\}$. When eliminated components exist, the separation between the fixed subspace and the kernel is one.

Proof. The projectors P_m are mutually orthogonal. In the full apparatus, the collection obtained by adding $P_\perp$ sums to I . Each summand $A \mapsto P_m A P_m$ is completely positive: after any matrix amplification it acts by congruence with $I \otimes P_m$. The same holds for $P_\perp$. The sum retains that property. The cyclicity of the trace and the sum of the projectors prove trace preservation on each domain. Orthogonality

annihilates the cross terms upon iteration of the map, so it is idempotent. Its fixed elements are precisely the stated blocks, and the map is bimodular with respect to that subalgebra. Moreover, $\text{Tr}(A^*P_mBP_m) = \text{Tr}((P_mAP_m)^*B)$, which gives self-adjointness for Hilbert–Schmidt. A self-adjoint projection has only the spectral values zero and one when both subspaces are nonzero. If only the fixed subspace remains, the map is the identity. $\square$

The sum $\sum_m P_mAP_m$, applied without the complement to the full apparatus, sends I to P . Its natural unit belongs to the pointer algebra. Formula (60.9) also retains the auxiliary block; an orthonormal resolution of that block provides, when desired, diagonalization of the entire apparatus.

If U is a unitary transformation of the apparatus and $P'_m = UP_mU^*$, then

$$\mathcal{E}_{P'}(UAU^*) = U\mathcal{E}_P(A)U^*.$$

The equality follows by substituting each projector in the sum. Likewise, for $A \in \mathcal{B}(\mathcal{P}_n)$, the square (60.5) implies

$$\mathcal{D}_{P_{n+1}}(W_n^0 A (W_n^0)^*) = W_n^0 \mathcal{D}_{P_n}(A) (W_n^0)^*.$$

Indeed, each matrix unit $|a_m\rangle\langle a_{m'}|$ is transported to the unit corresponding to the distinct registers $\hat{F}_n(m), \hat{F}_n(m')$; the diagonal reader retains exactly the terms with $m = m'$.

60.6 Factorization of the response through the effective boundary

Let $\mathcal{X}_\infty^{\text{enr}}$ be the domain of enriched compatible states, and let $b : \mathcal{X}_\infty^{\text{enr}} \rightarrow B_\infty$ be its global boundary trace. We denote by $B_{\text{ef}} = \text{im } b$ the effectively expressed boundary. For a response space $\mathcal{V}_v$, consider $I_v : \mathcal{X}_\infty^{\text{enr}} \rightarrow \mathcal{V}_v$.

Theorem 60.10 (Criterion for relational response). *There is a unique map $\bar{I}_v : B_{\text{ef}} \rightarrow \mathcal{V}_v$ such that $I_v = \bar{I}_v \circ b$ if and only if*

$$b(x) = b(x') \implies I_v(x) = I_v(x'). \quad (60.10)$$

Uniqueness extends to all of B_∞ when b is surjective.

Proof. A composition through b is constant on each fibre. Conversely, for $\beta \in B_{\text{ef}}$, define $\bar{I}_v(\beta) = I_v(x)$ with $b(x) = \beta$. Condition (60.10) guarantees independence of the representative; every value of $\bar{I}_v$ is determined by some state in that fibre. If b is surjective, $B_{\text{ef}} = B_\infty$. Otherwise, the composition equation determines only the values on B_{ef} . $\square$

This criterion is set-theoretic. For a continuous factorization, one additionally retains the topologies and the quotient-map condition on $b : \mathcal{X}_\infty^{\text{enr}} \rightarrow B_{\text{ef}}$; under this condition, continuity of I_v is equivalent to continuity of its factor. This clarification separates the determination of a readout from its regularity properties.

Let $\ell_v : \mathcal{X}_\infty^{\text{enr}} \rightarrow \mathcal{L}_v$ be the restriction accessible to a local neighbourhood. Effective global dependence is verified by two states satisfying

$$\ell_v(x) = \ell_v(x'), \quad I_v(x) \neq I_v(x'). \quad (60.11)$$

If I_v factors through b , these two states have distinct boundaries. Moreover, the response cannot be written as a function of ℓ_v , since that function would assign the same value to both states. The difference between boundaries thus acquires a verifiable operator effect.

60.7 Ambivalence weights and self-inertial response

For the general formula we consider positive readers A_∂, V_∂ and an energy reader Q defined on the declared effective boundary domain. We next make explicit a realization on $B_{\text{av}} \subseteq B_{\text{ef}}$. Memory remains as a coordinate $M_\partial(\beta)$ of the boundary. The ambivalence hinge provides the two orientations

$$r_{\text{av}} = \frac{\pi_{\text{HMT}}}{\varphi_{\text{HMT}}^2}, \quad r_{\text{av}}^J = \frac{\varphi_{\text{HMT}}}{\pi_{\text{HMT}}^2}.$$

The superscript J expresses interchange of the generating pair.

Normalized realization of the boundary readers. The APP-TRIT-TPK modal incidence, developed in the regional construction, provides the path envelope with matrix

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

The positive vector $v = (1, \varphi_{\text{HMT}})^\top$ satisfies $F_{\text{av}}v = \varphi_{\text{HMT}}v$. Its normalized transition and stationary weights are

$$P_{ij} = \frac{(F_{\text{av}})_{ij}v_j}{\varphi_{\text{HMT}}v_i}, \quad P = \begin{pmatrix} 0 & 1 \\ \varphi_{\text{HMT}}^{-2} & \varphi_{\text{HMT}}^{-1} \end{pmatrix}, \quad (\mu_{\text{ar}}, \mu_{\text{vol}}) = \frac{(1, \varphi_{\text{HMT}}^2)}{1 + \varphi_{\text{HMT}}^2}. \quad (60.12)$$

The rows of P sum to one because $\varphi^{-2} + \varphi^{-1} = 1$; the stationary equation $\mu_{\text{ar}} = \mu_{\text{vol}}\varphi^{-2}$, together with normalization, determines the stated weights. This Parry measure weights the paths of the envelope; the chronological frequency (1/3, 2/3) counts the instants of the modal calendar and retains its distinct function.

The common-memory-amplitude realization is defined on the sector $\mathcal{X}_{\text{av}} \subset \mathcal{X}_\infty^{\text{enr}}$ in which the readers of the two sheets have the factorization

$$\mathcal{L}_{\text{vol}}(x) = \mu_{\text{vol}}\mathcal{M}(x), \quad \mathcal{L}_{\text{ar}}(x) = \pi_{\text{HMT}}\mu_{\text{ar}}\mathcal{M}(x), \quad \mathcal{M}(x) > 0. \quad (60.13)$$

Both values belong to the same positive memory-amplitude line. The areal representation carries the regional mark π_{HMT} of the return. The dimensionless readout is obtained by dividing both sections by $\mathcal{L}_{\text{vol}}(x)$. In this way, sums of weights are taken between quantities of the same type; a possible realization as physical areas and volumes is composed afterwards with their respective dimensional transducers.

Proposition 60.11 (Descent of the normalized readers and the energy clock). *On $B_{\text{av}} = b(\mathcal{X}_{\text{av}})$, the rules*

$$A_\partial(b(x)) = \frac{\mathcal{L}_{\text{ar}}(x)}{\mathcal{L}_{\text{vol}}(x)} = r_{\text{av}}, \quad V_\partial(b(x)) = 1 \quad (60.14)$$

define two positive readers independent of the representative. With an orientation a of the action section fixed, the decision energy of (45.6) determines on states

$$q_a(x) = \frac{h_a(x)}{108t_0(x)} = \frac{\hbar_a(x)}{t_P(x)} \in \mathcal{L}_E^+, \quad t_P = \frac{54}{\pi_{\text{HMT}}}t_0. \quad (60.15)$$

We write

$$b_{\text{av}} := b|_{\mathcal{X}_{\text{av}}} : \mathcal{X}_{\text{av}} \longrightarrow B_{\text{av}}.$$

There is a unique reader $Q : B_{\text{av}} \rightarrow \mathcal{L}_E^+$ such that $q_a|_{\mathcal{X}_{\text{av}}} = Q \circ b_{\text{av}}$ if and only if, for any $x, x' \in \mathcal{X}_{\text{av}}$,

$$b(x) = b(x') \implies \frac{h_a(x)}{t_0(x)} = \frac{h_a(x')}{t_0(x')}. \quad (60.16)$$

In particular, a chart with fixed action and duration satisfies this condition and expresses $Q = h_a/(108t_0)$.

Proof. The quotient of (60.13) cancels the common amplitude and equals $\pi_{\text{HMT}}\mu_{\text{ar}}/\mu_{\text{vol}} = \pi_{\text{HMT}}/\varphi_{\text{HMT}}^2$. The result is the same for all representatives of a fibre of b_{av} within $\mathcal{X}_{\text{av}}$, and it is positive. Dividing both readers by the same positive factor preserves the weighted quotients in which they occur. For the clock, the factorization criterion of Theorem 60.10, applied to q_a , is exactly equivalent to (60.16). Fixed sections satisfy it; so do variable sections whose quotient descends through b_{av} . Uniqueness follows from the definition of B_{av} as the effective image. $\square$

The value of the reader Q belongs to the energy line $\mathcal{L}_E = \mathcal{L}_{\text{act}} \otimes \mathcal{L}_T^*$. With E_1, E_2 in that same line, $E_1 E_2 / Q^2$ is dimensionless and independent of the positive energy basis. The symbol Q in this section denotes exclusively that clock energy; the heat quantity in the Landauer realization belongs to the corresponding thermal balance. The common-amplitude factorization specifies a concrete realization of the general readers. Every other domain retains its boundary maps and its own verification of descent. The full memory remains in the enriched history and in its boundary representation, even though it cancels in the normalized quotient.

Define

$$w_A(\beta) = \frac{A_{\partial}(\beta)r_{\text{av}}}{A_{\partial}(\beta)r_{\text{av}} + V_{\partial}(\beta)r_{\text{av}}^J}, \quad w_V(\beta) = 1 - w_A(\beta).$$

In realization (60.14), the weights admit the exact evaluation

$$w_A = \frac{r_{\text{av}}^2}{r_{\text{av}}^2 + r_{\text{av}}^J} = \frac{\pi_{\text{HMT}}^4}{\pi_{\text{HMT}}^4 + \varphi_{\text{HMT}}^5}, \quad w_V = \frac{\varphi_{\text{HMT}}^5}{\pi_{\text{HMT}}^4 + \varphi_{\text{HMT}}^5}. \quad (60.17)$$

The substitution retains both factors of the definition: the normalized areal reader and its subsequent weighting by r_{av} . The generation and regional marking of the return are performed once. Multiplying numerator and denominator by $\varphi_{\text{HMT}}^4 \pi_{\text{HMT}}^2$ gives the final expression. These weights remain constant in the common-amplitude sector. With fixed projectors and energies, a variation in the response may arise from the reader Q ; the witness of global dependence additionally requires states with the same local restriction and different responses as in (60.11).

Let P_A, P_V be complementary orthogonal projections on a response space $\mathcal{H}_v$, and let $E_1, E_2 > 0$ be two internal energies fixed for the comparison. The response is

$$\bar{I}_v(\beta) = \frac{E_1 E_2}{Q(\beta)^2} (w_A(\beta)P_A + w_V(\beta)P_V), \quad I_v = \bar{I}_v \circ b. \quad (60.18)$$

Energies that are explicit readers of β are also admitted; in that case they are retained as such in the formula.

Proposition 60.12 (Positivity and covariance of the response). *Response (60.18) is self-adjoint, positive definite, and constant on the fibres of b . Its eigenvalues on the nonzero sectors are*

$$\lambda_A = E_1 E_2 Q^{-2} w_A, \quad \lambda_V = E_1 E_2 Q^{-2} w_V.$$

The full interchange $(A_{\partial}, r_{\text{av}}, P_A) \leftrightarrow (V_{\partial}, r_{\text{av}}^J, P_V)$ permutes the two summands and preserves the operator. A unitary transport of the response space conjugates that operator and preserves its spectrum with multiplicities.

Proof. Positivity of the readers gives $0 < w_A, w_V < 1$. For $z = z_A + z_V$, with $z_A = P_A z$, $z_V = P_V z$,

$$\langle z, \bar{I}_v z \rangle = \frac{E_1 E_2}{Q^2} (w_A \|z_A\|^2 + w_V \|z_V\|^2) > 0$$

if $z \neq 0$. The orthogonal decomposition proves the spectral expressions. Exclusive dependence on β gives constancy on fibres and the factorization. Interchanging the triples simultaneously interchanges numerators, weights, and projectors, thereby preserving the sum. For a unitary transport U , the substitution $P_A \mapsto U P_A U^*$, $P_V \mapsto U P_V U^*$ produces $U \bar{I}_v U^*$, with the same spectrum. $\square$

An interchange implemented within the same space by a unitary sending P_A to P_V requires both sectors to have equal dimension. Interchanging the two triples as labels in the formula is valid with the respective dimensions of each sector.

Condition (60.11) requires comparison of the responses. For example, simultaneously multiplying A_∂, V_∂ by the same positive factor preserves both weights; if E_1, E_2, Q are also preserved, the response remains identical. By contrast, with equal weights and energies, replacing Q by qQ , $q > 0$, multiplies the response by q^{-2} . If two states with the same local restriction realize that change with $q \neq 1$, they provide the global witness. These are operator separation conditions; the existence of the states is verified in the boundary domain being realized.

60.8 Connection of the joint state and expressed acceleration

In a differentiable realization of the joint state $\Gamma = (x, \mathcal{O}, m)$, let N be its realization space, ∇^{tot} its connection, and $\pi_S : N \rightarrow N_S$ a differentiable projection onto the subsystem, equipped with connection ∇^S . The connection and projection retain the provenance of the transports and readers they realize. The observed frame is obtained through

$$\mathfrak{F}_\mathcal{O}(\Gamma) = \text{Res}_\mathcal{O}(\Gamma, \nabla^{\text{tot}}).$$

For a twice-differentiable trajectory $\Gamma(t)$, write $\gamma_S = \pi_S \circ \Gamma$.

Definition 60.13 (Self-inertia and projection defect). A trajectory is self-inertial with respect to the declared connection when $\nabla_\Gamma^{\text{tot}} \dot{\Gamma} = 0$. The defect of its projection is

$$\mathcal{K}_{\pi_S}(\dot{\Gamma}, \dot{\Gamma}) = \nabla_{\dot{\gamma}_S}^S \dot{\gamma}_S - d\pi_S(\nabla_\Gamma^{\text{tot}} \dot{\Gamma}). \quad (60.19)$$

Proposition 60.14 (Subsystem acceleration). *For a self-inertial trajectory,*

$$a_S = \nabla_{\dot{\gamma}_S}^S \dot{\gamma}_S = \mathcal{K}_{\pi_S}(\dot{\Gamma}, \dot{\Gamma}).$$

In coordinates x^B on N and y^a on N_S , the defect is quadratic in velocity, with components

$$(\mathcal{K}_{\pi_S})^a_{BC} = \partial_B \partial_C \pi_S^a + (\Gamma^S)^a_{bc} \partial_B \pi_S^b \partial_C \pi_S^c - \partial_D \pi_S^a (\Gamma^{\text{tot}})^D_{BC}. \quad (60.20)$$

Proof. The chain rule gives $\ddot{\gamma}_S^a = \partial_B \pi_S^a \ddot{\Gamma}^B + \partial_B \partial_C \pi_S^a \dot{\Gamma}^B \dot{\Gamma}^C$. Adding the contribution of ∇^S and subtracting the image of the total acceleration cancels the terms containing $\ddot{\Gamma}$. The remaining terms are (60.20) contracted with $\dot{\Gamma}^B \dot{\Gamma}^C$. When the total acceleration vanishes, the stated equality follows. $\square$

The geometric response factors through b when the connection, the projection, and the kinematic datum used in the evaluation are determined by $\beta = b(x)$, or when those kinematic data are held fixed in the comparison. Under these conditions, (60.20) gives the same result for any representatives of a fibre; Theorem 60.10 constructs its unique factor on B_{ef} . If the kinematic datum varies, it forms part of the domain of the response being compared.

The self-inertial formulation describes a geodesic trajectory of the joint state and an effective acceleration of its subsystem due to the projection defect. The gravitational realization subsequently identifies that connection, frame, and response with their physical observables. Observer covariance, memory inscription, and factorization through the boundary thus retain complementary functions: the first determines the readout class, the second retains the genealogy, and the third specifies the relational dependence of the response.

61 Bidirectional propagation and the boundary absorption condition

APP-TRIT-TPK transport retains the route orientation and the memory that distinguishes its outward passage from its return. On this antecedent, three successive operations are constructed: the action of an enriched history, the propagation of a source through the realized transports, and the imposition of a common condition at the boundary. The inscription of the register in an apparatus space, established in Section 60.3, makes it possible to express these operations through linear maps without confusing the observable representation with the state that retains its antecedents.

The temporal direction, sheet conjugation, and spatial reflection have distinct domains. Their relations are first fixed on the register; the intertwining used by a dynamical realization are then specified. This order determines which information remains in the propagators and which condition is introduced by the choice of an absorbing boundary.

61.1 Route action and register involutions

A finite route contains its positions, cardinal directions $(d_1, \dots, d_N)$, sheets $(\mu_1, \dots, \mu_N)$, with $\mu_t \in \{\Sigma, \Pi\}$, and the increments of its enriched register. For $\lambda \geq 0$, define

$$\mathcal{A}_\lambda(\gamma) = \sum_{t=2}^N (1 - \delta_{d_t, d_{t-1}} + \lambda(1 - \delta_{\mu_t, \mu_{t-1}})). \quad (61.1)$$

The first summand counts changes of direction; the second counts changes of sheet with the declared weight. For $N \leq 1$, the sum is zero. The general action of the sector additionally retains its boundary and memory terms:

$$\mathcal{S}[\gamma] = \sum_{a \in \gamma} \mathcal{L}_a + \mathcal{B}_{\text{in}}(\partial_- \gamma) + \mathcal{B}_{\text{out}}(\partial_+ \gamma) + \mathcal{M}(\gamma). \quad (61.2)$$

The terms in this expression specify a variational class with typed endpoints and retained memory. The dynamics of the sector determines $\mathcal{L}_a$, the admissible conditions, and the term $\mathcal{M}$.

On the registers we define $\mathcal{C}_r$ as the global interchange of sheets; $\mathcal{P}_r$ as a reflection of the spatial chart and its directions; and $\mathcal{T}_r$ as the reversal of temporal order, the replacement of each direction by its opposite, and the interchange of endpoints. The cardinal reflection is an involutive permutation. The reversal of the signed register is $C_M(a_1, \dots, a_N) = (-a_N, \dots, -a_1)$. The sheets, quotients, and memory are transported with the full route; the scalar readout of a word is a subsequent operation.

Proposition 61.1 (Invariance of the local route action). *On the preceding register domain,*

$$\mathcal{A}_\lambda(\gamma) = \mathcal{A}_\lambda(\mathcal{C}_r \gamma) = \mathcal{A}_\lambda(\mathcal{P}_r \gamma) = \mathcal{A}_\lambda(\mathcal{T}_r \gamma) = \mathcal{A}_\lambda(\mathcal{C}_r \mathcal{P}_r \mathcal{T}_r \gamma).$$

Proof. A global permutation of the direction alphabet preserves equality between consecutive directions. Interchanging the two sheets preserves equality between consecutive sheets. Reversing the order establishes a bijection between adjacent pairs, and replacement by opposite directions also preserves their equalities. Each operation preserves the two counts in (61.1); their composition preserves them. $\square$

The proposition determines the symmetry of the local functional. For a transformation also to act on a dynamical problem with fixed boundaries, it must transport its class of admissible extensions and the other three terms of (61.2). In particular, two routes with the same final representation may retain distinct variational classes because of their memory.

An interaction between two sections realized on a boundary B uses a coupling map

$$\mathcal{J}_B : E_B^{(1)} \otimes E_B^{(2)} \longrightarrow E_B^{(12)}$$

and a term $\mathcal{S}_{\text{int}}$ in the same action. Its covariance is expressed by

$$\rho_{12}(g) \mathcal{J}_B = \mathcal{J}_B(\rho_1(g) \otimes \rho_2(g)).$$

When the histories are composed, the register of the shared boundary is counted once; the memories on both sides are transported to the composite section. The coupling and this covariance condition are data of the concrete interaction, distinct from the register involutions.

61.2 Reversal of the realized evolution

The realization of the register provides a real Hilbert space $\mathcal{H}_{\mathbb{R}}$, an orthogonal complex structure J_E , with $J_E^2 = -I$, and a self-adjoint Hamiltonian H . Its domain $\mathcal{D}(H)$ is invariant under J_E ; we assume $HJ_E = J_EH$ on that domain. Let $\mathcal{C}$ be an orthogonal involution that preserves $\mathcal{D}(H)$ and satisfies

$$\mathcal{C}J_E\mathcal{C}^{-1} = -J_E, \quad \mathcal{C}H\mathcal{C}^{-1} = H. \quad (61.3)$$

The internal action constant $\hbar_{\text{int}} > 0$ retains its prior generation; here it normalizes the evolution group.

Theorem 61.2 (Temporal conjugation of the group). *The generator $A = -J_EH/\hbar_{\text{int}}$, with domain $\mathcal{D}(H)$, is skew-adjoint. Its orthogonal group, unitary with respect to J_E , satisfies*

$$U(t) = \exp(-J_EHt/\hbar_{\text{int}}), \quad \mathcal{C}U(t)\mathcal{C}^{-1} = U(-t). \quad (61.4)$$

Proof. Orthogonality and $J_E^2 = -I$ imply $J_E^* = -J_E$. Domain invariance and commutation with H allow H to be treated as a complex self-adjoint operator on the space defined by J_E . By functional calculus, $-J_EH/\hbar_{\text{int}}$ is skew-adjoint and generates the stated group. In finite dimension, the same statements follow from the exponential series and $A^* = -A$. The hypotheses on $\mathcal{C}$ give $\mathcal{C}A\mathcal{C}^{-1} = -A$. Conjugating the group, or its functional calculus, produces $\exp(-tA) = U(-t)$. $\square$

The map realizing the signed involution C_M as $\mathcal{C}$ preserves the temporal orientation and the preceding domain. Charge conjugation in a field representation additionally preserves its own characters and operators. Equality (61.4) fixes the temporal reversal of the group under the stated conditions.

61.3 Propagators of the chronological channel of a route

The isometric inscription of the register admits the unitary extensions of Theorem 60.7. Fix a bilateral realization of a chronological channel with fibres $\mathcal{H}_n$, $n \in \mathbb{Z}$, and unitary transports $U_n : \mathcal{H}_n \rightarrow \mathcal{H}_{n+1}$. The choice includes auxiliary spaces when required; it retains the route and its previous registers. Denote by $U(n, k)$ the transport from k to n , with

$$U(k, k) = I, \quad U(n, k)U(k, l) = U(n, l), \quad U(n, k)^* = U(k, n).$$

For $n > k$, it is the product $U_{n-1} \cdots U_k$; for $n < k$, it is the inverse of the transport from n to k .

The source space $\mathcal{J}_c$ contains finitely supported sections. The response space $\mathcal{F}$ contains all sections of these fibres. The quadratic action of the route, in the internal units of this realization, uses

$$(D\psi)_n = \psi_{n+1} - U_n\psi_n, \quad E[\psi] = \sum_n \|(D\psi)_n\|^2$$

for finitely supported sections. The kinetic operator with sign convention $L = -D^*D$ is

$$(L\psi)_n = U_n^*\psi_{n+1} - 2\psi_n + U_{n-1}\psi_{n-1}. \quad (61.5)$$

Indeed, $(D^*z)_n = z_{n-1} - U_n^*z_n$, and substitution produces this formula. The variation of E is $-2 \operatorname{Re}\langle \delta\psi, L\psi \rangle$.

Theorem 61.3 (Exact Green operators for the covariant difference). *Writing $x_+ = \max(x, 0)$, the maps*

$$(G_{\text{ret}}j)_n = \sum_k (n-k)_+ U(n, k) j_k, \quad (61.6)$$

$$(G_{\text{adv}}j)_n = \sum_k (k-n)_+ U(n, k) j_k \quad (61.7)$$

from $\mathcal{J}_c$ to $\mathcal{F}$ satisfy

$$LG_{\text{ret}}j = LG_{\text{adv}}j = j.$$

The retarded response vanishes before and at the first position of the source; the advanced response vanishes after and at the last. They are the unique solutions that vanish sufficiently far in the respective direction. For finitely supported ψ , we also have $G_{\text{ret}}L\psi = G_{\text{adv}}L\psi = \psi$. With respect to pairing with finitely supported sources, $G_{\text{adv}} = G_{\text{ret}}^\dagger$.

Proof. Each sum has as many terms as there are source positions. Composition of the transports reduces the application of L to the scalar identity

$$(n+1-k)_+ - 2(n-k)_+ + (n-1-k)_+ = \delta_{nk}.$$

The function $(k-n)_+$ satisfies the same identity. This proves both equations and the support statements. For uniqueness, a homogeneous solution that vanishes at two consecutive positions is determined to be zero at the next by (61.5); the recurrence, using invertible transports, extends in both directions. The difference of two solutions with the same causal support vanishes sufficiently far away and is therefore zero. Applying this argument to ψ and $G_{\text{ret}}L\psi$, or to the advanced case, gives the right-inverse identities on finite support. Finally, for two compactly supported sources, the double sum in the pairing is finite, and

$$((k-n)_+ U(k, n))^* = (k-n)_+ U(n, k).$$

Interchanging the indices proves the stated adjoint relation. $\square$

The responses belong to $\mathcal{F}$; in general they grow linearly outside the source and do not belong to ℓ^2 . The finite-support domain and its pairing specify the Green adjoint relation used here. The result constructs propagation along the chronological channel from its action; a spatial-field realization additionally retains its own operator, geometry, and propagation domain.

Let C_0 be a linear or antilinear isometric involution on $\mathcal{H}_0$. Transport from the reference fibre constructs

$$C_n = U(-n, 0)C_0U(0, n) : \mathcal{H}_n \longrightarrow \mathcal{H}_{-n}, \quad (\mathcal{C}\psi)_n = C_{-n}\psi_{-n}.$$

We have $C_{-n}C_n = I$ and $C_{n+1}U_n = U_{-n-1}^*C_n$, by composition. In particular, $\mathcal{C}^2 = I$, and the change $(n, k) \mapsto (-n, -k)$ in the preceding sums proves

$$\mathcal{C}L = L\mathcal{C}, \quad \mathcal{C}G_{\text{ret}}\mathcal{C}^{-1} = G_{\text{adv}}. \quad (61.8)$$

When C_0 is the declared realization of the register involution, these formulas transport that involution to the entire channel, without identifying phase return with a memory reset.

61.4 Causal interchange and decomposition of the response

In a field realization, let $L : \mathcal{D}(L) \subset \mathcal{H}_\Phi \rightarrow \mathcal{H}_J$ be an operator with unique retarded and advanced Green operators on the domains of admissible sources. Denote by $J^+(S)$ and $J^-(S)$ the future and past causal domains of a support S . The conditions are

$$\begin{aligned} LG_{\text{ret}} &= LG_{\text{adv}} = I, \\ \text{supp}(G_{\text{ret}}j) &\subset J^+(\text{supp } j), \\ \text{supp}(G_{\text{adv}}j) &\subset J^-(\text{supp } j). \end{aligned}$$

Let C_Φ, C_J be isometric involutions, both linear or both antilinear, that transport these domains, reverse their causal orientations, and satisfy $LC_\Phi = C_JL$. A Green pairing is retained for which $G_{\text{adv}} = G_{\text{ret}}^\dagger$.

Proposition 61.4 (Conjugation of the Green operators and parities). *Under the preceding conditions,*

$$C_\Phi G_{\text{ret}} C_J^{-1} = G_{\text{adv}}.$$

On the real space of response operators, define

$$\Theta(K) = C_\Phi K C_J^{-1}, \quad P_\pm(K) = \frac{1}{2}(K \pm \Theta(K)).$$

These are complementary projections for the involution Θ , and

$$G_{\text{sym}} = \frac{1}{2}(G_{\text{ret}} + G_{\text{adv}}), \quad G_{\text{rad}} = \frac{1}{2}(G_{\text{ret}} - G_{\text{adv}}) \quad (61.9)$$

satisfy $\Theta(G_{\text{sym}}) = G_{\text{sym}}$, $\Theta(G_{\text{rad}}) = -G_{\text{rad}}$, $G_{\text{sym}}^\dagger = G_{\text{sym}}$, and $LG_{\text{rad}} = 0$. The support of the symmetric response is contained in the union of the two causal domains.

Proof. The conjugated operator solves the source equation, since

$$LC_\Phi G_{\text{ret}} C_J^{-1} = C_J LG_{\text{ret}} C_J^{-1} = I.$$

Reversal of orientation takes its support to the advanced domain; uniqueness identifies it with G_{adv} . The involutions give $\Theta^2 = I$; expanding $(I \pm \Theta)^2$ and $(I + \Theta)(I - \Theta)$ proves the statements about projections. The Green adjoint relation proves symmetry, and the difference of the two source equations gives $LG_{\text{rad}} = 0$. The sum of two responses has support contained in the union of their supports. $\square$

If the involutions are antilinear, Θ is linear over real scalars; this is the operator space used in the proposition. Conjugation of the temporal group, the support condition, and the Green adjoint relation are three distinct properties, transmitted by the corresponding realization maps.

The causal reduction of the bilateral update, developed in the section on conservation and recovery, identifies another precise origin of retardation: elimination of a memory component that continues to participate in the joint evolution. Its discrete kernel preserves the order of interactions and the contribution of the initial memory. The Green operators constructed here instead solve a propagation equation with support conditions. The two constructions can be composed once the realization map identifying their domains and operators is specified; their respective causality conditions remain explicit.

61.5 Projection of absorbed currents and variation of the action

Let $\gamma_\partial : \mathcal{H}_\Phi \rightarrow \mathcal{H}_\partial$ be the boundary reader, distinct from the route γ . Define

$$\mathcal{A}_\partial = \gamma_\partial G_{\text{rad}}.$$

The finite window fixes the source space and the boundary reader; the Green operators retain their response domain. At a finite level with fixed inner products, its Moore–Penrose inverse determines

$$P_{\text{abs}} = I - \mathcal{A}_\partial^\dagger \mathcal{A}_\partial. \quad (61.10)$$

Theorem 61.5 (Absorption condition and boundary response). *P_{abs} is the orthogonal projection onto $\ker \mathcal{A}_\partial$. For any source j , the source $j_{\text{abs}} = P_{\text{abs}} j$ satisfies*

$$\gamma_\partial G_{\text{ret}} j_{\text{abs}} = \gamma_\partial G_{\text{adv}} j_{\text{abs}}. \quad (61.11)$$

It is the absorbed source closest to j for the declared inner product. Nonzero absorbed sources exist exactly when $\text{rank } \mathcal{A}_\partial < \dim \mathcal{H}_J$.

Proof. The orthogonal decomposition $\mathcal{H}_J = \ker \mathcal{A}_\partial \oplus \text{im } \mathcal{A}_\partial^*$ shows that $\mathcal{A}_\partial^+ \mathcal{A}_\partial$ is the identity on the second summand and zero on the first. Its complement is the stated projector. By definition, $2\mathcal{A}_\partial j_{\text{abs}} = \gamma_\partial(G_{\text{ret}} - G_{\text{adv}})j_{\text{abs}} = 0$. For $k \in \ker \mathcal{A}_\partial$, the orthogonal decomposition gives

$$\|j - k\|^2 = \|(I - P_{\text{abs}})j\|^2 + \|P_{\text{abs}}j - k\|^2.$$

The unique minimum corresponds to $k = P_{\text{abs}}j$. The final equivalence is the rank–nullity theorem. $\square$

In the source–field pairing, the symmetric action is $\mathcal{S}[j] = \frac{1}{2}\langle j, G_{\text{sym}}j \rangle$. The declared adjoint relation makes this quantity real. For real variations,

$$d\mathcal{S}[j](h) = \text{Re}\langle h, G_{\text{sym}}j \rangle.$$

The action restricted to absorbed sources is written on the ambient space as

$$\mathcal{S}_{\text{abs}}[j] = \frac{1}{2}\langle P_{\text{abs}}j, G_{\text{sym}}P_{\text{abs}}j \rangle, \quad d\mathcal{S}_{\text{abs}}[j](h) = \text{Re}\langle P_{\text{abs}}h, G_{\text{sym}}P_{\text{abs}}j \rangle. \quad (61.12)$$

When the pairing identifies field and source through Riesz, the ambient gradient is $P_{\text{abs}}G_{\text{sym}}P_{\text{abs}}j$. The field response remains $G_{\text{sym}}j_{\text{abs}}$; the projector in the gradient expresses the restriction on variations. The formulas are proved by expanding the quadratic term in $j + th$ and using self-adjointness. A material term $\mathcal{S}_{\text{matter}}$ adds its own differential.

61.6 Explicit absorbing condition in a chronological window

The channel of Subsection 61.3 makes it possible to evaluate this condition without introducing a target response. Let a source be supported on $0, \dots, N$, with $N \geq 1$. Transport its components to the reference fibre:

$$\hat{j}_k = U(0, k)j_k, \quad M_0(j) = \sum_{k=0}^N \hat{j}_k, \quad M_1(j) = \sum_{k=0}^N k\hat{j}_k.$$

The identity $(n - k)_+ - (k - n)_+ = n - k$ gives

$$(G_{\text{ret}} - G_{\text{adv}})j|_n = U(n, 0)(nM_0(j) - M_1(j)). \quad (61.13)$$

Proposition 61.6 (Absorption and two transported moments). *If the trace evaluates the field at two distinct positions a, b , equality of the advanced and retarded traces is equivalent to $M_0(j) = M_1(j) = 0$. In a fibre of dimension d , the space of such sources has dimension $(N - 1)d$. For $N \geq 2$, it contains the nonzero sources with transported components $(z, -2z, z, 0, \dots, 0)$, $z \neq 0$.*

Proof. Invertibility of $U(a, 0)$ and $U(b, 0)$ reduces the boundary condition to $aM_0 - M_1 = 0$ and $bM_0 - M_1 = 0$. Subtracting the equations gives $(a - b)M_0 = 0$, and then $M_1 = 0$. The two-moment map is surjective: for prescribed values (z_0, z_1) , it suffices to take $\hat{j}_0 = z_0 - z_1$, $\hat{j}_1 = z_1$, and all other components zero. Its rank is $2d$; its nullity is $(N + 1)d - 2d$. The indicated source has both moments zero. $\square$

In the transported frame, write

$$B_N = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 0 & 1 & \cdots & N \end{pmatrix}.$$

The absorbing projector is evaluated by

$$P_N = [I - B_N^T(B_N B_N^T)^{-1}B_N] \otimes I_{\mathcal{H}_0}. \quad (61.14)$$

The two rows of B_N are independent for $N \geq 1$, so the inverse exists. Unitary transport between frames carries this projector to the original fibres. For $N = 2$,

$$P_2 = \frac{1}{6} \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{pmatrix} \otimes I_{\mathcal{H}_0}.$$

This construction produces a nonzero absorbed class and its projection operation in the chronological channel. Its amplitude and its identification with a material current retain the rules of the corresponding physical sector.

61.7 Refinement and continuity of the boundary condition

Let $R_m^\Phi, R_m^J, R_m^\partial$ be the refinement maps for fields, sources, and boundaries. Assume that they intertwine both Green operators and the trace. Composition gives

$$R_m^\Phi G_{\text{sym},m} = G_{\text{sym},m+1} R_m^J, \quad R_m^\partial \mathcal{A}_{\partial,m} = \mathcal{A}_{\partial,m+1} R_m^J. \quad (61.15)$$

The second equality carries an absorbed source to another absorbed source. Naturality of the projectors additionally preserves the orthogonal structure of the realizations.

Proposition 61.7 (Naturality of the absorbing projection). *For a source isometry $R = R_m^J$, we have $P_{\text{abs},m+1} R = R P_{\text{abs},m}$ exactly when*

$$R(\ker \mathcal{A}_{\partial,m}) \subset \ker \mathcal{A}_{\partial,m+1}, \quad R((\ker \mathcal{A}_{\partial,m})^\perp) \subset (\ker \mathcal{A}_{\partial,m+1})^\perp.$$

Under these conditions at all levels, the projectors define an orthogonal projection on the completion of the isometric inductive limit of the source spaces.

Proof. Decompose each source into its components in the kernel and its orthogonal complement. The two inclusions imply that applying the projection before or after refinement respectively retains the first component and annihilates the second. Conversely, the operator equality applied to each component gives the two inclusions. In the inductive limit, the equality makes it possible to define an operator on the union of the levels independently of the representative. All projectors have norm at most one, so the operator extends to the completion. Its self-adjointness and idempotence identities, valid on the dense union, pass by continuity to the completion. $\square$

To extend a Green response to a completion as well, one retains its domains and the corresponding continuity estimates. Chronological propagation on finite support and factorization of the response through the boundary image, proved in 60.10, thus retain their respective domains during refinement.

The physical realization of emission and absorption composes these operators with the material currents, action, and causal conditions of the sector. The intertwining must preserve the route involution, the source–field pairing, the transported action, and the boundary conditions at each level. The preceding construction provides reversal of the group, the Green operators of the chronological channel, the temporal decomposition, and an explicit absorbing projection; spatial realizations use their own maps and causality conditions.

62 Action realization and metrological comparison

62.1 Comparison of the reduced Planck constant

The International System in force since 20 May 2019 fixes $h = 6.62607015 \times 10^{-34}$ J s exactly [14]. Its reduced constant is therefore

$$\hbar_{\text{SI}} = \frac{6.62607015}{2\pi} 10^{-34} \text{ J s} = 1.0545718176461563912 \dots \times 10^{-34} \text{ J s}.$$

The ellipsis denotes the expansion of an exact expression, not an experimental uncertainty in h in the current SI. This reference value is used here after evaluation of the HMT readers; its function is comparative.

The article's two return coordinates are

$$\begin{aligned} H_5 &= 1.0545718176461544696 \dots, \\ \eta_{\text{ret}} &= 1.0545718169150390611 \dots. \end{aligned}$$

Under the identification of $\mathcal{U}_S$ with J s, the relative difference between the preceding section and $\hbar_{\text{SI}}$ is approximately -1.82×10^{-15} . For the subsequent section, it is approximately -6.93×10^{-10} . The second difference contains the return effect used by the electronic corrector; it does not equal the evaluation error of the first section.

The precision of these comparisons allows the quantitative proximity of the construction of H_5 to be recognized while distinguishing proximity from exact identity. In the present manuscript, the displayed expressions do not provide an exact equality with $h/(2\pi)$. The result proved for Σ_H is its decade -34 ; the analytic result for H_5 is the evaluation of the stated character. The physical identification is examined in this chart and with the explicit differences just given. The SI value does not participate in the definition of the determinantal norm or in the regional recurrence.

63 Covariance, normal form, and Gaussian spectrum

The positive action section $\hbar_a$ and the preceding canonical realization determine the normalization used here. The correspondence between covariance, spectrum, purity, and entropy is proved for a finite number of modes; Fock space retains all its occupations. Symplectic reduction and the Gaussian formulas are established results reproduced here to make their application to memory transport self-contained.

63.1 Canonical coordinates and characteristic function

On $L^2(\mathbb{R}^d)$, let $\hat{q}_j = \hat{Q}_j/\sqrt{\hbar_a}$, $\hat{p}_j = \hat{P}_j/\sqrt{\hbar_a}$, and $\hat{z} = (\hat{q}_1, \hat{p}_1, \dots, \hat{q}_d, \hat{p}_d)$. On Schwartz space,

$$[\hat{z}_j, \hat{z}_k] = iJ_{jk}I, \quad J = \bigoplus_{j=1}^d \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

This convention fixes the sign of the canonical form. The Weyl operators $\mathcal{W}(\xi) = \exp(i\xi^\top \hat{z})$ are defined through the self-adjoint closures of the indicated linear combinations. For a centred state ρ with finite second moments, the normalized covariance and its characteristic function are

$$W_{jk} = \text{Tr } \rho(\hat{z}_j \hat{z}_k + \hat{z}_k \hat{z}_j), \quad \chi_\rho(\xi) = \text{Tr}(\rho \mathcal{W}(\xi)), \quad V = \frac{\hbar_a}{2} W.$$

The state is centred Gaussian when

$$\chi_\rho(\xi) = \exp(-\xi^\top W \xi / 4). \quad (63.1)$$

The expectation of $(c^\top \hat{z})^\dagger (c^\top \hat{z})$ is $c^\dagger (W + iJ)c/2$; hence $W + iJ \geq 0$. Moreover, $W > 0$: if v is real and $v^\top W v = 0$, positivity implies $(W + iJ)v = 0$, hence $Jv = 0$ and $v = 0$.

Lemma 63.1 (Uniqueness of the characteristic function). *Two trace-class operators with the same characteristic function are equal.*

Proof. Regrouping positions and momenta as $\xi = (u, v)$,

$$(\mathcal{W}(u, v)f)(x) = e^{iu \cdot (x+v/2)} f(x+v).$$

For a finite-rank operator with Schwartz vectors and kernel K_A ,

$$\text{Tr}(A\mathcal{W}(u, v)) = \int K_A(x + v/2, x - v/2) e^{iu \cdot x} dx.$$

Plancherel and the change of variables with absolute Jacobian one give

$$\|A\|_{\text{HS}}^2 = (2\pi)^{-d} \int_{\mathbb{R}^{2d}} |\text{Tr}(A\mathcal{W}(\xi))|^2 d\xi.$$

The preceding operators are dense in trace class. Convergence in that norm implies Hilbert–Schmidt convergence and uniform convergence of the characteristic functions, because their difference is bounded by $\|A - B\|_1$. The equality extends in L^2 . Applying it to the difference of the operators proves the assertion. $\square$

63.2 Symplectic reduction of the positive form

Theorem 63.2 (Williamson normal form). *For a real symmetric matrix $W > 0$ of order $2d$, there exist $S \in \text{Sp}(2d, \mathbb{R})$ and $\nu_j > 0$ such that*

$$W = SDS^\top, \quad D = \bigoplus_j \nu_j I_2.$$

The eigenvalues of JW are $\pm i\nu_j$. The condition $W + iJ \geq 0$ is equivalent to $\nu_j \geq 1$ for every j .

Proof. The matrix $B = W^{1/2}JW^{1/2}$ is antisymmetric and invertible. If $-B^2u = \nu^2u$ and $\|u\| = 1$, the vector $v = -Bu/\nu$ has unit norm and is orthogonal to u ; it satisfies $Bu = -\nu v$, $Bv = \nu u$. The complement of this pair is invariant. Iteration produces an orthogonal matrix O with $O^\top BO = JD$. Defining $S = W^{1/2}OD^{-1/2}$,

$$S^\top JS = J, \quad SDS^\top = W.$$

The similarity $W^{1/2}(JW)W^{-1/2} = B$ determines the values ν_j , with multiplicity. Congruence by S^{-1} transforms $W + iJ$ into $D + iJ$; each block has eigenvalues $\nu_j + 1$ and $\nu_j - 1$. $\square$

Lemma 63.3 (Unitary implementation in a finite number of modes). *Every symplectic matrix S admits a unitary U_S satisfying $U_S^\dagger \mathcal{W}(\xi)U_S = \mathcal{W}(S^\top \xi)$. The conjugation $\rho \mapsto U_S \rho U_S^\dagger$ transports covariance by $W \mapsto SWS^\top$.*

Proof. In the polar decomposition $S = OP$, the identity $J^{-1}(S^\top S)J = (S^\top S)^{-1}$, applied to the positive square root through spectral calculus, proves $PJP = J$. Then O is orthogonal symplectic. The eigenspaces of P are paired by J with eigenvalues λ, λ^{-1} ; the eigenspace of eigenvalue one is J -invariant. Hence an orthogonal symplectic matrix K reduces P to $\bigoplus_j \text{diag}(e^{r_j}, e^{-r_j})$.

The dilation is implemented by

$$(\mathcal{D}_r f)(x) = e^{-\frac{1}{2} \sum_j r_j} f(e^{-r_1} x_1, \dots, e^{-r_d} x_d).$$

A change of variables proves its unitarity and the conjugation of $(\hat{q}_j, \hat{p}_j)$ to $(e^{r_j} \hat{q}_j, e^{-r_j} \hat{p}_j)$. An orthogonal symplectic transformation commutes with J and corresponds to a matrix $u \in U(d)$ on $a_j = (\hat{q}_j + i\hat{p}_j)/\sqrt{2}$. Its second quantization $\Gamma(u) = \bigoplus_{n \geq 0} u^{\otimes n}$ is unitary on $\mathcal{F}_s(\mathbb{C}^d)$, and the identity on finite symmetric products gives the transformation of the a_j .

The Hermite basis identifies this space with $L^2(\mathbb{R}^d)$. To justify its completeness, if f is orthogonal to all polynomials multiplied by $e^{-|x|^2/2}$, the entire function $F(z) = \int f(x) e^{-|x|^2/2} e^{z \cdot x} dx$ has all derivatives zero at the origin. Thus $F = 0$; restricting to $z = i\xi$, Fourier uniqueness gives $f = 0$. Normalizations and orthogonality follow from creation and annihilation. Composition of the preceding implementations produces U_S . The Weyl equality implies $\chi_{U_S \rho U_S^\dagger}(\xi) = \chi_\rho(S^\top \xi)$, proving the covariance law. $\square$

63.3 Calculation of the spectrum of the isotropic density operator

In one mode, the normalized coherent states are

$$|z\rangle = e^{-|z|^2/2} \sum_{n \geq 0} \frac{z^n}{\sqrt{n!}} |n\rangle.$$

Their position-space wavefunction, $\pi^{-1/4} \exp[-(x - q_0)^2/2 + ip_0(x - q_0/2)]$, where $q_0 = \sqrt{2}\Re z$, $p_0 = \sqrt{2}\Im z$, gives by integration

$$\langle z | \mathcal{W}(u, v) | z \rangle = e^{-(u^2 + v^2)/4} e^{i(uq_0 + vp_0)}.$$

For $s > 0$, the integral

$$\rho^{(s)} = \frac{1}{\pi s} \int_{\mathbb{C}} e^{-|z|^2/s} |z\rangle \langle z| d^2 z$$

converges in trace class, is positive, and has trace one. Angular integration cancels its entries off the occupation diagonal; radial integration gives

$$\langle n | \rho^{(s)} | n \rangle = \frac{s^n}{(1+s)^{n+1}}, \quad \chi_{\rho^{(s)}}(u, v) = e^{-(2s+1)(u^2+v^2)/4}.$$

With $\nu = 2s + 1$, the unique Gaussian state of covariance νI_2 is, by Lemma 63.1,

$$\rho_\nu = (1-t) \sum_{n \geq 0} t^n |n\rangle\langle n|, \quad t = \frac{\nu-1}{\nu+1}. \quad (63.2)$$

The case $\nu = 1$ is the projector onto the vacuum.

Theorem 63.4 (Spectrum, purity, and entropy). *A centred Gaussian state with covariance W and symplectic eigenvalues $\nu_1, \dots, \nu_d$ is unitarily equivalent to $\bigotimes_j \rho_{\nu_j}$. Its eigenvalues are*

$$\lambda_{\mathbf{n}} = \prod_j (1-t_j) t_j^{n_j}, \quad t_j = \frac{\nu_j-1}{\nu_j+1}, \quad \mathbf{n} \in \mathbb{N}^d.$$

With $0 \log 0 = 0$,

$$\text{Tr } \rho^2 = \prod_j \nu_j^{-1} = (\det W)^{-1/2}, \quad (63.3)$$

$$-\text{Tr}(\rho \log \rho) = \sum_j \left[\frac{\nu_j+1}{2} \log \frac{\nu_j+1}{2} - \frac{\nu_j-1}{2} \log \frac{\nu_j-1}{2} \right]. \quad (63.4)$$

Proof. Williamson gives $W = SDS^\top$. The product of the isotropic density operators has characteristic function $e^{-\xi^\top D \xi/4}$. Unitary implementation and Weyl uniqueness prove the equivalence and spectral formula. In one mode,

$$\sum_{n \geq 0} (1-t)^2 t^{2n} = \frac{1-t}{1+t} = \nu^{-1},$$

and

$$-\sum_{n \geq 0} (1-t) t^n \log((1-t)t^n) = -\log(1-t) - \frac{t}{1-t} \log t.$$

Substitution of $t = (\nu-1)/(\nu+1)$ proves the formulas; the limit at $t = 0$ is zero. In the finite product, Tonelli justifies summing the nonnegative entropy terms, and $\det W = \prod_j \nu_j^2$ gives the determinantal expression. $\square$

If the state is the reduction of a pure bipartite vector, its eigenvalues are the squares of the Schmidt coefficients. The singular-value decomposition of the Hilbert–Schmidt operator defined by that vector proves equality of the nonzero spectra of the two reductions. The calculated entropy is then the entanglement entropy.

63.4 Preparation and transport of the two covariances

The subsequent application uses two identical pure Gaussian inputs. In general, for $M \in \text{Sp}(2d, \mathbb{R})$, their covariance is $V_0 = (\hbar_a/2)MM^\top$, and the joint input has covariance $V_0 \oplus V_0$. The physical transport is obtained by conjugating U_H by $M \oplus M$. In the balanced chart, the normalized input covariance is I_{4d} ; with

$$T = \frac{8I+H}{9}, \quad D = \frac{\sqrt{8}(H-I)}{9},$$

the first reduction has

$$W = TT^\top + DD^\top = \frac{8I+HH^\top}{9}.$$

Its characteristic function is obtained by setting the arguments of the second group of modes to zero. It therefore remains Gaussian, and the preceding theorems compute its entire spectrum without truncating the occupation register.

64 Physical actions and the spectrum of reduced memory

64.1 Genealogy of the dimensional sections

The coordinates π, φ, e, α are realizations of the correlated outputs constructed in Sections 11 and 15. This section uses $\alpha = \alpha_A$, with the orientation and periodization of Definition 11.5:

$$\alpha = \pi + e - \varphi - 4 - \kappa_{\text{per}}, \quad \kappa_{\text{per}} = \frac{N_K}{1000^{12} - 1}, \quad N_K = 234543140729659824621058914794146601. \quad (64.1)$$

Identity (11.7) follows from the recurrence with compatible carry and its telescoping, proved in Proposition 11.2. Period twelve corresponds to the register: a window of twelve blocks with a zero boundary condition and a compatible infinite section are different objects. Equality (64.1) uses the latter. The ordered register K preserves its incidences in addition to admitting the scalar reading κ_{per} .

The action reader, constructed in Section 12, determines

$$\begin{aligned} H_5 &= \frac{1}{2}\sqrt{1000\alpha/\varphi} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \\ D_A &= \exp(-100\pi\alpha/9), \\ \mathcal{C}_\pi &= 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \\ \eta_{\text{ret}} &= H_5 - \frac{90}{\pi}\mathcal{C}_\pi, \\ \hbar_{\text{pre}} &= H_5\mathcal{A}_0, \quad \hbar_{\text{ret}} = \eta_{\text{ret}}\mathcal{A}_0, \quad R_{\text{act}} = H_5/\eta_{\text{ret}}, \end{aligned} \quad (64.2)$$

Here $\mathcal{A}_0 = 10^{-34}\mathcal{U}_S$. The decade comes from the determinantal reader of $\varphi\alpha^{16}$, whose construction and bounds appear in Theorem 12.3; H_5 and η_{ret} are the mantissas of the two positive sections of reduced action. In each section $a \in \{\text{pre}, \text{ret}\}$, we have $h_a = 2\pi\hbar_a$.

The construction of the central electron, developed in Section 15, determines the coordinate

$$\begin{aligned} \Delta_4 &= \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729}\right), \\ B_e &= \frac{\sqrt{3}}{4} [\exp(\varphi/\pi^2) - 22\alpha^3] (1 + 15\Delta_4), \\ \kappa_e &= 80 - 54\alpha + 6\Delta_4, \quad \varepsilon_e^\beta = B_e R_{\text{act}}^{\kappa_e}. \end{aligned} \quad (64.3)$$

The prefactor $\sqrt{3}/4$ comes from the norm of the commutator of the sum and product projectors on the fibres specified in (15.6). The central route and its registers determine the return coefficients in (15.37) and (15.38). The dimensional realization is $E_e^\beta = \varepsilon_e^\beta \mathcal{U}_E$, $m_e^\beta = E_e^\beta/c^2$, with the energy chart of (15.41).

The dodecaphasic clock determines the energy basis of (15.42):

$$\omega_{108} = \frac{2\pi}{108t_0}, \quad E_{\text{clk}} = \hbar_{\text{ret}}\omega_{108}, \quad b_e = B_e \frac{\mathcal{U}_E}{E_{\text{clk}}}.$$

Substitution of (64.3), preserving the same physical energy, proves

$$\boxed{\frac{m_e^\beta c^2}{\hbar_{\text{ret}}} = b_e R_{\text{act}}^{\kappa_e} \omega_{108}, \quad \frac{m_e^\beta c^2}{\hbar_{\text{pre}}} = b_e R_{\text{act}}^{\kappa_e-1} \omega_{108}.} \quad (64.4)$$

Division by h_a , rather than $\hbar_a$, converts angular frequency into cyclic frequency. The conversion $\mathcal{U}_E/E_{\text{clk}}$ remains explicit: a coordinate expressed in MeV changes when its energy basis is replaced by the clock quantum.

64.2 Normalization of the Maxwell–Dirac action

The constitutive response in Section 39 produces the speed c , impedance Z , permittivity ε , and permeability μ , all positive, with $\varepsilon c = Z^{-1}$ and $\mu c = Z$. The positive charge section in (40.1) satisfies

$$e_a^2 = \frac{2\alpha\hbar_a}{Z} = 4\pi\alpha\varepsilon\hbar_a c. \quad (64.5)$$

Here $e_a > 0$ denotes the elementary charge of section a ; e in (64.1)–(64.3) denotes the Euler coordinate. Fix a homogeneous chart with positive constants $c, Z, \varepsilon, \mu, \hbar_a, e_a$, mass $m = m_e^\beta > 0$, and charge $q \in \{-e_a, e_a\}$; then $\sigma = q/e_a \in \{-1, 1\}$. In the Minkowski realization of signature $(+, -, -, -)$, with $x^0 = ct$, the Maxwell–Dirac action is

$$\mathcal{J} = \int d^4x \left[-\frac{F_{\mu\nu}F^{\mu\nu}}{4\mu c} + \bar{\psi} (i\hbar_a \gamma^\mu (\partial_\mu + iqA_\mu/\hbar_a) - mc) \psi \right]. \quad (64.6)$$

The component A_0 is the scalar potential divided by c , and $F = dA$. The spinor representation is fixed through the matrices g^I in (55.1), which satisfy $\{g^I, g^J\} = 2\eta^{IJ}I_4$ for $\eta = \text{diag}(-1, 1, 1, 1)$. Define $\Gamma^I = -ig^I$, so that $\{\Gamma^I, \Gamma^J\} = -2\eta^{IJ}I_4$, and, in this flat chart, $\gamma^\mu = \delta_I^\mu \Gamma^I$, $\bar{\psi} = \psi^\dagger \Gamma^0$. Signature, adjoint, and imaginary factor are thereby fixed jointly; the matrices γ^μ are distinguished from the scalar Holst parameter used later. Consider smooth fields and compactly supported variations, or boundary conditions that cancel the boundary terms. The action realizes the already generated coefficients and the material representation of Section 55.1.

Define

$$\ell_a = \frac{\hbar_a}{mc}, \quad X = x/\ell_a, \quad \Psi = \ell_a^{3/2}\psi, \quad a_\mu = \frac{e_a \ell_a}{\hbar_a} A_\mu, \quad f = da.$$

Then

$$\boxed{\frac{\mathcal{J}}{\hbar_a} = \int d^4X \left[-\frac{f_{\mu\nu}f^{\mu\nu}}{16\pi\alpha} + \bar{\Psi} (i\gamma^\mu (\partial_\mu + i\sigma a_\mu) - 1) \Psi \right]}. \quad (64.7)$$

Proof. The volume change contributes ℓ_a^4 , each derivative contributes ℓ_a^{-1} , and the product of fermionic fields contributes ℓ_a^{-3} . The coefficient of the mass term is $m c \ell_a / \hbar_a = 1$. The electromagnetic term contributes $\hbar_a / (4\mu c e_a^2)$. Since $\mu c = Z$ and $Ze_a^2 = 4\pi\alpha\hbar_a$, this coefficient is $1/(16\pi\alpha)$. Both terms and their normalizations are thereby determined. $\square$

Variation with respect to a_ν , followed by substitution of (64.1), gives

$$\partial_\mu f^{\mu\nu} = 4\pi (\pi + e - \varphi - 4 - \kappa_{\text{per}}) \sigma \bar{\Psi} \gamma^\nu \Psi. \quad (64.8)$$

The cancellation composes the action sections, electronic construction, and constitutive response; the regional and dodecaphasic reader determines the remaining coupling. Mass persists in the length ℓ_a , and action in the dimensional conversion. Compton normalization is a subsequent realization of these outputs. In the Coulomb realization of the same electron, the Bohr length is ℓ_a/α : substituting $e_a^2 = 4\pi\alpha\varepsilon\hbar_a c$ into $4\pi\varepsilon\hbar_a^2/(me_a^2)$ gives exactly this expression.

The deduction uses fixed homogeneous coefficients. For spacetime-dependent coefficients, the product rule introduces derivatives of the scales and coupling into the transformed equations. Variation of an action with variable parameters must preserve these terms.

64.3 Areal scale of gravitational action and entropy

The circular realization of the 108-step cycle, constructed in Section 43, transports the length $L = L_* \alpha^{16} 5759/23040$ from (42.2). With $t_0 = \pi L/(54c)$, it determines

$$t_P = \frac{54}{\pi} t_0, \quad G_a = \frac{c^5 t_P^2}{\hbar_a}, \quad \ell_P^2 = \frac{\hbar_a G_a}{c^3} = c^2 t_P^2. \quad (64.9)$$

The radius-matching condition of this realization is $G_a m_{108,a}/c^2 = R_{108} = L$, with $m_{108,a} = \hbar_a \omega_{108}/c^2$. It is the restriction of (42.4) to the unit mode; the chronological form retains the factor $(54/\pi)^2$. The reader uses the reduced radius $G_a m/c^2$. Reciprocity $G_{\text{pre}}/G_{\text{ret}} = R_{\text{act}}^{-1}$ preserves $\ell_P^2 = L^2$ between the two sections. The reference L_* , edge base u_L , and advance ct_0 are related by (42.5).

Section 49 constructs the cotetrad and connection from the transport already realized. In the action of Section 53, fix a nondegenerate cotetrad e^I , the orientation, signature $\eta = \text{diag}(-1, 1, 1, 1)$, and a metric connection ω , with curvature $F^{IJ} = d\omega^{IJ} + \omega_K^I \wedge \omega^{KJ}$. Take $\gamma \in \mathbb{R} \setminus \{0\}$ and hold $\gamma, \Lambda, c, G_a, \hbar_a$ constant during variation. Put $\Sigma^{IJ} = e^I \wedge e^J$ and $P_\gamma = \star + \gamma^{-1}I$; internal Lorentzian duality satisfies $\star^2 = -I$ on bivectors. Variations have compact support or boundary conditions that cancel the surface term. Substitution of (64.9) into (53.3) gives

$$\boxed{\frac{J_{\text{grav}}}{\hbar_a} = \frac{1}{16\pi\ell_P^2} \left[\int \Sigma_{IJ} \wedge (P_\gamma F)^{IJ} - 2\Lambda \int \text{vol}_e \right]}. \quad (64.10)$$

With the cotetrad and material fields fixed during variation of the connection, let σ_{IJ} be the material current defined by $\delta_\omega J_m = -\frac{1}{2} \int \delta\omega^{IJ} \wedge \sigma_{IJ}$, according to (53.4). The same substitution in Theorem 53.1 produces

$$D_\omega(P_\gamma \Sigma) = 8\pi\ell_P^2 \frac{\sigma}{\hbar_a}. \quad (64.11)$$

Proof. In section a , the original coefficient is $1/(2\kappa_a c)$, with $\kappa_a = 8\pi G_a/c^4$. Division by $\hbar_a$ converts it into $1/(16\pi\ell_P^2)$; the cosmological term retains its factor two. In the connection equation, $\kappa_a c \hbar_a = 8\pi\ell_P^2$ holds. Both identities use the same dimensional section. $\square$

The Barbero functional of Section 41 retains its complete expression. Its channels are $q_\pm = e^{-x \mp y}$, with $x = \pi A/180$, $y = \pi C^*/180$, where A and C^* are the angular coordinates expressed in degrees. The domain $x > |y|$ ensures $0 < q_\pm < 1$:

$$\gamma = \frac{180}{\pi} xy + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)], \quad S_m(q) = \frac{q^m}{1 - q^{3m}}. \quad (64.12)$$

Hexadic and octadic incidence fixes the multiplicities 90 and 120, and the oriented chain fixes the coefficients 12 and -1 , as proved by (41.1). The areal construction of Section 44 additionally gives

$$a_0 = \frac{1 + 2\cos(\pi/18)}{3} \frac{y}{6}, \quad \hat{\mathcal{A}} = \gamma a_0 \ell_P^2 (N_{90} + N_{120}). \quad (64.13)$$

The inversion $y \mapsto -y$ changes the signs of γ and a_0 ; their product is even. The operator P_γ , by contrast, retains its orientation dependence. Invariance of the areal product is thus distinguished from invariance of each operator coefficient. The areal spectrum is realized in the positive section $\gamma a_0 > 0$, with the occupation operators $N_{90}, N_{120} \geq 0$ constructed in Section 44.

For the spherical cosmological horizon in (59.31), of radius $R > 0$ and area $\mathcal{A}_H = 4\pi R^2$, fix a section a and abbreviate $\hbar = \hbar_a$, $G = G_a$ during this calculation and the following bounce calculation. Then

$$\frac{S_H}{k_B} = \frac{\mathcal{A}_H}{4\ell_P^2}, \quad T_{\text{cos}} = \frac{\hbar c}{2\pi k_B R}, \quad E_\Lambda = \frac{c^4 R}{2G}, \quad T_{\text{cos}} S_H = E_\Lambda. \quad (64.14)$$

The areal scale in (64.10), (64.11), and (64.14) is the same. The temperature corresponds to the spherical cosmological horizon; the normalization of the Schwarzschild horizon is a different realization. Varying R while holding $c, G, \hbar, k_B$ fixed, differentiation of (64.14) retains both terms: $T_{\cos} dS_H = 2 dE_\Lambda$ and $S_H dT_{\cos} = -dE_\Lambda$, as in (59.35). This gives the differential of an identity between states in that collection, not a replacement for the dynamical balance with other variable parameters.

64.3.1 An explicit dynamical application

Theorem 55.4 determines the following amplitude for a spinor state with conserved second moment $q_\varphi > 0$. Conservation is expressed through (55.18); Corollary 55.5 provides a concrete realization in the three-occupation sector:

$$s_0 = \frac{3\pi G \hbar^2}{2c^2 V_c^2} \frac{\gamma^2}{1 + \gamma^2} q_\varphi. \quad (64.15)$$

In the spatially flat realization with pressureless matter and $\Lambda = 0$, the constant $\rho_0 > 0$ is an energy density, $V_c > 0$ is the reference comoving volume, and $E_0 = \rho_0 V_c > 0$. The solution of Theorem 57.1 is

$$a_b^3 = s_0 / \rho_0, \quad \tau^2 = \frac{s_0 c^2}{6\pi G \rho_0^2}, \quad a(t) = a_b \left(1 + \frac{(t - t_b)^2}{\tau^2} \right)^{1/3}.$$

Restoring c in (56.2) uses the coefficient $8\pi G/(3c^2)$ when the density is an energy density. In particular, $(\dot{a}/a)^2 = (8\pi G/(3c^2))(\rho_0 a^{-3} - s_0 a^{-6})$. Substitution of (64.15) into the solution parameters proves

$$\left(\frac{\tau E_0}{\hbar} \right)^2 = \frac{q_\varphi}{4} \frac{\gamma^2}{1 + \gamma^2}, \quad \mathcal{V}_b = \frac{3\pi}{2} q_\varphi \frac{\gamma^2}{1 + \gamma^2} \ell_P^2 \frac{\hbar c}{E_0}. \quad (64.16)$$

Here $\mathcal{V}_b = V_c a_b^3$ is the physical volume at the bounce. The quantities G and V_c cancel in the first equality; E_0 remains as material energy. In the conserved sector $\Lambda^3 \mathbb{C}^4$, Proposition 55.3 proves $\hat{Q}_{\text{sp}} = 4I$, so $q_\varphi = 4$ for every normalized state supported in that sector, and $\tau E_0 / \hbar = |\gamma| / \sqrt{1 + \gamma^2}$. This yields a dynamical scale for the specified spinor state and cosmological realization.

64.4 Prolongation of memory transport: quantum realization

The quantum realization uses the reversible coupling of Section 34. First consider a real canonical plane with alternating matrix $\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The nonadic partition determines

$$Q = \frac{1}{3} \begin{pmatrix} \sqrt{8}I & -I \\ I & \sqrt{8}I \end{pmatrix}, \quad U_H = Q^\top \text{diag}(I, H)Q = \frac{1}{9} \begin{pmatrix} 8I + H & \sqrt{8}(H - I) \\ \sqrt{8}(H - I) & I + 8H \end{pmatrix}. \quad (64.17)$$

The operator $H \in \text{Sp}(2, \mathbb{R})$ transports this plane. The matrix Q preserves the Euclidean inner product and the form $\Omega \oplus \Omega$; hence U_H is symplectic. The first component defines the observed system and the second the memory register; the complete transformation acts on both.

The ordered loop constructed in Section 35 is

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad H_n = C^{-1} P^{-n} C P^n = \begin{pmatrix} 1 + n & n^2 \\ n & n^2 - n + 1 \end{pmatrix}. \quad (64.18)$$

For every $n \in \mathbb{Z}$, $\det H_n = 1$ holds. In particular, for $n = 1$,

$$H_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = F_1^2, \quad F_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = P F_{\text{av}} P^{-1}. \quad (64.19)$$

Its spectrum is $\{\varphi^2, \varphi^{-2}\}$, by subsequent recognition of the already generated golden mode. The integer n specifies the oriented number of iterations in the chosen loop. The transport collection is determined algebraically; its realization as a physical protocol additionally preserves the corresponding preparation and evolution data.

64.4.1 Initial state, transport, and reading

Proposition 36.1 constructs the Gaussian realization of the action ellipse. In the positive section $\hbar_a > 0$, with $A > 0$ and $|C^*/A| < 1$, its covariance is

$$V_\eta = \frac{\hbar_a}{2} M_\eta M_\eta^\top, \quad M_\eta = \text{diag}(e^{\eta/2}, e^{-\eta/2}), \quad \eta = \text{artanh}(C^*/A). \quad (64.20)$$

In the Schrödinger representation on $L^2(\mathbb{R}, dQ)$, with $\hat{Q}$ multiplication by Q and $\hat{P} = -i\hbar_a d/dQ$, the relation $[\hat{Q}, \hat{P}] = i\hbar_a I$ holds on Schwartz space. The normalized pure state is

$$\psi_\eta(Q) = (\pi\hbar_a e^\eta)^{-1/4} \exp[-Q^2/(2\hbar_a e^\eta)].$$

The preparation is the product $\psi_\eta \otimes \psi_\eta$. The transport in (64.17) is expressed in the same covariance chart by

$$\tilde{U} = (M_\eta \oplus M_\eta) U_H (M_\eta^{-1} \oplus M_\eta^{-1}). \quad (64.21)$$

Lemma 63.3 provides a unitary implementation of this symplectic matrix in two modes. Implementations differing by a global phase produce the same conjugation of the state. Covariance transforms by $V \mapsto \tilde{U} V \tilde{U}^\top$; the joint state remains pure, and its partial trace over the second mode determines the visible reading, according to the construction in Section 63.

Theorem 64.1 (Reduced mixedness from relative deformation). *The covariance of the visible reading is*

$$V_{\text{vis}} = \frac{\hbar_a}{2} M_\eta \frac{8I + HH^\top}{9} M_\eta^\top. \quad (64.22)$$

Its normalized symplectic eigenvalue is

$$\nu(H) := \frac{2\sqrt{\det V_{\text{vis}}}}{\hbar_a} = \sqrt{1 + \frac{8}{81} [\text{tr}(HH^\top) - 2]}. \quad (64.23)$$

Proof. Let $T = (8I + H)/9$ and $D = \sqrt{8}(H - I)/9$. The two initial modes are independent and have the same covariance. In the balanced chart of the initial state, the normalized visible covariance is $TT^\top + DD^\top$. Expansion gives

$$\begin{aligned} TT^\top + DD^\top &= \frac{64I + 8(H + H^\top) + HH^\top + 8[HH^\top - H - H^\top + I]}{81} \\ &= \frac{8I + HH^\top}{9}. \end{aligned}$$

This proves (64.22). Since $\det M_\eta = 1$ and $\det(HH^\top) = 1$, $\det(8I + HH^\top) = 65 + 8 \text{tr}(HH^\top)$. Division by 81 proves (64.23). The two positive eigenvalues of $HH^\top$ are reciprocal; their sum is at least two. Consequently, $\nu \geq 1$, with equality exactly when $HH^\top = I$. $\square$

The scale $\hbar_a$ and deformation η are retained until the determinant is evaluated. Their cancellation is covariant because state and operator are transported jointly. The expression $HH^\top$ uses the Euclidean inner product of the initial state's balanced chart; a chart change also transports this metric inner product.

An orthogonal symplectic transport H may have $D \neq 0$ while preserving the isotropic Gaussian product, with $\nu = 1$. The operator complement and state entanglement therefore have distinct criteria. In this preparation, non-isometric relative deformation characterizes exactly the case $\nu > 1$.

64.4.2 Exact evaluation of the golden loop

For (64.18),

$$\text{tr}(H_n H_n^\top) - 2 = n^2(2n^2 - 2n + 5), \quad \nu_n^2 = 1 + \frac{8}{81}n^2(2n^2 - 2n + 5). \quad (64.24)$$

The identity follows by summing the squares of the four entries. For $n = 1$, the sum is seven and

$$\nu_1 = \frac{11}{9}, \quad \text{Tr} \rho_{\text{vis}}^2 = \frac{9}{11}, \quad \bar{n}_W = \frac{1}{9}, \quad \frac{S_{\text{vis}}}{k_B} = \frac{10}{9} \log 10 - \log 9. \quad (64.25)$$

Theorem 63.4 proves that a centred one-mode Gaussian state with symplectic eigenvalue ν is unitarily equivalent to the geometric density operator in (63.2), whose mean occupation is $\bar{n}_W = (\nu - 1)/2$. For $\nu = 11/9$, its eigenvalues are

$$\lambda_j = \frac{9}{10} 10^{-j}, \quad j = 0, 1, 2, \dots \quad (64.26)$$

The series sums to one. The sum of its squared terms is $(9/10)^2/(1 - 10^{-2}) = 9/11$. Finally,

$$-\sum_j \lambda_j \log \lambda_j = -\log(9/10) + \log 10 \sum_j j \lambda_j = \frac{10}{9} \log 10 - \log 9.$$

The Gaussian reduction and its spectrum are proved in Section 63; the two preceding series evaluate exactly the purity and entropy of this realization. Since the joint state is pure, S_{vis}/k_B is also its dimensionless entanglement entropy. The global transformation is unitary, whereas restriction to the first mode has positive entropy.

The result corresponds to the specified preparation and transport. Identifying it with a vacuum measurement requires experimental realization of that protocol. The entropy of a reduction and transferred heat remain distinct quantities.

64.4.3 Composition with the thermal transducer

In the Williamson normal representation, choose a frequency $\omega > 0$ and define the energy reader $\hat{H}_W = \hbar_a \omega (\hat{N} + \frac{1}{2})$. The frequency is part of this energy realization; covariance determines the populations. With this choice, (64.26) is a Gibbs state whose population ratio satisfies

$$e^{-\hbar_a \omega / (k_B T_W)} = \frac{1}{10}, \quad T_W = \frac{\hbar_a \omega}{k_B \log 10}.$$

If the clock frequency of the same section is taken, $\omega = 1/t_P = 2\pi/T_{108}$, the thermal-transducer identity (45.8), $\hbar_a/t_P = k_B \Theta_{\text{clk},a} \log 3$, gives

$$\frac{T_W}{\Theta_{\text{clk},a}} = \frac{\log 3}{\log 10}. \quad (64.27)$$

The action section and thermal transducer cancel in the ratio. The index a of $\Theta_{\text{clk},a}$ preserves compatibility with the section $\hbar_a$, as in (45.10). The temperature T_W is the Gibbs-equivalent temperature for the declared Hamiltonian; the isolated unitary evolution retains its reversible character. This spectral identification determines a thermal ratio and does not postulate thermalization dynamics.

64.5 Extension to arbitrary finite canonical dimension

Let $H \in \text{Sp}(2d, \mathbb{R})$, with finite $d \geq 1$. Prepare two pure, centred, identical Gaussian states of d modes each, with covariance $V_0 = (\hbar_a/2)MM^\top$, where $M \in \text{Sp}(2d, \mathbb{R})$. Section 63 specifies this preparation and the joint chart transport by $M \oplus M$. In the balanced chart, the same coupling U_H produces $W = (8I + HH^\top)/9$. The product $A = HH^\top$ is positive, symmetric, and symplectic.

There exists an orthonormal symplectic basis in which

$$A = \text{diag}(e^{2r_1}, e^{-2r_1}, \dots, e^{2r_d}, e^{-2r_d}), \quad r_j \geq 0.$$

The identity $A\Omega A = \Omega$, with $\Omega = \bigoplus_{j=1}^d \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, implies that Ω takes the eigenspace of λ to that of λ^{-1} . Paired orthonormal bases are chosen in these spaces. The eigenspace of $\lambda = 1$ is Ω -invariant and admits orthonormal canonical pairs. Their union constructs the required basis.

The matrix W is diagonal in that basis. The products of each pair determine its symplectic eigenvalues:

$$\boxed{\nu_j^2 = \frac{(8 + e^{2r_j})(8 + e^{-2r_j})}{81} = 1 + \frac{32}{81} \sinh^2 r_j.} \quad (64.28)$$

Theorem 63.4 composes partial-trace reduction with these symplectic eigenvalues:

$$\boxed{\text{Tr} \rho_{\text{vis}}^2 = \prod_{j=1}^d \nu_j^{-1}, \quad \frac{S_{\text{vis}}}{k_B} = \sum_{j=1}^d \left[\frac{\nu_j + 1}{2} \log \frac{\nu_j + 1}{2} - \frac{\nu_j - 1}{2} \log \frac{\nu_j - 1}{2} \right].} \quad (64.29)$$

Adopt $0 \log 0 = 0$ by continuity. The proof of Theorem 63.4 unitarily transforms the state into a product of d geometric density operators: the purity of the product is the product of the purities, and its entropy is the sum of the entropies. The result holds for every finite d and every transport H in the domain, retaining all occupations of Fock space.

For the loop (64.19), $r = 2 \log \varphi$ and $\sinh^2 r = 5/4$, reproducing (64.25). Thus the transport spectrum and nonadic partition $8 + 1$ determine the spectrum of the reduced reading; the action sections fix its dimensional realization.

The dimension of an exceptional realization and its transport come from the corresponding constructor. Equation (64.28) determines the spectral law after realizing its canonical modes and the specified Gaussian preparation.

64.5.1 Operator identity between memory and complex structure

In the balanced canonical chart, let $J = \Omega$, with $J^2 = -I$ and $J^\top = -J$. This matrix realizes the canonical complex structure of phase space, and $D = \sqrt{8}(H - I)/9$ is the memory block. For the same preparation and chart transport as in Section 63, we have

$$\boxed{-(JW)^2 = I + [D, J][D, J]^\top, \quad W = \frac{2}{\hbar_a} M^{-1} V_{\text{vis}} M^{-\top}.} \quad (64.30)$$

Proof. The symplectic identities $HJH^\top = J$ and $H^\top JH = J$, together with $A = HH^\top$, give $JAJ^\top = A^{-1}$. If $C = HJ - JH$, multiplication of the four summands produces

$$CC^\top = A + JAJ^\top - HJH^\top J^\top - JHJ^\top H^\top = A + A^{-1} - 2I.$$

On the other hand,

$$-(JW)^2 = \frac{(8I + A^{-1})(8I + A)}{81} = I + \frac{8}{81}(A + A^{-1} - 2I).$$

Since $[D, J] = \sqrt{8} C/9$, we obtain (64.30). The identity holds in every finite canonical dimension and for every H in the domain, including transports that mix the modes. $\square$

The eigenvalues of this operator are the ν_j^2 , each with multiplicity two. Formula (63.3) then expresses purity directly in terms of the memory block:

$$\boxed{\text{Tr } \rho_{\text{vis}}^2 = \det(I + [D, J][D, J]^\top)^{-1/4}}. \quad (64.31)$$

In one mode,

$$\nu^2 = 1 + \frac{1}{2} \|[D, J]\|_F^2. \quad (64.32)$$

In this protocol, D commutes with J exactly when the reduced reading preserves purity. For the loop H_1 , $\|[D, J]\|_F^2 = 80/81$, and (64.32) gives $\nu^2 = 121/81$, yielding the purity $9/11$.

The commutator uses the complex structure compatible with the initial covariance and its metric inner product. A canonical change jointly transports all three data; the transpose in (64.30) refers to the balanced chart. The prior construction of the imaginary element in the formal kernel precedes this representation, which specifies its canonical action.

The joint evolution preserves purity through its unitary implementation; the commutator determines the loss of purity of the partial reading. The information of the joint state remains in the two factors and their correlations. The formula determines the purity of the reduction, which is a nonlinear functional of its density operator.

The real decomposition relative to J distinguishes its complex-linear and complex-antilinear components:

$$D_{\text{lin}} = \frac{D - JDJ}{2}, \quad D_{\text{anti}} = \frac{D + JDJ}{2}.$$

The first component commutes with J and the second anticommutes with J . The identities $[D, J] = 2D_{\text{anti}}J$ and $JJ^\top = I$ imply

$$\boxed{-(JW)^2 = I + 4D_{\text{anti}}D_{\text{anti}}^\top, \quad \text{Tr } \rho_{\text{vis}}^2 = \det(I + 4D_{\text{anti}}D_{\text{anti}}^\top)^{-1/4}}. \quad (64.33)$$

The complex-antilinear component of the memory block, defined relative to the complex structure of the initial state, determines exactly the purity and mixedness spectrum of this protocol. The complex-linear component can transport memory while preserving purity. The distinction concerns linearity relative to J ; combinatorial complexity and the arithmetic classification of constants are properties of other objects.

65 Bilateral memory, relative holonomy, and oriented recovery

Bilateral conservation acquires operational content when the channel recording composition is retained alongside the visible state. The exceptional and temporal selection of K , developed above, determines the record; its arithmetic and incidence readers transmit distinct relations of the same enriched state. Here we make explicit how an ordering of operations leaves a recoverable response even when multiplicities agree and phase returns. The antecedent is APP-TRIT-TPK generation, with both projections, the complete prolongation, and dodecaphasic orientation. Nonadic holonomy Γ_9 belongs to that enriched dynamics. The relative commutator constructed in this section belongs to a specified operator realization; its domain, conserved form, and reader remain distinguished.

65.1 Unitary composition with bilateral archive

For a unitary transport C on a Hilbert space $\mathcal{H}$, let

$$T_C = \frac{8I + C}{9}, \quad D_C = \frac{\sqrt{8}(C - I)}{9}, \quad R_C = \frac{I + 8C}{9}, \quad \mathbb{U}_C = \begin{pmatrix} T_C & D_C \\ D_C & R_C \end{pmatrix}.$$

Theorem 65.1 (Bilateral representation and complete composition). *The operator $\mathbb{U}_C$ is unitary and satisfies*

$$\mathbb{U}_{C_2} \mathbb{U}_{C_1} = \mathbb{U}_{C_2 C_1}, \quad T_{C_2} T_{C_1} + D_{C_2} D_{C_1} = \frac{8I + C_2 C_1}{9}.$$

For a state with initially zero archive,

$$\|T_C v\|^2 + \|D_C v\|^2 = \|v\|^2, \quad v = T_C^* T_C v + D_C^* D_C v.$$

Proof. The matrices

$$P_0 = \frac{1}{9} \begin{pmatrix} 8 & -\sqrt{8} \\ -\sqrt{8} & 1 \end{pmatrix}, \quad P_1 = \frac{1}{9} \begin{pmatrix} 1 & \sqrt{8} \\ \sqrt{8} & 8 \end{pmatrix}$$

are complementary orthogonal projectors: multiplication verifies $P_i^2 = P_i$, $P_0 P_1 = 0$, and $P_0 + P_1 = I_2$. Since $\mathbb{U}_C = P_0 \otimes I + P_1 \otimes C$, its adjoint product is the identity, and composition multiplies only the factors C . The first component of the product gives the bilateral formula. Applying unitarity to $(v, 0)$ proves norm conservation; its upper-left block is $T_C^* T_C + D_C^* D_C = I$, which gives the recovery map. $\square$

Composition order is distributed exactly:

$$[T_{C_2}, T_{C_1}] = \frac{[C_2, C_1]}{81}, \quad [D_{C_2}, D_{C_1}] = \frac{8[C_2, C_1]}{81}.$$

Adding both contributions gives $[C_2, C_1]/9$. The factors 1 and 8 compare these operator terms; their energy interpretations depend on the form and reader. The archive retains the contribution that composition of visible contractions alone would omit.

65.2 Quarter-turn plane in the dodecaphasic record

Let $C_{12} e_j = e_{j+1}$, with indices modulo 12, be the calendar permutation. The projector onto period four and its antiperiodic component are

$$P_4 = \frac{I + C_{12}^4 + C_{12}^8}{3}, \quad Q = P_4 \frac{I - C_{12}^6}{2}.$$

Define

$$u_0 = (1, 0, -1, 0, 1, 0, -1, 0, 1, 0, -1, 0)^T, \quad v_0 = C_{12} u_0, \quad W = \frac{1}{\sqrt{6}}(u_0, v_0).$$

Proposition 65.2 (Exact quarter-turn realization).

$$W^T W = I_2, \quad W W^T = Q, \quad C_{12} W = W J_0, \quad C_{12}^3 W = -W J_0, \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Proof. The vectors u_0, v_0 each have six entries of modulus one and disjoint supports, so they are orthogonal with norm $\sqrt{6}$. Moreover, $C_{12} v_0 = -u_0$. The range of Q consists precisely of period-four vectors $(a, b, -a, -b)$ repeated three times; it is therefore two-dimensional and equals the image of W . The remaining identities follow by applying the operators to both columns. $\square$

65.2.1 Calendar recovery and ternary carry

The extraction of the quarter-turn plane admits an exact description of the temporal information retained. We use the calendar of 108 positions and the dodecaphasic extension over the nonadic phase constructed earlier. For this description we use zero-based representatives:

$$\theta = r + 9\beta, \quad 0 \leq r < 9, \quad 0 \leq \beta < 12.$$

The position enumerated from 1 to 9 in the record is $r + 1$ here. The enriched update precedes this finite calendar and also retains the route, incidences, and accumulated memory. On the plane already constructed, $C_{12}^\beta W = W J_0^\beta$; its reading therefore depends on $b = \beta \bmod 4$.

Proposition 65.3 (Ternary fibre of the quarter-turn reading). *The pair consisting of the nonadic phase r and the planar phase b determines a quotient of 36 calendar positions. Its composition law is*

$$(r, b)(r', b') = \left(r + r' \bmod 9, b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor \bmod 4 \right).$$

Each reading has three preimages in $\mathbb{Z}/108\mathbb{Z}$. The additional component $k = \beta \bmod 3$ recovers the dodecaphasic coordinate through

$$\beta = 9b + 4k \pmod{12}.$$

Proof. Adding two representatives gives $\theta + \theta' = r + r' + 9(\beta + \beta')$. Euclidean division of $r + r'$ by 9 produces the carry in the formula. The reading (r, b) is equivalent to the residue $\theta \bmod 36$, since $\theta \equiv r + 9b \pmod{36}$. Its kernel is $\{0, 36, 72\}$, so each fibre has three elements. Finally, $9b + 4k$ is congruent to b modulo 4 and to k modulo 3; these two residues determine a unique element modulo 12. $\square$

Proposition 65.4 (Localization of the non-split extension). *The decomposition*

$$\theta \mapsto (j, z) = (\theta \bmod 4, \theta \bmod 27), \quad \theta = 81j + 28z \pmod{108},$$

identifies the calendar with $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/27\mathbb{Z}$. The nonadic projection is $(j, z) \mapsto z \bmod 9$, and the dodecaphasic inclusion $\beta \mapsto 9\beta$ takes the form

$$\beta \mapsto (\beta \bmod 4, 9\beta \bmod 27).$$

The obstruction to splitting the full extension is concentrated in

$$0 \longrightarrow \mathbb{Z}/3\mathbb{Z} \longrightarrow \mathbb{Z}/27\mathbb{Z} \longrightarrow \mathbb{Z}/9\mathbb{Z} \longrightarrow 0.$$

Proof. The coefficients 81 and 28 have residues $(1, 0)$ and $(0, 1)$ modulo $(4, 27)$, respectively. This verifies the inverse and the two specified maps. The ternary sequence does not split: a product $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}$ has exponent 9, whereas $\mathbb{Z}/27\mathbb{Z}$ has exponent 27. By contrast, the projection of the 36-position quotient onto $\mathbb{Z}/9\mathbb{Z}$ admits the homomorphic section $n \mapsto 28n \pmod{36}$. Thus, the reduction removes precisely this ternary obstruction. $\square$

The distinction is already visible at $\theta = 0, 36, 72$: all three positions give $r = b = 0$, whereas z takes the values 0, 9, 18, respectively. Recovering the calendar requires retaining the fibre identified by the plane. Moreover, the global quaternary residue satisfies $j = r + b \pmod{4}$; the planar phase b is transported together with the nonadic phase and its carry.

This locates the scope of the quarter turn: it organizes an oriented representation of the record and, together with the nonadic residue, recovers exactly its quotient of 36 positions. The additional ternary coordinate restores the 108 positions. The memory of the enriched dynamics in turn retains the continuation between complete calendar returns. The periodicity of a representation and the persistence of the recorded trajectory therefore belong to distinct readings of the same transport. The plane is extracted from the already defined permutation. Its action form is received from the generated angular pair, with A and C^* expressed in the same unit, $|C^*/A| < 1$, and $\theta \in \mathbb{R}$:

$$\xi = C^*/A, \quad \zeta = \operatorname{artanh} \xi, \quad S_\zeta = \operatorname{diag}(e^{\zeta/2}, e^{-\zeta/2}), \quad U_\zeta(\theta) = S_\zeta e^{-\theta J_0} S_\zeta^{-1} = \begin{pmatrix} \cos \theta & e^\zeta \sin \theta \\ -e^{-\zeta} \sin \theta & \cos \theta \end{pmatrix}.$$

Each U_ζ preserves $g_\zeta = S_\zeta^{-T} S_\zeta^{-1}$ and has determinant one. Its conserved form is specified before comparing transports with different anisotropy.

Operator products act from right to left.

65.3 Relative commutator and action recovery

Theorem 65.5 (Commutation defect of conjugate forms). *Let $U = U_{\zeta_s}(\theta)$, $V = U_{\zeta_r}(\phi)$, and $H = UVU^{-1}V^{-1}$. With $\delta = \zeta_s - \zeta_r$ and $\chi = \sinh \delta \sin \theta \sin \phi$,*

$$\det H = 1, \quad \operatorname{tr} H = 2 + 4\chi^2, \quad \lambda_\pm(H) = (\sqrt{1 + \chi^2} \pm |\chi|)^2.$$

Moreover, $H = I$ precisely when $\chi = 0$.

Proof. Multiplication gives $\operatorname{tr}(UV) = 2(\cos \theta \cos \phi - \cosh \delta \sin \theta \sin \phi)$. For determinant-one matrices, Cayley–Hamilton gives $A^{-1} = (\operatorname{tr} A)I - A$. Substituting this expression for both inverses yields

$$\operatorname{tr}(UVU^{-1}V^{-1}) = a^2 + b^2 + c^2 - abc - 2, \quad a = \operatorname{tr} U, \quad b = \operatorname{tr} V, \quad c = \operatorname{tr}(UV).$$

Trigonometric substitution reduces this expression to $2 + 4\chi^2$. Solving $\lambda^2 - (2 + 4\chi^2)\lambda + 1 = 0$ gives the eigenvalues. If $\chi = 0$, either the forms coincide or one transport is $\pm I$; both cases commute. The converse follows from the trace. $\square$

For conjugate quarter turns $U_+ = U_\zeta(\pi/2)$, $U_- = U_{-\zeta}(\pi/2)$, the inverses are $-U_\pm$. Multiplication gives

$$H = (U_+ U_-)^2 = \operatorname{diag}(e^{4\zeta}, e^{-4\zeta}), \quad \operatorname{tr} H - 2 = \frac{16\xi^2}{(1 - \xi^2)^2}. \quad (65.1)$$

With the outputs $A = 1000\alpha$, $C^* = 2(\eta_{\text{ret}} + \alpha)$, both in degrees, this gives

$$\rho_{\text{lazo}} = e^{4\zeta} = \left(\frac{501\alpha + \eta_{\text{ret}}}{499\alpha - \eta_{\text{ret}}} \right)^2, \quad 0 < \eta_{\text{ret}} < 499\alpha. \quad (65.2)$$

Its inverse, retaining the oriented axis, is

$$\eta_{\text{ret}} = \alpha \left[500 \frac{\sqrt{\rho_{\text{lazo}}} - 1}{\sqrt{\rho_{\text{lazo}}} + 1} - 1 \right], \quad \hbar_{\text{ret}} = s_0 \eta_{\text{ret}}.$$

The proof uses $e^{2\zeta} = (1 + \xi)/(1 - \xi)$ and solves for ξ . The previously constructed $s_0 = 10^{-34}\mathcal{U}_S$ is retained. Under conjugation, the axis label is transported; the trace alone retains $|\zeta|$. The ratio ρ_{lazo} denotes this relative response, while π/φ^2 retains its distinct definition as the marked regional ratio.

The positive branch corresponds exactly to $\rho_{\text{lazo}} > (501/499)^2$. Under orientation reversal, $\rho_{\text{lazo}} \mapsto \rho_{\text{lazo}}^{-1}$, the axis label is transported before applying the recovery map of that branch.

65.4 Dodecaphasic lift and compatible amplification

Define on the twelve coordinates

$$\tilde{S}_\zeta = I - Q + WS_\zeta W^T, \quad L_\pm = \tilde{S}_{\pm\zeta} C_{12}^3 \tilde{S}_{\pm\zeta}^{-1}.$$

Proposition 65.6 (Restoration of the commutator in the full record).

$$L_+ L_- L_+^{-1} L_-^{-1} = I - Q + WHW^\top.$$

Proof. The operators Q and C_{12} commute. On $\text{im } Q$, the isometry W identifies C_{12}^3 with $-J_0$, and $L_\pm$ with $U_\pm$. On the complement, both $L_\pm$ equal C_{12}^3 , so their commutator is the identity. The orthogonal sum of both restrictions gives the formula. $\square$

For the specified amplification $\mathcal{H}_m = \mathbb{R}^{12} \otimes (\mathbb{R}^9)^{\otimes m}$, let

$$\tilde{H} = I - Q + WHW^\top, \quad H_m = \tilde{H} \otimes I_{9^m}, \quad J_m v = v \otimes u_9, \quad u_9 = (1, \dots, 1)/3.$$

Then J_m is isometric and $H_{m+1} J_m = J_m H_m$. Operator norm remains fixed under amplification by an identity, so the compatible action extends to the Hilbert limit. The preceding proofs are retained in this tensor refinement, with projectors and reader transported.

65.5 Chronological information and energy reading

The loop power is $H^n = \text{diag}(\rho_{\text{lazo}}^n, \rho_{\text{lazo}}^{-n})$. It preserves QP and $dQ \wedge dP$. For $\rho_{\text{lazo}} \neq 1$, a preparation with $Q_0 \neq 0$ and a known axis recovers $n = \log(Q_n/Q_0)/\log \rho_{\text{lazo}}$. Preparation and detector scale belong to the domain of this recovery map.

Allow $n \in \mathbb{Z}$, including iteration of the inverse loop. When $Q_0 = 0$ and $P_0 \neq 0$, the other component gives

$$n = -\frac{\log |P_n/P_0|}{\log \rho_{\text{lazo}}}.$$

The two orderings $UU^{-1}VV^{-1}$ and $UVU^{-1}V^{-1}$ contain the same four multiplicities, but their responses are I and H . The ordered archive retains a distinction removed by commutative counting. For a detector set $\mathcal{D}$, two histories are operationally equivalent when all their readings in $\mathcal{D}$ agree. Responses on a complete basis determine the linear operator; recovery of an enriched history additionally requires its remaining records.

The reference form fixes the energy meaning. For $g_{\text{ref}} = \text{diag}(e^{-\zeta r}, e^{\zeta r})$, the diagonal loop's quadratic-energy gains along the marked axes are

$$G_Q = e^{8\zeta}, \quad G_P = e^{-8\zeta}, \quad \mathcal{O} = \frac{1}{4} \log(G_Q/G_P) = 4\zeta.$$

These gains are the squares of the state dilations. H and H^{-1} have the same trace, whereas $\mathcal{O}$ retains the oriented sign. For $\zeta \neq 0$, U_+ and U_- have no common positive metric: if both preserved one, their commutator would also preserve it and all its eigenvalues would have modulus one, contradicting (65.1). Each individual transport preserves its own $g_{\pm\zeta}$.

The algebraic identity of bilateral composition remains valid for these operators. Joint unitary conservation additionally requires a common preserved positive form. Relative holonomy preserves area and its own recovery data.

A passive transport $T_n = S_{\zeta_{n+1}} R(-\theta_n) S_{\zeta_n}^{-1}$ telescopes: its product is $S_{\zeta_N} R(-\sum \theta_n) S_{\zeta_0}^{-1}$, equal to the identity when shape and total phase return. The relative commutator instead compares effective operations against a common reference. Attributing an energy exchange to that response requires including the system, memory, and device. Area conservation and operator reversibility are proved independently of that attribution.

The result assembles three levels of recovery: the bilateral archive completes visible composition; the dodecaphasic calendar determines an exact operational plane; and the oriented loop reader recovers angular contrast and action with their antecedents. This articulation extends the coupling role of K without replacing its selection by a later inversion. The distinction between phase, composition order, conserved form, and recoverable memory is expressed by explicit operators and recovery maps.

66 Synthesis of the results and consequences

The generating structure preserves operations and history before its numerical realization. In realized transport, this conservation is formulated as an identity of forms. A partial reading distributes charge between the visible component and the complement; including both defines a recoverable transformation. Refinement prolongs the same identity without identifying different depths or replacing the register by its intensity.

66.1 Conservation, reconstruction, and dimension

The equality $\mathcal{U}^*(b' \oplus b')\mathcal{U} = b \oplus b$ brings reading and memory together in a reversible operation. For compressions with a separate register, telescoping preserves the positive terminal component and the complete series of complements. For a fixed unitary holonomy, the terminal component is the projection onto its fixed-vector space; for a variable sequence, it is determined by the strong limit of the Gram operators.

Dodecaphasic memory allows the centred register to be recovered through two specified observations, with inverse norm $9/4$. Incidence transports this recovery to the dimensional projector. Its exterior prolongation preserves each degree through Gram determinants and requires the inclusion of mixed components. Thus the relations between length, area, and volume are expressed through maps defined on the same register, with their respective invariants.

66.2 Action and physical consequences

Action, clock, and energy basis retain their types throughout composition. The electronic ratio $m_e c^2 / \hbar$ is expressed through the clock frequency and electronic character, preserving the basis-conversion factor. Maxwell–Dirac normalization removes the chosen dimensional scales and leaves the dimensionless coupling. Gravitational transport preserves the areal scale $\hbar G / c^3$; the homogeneous bounce solution retains its conditions on matter, torsion, and cosmological constant.

The Noether relation is manifested when the transported observable is a charge of the declared action. Change of form adds a connection term to the Hamiltonian generator and preserves oriented action exactly. The holonomy of a return also preserves the information distinguishing phase return from return of the complete state.

66.3 Memory and quantum spectrum

Identity (1.2) determines the symplectic spectrum of the reduced covariance through the antilinear component of the memory operator. For the loop

$$H_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad W_1 = \frac{1}{9} \begin{pmatrix} 13 & 3 \\ 3 & 10 \end{pmatrix},$$

we obtain $\det W_1 = 121/81$, symplectic eigenvalue $11/9$, and purity $9/11$. The spectrum of the reduced density operator is

$$\lambda_j = \frac{9}{10} 10^{-j}, \quad j = 0, 1, 2, \dots,$$

and its entropy satisfies

$$\frac{S}{k_B} = \frac{10}{9} \log 10 - \log 9.$$

These figures come from the transport and state fixed in the theorem. Spectral equivalence with a Gibbs density operator assigns a temperature to the chosen Hamiltonian; that equivalence retains its distinction from a dynamical thermalization process.

66.4 Foundational content

The articulation between number, incidence, and dynamics resides in the conservation of a common antecedent and in the operations relating its realizations. The article makes this articulation explicit through reconstructions, bounds, spectra, and actions. Generation of values and transport of symmetries retain the same provenance; their observable consequences are calculated subsequently with the reader and state that actually participate. This order provides a precise mathematical formulation of dynamical memory and its physical realizations.

κ_{ag}

Arithmetic–geometric coupling: bilateral conservation and reversible refinement

Oriented register, ternary norm, reversible refinement and bilateral conservation and recovery law.

Proportion, refinement, and restoration

The holographic extreme and mean ratio here brings together arithmetic completion and operator recovery. Refinement with carry redistributes a unit among components and preserves the normalized sum. Euclidean division, applied with the declared depth and carry convention, recovers the previous labels. Conservation and the inverse thus receive concrete operations.

The ternary norm supplies the integer seventy-three and its algebraic and lattice interpretation. The relation to decimal completion is expressed through exact identities. The bilateral balance extends the analysis to two isometric transports between spaces of arbitrary dimension. The cross terms cancel in a weighted combination and restore the initial squared norm.

The nonadic specialization determines the initial allocation of weights: seventy-two over seventy-three and one over seventy-three. An orthogonal mixing then produces the two normalized channels. When the transports admit the declared relative unitary operator, the effective reader combines the observable with its conjugate; its invariance is then equivalent to commutation with that relative operator. Transport of the complete state simultaneously preserves the relations that each observable allows to be evaluated.

The archiving policy prolongs this balance. Successive memories and the terminal component maintain the whole at finite depth. The uniform estimate of the terminal component makes it possible to obtain an infinite archive recoverable through the adjoint of the isometric operator. The bounds describe the stability of this recovery as depth increases.

The coupling constant participates in the same construction. Its orbit and Gram form make it possible to identify operators; the archive isometry preserves these responses and their bounds. Label refinement lifts to a unitary transport and is composed with the bilateral balance. Unity thus preserves its arithmetic content, its linear realization, and the possibility of restoration. This is the operational culmination of the guiding argument: a generated structure admits transformations that redistribute its reading and preserve the data for its recovery.

67 The holographic arithmetic-geometric coupling constant

The dodecaphasic register links two realizations of the same state: the positional reading that enters into arithmetic closure and the reading that preserves incidence, orientation, and direction. The following development determines the operations made possible by the register already produced, the information retained by its memories, and the errors under which its relations can be recovered. Its reference function admits a precise formulation: the phase orbit of the register is a complete basis of its carrier, and isometric memory preserves exactly the Gram matrix of that basis.

The starting point remains the composition

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{enriched state} \longrightarrow \text{joint discrete structure of the continuum.}$$

APP supplies the sheets and their remainders and quotients; TRIT preserves regime and orientation; TPK selects, transports, and updates while preserving carry, route, boundary, survival, and memory. The five constructions of the continuum belong to this joint production, developed in the preceding body. The work here takes place after it, on its dodecaphasic integer representation and its typed readers. Neither the subsequent linear carrier nor a decoder's candidate set replaces the domain of generated histories.

67.1 Object, chart, and production of the register

Definition 9.1 fixes the generated object and its readers. We now develop its reversible encoding, its capacity for operator identification, and the stability of its incidence realizations.

The designation expresses a constitutive function of the object, not an automatic identification with a physical interaction coefficient. Canonicity refers to the declared profile, origin, and orientation. A chart transformation and an update that incorporates new events are not the same operation. Nor is it asserted that the same numerical vector must be a necessary invariant of every system.

The calendar of one hundred and eight positions consists of twelve nonadic windows. The extension

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0$$

preserves phase carry; its non-splitting and the distinction between phase return and state return are proved in proposition 9.2. In each window, the register of a history retains the subroute, its orientation, the carry, and the incidence register. The causal order of extraction is

$$x_{\text{term}} \longmapsto \mathcal{R}_{12}(x_{\text{term}}) \longmapsto (b^{90}, b^{120}, Q_{\text{TPK}}) \longmapsto U_{\text{sgn}} \longmapsto K. \quad (67.1)$$

The regional coordinates and the signed register are two readings of the same terminal state; it is not postulated that the visible ternary matrix alone determines the signed coordinate.

To specify the final composition explicitly, the channels produced are

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \quad Q_{\text{TPK}} = 6263. \end{aligned}$$

Let $B_r = \sum_{j=0}^3 b_{r+3j}^{120}$ and $d_{r,j} = b_{r+3j}^{90}$, for $r = 1, 2, 3$. The harmonic reader gives

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2,$$

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}).$$

In this compatible output, $(B_1, B_2, B_3) = (972, -1073, 101)$, $(s_1, s_2, s_3) = (2378, 1406, 2479)$, and

$$U^{(1)} = (2378, -452, -668, -322),$$

$$U^{(2)} = (1406, 998, -204, -28),$$

$$U^{(3)} = (2479, -551, -371, -997).$$

Division by three belongs to the reader's three orbits, and the subsequent division by four to the Hadamard inverse; these are not coefficients fitted to a physical output. On the orbits (1, 4, 7, 10), (2, 5, 8, 11), (3, 6, 9, 12), we apply

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I.$$

The products $H_4 U^{(r)}$ are, respectively,

$$(936, 2916, 2484, 3176), \quad (2172, 2636, 232, 584), \quad (560, 3296, 3656, 2404).$$

Their divisibility by four verifies membership in the Hadamard sublattice. After dividing and restoring the natural order, we obtain

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \quad \mathbf{1}^* K = 6263. \quad (67.2)$$

The inversion and its congruences, including the index 4096 of the signed image, are proved in lemma 9.10 and proposition 9.12. Reconstruction is unique because $H_4^{-1} = H_4/4$, not because a register close to a known value of alpha was sought.

To avoid a collision of symbols, we shall write $\Delta_j = I - S^j$ for the antecedent integer extractors, with $(Sx)_i = x_{i+1}$ and indices modulo twelve. Then

$$\Delta_3 K = b^{90}, \quad \Delta_4 K = b^{120}, \quad Q(K) = 6263.$$

The extractor (Δ_3, Δ_4, Q) is injective, with invariant factors 1, ..., 1, 12, according to proposition 9.13. Its integral-image conditions and accumulation law remain those of the antecedent register. The normalized memory complements to be denoted by D_j are different operators, derived from those shifts and with a declared metric codomain.

67.2 Periodic encoding, closure, and exact inversion

Fix $B = 1000$, twelve positions, and

$$N(k) = \sum_{j=1}^{12} k_j B^{12-j}, \quad d_B = B^{12} - 1, \quad \kappa(k) = \frac{N(k)}{d_B}.$$

In the canonical chart,

$$\begin{aligned} \kappa_{\text{ag}} &= \frac{234543140729659824621058914794146601}{10^{36} - 1}, \\ \alpha_A &= A_{\text{reg}} - \kappa_{\text{ag}}, \quad A_{\text{reg}} = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4. \end{aligned} \quad (67.3)$$

The three regional coordinates are expressed before acting in this reader; their common origin with the dodecaphasic closure does not require conflating the evaluation order of the correlated coordinates. The equality is a subsequent reader of the generated state. Rearranging it once its outputs are known is a legitimate inverse operation, distinct from using a target value to produce the register.

Proposition 67.1 (Reversibility and radius of the encoding). *On the domain $\{0, \dots, 999\}^{12}$, the fraction $\kappa(k)$, with declared base and length, determines k exactly. The same holds for the exact pair $(A_{\text{reg}}, \alpha_A)$ when it satisfies (67.3). If*

$$|\tilde{\kappa} - \kappa(k)| < \frac{1}{2d_B},$$

rounding $d_B \tilde{\kappa}$ to the nearest integer recovers $N(k)$ without error. If the two terms of the closure are approximated separately, it suffices that the sum of their absolute errors be less than $1/(2d_B)$.

Proof. Multiplication by d_B recovers the integer $N(k)$. Twelve Euclidean divisions by B , retaining the remainders and the number of positions, recover its blocks, including leading zero blocks. The error hypothesis places the true integer at a distance strictly less than half a unit. For the closure, the triangle inequality is applied to the error in the difference. Two consecutive integers show that the radius cannot be increased uniformly over the entire code domain: their midpoint is ambiguous. $\square$

If a reduced fraction p/q is received, one computes $N = d_B p/q$ and checks integrality and $0 \leq N \leq d_B$. The reduced denominator does not replace the base, length, and chart metadata. The code contains 10^{36} words and requires $\lceil \log_2(10^{36}) \rceil = 120$ bits in a fixed-length binary representation. Twelve ten-bit blocks also occupy 120 bits: no additional compression, authentication, or cryptographic integrity has been proved. An exact fraction or the integer N avoids losing the precision needed for this inversion.

The periodic reading is not to be confused with the truncation N/B^{12} . If $\kappa_j = \kappa(S^j K)$, positional division gives

$$\kappa_{j+1} = 1000\kappa_j - K_{j+1}, \quad K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{6263}{999}. \quad (67.4)$$

Thus the scalar orbit preserves all phase differences and makes it possible to read $B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\}$. The reversibility of these representations does not imply that a scalar reconstructs the entire enriched history that produced the register.

67.3 The complete orbit as a reference for its carrier

In $V = \mathbb{R}^{12}$, with its canonical Euclidean inner product, consider

$$O_K = (K, SK, \dots, S^{11}K) : \mathbb{R}^{12} \longrightarrow V.$$

The shift S acts on positions of the register. Periodicity alone does not identify it with the full TPK evolution or with a subsequent exceptional symmetry.

Theorem 67.2 (Complete cyclic reference and exact Gram matrix). *The matrix O_K is invertible. Its Gram matrix has the eigenvalues*

value	multiplicity
$39225169 = 6263^2$	1
$1248025 - 345420\sqrt{3}$	2
589609	2
1407529	2
1053157	2
$1248025 + 345420\sqrt{3}$	2
697225	1

and satisfies

$$589609\|c\|^2 \leq \|O_K c\|^2 \leq 39225169\|c\|^2, \quad \|O_K^{-1}\| = \frac{1}{\sqrt{589609}}, \quad \text{cond}(O_K) = \frac{6263}{\sqrt{589609}}. \quad (67.5)$$

The orbit of the centred register has rank eleven.

Proof. The entries of the Gram matrix are the autocorrelations $\langle K, S^j K \rangle$. In the order $j = 0, \dots, 11$, their values are

$$(4251257, 2999323, 3163385, 3287920, 3216553, 3344743, \\ 2950064, 3344743, 3216553, 3287920, 3163385, 2999323).$$

Since S is orthogonal, the Gram matrix is circulant. Substituting the twelve characters $\exp(2\pi im/12)$ into that row gives the real table above. All values are positive. The smaller of the two values involving

$\sqrt{3}$ exceeds 589609, because $658416 > 345420\sqrt{3}$, an inequality verified by squaring its two positive sides. The other values in the table are greater than or equal to 589609. The uniform mode has value $(\sum K_i)^2 = 6263^2$; the triangle inequality for the Fourier transform, using $K_i \geq 0$, bounds all the others by that same value. The spectral theorem gives (67.5). Centring the register removes only the uniform mode; the remaining eleven retain the nonzero values in the table. $\square$

The determinant, with the column order and the orientation of S fixed here, is

$$\det O_K = 5483119259214020183850397930008283625.$$

The absence of a proper period of the vector would not, by itself, prove invertibility: the decisive fact is that none of its Fourier modes vanishes. The condition number is approximately 8.15643. If K is normalized to unit norm, the extreme eigenvalues of the Gram matrix are divided by $\|K\|^2 = 4251257$; the amplitude of its components must not be confused with an intrinsic improvement over every reference of equal norm.

Corollary 67.3 (Polar separation of form and orientation). *Let*

$$G_K = O_K^* O_K, \quad P_K = G_K^{1/2}, \quad F_K = O_K G_K^{-1/2}.$$

Then P_K is positive definite, F_K is orthogonal, and $O_K = F_K P_K$. An isometric realization $J : V \rightarrow \mathcal{Y}$ preserves G_K and P_K , and transports the isometric part to JF_K .

Proof. Positivity of the Gram matrix allows its square root and inverse to be defined through the spectral theorem. The identity $F_K^* F_K = G_K^{-1/2} G_K G_K^{-1/2} = I$ proves the assertion. Moreover, $(JO_K)^*(JO_K) = O_K^* O_K$. $\square$

This polar decomposition is defined from the complete Gram matrix: it is not obtained solely from the list of its eigenvalues or from the norm of K . The symbol P_K in this corollary is not the rank-ten dimensional projector to be constructed below. This separation is a consequence of the cyclic reference, not an identification of all its readers.

67.4 Operator identification and relative holonomy

Theorem 67.4 (Twelve general responses or one cyclic response). *For any linear operator $H : V \rightarrow V$, its twelve responses $Y = (HK, HSK, \dots, HS^{11}K)$ determine*

$$H = YO_K^{-1}.$$

If $HS = SH$, a single complete vector-valued response $y = HK$ determines

$$H = O_y O_K^{-1}, \quad H = \sum_{j=0}^{11} h_j S^j, \quad h = O_K^{-1} y. \quad (67.6)$$

With errors E in the response matrix or δy in the vector, respectively,

$$\|\hat{H} - H\| \leq \frac{\|E\|}{\sqrt{589609}}, \quad (67.7)$$

$$\|\hat{h} - h\| \leq \frac{\|\delta y\|}{\sqrt{589609}}, \quad \|\hat{H} - H\| \leq \sqrt{\frac{12}{589609}} \|\delta y\|. \quad (67.8)$$

Proof. The first equality is $Y = HO_K$. If H commutes with S , then $HS^j K = S^j y$, so that $Y = O_y$. Commutation with the cycle determines all columns of H from any one of them; the twelve powers $I, S, \dots, S^{11}$ are independent and span that space of operators. Hence $y = O_K h$, which gives the second formula. The first two bounds use $\|O_K^{-1}\| = 1/\sqrt{589609}$. The last uses $\|O_{\delta y}\| \leq \|O_{\delta y}\|_F = \sqrt{12}\|\delta y\|$. $\square$

A response here means a vector of twelve components, not a scalar measurement. For example, if $H = 2I + 3S - S^5$, the response $y = HK$ returns exactly the coefficients 2, 3, -1 in their positions through (67.6); the inversion procedure needs y, K, S , not the filter coefficients. By contrast, the nonzero operator whose first row is $(543, -234, 0, \dots, 0)$ and whose remaining rows are zero annihilates K . A single response does not identify an arbitrary operator.

The reference function admits an exact generalization. On a cycle of n positions, $H \mapsto Hg$ identifies all operators commuting with the cycle if and only if $O_g = (g, Sg, \dots, S^{n-1}g)$ is invertible. Sufficiency is the preceding proof. If O_g is singular, a nonzero vector in its kernel gives a polynomial of degree less than n that annihilates g ; the operator is nonzero because the powers of the cycle are independent. The unit impulse also has an orthonormal orbit and serves as a reference. The specific consequence for K is that the same object already used in closure and incidence satisfies this condition, without having been selected anew to optimize the test. No identification of nonlinear dynamics or arbitrary hidden states follows; an application to a linearization requires that the linearization have been defined and justified.

Corollary 67.5 (Recovery of relative holonomy from the complement). *Let H_B be known and invertible, and let H_C be another transport of the same type. The complement*

$$D_\gamma = \frac{\sqrt{8}}{9}(H_C - H_B)$$

determines

$$H_B^{-1}H_C = I + \frac{9}{\sqrt{8}}H_B^{-1}D_\gamma.$$

If D_γ commutes with S , in particular if both transports do, $y = D_\gamma K$ suffices to recover

$$H_B^{-1}H_C = I + \frac{9}{\sqrt{8}}H_B^{-1}O_y O_K^{-1}. \quad (67.9)$$

The error in this reconstruction is bounded by

$$\frac{9}{\sqrt{8}}\|H_B^{-1}\|\sqrt{\frac{12}{589609}}\|\delta y\|.$$

Proof. Solve for H_C in the definition of the complement and apply theorem 67.4. The bound follows from submultiplicativity. $\square$

For $H_B = S^3$ and $H_C = S^4$, the rational datum $y/\sqrt{8} = (S^4 - S^3)K/9$ makes it possible to reconstruct the relative holonomy S without supplying H_C to the inversion procedure. If H_B is unitary, the error factor is approximately 0.01435510. Without commutation, the twelve general responses are retained, rather than a fictitious symmetry hypothesis. The selection of a physical loop and its representation still belong to its generating dynamics; this protocol does not replace them.

67.5 Complete memory preserves the reference and the observables

Let C_n be unitary operators on a Hilbert space H ,

$$T_n = \frac{8I + C_n}{9}, \quad D_n = \frac{\sqrt{8}(C_n - I)}{9}, \quad R_0 = I, \quad R_{n+1} = T_n R_n.$$

This formulation includes nonstationary sequences. It does not require commutation between C_n at different stages.

Theorem 67.6 (Isometric register, terminal component, and transport). *For every N ,*

$$R_N^* R_N + \sum_{n=0}^{N-1} R_n^* D_n^* D_n R_n = I.$$

The positive strong limit $A_\infty = \lim_N R_N^ R_N$ exists, and*

$$\mathcal{V}x = (A_\infty^{1/2}x, (D_n R_n x)_{n \geq 0}) \quad \text{satisfies} \quad \mathcal{V}^* \mathcal{V} = I. \quad (67.10)$$

Its image $\mathcal{M}$ is closed, and the inverse on it is $\mathcal{V}^$, of norm one. In the finite case, the first component is replaced by $R_N x$ and the N complements are retained. For any bounded operator A ,*

$$A^\mathcal{V} = \mathcal{V} A \mathcal{V}^*|_{\mathcal{M}}$$

preserves products, adjoints, norm, and spectrum on that image, and

$$\langle \mathcal{V}x, A^\mathcal{V} \mathcal{V}y \rangle = \langle x, Ay \rangle.$$

Proof. Expanding the two terms of one stage gives $T_n^* T_n + D_n^* D_n = I$. Hence $R_n^* R_n - R_{n+1}^* R_{n+1} = R_n^* D_n^* D_n R_n$, and the telescoping sum proves the finite identity. The sequence of positive operators $R_N^* R_N$ is decreasing and bounded below; its strong limit exists. Taking inner products with x , the sum of the squared norms of the complements is $\|x\|^2 - \langle x, A_\infty x \rangle$. This proves that the series belongs to the Hilbert direct sum and that $\mathcal{V}$ is an isometry. An isometry has closed image and an adjoint inverse on it. Inserting $\mathcal{V}^* \mathcal{V} = I$ into the compositions proves the transport properties. Neither convergence of $R_N x$ nor the assertion that A_∞ is a projector has been assumed. $\square$

Applying this result to the carrier of the register,

$$(\mathcal{V} O_K)^* (\mathcal{V} O_K) = O_K^* O_K. \quad (67.11)$$

The twelve directions and all their correlations are preserved, with the same bounds as in theorem 67.2, at any depth of complete memory. It is not asserted that the entire continuum has dimension twelve: this twelve-dimensional reference is preserved in its image. The correct operators there are $S^\mathcal{V} = \mathcal{V} S \mathcal{V}^*$ and $H^\mathcal{V} = \mathcal{V} H \mathcal{V}^*$. An ambient shift of storage slots is not presumed to be $S^\mathcal{V}$.

If $z = \mathcal{V}y + \delta z$, the estimate

$$\hat{h} = O_K^{-1} \mathcal{V}^* z$$

retains the first bound of (67.8) with $\|\delta z\|$ in place of $\|\delta y\|$. Depth does not worsen this constant because the isometry and its terminal component are preserved. A partial selection of channels has a different Gram matrix and requires its own bounds. The exterior powers $\bigwedge^p \mathcal{V}$ preserve Gram determinants; finite-rank traces and spectral measures of the transported observables are also preserved. Squared norms are not thereby interpreted automatically as Born probabilities: that reading requires its subsequent realization.

67.6 Exceptional incidence and selected direction

Let $e = \mathbf{1}$, $\Pi = I - ee^*/12 = P_{11}$. In the icosahedral chart A_5/C_5 already constructed, the orthogonal representation ρ and the three-dimensional character fix

$$P_3 = \frac{3}{60} \sum_{g \in A_5} \chi_3(g^{-1}) \rho(g), \quad P_3^* = P_3 = P_3^2, \quad \text{rank } P_3 = 3, \quad P_3 e = 0. \quad (67.12)$$

This projector is received together with its representation and markings; it is not chosen to fit a direction to the register. Put

$$k = \Pi K, \quad u = P_3 k, \quad E_u = \frac{uu^*}{\|u\|^2}, \quad Q_K = \Pi - E_u = P_{10}(K).$$

The exact evaluations are

$$\|k\|^2 = \frac{11789915}{12}, \quad \|u\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20}, \quad (67.13)$$

$$\eta_K := \frac{\|u\|^2}{\|k\|^2} = \frac{3(6638585 + 2275584\sqrt{5})}{5 \cdot 11789915} \simeq 0.5967954228259091. \quad (67.14)$$

In particular, $u \neq 0$. Since $u^*k = \|u\|^2$,

$$E_u k = u, \quad Q_K k = k - u, \quad k = u + Q_K k, \quad \|k\|^2 = \|u\|^2 + \|Q_K k\|^2. \quad (67.15)$$

The reduction $12 \rightarrow 11 \rightarrow 10$ removes first the uniform direction and then a selected line. It does not remove all of $\text{im } P_3$: in $\text{im } Q_K$, the two directions of that subspace orthogonal to u remain. These dimensions are those of the algebraic spaces written above; identifying them with physical dimensions requires their realization maps.

Proposition 67.7 (Exact transport of direction and descriptor). *For a complete isometry $\mathcal{V}$, let $\Pi^\mathcal{V} = \mathcal{V}\Pi\mathcal{V}^*$, $P_3^\mathcal{V} = \mathcal{V}P_3\mathcal{V}^*$, and $u^\mathcal{V} = \mathcal{V}u$. On $\mathcal{M}$,*

$$Q_K^\mathcal{V} = \mathcal{V}Q_K\mathcal{V}^* = \Pi^\mathcal{V} - \frac{u^\mathcal{V}(u^\mathcal{V})^*}{\|u^\mathcal{V}\|^2}, \quad \text{rank } Q_K^\mathcal{V} = 10, \quad \eta_K^\mathcal{V} = \eta_K.$$

Moreover, $\mathcal{V}^*u^\mathcal{V} = u$ and $\mathcal{V}^*Q_K^\mathcal{V}\mathcal{V} = Q_K$.

Proof. Substitute the definitions and $\mathcal{V}^*\mathcal{V} = I$. Norms and rank are preserved by the isometry. The identity on the image is $\mathcal{V}\mathcal{V}^*$, not the identity of a larger ambient storage space. Extension by zero outside the image adds no physical directions. $\square$

The complete incidence screen provides another exact realization. If $\mathcal{I}$ is the incidence operator already constructed and $E_{\text{inc}} = \mathcal{I}(e^\perp)$, then

$$F : V \longrightarrow Y = \mathbb{R} \oplus E_{\text{inc}}, \quad Fx = \left(\mu(x), \frac{\mathcal{I}\Pi x}{6} \right), \quad \mu(x) = \frac{\mathbf{1}^*x}{12}, \quad \langle (\mu, y), (\nu, z) \rangle_Y = 12\mu\nu + \langle y, z \rangle \quad (67.16)$$

is a surjective isometry, since $\Pi\mathcal{I}^*\mathcal{I}\Pi = 36\Pi$. For the raw incidence of the 132 hexads, the complete identity is $\mathcal{I}^*\mathcal{I} = 36I + 30\mathbf{1}\mathbf{1}^*$; centring removes exactly its uniform component. The incidence image has dimension eleven; Y is not all of $\mathbb{R} \oplus \mathbb{R}^{132}$. Applying F term by term to memory preserves both the Gram matrix and the preceding operators and their domains. This composition introduces no Monster action on K .

67.7 Continuous stability and the need to preserve correlations

Theorem 67.8 (Direction error from complete memory). *Let $z = \mathcal{V}K + \delta$, $\|\delta\| \leq \varepsilon$, and reconstruct $\hat{K} = \mathcal{V}^*z$, $\hat{k} = \Pi\hat{K}$, $\hat{u} = P_3\hat{k}$. All three errors are less than or equal to ε . If $\varepsilon < \|u\|$,*

$$\|Q_{\hat{K}} - Q_K\| \leq \frac{\varepsilon}{\|u\|}.$$

If $\varepsilon < \|k\|$,

$$|\eta_{\hat{K}} - \eta_K| \leq \frac{\varepsilon}{\|k\|}.$$

Proof. The adjoint of the isometry and the projectors have norm one. Noise orthogonal to $\mathcal{M}$ disappears upon applying $\mathcal{V}^*$. For two nonzero lines, the norm of the difference of their projectors is the sine of the angle between them. The distance from u to the line of $\hat{u}$ does not exceed $\|u - \hat{u}\|$, so this sine is less than or equal to $\varepsilon/\|u\|$. The hypothesis prevents $\hat{u} = 0$.

For the second assertion, normalize k and $\hat{k}$. The difference of their rank-one projectors has eigenvalues $\pm \sin \theta$ in their plane. When paired with $0 \leq P_3 \leq I$, its absolute value does not exceed $\sin \theta$: in an eigenvector basis one obtains $\sin \theta(a - b)$, with $a, b \in [0, 1]$. The same distance argument gives $\sin \theta \leq \varepsilon/\|k\|$. If the lines coincide, the difference of the ratios is zero. $\square$

Here $\|u\| \simeq 765.733$, so the factor for the projector is approximately 0.00130594. In the reader with only two memories, the inverse has norm $9/4$ and the corresponding factor is $9/(4\|u\|) \simeq 0.00293836$. These are different observed data; the difference is not a cost-free improvement of the same channel. The integer correction in theorem 68.5 will allow the register to be recovered first without error within its radius, instead of propagating a continuous error to the projector.

Proposition 67.9 (Cross blocks are part of the observable). *Write $\mathcal{V}x = (W_a x)_a$, including the terminal component. Then the (a, b) block of $A^\mathcal{V}$ is*

$$(A^\mathcal{V})_{ab} = W_a A W_b^*.$$

Its quadratic form is computed as a limit of finite truncations

$$\langle y, A^\mathcal{V} y \rangle = \lim_F \sum_{a, b \in F} \langle y_a, W_a A W_b^* y_b \rangle.$$

General conservation is not obtained by discarding all terms $a \neq b$.

Proof. The block formula follows from multiplying $\mathcal{V}A^\mathcal{V}$. The truncation projectors converge strongly to the identity and $A^\mathcal{V}$ is bounded, proving the limit without asserting absolute convergence of an arbitrary double series. For a counterexample, take $H = \mathbb{R}$, $C = -I$, $T = 7/9$, $D = -2\sqrt{8}/9$, $\mathcal{V}x = (Tx, Dx)$, and $A = I$. The diagonal part, evaluated at $\mathcal{V}x$, equals

$$(T^4 + D^4)x^2 = \frac{3425}{6561}x^2.$$

The cross terms contribute $3136x^2/6561$, and the sum is x^2 , as required by the isometry. $\square$

67.8 Discrete correction before incidence selection

The canonical two-memory readers are

$$T_j = \frac{8I + S^j}{9}, \quad D_j = \frac{\sqrt{8}(S^j - I)}{9}, \quad Mx = (D_3x, D_4T_3x).$$

The second memory acts after the first compression; the first complement is not reinjected in this protocol. The general theorem in theorem 68.5 will prove, without using the register as a target, the strict radii

$$r_0 = \frac{2\sqrt{146}}{81}, \quad r_q = \frac{4\sqrt{73}}{81}.$$

The first radius recovers ΠK from MK , modulo uniform integer translation. The second recovers K in full when $q = \mathbf{1}^*K$ is also preserved. If a real-valued charge is observed with error less than $1/2$, rounding first recovers its integer charge channel. None of these procedures requires the terminal component T_4T_3K .

The high-block selection in the marked chart is

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\}. \quad (67.17)$$

The fourth component is exactly at the threshold. This circumstance separates continuous stability from discrete stability.

Proposition 67.10 (Zero continuous margin and integer restoration). *The direct selector in (67.17), applied to the real-valued reconstruction of the memories, has no positive Euclidean stability radius at MK , even with exact charge. By contrast, prior integer decoding with error less than r_q restores B_K exactly without changing the threshold.*

Proof. Let L be the least-squares inverse, satisfying $LM = \Pi$. For $t > 0$, put

$$e_t = -tM\Pi e_4, \quad \tilde{K}_t = \frac{q}{12}\mathbf{1} + L(MK + e_t) = K - t\Pi e_4.$$

We have $\|e_t\| \rightarrow 0$, but the fourth component is $729 - 11t/12 < 729$. For small t , only this membership changes. The second assertion follows by first recovering K exactly through theorem 68.5 and then applying the same selector. $\square$

If the remaining marked data $P_\pi, H_e, H_\varphi, H_{APP}$ are kept exact or transported according to their law, uniqueness of the exceptional construction returns the same flag $B_K \subset H_\alpha^-$ and the same marked origin. Correcting the register is not equivalent to correcting independent errors in these markings. The witness

$$K' = K - e_4 + e_6, \quad Q(K') = 6263, \quad B_{K'} = \{6, 9, 10\}$$

also belongs to $\{0, \dots, 999\}^{12}$ and satisfies $\|MK' - MK\|^2 = 4672/6561$. The midpoint of the two memories is at distance r_q from each. This proves that the strict radius is uniformly optimal on the fixed-charge algebraic domain, not that K' has a constructed HMT terminal history.

Without charge, $M(K - \mathbf{1}) = MK$ and the high-block selections differ. The two memories can recover Q_K and η_K exactly, since these ignore the mean, without recovering that absolute selector. The condition $P_3\Pi K \neq 0$ remains necessary for the projector; integer correction does not remove this defining condition. A concrete finite test uses $z = MK/\sqrt{8}$ and adds $1/10$ to its first component. The error in the norm of M is $\sqrt{8}/10 < r_q$. The floor-and-ceiling algorithm in proposition 68.6 leaves 924 candidates of charge 6263 and recovers K and B_K exactly.

Applying the isometry (67.16) gives $M_{\text{inc}} = (F \oplus F)M$. All distances between words and all noise norms are preserved, so the same radii and decoders apply in $Y \oplus Y$. This is the quantitative connection between memory and incidence; it does not depend on attributing an additional exceptional symmetry to the shift S .

67.9 Covariance, refinement, and variational charge

If G is orthogonal and $K' = GK$, $\Pi' = G\Pi G^*$, and $P'_3 = GP_3G^*$ are transported jointly, then

$$u' = Gu, \quad Q'_{K'} = GQ_KG^*, \quad \eta'_{K'} = \eta_K.$$

In a fixed chart, $K \mapsto aK + b\mathbf{1}$, $a \neq 0$, leaves Q_K and η_K invariant; it does not leave the entire object invariant. For example, $\kappa(K + \mathbf{1}) - \kappa(K) = 1/999$, and the reading α_A would change by $-1/999$ if the regions were held fixed. This is an algebraic counterexample to identifying the descriptor with the entire register, not an assertion that this update is admissible as an HMT history.

Nor can one move only the register and arbitrarily keep the projector fixed. The finite formula (67.12) gives exactly

$$\|P_3S^3K\|^2 = \frac{369035}{4} - \frac{281861\sqrt{5}}{10}, \quad \|P_3S^4K\|^2 = \frac{1327717}{4} + \frac{694618\sqrt{5}}{5}.$$

These differ from (67.13); in particular, $[P_3, S^3]$ and $[P_3, S^4]$ do not vanish. Transporting P_3 by the same conjugation does preserve η_K .

Decoding is also covariant. For a permutation R ,

$$S' = RSR^{-1}, \quad M' = (R \oplus R)MR^{-1}, \quad M'R = (R \oplus R)M.$$

The lattice, charge, and distances are preserved. In the uniqueness regime, the decoder output transforms by R ; outside it, the set of minimizers is transported without arbitrarily breaking a tie. A general orthogonal frame also requires transporting $\mathbb{Z}^{12}$ to $R\mathbb{Z}^{12}$ and the uniform direction to $R\mathbf{1}$. Rounding in the old lattice after changing the chart does not represent the same problem.

Proposition 67.11 (Naturality under refinement with local frames). *Let z be a cylinder with successors z' , compatible positive weights $\sum_{\tau z'=z} \mu(z') = \mu(z)$, and orthogonal frames G_z . The refinement*

$$\mathcal{R}_n(e_z \otimes x) = \sum_{\tau z'=z} \sqrt{\frac{\mu(z')}{\mu(z)}} e_{z'} \otimes G_{z'} G_z^* x$$

is isometric. For the projector field $Q_z = G_z Q_K G_z^*$ and its block operator Q_n ,

$$Q_{n+1} \mathcal{R}_n = \mathcal{R}_n Q_n, \quad \mathcal{R}_n^* Q_{n+1} \mathcal{R}_n = Q_n.$$

The identities compose between levels and pass to the inductive limit.

Proof. The successors have orthogonal supports and the weights sum to one; the frame changes are orthogonal. In each successor, $Q_{z'} G_{z'} G_z^* = G_{z'} G_z^* Q_z$, proving the first square. Multiplication by the adjoint of the refinement gives the second. Compatibility under composition defines the operator on the limit of the isometric inclusions. The same argument applies to E_u . $\square$

The statement preserves the representation of a prefix under compatible refinements. New events may produce another register; temporal constancy of K does not follow. The ranks of fields on tensor fibres are those of these fibres, not a count of physical dimensions.

Proposition 67.12 (A transport charge with an explicit action). *Let $U_n : E_n \rightarrow E_{n+1}$ be invertible transports. The auxiliary discrete action with cotangent variables is*

$$\mathcal{A}(q, p) = \sum_n p_{n+1}^* (q_{n+1} - U_n q_n).$$

Finitely supported variations give $q_{n+1} = U_n q_n$ and $p_n = U_n^* p_{n+1}$. If $G_0 = I$, $G_{n+1} = U_n G_n$, $A_0 = Q_K$, and $A_n = G_n A_0 G_n^{-1}$, the quantity

$$J_n = p_n^* A_n q_n$$

is independent of n . The action is invariant under the continuous group $q_n \mapsto \exp(tA_n)q_n$, $p_n \mapsto \exp(-tA_n^*)p_n$.

Proof. The indicated equations follow by varying p_{n+1} and q_n . The definition of A_n implies $A_{n+1}U_n = U_n A_n$. Hence

$$p_{n+1}^* A_{n+1} q_{n+1} = p_{n+1}^* U_n A_n q_n = p_n^* A_n q_n.$$

The same intertwining holds for the exponentials and makes every term of the action invariant. It is not required that A_n be antisymmetric: the covector transformation is contragredient. $\square$

For orthogonal transports one may choose $p_n = q_n$; then $J_n = \|A_n q_n\|^2$, and the complement provides the remaining charge. In a symplectic realization without a positive form, the pairing written above is preserved and is not called positive energy. If the fibres are archived through isometries $\mathcal{V}_n$, the definitions

$$\hat{U}_n = \mathcal{V}_{n+1} U_n \mathcal{V}_n^*, \quad \hat{A}_n = \mathcal{V}_n A_n \mathcal{V}_n^*, \quad \hat{q}_n = \mathcal{V}_n q_n,$$

with $\hat{p}_n(y) = p_n(\mathcal{V}_n^* y)$ preserve the intertwining and the same charge on the images. This composition provides a concrete continuous action and symmetry for a conservation law; it attributes no automatic Noether currents to a finite group, nor does it identify the auxiliary action with every physical action.

67.10 Encoding, operator identification, and incidence stability

The contribution assembled here is an operational chain: the generated register has a reversible encoding; its orbit is a complete reference; terminal component and complements preserve this reference and the transported observables; the integer separation of two memories allows incidence and direction to be restored even when the real-valued selector has zero margin. The proofs concerning linearity, projection, and lattices are used as the language of subsequent realization, without changing the production of the object.

Each hypothesis has a witness that prevents its inadvertent extension. An isolated response does not identify an operator that fails to commute with the cycle; only complete memory automatically preserves the original Gram matrix; cross terms cannot be omitted; absolute selection requires charge; the projector requires a nonzero direction; noise bounds depend on normalization; a chart transformation requires transporting the form, lattice, and readers. A perturbation of a reading does not become an admissible history by decree. These delimitations preserve the functions of K , without reducing it to twelve disconnected numbers or promoting a canonical property to universal invariance.

68 Ternary algebraic norm and optimal recovery radii

The recovery radius of the two memories contains the factor $\sqrt{73}$. Its origin can be traced without introducing a target magnitude: it comes from the Gram matrix of the nonadic reader and, in another realization of the ternary cycle already constructed, is the scale factor of a similitude. Its square, 73, is the index of the rank-two lattice inclusion. These occurrences are brought together through explicit operations; they are not identified through a decimal resemblance to alpha. The subsequent comparison with $10^4\alpha_A$ will be treated as an inverse problem, with its data, permitted transformations, and exact conclusions.

The starting material preserves its APP-TRIT-TPK production and the subsequent point of departure described in section 67.1. The integer register, orientation of the shifts, charge, and readers are received before distances are calculated. The canonical case uses $8 + 1$ and denominator nine. The parameter $a > 0$ introduced below defines a subsequent algebraic collection, not a new seed or a physical parameter selected by alpha. The errors belong to the reading channel; they are not identified with admissible events or histories.

68.1 A two-memory collection and its Gram matrix

In $V = \mathbb{R}^{12}$, let $(Sx)_i = x_{i+1}$, with indices modulo twelve. For $a > 0$ and $j = 3, 4$, define

$$T_{j,a} = \frac{aI + S^j}{a+1}, \quad D_{j,a} = \frac{\sqrt{a}(S^j - I)}{a+1}, \quad M_a x = (D_{3,a}x, D_{4,a}T_{3,a}x). \quad (68.1)$$

The second memory is produced after the first compression. Archiving both is not equivalent to reinjecting the first complement into a reversible two-channel evolution. Let

$$p(a) = a^2 + a + 1, \quad A(a) = a(a^2 + 1), \quad B(a) = a^2, \quad \Pi = I - \frac{\mathbf{1}\mathbf{1}^*}{12}, \quad P_u = I - \Pi.$$

Proposition 68.1 (Balance and difference form). *For every $a > 0$,*

$$T_{j,a}^* T_{j,a} + D_{j,a}^* D_{j,a} = I, \quad G_a := M_a^* M_a = I - (T_{4,a} T_{3,a})^* (T_{4,a} T_{3,a}),$$

and

$$(a+1)^4 G_a = 4ap(a)I - A(a)(S^3 + S^4 + S^8 + S^9) - B(a)(S + S^5 + S^7 + S^{11}). \quad (68.2)$$

Equivalently,

$$(a+1)^4 \|M_a h\|^2 = A(a) \sum_i ((h_i - h_{i+3})^2 + (h_i - h_{i+4})^2) + B(a) \sum_i ((h_i - h_{i+1})^2 + (h_i - h_{i+5})^2). \quad (68.3)$$

Proof. Orthogonality of S gives

$$T_{j,a}^* T_{j,a} = \frac{(a^2 + 1)I + a(S^j + S^{-j})}{(a+1)^2}, \quad D_{j,a}^* D_{j,a} = \frac{a(2I - S^j - S^{-j})}{(a+1)^2}.$$

Their sum is I ; applying the identity to the two stages cancels the intermediate term $T_{3,a}^* T_{3,a}$. Since all operators are polynomials in S , the product of the two individual Gram matrices has diagonal $(a^2 + 1)^2$, single terms of weight $a(a^2 + 1)$, and cross terms of weight a^2 . The identity $(a+1)^4 - (a^2 + 1)^2 = 4a(a^2 + a + 1)$ gives (68.2). Expanding the squared differences gives (68.3); each unoriented edge of each step appears exactly once, and the weighted degree is $4A + 4B = 4ap(a)$. $\square$

At $a = 8$ we obtain

$$6561G_8 = 2336I - 520(S^3 + S^4 + S^8 + S^9) - 64(S + S^5 + S^7 + S^{11}), \quad (68.4)$$

whose first row is $(2336, -64, 0, -520, -520, -64, 0, -64, -520, -520, 0, -64)/6561$. Also,

$$T_{4,8}T_{3,8} = \frac{64I + 8S^3 + 8S^4 + S^7}{81}.$$

The four distinct shifts produce the diagonal contribution

$$1 - \frac{64^2 + 8^2 + 8^2 + 1}{81^2} = \frac{2336}{6561} = \frac{32(8^2 + 8 + 1)}{9^4}.$$

Theorem 68.2 (Kernel, spectrum, and complete inverse). *For every $a > 0$, $\ker M_a = \mathbb{R}\mathbf{1}$. The eigenvalues of G_a , indexed by the twelfth-root modes m , are*

m	$\lambda_m(G_a)$	multiplicity
0	0	1
3, 9	$2a/(a+1)^2$	2
4, 8	$3a/(a+1)^2$	2
6	$4a/(a+1)^2$	1
1, 5, 7, 11	$a(5a^2 + 4a + 5)/(a+1)^4$	4
2, 10	$a(7a^2 + 2a + 7)/(a+1)^4$	2

In particular,

$$L_a^{\text{rec}} = (G_a + P_u)^{-1}M_a^*, \quad L_a^{\text{rec}}M_a = \Pi, \quad \|L_a^{\text{rec}}\| = \frac{a+1}{\sqrt{2a}}. \quad (68.5)$$

Proof. If $M_a h = 0$, the first memory implies $S^3 h = h$, hence $T_{3,a} h = h$; the second implies $S^4 h = h$. Since $S = S^4(S^3)^{-1}$, the vector is uniform. The converse is immediate. In mode m , the balance gives

$$\lambda_m(G_a) = 1 - \frac{[a^2 + 1 + 2a \cos(\pi m/2)][a^2 + 1 + 2a \cos(2\pi m/3)]}{(a+1)^4}.$$

The twelve cosine pairs give the table. Relative to $2a/(a+1)^2$, the differences of the last two values are

$$\frac{3a(a^2 + 1)}{(a+1)^4} > 0, \quad \frac{a(5a^2 - 2a + 5)}{(a+1)^4} > 0.$$

The other positive values are larger multiples. Hence the minimum on $\mathbf{1}^\perp$ is $2a/(a+1)^2$. The operator $G_a + P_u$ equals one on the uniform direction and is positive and invertible on its complement. The formula displayed is the least-squares inverse, and its norm is the reciprocal of the smallest nonzero singular value of M_a . $\square$

At $a = 1$, $T_{3,1} = (I + S^3)/2$ is singular, but (68.5) remains valid. Singularity of an individual compression does not eliminate recovery through its two memories. At $a = 8$, the norm of the inverse is $9/4$.

68.2 Minimum distances: all integer registers

The algebraic decoding domains are

$$\mathcal{C}_0(a) = M_a \mathbb{Z}^{12}, \quad \Lambda_q = \{k \in \mathbb{Z}^{12} : \mathbf{1}^* k = q\}, \quad \mathcal{C}_q(a) = M_a \Lambda_q.$$

The first image is identified with the lattice $\Pi \mathbb{Z}^{12}$ or with classes modulo $\mathbb{Z}\mathbf{1}$; the second preserves a fixed integer charge. These are domains containing the integer representations of the indicated type. Their use does not assert that every candidate corresponds to a produced HMT history.

Lemma 68.3 (Minimum cut of the memory graph). *Every nonempty proper cut of the twelve positions has at least four edges of steps 1, 5 and at least four of steps 3, 4. Its weight in (68.3) is therefore at least $4A(a) + 4B(a) = 4ap(a)$; a cut consisting of one position attains this bound.*

Proof. Each of steps 1 and 5 forms a connected cycle of twelve positions; each cycle crosses a nontrivial cut at least twice. For steps 3, 4, if the set is invariant under neither, each collection contributes at least two edges. If it is invariant under S^3 , it is a union of classes modulo three; each of the four triangles of S^4 crosses it twice, giving eight edges. If it is invariant under S^4 , it is a union of classes modulo four; each of the three length-four cycles of S^3 crosses it at least twice, giving at least six. Simultaneous invariance would force invariance under S , which is impossible for a nonempty proper set. All weights are positive; the cut consisting of a single position has exactly four edges of each weight. $\square$

Theorem 68.4 (Exact distances and fixed-charge minimizers). *For every $a > 0$ and every integer charge q ,*

$$d_0(a)^2 = \min_{h \in \mathbb{Z}^{12}} \|M_a h\|^2 = \frac{4ap(a)}{(a+1)^4}, \quad d_q(a)^2 = \frac{8ap(a)}{(a+1)^4}. \quad (68.6)$$

In the first minimum, h ranges over $\mathbb{Z}^{12}$; the second is the minimum distance between distinct words of $\mathcal{C}_q(a)$. The differences attaining the second minimum are exactly $e_i - e_j$ with $j - i \equiv 2, 6, 10 \pmod{12}$.

Proof. For a nonconstant integer vector h , $(h_i - h_j)^2 \geq |h_i - h_j|$. Each absolute difference decomposes as a sum of cuts of the level sets $\{i : h_i > t\}$. There is at least one nontrivial level; by lemma 68.3, the weighted energy is at least $4ap(a)$. The vector e_i attains this energy. Division by $(a+1)^4$ proves the first minimum over all integer vectors, not only over indicators of subsets.

For fixed charge, the difference $h \neq 0$ is integral and has zero sum. Its squared norm is even because $h_i^2 \equiv h_i \pmod{2}$. If $\|h\|^2 \geq 4$, the spectral minimum gives

$$\|M_a h\|^2 \geq \frac{8a}{(a+1)^2} > \frac{8ap(a)}{(a+1)^4},$$

since $(a+1)^2 - p(a) = a > 0$. Only $\|h\|^2 = 2$ remains, that is, $h = e_i - e_j$. Its energy is $2((G_a)_{ii} - (G_a)_{ij})$. The off-diagonal entries are zero at separations 2, 6, 10, negative with weight $B(a)/(a+1)^4$ at 1, 5, 7, 11, and negative with weight $A(a)/(a+1)^4$ at 3, 4, 8, 9. The diagonal is $4ap(a)/(a+1)^4$; this proves the minimum and the exhaustive classification over the entire half-line $a > 0$. $\square$

Theorem 68.5 (Strict correction radii of the memories). *For a datum $y = M_a k + e$, with error measured in the joint norm of the two memories, the optimal uniform radii are*

$$r_0(a) = \frac{\sqrt{ap(a)}}{(a+1)^2}, \quad r_q(a) = \frac{\sqrt{2ap(a)}}{(a+1)^2}. \quad (68.7)$$

If $\|e\| < r_0(a)$, the nearest neighbour in $\mathcal{C}_0(a)$ recovers Πk exactly. With known charge q and $\|e\| < r_q(a)$, the nearest neighbour in $\mathcal{C}_q(a)$ recovers k exactly.

Proof. The lattice $\Pi\mathbb{Z}^{12} \subset (1/12)\mathbb{Z}^{12}$ is discrete, and M_a is bounded below on $\mathbf{1}^\perp$. Its images are discrete and admit a nearest neighbour. If c' is another codeword and d is the corresponding minimum distance, $\|y - c'\| \geq d - \|e\| > \|e\|$ when $\|e\| < d/2$. The minimum is unique, and the preimage is unique in each indicated domain. The midpoint of two words at minimum distance lies at $d/2$ from both: no decoder can uniformly distinguish both inputs from that same datum. Optimality refers to the complete algebraic domains, not to an additional subset of histories that might have greater separation. $\square$

In the canonical case,

$$r_0(8) = \frac{2\sqrt{146}}{81} \simeq 0.298346814162829, \quad r_q(8) = \frac{4\sqrt{73}}{81} \simeq 0.421926110879878. \quad (68.8)$$

The 4 comes from $\sqrt{2} \cdot 8$ when taking half the distance between fixed-charge registers; $73 = 8^2 + 8 + 1$ comes from the Gram matrix; $81 = 9^2$ comes from the two nonadic normalizations. The same product 9×9 counts the APP cells, but these two roles are not identified without their typed map. Nor does the factor four alone prove an assertion about spatial directions.

If the memories are stored as $z = M_a k / \sqrt{a}$, the noise norm changes. At $a = 8$, the radii for this rational datum are $r_0 / \sqrt{8} = \sqrt{73}/81$ and $r_q / \sqrt{8} = \sqrt{146}/81$. Using the radii in (68.8) without this conversion changes the problem. The application to Q_K , η_K , B_K , and the incidence screen has been proved in section 67.8; in particular, the exact threshold $K_4 = 729$ does not prevent its discrete restoration.

68.3 Finite decoder and parameter duality

Proposition 68.6 (Floor and ceiling without a target table). *With known charge, below radius $r_q(a)$, it suffices to examine at most $2^{12} = 4096$ integer candidates. Without charge, below $r_0(a)$, it suffices to examine at most $12 \cdot 2^{12} = 49152$ candidates distributed among twelve charge residues. The residual minimum returns k or its uniform class, respectively.*

Proof. With charge q , form $\tilde{k} = L_a^{\text{rec}} y + (q/12)\mathbf{1}$. Then

$$\|\tilde{k} - k\| < \|L_a^{\text{rec}}\| r_q(a) = \frac{\sqrt{p(a)}}{a+1} < 1.$$

Each true integer coordinate belongs to $\{\lfloor \tilde{k}_i \rfloor, \lceil \tilde{k}_i \rceil\}$. These products are enumerated, those whose sum differs from q are removed, and $\|M_a k' - y\|$ is minimized. The correct candidate is present, and theorem 68.5 ensures its uniqueness. Coordinates that are already integers reduce the number of candidates.

Without charge, write $\rho \in \{0, \dots, 11\}$ for the charge residue. Each class modulo $\mathbb{Z}\mathbf{1}$ has a unique representative whose sum is ρ , obtained by subtracting $\lfloor q/12 \rfloor \mathbf{1}$. For each ρ , form $\tilde{k}_\rho = L_a^{\text{rec}} y + (\rho/12)\mathbf{1}$. At the correct residue,

$$\|\tilde{k}_\rho - k_\rho\| < \|L_a^{\text{rec}}\| r_0(a) = \frac{\sqrt{p(a)}}{\sqrt{2}(a+1)} < 1.$$

Apply floor and ceiling, retain the sum ρ , and minimize the residual across the twelve groups. Uniqueness of the code class gives a unique centred output. These procedures use the reader and integrality; they do not receive K as a comparison table. $\square$

For rational a , rational arithmetic can be retained by writing $R_a = M_a / \sqrt{a}$, $z = y / \sqrt{a}$. The centred reconstruction is

$$(R_a^* R_a + P_u)^{-1} R_a^* z = L_a^{\text{rec}} y.$$

The residual can be compared through its rational squares. These guarantees concern twelve coordinates: they assert neither polynomial complexity for arbitrary dimension, cryptographic authentication, nor quantum advantage. The perturbed example in section 67.8 illustrates the canonical algorithm.

Proposition 68.7 (Reciprocal duality and maximum radius). *The substitution $a \mapsto 1/a$ preserves the Gram matrix, distances, radii, and norm of the inverse. We have*

$$D_{j,1/a} = D_{j,a}, \quad T_{j,1/a} = S^j T_{j,a}^*.$$

The maximum radii of this collection are attained only at $a = 1$:

$$r_q(a) \leq \frac{\sqrt{6}}{4}, \quad r_0(a) \leq \frac{\sqrt{3}}{4}.$$

Proof. The two operator identities follow by multiplying numerator and denominator by a . The individual Gram matrices and formula (68.1) are invariant under reciprocity. Put $t = a/(a+1)^2 \in (0, 1/4]$. Then

$$r_q(a)^2 = 2t(1-t), \quad r_0(a)^2 = t(1-t).$$

Both expressions increase for $0 < t \leq 1/4$, and $t = 1/4$ is equivalent to $a = 1$. Reciprocity leaves t fixed. As a tends to zero or infinity, t and the radii tend to zero. $\square$

Equality of Gram matrices does not mean that $M_a = M_{1/a}$: the second complement includes a different compression. The endpoints $a = 0$ and $a = \infty$ do not belong to the theorem's domain. Optimizing the collection at $a = 1$, or using the reciprocal representative $1/8$, does not retroactively change the canonical origin $a = 8$.

68.4 The norm on the ternary augmentation module

The cycle $r \mapsto 4r \pmod{9}$, with orbits $(1, 4, 7)$ and $(2, 8, 5)$, supplies the augmentation module whose chart and form are

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad C^2 + C + I = 0, \quad C^T G_{A_2} C = G_{A_2}.$$

The action and its contragredient orientation belong to the antecedent ternary fibres. On this already constructed module, we compose

$$Q_-(a) = aI - C, \quad Q_+(a) = (a+1)I + C.$$

Proposition 68.8 (Ternary similitude and integer index). *For $a > 0$,*

$$Q_- Q_+ = p(a)I, \quad Q_-^{\dagger G_{A_2}} = Q_+, \quad Q_-^T G_{A_2} Q_- = p(a)G_{A_2}, \quad \det Q_- = p(a).$$

At $a = 8$, the two conjugate charts are

$$Q_- = \begin{pmatrix} 8 & 1 \\ -1 & 9 \end{pmatrix}, \quad Q_+ = \begin{pmatrix} 9 & -1 \\ 1 & 8 \end{pmatrix}, \quad Q_- Q_+ = 73I.$$

Each integer quotient is cyclic of order 73. On the quotient of Q_- , C acts by 8; on that of Q_+ , by $-9 \equiv 64 \equiv 8^{-1} \pmod{73}$.

Proof. Expanding the product and substituting $C^2 = -C - I$ gives $p(a)I$. Invariance of the form implies $C^{\dagger G_{A_2}} = C^{-1} = C^2 = -I - C$, from which the adjoint follows. Composing product and adjoint gives the similitude. The determinant of $\begin{pmatrix} a & 1 \\ -1 & a+1 \end{pmatrix}$ is $p(a)$. At $a = 8$, the greatest common divisor of the entries of each matrix is one and its determinant is 73; the Smith normal form is $\text{diag}(1, 73)$. The relation $8I - C = 0$ in the first quotient and $9I + C = 0$ in the second gives the indicated actions. Since $8^3 \equiv 1 \pmod{73}$ and $8 \not\equiv 1$, both actions have order three and opposite orientations. $\square$

Thus $\sqrt{73}$ is a similitude scale factor in the inherited ternary metric, not merely a number inside a radius. Its subsequent complex recognition is $|8 - \omega|^2 = 73$, with $\omega^2 + \omega + 1 = 0$. The complex unit describes the realization of the operator; it was not used to select the register. These equalities do not identify the operator Q_- with K , with the entire TPK dynamics, or with the cokernel of the action-scale reader.

68.5 The complete cycle and the origin of the fixed term

On the twelve positions, let $U = S^4$. There are now four cycles of length three. Their projectors are

$$P_0 = \frac{I + U + U^2}{3}, \quad P = I - P_0.$$

The rank of P_0 is four and that of P is eight. In particular, $P_0 \neq P_u$, since the latter preserves only the global uniform direction; nor is P to be confused with the exceptional projector P_3 . We shall now use $Q_{-,a} = aI - U$ and $Q_{+,a} = (a+1)I + U$ for the operators of the complete cycle.

Theorem 68.9 (Ternary Gram matrix with a fixed sector). *We have*

$$Q_{-,a}^* Q_{-,a} = (a-1)^2 P_0 + p(a)P = p(a)I - 3aP_0. \quad (68.9)$$

For $v \neq 0$, let

$$w(v) = \frac{\|P_0 v\|^2}{\|v\|^2}, \quad s_{-,a}(v) = \frac{\|Q_{-,a} v\|^2}{\|v\|^2}.$$

Then $s_{-,a}(v) = p(a) - 3aw(v)$. In the canonical case,

$$Q_-^* Q_- = 49P_0 + 73P = 73I - 24P_0, \quad s(v) = 73 - 24w(v) \in [49, 73]. \quad (68.10)$$

Proof. From $U^3 = I$ and $U^* = U^2$ it follows that P_0 is the orthogonal projector onto $\ker(U - I)$. Expanding the Gram matrix,

$$Q_{-,a}^* Q_{-,a} = (a^2 + 1)I - a(U + U^2).$$

The relation $U + U^2 = 3P_0 - I$ gives (68.9). Evaluating on v and dividing by its squared norm proves the remaining identities. $\square$

The term 24 is exactly $3 \cdot 8$: it is the difference $p(8) - (8-1)^2$ between the augmentation reading and the fixed reading. It already appears in a single cycle of three positions. For m cycles, the fixed rank is m , the dimension is $3m$, and

$$\text{tr}(Q_{-,a}^* Q_{-,a}) = 3m(a^2 + 1).$$

At $m = 4, a = 8$, the trace is 780 and the total trace correction is $24 \cdot 4 = 96$. The numerical equality $24 = 2 \cdot 12$ is not the operator origin of the coefficient.

The two complete charts also distinguish their determinants. On a single cycle,

$$\det(8I - U) = 8^3 - 1 = 511 = 7 \cdot 73, \quad \det(9I + U) = 9^3 + 1 = 730 = 10 \cdot 73.$$

The fixed factor is 7 and 10, respectively; the augmentation module contributes 73. On the twelve positions, $\det(8I - U) = 511^4 = 68184176641$. The identity

$$73 = 8^2 + 8 + 1 = 9^2 - 9 + 1 = \frac{9^3 + 1}{9 + 1} = \frac{729 + 1}{10}. \quad (68.11)$$

is preserved. Its interpretation as an index and determinant factor precedes any subsequent completion or refinement reading.

68.6 Six received states and conservation of the complete reading

We again fix $a = 8$ and omit that subscript in the readers. The six objects being compared are

$$K, \quad k = \Pi K, \quad m_0 = D_3 K, \quad m_1 = D_4 T_3 K, \quad u = P_3 k, \quad q_K = Q_K K = k - u.$$

None is selected from the target of the test. To give the evaluations without replacing them by decimals, write $a_v = \|v\|^2$, $b_v = \|P_0 v\|^2$. We obtain

v	a_v	b_v	
K	4251257	10684363/3	
k	11789915/12	1170761/4	
m_0	15413392/81	16838032/243	(68.12)
m_1	1105759232/6561	0	
u	$1327717/4 + (568896/5)\sqrt{5}$	$3029657/20 + (145120/3)\sqrt{5}$	
q_K	$1951691/3 - (568896/5)\sqrt{5}$	$2292404/15 - (145120/3)\sqrt{5}$	

The table is obtained by applying $P_0 = (I + S^4 + S^8)/3$, the formulas for $D_3, D_4 T_3$, and the finite character projector (67.12) to (67.2). All its entries belong to $\mathbb{Q}(\sqrt{5})$; the first four are rational. In particular, the first three weights and readings are

$$\begin{aligned} w(K) &= \frac{10684363}{12753771}, & s(K) &= \frac{224866857}{4251257}, \\ w(k) &= \frac{3512283}{11789915}, & s(k) &= \frac{776369003}{11789915}, \\ w(m_0) &= \frac{1052377}{2890011}, & s(m_0) &= \frac{61904585}{963337}. \end{aligned}$$

For u and q_K , the exact expressions are $w = b_v/a_v$, $s = 73 - 24b_v/a_v$, with the numerators and denominators in (68.12). The subsequent evaluations are

v	$w(v)$, approximately	$s(v)$, approximately
K	0.8377414805	52.89420446705527
k	0.2979057101	65.85026295779062
m_0	0.3641429046	64.26057028848679
m_1	0	73 exact
u	0.4428244530591254	62.37221312658099
q_K	0.1127385160275117	70.29427561533972

The equality for the second memory is forced: $P_0 D_4 T_3 = \sqrt{8} P_0 (U - I) T_3 / 9 = 0$, while $\|m_1\|^2 > 0$. It is not an evaluational coincidence.

Proposition 68.10 (Complete memory does not alter the fixed weight). *The map*

$$V_2 v = (t(v), m_0(v), m_1(v)) = (T_4 T_3 v, D_3 v, D_4 T_3 v)$$

is isometric and intertwines P_0 and $Q_- = 8I - U$ with their diagonal copies. It separately preserves $\|v\|^2$, $\|P_0 v\|^2$, and $\|Q_- v\|^2$. Hence the joint readings w and s do not change.

Proof. The two-stage balance gives $V_2^* V_2 = I$. Each channel is a polynomial in S , so it commutes with U, P_0, Q_- . Applying the isometry to $v, P_0 v$, and $Q_- v$ gives the three norm equalities. After division, the total reading is the average of the channel readings weighted by their squared norms, not their unweighted arithmetic mean. Zero channels do not contribute to the total ratio. $\square$

The terminal component explicitly confirms this partition. For K ,

$$\begin{aligned} \|t(K)\|^2 &= \frac{25538253193}{6561}, & \|P_0 t(K)\|^2 &= \frac{848595371}{243}, \\ w(t(K)) &= \frac{22912075017}{25538253193}, & s(t(K)) &= \frac{1314402682681}{25538253193} \simeq 51.46799480558349. \end{aligned}$$

The sum of the three squared norms is 4251257, and that of the fixed-sector norms is $(848595371 + 16838032)/243 = 10684363/3$. For $k = \Pi K$,

$$\|t(k)\|^2 = \frac{16367568169}{26244}, \quad \|P_0 t(k)\|^2 = \frac{217142795}{972}.$$

The complements do not change under centring, and their sums recover the norms in the second row of (68.12). Discarding the terminal component or a complement alters the partial reading; retaining them introduces no hidden correction.

In a complete infinite memory, the same conclusion holds when the channels commute with U : the positive limit and its square root also commute with it. Without this commutation, the transport $\mathcal{V}P_0\mathcal{V}^*$ is used on the image, in accordance with theorem 67.6, not an assumed diagonal copy.

68.7 Inversion, charts, and bilateral balance

Proposition 68.11 (What inversion or a frame change preserves). *Replacing U by U^{-1} exchanges each $Q_{\pm,a}$ with its adjoint and preserves P_0 , their Gram matrices, singular values, determinants, and induced distances. For every $v \neq 0$, it preserves $w(v)$ and $s_{-,a}(v)$. This is not the same as inverting $Q_{-,a}$, whose inverse, for $a \neq 1$, is*

$$Q_{-,a}^{-1} = \frac{1}{a-1}P_0 + \frac{1}{p(a)}Q_{+,a}P. \quad (68.13)$$

*For an invertible chart R , transporting $U' = RUR^{-1}$, $v' = Rv$, and $g' = (R^{-1})^*R^{-1}$ preserves these norm ratios in the metric g' .*

Proof. Since $U^* = U^{-1} = U^2$, inversion produces the adjoints. The operators are normal polynomials in U ; their Gram matrices depend on $U + U^2$, which does not change. The displayed inverse is verified sector by sector: $Q_{-,a}$ acts by $a - 1$ on the fixed sector, and its product with $Q_{+,a}$ equals $p(a)I$ on the augmentation sector. For the chart, $Q'v' = RQv$ and $R^*g'R = I$; numerator and denominator recover exactly their original values. The adjoint with respect to g' is also the transport of the original adjoint. $\square$

Reversing the order of the twelve positions satisfies $RSR^{-1} = S^{-1}$ and preserves P_0 . The translations S^j , including the antipodal translation S^6 , commute with U ; a global negation also leaves the norms unchanged. None of these operations alone turns 73 into a different reading.

With the Euclidean metric fixed, an orthogonal frame applied only to the state preserves $s_{-,a}(v)$ for all v if and only if $R^*P_0R = P_0$, by (68.9) and polarization. The group of these transformations is $O(\text{im } P_0) \times O(\text{im } P) \cong O(4) \times O(8)$, larger than the commutant of U . By contrast, a state filter $c_0P_0 + c_1P$, when its output is nonzero, produces

$$w' = \frac{|c_0|^2w}{|c_0|^2w + |c_1|^2(1-w)}. \quad (68.14)$$

It can change the reading. Choosing these coefficients from a target is a declared fitting of the filter, not a change of chart for the same reading.

Theorem 68.12 (Bilateral balance and explicit recovery). *For every $a > 0$, the complete cycle satisfies*

$$\begin{aligned} Q_{+,a} - Q_{-,a}^* &= 3P_0, & Q_{+,a}Q_{-,a} &= p(a)I - 3P_0, \\ Q_{+,a}^*Q_{+,a} &= p(a)I + 3(a+1)P_0, \end{aligned} \quad (68.15)$$

and

$$(a+1)Q_{-,a}^*Q_{-,a} + aQ_{+,a}^*Q_{+,a} = (2a+1)p(a)I. \quad (68.16)$$

The map

$$W_a v = \frac{(\sqrt{a+1}Q_{-,a}v, \sqrt{a}Q_{+,a}v)}{\sqrt{(2a+1)p(a)}} \quad (68.17)$$

is isometric and is inverted on its image by W_a^* , of norm one. For the unnormalized channels $x = Q_{-,a}v$, $y = Q_{+,a}v$,

$$v = \frac{x+y}{2a+1}, \quad Uv = \frac{ay - (a+1)x}{2a+1}. \quad (68.18)$$

A pair (x, y) belongs to this unnormalized image if and only if $Q_{+,a}x = Q_{-,a}y$.

Proof. The expression $Q_+ - Q_-^*$ contains $I + U + U^2 = 3P_0$. Expanding Q_+Q_- , $Q_+^*Q_+$, and using $U^2 = 3P_0 - I - U$ gives (68.15). The weighted combination cancels the P_0 terms in (68.9) and (68.15); this proves $W_a^*W_a = I$. Since $Q_- + Q_+ = (2a + 1)I$, the sum of the channels gives v ; their combination $ay - (a + 1)x$ gives $(2a + 1)Uv$.

The image condition is necessary because the two operators commute. If it holds, define $v = (x + y)/(2a + 1)$. Then

$$Q_-v = \frac{Q_-x + Q_-y}{2a + 1} = \frac{(Q_- + Q_+)x}{2a + 1} = x,$$

and similarly $Q_+v = y$. The adjoint inverse formula is that of any isometry; for the raw pair, the linear inverse $(x, y) \mapsto (x + y)/(2a + 1)$ has norm $\sqrt{2}/(2a + 1)$ on the product space. $\square$

At $a = 8$, the fixed and augmentation sectors have readings (49, 100) and (73, 73), respectively, and

$$9\|Q_-v\|^2 + 8\|Q_+v\|^2 = 17 \cdot 73\|v\|^2 = 1241\|v\|^2, \quad v = \frac{Q_-v + Q_+v}{17}.$$

For $s_{+,a}(v) = \|Q_{+,a}v\|^2/\|v\|^2$,

$$s_{+,a} = p(a) + 3(a + 1)w, \quad (a + 1)s_{-,a} + as_{+,a} = (2a + 1)p(a).$$

Cancellation holds even with a fixed component. The bilateral composition is a formalization on antecedent operators, not an assertion that every TPK return is automatically this pair.

If $\mathcal{V} : H \rightarrow \mathcal{M}$ is a complete isometric memory, $U^\mathcal{V} = \mathcal{V}U\mathcal{V}^*$ satisfies $(U^\mathcal{V})^3 = \Pi_{\mathcal{M}}$, where $\Pi_{\mathcal{M}} = \mathcal{V}\mathcal{V}^*$ is the identity on the image. There the transported Gram matrix is

$$(Q_{-,a}^\mathcal{V})^*Q_{-,a}^\mathcal{V} = p(a)\Pi_{\mathcal{M}} - 3a\mathcal{V}P_0\mathcal{V}^*.$$

It preserves the readings and bilateral recovery. Using diagonal copies of U is legitimate only when the channels intertwine that action; otherwise transport by $\mathcal{V}$ is the correct definition on the image.

68.8 The alpha datum and collections of integer iterations

The complete alpha reader, with carry, supplies the block prefix

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

The zero-boundary window ending in 663 is a different object; it is not interchanged with the complete output. The inherited closed interval suffices for the test:

$$\frac{7297352569283800997285105472380662}{10^{36}} \leq \alpha_A \leq \frac{7297352569283800997285105472380663}{10^{36}}. \quad (68.19)$$

The constant is not identified with either of its rational endpoints. Put $s_\alpha = 10^4\alpha_A$, so that

$$\begin{aligned} \ell &\leq s_\alpha \leq h, \\ \ell &:= 72.97352569283800997285105472380662, \\ h &:= 72.97352569283800997285105472380663. \end{aligned} \quad (68.20)$$

The inverse condition for the fixed reader is exactly

$$s(v) = s_\alpha \iff w(v) = \frac{73 - s_\alpha}{24} \simeq 0.00110309613175. \quad (68.21)$$

None of the six states in section 68.6 satisfies this condition. The second memory has $s = 73$, whereas the other five have $s < 72$. This last inequality is verified exactly as $24b_v > a_v$. In the three rational rows, multiplying integers suffices; for u and q_K , the differences are

$$24b_u - a_u = \frac{66073183}{20} + \frac{5235904}{5}\sqrt{5} > 0,$$

and satisfy, in each row,

$$\frac{A49^N}{A49^N + B73^N} > w_H > w_L > \frac{A49^{N+1}}{A49^{N+1} + B73^{N+1}}.$$

These are integer inequalities: for example, if $w_H = p_H/q_H$, the first is equivalent to $(q_H - p_H)A49^N > p_H B73^N$, and the last is verified analogously with w_L . Substitution of the three rows and the displayed integers proves the strict crossings without rounding alpha. Strict monotonicity in theorem 68.13 excludes all preceding and subsequent indices; between N and $N + 1$ there is no other integer. For m_1 , $w_0 = 0$ remains zero by (68.22). $\square$

For numerical reference, the readings on either side of the crossings are

v	s_N	s_{N+1}
K	72.97136264842803	72.98077012438719
k	72.96167997982852	72.97426483341773
m_0	72.96527892867391	72.97668298511348

The proof covers all $n \geq 0$, not merely a finite sample. The opposite orientation gives the same crossings; positive powers of Q_+ and inverse powers of Q_- decrease the initial readings, already below 72, and likewise do not reach the target in these three collections. The mixed products $Q_-^n Q_+^m$ form a different, two-index class, with weight ratio $(49/73)^n (100/73)^m w_0 / (1 - w_0)$; they are not excluded by the preceding one-parameter proof, nor are indices chosen here to fit alpha.

68.9 Angular reciprocity and the existing regional correction

The chart changes in proposition 68.11 include the reciprocity and elliptic deformation already constructed. In a plane realized by M_η , the transported form is $M_\eta^{-T} M_\eta^{-1}$; similarity does not change the eigenvalues of the cycle. Retaining the previous Euclidean inner product after changing the chart would define a different observable. The involution $q_+ \leftrightarrow q_-$ preserves A , reverses C^* , and sends η_{el} to $-\eta_{\text{el}}$. The polar relation of the ellipse $\tan \phi = e^{-\eta_{\text{el}}} \tan \theta$ changes the parametrization of the curve, not the spectral phase of a similar transport. It adds no small phase to S^4 .

Incidence conjugation has a different type. Its block law preserves

$$z_m^\dagger = z_{\rho(m)} + 100 \mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20 c_{\rho(m)}, \quad c_m = [C_m]_{10},$$

with the boundary terms before decimal reduction. In general, it is not an isometry of the block vector. Applying it to the register requires its connection to the incidence register and its endpoints; it is not replaced by an invented angular correction.

Proposition 68.15 (Regional correction of the counter-angle). *The antecedent action and angle coordinates satisfy*

$$A_{\text{deg}} = 1000\alpha_A, \quad C_{\text{ret,deg}}^* = 2(\eta_{\text{ret}} + \alpha_A), \quad C_{\text{pre,deg}}^* = 2(H_5 + \alpha_A),$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} C_\pi(\alpha_A), \quad C_\pi(\alpha_A) = 169\alpha_A^6 + \frac{D_A \alpha_A^7}{1 - D_A \alpha_A}, \quad D_A = \exp(-100\pi\alpha_A/9).$$

The difference in radians is exactly

$$\theta_{C,\text{pre}} - \theta_{C,\text{ret}} = C_\pi(\alpha_A).$$

Proof. The difference in degrees is $2(H_5 - \eta_{\text{ret}}) = 180 C_\pi / \pi$. Multiplication once by $\pi/180$ gives the result. $\square$

Here π and α_A are the coordinates already generated; H_5 and the remaining readers retain their antecedent definitions, not a recalibration. The subsequent evaluations are

coordinate	radians, approximately
θ_A	0.127362829013
θ_C	0.0370662264658
$\theta_C/6$ in each antipodal pair	0.00617770441097
$\mathcal{C}_\pi$	$2.55207421805 \cdot 10^{-11}$

The channels $q_\pm$ are positive real attenuations; their exponent does not automatically become a unitary phase through a factor i . No additional divisors are introduced to approximate the scales of the following test.

68.10 Angular inverse test on the twelve positions

For a general target $s \in [49, 81]$, the principal orientation of the pure sector is obtained from

$$|8 - e^{i\theta}|^2 = 65 - 16 \cos \theta, \quad \theta(s) = \arccos \frac{65 - s}{16} \in [0, \pi].$$

In particular, the subsequent datum s_α determines

$$\theta_\alpha = \theta(s_\alpha), \quad \delta_\alpha = \frac{2\pi}{3} - \theta_\alpha.$$

This formula solves an inverse problem using an output already produced; it does not claim to produce alpha anew.

Lemma 68.16 (Angular structure of the ternary complement). *The operator $J = (U - U^2)/\sqrt{3}$ satisfies*

$$J^* = -J, \quad J^2 = -P, \quad JB_0 = B_0J = 0, \quad JP = PJ = J, \quad U = B_0 - \frac{1}{2}P + \frac{\sqrt{3}}{2}J.$$

Proof. The adjoint exchanges U, U^2 . Moreover, $(U - U^2)^2 = U + U^2 - 2I = -3P$, and $(I + U + U^2)(U - U^2) = 0$. The remaining relations follow from $P = I - B_0$; adding $(U + U^2)/2$ and $(U - U^2)/2$ gives the final expression. $\square$

Theorem 68.17 (Angular collection and preserved properties). *For $\theta \in \mathbb{R}$, define*

$$U_\theta = B_0 + \cos \theta P + \sin \theta J, \quad Q_{-, \theta} = 8I - U_\theta.$$

We have

$$\begin{aligned} U_\theta^* U_\theta &= I, & U_\theta U_\varphi &= U_{\theta+\varphi}, & U_{2\pi/3} &= U, \\ Q_{-, \theta}^* Q_{-, \theta} &= 49B_0 + (65 - 16 \cos \theta)P. \end{aligned} \tag{68.24}$$

The readers $T_\theta = (8I + U_\theta)/9$, $D_\theta = \sqrt{8}(U_\theta - I)/9$ preserve $T_\theta^ T_\theta + D_\theta^* D_\theta = I$. By contrast,*

$$U_\theta^3 = I \iff 3\theta \in 2\pi\mathbb{Z}, \quad \det Q_{-, \theta} = 7^4(65 - 16 \cos \theta)^4.$$

Proof. The relations in lemma 68.16 cancel the terms between sectors. On the complement, $J^2 = -I$, so that $U_\theta^* U_\theta = B_0 + (\cos^2 \theta + \sin^2 \theta)P = I$. Multiplying two expressions and using the addition formulas gives the composition law. The expression for U in the lemma gives the ternary value. Since $U_\theta + U_\theta^* = 2B_0 + 2 \cos \theta P$, expanding the Gram matrix gives (68.24). The expansions

$$81T_\theta^* T_\theta = 65I + 8(U_\theta + U_\theta^*), \quad 81D_\theta^* D_\theta = 16I - 8(U_\theta + U_\theta^*)$$

prove the balance. The composition law gives $U_\theta^3 = B_0 + \cos(3\theta)P + \sin(3\theta)J$, which is the identity exactly under the indicated condition.

For the determinant, $Q_{-, \theta}$ acts by 7 on the four fixed dimensions. On the complement, a unit vector e and Je form an orthonormal pair; their orthogonal complement is stable under J . Repeating this yields four planes, on each of which the determinant is $(8 - \cos \theta)^2 + \sin^2 \theta = 65 - 16 \cos \theta$. Multiplying the contributions proves the formula. $\square$

Corollary 68.18 (Inverse solution and separation of corrections). *On im P , the reader Q_{-, θ_α} has quadratic ratio exactly s_α . The orientations $\pm \theta_\alpha$ give the same ratio, and*

$$73 - s_\alpha = 8(1 - \cos \delta_\alpha + \sqrt{3} \sin \delta_\alpha).$$

We have $U_{\theta_\alpha}^3 \neq I$, and $\delta_\alpha \neq \mathcal{C}_\pi(\alpha_A)$.

Proof. The definition of the phase solves for the complementary coefficient in (68.24); cosine is even. Substituting $\theta_\alpha = 2\pi/3 - \delta_\alpha$ gives the stable identity for the difference. Since $72 < s_\alpha < 73$, $\pi/2 < \theta_\alpha < 2\pi/3$ and $0 < 3\delta_\alpha < \pi/2$; hence $3\theta_\alpha$ is not a multiple of 2π .

The separation of corrections can be proved without relying on approximate figures. The interval gives $73 - s_\alpha > 0.026$. If $\delta_\alpha \leq 10^{-3}$, the inequalities $1 - \cos t \leq t^2/2$, $\sin t \leq t$, and $\sqrt{3} < 2$ would give

$$73 - s_\alpha \leq 8(10^{-6}/2 + 2 \cdot 10^{-3}) = 0.016004,$$

a contradiction. Thus $\delta_\alpha > 10^{-3}$. On the other hand, $0 < \alpha_A < 10^{-2}$, $0 < D_A < 1$ imply

$$0 < \mathcal{C}_\pi(\alpha_A) < 169 \cdot 10^{-12} + \frac{10^{-14}}{1 - 10^{-2}} < 170 \cdot 10^{-12} < 10^{-7}.$$

The two quantities are distinct. □

The diagnostic evaluation gives

$$\delta_\alpha \simeq 0.00190956706826 \text{ rad} \simeq 0.109410133709^\circ, \quad \theta_\alpha \simeq 119.890589866291^\circ.$$

The regional correction is smaller by a factor of approximately $7.48 \cdot 10^7$. The inverse collection preserves orthogonality and balance, but loses order three at that phase and changes the original determinant. Its real chart alone provides neither an integer matrix nor the lattice quotient of index 73. Solving for the prescribed datum does not prove $73 = 10^4 \alpha_A$.

Proposition 68.19 (Reachability with the state preserved). *For $v \neq 0$, with $w = w(v)$ fixed,*

$$s_\theta(v) = 49w + (65 - 16 \cos \theta)(1 - w).$$

If $w < 1$, its image is exactly $[49, 81 - 32w]$. Hence a phase attaining s_α exists if and only if

$$w \leq \frac{81 - s_\alpha}{32} \simeq 0.250827322099.$$

When it exists, its principal orientation is

$$\theta_v = \arccos \left[\frac{1}{16} \left(65 - \frac{s_\alpha - 49w}{1 - w} \right) \right].$$

If $w = 1$, the reading is always 49 and does not reach the target.

Proof. Evaluating (68.24) and dividing by $\|v\|^2$ gives the formula. Cosine ranges over $[-1, 1]$, so the complementary coefficient ranges over $[49, 81]$. Its combination with the fixed weight gives the entire interval, without gaps. Solving the equality and checking the domain of arccosine gives the condition and the phase. The case $w = 1$ is evaluated without dividing by $1 - w$. □

Corollary 68.20 (Four impossibilities and two possible phases). *Keeping the state and fixed sector unchanged, the angular collection does not attain s_α on K, k, m_0, u . It does attain it on m_1 and q_K .*

Proof. From the exact table (68.12), we obtain

v	$32b_v - 9a_v$
K	$227115677/3$
k	$2094607/4$
m_0	$122655440/243$
u	$37201759/20 + (7859008/15)\sqrt{5}$

All entries are positive. Thus $w > 9/32$, and the maximum $81 - 32w$ is less than $72 < s_\alpha$. For m_1 , $b_{m_1} = 0$ and $a_{m_1} > 0$, so the interval is $[49, 81]$. For q_K ,

$$a_{q_K} - 4b_{q_K} = \frac{588839 + 1195712\sqrt{5}}{15} > 0.$$

Hence $w(q_K) < 1/4$ and its maximum is greater than $73 > s_\alpha$. In both cases, the target belongs to the reachable interval. $\square$

The evaluations of the maxima and, where they exist, the phases are

v	$\max_\theta s_\theta(v)$	θ_v
K	54.1922726227	does not exist
k	71.4670172771	does not exist
m_0	69.3474270513	does not exist
u	66.8296175021	does not exist
m_1	81	119.8905898663°
q_K	77.3923674871	133.5296853010°

The phase of the pure sector does not apply indiscriminately to a mixed state: its reading would be $49w + s_\alpha(1 - w) = s_\alpha - (s_\alpha - 49)w$. The original readers of V_2 are polynomials in S , as are P_0, P, J, U_θ . They therefore also intertwine the angular collection, and $s_\theta(V_2v) = s_\theta(v)$. Archiving the complements preserves the joint reading but does not remove the reachability restrictions.

68.11 Result of the composition and scope of the tests

The factor 73 has been determined by three compatible and distinct operations: the Gram matrix of two nonadic readers, the similitude of the ternary augmentation module, and the integer determinants of the two conjugate charts. The bilateral balance preserves the complete state and admits explicit inversion; complete memory preserves the transported readings, whereas selecting a channel can change their weights. The correction radii are uniform over all integer registers in the algebraic domain and throughout the collection $a > 0$, with cut and parity proofs that do not depend on sampling.

The inverse test adds verifiable conditions, not a hidden calibration. Orientation reversal and metric covariance do not change the reading. The specified integer iterations are resolved for all their indices by an exact crossing and monotonicity. Angular variation produces a collection that realizes the target on the permitted states, but the datum s_α selects its phase and the original ternary index is lost. Identifying it with a prior realization of Γ_9 would require a typed composition on the enriched state; this test neither provides that identification nor declares its absence throughout the corpus.

The new formalizations assemble antecedent operators and questions; no claim of worldwide priority is made for encoding, the use of cyclic vectors, or polar decomposition. The decimal figures presented are subsequent evaluations of exact formulas. The conditions of nonvanishing, charge, metric, lattice, and normalization are part of the result, and their counterexamples remain alongside the proofs. The function of the holographic arithmetic-geometric coupling constant is thereby preserved: a produced object that articulates readers, memory, and incidence, without turning every numerical coincidence into a new physical law.

69 Conservative refinement of unity and carry memory

Dimensional completion and the record of operations admit an explicit arithmetic composition. The APP-TRIT-TPK architecture first provides the enriched state, the oriented readings, and the completions of unity. From these outputs we construct an affine transport that preserves the normalized sum and permits recovery, by Euclidean division, of every finite word of canonical carries. Its limit has cylinders of controlled diameter; boundary marks distinguish the histories that a scalar evaluation identifies.

The central block, its concatenations, and the partition $1000 = 729 + 243 + 27 + 1$ are antecedents of this construction. The composition of these objects with the affine transport, the residue inverse, and its prolongation constitutes the formalization developed in this section. The map acts on a declared arithmetic domain. Each realization on a particular enriched history also retains the prospective rule that produces its carries.

69.1 From the quadratic norm to cubic completion

Let $b \geq 1$ be an integer and $B = b + 1 \geq 2$. Define

$$p_b = b^2 - b + 1, \quad q_b = B^2 - p_b = 3b. \quad (69.1)$$

Proposition 69.1 (Transfer of the completion). *The identities*

$$Bp_b = b^3 + 1, \quad Bq_b + 1 = B^3 - b^3 \quad (69.2)$$

determine the transformation

$$(p_b, q_b) \mapsto (Bp_b - 1, Bq_b + 1) = (b^3, B^3 - b^3). \quad (69.3)$$

The sum changes from B^2 to B^3 , while the sum of the coordinates divided by their respective scales remains equal to one.

Proof. The factorization $b^3 + 1 = (b + 1)(b^2 - b + 1)$ yields the first identity. The second follows from $(b + 1)^3 - b^3 = 3b^2 + 3b + 1$. Adding them proves the change of scale and, after division by B^3 , the conservation of normalized unity. $\square$

For $b = 9$,

$$(73, 27) \mapsto (10 \cdot 73 - 1, 10 \cdot 27 + 1) = (729, 271). \quad (69.4)$$

The complement retains the decomposition $271 = 243 + 27 + 1$. In the preceding completion, the last contribution corresponds to the added apex. The correction $(-1, +1)$ has zero sum, and multiplication by ten changes the scale from one hundred to one thousand. For the full collection, the transport of the proportions is

$$\left(\frac{p_b}{B^2}, \frac{q_b}{B^2}\right) \mapsto \left(\frac{p_b}{B^2} - \frac{1}{B^3}, \frac{q_b}{B^2} + \frac{1}{B^3}\right). \quad (69.5)$$

The quantity preserved by this identity is a linear sum. Quadratic forms and physical quantities have their own readers, which will subsequently be composed with this transport.

69.2 Concatenations of the central block and restoration of unity

For $b \geq 2$, the digits $b - 2$ and 2 belong to the alphabet in base $B = b + 1$. Consider the algebraic collection

$$B_c^{(b)} = \begin{pmatrix} b-2 & 2 \\ 2 & b-2 \end{pmatrix}, \quad C_B(u, v) = Bu + v. \quad (69.6)$$

The case $b = 9$ recovers the documented central block $\begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$, with eigenvalues 9 and 5. Its concatenations satisfy $72 + 27 = 77 + 22 = 99$. Concatenation, spectrum, and central position are distinct readings of the same antecedent and retain their domains.

Applying the reader to the rows of (69.6) gives

$$C_B(b-2, 2) = B(b-2) + 2 = b(b-1) = p_b - 1, \quad (69.7)$$

$$C_B(2, b-2) = 2B + (b-2) = 3b = q_b. \quad (69.8)$$

The sum is $B^2 - 1$. Restoring one unit in the first coordinate and performing the subsequent transfer yields

$$(p_b - 1, 3b) \xrightarrow{+(1,0)} (p_b, 3b) \xrightarrow{L_1} (b^3, B^3 - b^3). \quad (69.9)$$

In particular,

$$(72, 27) \mapsto (73, 27) \mapsto (729, 271). \quad (69.10)$$

The first arrow increases the sum by one. The second changes the scale and redistributes one unit through a zero-sum correction. To represent the contribution explicitly before incorporating it, the triple $(p_b - 1, 3b, 1)$ already has sum B^2 , and the map $(x, y, r) \mapsto (x + r, y, 0)$ preserves that sum. Its inversion requires retaining the datum $r = 1$: on arbitrary triples the map is not injective.

The restoration is thus recorded without attributing a closed conservation law to the pair before the unit is incorporated. The collection in b algebraically extends the given central case; the selection of atlas states remains that established by APP-TRIT-TPK. The composition (69.9) determines a carry equal to one in that transition, while leaving the prospective specification of subsequent operations open.

69.3 Affine transport and the positivity domain

Definition 69.2 (Transport with carry). For an integer base $B \geq 2$ and $c \in \mathbb{Z}$, define

$$L_c(x, y) = (Bx - c, By + c). \quad (69.11)$$

The integer c specifies the oriented transfer between the two coordinates. A positive sign reduces the first relative to Bx and increases the second by the same amount.

Proposition 69.3 (Inverse and positivity). *In the subsequent affine realization $\mathbb{R}^2$, with B, c fixed,*

$$L_c^{-1}(X, Y) = \left(\frac{X + c}{B}, \frac{Y - c}{B} \right), \quad X + Y = B(x + y). \quad (69.12)$$

The preimage is integral exactly when $X + c$ and $Y - c$ are divisible by B . For $x, y \geq 0$, the image is nonnegative exactly when

$$-By \leq c \leq Bx. \quad (69.13)$$

Proof. Direct substitution proves both identities in (69.12) and the integrality criterion. The inequalities $Bx - c \geq 0$, $By + c \geq 0$ are equivalent to (69.13). $\square$

For the canonical alphabet $c \in \{0, \dots, B-1\}$, all prolongations are admissible if $x \geq 1$, $y \geq 0$. At the boundary $x = 0$, nonnegativity permits only $c = 0$. The positive domain and the general integer domain are thus distinguished by an explicit condition.

69.4 Canonical recovery through the residue

Theorem 69.4 (Canonical inverse by residue). *Let $M \geq 1$ be an integer and let $X, Y \geq 0$ be integers such that $X + Y = BM$. There is a unique admissible triple (x, y, c) satisfying*

$$x + y = M, \quad 0 \leq c \leq B - 1, \quad L_c(x, y) = (X, Y).$$

It is obtained from

$$c = Y \bmod B = (-X) \bmod B, \quad y = \frac{Y - c}{B}, \quad x = \frac{X + c}{B}. \quad (69.14)$$

Proof. Every preimage satisfies $Y \equiv c \pmod{B}$. The representative in $\{0, \dots, B - 1\}$ is unique. The congruence $X + Y \equiv 0 \pmod{B}$ implies the divisibility of $X + c$. Euclidean division of Y gives $y \geq 0$, and $X + c \geq 0$ gives $x \geq 0$. If $c > 0$, the case $x = 0$ would require $X = -c < 0$, so the resulting triple belongs to the admissible domain. Its sum is M , and substitution into (69.11) recovers (X, Y) . Uniqueness of the residue and the two quotients proves uniqueness of the triple. $\square$

The integer pair and its scale contain the last canonical carry. Its recovery follows from the residue and does not require retaining another independent copy of that carry. There are $M + 1$ triples with $c = 0$ and M triples in each of the remaining $B - 1$ classes: the total is $BM + 1$, equal to the number of nonnegative pairs with sum BM .

For a general integer carry, write $c = r + B\ell$, with $0 \leq r \leq B - 1$. The residue determines r , whereas complete reconstruction also uses the lift ℓ :

$$y = \frac{Y - r}{B} - \ell, \quad x = \frac{X + r}{B} + \ell. \quad (69.15)$$

Retaining the lift distinguishes the signed carry from its residual reduction.

69.5 Variable iteration and complete finite memory

Fix $(x_0, y_0) \in \mathbb{Z}^2$, with sum $M_0 > 0$, and a sequence of operations $c_1, c_2, \dots$. Define

$$(x_n, y_n) = L_{c_n} \cdots L_{c_1}(x_0, y_0), \quad C_n = \sum_{j=1}^n c_j B^{n-j}, \quad C_0 = 0. \quad (69.16)$$

The carries are supplied by the corresponding operational rule; the limit value does not participate in their selection.

Theorem 69.5 (Exact accumulated record). *For every $n \geq 0$,*

$$\begin{aligned} x_n &= B^n x_0 - C_n, & y_n &= B^n y_0 + C_n, \\ x_n + y_n &= B^n M_0, & C_{n+1} &= B C_n + c_{n+1}. \end{aligned} \quad (69.17)$$

Proof. For $n = 0$, the identities are immediate. If they hold at n , applying $L_{c_{n+1}}$ multiplies all previous contributions by B and adds c_{n+1} to the accumulated carry. This gives the recurrence and, by substitution, both coordinates of the next stage. Their sum eliminates C_n . $\square$

When the carries are canonical,

$$0 \leq C_n \leq B^n - 1. \quad (69.18)$$

In particular, $x_0 \geq 1, y_0 \geq 0$ guarantee $x_n \geq B^n(x_0 - 1) + 1$ and $y_n \geq 0$ at every stage.

Corollary 69.6 (Recovery of the complete word). *Given B, n and the final pair, a finite canonical history recovers its input and all its carries through*

$$C_n = y_n \bmod B^n, \quad y_0 = \left\lfloor \frac{y_n}{B^n} \right\rfloor, \quad x_0 = \frac{x_n + C_n}{B^n}, \quad (69.19)$$

$$c_j = \left\lfloor \frac{C_n}{B^{n-j}} \right\rfloor \bmod B, \quad 1 \leq j \leq n. \quad (69.20)$$

Proof. The bound (69.18) makes $y_n = B^n y_0 + C_n$ a Euclidean division. Its quotient and residue give y_0, C_n ; the first coordinate gives x_0 . Successive divisions of C_n by powers of B recover the digits. Equivalently, apply (69.14) from the last step to the first. $\square$

The depth retains the leading zeros of the word, even though they do not change the integer C_n . If M_0 is stored instead of n , the exact equality $(x_n + y_n)/M_0 = B^n$ recovers the depth. Normalizing the pair to sum to one removes that scale counter unless it is retained by another reading.

69.6 Block composition and compatibility with the partition

A block of n operations, with accumulated carry C , acts by

$$L_C^{[n]}(x, y) = (B^n x - C, B^n y + C). \quad (69.21)$$

The composition of two blocks satisfies

$$L_D^{[m]} \circ L_C^{[n]} = L_{B^m C + D}^{[n+m]}. \quad (69.22)$$

Indeed, substituting the first image into the second multiplies its accumulated carry by B^m and adds D . The law

$$(n, C), (m, D) \mapsto (n + m, B^m C + D)$$

is associative by composition of maps. It expresses the contribution of the first block at the scale of the second and preserves the output when the same ordered events are regrouped. Adding new events changes the accumulated carry according to this same law.

69.7 Convergence and quantitative control of the reading

Write $M_n = B^n M_0$ and

$$f_n = \frac{x_n}{M_n}, \quad g_n = \frac{y_n}{M_n}, \quad \theta_n = \frac{C_n}{B^n} = \sum_{j=1}^n \frac{c_j}{B^j}. \quad (69.23)$$

The exact identities are

$$f_n = \frac{x_0}{M_0} - \frac{\theta_n}{M_0}, \quad g_n = \frac{y_0}{M_0} + \frac{\theta_n}{M_0}, \quad f_n + g_n = 1. \quad (69.24)$$

Theorem 69.7 (Convergence of bounded carries). *If $|c_j| \leq C$ for every j , the series $\theta = \sum_{j \geq 1} c_j B^{-j}$ converges absolutely. The readings f_n, g_n converge to f, g , with $f + g = 1$, and*

$$|f - f_n| = |g - g_n| \leq \frac{C}{M_0(B-1)B^n}. \quad (69.25)$$

Proof. The absolute tail satisfies

$$\sum_{j > n} \frac{|c_j|}{B^j} \leq C \sum_{j > n} B^{-j} = \frac{C}{(B-1)B^n}.$$

Convergence of the geometric series and (69.24) prove all the assertions. $\square$

For carries of both signs, a sufficient condition for positivity at every stage is $x_0, y_0 \geq C/(B-1)$. For the canonical alphabet, $x_0 \geq 1, y_0 \geq 0$ suffices. In this second case, $0 \leq \theta \leq 1$, f_n decreases and g_n increases, and

$$0 \leq f_n - f = g - g_n \leq \frac{1}{M_0 B^n}. \quad (69.26)$$

The sum remains fixed while the individual proportions evolve. The finite history remains recoverable when the integer coordinates and the scale are retained.

69.8 Compatible cylinders and boundary multiplicity

Consider the full canonical alphabet and an input $x_0 \geq 1, y_0 \geq 0$. A prefix of depth n , with accumulated carry C_n , admits exactly the limit values

$$I_n(C_n) = \left[\frac{C_n}{B^n}, \frac{C_n + 1}{B^n} \right] \quad (69.27)$$

of the reading θ .

Theorem 69.8 (Nested cylinders and distinction of histories). *The B successor cylinders of $I_n(C_n)$ cover the predecessor and overlap only at endpoints. The cylinders of an infinite history have diameter B^{-n} and a singleton intersection. Two different canonical histories produce the same value if and only if, at their first differing position, their digits differ by one and the subsequent tails are, respectively, entirely $B - 1$ for the smaller digit and entirely zero for the larger digit.*

Proof. The successor associated with c has endpoints $(BC_n + c)/B^{n+1}$ and $(BC_n + c + 1)/B^{n+1}$. These expressions establish inclusion, coverage, and overlap only at endpoints. The tail of an expansion ranges over the interval $[0, B^{-n}]$; successively choosing one of the successors produces an expansion of every point of that interval. The nested closed intervals have diameters tending to zero, proving existence and uniqueness of the intersection for each history.

Let k be the first differing position and let $d_k > c_k$. If the readings coincide,

$$\frac{d_k - c_k}{B^k} = \sum_{j>k} \frac{c_j - d_j}{B^j} \leq \sum_{j>k} \frac{B - 1}{B^j} = \frac{1}{B^k}.$$

The positive integer difference $d_k - c_k$ must equal one. Equality in the tail bound requires $c_j = B - 1, d_j = 0$ for every $j > k$. Conversely, those tails differ by exactly B^{-k} , which compensates for the first digit. This proves the complete characterization. $\square$

The image of the cylinder for f has diameter $1/(M_0 B^n)$, matching (69.26). The histories $(1, 0, 0, \dots)$ and $(0, B - 1, B - 1, \dots)$ both produce $\theta = 1/B$, while retaining distinct prefixes and finite integer states. A boundary mark retains both provenances. A convention excluding tails eventually constant at $B - 1$ selects one expansion for $\theta < 1$; the endpoint $\theta = 1$ is retained separately, since in the fractional alphabet it corresponds to the constant tail $B - 1$.

The history of operations, the cylinder system, and the limit value are objects related by these maps. The uniqueness of a history's reading coexists with multiplicity of histories at boundary values.

69.9 Signed carries and prospective specification

Allowing carries outside the canonical alphabet requires retaining their lifts. Already in two steps, the words

$$(c_1, c_2) = (1, -1), \quad (d_1, d_2) = (0, B - 1)$$

produce $C_2 = B - 1$ and the same final pair. Both are bounded by $B - 1$ and can preserve positivity, for example starting from $(73, 27)$ in base ten. The signed word, or the lifts in (69.15), distinguishes that information. The convergence theorem remains valid for both words; the canonical inverse, however, has the domain specified in its statement.

The transition (69.4) fixes $c_1 = 1$. An additional rule imposing $c_j = 1$ for every j gives

$$\theta_n = \frac{1 - B^{-n}}{B - 1}, \quad \theta = \frac{1}{B - 1}. \quad (69.28)$$

In base ten, starting from $(73, 27)$, the sequence is

$$(73, 27) \mapsto (729, 271) \mapsto (7289, 2711) \mapsto (72889, 27111),$$

and the first fraction converges to

$$\frac{73}{100} - \frac{1}{900} = \frac{164}{225} = 0,728888 \dots \quad (69.29)$$

The conclusion belongs to that additional constant-carry rule. With only $c_1 = 1$ and nonnegative canonical prolongations, the first-step cylinder yields exactly

$$\frac{b^3 - 1}{B^3} \leq f \leq \frac{b^3}{B^3}. \quad (69.30)$$

In base ten it is $[0,728, 0,729]$. Changing the sign or alphabet changes this collection and requires retaining the corresponding additional memory. The occurrence of 729 in the completion determines the arithmetic transition written above; identification with another reader of constants also uses its effective generation rule and intertwining map.

69.10 Operational content and transition to amplitudes

The resulting procedure comprises updating through (69.11), local residue checking, reconstruction of every canonical word through (69.20), control of the limit reading through (69.25), and regrouping of events through (69.22). Each operation retains the order and scale counter used by its inverse.

Conservation of the sum and quadratic conservation have different forms. For a normalized transfer $\delta = c/(BM)$,

$$(f - \delta)^2 + (g + \delta)^2 - (f^2 + g^2) = 2\delta(g - f) + 2\delta^2. \quad (69.31)$$

This expression determines the quadratic variation and is generally nonzero. The connection with an isometry will be obtained on the amplitudes of the admissible labels, through the residue bijection of section 71. Arithmetic transport and its unitary realization are thus composed while retaining their respective quantities, rather than being identified through a scalar coincidence.

70 Bilateral conservation and arithmetic–geometric transduction

The holographic constant of arithmetic–geometric coupling denotes the generated record K , with its ordered windows, phase origin, orientation, and incidences. Its production through the signed record and Hadamard inversion is developed in section 9. The rational reading κ , the orbital frame O_K , and the operators transporting its realizations are different objects. This distinction allows their functions to be composed: generation of the record, arithmetic encoding, information transport, and recovery of geometric readings.

The regional coordinates and α are supplied by the APP–TRIT–TPK genealogy, by the same enriched state and its joint discrete structure of the continuum. The following constructions act on those subsequent realizations. Their additional content is dimension-independent bilateral conservation, a reversible extension admitting incoming memory, and an archiving policy with uniformly controlled recovery.

70.1 Bilateral law for two isometric transports

Let H, H' be Hilbert spaces and $\mathcal{B}, \mathcal{C} : H \rightarrow H'$ linear isometries, so that $\mathcal{B}^* \mathcal{B} = \mathcal{C}^* \mathcal{C} = I_H$. Calligraphic type distinguishes these transports from the arithmetic base of refinement. For $a > 0$, define

$$\begin{aligned} p &= a^2 + a + 1 = a(a + 1) + 1, \\ N &= (2a + 1)p = (a + 1)^3 + a^3, \\ Q_- &= a\mathcal{B} - \mathcal{C}, \quad Q_+ = (a + 1)\mathcal{B} + \mathcal{C}. \end{aligned} \tag{70.1}$$

Theorem 70.1 (Balance of two isometric realizations). *For the maps (70.1),*

$$(a + 1)Q_-^* Q_- + aQ_+^* Q_+ = NI_H. \tag{70.2}$$

The identity holds in every dimension, including infinite-dimensional Hilbert spaces, and admits rectangular transports with a common domain and codomain.

Proof. Write $X = \mathcal{B}^* \mathcal{C} + \mathcal{C}^* \mathcal{B}$. The isometric property of both transports gives

$$\begin{aligned} Q_-^* Q_- &= (a^2 + 1)I - aX, \\ Q_+^* Q_+ &= ((a + 1)^2 + 1)I + (a + 1)X. \end{aligned}$$

The weights $a + 1$ and a cancel the terms involving X exactly. The remaining coefficient is

$$(a + 1)(a^2 + 1) + a((a + 1)^2 + 1) = 2a^3 + 3a^2 + 3a + 1 = N.$$

The compositions involve bounded operators, and the calculation uses only their products and adjoints. The proof therefore also applies in infinite dimension. $\square$

For $a = 8$, supplied by the preceding nonadic partition,

$$9\|Q_- v\|^2 + 8\|Q_+ v\|^2 = (9^3 + 8^3)\|v\|^2 = 17 \cdot 73 \|v\|^2. \tag{70.3}$$

The factor $73 = 8 \cdot 9 + 1$ normalizes this general identity. In the ternary module it is also realized as a determinant and a similarity factor; each of these roles retains its own hypotheses. The balance (70.2) uses the isometries and admits both finite-order transports and transports without that property.

70.2 Factorization, recovery, and channel compatibility

Define

$$E_a v = \frac{1}{\sqrt{p}} \begin{pmatrix} \sqrt{a(a+1)} \mathcal{B}v \\ \mathcal{C}v \end{pmatrix}, \quad R_a = \frac{1}{\sqrt{2a+1}} \begin{pmatrix} \sqrt{a} & -\sqrt{a+1} \\ \sqrt{a+1} & \sqrt{a} \end{pmatrix}. \quad (70.4)$$

The matrix R_a acts on the two channels and is understood to be tensored with $I_{H'}$. One verifies $E_a^* E_a = I_H$, $R_a^* R_a = I_{H' \oplus H'}$, and

$$W_a v = R_a E_a v = \frac{1}{\sqrt{N}} \begin{pmatrix} \sqrt{a+1} Q_- v \\ \sqrt{a} Q_+ v \end{pmatrix}, \quad W_a^* W_a = I_H. \quad (70.5)$$

When $a = 8$, the initial allocation has weights $72/73$ and $1/73$; the second operation is a rotation between the channels. This factorization expresses the role of $72 + 1$ within the total norm, distinguishing it from the linear restoration in (69.9).

Proposition 70.2 (Bilateral recovery). *The unnormalized channels $x = Q_- v$, $y = Q_+ v$ determine*

$$\mathcal{B}v = \frac{x+y}{2a+1}, \quad \mathcal{C}v = \frac{ay - (a+1)x}{2a+1}, \quad v = \mathcal{B}^* \frac{x+y}{2a+1}. \quad (70.6)$$

From the normalized channels one recovers $v = W_a^* W_a v$, with $\|W_a^*\| = 1$ when $H \neq \{0\}$.

Proof. Adding the definitions of $Q_{\pm}$ eliminates $\mathcal{C}v$. The combination $aQ_+ - (a+1)Q_-$ eliminates $\mathcal{B}v$. The first isometry has left inverse $\mathcal{B}^*$, which proves the formulas. The final assertion follows from (70.5): the adjoint of an isometry has norm one. $\square$

The image $\text{im } W_a$ is closed, and its orthogonal projector is $W_a W_a^*$. A normalized pair is compatible with a unique input exactly when it belongs to that image; its orthogonal component quantifies the compatibility defect.

If $\mathcal{B} = I$ and $\mathcal{C} = U$ is unitary, the two $Q_{\pm}$ are polynomials in the same operator. For the unnormalized channels, the criterion simplifies to

$$Q_+ x = Q_- y. \quad (70.7)$$

Indeed, this equality is equivalent to $U(x+y) = ay - (a+1)x$. Defining $v = (x+y)/(2a+1)$, substitution gives $Q_- v = x$, $Q_+ v = y$. Necessity follows from commutation of the two polynomials. This simplified form belongs to the stated unitary domain; the image criterion covers general transports between different spaces.

70.3 Transformation of readers and relative symmetry

Theorem 70.3 (A common reader in both channels). *For every bounded self-adjoint operator F on H' ,*

$$W_a^* \text{diag}(F, F) W_a = \frac{a(a+1)\mathcal{B}^* F \mathcal{B} + \mathcal{C}^* F \mathcal{C}}{a(a+1) + 1}. \quad (70.8)$$

If $\mathcal{B}$ is unitary and $\mathcal{C} = \mathcal{B}R$, with R unitary, the expression, for $G = \mathcal{B}^ F \mathcal{B}$, is*

$$\mathcal{J}_a(G) = \frac{a(a+1)G + R^* G R}{a(a+1) + 1}. \quad (70.9)$$

One has $\mathcal{J}_a(G) = G$ if and only if $GR = RG$.

Proof. Expanding $(a+1)Q_-^* F Q_- + aQ_+^* F Q_+$ cancels the terms $\mathcal{B}^* F \mathcal{C} + \mathcal{C}^* F \mathcal{B}$. The coefficients of the remaining terms are $a(a+1)(2a+1)$ for $\mathcal{B}^* F \mathcal{B}$ and $2a+1$ for $\mathcal{C}^* F \mathcal{C}$. Division by N proves (70.8). Substituting $\mathcal{C} = \mathcal{B}R$ gives (70.9). Equality with G is equivalent to $R^* G R = G$, and unitarity of R makes this equivalent to $GR = RG$. $\square$

The reader $F = I$ recovers the norm. For a general reader, the identity determines its transformation, and the coincidence $\mathcal{B}^*F\mathcal{B} = \mathcal{C}^*F\mathcal{C} = G$ guarantees its conservation in every state. In the canonical case,

$$\mathcal{T}_8(G) = \frac{72G + R^*GR}{73}. \quad (70.10)$$

This formula relates conservation and symmetry through a concrete commutator. The observable that transports an operator A of the domain exactly to the complete image is $W_aAW_a^*$; its ambient identity is $W_aW_a^*$. The diagonal observable in (70.8) is a different choice, whose difference is calculated by the same theorem.

70.4 Reversible updating with incoming memory

Now let $\mathcal{B} = I$, $\mathcal{C} = U$, with U unitary on the same space, and write

$$A = \frac{\sqrt{a+1}}{\sqrt{N}}Q_-, \quad D = \frac{\sqrt{a}}{\sqrt{N}}Q_+. \quad (70.11)$$

The operators A, D are normal and commute with each other and with their adjoints, since they are polynomials in U and U^* .

Theorem 70.4 (Unitary extension). *The update*

$$\mathcal{R}_a = \begin{pmatrix} A & -D^* \\ D & A^* \end{pmatrix} : H \oplus H \rightarrow H \oplus H \quad (70.12)$$

is unitary. Its first column is W_a , and an input (v, m) preserves $\|v\|^2 + \|m\|^2$.

Proof. The balance gives $A^*A + D^*D = I$. Normality yields $AA^* + DD^* = I$. The cross products of $\mathcal{R}_a^*\mathcal{R}_a$ and $\mathcal{R}_a\mathcal{R}_a^*$ vanish by the preceding commutation relations; both diagonal blocks are the identity. The first column reproduces (70.5), and conservation of the joint norm is the unitary identity applied to (v, m) . $\square$

For data $z = (z_1, z_2)$, the explicit inverse is

$$v_{\text{rec}} = A^*z_1 + D^*z_2, \quad m_{\text{rec}} = -Dz_1 + Az_2. \quad (70.13)$$

The hypothesis of zero incoming memory is checked through $m_{\text{rec}} = 0$. If e is a perturbation of the data, unitarity gives

$$\|e\|^2 = \|e_{\text{rec}}\|^2 + \|e_{\text{mem}}\|^2. \quad (70.14)$$

The second component detects the part incompatible with that input. An error belonging to $\text{im } W_a$ has zero memory residual and changes the recovered input. Correction of the integer record also uses its discrete separation distances.

70.5 Causal reduction of the bilateral update

The preceding unitary update realizes the transport of the state and its memory in a common space. Observation of the first component admits an equivalent description that explicitly retains the contribution of the record. This description follows from the transport already constructed, with the same coefficients and orientation: memory removed from the list of variables reappears as dependence on earlier states.

Fix the operators A, D of (70.11) and a realization $H \oplus H$. For $n \geq 0$, let

$$v_{n+1} = Av_n - D^*m_n, \quad m_{n+1} = Dv_n + A^*m_n. \quad (70.15)$$

The joint update is $\mathcal{R}_a$ and preserves $\|v_n\|^2 + \|m_n\|^2$.

Proposition 70.5 (Exact elimination of the auxiliary record). *For every $n \geq 0$, interpreting the empty sum as zero,*

$$m_n = (A^*)^n m_0 + \sum_{k=0}^{n-1} (A^*)^{n-1-k} D v_k, \quad (70.16)$$

$$v_{n+1} = A v_n - D^* (A^*)^n m_0 + \sum_{k=0}^{n-1} \mathcal{K}_{n-1-k} v_k, \quad (70.17)$$

$$\mathcal{K}_j = -D^* (A^*)^j D \quad (j \geq 0). \quad (70.18)$$

The reduced equation, together with (70.16), is equivalent to the joint recurrence for the same data (v_0, m_0) .

Proof. The first identity holds for $n = 0$. If it holds for n , the second equation in (70.15) gives

$$m_{n+1} = (A^*)^{n+1} m_0 + \sum_{k=0}^{n-1} (A^*)^{n-k} D v_k + D v_n,$$

which is the identity at $n + 1$. Substitution at depth n into the first equation proves (70.17). Conversely, if v_n satisfies the reduced equation and m_n is defined by (70.16), the same difference of sums proves $m_{n+1} = D v_n + A^* m_n$; substitution recovers the other equation. $\square$

The kernel $\mathcal{K}_j$ describes a transfer into the record, j updates within it, and re-entry into the observed component. The term $-D^* (A^*)^n m_0$ preserves the initial memory. Omitting it corresponds to the particular case $m_0 = 0$, rather than an identity of the general transport. Equation (70.17) involves only the current and earlier states. The causal character of this reduction follows from the order of the joint recurrence.

The normalization of the balance also provides an explicit estimate. Since $Q_- = aI - U$, we have $\|A\|^2 \leq r_a = (a+1)^3 / ((a+1)^3 + a^3) < 1$, whereas $D^* D \leq I$. Consequently,

$$\|\mathcal{K}_j\| \leq r_a^{j/2}, \quad \sum_{j \geq 0} \|\mathcal{K}_j\| \leq \frac{1}{1 - \sqrt{r_a}}. \quad (70.19)$$

This bound controls the weight of memory in the reduced equation; the norm of the joint state remains conserved by the unitary update.

The proposition is the temporal realization of the Schur reduction in proposition 2.19. For the formal series $V(z) = \sum_{n \geq 0} v_n z^n$ and $M(z) = \sum_{n \geq 0} m_n z^n$, the recurrences yield $V = v_0 + zAV - zD^*M$ and $M = m_0 + zDV + zA^*M$. Eliminating M ,

$$V(z) = [I - zA + z^2 D^* (I - zA^*)^{-1} D]^{-1} [v_0 - zD^* (I - zA^*)^{-1} m_0]. \quad (70.20)$$

The formal inverses exist because their constant terms are identities; the state series and the joint resolvent also converge for $|z| < 1$. The initial source and the effective operator transform together. A readout that also acts on m_n retains its contribution through (70.16), in accordance with (2.31).

Re-entry here updates an incoming record. By contrast, the archiving policy below retains each complement in a distinct coordinate. Each operation has its own conservation law. The kernel $\mathcal{K}_j$, Green propagators, and a dimensional force have different types: the first eliminates a state component, Green propagators solve source equations with support conditions, and the force adds a dynamical realization and its units.

70.6 Nonstationary archiving and uniform exhaustion of the terminal

At each stage let H_n, H_{n+1} be Hilbert spaces and let $\mathcal{B}_n, \mathcal{C}_n : H_n \rightarrow H_{n+1}$ be isometries. The parameter $a > 0$ remains fixed. Let A_n, D_n be the normalized blocks of (70.5); continuation proceeds through the first and the second is recorded:

$$R_0 = I, \quad R_{n+1} = A_n R_n, \quad m_n = D_n R_n v. \quad (70.21)$$

This policy has a different domain from memory reintroduction in (70.12); each retains its own update.

Lemma 70.6 (Uniform transfer). *At every stage,*

$$\|A_n\|^2 \leq r_a := \frac{(a+1)^3}{(a+1)^3 + a^3} < 1, \quad D_n^* D_n \geq (1 - r_a)I, \quad \|R_n\|^2 \leq r_a^n. \quad (70.22)$$

Proof. The triangle inequality gives $\|a\mathcal{B}_n - \mathcal{C}_n\| \leq a + 1$. Normalization yields the first bound. The difference $N - (a+1)^3 = a^3 > 0$ proves $r_a < 1$. The local balance implies $D_n^* D_n = I - A_n^* A_n$, and multiplying the bounds for the continued channels gives the estimate for R_n . $\square$

Theorem 70.7 (Complete memory at arbitrary depth). *The finite map*

$$Z_N v = (R_N v, D_0 v, D_1 R_1 v, \dots, D_{N-1} R_{N-1} v) \quad (70.23)$$

is isometric. The infinite archive

$$Z_\infty v = (D_n R_n v)_{n \geq 0} \quad (70.24)$$

belongs to the Hilbert direct sum of the output spaces and satisfies

$$\|v\|^2 = \sum_{n \geq 0} \|m_n\|^2, \quad v = \sum_{n \geq 0} R_n^* D_n^* m_n, \quad Z_\infty^* Z_\infty = I. \quad (70.25)$$

The inverse series converges in norm and the adjoint has norm one.

Proof. The identity $A_n^* A_n + D_n^* D_n = I$, conjugated by R_n , gives

$$R_n^* R_n - R_{n+1}^* R_{n+1} = R_n^* D_n^* D_n R_n.$$

Summing from zero to $N - 1$ yields

$$I = R_N^* R_N + \sum_{n < N} R_n^* D_n^* D_n R_n. \quad (70.26)$$

This is the Gram identity for Z_N . By (70.22), the first term tends to zero in operator norm. The limit identity proves square summability of the archive and its isometric property. The adjoint of this isometry is bounded; the truncations of any square-summable archive converge in the Hilbert direct sum, and their images under the adjoint prove convergence of the series in (70.25). Applying it to the exact data recovers v . $\square$

With only the first N complements, the truncated adjoint has the exact residual

$$v - \sum_{n < N} R_n^* D_n^* m_n = R_N^* R_N v, \quad \left\| v - \sum_{n < N} R_n^* D_n^* m_n \right\| \leq r_a^N \|v\|. \quad (70.27)$$

If the joint data error is at most ε , the total bound is $\varepsilon + r_a^N \|v\|$. With Z_N , including its terminal, or with Z_∞ , the adjoint recovers the input with error at most ε .

The preceding policy exhausts the terminal through the proven uniform bound. The earlier evolutions with a positive terminal $A_\infty \neq 0$ retain the component $A_\infty^{1/2} v$ in their archive. Arbitrary depth and elimination of the terminal are therefore different conclusions, each retaining its respective hypotheses.

Proposition 70.8 (Inverse of finite memories without the terminal). *For $N \geq 1$, let $M_N v = (D_n R_n v)_{n < N}$. Then*

$$M_N^* M_N = I - R_N^* R_N \geq (1 - r_a^N) I, \quad (70.28)$$

and the least-squares inverse

$$L_N = (I - R_N^* R_N)^{-1} M_N^*, \quad L_N M_N = I, \quad \|L_N\| \leq (1 - r_a^N)^{-1/2} \quad (70.29)$$

recovers the input exactly from exact data.

Proof. The Gram identity is (70.26); the bound follows from (70.22). The positive operator is invertible by its strict lower bound. Composition proves $L_N M_N = I$, and $L_N L_N^* = (M_N^* M_N)^{-1}$ proves the stated norm bound. $\square$

This finite inverse removes the bias in (70.27); its noise amplification is specified by (70.29). The truncated adjoint, finite inverse, and adjoint of the complete archive are thus three operators with distinct behavior.

For $a = 8$, the general bound is $r_a = 729/1241$. In the case $\mathcal{B} = I$, $\mathcal{C} = S^4$, with S the cycle of twelve positions, the fixed-space projector $P_0 = (I + S^4 + S^8)/3$ gives

$$A^* A = \frac{a+1}{(2a+1)p} ((a-1)^2 P_0 + p(I - P_0)). \quad (70.30)$$

Since $p - (a-1)^2 = 3a > 0$, its norm is $(a+1)/(2a+1)$. At $a = 8$, this is $9/17$; the eigenvalue of the fixed sector is $441/1241$. These rates belong to the declared transports and policy.

If both channels are refined in a tree, every complete level is also isometric by composition. Summing the norms of all complete layers would count the same initial norm repeatedly. Conservation is formulated through the refinement maps between levels or through the archive of complements in (70.24).

70.7 Geometric forms and conservation under transport

Let b, b' be nondegenerate symmetric or alternating forms on the corresponding spaces, and suppose that $\mathcal{B}^* b' \mathcal{B} = \mathcal{C}^* b' \mathcal{C} = b$. The same bilinear expansion proves

$$(a+1)Q_-^* b' Q_- + aQ_+^* b' Q_+ = Nb. \quad (70.31)$$

Indeed, the mixed terms $\mathcal{B}^* b' \mathcal{C} + \mathcal{C}^* b' \mathcal{B}$ cancel with the same weights; the pure terms sum to Nb . The transported form is thus preserved without requiring positivity. The contraction and noise bounds of the preceding subsection, by contrast, use positive Hilbert norms. A Lorentzian or alternating form retains (70.31), with its own geometric interpretations, and does not replace positivity in those inequalities.

70.8 Channel correlation and an inverse test of a reading

For $v \neq 0$, define

$$\chi(v) = \frac{\operatorname{Re} \langle \mathcal{B}v, \mathcal{C}v \rangle}{\|v\|^2}, \quad -1 \leq \chi(v) \leq 1. \quad (70.32)$$

The bound follows from Cauchy–Schwarz applied to the two isometric images. The normalized readings are

$$s_- = a^2 + 1 - 2a\chi, \quad s_+ = (a+1)^2 + 1 + 2(a+1)\chi. \quad (70.33)$$

For $a = 8$,

$$s_- = 65 - 16\chi, \quad s_+ = 82 + 18\chi, \quad 9s_- + 8s_+ = 1241. \quad (70.34)$$

The two individual norms contain a common real correlation. The balance is an exact relation between them, not two independent arithmetic determinations of a constant.

Using the previous output α_A in the inverse test $s_- = 10^4 \alpha_A$ determines the complementary reading

$$s_+ = 73 + \frac{9}{8}(73 - 10^4 \alpha_A) \simeq 73,029783595557. \quad (70.35)$$

The equality follows by substitution into (70.34). The weighted average $(9s_- + 8s_+)/17$ remains exactly 73. The comparison preserves the prior genealogy of α_A : it is a condition of the inverse test, not another independent producer of that output. A unitary change can alter χ and both readings while preserving the balance, even if a particular ternary-order relation changes.

70.9 Cyclic frame and transport of responses

On the carrier of the generated record, let $H = \mathbb{R}^{12}$, $(Sx)_i = x_{i+1}$, with cyclic indices, and

$$O_K = (K, SK, \dots, S^{11}K), \quad 589609I \leq O_K^* O_K \leq 39225169I. \quad (70.36)$$

These are the bounds for the cyclic frame of the canonical record, proved in theorem 67.2. Let Z be one of the complete isometries Z_N, Z_∞ , or a composition with the preceding archives that retains their terminals and complements.

Theorem 70.9 (Preservation of frame conditioning). *If $x = O_K \beta$, the datum $z = Zx$ determines*

$$\beta = O_K^{-1} Z^* z, \quad \|\delta \beta\| \leq \frac{\|\delta z\|}{\sqrt{589609}}. \quad (70.37)$$

The twelve columns of ZO_K form a basis of its image, even when the archive has infinitely many coordinates.

Proof. The equality $Z^* Z = I$ gives $(ZO_K)^*(ZO_K) = O_K^* O_K$. The Gram bounds are preserved, and O_K is invertible by the strict lower bound. Composing their inverses proves recovery, and the norm bound $\|O_K^{-1}\| \leq 589609^{-1/2}$, together with $\|Z^*\| = 1$, gives stability. The number of columns and their independence determine the dimension of the image, without identifying it with the entire storage space. $\square$

For an operator H_0 commuting with S , the complete vector response $y = H_0 K$ determines

$$H_0 = O_y O_K^{-1}. \quad (70.38)$$

Indeed, commutation implies $H_0 O_K = (y, Sy, \dots, S^{11}y) = O_y$. If only $z = Zy + e$ is received, define $\hat{y} = Z^* z$ and $\hat{H}_0 = O_{\hat{y}} O_K^{-1}$. Then

$$\|\hat{H}_0 - H_0\| \leq \sqrt{\frac{12}{589609}} \|e\|. \quad (70.39)$$

The orbital matrix of δy has twelve columns of norm $\|\delta y\|$, so its operator norm is at most $\sqrt{12}\|\delta y\|$. This and (70.36) prove (70.39). For a general operator, the protocol requires the twelve responses $H_0 S^j K$; a response in the cyclic case means its twelve components, not a scalar measurement.

In the transported image, the shift is $S_Z = ZS Z^*$, and satisfies $S_Z^j Z = ZS^j$. This equality determines phase advance in the archive; a shift of slots in the ambient space represents that advance only when it coincides with the transported operator.

70.10 Positional reading and scalar precision

Fix $B_0 = 1000$, $d = B_0^{12} - 1$, and the row

$$\ell = \frac{(B_0^{11}, B_0^{10}, \dots, 1)}{d}. \quad (70.40)$$

Its linear extension acts on H , and on the domain of words $\kappa(K) = \ell K$. The representative of the transported reader is $Z\ell^*$, so

$$\kappa(K) = \langle Z\ell^*, ZK \rangle, \quad |\delta\kappa| \leq \|\ell\| \|\ell\|. \quad (70.41)$$

The geometric sum gives

$$\|\ell\|^2 = \frac{B_0^{12} + 1}{(B_0^2 - 1)(B_0^{12} - 1)}. \quad (70.42)$$

Indeed, the numerator of the sum of squares is $(B_0^{24} - 1)/(B_0^2 - 1)$, which, when divided by $(B_0^{12} - 1)^2$, yields (70.42).

With only N complements and the truncated adjoint, there is an additional error of at most $\|\ell\| r_a^N \|K\|$. The finite inverse (70.29) removes that bias and retains its explicit noise-amplification bound.

On the domain $\{0, \dots, 999\}^{12}$, the exact reading recovers $N_K = d\kappa$ and its twelve blocks through successive Euclidean divisions. If

$$|\tilde{\kappa} - \kappa| < \frac{1}{2d}, \quad (70.43)$$

rounding $d\tilde{\kappa}$ to the nearest integer recovers N_K : its distance from the true integer is less than half a unit. The base, length, origin, and orientation accompany that representation. The scalar threshold in (70.43) is distinct from the vector bounds for memory records.

70.11 Incidence, charge, and exact recovery of the record

Let $\mathbf{1} = (1, \dots, 1)^T$ and $\Pi = I - \mathbf{1}\mathbf{1}^*/12$. The incidence isometry of section 28.1 is

$$Fv = (\mu(v), \mathcal{I}\Pi v/6), \quad \mu(v) = \frac{\mathbf{1}^*v}{12}, \quad \|(\mu, y)\|_Y^2 = 12\mu^2 + \|y\|^2. \quad (70.44)$$

Its codomain is its declared image. The Gram identity $\mathcal{I}^*\mathcal{I} = 36I + 30\mathbf{1}\mathbf{1}^*$ gives $\|\mathcal{I}\Pi v\|^2 = 36\|\Pi v\|^2$, and $12\mu(v)^2 + \|\Pi v\|^2 = \|v\|^2$, hence $F^*F = I$. Thus, from $z = Zv + e$,

$$\widehat{Fv} = FZ^*z, \quad \|FZ^*e\|_Y \leq \|e\|. \quad (70.45)$$

The unnormalized incidence $\mathcal{I}\Pi v$ has error at most $6\|e\|$; the signed record $H_{12}v$, whose normalization satisfies $H_{12}^*H_{12} = 4I$, has error at most $2\|e\|$.

The direction $u = P_3\Pi K$ and the projector $P_{10} = \Pi - uu^*/\|u\|^2$, with $u \neq 0$, are recovered with their previously defined readers and chart. The absolute selection $\{j : K_j \geq 729\}$ has a discontinuity at $K_4 = 729$. Integer recovery is performed before applying that threshold.

Proposition 70.10 (Decoding after the complete archive). *Let K be integral and let $z = ZK + e$, with $Z^*Z = I$. If $\|e\| < 1/2$, rounding each coordinate of Z^*z recovers K exactly. Composition then restores κ , the signatures, and the geometric readers defined on that record.*

Proof. Contractivity of Z^* gives $\|Z^*z - K\| \leq \|e\| < 1/2$. Each coordinate error is bounded by the joint norm, so the unique integer coordinate at distance less than half a unit is the original one. Once K has been recovered, evaluating any of its exact readers restores its value. $\square$

In particular, the support $\{4, 6, 9, 10\}$ is recovered, including its boundary component. The other marks involved in an exceptional flag are retained by their own channels: recovery of K corrects exactly the information for which it supplies sufficient arguments.

70.12 Preservation of the conditioning of two memories

Retain the preceding readers

$$T_j = \frac{8I + S^j}{9}, \quad D_j = \frac{\sqrt{8}}{9}(S^j - I), \quad Mv = (D_3v, D_4T_3v). \quad (70.46)$$

The D_j in this equation are the preceding complements; the second bilateral block D of (70.11) retains its own definition. One has $\ker M = \mathbb{R}\mathbf{1}$ and an inverse L_M of norm $9/4$ on the centered sector, as proved in section 35.

Let $E = W_a \oplus W_a$ on the space of the two memories. Its isometric property implies

$$(EM)^*(EM) = M^*M, \quad \ker(EM) = \mathbb{R}\mathbf{1}. \quad (70.47)$$

For $z = EMv + e$ and exact charge $q = \mathbf{1}^*v$, the reconstruction is

$$\hat{v} = \frac{q}{12}\mathbf{1} + L_ME^*z, \quad \|\hat{v} - v\| \leq \frac{9}{4}\|e\|. \quad (70.48)$$

The proof consists in applying $E^*E = I$, using $L_MM = \Pi$ and $\|E^*\| = 1$. If the charge has error δq , the uniform and centered sectors are orthogonal, and

$$\|\hat{v} - v\|^2 \leq \frac{|\delta q|^2}{12} + \frac{81}{16}\|e\|^2. \quad (70.49)$$

The distances between all words in the integer image are preserved exactly by E . Therefore, the recovery radii of theorem 68.5 remain

$$r_0 = \frac{2\sqrt{146}}{81} \quad \text{without charge}, \quad r_q = \frac{4\sqrt{73}}{81} \quad \text{with exact charge}. \quad (70.50)$$

Within the first radius, ΠK is recovered, and with it the direction and its projector whenever they are defined. Within the second, with q retained, one recovers K and the absolute threshold selector. Subsequent composition with another complete isometry, finite or infinite, preserves the same guarantees: its adjoint is applied before the decoder. The radius is always expressed in the joint data norm with the fixed normalization.

This composition redistributes the information in M without degrading its conditioning or creating the uniform channel that M eliminates. Gram preservation distinguishes this property from a numerical improvement obtained by changing the reader or the noise scale.

70.13 Orbital norm and polar normalization of the frame

The preceding inversion satisfies $K = H_{12}U_{\text{sgn}}/4$, $H_{12}^*H_{12} = 4I$. Therefore

$$\|K\|^2 = \frac{1}{4}\|U_{\text{sgn}}\|^2 = 4251257, \quad \|U_{\text{sgn}}\|^2 = 17005028. \quad (70.51)$$

Each shift S^j is orthogonal and preserves $\nu = \sqrt{4251257}$. Hence

$$\text{tr}(O_K^*O_K) = 12\nu^2 = 51015084. \quad (70.52)$$

Division by ν normalizes each orbital column to norm one and preserves its order. It is uniform over the orbit and under its isometric transports. Different records retain their own norms, and the normalized columns still have a nonscalar Gram matrix: its eigenvalues are the preceding ones divided by 4251257.

Operator normalization uses the complete Gram matrix:

$$G_K = O_K^*O_K, \quad F_K = O_K G_K^{-1/2}, \quad F_K^*F_K = I. \quad (70.53)$$

The inverse square root exists by (70.36). The equality is verified by multiplying the Gram matrix on both sides by $G_K^{-1/2}$. Thus ZF_K is isometric at any depth. The complete recovery of the frame is

$$O_K = F_K G_K^{1/2}. \quad (70.54)$$

The pair (F_K, G_K) preserves the complete information. The unitary factor retains an oriented orthonormal frame in the fixed charts, whereas the amplitudes also reside in G_K . Multiplying K by a positive scalar preserves F_K and changes G_K ; K and $-K$ share a Gram matrix and have opposite polar factors, with different linear readings κ . Each example shows why an isolated factor cannot replace the pair.

Shifting the origin preserves ν , but in the fixed positional chart

$$\kappa(SK) = 1000\kappa(K) - K_1. \quad (70.55)$$

This follows by shifting the positional numerator and using $d = 1000^{12} - 1$. To preserve the same reading under a change of chart, the row ℓ is also transported by the inverse of that change. A general unitary chart can leave the integer-block lattice; decoding by Euclidean division is performed in the recovered block chart.

70.14 An exact example of transduction and error correction

Take $a = 8$, $U = S^4$, and the canonical record

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601). \quad (70.56)$$

Denoting its two blocks of six coordinates by $(v)_{1:6}$ and $(v)_{7:12}$, the unnormalized readings are

$$\begin{aligned}(Q_-K)_{1:6} &= (1213, 3520, 499, 5774, 4358, 5798), \\(Q_-K)_{7:12} &= (4822, -137, 7078, 5809, 1028, 4079), \\(Q_+K)_{1:6} &= (2765, 5711, 1881, 6619, 6845, 8210), \\(Q_+K)_{7:12} &= (5735, 1123, 8460, 7689, 1454, 6138).\end{aligned}\tag{70.57}$$

Their sum, divided by 17, recovers every block of K . The matrix identity $9Q_-^*Q_- + 8Q_+^*Q_+ = 1241I$ precedes its evaluation on the particular vector.

To introduce a controlled error, the two memories are rationalized as $z_0 = MK/\sqrt{8}$, and $1/10$ is added to the first coordinate. Each block of twelve components is then transported through the two bilateral channels. The sum inverse recovers the perturbed memory data, and the charge decoder receives $q = 6263$ together with those data, without receiving the record sought. The norm of the error in the original memories is $\sqrt{8}/10 < r_q$. Isometric normalization preserves that norm; the unnormalized channels merely facilitate exact rational calculation.

In this instance, there are 924 candidates from floor-and-ceiling reconstruction that satisfy the charge. The minimum rationalized squared distance is $1/100$, and the decoder recovers (70.56). Uniqueness follows from the strict radius (70.50), not merely from the size of the enumeration. The recovered arithmetic reading is

$$\kappa = \frac{234543140729659824621058914794146601}{9999999999999999999999999999999999}, \quad (70.58)$$

and the incidence selection is again $\{4, 6, 9, 10\}$, with the same threshold 729. The example introduces noise before transport. The adjoint contractivity proofs also cover arbitrary subsequent errors whose joint norm lies within the declared radius.

The combined operational capability is to transport and recover a generated record, preserve the conditioning of its responses, and restore its arithmetic and incidence readings before applying discontinuous selections. Its effective premises are the prior production of K , the invertible cyclic frame, the bilateral balance, the discrete separation of memories, and the archiving theorem. The twelve-component record retains the scope of its readers; the complete enriched history retains its other records and dependencies.

71 Unitary lift of refinement and transport of readers

The residue bijection of theorem 69.4 and the bilateral balance of theorem 70.1 admit an explicit composition. An admissible arithmetic triple, containing a pair and its carry, determines a refined pair. The linearization of this bijection on orthonormal bases provides a unitary transport; two such transports can act as the two realizations of the bilateral balance.

The realization is established downstream of the objects generated by APP-TRIT-TPK. Its combinatorial domain comprises the pairs and carries satisfying the conditions stated below. The norm preserved by its linearization is the sum of squared amplitudes over those labels; the arithmetic conservation law for $L_c(x, y) = (Bx - c, By + c)$ remains the sum of the two coordinates after rescaling. The composition relates the two conservation laws through a concrete map.

71.1 Bijection of admissible arithmetic states

Let $B \geq 2$ and $M \geq 1$ be integers. Define

$$\mathcal{D}_{B,M} = \{(x, y, c) \in \mathbb{Z}_{\geq 0}^2 \times \{0, \dots, B-1\} : x + y = M, c \leq Bx\}, \quad (71.1)$$

$$\mathcal{E}_{BM} = \{(X, Y) \in \mathbb{Z}_{\geq 0}^2 : X + Y = BM\}. \quad (71.2)$$

The condition $c \leq Bx$ preserves the nonnegativity of the first output coordinate. For $x = 0$, it permits only $c = 0$; for $x \geq 1$, it permits the entire canonical alphabet.

Theorem 71.1 (Exact correspondence of labels). *The map*

$$\ell_{B,M} : \mathcal{D}_{B,M} \longrightarrow \mathcal{E}_{BM}, \quad (x, y, c) \longmapsto (Bx - c, By + c) \quad (71.3)$$

is bijective. Both sets have cardinality $BM + 1$. Its inverse is given by

$$c = Y \bmod B, \quad y = \frac{Y - c}{B}, \quad x = \frac{X + c}{B}. \quad (71.4)$$

Proof. The domain condition and the identity $Bx - c + By + c = BM$ ensure that the image belongs to $\mathcal{E}_{BM}$. Given a pair in this set, the residue c is the unique element of the alphabet satisfying $Y \equiv c \pmod{B}$. Since $X + Y$ is divisible by B , so is $X + c$. Euclidean division of Y gives $y \geq 0$, and $X + c \geq 0$ gives $x \geq 0$. If $c > 0$, $x = 0$ would require $X = -c < 0$, so admissibility is preserved. The preimage coordinates sum to M , and substitution verifies the original image. Uniqueness of the residue and quotients proves the bijection.

There are $M + 1$ labels with $c = 0$ and M with each positive carry; hence the cardinality is $(M + 1) + (B - 1)M = BM + 1$, equal to that of the output. $\square$

The proof makes the boundary endpoint of the domain explicit, a condition necessary for equality of cardinalities. The new carry label is a typed part of the input to refinement.

71.2 Unitarity on label amplitudes

Take the real or complex Hilbert spaces with counting measure

$$\mathcal{H}_D = \ell^2(\mathcal{D}_{B,M}), \quad \mathcal{H}_E = \ell^2(\mathcal{E}_{BM}), \quad (71.5)$$

and their labelled orthonormal bases. Define linearly

$$J_{B,M}|x, y, c\rangle = |Bx - c, By + c\rangle. \quad (71.6)$$

We write J when the parameters are fixed.

Proposition 71.2 (Unitary lift). *The operator $J : \mathcal{H}_D \rightarrow \mathcal{H}_E$ is unitary: $J^*J = I_D, JJ^* = I_E$. The adjoint recovers each label through (71.4).*

Proof. The result in theorem 71.1 ensures that J maps two complete orthonormal bases bijectively onto one another. For $\psi = \sum_{d \in \mathcal{D}} \psi_d |d\rangle$,

$$\|J\psi\|^2 = \sum_{e \in \mathcal{E}} |\psi_{\ell_{B,M}^{-1}(e)}|^2 = \sum_{d \in \mathcal{D}} |\psi_d|^2 = \|\psi\|^2.$$

Surjectivity proves the two unitary identities, and the inverse bijection describes the adjoint on each basis vector. $\square$

The label $|73, 27, 1\rangle$ has norm one; the integers 73, 27, 1 describe its arithmetic values, not its amplitudes. The construction represents reversible processing of labels or amplitudes. A concrete physical realization also retains the protocol that prepares and transports those amplitudes.

71.3 Diagonal readers of proportion and carry

For a real-valued function h on a finite set, let M_h denote its diagonal operator: each basis vector is multiplied by the value of h at its label. The bijection satisfies

$$J^* M_h J = M_{h \circ \ell_{B,M}}. \quad (71.7)$$

This follows by acting on every basis vector.

Define on the output

$$f_E(X, Y) = \frac{X}{BM}, \quad g_E(X, Y) = \frac{Y}{BM}, \quad \gamma_E(X, Y) = Y \bmod B, \quad (71.8)$$

and on the input $f_D = x/M, g_D = y/M, \gamma_D = c$. Identity (71.7) yields

$$\begin{aligned} J^* M_{f_E} J &= M_{x/M - c/(BM)}, \\ J^* M_{g_E} J &= M_{y/M + c/(BM)}, \\ J^* M_{\gamma_E} J &= M_c. \end{aligned} \quad (71.9)$$

The first two readers transport the transfer of a unit. Since $f_E + g_E = 1$, their sum is I_E and its compression is I_D . The third reader preserves the carry exactly as an integer observable, with its residue representation in the output.

71.4 Complete paths at arbitrary depth

Let $\mathcal{P}_n(B, M)$ be the set of paths of length n starting from all nonnegative pairs with sum M and, at each stage, satisfying the conditions of (71.1) for the corresponding sum. A path retains its initial pair and its n admissible carries. Let Φ_n be the map assigning to each path its final pair.

Theorem 71.3 (Recovery of paths at every finite depth). *For every $n \geq 0$,*

$$\Phi_n : \mathcal{P}_n(B, M) \longrightarrow \mathcal{E}_{B^n M} \text{ is bijective,} \quad |\mathcal{P}_n(B, M)| = B^n M + 1. \quad (71.10)$$

Its linearization

$$J_n : \ell^2(\mathcal{P}_n(B, M)) \longrightarrow \ell^2(\mathcal{E}_{B^n M}), \quad J_n |\pi\rangle = |\Phi_n(\pi)\rangle \quad (71.11)$$

is unitary.

Proof. From a pair with sum $B^n M$, the residue in (71.4) recovers the last carry and a unique preceding pair with sum $B^{n-1} M$. Repeating this n times produces a unique path whose input has sum M . The forward composition recovers the original endpoint, proving bijectivity and the inverse. The count also follows from the inputs: the M pairs with $x_0 \geq 1$ each admit B^n words, whereas $(0, M)$ admits only the all-zero word. The total is $MB^n + 1$. The bijective transformation of orthonormal bases proves the unitarity of J_n . $\square$

Once the depth is known, the readers on the final pair are

$$c_j = \left\lfloor \frac{Y}{B^{n-j}} \right\rfloor \bmod B, \quad y_0 = \left\lfloor \frac{Y}{B^n} \right\rfloor, \quad x_0 = M - y_0, \quad 1 \leq j \leq n. \quad (71.12)$$

These functions are diagonal and recover the complete memory of the finite operations. Compatibility with prolongation is

$$\Phi_{n+1}(\pi, c) = L_c(\Phi_n(\pi)). \quad (71.13)$$

Passing from n to $n + 1$ incorporates the new admissible carry label. The spaces increase in dimension, and the boundary label admits only $c = 0$; the domain of the next step therefore retains admissibility instead of replacing it with an unrestricted product with B digits.

71.5 Fixed input and exact image

If an initial pair (x_0, y_0) is fixed, with $x_0 \geq 1$, there are B^n paths, and their image is exactly

$$\{(B^n x_0 - C, B^n y_0 + C) : 0 \leq C \leq B^n - 1\}. \quad (71.14)$$

The result in corollary 69.6 proves that the B^n accumulated values are distinct and recover the respective words. The linearization is unitary onto the subspace spanned by this image. If $x_0 = 0$, the image consists of a single label.

The cylinders of theorem 69.8 correspond to a fixed input. The full domain of (71.10) comprises all pairs with sum M . Both constructions have explicit inverses, but their images and cardinalities differ. Arbitrary depth expresses the validity of the proof for every n ; the limit of the readout and its boundaries remain those established in section 69.7.

71.6 Bilateral coupling of refinement

Let R be a previously declared permutation of $\mathcal{D}_{B,M}$, linearized unitarily on $\mathcal{H}_D$. Define the two transports

$$\mathcal{B} = J, \quad \mathcal{C} = JR : \mathcal{H}_D \longrightarrow \mathcal{H}_E. \quad (71.15)$$

The permutation is part of the specification of the procedure and is fixed before an output readout is evaluated. For $a > 0$, $p = a^2 + a + 1$, $N = (2a + 1)p$,

$$Q_- = J(aI - R), \quad Q_+ = J((a + 1)I + R). \quad (71.16)$$

Proposition 71.4 (Distribution and recovery of refined amplitudes). *The operators (71.16) satisfy*

$$(a + 1)Q_-^* Q_- + aQ_+^* Q_+ = NI_D, \quad (71.17)$$

and the map

$$W_a \psi = \frac{1}{\sqrt{N}} (\sqrt{a + 1} Q_- \psi, \sqrt{a} Q_+ \psi) \quad (71.18)$$

is isometric. For the unnormalized channels $u = Q_- \psi$, $v = Q_+ \psi$,

$$J\psi = \frac{u + v}{2a + 1}, \quad \psi = J^* \frac{u + v}{2a + 1}. \quad (71.19)$$

Proof. The two maps $\mathcal{B}, \mathcal{C}$ are unitary. The cross terms in the Gram operators of Q_- and Q_+ have coefficients $-a$ and $a + 1$, respectively. The balance weights cancel them, while the diagonal terms sum to N , as in theorem 70.1. Normalization proves the isometry. The sum of the channels is $(2a + 1)J\psi$, and the unitarity of J gives the inverse. $\square$

The input is recovered directly from the normalized channels by W_a^* , with inverse norm one; this adjoint already incorporates J^* . The alternative formula (71.19) separates recovery of $J\psi$ by summation from application of J^* , whose action on labels is computed through residues. Reversible arithmetic refinement and bilateral transport of amplitudes are thus composed.

Proposition 71.5 (Compatibility in the refined space). *An unnormalized pair (u, v) belongs to the image of $\psi \mapsto (Q_- \psi, Q_+ \psi)$ if and only if*

$$[(a+1)I_E + JRJ^*]u = [aI_E - JRJ^*]v. \quad (71.20)$$

Proof. The equality is equivalent to $JRJ^*(u+v) = av - (a+1)u$. If it holds, define $\psi = J^*(u+v)/(2a+1)$. Substituting into (71.16), this identity gives $Q_- \psi = u$ and $Q_+ \psi = v$. Necessity follows by inserting the actual channels. All compositions act on $\mathcal{H}_E$, so their types are well defined. $\square$

71.7 Law of observables on the two channels

Theorem 71.6 (Bilateral compression of a refined reader). *For $F = F^*$ on $\mathcal{H}_E$, write $F_J = J^*FJ$. Then*

$$W_a^* \text{diag}(F, F) W_a = \frac{a(a+1)F_J + R^*F_J R}{p}. \quad (71.21)$$

The reader F_J is preserved exactly if and only if it commutes with R .

Proof. Expanding $(a+1)Q_-^*FQ_- + aQ_+^*FQ_+$ cancels the mixed terms. The coefficient of $\mathcal{B}^*F\mathcal{B}$ is $a(a+1)(2a+1)$; that of $\mathcal{C}^*F\mathcal{C}$ is $2a+1$. Dividing by N , substituting $\mathcal{C} = JR$, and using $F_J = J^*FJ$ proves the formula. Equality with F_J is equivalent to $R^*F_J R = F_J$, or to $[F_J, R] = 0$ by unitarity. $\square$

In the case $a = 8$, the weights are $72/73$ and $1/73$. The identity reader recovers the norm. A nonconstant reader may change even when the norm is preserved; its variation is computed by (71.21). The operator that preserves an original reader A exactly on the full image is $W_a A W_a^*$, whose image identity is $W_a W_a^*$. Reading $\text{diag}(F, F)$ corresponds to the specific compression in the theorem and does not automatically replace that transported observable.

71.8 Arithmetic example in dimension one thousand and one

Take $B = 10$, $M = 100$. The label spaces have dimension 1001, and

$$d_1 = (73, 27, 1) \mapsto e_1 = (729, 271). \quad (71.22)$$

The diagonal readers give

$$f_E(e_1) = \frac{729}{1000} = \frac{73}{100} - \frac{1}{1000}, \quad g_E(e_1) = \frac{271}{1000}, \quad \gamma_E(e_1) = 1. \quad (71.23)$$

Declare R to be the transposition of the admissible labels $d_1 = (73, 27, 1)$ and $d_2 = (73, 27, 2)$, leaving all others fixed. The second image is $e_2 = (728, 272)$. The transposition rule has been fixed on the labels before any comparison with a constant.

For $a = 8$ and $\psi = |d_1\rangle$,

$$Q_- \psi = 8|e_1\rangle - |e_2\rangle, \quad Q_+ \psi = 9|e_1\rangle + |e_2\rangle. \quad (71.24)$$

The squared norms are 65 and 82, and

$$9 \cdot 65 + 8 \cdot 82 = 585 + 656 = 1241 = 17 \cdot 73. \quad (71.25)$$

The sum of the channels is $17|e_1\rangle$. The inverse (71.19) and the residue recover $|d_1\rangle$, including the carry one.

If $F = M_{f_E}$ is evaluated diagonally on both normalized channels, theorem 71.6 gives

$$\begin{aligned} \langle W_8 \psi, \text{diag}(F, F) W_8 \psi \rangle &= \frac{72 \cdot 729 + 728}{73 \cdot 1000} \\ &= \frac{729}{1000} - \frac{1}{73000}. \end{aligned} \quad (71.26)$$

The complementary readout is one minus this quantity. For the diagonal carry reader, the result is

$$\frac{72 \cdot 1 + 2}{73} = \frac{74}{73}. \quad (71.27)$$

The original carry remains exactly recoverable: the observable $W_8 M_c W_8^*$ has eigenvalue one on $W_8 |d_1\rangle$. The value $74/73$ is the response of the diagonal reader applied to both channels, whose relative transport is specified by (71.21). The example distinguishes preservation of information from variation in a particular readout.

71.9 Relation to the generated register and domains of application

The unitary J acts on $BM + 1$ labels, whereas the realization of K has twelve coordinates. In the example, 1001 counts refined pairs with sum one thousand, and twelve counts windows of the produced register. Both dimensions retain their specific combinatorial origin.

A particular composition between these two carriers can be formulated through a materially specified selector of twelve distinct paths, with its provenance and operators. If this selector provides an isometry $J_{12} : \mathbb{R}^{12} \rightarrow \ell^2(\mathcal{D}_{B,M})$, then JJ_{12} is isometric, and the intertwining of readers is checked on it. This is the hypothesis of that additional composition, not an implicit identification based on the number of coordinates. The bilateral maps already defined directly on K in section 70 retain their full domains without using that additional selector.

The result of this section brings together an arithmetic bijection, its unitary lift, intertwined diagonal readers, recovery of paths at every finite depth, and an explicit bilateral composition. The path domain is the one defined in (71.10). Applications to histories filtered by TPK retain their survival and prolongation rules; the label correspondence proves the declared transport here without replacing those operators with an unrestricted arithmetic domain.

72 Dimensional unity, relative symmetry, and recovery of structure

The holographic extreme and mean ratio articulates conservation of unity under evolution with its arithmetic and dimensional realizations. The composition developed here makes this principle precise through conserved quantities, domains, and inverses. The enriched state generated by APP-TRIT-TPK precedes the numerical and geometric readouts; its operations preserve the orientation, carry, incidence, and memory used by each realization.

72.1 Arithmetic conservation, unitarity, and isometric decomposition

Arithmetic refinement preserves normalized unity:

$$(x, y, c) \longmapsto (Bx - c, By + c), \quad \frac{Bx - c}{B(x + y)} + \frac{By + c}{B(x + y)} = 1. \quad (72.1)$$

The residue inverse recovers the canonical carry and the preceding pair. Block composition preserves the contributions of events with their scale, and successive divisions recover every finite history. The limiting readout is controlled by decreasing cylinders; boundary marks retain the distinction between histories sharing a scalar evaluation.

The lift on labels transforms that bijection into a unitary J . It preserves the norm of amplitudes and transports the fraction and carry readers through $J^*M_hJ = M_{h \circ \ell}$. The quantity $x^2 + y^2$ and the norm of $|x, y, c\rangle$ belong to different realizations. The proof of unitarity establishes the precise link between arithmetic reversibility and preservation of amplitudes.

The third conservation law corresponds to two isometric realizations:

$$(a + 1)Q_-^*Q_- + aQ_+^*Q_+ = (2a + 1)(a^2 + a + 1)I. \quad (72.2)$$

Cancellation of the mixed terms allows bilateral distribution and its norm-one reconstruction. The law holds for the transports on the domain specified in theorem 70.1. Its specialization to the ternary cycle provides additional norm and index properties; the general law remains valid for isometries without that finite order.

The chain formed by the three maps is effective: the bijection produces J ; a previously declared permutation produces the second transport JR ; the two transports satisfy the balance, and their channels recover the input through (71.19). The example in section 71.8 shows the complete procedure with (73, 27, 1), (729, 271), and 1001 admissible labels. The numbers in that realization are not introduced as substitutes for the structure generating the register K .

72.2 Complete memory and the logical contribution to the Landauer bound

The vanishing logical contribution to the Landauer bound, established in (45.4), applies to the recoverable transports constructed in this article. The bijection of theorem 71.1 preserves the carry through (71.4); its prolongation preserves the path through theorem 71.3. The archives of theorem 70.7 preserve the amplitudes and correlations needed to invert the transport.

Corollary 72.1 (Recovery and spectral conservation of the archive). *Let $Z : H \rightarrow \mathcal{K}$ be one of the complete archives Z_N, Z_∞ of theorem 70.7, or the unitary lift J_n of theorem 71.3, on its respective domain. For a finite-rank density operator ρ on H , the output $\rho_Z = Z\rho Z^*$ satisfies*

$$\text{Tr } \rho_Z = 1, \quad Z^* \rho_Z Z = \rho, \quad (72.3)$$

$$s(\rho_Z) = s(\rho), \quad s(\rho) = -\text{Tr}(\rho \log \rho), \quad (72.4)$$

with $0 \log 0 = 0$. If X is a random variable on the finite paths of theorem 71.3, then

$$H(X \mid \Phi_n(X)) = 0. \quad (72.5)$$

Proof. Write the finite spectral decomposition $\rho = \sum_j p_j |v_j\rangle\langle v_j|$, with the v_j orthonormal, $p_j > 0$, and $\sum_j p_j = 1$. The identity $Z^*Z = I$ implies that the Zv_j are orthonormal. Thus, $\rho_Z = \sum_j p_j |Zv_j\rangle\langle Zv_j|$ has the same nonzero eigenvalues, proving the trace and entropy identities. Multiplication by Z^* and Z recovers ρ . Finally, Φ_n^{-1} recovers each path from its final pair; the conditional distribution is concentrated on a single path and its entropy is zero. $\square$

Conservation pertains to the complete archive. A visible readout q , accompanied by memory M , recovers the state under the hypothesis of (45.3); the information not expressed by q remains in $H(M(X) | q(X))$. Discarding a complement or a necessary terminal defines another operation, whose recoverability is determined by the information actually retained. The preceding spectral identity preserves the distinction between entropy of the complete state, entropy of a partial reduction, and heat transferred.

In the thermal model of Section 45, with degenerate logical states, an isothermal environment, and accounting for auxiliary registers, the term associated with erasure is $k_B \Theta H(X | F(X))$, for a deterministic logical map F . Its contribution vanishes for the preceding recoverable transports; resetting a uniform TRIT satisfies $Q_{\text{reset}} \geq k_B \Theta \log 3$. Preparation, control, finite operating speed, and coupling to the environment retain their own physical balances. The total heat of an implementation requires combining these contributions with the logical operation.

72.3 Conservation of an observable and its relative symmetry

Norm balance is the identity case of a more informative operator law. When the transports are unitary, with relative operator R , the effective reader is

$$\mathcal{T}_a(G) = \frac{a(a+1)G + R^*GR}{a(a+1) + 1}. \quad (72.6)$$

The observable remains fixed exactly when $[G, R] = 0$. In the canonical case,

$$\mathcal{T}_8(G) - G = \frac{R^*GR - G}{73}. \quad (72.7)$$

This identity determines both conservation and variation: the defect measures the difference between two realizations of the same reader, weighted by the split $72 + 1$.

Complete memory also transports the observable through $A \mapsto ZAZ^*$ on $\text{im } Z$. Products, adjoints, and pairings are preserved on that image. The diagonal channel readout and the transported observable are different operations; the carry example shows the difference between $74/73$ for the former and eigenvalue one for the latter. The information remains recoverable even if a particular diagonal readout changes value.

The variational charges of proposition 35.1 retain their foundation in an explicit action and intertwining. For $A_{n+1}U_n = U_nA_n$, the transport action gives the charge $p_n^*A_nq_n$. Its conservation and the norm identity are compatible results with different hypotheses. Likewise, preservation of an alternating or Lorentzian form in (70.31) retains its geometric meaning; the noise and contraction estimates use the Hilbert positivity declared in their proofs.

72.4 Role of the register in transduction

The holographic arithmetic–geometric coupling constant has a role defined by its compositions. The register K is obtained beforehand by oriented and Hadamard extraction. Its readout κ is reversible with the base, length, and origin fixed. Its frame O_K identifies cyclic responses because its Gram matrix has a strictly positive lower bound. Incidence and geometric direction use their declared readers on the same register.

The archive isometry preserves the Gram matrix of that frame and therefore retains the recovery bounds at every depth. The complete register admits norm-one inversion; the two partial memories

retain their factor $9/4$ and their uniform kernel. The additional charge distinguishes recovery of the absolute register from that of its centred component. Integer decoding subsequently recovers the incidence selection even at a threshold lying exactly on a coordinate.

This capacity is more specific than preserving a total quantity: it allows recovery of operations and relations acting on a generated object. Its guarantees rest on the Gram matrix, the distances in the integer image, and the proven inverses, not on retaining another copy of the input by definition. At the same time, the finite number of windows of K remains distinct from the complete enriched history; the other registers and accompanying marks retain their own transports.

The orbit and its isometric transports preserve the norm $\nu = \|K\|$. Normalizing its columns to norm one requires retaining ν to invert that operation. Polar normalization uses the pair (F_K, G_K) . The formula $O_K = F_K G_K^{1/2}$ preserves orientation and amplitudes. The positional reader must be transported with the change of chart, since the common orbital norm does not make κ invariant under an isolated renumbering of its components.

72.5 Depth, dimensions, and physical realizations

The bilateral policy that continues one channel and archives the other has uniform exhaustion of the terminal. Its limit is an isometry into the sum of all memories, and its inverse is the convergent adjoint series in (70.25). A finite number of complements has two distinct reconstructions: the truncated adjoint retains an explicit bias, and the least-squares inverse eliminates it with the computed noise amplification. The preceding evolutions with a positive terminal retain that component in the complete archive.

The dimension of the label space, the rank of a projector, and the dimension of a physical realization are related through their concrete maps. The unitary on 1001 labels and the twelve-window register are two explicit realizations of different domains. The exterior conservation law of the preceding development extends the Gram products to every finite degree, including the mixed contributions between terminal and memories. The exceptional and physical realizations remain linked to their already constructed representations and operators; numerical ranks alone do not replace that link.

72.6 Synthesis of conservation under evolution

The physical actions and the Gaussian protocol of the manuscript retain the couplings, forms, units, and preparations involved in their proofs. The bilateral balance added here extends the collection of conservative transports and their readouts; its application to a physical realization is established by composing the operators of that realization without replacing them with a coincidence of normalization.

The unity of the development thus resides in a proven sequence: generation and completion, refinement with memory, unitary realization, relative symmetry of the reader, and recovery of structure. The holographic extreme and mean ratio thereby receives an exposition in which change of scale, incidence, conservation, and transduction have explicit operations and jointly retain the information needed to invert their readouts.

Arithmetic formalization of the bilateral balance

The nonadic specialization of the bilateral law admits an independent arithmetic formalization. The point of application comes after the construction of its coefficients and transports: the base-ten partition fixes $a = 8$, and the two registers are $8x - y$ and $9x + y$. On this domain, Lean 4.21.0 verifies seven theorems using its `Std` library. The formal sources and execution procedure are provided as a supplement. This addendum states their mathematical content and relates it to the operator proofs in the article.

Polynomial identity and conservation of the norm

For all $x, y \in \mathbb{Z}$, we have

$$9(8x - y)^2 + 8(9x + y)^2 = 17(72x^2 + y^2).$$

Expanding the two squares cancels the cross terms. When $y^2 = x^2$, the same equality gives

$$9(8x - y)^2 + 8(9x + y)^2 = 17 \cdot 73 x^2.$$

These are the first two formalized theorems. The first expresses a balance for arbitrary integer inputs; the second applies the explicit norm-conservation hypothesis. The integer $73 = 72 + 1$ occupies exactly the place it has in theorem 70.1.

Arbitrary finite dimension

Let $n \in \mathbb{N}$ and $x, y : \mathbb{N} \rightarrow \mathbb{Z}$. Define

$$S_n(x) = \sum_{j < n} x_j^2, \quad E_n(x, y) = \sum_{j < n} (9(8x_j - y_j)^2 + 8(9x_j + y_j)^2).$$

The next two theorems establish

$$E_n(x, y) = 17(72S_n(x) + S_n(y))$$

and, under the hypothesis $S_n(y) = S_n(x)$,

$$E_n(x, y) = 17 \cdot 73 S_n(x).$$

The formal proof proceeds by induction on n . The initial case is the empty sum; the induction step adds one coordinate and applies the polynomial identity already proved. The universal quantifier over n includes all finite dimensions. The hypothesis uses the total sum of squares, so it allows redistributions among coordinates.

Recovery of the inputs

Write $q_- = 8x - y$ and $q_+ = 9x + y$. The fifth and sixth theorems are the exact equalities

$$q_- + q_+ = 17x, \quad 8q_+ - 9q_- = 17y.$$

On the image of the map $(x, y) \mapsto (q_-, q_+)$, both left-hand sides are multiples of 17. Division recovers the two inputs. The seventh theorem states their injectivity consequence:

$$\left. \begin{array}{l} 8x - y = 8x' - y', \\ 9x + y = 9x' + y' \end{array} \right\} \implies x = x' \text{ and } y = y'.$$

Cancellation in $\mathbb{Z}$ justifies this conclusion. The formal content coincides with recovery of the pair of registers before they are normalized as channels of a metric realization.

Relation between the proofs

The formalization uses arbitrary integer variables and mathematical induction. It therefore constitutes a proof of the preceding statements, distinct from checking a finite set of examples. The system accepts the proof terms without added axiomatic declarations or deferred proofs. The supplement also records the logical dependencies reported by the tool for each declaration.

The operator law in the body of the article has a broader domain: isometries between Hilbert spaces, an archive of arbitrary depth and conjugate readers. Its proofs are found in theorem 70.1 and sections 72.3 and 72.5. The Lean proof included here certifies the stated arithmetic specialization, with its quantifiers and starting structure. The provenance of the coefficients, generation of the register and exceptional realizations retain the earlier proofs on which they depend.

$$K \longleftrightarrow \Lambda_t \longleftrightarrow \Lambda_t^*$$

Recoverable geometry and reciprocity of coupling

Positive deformation, lattice duality, polar structure, and oriented recovery of action.

The function of the holographic arithmetic–geometric coupling constant becomes more precise through the transformations determined by its register. Its positional reading preserves twelve ordered components; its incidence realization distinguishes a direction; the bilateral archive allows that direction to be reconstructed. Composing these operations produces a collection of geometries whose fundamental volume remains constant and whose inversion yields the dual lattice. The register therefore participates both in the constitution of the geometry and in its recovery.

This development preserves the generative order established in the article. The additive and multiplicative evaluations of APP, the orientation of TRIT, and the selection, transport, and memory operations of TPK produce the enriched state and its joint discrete structure of the continuum. The regional coordinates and exceptional incidence are readings of that state. The vector, lattice, and hyperbolic realizations that follow receive those outputs together with their charts and orientations. The genealogy is preserved under a change of representation.

The distinction between scale and shape acquires an exact expression here. A deformation can expand certain directions and contract others while preserving the total product of its factors. In the action ellipse, the product of the semiaxes preserves the action section. In the twelve radii determined by K , compensation preserves the lattice covolume. In the mass operator, a decomposition separates the uniform factor from the relative transformation of determinant one. Each conservation law has its own object and proof; the readers relate them without conflating their domains.

Radial reciprocity connects two complementary constructions. In the chart of the spectral double circle, inverting the radius is exactly equivalent to reflecting the complex coordinate across the line of real part $1/2$. In the modular chart of the ellipse, exchanging the semiaxes inverts the radius and changes the sign of the Klein coordinate. Both correspondences admit inverse formulas. The orientation register distinguishes the conjugate reading from a modification of the physical action section.

Substituting the complete expressions for the angular readers reveals a specific arithmetic cancellation: the linear term of the pre-return action section cancels against the corresponding term of the counter-angle. The regional continuation remains intact. The angular contrast, the radius, and the hyperbolic coordinate are then expressed through the same representations that enter into α , with the coupling constant within its dodecaphasic closure. Conversely, the oriented radial coordinate recovers the return section and allows the complete electronic expression to be rewritten.

The spectral realization, in turn, distinguishes two classes of information. Lattice duality imposes a functional equation; the covolume fixes the poles and their residues; deformation changes the entire part of the completed function. The archive recovers the direction determining that variation and provides explicit bounds on reconstruction error. Conservation and reversibility are thus expressed through operations encompassing the register, the geometry, and their analytic readings.

73 Positive deformation and lattice duality of the generated register

73.1 Coupling direction and unimodular flow

The extraction of the register, its rational reading, and the incidence projectors have been constructed in the preceding sections. The base-thousand convention, phase origin, carry boundary, and chart of twelve positions are retained. In that chart,

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), \quad u = P_3 P_{11} K,$$

$$P_{11} = I - \frac{1}{12} \mathbf{1} \mathbf{1}^\top, \quad u \perp \mathbf{1}, \quad \|u\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

The finite definition of P_3 , the order of its classes, and the exact evaluation of this norm are given in Lemma 16.10. The following realization uses that generated object. We write

$$\rho = \frac{\pi_{\text{HMT}}}{\varphi_{\text{HMT}}^2} > 0, \quad h_i = (\log \rho) \frac{u_i}{\|u\|}, \quad r_i(t) = e^{th_i}, \quad g_t = \text{diag}(r_i(t)). \quad (73.1)$$

The regional coordinates are evaluated before this composition. The parameter $t \in \mathbb{R}$ indexes a mathematical deformation; its definition identifies it with neither the TPK clock nor a refinement depth.

Proposition 73.1 (Scale compensation). *The flow satisfies*

$$g_t g_s = g_{t+s}, \quad g_0 = I, \quad g_t^{-1} = g_{-t}, \quad \det g_t = \prod_i r_i(t) = 1.$$

Proof. Orthogonality to the uniform mode gives $\sum_i h_i = 0$. Multiplication of exponentials proves the group law and inversion. The determinant is $\exp(t \sum_i h_i) = 1$. Every diagonal factor is positive. $\square$

The conservation law is multiplicative and global: each radius may vary. In logarithmic coordinates, the vanishing sum is preserved. This distinction will provide the geometric interpretation of the transverse projector of rank ten.

73.2 Exact reciprocity of the double circle

Consider the complex chart and its inverse

$$z(s) = \frac{s-1}{s}, \quad s(z) = \frac{1}{1-z}.$$

For $r > 0$, $z = re^{i\theta} \neq 1$, define $s(r, \theta) = (1 - re^{i\theta})^{-1}$.

Proposition 73.2 (Conjugacy of the spectral reflection). *The following identities hold:*

$$s(r^{-1}, \theta) = 1 - \overline{s(r, \theta)}, \quad (73.2)$$

$$\Re s(r, \theta) - \frac{1}{2} = \frac{1 - r^2}{2|1 - re^{i\theta}|^2}. \quad (73.3)$$

The unit circle, apart from the excluded point $z = 1$, corresponds to the line $\Re s = 1/2$.

Proof. Radial inversion at fixed angle is $z' = 1/\bar{z}$. Hence

$$\frac{1}{1-z'} = \frac{\bar{z}}{\bar{z}-1} = 1 - \frac{1}{1-\bar{z}}.$$

Moreover, $(1-z)^{-1} = (1-\bar{z})/|1-z|^2$. Subtracting $1/2$ from its real part gives (73.3). $\square$

The relation to the radii of (73.1) is explicit: $r_i(-t) = r_i(t)^{-1}$. Equality of products preserves volume, whereas the proposition identifies an involution between complex charts. Locating the zeros of a function requires the properties of that function; it does not follow from the unit product of radii.

Three circular realizations in the corpus should be distinguished. The spectral chart maps $s = 1/2 + i\gamma$ to $z_\gamma = (\gamma + i/2)/(\gamma - i/2)$. The dual circle of integer memory maps an integer n to the character z^n . Nonadic compression maps z to $(8 + z)/9$, a circle with centre $8/9$ and radius $1/9$, tangent to the unit circle at 1. The modal ratio $5/9$, by contrast, belongs to the eigenvalues 9, 5 of the APP block $\begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$. Their functions remain distinct when they are composed.

73.3 Positroid strata and balanced invariants

Let X be a real matrix of rank k , with twelve columns, and let $\Delta_I(X)$ denote its maximal minor with columns I . The right action satisfies

$$\Delta_I(Xg_t) = e^{t \sum_{i \in I} h_i} \Delta_I(X). \quad (73.4)$$

Diagonal positivity preserves all signs and all vanishing minors. Thus, for $1 \leq k \leq 11$, a trajectory starting in $\text{Gr}_{\geq 0}(k, 12)$ remains in the same positroid stratum. The stratification by minors is used here as a downstream realization; see Postnikov's general construction [27].

Proposition 73.3 (Positive duality of complementary ranks). *Let $A_\pm = \text{diag}(1, -1, 1, -1, \dots)$ and $\mathfrak{D}(V) = V^\perp A_\pm$, with the positive orientation of its complementary minors. Then*

$$\mathfrak{D}(Vg_t) = \mathfrak{D}(V)g_{-t}.$$

Proof. Since g_t is diagonal and self-adjoint, $(Vg_t)^\perp = V^\perp g_t^{-1}$. The matrix $A_\pm$ commutes with g_t . For the signs of the minors, orthogonal complementation introduces, up to a global sign, $(-1)^{\sum_{j \in J} j}$. Multiplication of the columns by $(-1)^{j-1}$ leaves a sign independent of J ; a single global orientation removes it. The complementary minors are then nonnegative. $\square$

If $\mathbf{1}_I + \mathbf{1}_J = \mathbf{1}_L + \mathbf{1}_M$ and the denominator is nonzero, coordinatewise cancellation of the exponents proves

$$\frac{\Delta_I(Xg_t)\Delta_J(Xg_t)}{\Delta_L(Xg_t)\Delta_M(Xg_t)} = \frac{\Delta_I(X)\Delta_J(X)}{\Delta_L(X)\Delta_M(X)}. \quad (73.5)$$

In rank two this recovers the four-index ratios and their exchange relations. By contrast, the product of complementary minors of the two paired trajectories receives the same factor twice and may vary. Nor does this action make the circulant frame O_K , which is invertible for recovery, totally nonnegative: its minors retain the signs established in its own realization.

73.4 Covolume, dual lattice, and ternary symmetry

Let Λ be the marked Leech lattice constructed earlier, self-dual and of covolume one in the metric of its twelve orthogonal planes of type A_2 . The marked isometry

$$\sigma = \bigoplus_{i=1}^{12} P_i C P_i^{-1}, \quad C^2 + C + I = 0,$$

preserves the gluing and Λ , has order three, and has no nonzero fixed vectors. The P_i are plane frames, distinct from the projectors P_3, P_{11} . Define

$$G_t = \bigoplus_{i=1}^{12} r_i(t) I_{A_{2,i}}, \quad \Lambda_t = G_t \Lambda. \quad (73.6)$$

Theorem 73.4 (Reciprocal lattice deformation). *For every $t \in \mathbb{R}$,*

$$\text{covol}(\Lambda_t) = 1, \quad \Lambda_t^* = \Lambda_{-t}, \quad \sigma\Lambda_t = \Lambda_t.$$

Every isometry of Λ that commutes with G_t also preserves Λ_t .

Proof. Each radius acts on two coordinates; hence $\det G_t = \prod_i r_i(t)^2 = 1$. For an invertible operator G , a vector y belongs to $(G\Lambda)^*$ exactly when $G^*y \in \Lambda^*$. Therefore $(G\Lambda)^* = G^{-*}\Lambda^*$. Applying self-duality, self-adjointness, and inversion gives $\Lambda_t^* = G_{-t}\Lambda$. Finally, G_t is scalar on each plane and commutes with σ , whence $\sigma G_t\Lambda = G_t\sigma\Lambda = G_t\Lambda$. The same calculation applies to the other commuting isometries. The operator σ is unchanged, so it retains its order and its absence of fixed vectors. $\square$

The point $t = 0$ recovers the original lattice together with its integrality, evenness, self-duality, and exceptional realization. The full collection consists of unimodular Euclidean lattices in the sense of covolume one; away from the original point, neither integrality nor a Moonshine realization is presumed. The theorem identifies precisely the ternary symmetry that persists and the duality relating the two directions of deformation.

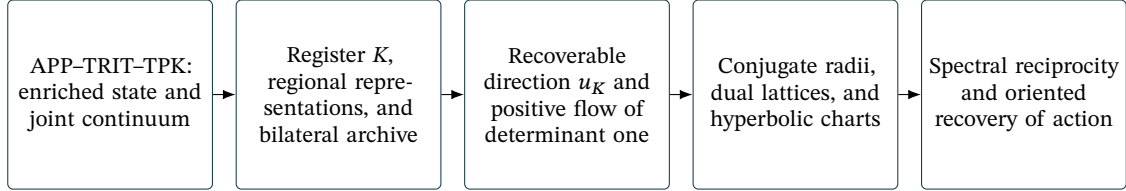


Figure 16: Dependencies of the extension. The arrows represent compositions specified in the proofs; recovery preserves the charts and orientation.

74 Split form, integrality, and lattice energy

The deformation in Theorem 73.4 starts from the marked lattice Λ , the direction $u = P_3 P_1 K$, and the previously generated regional coordinates. Let E be its twenty-four-dimensional real space, and let $\langle \cdot, \cdot \rangle_B$ denote the positive metric of the twelve orthogonal planes of type A_2 . We retain the normalization of (73.1) and (73.6):

$$\kappa = \frac{\log \rho}{\|u\|}, \quad h_i = \kappa u_i, \quad G_t = \bigoplus_{i=1}^{12} e^{th_i} I_{A_{2,i}}, \quad \Lambda_t = G_t \Lambda. \quad (74.1)$$

The operator G_t is self-adjoint for B , and $G_t^{-1} = G_{-t}$. If the same collection is written with the unnormalized parameter, $D_s = \bigoplus_i e^{su_i} I_{A_{2,i}}$, the exact relation is $G_t = D_{\kappa t}$. The deformation parameter and the TPK clock retain their respective domains.

74.1 Arithmetic variation of the Euclidean projection

The ordered chart in Lemma 16.10 determines, through the finite sum (16.23),

$$u_1 = -\frac{495}{4} - \frac{233}{10}\sqrt{5}, \quad \sum_{i=1}^{12} u_i = 0. \quad (74.2)$$

The regional normalization satisfies $\rho = \pi_{\text{HMT}}/\varphi_{\text{HMT}}^2 > 1$: in the realization already established, $\pi_{\text{HMT}} > 3$ and $\varphi_{\text{HMT}}^2 = (3 + \sqrt{5})/2 < 3$. Thus $\kappa > 0$. These comparisons concern the regional outputs after their generation.

Proposition 74.1 (Local witness for variation of integrality). *There exists $\delta > 0$ such that, for $0 < |t| < \delta$, the lattice Λ_t contains a vector with noninteger squared norm. Consequently, its Euclidean metric lacks integrality and an isometry with the Leech lattice.*

Proof. The marked vector in (16.28) is $v_\alpha = 4\rho_1 + \sum_{j=2}^{12} \rho_j$, with $\|\rho_j\|_B^2 = 2$ in mutually orthogonal planes and $v_\alpha^2 = 54$. The neighbour construction (16.29) gives $v_\alpha/3 \in \Lambda$. Its transported norm is

$$f(t) = \left\| G_t \frac{v_\alpha}{3} \right\|_B^2 = \frac{2}{9} \left(16e^{2th_1} + \sum_{j=2}^{12} e^{2th_j} \right). \quad (74.3)$$

The zero sum and (74.2) give

$$\begin{aligned} f(0) &= 6, \\ f'(0) &= \frac{4}{9} \left(16h_1 + \sum_{j=2}^{12} h_j \right) = \frac{20}{3} \kappa u_1 = \kappa \left(-825 - \frac{466}{3}\sqrt{5} \right) \neq 0. \end{aligned} \quad (74.4)$$

Continuity of f' provides an interval $(-\delta, \delta)$ on which f is strictly monotone. Reducing δ also gives $11/2 < f(t) < 13/2$. For $0 < |t| < \delta$, monotonicity yields $f(t) \neq f(0) = 6$; six is the only integer in that interval. Thus $G_t(v_\alpha/3)$ has noninteger squared norm. An integral lattice has integer squared norms, and isometries preserve those norms. Integrality of the Leech lattice proves the final conclusion. $\square$

The deformation preserves covolume and ternary symmetry while varying its arithmetic metric structure. Reciprocity $\Lambda_t^* = \Lambda_{-t}$ uses self-adjointness of the flow and initial self-duality. An ambient permutation interchanging expansions and contractions acts on the glued lattice when it also preserves the code and the neighbour from which that lattice is constructed.

74.2 Integral self-duality for the split-signature form

Simultaneous consideration of both projections uses the space $E \oplus E$, with bilinear form

$$\mathcal{B}((x, p), (x', p')) = \langle x, p' \rangle_B + \langle p, x' \rangle_B. \quad (74.5)$$

Define

$$\begin{aligned} \mathcal{G}_t &= G_t \oplus G_{-t}, \\ \Gamma_t &= \mathcal{G}_t(\Lambda \oplus \Lambda) = \{(G_t \lambda, G_{-t} \mu) : \lambda, \mu \in \Lambda\}. \end{aligned} \quad (74.6)$$

Theorem 74.2 (Integrality, evenness, and self-duality of the split lattice). *The form $\mathcal{B}$ has signature $(24, 24)$. For every $t \in \mathbb{R}$, $\mathcal{G}_t$ is an isometry of $\mathcal{B}$, and Γ_t is an integral, even, self-dual lattice with respect to that form.*

Proof. The coordinates

$$p_L = \frac{x + p}{\sqrt{2}}, \quad p_R = \frac{x - p}{\sqrt{2}}$$

give an invertible linear change, with $x = (p_L + p_R)/\sqrt{2}$ and $p = (p_L - p_R)/\sqrt{2}$. They satisfy

$$\mathcal{B}((x, p), (x, p)) = \|p_L\|_B^2 - \|p_R\|_B^2.$$

Positivity of B in dimension twenty-four gives signature $(24, 24)$. Self-adjointness of G_t and the identity $G_t G_{-t} = I$ imply

$$\langle G_t x, G_{-t} p' \rangle_B = \langle x, G_t G_{-t} p' \rangle_B = \langle x, p' \rangle_B.$$

The other cross product satisfies the same equality. Hence $\mathcal{B}(\mathcal{G}_t \xi, \mathcal{G}_t \eta) = \mathcal{B}(\xi, \eta)$.

On $\Gamma_0 = \Lambda \oplus \Lambda$, the cross products are integers because Λ is integral, and

$$\mathcal{B}((\lambda, \mu), (\lambda, \mu)) = 2\langle \lambda, \mu \rangle_B \in 2\mathbb{Z}.$$

This proves integrality and evenness. If (x, p) belongs to the dual of Γ_0 with respect to $\mathcal{B}$, its pairings with $(\lambda, 0)$ and $(0, \mu)$, for all $\lambda, \mu \in \Lambda$, require respectively $p, x \in \Lambda^* = \Lambda$. Conversely, these two memberships ensure that every pairing is integral. Thus $\Gamma_0^{*\mathcal{B}} = \Gamma_0$. The isometry $\mathcal{G}_t$ transports all three properties, giving $\Gamma_t^{*\mathcal{B}} = \mathcal{G}_t \Gamma_0^{*\mathcal{B}} = \Gamma_t$. $\square$

The metric determines the domain of each conclusion. The cross products of $\mathcal{B}$ preserve integrality of Γ_t , including at the parameter values of Proposition 74.1. The lattices Γ_t , equipped with that split form, are mutually isometric. Their Euclidean projections Λ_t and Λ_{-t} , equipped with B , vary with the parameter.

Proposition 74.3 (Exchange of reciprocal projections). *The involution $\mathcal{J}(x, p) = (p, x)$ satisfies*

$$\mathcal{J}\Gamma_t = \Gamma_{-t}, \quad \mathcal{J}\mathcal{G}_t\mathcal{J} = \mathcal{G}_{-t}.$$

In the coordinates (p_L, p_R) , it fixes p_L and reverses the sign of p_R .

Proof. The exchange transforms $(G_t \lambda, G_{-t} \mu)$ into $(G_{-t} \mu, G_t \lambda)$. As λ, μ range over Λ , the image ranges over Γ_{-t} . Conjugation interchanges the two blocks of $\mathcal{G}_t$, producing $\mathcal{G}_{-t}$. Finally, $x + p$ is preserved and $x - p$ changes sign. $\square$

This involution acts on the complete lattice construction. A subsequent physical representation must also transport the operators and identification data proper to that realization.

74.3 Positive operator and unitary equivalence of domains

The energy of the two projections is defined on the fixed lattice by

$$E_t^{\text{lat}}(\lambda, \mu) = \|G_t \lambda\|_B^2 + \|G_{-t} \mu\|_B^2, \quad \lambda, \mu \in \Lambda. \quad (74.7)$$

It is nonnegative, vanishes exactly at $(0, 0)$, and satisfies $E_t^{\text{lat}}(\lambda, \mu) = E_{-t}^{\text{lat}}(\mu, \lambda)$. Exchange of the two contributions expresses, on the twelve marked planes, the reciprocity of a sum of oppositely scaled terms. For $M_h = \max_i |h_i|$, the orthogonal decomposition into planes gives

$$e^{-2|t|M_h} \|v\|_B^2 \leq \|G_t v\|_B^2 \leq e^{2|t|M_h} \|v\|_B^2. \quad (74.8)$$

Indeed, each summand $\|v_i\|_B^2$ is multiplied by e^{2th_i} , which lies between these bounds.

Theorem 74.4 (Self-adjointness, compact resolvent, and unitary duality). *On $\ell^2(\Lambda \oplus \Lambda)$, the operator*

$$(H_t^{\text{lat}} \psi)(\lambda, \mu) = E_t^{\text{lat}}(\lambda, \mu) \psi(\lambda, \mu)$$

with maximal domain

$$\mathcal{D}(H_t^{\text{lat}}) = \left\{ \psi \in \ell^2(\Lambda \oplus \Lambda) : \sum_{\lambda, \mu} (E_t^{\text{lat}}(\lambda, \mu))^2 |\psi(\lambda, \mu)|^2 < \infty \right\} \quad (74.9)$$

is positive and self-adjoint with compact resolvent. The map $U\psi(\lambda, \mu) = \psi(\mu, \lambda)$ is unitary and satisfies

$$U\mathcal{D}(H_t^{\text{lat}}) = \mathcal{D}(H_{-t}^{\text{lat}}), \quad UH_t^{\text{lat}}U^{-1} = H_{-t}^{\text{lat}}. \quad (74.10)$$

Proof. Finitely supported functions belong to the domain and form a dense subspace. The energy is real. If ψ belongs to the adjoint domain, evaluating its functional against each singleton-supported orthonormal basis vector forces the corresponding adjoint coordinate to be $E_t^{\text{lat}}(\lambda, \mu) \psi(\lambda, \mu)$. Membership of this sequence in ℓ^2 is exactly condition (74.9). Conversely, that condition permits pairing with every function in the domain by Cauchy–Schwarz. The adjoint domain and action coincide with those of H_t^{lat} , proving self-adjointness. Moreover,

$$\langle \psi, H_t^{\text{lat}} \psi \rangle = \sum_{\lambda, \mu} E_t^{\text{lat}}(\lambda, \mu) |\psi(\lambda, \mu)|^2 \geq 0.$$

By (74.8),

$$E_t^{\text{lat}}(\lambda, \mu) \geq e^{-2|t|M_h} (\|\lambda\|_B^2 + \|\mu\|_B^2).$$

Discreteness of $\Lambda \oplus \Lambda$ implies that every energy sublevel set is finite. The resolvent $(H_t^{\text{lat}} + I)^{-1}$ is diagonal, with coefficients $(1 + E_t^{\text{lat}})^{-1}$. Its restriction to the sublevel $E_t^{\text{lat}} \leq R$ has finite rank, and the remainder has norm at most $(1 + R)^{-1}$. The resolvent is therefore a norm limit of finite-rank operators and is compact.

Index exchange preserves the ℓ^2 norm and satisfies $U^{-1} = U$. Replacing (λ, μ) by (μ, λ) in the sum defining the domain and using the energy equality gives the first identity in (74.10). On that domain,

$$(UH_t^{\text{lat}}U^{-1}\psi)(\lambda, \mu) = E_t^{\text{lat}}(\mu, \lambda) \psi(\lambda, \mu) = E_{-t}^{\text{lat}}(\lambda, \mu) \psi(\lambda, \mu),$$

which proves the operator identity together with its domains. $\square$

The three preservation laws are thus typed. The Euclidean metric preserves covolume and ternary symmetry and relates opposite parameters through the dual lattice. The split form preserves integrality, evenness, and self-duality of the doubled lattice. The operator realization preserves energy through a unitary equivalence that transports its domain. The theta reader in Theorem 75.1 expresses reciprocity in the convergent sum of lengths, and Mellin continuation preserves its polar structure by Theorem 75.2.

75 Theta–Mellin reciprocity and conservation of the polar structure

75.1 Periodization of the generated lattice

The lattice Λ_t of Theorem 73.4 has an analytic reading that preserves its dependence on the register. For $x > 0$, define

$$\Theta_t(x) = \sum_{v \in \Lambda} \exp(-\pi_{\text{HMT}} x \|G_t v\|^2). \quad (75.1)$$

In this realization, the previously generated coordinate π_{HMT} is identified with the circular normalization of Fourier analysis. That identification follows the construction of the lattice and the flow. To simplify the analytic notation, we shall use π with this meaning.

On every compact interval of t , the singular values of G_t are bounded above and below by positive constants. The lattice has $O(R^{24})$ points in a ball of radius R . Comparison over annuli with a Gaussian proves absolute and uniform convergence of (75.1) and its derivatives when x remains in a compact subset of $(0, \infty)$. The mode $v = 0$ contributes exactly one.

Theorem 75.1 (Reciprocity of the theta reading). *For every $x > 0$ and $t \in \mathbb{R}$,*

$$\Theta_t(x) = x^{-12} \Theta_{-t}(1/x). \quad (75.2)$$

Proof. With the convention $\widehat{f}(\xi) = \int f(y) e^{-2\pi i \langle y, \xi \rangle} dy$, the Gaussian $f_x(y) = e^{-\pi x \|y\|^2}$ in dimension twenty-four has transform $\widehat{f}_x(\xi) = x^{-12} e^{-\pi \|\xi\|^2/x}$. Periodize over Λ_t . Integrating the periodization over a fundamental domain shows that the Fourier coefficient indexed by $\xi \in \Lambda_t^*$ is $\widehat{f}_x(\xi) / \text{covol}(\Lambda_t)$. Both Gaussian series converge absolutely; evaluating the expansion at the origin gives

$$\sum_{v \in \Lambda_t} f_x(v) = \text{covol}(\Lambda_t)^{-1} \sum_{\xi \in \Lambda_t^*} \widehat{f}_x(\xi).$$

Substitute $\text{covol}(\Lambda_t) = 1$ and $\Lambda_t^* = \Lambda_{-t}$. The proof retains the constant mode on both sides. $\square$

The power x^{-12} comes from rank twenty-four. Inversion of the parameter follows from the dual lattice; it is not added as a symmetry independent of the reading.

75.2 Meromorphic continuation and functional equation

For $\Re s > 12$, let

$$\mathcal{E}_t(s) = \sum_{v \in \Lambda \setminus \{0\}} \|G_t v\|^{-2s}, \quad \mathcal{L}_t(s) = \pi^{-s} \Gamma(s) \mathcal{E}_t(s). \quad (75.3)$$

The lattice-point count above proves absolute convergence in that half-plane. Applying the gamma integral term by term yields

$$\mathcal{L}_t(s) = \int_0^\infty (\Theta_t(x) - 1) x^{s-1} dx.$$

Theorem 75.2 (Polar decomposition and bilateral reciprocity). *The function $\mathcal{L}_t$ extends meromorphically through*

$$\begin{aligned} \mathcal{L}_t(s) &= \frac{1}{s-12} - \frac{1}{s} \\ &\quad + \int_1^\infty [(\Theta_t(x) - 1)x^{s-1} + (\Theta_{-t}(x) - 1)x^{11-s}] dx. \end{aligned} \quad (75.4)$$

Its only poles are simple, at $s = 0, 12$, with residues $-1, +1$, respectively. It satisfies

$$\mathcal{L}_t(s) = \mathcal{L}_{-t}(12-s), \quad \Xi_t(s) := s(s-12)\mathcal{L}_t(s) = \Xi_{-t}(12-s), \quad (75.5)$$

and Ξ_t is entire.

Proof. Split the integral at $x = 1$. For $0 < x < 1$, theta reciprocity gives

$$\Theta_t(x) - 1 = x^{-12} - 1 + x^{-12}(\Theta_{-t}(1/x) - 1).$$

The first two terms integrate to $1/(s - 12) - 1/s$. In the third, the change of variable $y = 1/x$ gives $\int_1^\infty (\Theta_{-t}(y) - 1)y^{11-s} dy$. The positive minimum length of the lattice ensures exponential decay of $\Theta_t(x) - 1$ as $x \rightarrow \infty$. On every compact set of s , that decay dominates every power of x and $\log x$; the integrals and their complex derivatives therefore converge uniformly and define entire functions. The residues are read from the rational part. Simultaneously replacing (t, s) by $(-t, 12 - s)$ exchanges the integrals and leaves the rational part invariant. Multiplication by $s(s - 12)$ removes both poles. $\square$

The centre $s = 6$ follows from the rank and the exponent $-2s$ adopted in (75.3). This collection is the Epstein completion of the deformed lattices. Its spectral variable and normalization remain distinct from those of the Riemann zeta function; the relation proved between the two constructions is the reciprocity of their circular and lattice realizations.

Corollary 75.3 (Polar conservation and antisymmetric component). *The functions*

$$\mathcal{L}_t(s) - \mathcal{L}_0(s), \quad \mathcal{A}_t(s) = \frac{\mathcal{L}_t(s) - \mathcal{L}_{-t}(s)}{2}$$

are entire. Moreover,

$$\mathcal{A}_t(12 - s) = -\mathcal{A}_t(s), \quad \mathcal{A}_t(6) = 0.$$

Proof. The rational part of (75.4) is independent of t , so it cancels in both differences. The functional equation changes the sign of the second. Evaluation at its fixed point gives $2\mathcal{A}_t(6) = 0$. $\square$

The deformation therefore resides in an entire difference and preserves the polar data exactly. The central zero belongs to the bilateral difference, without imposing a zero on each $\mathcal{L}_t$. The poles described are singularities of a Mellin transform; the physical connection with the mass operator is established later through the action section, with its own variable and reader.

75.3 Stable reconstruction from the bilateral archive

For the unitary cyclic shift S , the preceding recovery law provides

$$T_j = \frac{8I + S^j}{9}, \quad D_j = \frac{\sqrt{8}(S^j - I)}{9}, \quad j = 3, 4, \\ m = (D_3K, D_4T_3K) = MK, \quad LM = P_{11}, \quad \|L\| = \frac{9}{4}. \quad (75.6)$$

Consequently,

$$u = P_3Lm. \quad (75.7)$$

With the incidence chart, the marked lattice, and the regional ratio ρ fixed, this identity recovers successively $u/\|u\|$, g_t , G_t , Λ_t , Θ_t , and $\mathcal{E}_t$. Reconstructing this collection requires neither the terminal T_4T_3K nor the register's uniform charge. The geometric antecedents remain among the fixed data.

Theorem 75.4 (Geometric and spectral stability of recovery). *Suppose that $m' = m + e$, $\|e\| \leq \varepsilon$, and $9\varepsilon/4 < \|u\|$. Let $u' = P_3Lm'$, and reconstruct the flows with $u'/\|u'\|$, leaving ρ, P_3, L and the charts unchanged. If $a = |t \log \rho|$, then*

$$\|g'_t - g_t\|, \|G'_t - G_t\| \leq \frac{9}{2} ae^a \frac{\varepsilon}{\|u\|}. \quad (75.8)$$

With $H_K = \text{diag}(h_i)$, $b = |t|\|H'_K - H_K\|$, we have

$$b \leq \frac{9a\varepsilon}{2\|u\|},$$

$$\Theta_t(e^{2b}x) \leq \Theta'_t(x) \leq \Theta_t(e^{-2b}x), \quad (75.9)$$

$$e^{-2\sigma b} \mathcal{E}_t(\sigma) \leq \mathcal{E}'_t(\sigma) \leq e^{2\sigma b} \mathcal{E}_t(\sigma), \quad \sigma > 12. \quad (75.10)$$

Proof. The norm of the recovery operator gives $\|u' - u\| \leq 9\varepsilon/4$, ensuring $u' \neq 0$. The triangle inequality and the reverse triangle inequality for norms imply

$$\left\| \frac{u'}{\|u'\|} - \frac{u}{\|u\|} \right\| \leq \frac{2\|u' - u\|}{\|u\|}.$$

Every exponent lies between $-a$ and a . Applying the mean value theorem to the exponential and taking the diagonal operator norm proves (75.8); repeating each entry twice does not change that norm. The same estimate before exponentiation gives the bound on b .

Since both generators are diagonal in the same chart,

$$e^{-2b}\|G_t v\|^2 \leq \|G'_t v\|^2 \leq e^{2b}\|G_t v\|^2.$$

Applying the decreasing exponential and summing proves (75.9). Raising to $-\sigma$ and summing within the domain of convergence proves (75.10). The bounds apply to all vectors, not to a finite truncation of the series. $\square$

The error ε is measured in the normalized memories of (75.6). If the rational records $z = m/\sqrt{8}$ are retained, their error is multiplied by $\sqrt{8}$ before applying the estimate. Independent errors in the regional ratio or the charts require their own bounds. The result recovers a collection of geometries and analytic readings, with quantitative control of the archive that determines it.

75.4 Pell dilation, Catalan moment and oriented return

The preceding parametric moment $\mathcal{C}_2$ retains the depth variable; write $G_{\text{Cat}} = \mathcal{C}_2(1)$. Its following specialization receives the scale of the Pell spectral screw

$$V^{-1}SV = \lambda_P iS, \quad \lambda_P = 2 + \sqrt{3}, \quad \rho = \lambda_P^{-1} = 2 - \sqrt{3}.$$

The covariance law is the operator hypothesis of this specialization; it is understood on the common invariant domain of S and the invertible transport V . The unit λ_P belongs to $\mathbb{Q}(\sqrt{3})$; the unit $3 + 2\sqrt{2}$ of the compression realization belongs to $\mathbb{Q}(\sqrt{2})$ and retains its own reader. On its scalar transverse orbit,

$$z_n = z_0(\lambda_P i)^n, \quad r_n = r_0 \lambda_P^n, \quad \theta_n = \theta_0 + \frac{\pi_{\text{HMT}} n}{2}.$$

For $z_0 \neq 0$, eliminating n gives the suspension

$$r(\theta) = r_0 \exp\left(\frac{2 \log \lambda_P}{\pi_{\text{HMT}}}(\theta - \theta_0)\right).$$

The inverse branch has multiplier $-i\rho$. This scale realization receives the screw as its operator antecedent and retains its orientation.

Proposition 75.5 (Spectral reading and evaluation at the contractive scale). *On $\mathcal{H} = \ell^2(\mathbb{N}_{>0})$, let*

$$N\delta_n = n\delta_n, \quad U_4\delta_n = i^n\delta_n, \quad Q_4 = \frac{U_4 - U_4^*}{2i}, \quad C_4 = \frac{U_4 + U_4^*}{2}.$$

For $0 < r < 1$ and $v \in \mathbb{C}$, the traces

$$F_v(r) = \text{Tr}(N^{-\nu} r^N Q_4), \quad H_v(r) = \text{Tr}(N^{-\nu} r^N C_4)$$

converge absolutely and locally uniformly in (v, r) . One has $F_2(r) = \mathcal{C}_2(r)$ and

$$\mathcal{C}_2(\rho) = \frac{2}{3}G_{\text{Cat}} - \frac{\pi_{\text{HMT}}}{12} \log \lambda_P, \tag{75.11}$$

$$\mathcal{D}\mathcal{C}_2(\rho) = \frac{\pi_{\text{HMT}}}{12}, \quad \mathcal{D}^2\mathcal{C}_2(\rho) = \frac{1}{4}. \tag{75.12}$$

Proof. The diagonal basis gives the series with coefficients $\sin(n\pi/2)$ and $\cos(n\pi/2)$. On a compact set of ν and with $r \leq r_0 < 1$, their absolute values are bounded by $n^M r_0^n$ for some M , a summable series. Selection of the odd indices in the first trace produces exactly $\mathcal{C}_2(r)$.

The tangent subtraction formula gives $\rho = \tan(\pi_{\text{HMT}}/12)$, as a subsequent angular recognition of the Pell unit. For $0 < \theta < \pi/2$, integration by parts of the moment gives

$$\mathcal{C}_2(\tan \theta) = \theta \log(\tan \theta) - \int_0^\theta \log(\tan x) dx.$$

The boundary term at zero vanishes. Subtracting the Fourier series of $\log(2 \sin x)$ and $\log(2 \cos x)$, and integrating with Abel regularization, gives

$$-\int_0^\theta \log(\tan x) dx = \sum_{\substack{n \geq 1 \\ 2 \nmid n}} \frac{\sin(2n\theta)}{n^2}.$$

The regularization converges in the integral because the logarithmic singularity at zero is integrable; the integrated series is dominated by $\sum n^{-2}$.

For odd n , let $\chi_{-4}(n) = (-1)^{(n-1)/2}$. The identity, checked in the six odd residue classes modulo twelve, is

$$2 \sin(n\pi_{\text{HMT}}/6) = \chi_{-4}(n) + 3 \mathbf{1}_{3|n} \chi_{-4}(n/3).$$

Its sum with weight n^{-2} gives $\frac{1}{2}(1 + 3/9)G_{\text{Cat}} = 2G_{\text{Cat}}/3$. Taking $\theta = \pi_{\text{HMT}}/12$ and $\log \rho = -\log \lambda_P$ yields the first evaluation. The others follow from $\mathcal{D}\mathcal{C}_2(r) = \arctan r$, $\mathcal{D}^2\mathcal{C}_2(r) = r/(1 + r^2)$ and $\rho + \rho^{-1} = 4$. $\square$

The real component of the same trace retains

$$H_2(r) = \frac{1}{4} \sum_{k \geq 1} \frac{(-r^2)^k}{k^2}, \quad \mathcal{D}H_2(r) = -\frac{1}{2} \log(1 + r^2), \quad \mathcal{D}^2H_2(r) = -\frac{r^2}{1 + r^2}.$$

These formulas follow by selecting the even indices and differentiating the absolutely convergent series. Hence

$$\text{Tr}(N^{-2}r^N U_4) = H_2(r) + i\mathcal{C}_2(r).$$

The designation $\text{Li}_2(z) = \sum_{n \geq 1} z^n/n^2$ subsequently recognizes this trace as $\text{Li}_2(ir)$. In particular,

$$\text{Tr}(N^{-2}\rho^N U_4) = \frac{1}{4} \text{Li}_2(-\rho^2) + i \left(\frac{2}{3} G_{\text{Cat}} - \frac{\pi_{\text{HMT}}}{12} \log \lambda_P \right).$$

Replacing U_4 by U_4^* preserves the real part and reverses the imaginary part.

Proposition 75.6 (Composition of depth and orientation). *For $0 < r, s < 1$, $a, b \in \mathbb{Z}/4\mathbb{Z}$ and $m \geq 0$, let $B(r, a) = r^N U_4^a$. Then*

$$B(r, a)B(s, b) = B(rs, a + b), \quad B(r, 1)^m = r^{mN} U_4^m.$$

In particular, $B(r, 1)^4 = r^{4N} \neq I$ for $0 < r < 1$.

Proof. Both factors are diagonal in the same basis. On δ_n , their product multiplies by $r^n i^{an} s^n i^{bn} = (rs)^n i^{(a+b)n}$. The power formula follows by induction, and $U_4^4 = I$ gives the fourth return. The factor r^{4N} remains distinct from the identity. $\square$

The power reader of the inverse multiplier $-i\rho$ is $B(\rho, -1)$. Its imaginary moment has the opposite orientation to that of $B(\rho, 1)$, whereas the damped depth is the same. The transport V , observable $\mathcal{S}$ and reader U_4 retain their respective spaces: the preceding composition concerns the orbit multipliers and their diagonal reading. Phase returns after four steps, while accumulated contraction remains recorded.

76 Hyperbolic geometry of the angular pair and recovery of action

76.1 Radion, action ellipse, and marked charts

The oriented section of phase, action, and memory termed the *radion* jointly preserves the angular increment $\theta = 1$, the signed action $\sigma\hbar_{\text{ret}}$, membership in the revolution $2\pi_{\text{HMT}}$, the sheet, the nonadic return, and the refinement cylinder. Its unit gluing is expressed in the continuum chart by $1 = S_E - \Phi_T + \varepsilon_1$. The projection retaining only $\theta = 1$ produces the ordinary radian; the enriched section also preserves the register needed to transport and invert its other readings. In this section, *radius* denotes a positive shape coordinate, distinct from that enriched object.

To specify the terms of the gluing identity, the dodecaphase wheel has centres $30m - 10$ in its degree chart. The second phase contributes 50, and the representation $A = 1000\alpha$ determines

$$S_E = \frac{2\pi}{360}(50 + 1000\alpha).$$

The active sector of the tritic selector comprises $H_+ = \{1, 4, 7\}$ and $H_- = \{2, 5, 8\}$. The fifteen unordered pairs, read at the six active positions, produce $N_6 = 6\binom{6}{2} = 90$; their extension to eight positions produces $N_8 = 8\binom{6}{2} = 120$. The two sheets of the active sector, normalized by the ambient fibre of 729 elements, determine

$$\rho_{\text{tw}} = \frac{2N_6}{729} = \frac{20}{81}, \quad \Phi_T = \frac{20}{81}\Delta_4, \quad S_T = 1 + \Phi_T, \quad \varepsilon_1 = S_T - S_E,$$

where Δ_4 is the torsional coordinate in (77.1). The subtraction retains the preceding sections and directly proves $S_E - \Phi_T + \varepsilon_1 = S_E - \Phi_T + 1 + \Phi_T - S_E = 1$. The coefficient comes from the census and the two sheets, not from the value of the final difference.

The angular representations of the preceding sections satisfy, in the chart under consideration,

$$A > 0, \quad |C^*| < A, \quad \xi = \frac{C^*}{A}, \quad \eta_{\text{el}} = \text{artanh } \xi.$$

The notation η_{el} distinguishes elliptic anisotropy from the return coefficient η_{ret} . For a positive action section $\hbar$, the semiaxes are

$$a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}, \quad a_S b_S = 2\hbar. \quad (76.1)$$

The product fixes the action per revolution, whereas the ratio fixes the shape. Define

$$R_\star = \sqrt{b_S/a_S} = e^{-\eta_{\text{el}}/2}, \quad \tau_{\text{el}} = iR_\star^2.$$

Theorem 76.1 (Compatibility of the radial, Poincaré, and Klein charts). *Let*

$$w = \frac{\tau_{\text{el}} - i}{\tau_{\text{el}} + i}, \quad x_K = \frac{2w}{1 + w^2}.$$

On the real diameter of the Poincaré disk and its image in the Klein disk,

$$w = \frac{R_\star^2 - 1}{R_\star^2 + 1} = -\tanh \frac{\eta_{\text{el}}}{2}, \quad (76.2)$$

$$x_K = \frac{R_\star^4 - 1}{R_\star^4 + 1} = -\tanh \eta_{\text{el}} = -\frac{C^*}{A}. \quad (76.3)$$

All these transformations are invertible on their domains; in particular, $R_\star^2 = (1 + w)/(1 - w)$.

Proof. Substitute $\tau = iR_\star^2$ into the fractional transformation. The simplified ratio has denominator $R_\star^2 + 1 > 0$ and $-1 < w < 1$. Substituting this ratio into $2w/(1 + w^2)$ and multiplying numerator and denominator by $(R_\star^2 + 1)^2$ gives $(R_\star^4 - 1)/(R_\star^4 + 1)$. The identity $R_\star^4 = e^{-2\eta_{\text{el}}}$ yields the hyperbolic tangent. Finally, $\tanh \eta_{\text{el}} = \xi$. The recovery maps follow by solving the ratios. $\square$

The sign in (76.3) corresponds to the chosen orientation of the diameter. The opposite orientation changes both signs. This is a hyperbolic coordinate of the marked rectangular chart: it retains the marking and is not confused with a modular form or with a coordinate on the modular quotient stripped of its orientation.

Exchanging the semiaxes acts by

$$(R_\star, \eta_{\text{el}}, \tau_{\text{el}}, w, x_K) \mapsto (R_\star^{-1}, -\eta_{\text{el}}, -1/\tau_{\text{el}}, -w, -x_K). \quad (76.4)$$

On this diameter, the hyperbolic metrics of curvature -1 satisfy

$$\frac{|d\tau|^2}{(\Im\tau)^2} = \frac{4dw^2}{(1-w^2)^2} = \frac{dx_K^2}{(1-x_K^2)^2} = d\eta_{\text{el}}^2 = 4\frac{dR_\star^2}{R_\star^2}. \quad (76.5)$$

These identities are verified by differentiating (76.2)–(76.3). In particular, $d(0, w) = |\eta_{\text{el}}|$ and $d(w, -w) = 2|\eta_{\text{el}}|$. The transformation $w \mapsto -w$ is a rotation by π on the entire disk and reflection about the centre on the generated diameter.

The normalization $(Q/a_S, P/b_S)$ of the ellipse's interior produces another realization of the Klein disk, with position coordinates. Its metric is

$$ds_K^2 = \frac{(1-r^2)(du^2 + dv^2) + (u du + v dv)^2}{(1-r^2)^2}, \quad r^2 = u^2 + v^2 < 1.$$

This physical chart and the marked shape chart retain distinct domains. On the focal branch $\eta_{\text{el}} \geq 0$, the eccentricity $e_f = \sqrt{1 - e^{-2\eta_{\text{el}}}}$ gives depth $u_f = \text{artanh } e_f$, whence $\eta_{\text{el}} = \log \cosh u_f$. Focal depth is not identified with anisotropy.



Figure 17: Two typed reciprocities. The first remains inside the disk and transports shape orientation; the second exchanges the interior and exterior of the unit circle. The plotted positions are schematic.

76.2 Complete substitution of the angular representations

The dodecaphasic closure already constructed expresses the same coordinate α through

$$\alpha = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{ag}}, \quad \kappa_{\text{ag}} = \frac{K_{\text{int}}}{1000^{12} - 1}.$$

The compatible base, phase origin, and carry boundary are retained. The article's complete analytic recurrence represents that correlated coordinate; its representation is not replaced by a finite jet. Abbreviating the HMT subscripts, the action readers of (12.6)–(12.9) are

$$\begin{aligned} H_5 &= \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \\ D_A &= e^{-100\pi\alpha/9}, \quad \mathcal{C}_\pi = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \\ \eta_{\text{ret}} &= H_5 - \frac{90}{\pi}\mathcal{C}_\pi, \quad A = 1000\alpha, \quad C^* = 2(\eta_{\text{ret}} + \alpha). \end{aligned} \quad (76.6)$$

The rational continuation retains every term of the regional recurrence. Its convergence follows without truncation: for $\alpha > 0$,

$$0 < \alpha e^{-100\pi\alpha/9} \leq \frac{9}{100\pi e} < 1,$$

since xe^{-cx} attains its maximum $1/(ce)$ at $x = 1/c$.

Proposition 76.2 (Structural cancellation in the angular contrast). *The generated ratio $\xi = C^*/A$ satisfies exactly*

$$\begin{aligned} \xi = & \frac{1}{\sqrt{1000\varphi\alpha}} + \frac{9\alpha}{8000} - \frac{\alpha^2}{900} + \frac{7\alpha^3}{24000} - \frac{\alpha^4}{27000} \\ & - \frac{9}{50\pi} \left(169\alpha^5 + \frac{D_A\alpha^6}{1 - D_A\alpha} \right). \end{aligned} \quad (76.7)$$

Proof. Start from $\xi = (\eta_{\text{ret}} + \alpha)/(500\alpha)$. The term $-\alpha$ in H_5 cancels against the $+\alpha$ of the counter-angle. Dividing each remaining term by 500α yields the expression: the square root gives $1/\sqrt{1000\varphi\alpha}$, and $90/500 = 9/50$. The regional continuation retains its complete denominator. $\square$

Consequently, $x_K = -\xi$ and $R_\star = ((1 - \xi)/(1 + \xi))^{1/4}$ are explicit expressions in the regional representations and the register. Substituting the closure of α completes the dependence on $\pi, \varphi, e, \kappa_{\text{ag}}$, without promoting metrological identifications to generators.

76.3 Recovery of Planck's action with orientation preserved

Theorem 76.3 (Radial recovery map for the return section). *In the positive chart, if $q = R_\star^4$, then*

$$\eta_{\text{ret}} = \alpha \frac{499 - 501q}{1 + q}, \quad (76.8)$$

$$\hbar_{\text{ret}} = \alpha \frac{499 - 501q}{1 + q} 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad (76.9)$$

$$R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5(1 + q)}{\alpha(499 - 501q)}. \quad (76.10)$$

The branch $\eta_{\text{ret}} > 0$ is equivalent to $q < 499/501$, within $q > 0$.

Proof. The angular identities give $\eta_{\text{ret}} = \alpha(500\xi - 1)$. Substitute $\xi = (1 - q)/(1 + q)$; the numerator is $499 - 501q$. Both the denominator and α are positive. The dimensional realization and the ratio are those of the action sections already constructed. The decade -34 and the unit $\mathcal{U}_S$ retain their preceding determinantal derivation. $\square$

This formula makes the Planck section within the counter-angle explicit. In terms of the action quantities, the angular relation is

$$C^* = 2 \left(\frac{\hbar_{\text{ret}}}{10^{-34} \mathcal{U}_S} + \alpha \right).$$

Division by the dimensional basis extracts the dimensionless coefficient; it does not add an action quantity to a dimensionless constant.

For the angular orientation $\varepsilon \in \{-1, +1\}$, define

$$C_\varepsilon^* = 2\varepsilon(\eta_{\text{ret}} + \alpha), \quad \xi_\varepsilon = C_\varepsilon^*/A, \quad q_\varepsilon = \frac{1 - \xi_\varepsilon}{1 + \xi_\varepsilon} > 0.$$

The recovery map on both charts is

$$\eta_{\text{ret}} = \alpha \left(500\varepsilon \frac{1 - q_\varepsilon}{1 + q_\varepsilon} - 1 \right). \quad (76.11)$$

Corollary 76.4 (Invariance of the conjugate recovery map). *The transformation $(\varepsilon, q) \mapsto (-\varepsilon, q^{-1})$ preserves η_{ret} , $\hbar_{\text{ret}}$, and R_{act} .*

Proof. The ratio $(1 - q)/(1 + q)$ changes sign when q is inverted. Its product with ε remains invariant. Apply (76.11), followed by the action representations. $\square$

If orientation is discarded and $\varepsilon = +1$ is incorrectly held fixed, the same expression gives $\eta_{\text{ret}}(q^{-1}) = -\eta_{\text{ret}}(q) - 2\alpha$. The register therefore distinguishes two algebraically different operations. Nor is the angular orientation ε automatically identified with the symplectic sign σ : exchanging the axes is antisymplectic, whereas the rotation that exchanges them preserves the area form.

76.4 Intertwining phase and anisotropy

Let

$$S_\eta = \text{diag}(e^{\eta/2}, e^{-\eta/2}), \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J_\eta = S_\eta J_0 S_\eta^{-1} = \begin{pmatrix} 0 & -e^\eta \\ e^{-\eta} & 0 \end{pmatrix}.$$

Then $\det S_\eta = 1$, $J_\eta^2 = -I$, and $C_\eta(\theta) = e^{\theta J_\eta} = S_\eta e^{\theta J_0} S_\eta^{-1}$. The law $S_{\eta+\delta} = S_\delta S_\eta$ proves the exact intertwining identity

$$C_{\eta+\delta}(\theta) S_\delta = S_\delta C_\eta(\theta). \quad (76.12)$$

The transformation preserves $dQ \wedge dP$ and carries the phase evolution to the new ellipse. With the normalization of (76.1), the swept action is $\hbar\theta$, or $\sigma\hbar\theta$ on the oriented sheet. The phase θ , the anisotropy η_{el} , and the parameter t of the flow of K thus retain distinct functions, related by explicit operators.

76.5 Radial geodesic and differential realization of the direction of K

Each radius in (73.1) determines a rectangular chart and a hyperbolic coordinate:

$$\tau_i(t) = i r_i(t)^2, \quad w_i(t) = \frac{r_i(t)^2 - 1}{r_i(t)^2 + 1} = \tanh(th_i).$$

Theorem 76.5 (Geodesic of compensated radii). *In the product of twelve hyperbolic diameters,*

$$d(w(s), w(t)) = 2|t - s| |\log \rho|, \quad \sum_i \text{artanh } w_i(t) = 0. \quad (76.13)$$

Composition of advances is given by

$$w_i(t + s) = \frac{w_i(t) + w_i(s)}{1 + w_i(t)w_i(s)}.$$

Proof. In coordinates $a_i = \log r_i$, the product metric is $4 \sum_i da_i^2$. The curve $a(t) = th$ is a straight line, with speed $2\|h\| = 2|\log \rho|$, and $\sum_i a_i = 0$. Each diameter is totally geodesic. Distance in the product is the square root of the sum of squared distances in its factors, which are $2|t - s||h_i|$. The vanishing-sum identity and the composition law follow from $\text{artanh } w_i = th_i$ and the addition formula for the hyperbolic tangent. $\square$

The regional ratio determines the speed of this realization; K determines its direction. If $\rho = 1$, the trajectory is constant. For $\rho \neq 1$, the space of radii with product one has dimension eleven, with tangent space $V_{11} = \mathbf{1}^\perp$, and the normal to the trajectory within it is

$$V_{10}(K) = V_{11} \cap u^\perp, \quad P_{10}(K) = P_{11} - \frac{uu^T}{\|u\|^2}.$$

The projector of Proposition 16.12 therefore acquires a differential realization: it removes the uniform scale and distinguishes the generated direction from its ten transverse variations. Its domain is this radial space, not the real twenty-four-dimensional space of all the disks.

With $D_t = \text{diag}(\text{sech}^2(th_i))$, its expression in coordinates w is $D_t P_{10} D_t^{-1}$. Indeed, D_t is the differential of $a \mapsto \tanh a$, and transforms the metric $4I$ into $4 \text{diag}((1 - w_i^2)^{-2})$. The transported projection is orthogonal for that metric.

76.6 Lattice lift and rectangular lift

The radii admit two realizations that should be retained explicitly. The operator $G_t = \bigoplus_i r_i I_2$ scales both axes of each A_2 plane simultaneously. It preserves the total covolume and commutes with the ternary isometry. The homothety of each plane preserves its internal complex modulus.

The chart $\tau_i = ir_i^2$ has another realization, of fixed area:

$$T_i = \text{diag}(r_i^{-1}, r_i), \quad (\omega_1, \omega_2) = (r_i^{-1}, ir_i), \quad \omega_2/\omega_1 = ir_i^2.$$

This is the elliptic lift with $\eta = -2 \log r_i$. This convention transforms physical periods. The fractional linear action of the same matrix on i would give i/r_i^2 ; to obtain ir_i^2 under that other convention, one uses T_i^{-1} . Distinguishing the actions prevents an inadvertent inversion.

Both realizations are reconstructed from the same radii, but the ternary commutation of the homothety does not transfer to the anisotropic deformation. The state and its readers allow the two geometries to be compared without identifying their operators.

77 Electron coordinate, operator scale, and reversibility of readings

77.1 Algebraic structure in the complete electronic expression

The expression in Section 15 is now composed with the radial recovery map, retaining its factors and dimensional realization. The electronic prefactor comes from the conditional projectors of addition and multiplication on $\ell^2(\{1, \dots, 9\}^3)$. On the corresponding principal plane,

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad \|[P, Q]\| = \frac{\sqrt{3}}{4}.$$

The equality follows because $[P, Q] = (\sqrt{3}/4) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Moreover,

$$S = 2P - I, \quad J = \frac{4}{\sqrt{3}}[P, Q], \quad S^2 = I, \quad J^2 = -I, \quad SJ = -JS.$$

Hence $n_{\pm} = (S \pm J)/2$ satisfy $n_{\pm}^2 = 0$. The block retains hyperbolic, elliptic, and nilpotent generators within the algebraic realization already derived. The coefficient $\sqrt{3}/4$ and the normalization $4/\sqrt{3}$ serve distinct functions.

Using the previously constructed coordinates and regional integers, define

$$\begin{aligned} \Delta_4 &= \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729}\right), \\ B_e &= \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4), \quad \kappa_e = 80 - 54\alpha + 6\Delta_4. \end{aligned} \quad (77.1)$$

Thus $\varepsilon_e^\beta = B_e R_{\text{act}}^{\kappa_e}$. The pre-return action section and its return determine

$$\delta_{\text{ret}} = H_5 - \eta_{\text{ret}} = \frac{90}{\pi} \mathcal{C}_\pi, \quad R_{\text{act}} = 1 + \frac{\delta_{\text{ret}}}{\eta_{\text{ret}}}.$$

Corollary 77.1 (Radial realization of the electronic coordinate). *On the branch $\alpha > 0$, $\eta_{\text{ret}} > 0$, $|\xi| < 1$,*

$$\begin{aligned} \varepsilon_e^\beta &= \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4) \\ &\quad \times \left[1 + \frac{90\mathcal{C}_\pi}{\pi\alpha} \frac{1 + R_\star^4}{499 - 501R_\star^4} \right]^{80 - 54\alpha + 6\Delta_4}. \end{aligned} \quad (77.2)$$

In the Klein coordinate of the marked chart,

$$R_{\text{act}} = 1 + \frac{\delta_{\text{ret}}}{\alpha(-500x_K - 1)}. \quad (77.3)$$

The corresponding domains are

$$\frac{1}{500} < \xi < 1, \quad 0 < R_\star^4 < \frac{499}{501}, \quad -1 < x_K < -\frac{1}{500}.$$

Proof. Substitute (76.8) into $R_{\text{act}} = 1 + \delta_{\text{ret}}/\eta_{\text{ret}}$, and then into the complete electronic expression (15.1). The Klein chart gives $\xi = -x_K$. The intervals follow from $\eta_{\text{ret}} = \alpha(500\xi - 1) > 0$ and the monotone correspondences already proved. $\square$

The substitution preserves the regional deficit, torsional normalization, return character, and incompatibility norm. It expresses the same coordinate through a recoverable geometric reader. The radius and α remain correlated by their generation; they are not independent parameters fitted to a mass. The dimensional realization remains

$$E_e^\beta = \varepsilon_e^\beta \mathcal{U}_E, \quad m_e^\beta = E_e^\beta / c_{\text{int}}^2.$$

The action decade already incorporated in the Planck sections is not multiplied again into a coordinate expressed in the energy chart.

A second exact form, useful for isolating the correction, is

$$\hat{\delta}_{\text{ret}} = \frac{90\mathcal{C}\pi}{\pi H_5}, \quad R_{\text{act}} = (1 - \hat{\delta}_{\text{ret}})^{-1}, \quad \varepsilon_e^\beta = B_e(1 - \hat{\delta}_{\text{ret}})^{-\kappa_e}.$$

Positivity of both sections ensures $1 - \hat{\delta}_{\text{ret}} > 0$. The three expressions are identities of the same return, not independent predictions.

77.2 Separation of scale and shape in a positive mass fibre

The operator law of return receives a finite positive fibre of routes together with its readers. To formulate its matrix consequence independently of species names, let $\mathcal{H}$ have dimension $d \geq 1$, let $B = B^*$ be the exponent operator, and let M^0 be positive definite. The return is the congruence

$$M^\beta = R_{\text{act}}^{B/2} M^0 R_{\text{act}}^{B/2}. \quad (77.4)$$

In the route basis of the diagonal reader, $Be_\gamma = \kappa_\gamma e_\gamma$ and $M^0 e_\gamma = m_e^{(0)} \chi_{\text{ref}}(\sigma_\gamma - \sigma_e, \eta_\gamma - \eta_e) e_\gamma$. The signatures and character are those of the route construction; the following theorem applies to each positive fibre without replacing its selector.

Theorem 77.2 (Uniform and unimodular decomposition of return). *If $R_{\text{act}} > 0$, define*

$$\bar{\kappa} = \frac{\text{tr } B}{d}, \quad B_0 = B - \bar{\kappa}I, \quad S_0 = R_{\text{act}}^{B_0/2}.$$

Then

$$M^\beta = R_{\text{act}}^{\bar{\kappa}} S_0 M^0 S_0, \quad \det S_0 = 1, \quad (77.5)$$

$$\det M^\beta = R_{\text{act}}^{\text{tr } B} \det M^0. \quad (77.6)$$

With $\hat{M} = M/(\det M)^{1/d}$,

$$\hat{M}^\beta = S_0 \hat{M}^0 S_0. \quad (77.7)$$

Proof. The scalar part of B commutes with B_0 . Functional calculus gives $R_{\text{act}}^{B/2} = R_{\text{act}}^{\bar{\kappa}/2} S_0$. In an orthonormal eigenbasis of B ,

$$\det S_0 = R_{\text{act}}^{\frac{1}{2} \sum_\gamma (\kappa_\gamma - \bar{\kappa})} = 1.$$

Multiplication of determinants proves (77.6). Dividing (77.5) by the d -th root of that determinant cancels the scalar factor. The proof does not require B to commute with M^0 ; it does require positive definiteness for normalization by the determinant. $\square$

In the joint diagonal basis of the source construction, the individual ratios are

$$q_\gamma = \frac{m_\gamma^\beta}{m_\gamma^{(0)}} R_{\text{act}}^{-\bar{\kappa}} = R_{\text{act}}^{\kappa_\gamma - \bar{\kappa}}, \quad \prod_\gamma q_\gamma = 1.$$

The formal inversion $R_{\text{act}} \mapsto R_{\text{act}}^{-1}$ inverts each q_γ . It differs from oriented radial reciprocity, which preserves R_{act} by (76.11). The congruence preserves the normalized determinant, not the individual eigenvalues or the sum of masses. The null sector lies outside this positive normalization. For $d = 1$, $B_0 = 0$: the electron's return lies entirely in the uniform factor and remains in (77.2).

Substitution of (77.3) also acts on (77.4) through functional calculus. If

$$q(\xi) = 1 + \frac{\delta_{\text{ret}}}{\alpha(500\xi - 1)} > 0,$$

then

$$M^\beta = e^{\frac{1}{2}(\log q(\xi))B} M^0 e^{\frac{1}{2}(\log q(\xi))B}.$$

The genealogy of each exponent and each multisection is preserved. The structure shared with the flow of K consists of removing a uniform component and then applying a deformation of determinant one; B_0 is not identified with $\text{diag}(u)$, nor R_{act} with ρ .

77.3 Circular compression and identification of transport

Theorem 77.3 (Bilateral recovery of a unitary operator). *Let C be unitary on a Hilbert space, and let*

$$T = \frac{8I + C}{9}, \quad D = \frac{\sqrt{8}(C - I)}{9}.$$

Then

$$T^*T + D^*D = I, \quad v = Tv - \frac{Dv}{\sqrt{8}}, \quad Cv = Tv + \sqrt{8}Dv. \quad (77.8)$$

Proof. Expansion gives

$$81T^*T = 65I + 8(C + C^*), \quad 81D^*D = 16I - 8(C + C^*).$$

The sum is $81I$. The other identities follow by taking linear combinations of the two definitions. At a circular eigenvalue $|z| = 1$, the balance is $1 - |(8 + z)/9|^2 = (8/81)|1 - z|^2$. $\square$

The archived pair recovers both the state and its image. For the invertible frame $O_K = (K, SK, \dots, S^{11}K)$ already constructed, the twelve responses form

$$Y = (CK, CSK, \dots, CS^{11}K), \quad C = YO_K^{-1}.$$

If $CS = SC$, a single vector response $y = CK$ determines all the columns: $Y = (y, Sy, \dots, S^{11}y)$. The general case requires twelve vector trials. Recovery does not reduce that information to a single scalar.

The real diagonal generator $H_K = \text{diag}(h_i)$ admits the unitary chart

$$C_K = (H_K + iI/2)(H_K - iI/2)^{-1}.$$

Each real eigenvalue is transformed into a number of modulus one. Moreover,

$$C_K - I = i(H_K - iI/2)^{-1}, \quad H_K = \frac{i}{2}(C_K + I)(C_K - I)^{-1}.$$

Both inverses exist because H_K is self-adjoint and its spectrum is real. The archive of (77.8) therefore recovers this finite realization of the direction of K . Its dimension twelve and its domain remain distinct from those of a spectral operator for ζ .

77.4 Hierarchy of invariants and scope of the proofs

The completion $1000 = 729 + 271$, with normalized reading $1 = 729/1000 + 271/1000$, retains its generative function. The subsequent conservation laws are organized according to the object transported:

Realization	Invariant	Recovery or reciprocity
Bilateral archive	Joint norm of reading and memory	State and image under the operator
Action ellipse	Product $a_S b_S = 2\hbar$	Radius and angular orientation
Flow of K	Product of twelve radii	Inversion $t \mapsto -t$
Deformed marked lattice	Covolume and ternary isometry	$\Lambda_t^* = \Lambda_{-t}$
Theta–Mellin reading	Poles and residues	Entire difference and functional equation
Positive mass fibre	Determinant after removing the common scale	Congruence by S_0

Conservation of volume does not imply conservation of each vector's norm. Polar conservation permits variation in the entire part. Recovery of action requires orientation to be retained. These distinctions express the operational content of evolutionary conservation: each transformation retains certain data, changes others, and preserves the register that allows the correspondence to be reconstructed.

The composition extends the function of K already demonstrated in the article. The register participates in the reading of α ; its direction is recovered from memory; that direction determines a radial geodesic and a reciprocal lattice deformation; the lattice collection has a stable spectral reading. The action section provides the connection to the electronic coordinate and to the positive congruence of the mass operator. This yields a chain of applications with explicit domains, inverses, and invariants.

The supplement distinguishes written proofs from formalized ones. `FlujoReciproco.lean` contains ten theorems on diagonal flow, centring, determinants, inversion, and the lift to twenty-four dimensions. `RadioKlein.lean` contains eleven rational theorems on the charts, bounds, reciprocities, and oriented recovery. The hyperbolic proofs, complete substitutions, separation of the mass operator, and theta-Mellin argument are given in the text; their formalization is not attributed to compilation of those two files. The documentary receipts record provenance and reproduction, with a scope distinct from that of a mathematical proof.

$\mathcal{C}$

Compatible recovery and collections of the continuum

Sufficient memory, naturality, measure and membership transport in the generated continuum.

From recovering a state to transporting a collection

Compatible recovery makes it possible to extend the exposition from one state to collections of states. An enriched reading uniquely determines its antecedents when its memory distinguishes the elements that share the same visible value. The inverse then restores the state on the image of that reading. Compatibility with refinement assembles the finite inverses into a reconstruction of complete histories.

Transporting collections requires preserving membership. A bijection of histories takes each collection to its image and each element to its corresponding element. Unions, intersections, and relative complements are transported through these operations. The restriction of the bijection to a collection preserves its cardinality. Memory theory thus acquires a precise link with operations on sets of histories.

Measure also requires the transport of weights. A symmetry compatible with the refinement tree preserves cylinder measure when it respects emissions and their multiplicities. Naturality expresses the agreement between transporting and truncating. The proofs that follow make these relations explicit on the objects defined in the article.

The specific study of the continuum hypothesis develops genealogical closure and its hierarchy of sets. Its cardinal result is formulated on that closure and its internal power set. The relation incorporated by this part is the recovery of states and the transport of collections that maintain the structure of the generated continuum. The cardinal statement preserves its specialized development and its expressly identified domain.

This articulation places the continuum within the conservation argument. The complete history, the collection to which it belongs, and the measure of its cylinder require related operations with different contents. The article brings together their compatibility conditions to show how the double projection is maintained in the passage from an individual reading to the organization of collections. The unity of the exposition resides in these maps and in the proofs that preserve their relations.

78 Carry and incidence registers and projective reconstruction of histories

The APP–TRIT–TPK construction fixes an enriched state before its numerical, incidence and operator realizations. The registers entering that construction are distinguished by the operation they allow to be inverted and by the relations they preserve. This section brings these distinctions together and composes recovery with refinement.

78.1 Carry, history, incidence and archive

The carry in Sections 69.4 and 69.5 is an integer arithmetic datum. Preserving it, together with depth and the final state, restores the antecedent through the explicit inverse (69.12). The transport history contains the ordered sequence of operations, sheets, orientations and returns. Within it, two stages with the same phase may have different accumulated registers. Holonomy acts on that history and on the frame in which its readings are performed.

The incidence register preserves the boundary relations used by the exceptional constructions. The ordered object K and its reading κ_{ag} enter with the normalizations and maps of Section 67.4. Sufficiency for recovering an operator is expressed there through the frame responses and the corresponding inversion. The archive memory of Section 70.6, in turn, is a sequence of components of a Hilbert space. The telescoping identity (70.26) preserves the norm between the remaining state and the archived components; the inverse (70.25) reconstructs the initial amplitude.

These four functions are composed while preserving their types. A carry integer, a transport word, an incidence and an archived amplitude constitute distinct data of the system. The recovery proofs specify which combination of them suffices for each reading. Figure 18 represents this functional hierarchy.

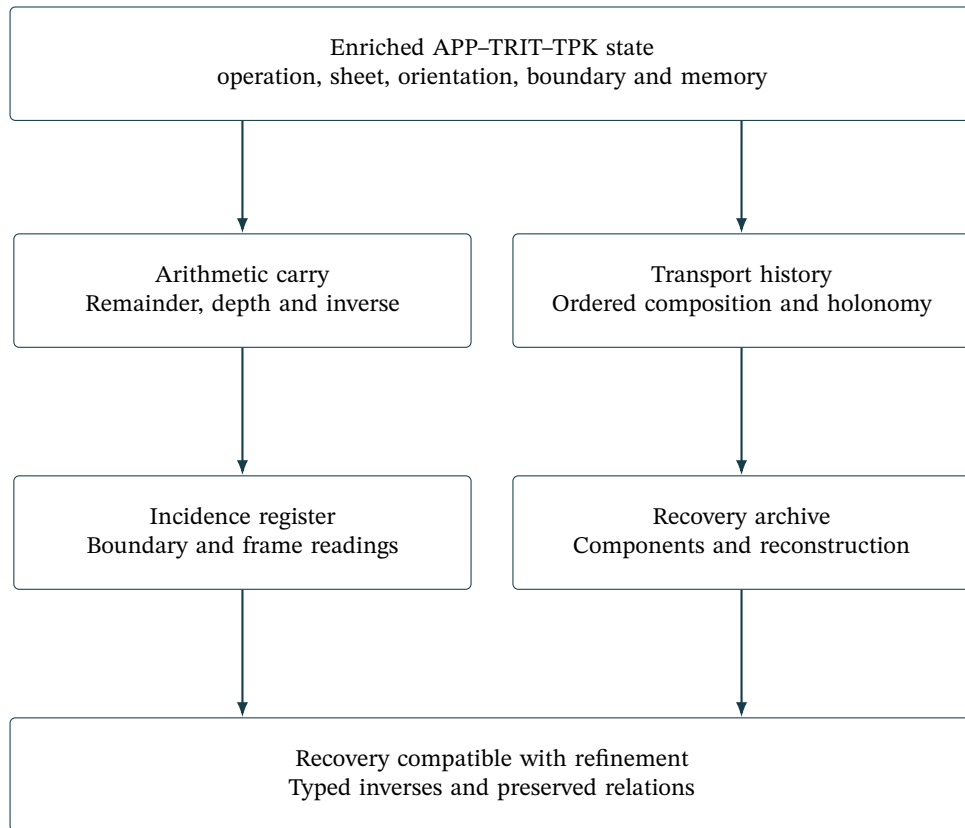


Figure 18: Organization of the registers by their role. The lower arrows indicate composition of recovery conditions, not identification of the four data types.

78.2 Refinement tree and local weights

Let H_k be the finite set of admissible prefixes of depth k , with truncations $p_{k+1,k}$, of the history system already constructed. An emission is counted with its multiplicity, and $d(\zeta) > 0$ denotes the number of emissions from ζ . The uniform seed distribution and the local weight $1/d(\zeta)$ define μ_k . The sum of the weights of the extensions of a prefix equals its weight: the measures are compatible and determine the cylinder measure μ . The propositions on fiber memory and naturality of the measure are proved below using these explicit definitions.

78.3 Enriched readers and fiber memory

The difference between history and value admits an operational criterion. At a finite level H_k , let $q_k : H_k \rightarrow Z_k$ be a reader and let $M_k : H_k \rightarrow \mathcal{M}_k$ be the retained register. The enriched reader is $\hat{q}_k = (q_k, M_k)$. Memory distinguishes states within the same reader fiber; its sufficiency is checked on the domain under consideration. This formulation incorporates into the article itself the complete development of the fiber information used below.

Proposition 78.1 (Recoverability at a finite level). *An inverse of $\hat{q}_k$ on its image exists if and only if $\hat{q}_k$ is injective. For a random variable Z with values in H_k and distribution μ_k , we have*

$$H(Z) = H(q_k(Z)) + H(Z | q_k(Z)).$$

If $\hat{q}_k$ is injective on the support of Z , then

$$H(Z | q_k(Z), M_k(Z)) = 0.$$

Here $H(Z) = -\sum_z \Pr(Z = z) \log \Pr(Z = z)$, with $0 \log 0 = 0$, is finite entropy in nats; conditional entropy is the average entropy of the conditional distributions.

Proof. An inverse recovers a unique antecedent, which requires injectivity; conversely, injectivity allows each expressed pair to be assigned its unique antecedent. Since $q_k(Z)$ is a deterministic function of Z , grouping the sum defining $H(Z)$ by fibers of q_k gives the chain rule. If the pair $(q_k(Z), M_k(Z))$ distinguishes the support, each corresponding conditional distribution is concentrated on a single state and its entropy is zero. $\square$

Not publishing a coordinate and erasing the register that allows it to be recovered are distinct operations. This result neither identifies entropy with the cardinality of the continuum nor introduces a thermal interpretation: its domain, reader and distribution are those just fixed.

78.4 Equivariance of the measure under tree symmetries

Proposition 78.2 (Joint transport of the tree and its measure). *Let $g = (g_k)_{k \geq 0}$ be a collection of bijections $g_k : H_k \rightarrow H_k$ commuting with truncations, fixing the root and transporting admissible emissions bijectively while preserving their multiplicities. Then*

$$d(g_k \zeta) = d(\zeta), \quad (g_k)_* \mu_k = \mu_k, \quad (g_\infty)_* \mu = \mu.$$

More generally, if μ_ζ is the law of continuations of a history $\zeta \in H_k$, we have

$$(g_\infty)_* \mu_\zeta = \mu_{g_k \zeta}.$$

Proof. The bijection of emissions preserves their number with multiplicities. The uniform seed distribution is also invariant. Each trajectory and its image therefore receive the same product of local weights $1/d(\zeta)$. Equality of measures follows at each level and on each cylinder. Coherence with truncations defines g_∞ , and uniqueness of the measure determined by cylinders proves equality in the limit. The same argument, starting at ζ , proves the assertion about continuations. $\square$

This gives the naturality of the tree and measure: a coherent change of representation transports the weights without requiring them to be chosen anew. Preservation of emissions and multiplicities is an essential hypothesis of the proposition, not a datum added after the construction.

78.5 Recovery of histories and transport of membership

Proposition 78.3 (Compatibility of inverses and collections). *Let H_k be the preceding levels and $Y_k = \hat{q}_k(H_k)$. Suppose that $\hat{q}_k : H_k \rightarrow Y_k$ is bijective and that there exist truncations $s_{k+1,k}$ such that*

$$s_{k+1,k} \hat{q}_{k+1} = \hat{q}_k p_{k+1,k}.$$

Then the finite inverses induce a bijection

$$\hat{q}_\infty : \varprojlim H_k \longrightarrow \varprojlim Y_k.$$

If $H = \varprojlim H_k$, $Y = \varprojlim Y_k$ and $\mathcal{T}(A) = \hat{q}_\infty[A]$, the map $\mathcal{T} : \mathcal{P}(H) \rightarrow \mathcal{P}(Y)$ is bijective and preserves unions, intersections, relative complements and membership. Each restriction $A \rightarrow \mathcal{T}(A)$ is a bijection.

Proof. Composing the compatibility identity with the inverses gives

$$p_{k+1,k} \hat{q}_{k+1}^{-1} = \hat{q}_k^{-1} s_{k+1,k}.$$

Therefore, a compatible collection $y = (y_k)$ determines the compatible collection $x = (\hat{q}_k^{-1}(y_k))$. The two coordinate maps are inverses. Direct image under a bijection has as its inverse direct image under the inverse bijection. For every collection $(A_i)_{i \in I}$ of subsets of H ,

$$\begin{aligned} \mathcal{T}\left(\bigcup_i A_i\right) &= \bigcup_i \mathcal{T}(A_i), \\ \mathcal{T}\left(\bigcap_i A_i\right) &= \bigcap_i \mathcal{T}(A_i), \\ \mathcal{T}(H \setminus A) &= Y \setminus \mathcal{T}(A). \end{aligned}$$

The first equality uses the definition of image. For the second and third, uniqueness of the antecedent allows the membership conditions to be transferred. Finally, $x \in A$ is equivalent to $\hat{q}_\infty(x) \in \mathcal{T}(A)$, which also proves the assertion about restrictions. $\square$

The route-refinement theorem 71.3 provides a realization of this compatibility on the arithmetic labels and their carries. The result distinguishes recovery of each state from preservation of operations on collections. At amplitude level, the unitary lift in 71.2 also transports the inner product. At probabilistic level, Proposition 78.2 preserves the weights when emissions are transported with their multiplicities.

78.6 Connection with the genealogical closure of the continuum

Section 79 constructs a genealogical closure $\mathfrak{G}$ and establishes the cardinal identity $2^{\aleph_0} = \aleph_1$ over its internal power set. The closure, generator code and realized history form part of its definition; its set-theoretic representation is established after the genealogical construction through the operations and proofs included in that section.

The present connection incorporates the proofs concerning memory, naturality and transport of collections used by that relation with conservation. The proof chain remains differentiated: the inverses recover states; their compatibility recovers histories; the bijection transports membership operations; the genealogical closure in Section 79 fixes the domain of the cardinal statement. Recovery and the organization of collections are thus integrated into the double projection with their respective objects, operations and conclusions.

79 Genealogical closure and cardinality of the generated continuum

This section brings together, in a contiguous proof, extension, double projection, and arbitrary depth; it then constructs the semantic closure, the uniform lift, the genealogical machinery, and the cardinal closure. The set-theoretic hierarchy is identified only at the end of the construction and cannot be read as its causal origin. The causal dependence is:

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_{\infty}^{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}} \longrightarrow m_{\infty} \longrightarrow \mathfrak{G}_{\text{HMT}}(h, m_{\infty}). \quad (\text{C.32})$$

First, the state, its extensions, the continuum, and the genealogical ledger are constructed. Then **that produced genealogy** is extended to all its hereditary objects by an explicit semantic rule. The subsequent identification with a known set-theoretic hierarchy selects neither the seeds, the operators, the genealogical ledger, the line, nor its projections. It does, however, permit the exact logical-cardinal strength of the adopted semantic rule to be assessed with complete transparency.

79.1 Extension of states and transfinite semantic extension

Let

$$\Sigma_{\text{HMT}} = (\dot{R}_{\text{APP}}, \dot{R}_{\text{TRIT}}, \dot{R}_{\text{TPK}}, \dot{R}_U, \dot{R}_{\Gamma_9}, \dot{R}_{\text{Ext}}, \dot{R}_{\rho}, \dot{R}_{\pi}, \dot{R}_{\varphi}, \dot{R}_e, \dot{R}_{\alpha}, \dot{R}_{\text{ar}}, \dot{R}_{\text{exc}}) \quad (\text{C.33})$$

be the countable **relational** signature whose effective code is h . Each $\dot{R}_S$ is the graph of the operator S on the codes where it is defined; no external function is assumed to be total within each level. The graphs $\dot{R}_{\pi}, \dot{R}_{\varphi}, \dot{R}_e, \dot{R}_{\alpha}$ are the output readers constructed by coinductive generation and dodecaphase closure; they do not take the conventional values of the constants as arguments. Their presence in Σ_{HMT} ensures that the closure preserves the full state: it simultaneously integrates the routes that express value and the incidence that extends toward the exceptional realization.

At each horizon, the state code records the APP–TRIT–TPK coordinates, the route, sheet, carry, boundary, and memory; at its representation stages it also records R_N, Φ_N, E_N, A_N , where A_N is the prefix of α_{HMT} produced by the same state. These coordinates do not select the branch from outside: they are part of the history being encoded. Once a primitive-recursive numbering of the finite coordinates of a state is fixed, every inverse history $x_{\infty} = (x_N)_{N \geq 1} \in X_{\infty}^{\text{enr}}$ determines the code

$$\text{code}(x_{\infty}) := \{(N, j, c) : (x_N)_j \text{ has code } c\} \subseteq \omega.$$

The coherence equations $\rho_{N+1, N}(x_{N+1}) = x_N$ make it possible to decode the history uniquely from that real. We write m_{∞} for the code of the realized history and $m_{\infty}(n)$ for its characteristic function under the fixed numbering. The operator code h and this complete ledger are combined, by means of a primitive-recursive pairing function, into the real

$$u_{h, m} := \{\langle 0, \langle n, h(n) \rangle \rangle : n < \omega\} \cup \{\langle 1, \langle n, m_{\infty}(n) \rangle \rangle : n < \omega\}. \quad (\text{C.33a})$$

The two primitive projections recover h and m_{∞} from $u_{h, m}$.

The first extension is the metric and operator extension of the finite states. For $N \geq 1$, let $\hat{\mathcal{X}}_{6N}^{\text{enr, raw}}$ be the set of raw enriched states of length $6N$, and define the surviving subsystem set-theoretically by

$$\begin{aligned} X_N &:= \hat{\mathcal{X}}_{6N}^{\text{enr, surv}} \\ &:= \left\{ x \in \hat{\mathcal{X}}_{6N}^{\text{enr, raw}} : \right. \\ &\quad \forall M > N \exists z \in \hat{\mathcal{X}}_{6M}^{\text{enr, raw}} \text{ such that} \\ &\quad \left. \text{tr}_{6M, 6N}(z) = x \wedge z \text{ satisfies every intermediate TPK transition} \right\}. \end{aligned}$$

and set

$$\rho_{N+1, N} := \text{tr}_{6N+6, 6N} \upharpoonright X_{N+1},$$

and

$$\text{Ext}_N(x) := \{y \in X_{N+1} : (x, y) \in \text{Ext}_{6N+6, 6N}\}.$$

The root level is included without any implicit exception:

$$X_0 := \{*\}, \quad \rho_{1,0} : X_1 \rightarrow X_0 \text{ is constant}, \quad \text{Ext}_0(*) := X_1.$$

The superscript *surv* denotes the pruned tree of states possessing compatible TPK descendants at every finite horizon. The TPK extension law established in the preceding sections proves that the emitted seeds belong to this tree; finite branching and Kőnig's lemma turn compatibility at every horizon into an infinite branch. In particular,

$$\forall N \geq 0 \quad \forall x \in X_N \quad \emptyset \neq \text{Ext}_N(x) \subseteq \rho_{N+1,N}^{-1}(x). \quad (\text{C.34})$$

By the same finitely branching tree lemma, every surviving seed has a compatible extension

$$x_\infty = (x_N)_{N \geq 1} \in X_\infty^{\text{enr}} := \varprojlim_N (X_N, \rho_{N+1,N}). \quad (\text{C.35})$$

The second extension acts **afterward** on the complete code of that realized history. It is not inferred from the mere existence of the finite operator Ext_N : at set-theoretic scale it formalizes the HMT principle that the only objects in the integral domain are those having a genealogical derivation from the state and its rules. The transfinite semantic extension is:

$$\begin{aligned} \mathfrak{G}_0(h, m_\infty) &:= \text{TC}(\{\omega, u_{h,m}, h, m_\infty\}), \\ \mathfrak{G}_{\beta+1}(h, m_\infty) &:= \text{Def}_{\Sigma_{\text{HMT}}}(\mathfrak{G}_\beta(h, m_\infty)), \\ \mathfrak{G}_\lambda(h, m_\infty) &:= \bigcup_{\beta < \lambda} \mathfrak{G}_\beta(h, m_\infty) \quad (\lambda \text{ limit}), \\ \mathfrak{G}_{\text{HMT}}(h, m_\infty) &:= \bigcup_{\beta \in \text{Ord}} \mathfrak{G}_\beta(h, m_\infty). \end{aligned} \quad (\text{C.36})$$

The Continuum Hypothesis and a target bijection are not introduced among the conditions defining the successor operator; the operator is given prospectively by

$$\text{Def}_{\Sigma_{\text{HMT}}}(M) := M \cup \left\{ \{x \in M : \langle M, \in, (\dot{R}_S \cap M^{k_S})_{S \in \Sigma_{\text{HMT}}} \rangle \models \varphi(x, \bar{p})\} : \ulcorner \varphi \urcorner \in \omega, \bar{p} \in M^{<\omega} \right\}. \quad (\text{C.37})$$

That is, it simultaneously expresses all subsets definable by a finite formula of the HMT relational language and parameters already produced by the genealogy. It consults neither a target real value, a conventional constant, a target cardinal, nor an external model. Formula (C.37) defines the transfinite closure over the previously constructed seeds, operators, enriched state, ledger, and double projection.

Lemma 79.1 (Joint coding of the HMT program and the realized ledger). *The pair formed by the operator code h and the ledger m_∞ is encoded by a single real $u_{h,m} \subseteq \omega$, and every formula of Σ_{HMT} is uniformly translated into a formula of the language $\{\in, u_{h,m}\}$.*

Proof. Both h and m_∞ are sequences of natural numbers: h encodes the signature, the finite tables, and the operator programs; m_∞ encodes one realized history, not the complete history space. Formula (C.33a) and its two decoders prove the first assertion. Each symbol in (C.33) denotes its **graph relation encoded in h** , never the external graph of all its possible evaluations. APP, TRIT, U_t , Γ_9 , the restrictions ρ , and the finite extensions have primitive-recursive graphs on integer codes and, therefore, Δ_0 formulas absolute between transitive levels. The two representations are expressed as graph relations through the formulas

$$\begin{aligned} \dot{R}_{\text{ar}}(x, y) &\iff \forall k \in \omega \exists N \forall n \geq N \quad |\mu_n(x) - y| < 2^{-k}, \\ \dot{R}_{\text{exc}}(x, c, z) &\iff \forall N \in \omega \quad z \upharpoonright N = E_N(x \upharpoonright N, c \upharpoonright N), \end{aligned} \quad (\text{C.37c}_0)$$

where μ_n and E_N are the rational/finite functions whose codes occur in h . Every atomic formula containing an HMT symbol is inductively replaced by its graph formula with parameter $u_{h,m}$. Connectives and quantifiers are preserved. This induction on syntactic complexity produces a translation $\varphi \mapsto \varphi^*$ such that, for every transitive level M of (C.36) containing the parameters,

$$\langle M, \in, (\dot{R}_S \cap M^{k_S})_S \rangle \models \varphi(\bar{a}) \iff M \models \varphi^*(\bar{a}; u_{h,m}). \quad (\text{C.37c})$$

Thus, (C.37) cannot consult an oracle, a target decimal table, or an uncoded relation that introduces uncountably many objects in a single step. $\square$

HMT semantics is now defined **independently of identifying it with $\mathfrak{G}$** . Let $\mathcal{T}_0^{\text{HMT}}$ be the set of canonical names of the elements of $\mathfrak{G}_0(h, m_\infty) = \text{TC}(\{\omega, u_{h,m}, h, m_\infty\})$. In particular, the initial level contains the realized ledger m_∞ , not every history in X_∞^{enf} as primitive parameters. The admissible terms are the well-founded trees obtained by

$$\begin{aligned} \mathcal{T}_{\beta+1}^{\text{HMT}} &:= \mathcal{T}_\beta^{\text{HMT}} \cup \{\langle \beta, \ulcorner \varphi \urcorner, t_0, \dots, t_{k-1} \rangle : t_i \in \mathcal{T}_\beta^{\text{HMT}}\}, \\ \mathcal{T}_\lambda^{\text{HMT}} &:= \bigcup_{\beta < \lambda} \mathcal{T}_\beta^{\text{HMT}}, \quad \mathcal{T}^{\text{HMT}} := \bigcup_{\beta \in \text{Ord}} \mathcal{T}_\beta^{\text{HMT}}. \end{aligned} \quad (\text{C.37a})$$

The value of a successor term is the subset of the preceding level defined by φ in the induced relational structure, with the values of $t_0, \dots, t_{k-1}$ as parameters. Before any set-theoretic comparison, define

$$\text{Ob}_{\text{sem}}^{\text{HMT}}(h, m_\infty) := \{\llbracket t \rrbracket : t \in \mathcal{T}^{\text{HMT}}\}. \quad (\text{C.37b})$$

This is the rule of **generative exhaustivity** adopted by HMT: every object in the integral domain must possess one of these derivation trees, and every admissible tree expresses an object. It is not a consequence of finite branching. The Continuum Hypothesis is not introduced into this rule; it is obtained through the proof developed in the article. Neither an enumeration of the reals nor a target bijection is introduced as data.

Theorem 79.2 (Equivalence between term semantics and the genealogical hierarchy). *The evaluation of the preceding terms satisfies*

$$\text{Ob}_{\text{sem}}^{\text{HMT}}(h, m_\infty) = \mathfrak{G}_{\text{HMT}}(h, m_\infty). \quad (\text{C.37d})$$

Proof. By transfinite induction. At the initial level, the two classes coincide. If every value of a term in $\mathcal{T}_\beta^{\text{HMT}}$ belongs to $\mathfrak{G}_\beta$, the semantics of a successor term is one of the subsets in (C.37). Conversely, every subset in (C.37) is given by a formula code and a finite tuple of parameters; by the induction hypothesis those parameters possess terms, and (C.37a) forms the term that denotes it. At a limit, both constructions take the same union. $\square$

The state encoder is compatible with truncations and sends every TPK extension to a successor term. Let $\mathcal{T}_N^{\text{st}} \subset \mathcal{T}^{\text{HMT}}$ be the subclass of terms encoding complete states of depth N , and let $\tau_{N+1,N} : \mathcal{T}_{N+1}^{\text{st}} \rightarrow \mathcal{T}_N^{\text{st}}$ delete the last block together with the corresponding truncation of the genealogical ledger, carry, and memory. If $J_N : X_N \rightarrow \mathcal{T}_N^{\text{st}}$ is the complete encoder, compatibility with Ext_N proves the well-typed identity

$$\forall x \in X_N \forall y \in \text{Ext}_N(x) \quad \tau_{N+1,N}(J_{N+1}(y)) = J_N(x). \quad (\text{C.37e}_0)$$

Define within the semantics

$$\text{Succ}_{N+1}^{\text{TPK}}(J_N(x)) := \{t \in \mathcal{T}_{N+1}^{\text{st}} : \exists y \in X_{N+1} (y \in \text{Ext}_N(x) \wedge t = J_{N+1}(y))\}.$$

Choose β such that $\mathfrak{G}_\beta$ contains x, X_N, X_{N+1} , the finite graphs of $\text{Ext}_N, J_N, J_{N+1}$, and their codes in h . Then the primitive-recursive formula for the graph of Ext_N gives

$$J_{N+1}[\text{Ext}_N(x)] = \text{Succ}_{N+1}^{\text{TPK}}(J_N(x)), \quad J_{N+1}[\text{Ext}_N(x)] \in \mathfrak{G}_{\beta+1}, \quad (\text{C.37e})$$

and both sides are formed by the graph formula of the same TPK extension. Equations (C.37e₀)–(C.37e) are the explicit semantic link: the subclass of successor terms is exactly the image of the admissible extensions, and its truncation commutes. The complete fiber of ρ is not identified with Ext_N , nor is Def identified with TPK merely by nomenclature.

Lemma 79.3 (Transitivity, axioms, and well-ordering of the genealogical model). *$\mathfrak{G}_{\text{HMT}}(h, m_\infty)$ is a transitive class, contains all ordinals, satisfies Zermelo–Fraenkel set theory (ZF), and possesses a definable global well-order; it therefore satisfies Choice and all axioms used in the subsequent cardinal comparison.*

Proof. Abbreviate $B = \mathfrak{G}_0$ and $u = u_{h,m}$. Induction on (C.36) proves simultaneously that every level is transitive and that $\mathfrak{G}_\alpha \in \mathfrak{G}_{\alpha+1}$; if all ordinals smaller than α are present, their set is definable at the corresponding level and expresses α . Hence the union contains all ordinals. Extensionality and Foundation are inherited from the ambient universe; Empty Set and Infinity belong to B . Once a level containing the parameters is chosen, the formulas defining $\{a, b\}$, $\bigcup a$, and subsets by Separation express them at a successor level.

For Collection–Replacement, let F be a functional relation definable in $\mathfrak{G}$ on the set a . The reflection scheme relative to the definable hierarchy $(\mathfrak{G}_\xi, \in, u_{h,m})_{\xi \in \text{Ord}}$, applied in the ambient metatheory to the finite list of formulas defining F , produces a level $\mathfrak{G}_\theta$ containing a and the parameters and correct for those formulas. The ranks of the witnesses in $F[a]$ are then bounded by θ ; Separation over $\mathfrak{G}_\theta$ forms the image in $\mathfrak{G}_{\theta+1}$. For Power Set, the internal set to be constructed is not presupposed. Given $a \in \mathfrak{G}_\gamma$, consider in the ambient metatheory the already existing set $\mathcal{P}^V(a)$. For every $b \in \mathcal{P}^V(a) \cap \mathfrak{G}$, let $r(b)$ be its first genealogical stage. Since $\mathfrak{G}$ is a definable class, Separation in the ambient metatheory first forms the set $\mathcal{P}^V(a) \cap \mathfrak{G}$. Replacement **in the ambient metatheory**, applied to that set, bounds the ordinals $r(b)$ by some θ . Consequently,

$$\mathcal{P}^\mathfrak{G}(a) = \{b \in \mathfrak{G}_\theta : b \subseteq a\},$$

and the right-hand side is expressed by Separation in $\mathfrak{G}_{\theta+1}$. It does not introduce into $\mathfrak{G}$ the subsets external to the semantics (C.37b). This is the noncircular rank-bounding argument.

Finally, each object is assigned its first level of birth and its least **derivation name**: a well-founded, finitely branching derivation tree (and, by (C.37a), finite for each individual name), labeled by a level ordinal, a formula code, and the tuple of names of its parameters. These names are ordered by transfinite recursion: first by the ordinal label, then by the formula code, and then lexicographically by the already ordered parameter names. The least name of each value defines a class relation $<_\mathfrak{G}$ that compares all objects. It is a definable global well-order and, by taking the least element of every nonempty set, proves Choice. $\square$

79.2 TPK lift of the ternary fiber and complete cylinder partition

The fiber of $729 = 3^6$ refinement subcylinders does not arise from the transfer quotient $\mathcal{K}_{\text{ph}}$. That operator is a subsequent representation of reduced memory and cannot generate histories. The covering is constructed directly in the system $(H_k, \rho_k, \text{Ext}_k)$ through the three one-turn extensions of the nonadic holonomy.

79.2.1 Hensel inversion in the arithmetic tower and realization by histories

Before lifting the branches to enriched histories, it is useful to recover the finite argument determining the capacity of the arithmetic tower. The complete proof for the matrix's unfiltered transducer is included; its domain is distinguished from the surviving tree.

Proposition 79.4 (Conjugacy of the unfiltered Hensel tower). *Let p be prime, $d \geq 1$ and $A \in \text{GL}_d(\mathbb{Z})$. For row vectors, taking residue representatives in $\{0, \dots, p-1\}^d$, define*

$$x_t A = u_t + p x_{t+1}, \quad u_t = x_t A \bmod p.$$

The map

$$\Psi_T : (\mathbb{Z}/p^T \mathbb{Z})^d \longrightarrow (\mathbb{F}_p^d)^T, \quad x_0 \longmapsto (u_0, \dots, u_{T-1})$$

is bijective, with inverse

$$x_0 \equiv \sum_{t=0}^{T-1} p^t u_t A^{-(t+1)} \pmod{p^T}.$$

The bijections commute with reductions and give a homeomorphism $\Psi : \mathbb{Z}_p^d \rightarrow (\mathbb{F}_p^d)^\mathbb{N}$, where $\mathbb{Z}_p := \varprojlim_T \mathbb{Z}/p^T \mathbb{Z}$. The update $x_t \mapsto x_{t+1}$ corresponds to deleting the first block.

Proof. The integer inverse of A exists because its determinant is ± 1 . Iterating $x_t = (u_t + p x_{t+1}) A^{-1}$ gives

$$x_0 = \sum_{t=0}^{T-1} p^t u_t A^{-(t+1)} + p^T x_T A^{-T}.$$

Modulo p^T , a word determines a unique initial class. Given an arbitrary word, setting $x_T = 0$ and reconstructing backwards produces a class emitting it, proving surjectivity. The same formula proves compatibility under truncation. Passage to the inverse limit gives the homeomorphism and the relation to the tape shift. $\square$

For $p = 3$ and $d = 6$, the block alphabet has $3^6 = 729$ elements. This arithmetic capacity is not identified by definition with the realizability of every block after every surviving prefix. The link to the Ext edges is precisely the role of Lemma 79.6. The two operations distinguished by the matrix are thus maintained: representation in the raw tower and realization under the APP-TRIT-TPK compatibilities.

79.2.2 Edge-by-edge realization of the three nonadic extensions

This subsection specifies the autonomous model generated by the TPK extension relation entering the cardinal proof. The existence of an edge is not inferred from a coincidence of types: its initial and final states, APP relation, fiber transport, and unary update of the ledger are constructed. The construction uses exclusively the two APP sheets, the TRIT selector, and the TPK composition Sel-Tra-Upd.

Complete cells and three balanced reductions. In the chart $D_9 = \{1, \dots, 9\}$, we always retain the complete APP cell

$$a_{p,q} := (p, q; R_{p,q}^\Sigma, q_{p,q}^\Sigma, R_{p,q}^\Pi, q_{p,q}^\Pi),$$

with

$$p + q = R_{p,q}^\Sigma + 9q_{p,q}^\Sigma, \quad pq = R_{p,q}^\Pi + 9q_{p,q}^\Pi.$$

The active sheet is recorded in a separate coordinate; $a_{p,q}$ is not split into a purported additive cell and a separate multiplicative cell. For $r \in D_9$, let

$$a(r) := \begin{cases} a_{1,9}, & r = 1, \\ a_{1,r-1}, & 2 \leq r \leq 9, \end{cases} \quad \tau(r) := M_{\text{ph}}(r), \quad m(r) := \tau(r) \in \{-1, 0, +1\}. \quad (79.1)$$

The additive residue of $a(r)$ supplies an APP representative of phase here; this does not make Σ the universally active sheet. The dynamic selector activates Σ for $r \in \{1, 4, 7\}$, activates Π for $r \in \{2, 5, 8\}$, and preserves both sheets without moving the cursor for $r \in \{3, 6, 9\}$. In all three cases, $R^\Sigma, q^\Sigma, R^\Pi, q^\Pi$ are transported jointly. We have

$$(m(1), \dots, m(9)) = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The three pairs ordered by the nonadic base are

$$\mathfrak{p}_{+1} := (1, 4), \quad \mathfrak{p}_{-1} := (2, 5), \quad \mathfrak{p}_0 := (3, 6). \quad (79.2)$$

These pairs are the normalized section taking the first two occurrences of each TRIT type under the shift $r \mapsto r + 3$; the third phase block 7, 8, 9 retains the turn-closure control. Both members belong to the same fiber of M_{ph} . Without introducing a target coefficient, the balanced reduction from the TRIT chapter gives

ε	$m(\mathfrak{p}_\varepsilon^{(1)}) + m(\mathfrak{p}_\varepsilon^{(2)})$	$r_3(2\varepsilon)$	$\chi(\varepsilon, \varepsilon)$
+1	$1 + 1 = (-1) + 3 \cdot 1$	-1	+1
0	$0 + 0 = 0 + 3 \cdot 0$	0	0
-1	$(-1) + (-1) = 1 + 3 \cdot (-1)$	+1	-1

(79.3)

Therefore,

$$\chi(\varepsilon, \varepsilon) = \frac{2\varepsilon - r_3(2\varepsilon)}{3} = \varepsilon. \quad (79.4)$$

The branch sign is thus the output of the balanced cocycle applied to two TRIT emissions produced by APP. Each pair supplies one of the three carry outputs. For clarity, we subsequently index the pair by $\varepsilon(\mathfrak{p}) := \chi(m(\mathfrak{p}^{(1)}), m(\mathfrak{p}^{(2)}))$; the symbol ε does not supply a free carry to Upd^{cal} .

The phase architecture, balanced lift, and memory transport are realized here through twenty-seven local incidences and their extension graphs. The normalized section fixes representatives of the fibers of M_{ph} ; the covering to be proved uses the existence of these representatives and remains invariant when they are replaced by another compatible section.

Typed data of the nine edges. Let $K_{\text{Carr}} := \langle \kappa_{\text{Carr}} \rangle_{\text{Ab}}$ be the carry module. The actions are typed by

$$H_{\text{Carr}} : \mathcal{P}_{\text{TPK}}^{\text{op}} \longrightarrow \text{End}(K_{\text{Carr}}), \quad Y_{\text{Carr}} : \mathcal{P}_{\text{TPK}} \longrightarrow \text{End}(K_{\text{Carr}}).$$

In this generated model, take the trivial actions $H_{\text{Carr}}(e) = Y_{\text{Carr}}(e) = \text{id}_{K_{\text{Carr}}}$: the active sheet and the route are retained as coordinates of the ledger itself and do not act on the free carry summand. Also let (M_d, J_d, d) be the graded direct system of memory. To avoid confusing two representatives identified by the colimit, the state also retains the degree as a coordinate, and the labeled carrier

$$\tilde{M} := \bigsqcup_{d \geq 1} (\{d\} \times M_d), \quad \text{gr}_M(d, \mu) = d.$$

The representation in the limit memory is given by

$$\begin{aligned} M &:= \varinjlim_d M_d, & J_d &: M_d \longrightarrow M, \\ u &:= J_d(u_d), & \iota_M &: \mathbb{Z} \longrightarrow M, & n &\longmapsto nu. \end{aligned}$$

Here J_d is the canonical colimit map, and the u_d form a compatible collection. Its free component permits the reader $\deg_u(nu) = n$. The configuration, phase, memory, carry, route, and survival projections will be typed on the calibrated carrier constructed recursively below; no ambient space will be imported for them. For every constructed edge e , the calibrated component $\text{Upd}_e^{\text{cal}} := \text{Mem}_e^{\text{cal}}$ will denote the third map of the factorization; its law is fixed field by field below and is not presupposed as the restriction of an ambient relation.

Fix the initial depth $k_0 = 1$, a complete configuration $q_\star \in \mathcal{Q}_{\text{obs}}$ of phase one, and

$$q_{n,r} := F_{\text{obs}}^{9n+r-1}(q_\star), \quad q_{n,10} := q_{n+1,1}.$$

Because the observable calendar is periodic, its value $q_{n,r}$ does not by itself distinguish occurrences separated by twelve turns. The calibrated model therefore uses the labeled lift

$$\tilde{q}_{n,r} := (q_{n,r}, n, r), \quad \tilde{q}_{n,10} := \tilde{q}_{n+1,1}, \quad \pi_{\text{obs}}(\tilde{q}_{n,r}) = q_{n,r}.$$

The lifted dynamics $\tilde{F}_{\text{obs}}(\tilde{q}_{n,r}) = \tilde{q}_{n,r+1}$ projects to F_{obs} ; the label does not alter the physical phase, but makes its levels of occurrence distinct. The copy of the cell at each level is labeled by

$$a_{n,r} := (n, r; a(r)), \quad (n, r)^+ := \begin{cases} (n, r+1), & 1 \leq r < 9, \\ (n+1, 1), & r = 9. \end{cases}$$

The label only prevents two occurrences of the same cell from being identified; the two Euclidean divisions and their four coordinates are those of $a(r)$. For turn $n \geq 0$ and phase $r \in D_9$, the symbol

$$e_{\varepsilon,n,r} : \tilde{q}_{n,r} \longrightarrow \tilde{q}_{n,r+1}, \quad r^+ := 1 + (r \bmod 9),$$

occupies level $d_{n,r} := k_n + r - 1$, where $k_n := k_0 + 9n$. Its local ledger contains the complete cell $a(r)$, the active sheet determined by $M_{\text{ph}}(r)$, the type $\tau(r)$, the membership mark in $\mathfrak{p}_\varepsilon$, and, upon reaching the second member of that pair, the certified identity (79.3). We define its operators solely from that ledger:

$$C_M(e_{\varepsilon,n,r}) = J_{d_{n,r}+1, d_{n,r}}, \quad \kappa_M(e_{\varepsilon,n,r}) = \mathbf{1}_{\{r=9\}} u_{d_{n,r}+1}, \quad (79.5)$$

$$\text{Carr}(e_{\varepsilon,n,r}) = d_{\varepsilon,r} \kappa_{\text{Carr}}, \quad d_{\varepsilon,r} := \mathbf{1}_{\{r=\mathfrak{p}_\varepsilon^{(2)}\}} \chi\left(m(\mathfrak{p}_\varepsilon^{(1)}), m(\mathfrak{p}_\varepsilon^{(2)})\right). \quad (79.6)$$

For identities and every admissible composition, also define

$$C_M(\text{id}_{\tilde{q}}; d) := \text{id}_{M_d}, \quad \kappa_M(\text{id}_{\tilde{q}}; d) := 0, \quad (79.7)$$

$$C_M(e_t \cdots e_1) := C_M(e_t) \circ \cdots \circ C_M(e_1), \quad \kappa_M(e_2 e_1) := C_M(e_2)(\kappa_M(e_1)) + \kappa_M(e_2). \quad (79.8)$$

The identity carries degree d because the same observable configuration may recur at different depths. Iterated, the final equality defines $\kappa_M(e_t \cdots e_1)$ for every length and is the cocycle law of the direct system; the inclusions satisfy $J_{d+2, d+1} J_{d+1, d} = J_{d+2, d}$. In addition, set $s(e_{\varepsilon,n,r}) = \top$. The following twenty-seven entries certify the coordinates distinguishing the three extensions; the universal transport fields are specified immediately afterward. The symbols *I* and *II* designate, respectively, the first retained emission and the second emission performing the reduction; a dash denotes a zero carry increment, not the absence of an edge.

r	$M_{\text{ph}}(r)$	$\mathfrak{p}_{+1}$ mark	$d_{+1,r}$	$\mathfrak{p}_0$ mark	$d_{0,r}$	$\mathfrak{p}_{-1}$ mark	$d_{-1,r}$	Δ mem
1	+1	I	0	–	0	–	0	0
2	–1	–	0	–	0	I	0	0
3	0	–	0	I	0	–	0	0
4	+1	II	+1	–	0	–	0	0
5	–1	–	0	–	0	II	–1	0
6	0	–	0	II	0	–	0	0
7	+1	–	0	–	0	–	0	0
8	–1	–	0	–	0	–	0	0
9	0	–	0	–	0	–	0	u

Thus, $d_{\varepsilon,r}$ is computed from two outputs of the selector and is distinct from the cell quotients q^Σ, q^Π . The ledger also preserves the marked pair and the balanced residue; in particular, the zero branch is not confused with an absence of event.

The carry composition law used by the TPK category is

$$\text{Carr}(\text{id}_{\tilde{q}}) = 0, \quad \text{Carr}(e_2 e_1) = H_{\text{Carr}}(e_1)(\text{Carr}(e_2)) + Y_{\text{Carr}}(e_2)(\text{Carr}(e_1)). \quad (79.9)$$

On identities and paths, they are extended functorially:

$$\begin{aligned} H_{\text{Carr}}(\text{id}) &= Y_{\text{Carr}}(\text{id}) = \text{id}, \\ H_{\text{Carr}}(e_t \cdots e_1) &= H_{\text{Carr}}(e_1) \circ \cdots \circ H_{\text{Carr}}(e_t), \\ Y_{\text{Carr}}(e_t \cdots e_1) &= Y_{\text{Carr}}(e_t) \circ \cdots \circ Y_{\text{Carr}}(e_1). \end{aligned}$$

In this model all these actions remain the identity, but their typing makes explicit which endomorphism acts on each summand of (79.9). On the calibrated edges, $H_{\text{Carr}} = Y_{\text{Carr}} = \text{id}$, so composition adds the increments already produced. Equation (79.9), not an identification between orientation and quotient, governs the accumulated ledger.

Independent construction of the transition tuples. A fiber state is written, in the order of its relevant fields, as

$$x = (a; q; \tau, o, h; b_{\text{ref}}; R^\Sigma, q^\Sigma, R^\Pi, q^\Pi; c; \text{Sig}; C, \partial C; p_{\text{APP}}, p_{\{-1,0,+1\}}; \lambda; \text{gr}_M; \text{mem}; s; \text{Led}).$$

The update notation $x[f \leftarrow v]$ changes only the indicated field. For $h \in \{(\Sigma, \Pi), (\Pi, \Sigma)\}$, define

$$h_r(h) = \begin{cases} (\Sigma, \Pi), & r \in \{1, 4, 7\}, \\ (\Pi, \Sigma), & r \in \{2, 5, 8\}, \\ h, & r \in \{3, 6, 9\}. \end{cases}$$

The complete sheet-and-arithmetic coordinate is abbreviated as

$$\hat{h}(x) := (h(x), R^\Sigma(x), q^\Sigma(x), R^\Pi(x), q^\Pi(x)).$$

Let $\hat{\mathcal{H}}_{\tilde{q},r}^{\text{cal}}$ be the set of quintuples

$$(h, R_{a_{n,r}}^\Sigma, q_{a_{n,r}}^\Sigma, R_{a_{n,r}}^\Pi, q_{a_{n,r}}^\Pi), \quad h \in \{(\Sigma, \Pi), (\Pi, \Sigma)\},$$

whose four arithmetic data are the two Euclidean divisions of the labeled APP cell $a_{n,r}$. This direct definition does not presuppose a state in a carrier not yet constructed. Let $\text{Hist}_{\tilde{q}}^{\text{cal}}$ be the set of admissible finite paths in the graph of labeled edges ending at $\tilde{q}$, including the identity, and let $\text{B}_{\tilde{q}}^{\text{cal}}$ be the set of their ordered endpoint sequences. Define

$$\text{S}_{\tilde{q}}^{\text{cal}} := \hat{\mathcal{H}}_{\tilde{q}}^{\text{cal}} \times \{o_{\rightarrow}, o_o, o_{\leftarrow}\} \times K_{\text{Carr}} \times \text{Hist}_{\tilde{q}}^{\text{cal}} \times \text{B}_{\tilde{q}}^{\text{cal}} \times \text{M}.$$

The calibrated signature is not postulated: it is the typed assembly map

$$\text{Sig}_{\tilde{q}}^{\text{cal}} : \hat{\mathcal{H}}_{\tilde{q}}^{\text{cal}} \times \{o_{\rightarrow}, o_o, o_{\leftarrow}\} \times K_{\text{Carr}} \times \text{Hist}_{\tilde{q}}^{\text{cal}} \times \text{B}_{\tilde{q}}^{\text{cal}} \times \text{M} \longrightarrow \text{S}_{\tilde{q}}^{\text{cal}}, \quad (\hat{h}, o, c, \lambda, \partial C, \mu) \longmapsto (\hat{h}, o, c, \lambda, \partial C, \mu).$$

Likewise, $\pi_{\mathcal{F}}$ denotes the projection that removes only the first two coordinates $(a; q)$ of the state tuple; it preserves $\tau, o, h, b_{\text{ref}}$, the four Euclidean data, carry, signature, cylinder, boundary, prefixes, route, memory degree, memory, survival, and ledger. The corresponding orientation is fixed by $o(+1) = o_{\rightarrow}$, $o(0) = o_o$, and $o(-1) = o_{\leftarrow}$. The calibrated APP fiber over $\tilde{q}_{n,r}$ contains the already defined labeled copy $a_{n,r}$. The initial state is fixed without free coordinates:

$$\begin{aligned} \hat{h}_{\star} &:= ((\Sigma, \Pi), R_{1,9}^\Sigma, q_{1,9}^\Sigma, R_{1,9}^\Pi, q_{1,9}^\Pi). \\ x_{\star} &:= (a_{0,1}; \tilde{q}_{0,1}; +1, o_{\rightarrow}, (\Sigma, \Pi); \perp; R_{1,9}^\Sigma, q_{1,9}^\Sigma, R_{1,9}^\Pi, q_{1,9}^\Pi; 0; \\ &\quad \text{Sig}_{\tilde{q}_{0,1}}^{\text{cal}}(\hat{h}_{\star}, o_{\rightarrow}, 0, \text{id}_{\tilde{q}_{0,1}}, \emptyset, J_{k_0}(0_{\text{M}_{k_0}})); \emptyset, \emptyset; (a_{0,1}), (+1); \text{id}_{\tilde{q}_{0,1}}; k_0; 0_{\text{M}_{k_0}}; \top; \emptyset). \end{aligned} \quad (79.10)$$

The tuple contains no indeterminate coordinates and satisfies $s(x_{\star}) = \top$. Let $G_0 := \{x_{\star}\}$. Assuming G_n has been constructed, fix $x \in G_n$, $\varepsilon \in \{-1, 0, +1\}$, and $x_{\varepsilon,0}(x) := x$. For $r = 1, \dots, 9$, first construct the elementary ledger

$$\begin{aligned} \ell_{\varepsilon,n,r} &:= (e_{\varepsilon,n,r}, a_{n,r}, a_{(n,r)^+}, \tilde{q}_{n,r}, \tilde{q}_{n,r+1}, M_{\text{ph}}(r), \\ &\quad h_r(h(x_{\varepsilon,r-1}(x))), o(M_{\text{ph}}(r)), \mathfrak{p}_{\varepsilon}, d_{\varepsilon,r}, \\ &\quad C_{\text{M}}(e_{\varepsilon,n,r}), \kappa_{\text{M}}(e_{\varepsilon,n,r}), \text{Carr}(e_{\varepsilon,n,r})), \end{aligned} \quad (79.11)$$

and set $\text{Led}_{\varepsilon,n,r}^{\text{new}} := \text{Led}(x_{\varepsilon,r-1}(x)) \frown \ell_{\varepsilon,n,r}$. Then construct the tuple

$$y_{\varepsilon,r}(x) := x_{\varepsilon,r-1}(x)[\tau \leftarrow M_{\text{ph}}(r), h \leftarrow h_r(h(x_{\varepsilon,r-1}(x))), o \leftarrow o(M_{\text{ph}}(r)), b_{\text{ref}} \leftarrow p_{\varepsilon}], \quad (79.12)$$

where selection leaves the cell, quotients, carry, route, memory, cylinder, boundary, survival, and ledger unchanged; the coordinate b_{ref} makes the chosen balanced reduction explicit. Membership in an ambient relation is not presupposed here.

Next, define $z_{\varepsilon,r}(x)$ field by field: its configuration is $\tilde{q}_{n,r+1}$; its complete cell is $a_{(n,r)+}$, with the four residues and quotients recomputed by the two Euclidean divisions; it preserves the selected sheet and orientation; and it performs the following six updates, in which only the common argument x is omitted:

$$\begin{aligned} C(z_{\varepsilon,r}) &:= C(y_{\varepsilon,r}) \frown e_{\varepsilon,n,r}, \\ \partial C(z_{\varepsilon,r}) &:= \partial C(y_{\varepsilon,r}) \frown (\tilde{q}_{n,r}, \tilde{q}_{n,r+1}), \\ p_{\text{APP}}(z_{\varepsilon,r}) &:= p_{\text{APP}}(y_{\varepsilon,r}) \frown a_{(n,r)+}, \\ p_{\{-1,0,+1\}}(z_{\varepsilon,r}) &:= p_{\{-1,0,+1\}}(y_{\varepsilon,r}) \frown \tau(r), \\ \text{Led}(z_{\varepsilon,r}) &:= \text{Led}_{\varepsilon,n,r}^{\text{new}}, \\ \lambda(z_{\varepsilon,r}) &:= e_{\varepsilon,n,r} \lambda(y_{\varepsilon,r}), \end{aligned} \quad (79.13)$$

and still retains the fields $c, \text{gr}_M, \text{mem}, s$ of $y_{\varepsilon,r}(x)$. Its signature, by contrast, is retyped at the target configuration through the transported presignature

$$\text{Sig}(z_{\varepsilon,r}(x)) := \text{Sig}_{\tilde{q}_{n,r+1}}^{\text{cal}} \left(\hat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(y_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)), \partial C(z_{\varepsilon,r}(x)), J_{d_{n,r}}(\text{mem}_{d_{n,r}}(y_{\varepsilon,r}(x))) \right). \quad (79.14)$$

Thus, and only in Tra^{cal} , the edge, its endpoints, the mark $(p_{\varepsilon}, r, d_{\varepsilon,r})$, and its cocycles are added to the incidence ledger.

The initial state satisfies

$$c(x_{\star}) = 0 = \text{Carr}(\text{id}_{\tilde{q}_{0,1}}) = \text{Carr}(\lambda(x_{\star})).$$

In the complete states $x_{\varepsilon,r-1}(x)$, and also after the selection $y_{\varepsilon,r}(x)$, the typed invariant $c(v) = \text{Carr}(\lambda(v))$ is preserved, since Sel^{cal} modifies neither field. Transport advances the route but leaves the carry pending; hence the pre-update tuple satisfies exactly

$$c(z_{\varepsilon,r}(x)) = c(y_{\varepsilon,r}(x)) = \text{Carr}(\lambda(y_{\varepsilon,r}(x))).$$

The map Upd^{cal} then restores $c(x_{\varepsilon,r}(x)) = \text{Carr}(\lambda(x_{\varepsilon,r}(x)))$ by means of the composition law, without attributing that invariant to the intermediate state z .

Finally, apply the unary function $\text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} = \text{Mem}_{e_{\varepsilon,n,r}}^{\text{cal}}$ to that tuple. Its result $x_{\varepsilon,r}(x)$ is given by

$$\begin{aligned} c(x_{\varepsilon,r}(x)) &= \text{Carr}(\lambda(z_{\varepsilon,r}(x))) \\ &= H_{\text{Carr}}(\lambda(y_{\varepsilon,r}(x)))(\text{Carr}(e_{\varepsilon,n,r})) + Y_{\text{Carr}}(e_{\varepsilon,n,r})(c(y_{\varepsilon,r}(x))), \\ \lambda(x_{\varepsilon,r}(x)) &= \lambda(z_{\varepsilon,r}(x)), \\ \text{gr}_M(x_{\varepsilon,r}(x)) &= d_{n,r} + 1, \\ \text{mem}_{d_{n,r}+1}(x_{\varepsilon,r}(x)) &= C_M(e_{\varepsilon,n,r}) \text{mem}_{d_{n,r}}(z_{\varepsilon,r}(x)) + \kappa_M(e_{\varepsilon,n,r}), \\ s(x_{\varepsilon,r}(x)) &= s(e_{\varepsilon,n,r}) \wedge s(z_{\varepsilon,r}(x)), \\ \text{Sig}(x_{\varepsilon,r}(x)) &= \text{Sig}_{\tilde{q}_{n,r+1}}^{\text{cal}} \left(\hat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(x_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)), \partial C(z_{\varepsilon,r}(x)), J_{d_{n,r}+1}(\text{mem}_{d_{n,r}+1}(x_{\varepsilon,r}(x))) \right). \end{aligned} \quad (79.15)$$

All fields not enumerated in (79.15) coincide with those of $z_{\varepsilon,r}(x)$. In particular, the signature is computed from the sheet, orientation, boundary, and route already fixed by Sel^{cal} and Tra^{cal} , and from the carry and memory computed in the preceding lines; it is not circularly defined from the final state. Neither ε , nor $d_{\varepsilon,r}$, nor u is an additional argument of Upd^{cal} : the function reads all three data from the already constructed edge.

Simultaneous induction of the levels. Once the preceding nine stages have been completed for an already constructed G_n , define immediately

$$\gamma_{\varepsilon, k_n}(x) := x_{\varepsilon, 9}(x), \quad G_{n+1} := \{\gamma_{\varepsilon, k_n}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}. \quad (79.16)$$

The base $G_0 = \{x_\star\}$, the intermediate tuples, and (79.16) constitute a single definition by recursion on n . Therefore, no subsequent carrier is used to assume the existence of G_n : the next level is introduced only after constructing all its edges from the preceding level.

TPK extension relation and effective membership. Let $\mathcal{E}_{\tilde{q}}^{\text{cal}}$ be the least labeled collection containing $x_\star$ and, at each step of the preceding recursion, the four tuples $x_{\varepsilon, r-1}(x), y_{\varepsilon, r}(x), z_{\varepsilon, r}(x), x_{\varepsilon, r}(x)$ in their source and target configurations. Also let $\mathcal{A}_{\tilde{q}, r}^{\text{cal}} := \{a_{n, r}\}$, and set

$$\tilde{\mathcal{Q}}_{\text{obs}} := \{\tilde{q}_{n, r} : n \geq 0, 1 \leq r \leq 9\}, \quad \mathcal{E}^{\text{cal}} := \bigsqcup_{\tilde{q}} \mathcal{E}_{\tilde{q}}^{\text{cal}}, \quad \mathcal{F}_{\tilde{q}}^{\text{cal}} := \pi_{\mathcal{F}}(\mathcal{E}_{\tilde{q}}^{\text{cal}}).$$

For each depth d , let $\mathcal{E}_d^{\text{cal}} := \text{gr}_M^{-1}(d) \subset \mathcal{E}^{\text{cal}}$. The explicit degree coordinate, rather than the mere membership of a representative in some image of the direct system, makes these carriers disjoint. On them, the projections are typed as

$$\begin{aligned} \text{cfg} : \mathcal{E}^{\text{cal}} &\longrightarrow \tilde{\mathcal{Q}}_{\text{obs}}, \\ g(x) &:= \text{fase}(\pi_{\text{obs}}(\text{cfg}(x))) \in C_9, \\ c : \mathcal{E}^{\text{cal}} &\longrightarrow K_{\text{Carr}}, \\ s : \mathcal{E}^{\text{cal}} &\longrightarrow \{\perp, \top\}, \\ \lambda : \mathcal{E}_{\tilde{q}}^{\text{cal}} &\longrightarrow \text{Hist}_{\tilde{q}}^{\text{cal}}, \\ \text{gr}_M : \mathcal{E}^{\text{cal}} &\longrightarrow \mathbb{N}_{\geq 1}, \\ \text{mem}_d : \mathcal{E}_d^{\text{cal}} &\longrightarrow M_d, \\ \text{mem}(x) &:= J_d(\text{mem}_d(x)) \in M \quad (x \text{ of depth } d). \end{aligned}$$

Here $\text{cfg}(x)$ is the second coordinate of the tuple, and fase reads the phase of the observable calendar. For the three carrier classes, fix the diagonals

$$\begin{aligned} \Delta_{\text{APP}, \tilde{q}} &:= \{(a, a) : a \in \mathcal{A}_{\tilde{q}}^{\text{cal}}\}, \\ \Delta_{\text{fib}, \tilde{q}} &:= \{(f, f) : f \in \mathcal{F}_{\tilde{q}}^{\text{cal}}\}, \\ \Delta_{\text{enr}, \tilde{q}} &:= \{(x, x) : x \in \mathcal{E}_{\tilde{q}}^{\text{cal}}\}. \end{aligned}$$

In particular,

$$a_{0,1} \in \mathcal{A}_{\tilde{q}_{0,1}}^{\text{cal}}, \quad x_\star \in \mathcal{E}_{\tilde{q}_{0,1}}^{\text{cal}}.$$

On these carriers, the following relations are defined by their complete graphs, and not through necessary implications:

$$\text{Sel}_{e_{\varepsilon, n, r}}^{\text{cal}} := \{(x_{\varepsilon, r-1}(x), y_{\varepsilon, r}(x)) : x \in G_n\}, \quad (79.17)$$

$$\text{Tra}_{e_{\varepsilon, n, r}}^{\text{cal}} := \{(y_{\varepsilon, r}(x), z_{\varepsilon, r}(x)) : x \in G_n\}, \quad (79.18)$$

$$\text{Ext}_{e_{\varepsilon, n, r}}^{\text{APP}, \text{cal}} := \{(a_{n, r}, a_{(n, r)^+})\}, \quad (79.19)$$

$$\text{Ext}_{e_{\varepsilon, n, r}}^{\text{fib}, \text{cal}} := \{(\pi_{\mathcal{F}} x_{\varepsilon, r-1}(x), \pi_{\mathcal{F}} x_{\varepsilon, r}(x)) : x \in G_n\}. \quad (79.20)$$

The synchronized enriched extension is

$$\text{Ext}_{e_{\varepsilon, n, r}}^{\text{enr}, \text{cal}} := \{(x_{\varepsilon, r-1}(x), x_{\varepsilon, r}(x)) : x \in G_n\}. \quad (79.21)$$

The third map is defined by

$$\text{graph}(\text{Upd}_{e_{\varepsilon, n, r}}^{\text{cal}}) := \{(z_{\varepsilon, r}(x), x_{\varepsilon, r}(x)) : x \in G_n\}, \quad (79.22)$$

where the second component is exactly the one in (79.15). Consequently,

$$\mathcal{U}_{e_{\varepsilon,n,r}}^{(1),\text{cal}} := \text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Tra}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Sel}_{e_{\varepsilon,n,r}}^{\text{cal}} = \{(x_{\varepsilon,r-1}(x), x_{\varepsilon,r}(x)) : x \in G_n\}. \quad (79.23)$$

Each operator is indexed by the edge $e_{\varepsilon,n,r}$ produced by the reduction $\mathfrak{p}_\varepsilon$, rather than by a trit supplied from outside. The coordinate b_{ref} also distinguishes their codomains. Therefore, $\text{Sel}_e^{\text{cal}}$, Tra^{cal} and Upd^{cal} are functions on their respective domains, whereas the union $\bigcup_{\varepsilon \in \{-1,0,+1\}} \mathcal{U}_{e_{\varepsilon,n,r}}^{(1),\text{cal}}$ deliberately preserves the three outputs of the trisection.

For $\bullet \in \{\text{APP}, \text{fib}, \text{enr}\}$, identities and compositions are not left implicit. Define

$$\text{Ext}_{\text{id}_{\tilde{q}}}^{\bullet,\text{cal}} := \Delta_{\bullet,\tilde{q}}, \quad (79.24)$$

$$\text{Ext}_{e_i \dots e_1}^{\bullet,\text{cal}} := \text{Ext}_{e_i}^{\bullet,\text{cal}} \circ \dots \circ \text{Ext}_{e_1}^{\bullet,\text{cal}} \quad (79.25)$$

for every admissible edge word. The two APP and fiber path closures are formed separately,

$$\text{Path}(\text{Ext}^{\text{APP},\text{cal}}), \quad \text{Path}(\text{Ext}^{\text{fib},\text{cal}}), \quad (79.26)$$

and Tra^{cal} synchronizes them edge by edge.

Finally, the levels of complete states and their truncations are fixed by the recursion itself:

$$H_{d_{n,r}}^{\text{cal}} := \{x_{\varepsilon,r-1}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}, \quad (79.27)$$

$$H_{d_{n,r}+1}^{\text{cal}} := \{x_{\varepsilon,r}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}, \quad (79.28)$$

$$\rho_{d_{n,r}+1,d_{n,r}}^{\text{cal}}(x_{\varepsilon,r}(x)) := x_{\varepsilon,r-1}(x). \quad (79.29)$$

The final ledger preserves $(n, r, \mathfrak{p}_\varepsilon)$, so the predecessor of the left-hand side is unique and ρ^{cal} is well-defined. For $b > a$, set $\rho_{b,a}^{\text{cal}} := \rho_{a+1,a}^{\text{cal}} \circ \dots \circ \rho_{b,b-1}^{\text{cal}}$. This map recovers the complete preceding state, including its signature; it does not purport to invert an overwritten coordinate without consulting the ledger.

Proposition 79.5 (Autonomous realization of the TPK through explicit transitions). *The collection*

$$\begin{aligned} \mathfrak{T}_{\text{TPK}}^{\text{cal}} := & ((\tilde{q}_{n,r})_{n,r}, \pi_{\text{obs}}, \tilde{F}_{\text{obs}}, M_{\text{ph}}, \\ & (\mathcal{A}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, (\mathcal{E}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, (\mathcal{F}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, \text{Ext}^{\text{APP},\text{cal}}, \text{Ext}^{\text{fib},\text{cal}}, \text{Ext}^{\text{enr},\text{cal}}, \rho^{\text{cal}}, \\ & \text{Sel}^{\text{cal}}, \text{Tra}^{\text{cal}}, \text{Upd}^{\text{cal}}, C_M, \kappa_M, \text{gr}_M, H_{\text{Carr}}, Y_{\text{Carr}}, \\ & \text{Sig}^{\text{cal}}, \pi_{\mathcal{F}}). \end{aligned}$$

is an enriched model of the TPK. In particular, every pair on the right-hand side of (79.23) is an effective TPK edge, not a consequence inferred from a merely necessary condition.

Proof. The labels (n, r) make the carrier copies disjoint and unambiguously fix each source and target. The relation $\text{Ext}^{\text{APP},\text{cal}}$ transports the complete cell and preserves the two Euclidean identities; the relation $\text{Ext}^{\text{fib},\text{cal}}$ relates the fibers of the preceding and subsequent complete states: it transports sheet, orientation, prefix, cylinder, boundary, and ledger, and incorporates the updated carry, memory, and signature. Selection does not modify the ledger, and transport concatenates exactly one entry and produces the target presignature (79.14). The update recomputes the signature by means of the typed map $\text{Sig}_{\tilde{q}}^{\text{cal}}$ from the already determined sheet, orientation, carry, route, boundary, and memory. The map Upd^{cal} is unary because the edge contained in $z_{\varepsilon,r}(x)$ determines its cocycles.

The identities are the diagonals, and the word extensions are the compositions in (79.25). Concatenation of prefixes and ledgers is associative; memory satisfies the cocycle law, and carry satisfies (79.9). Therefore, the two parenthesizations of three edges produce the same state. Formula (79.29) defines truncation on complete states, and uniqueness of the final ledger proves that it recovers the source in every field. These are the identity, source, target, composition, memory, carry, and truncation laws of the enriched TPK. In particular, the extension relations are determined as graphs of effective maps rather than as consequences of merely necessary conditions. $\square$

Each γ_{ε,k_n} comprises exactly nine applications of $\text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Tra}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Sel}_{e_{\varepsilon,n,r}}^{\text{cal}}$. Survival will be proved by means of a constructed tail and will not be used to define G_n .

Lemma 79.6 (Effective internal trisection of one turn). *For every $n \geq 0$, every $x \in G_n$, and every $\varepsilon \in \{-1, 0, +1\}$, the expression $\gamma_{\varepsilon,k_n}(x)$ is defined by nine TPK transitions, with its synchronized relations $\text{Ext}^{\text{APP,cal}}$ and $\text{Ext}^{\text{fib,cal}}$, and satisfies*

$$\rho_{k_n+9,k_n}^{\text{cal}} \gamma_{\varepsilon,k_n}(x) = x, \quad (79.30)$$

$$g(\gamma_{\varepsilon,k_n}(x)) = g(x), \quad J_{k_n+9}(\text{mem}_{k_n+9}(\gamma_{\varepsilon,k_n}(x))) = J_{k_n}(\text{mem}_{k_n}(x)) + u, \quad (79.31)$$

$$c(\gamma_{\varepsilon,k_n}(x)) = c(x) + \varepsilon \kappa_{\text{Carr}}. \quad (79.32)$$

The ordered ledger of the nine edges determines ε uniquely. The three images are therefore disjoint. Each endpoint belongs to G_{n+1} and again admits the three maps $\gamma_{-1,k_{n+1}}, \gamma_{0,k_{n+1}}, \gamma_{+1,k_{n+1}}$.

Proof. Formulas (79.1) traverse the nine phases once and return the copy $a_{n+1,1}$ of $a(1)$. At each step, the APP relation transports the complete sextuple; Sel^{cal} fixes the TRIT type from the phase mark r preserved in the state; and Tra^{cal} adds exactly one symbol to the prefix, one segment to the cylinder, and its ordered pair of endpoints to the boundary. By induction on r , the explicit tuples $y_{\varepsilon,r}(x)$ and $z_{\varepsilon,r}(x)$ satisfy the predicates of the indicated relations, and the unary update produces the state at the next level. The identity $\rho_{d_{n,r}+1,d_{n,r}}^{\text{cal}}(x_{\varepsilon,r}(x)) = x_{\varepsilon,r-1}(x)$ is the definition (79.29); its well-definedness follows from uniqueness of the final ledger. The composition of the nine truncation maps recovers x , proving (79.30) without separately inverting the overwritten fields.

The phase $9 \rightarrow 1$ occurs exactly once. By (79.5) and the cocycle law, the linear memory component transports the preceding memory, and its translational component adds exactly u in the direct limit. The compatibility $J_{d+1}J_{d+1,d} = J_d$ proves (79.31); the return concerns the phase, not the enriched state.

By (79.6), eight edges have zero increment, and the edge completing p_ε has the computed increment $\chi(\varepsilon, \varepsilon) \kappa_{\text{Carr}}$. The laws (79.9) and (79.4) give (79.32). The ledger preserves $(p_\varepsilon, 2\varepsilon, r_3(2\varepsilon), \chi(\varepsilon, \varepsilon))$; the three values of this ledger are distinct even when the scalar increment is zero. Hence the images are not identified, and the block decoder recovers a unique ε .

The edge definitions depend only on the new phase and concatenate, rather than replace, the prefix, cylinder, boundary, and ledger. They are also invariant under $J_d \text{ mem}_d \mapsto J_d \text{ mem}_d + nu$ and under translation of the input carry. Consequently, the same construction applies at the endpoint of any turn. For every finite state, the explicit sequence

$$x, \quad \gamma_{0,k_n}(x), \quad \gamma_{0,k_{n+1}}\gamma_{0,k_n}(x), \quad \dots$$

is a compatible infinite tail. This proves the survival of all constructed states and the repeated availability of the three branches, without postulating either property. Consequently, in accordance with the preceding level definitions, we may finally identify

$$H_{k_n}^{\text{cal}} := G_n \quad (79.33)$$

for every $n \geq 0$. □

The preceding closure distinguishes the three clocks involved. Nine emissions return the phase in C_9 and advance memory. Twelve turns, that is, 108 emissions, also return the position of the observable calendar; even then, neither the ledger nor memory is erased. The operator $\mathcal{K}_{\text{ph}}$ acts only afterward, as a recognition of reduced memory from the already constructed turns; it never generates the edges of $\text{Ext}^{\text{enr,cal}}$.

Let

$$v : \mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad v(0) = 0, \quad v(1) = +1, \quad v(2) = -1$$

be the balanced representative. For $b = (b_1, \dots, b_6) \in \mathbb{F}_3^6$, the block extension is the ordered composition of six turns

$$\Gamma_{b,k}^{[54]} := \gamma_{v(b_6),k+45} \circ \gamma_{v(b_5),k+36} \circ \gamma_{v(b_4),k+27} \circ \gamma_{v(b_3),k+18} \circ \gamma_{v(b_2),k+9} \circ \gamma_{v(b_1),k}. \quad (\text{C.39d})$$

At block level one always uses $k = k_{6N} = 1 + 54N$; consequently, each factor has the domain constructed by the preceding factor. Fix a canonical APP seed $x_\star \in H_1^{\text{cal}}$ and define recursively

$$\begin{aligned} Y_0 &:= \{x_\star\}, & Y_N &\subseteq H_{1+54N}^{\text{cal}}, \\ Y_{N+1} &:= \{\Gamma_{b,1+54N}^{[54]}(x) : x \in Y_N, b \in \mathbb{F}_3^6\}, \\ \rho_{N+1,N}^{\text{blk}} &:= \rho_{1+54(N+1),1+54N}^{\text{cal}} \upharpoonright Y_{N+1}, \\ \text{Ext}_N^{\text{cal}}(x) &:= \{\Gamma_{b,1+54N}^{[54]}(x) : b \in \mathbb{F}_3^6\}. \end{aligned} \tag{C.39e}$$

The notation $\rho_{b,a}^{\text{cal}}$ denotes the composition of all deletion maps between levels b and a . By construction, every extension in $\text{Ext}_N^{\text{cal}}(x)$ is a chain of fifty-four TPK edges, not a jump of the Markov quotient.

The reader is defined directly on the ledger, before invoking a representation through Γ_b . The preceding lemma provides a decoder

$$\text{dec}_9 : \{\text{generated nine-edge segments}\} \longrightarrow \{-1, 0, +1\},$$

which reads the TRIT pair and its reduction on the segment. If $\text{Led}_{j,i}^{(9)}(x)$ denotes the i -th nonadic segment of block j in the ledger of x , set

$$\beta_{\text{blk},N}(x) := \left((v^{-1} \text{dec}_9(\text{Led}_{j,i}^{(9)}(x)))_{i=1}^6 \right)_{j=0}^{N-1}. \tag{79.34}$$

For $N = 0$, the value is the empty word. This definition does not presuppose that the decomposition of x as a composition of maps Γ_b is unique.

We now specialize the preceding arithmetic conjugacy to $p = 3$, $d = 6$, and $A = I_6$. If $\pi_{N+1,N} : (\mathbb{Z}/3^{N+1}\mathbb{Z})^6 \rightarrow (\mathbb{Z}/3^N\mathbb{Z})^6$ is the canonical reduction, define

$$\Phi_N := \Psi_N^{-1} \circ \beta_{\text{blk},N} : Y_N \longrightarrow (\mathbb{Z}/3^N\mathbb{Z})^6. \tag{79.35}$$

This definition does not assign a coordinate after observing a branch: the reader $\beta_{\text{blk},N}$ is obtained from the already constructed TPK ledger, and Ψ_N^{-1} is the explicit inverse, fixed beforehand, of the unfiltered arithmetic tower.

Proposition 79.7 (Finite conjugacy and complete covering of the fiber). *For every $N \geq 0$,*

$$\beta_{\text{blk},N} : Y_N \xrightarrow{\cong} (\mathbb{F}_3^6)^N$$

is a bijection compatible with truncations, and for every $x \in Y_N$

$$\begin{aligned} \beta_{\text{blk},N+1}[\text{Ext}_N^{\text{cal}}(x)] &= \{\beta_{\text{blk},N}(x) \frown b : b \in \mathbb{F}_3^6\}, \\ \text{pref}_N \circ \beta_{\text{blk},N+1} &= \beta_{\text{blk},N} \circ \rho_{N+1,N}^{\text{blk}}. \end{aligned} \tag{C.40_0}$$

The maps Φ_N are bijections and satisfy the intertwining identity

$$\pi_{N+1,N} \circ \Phi_{N+1} = \Phi_N \circ \rho_{N+1,N}^{\text{blk}}. \tag{79.36}$$

Consequently, the tower of generated histories and the Hensel arithmetic tower are isomorphic as finite inverse systems.

Proof. The assertion is immediate for $N = 0$. Assume that it holds at level N . For each word $b \in \mathbb{F}_3^6$, the preceding lemma constructs an extension $\Gamma_{b,1+54N}^{[54]}(x)$. If $b \neq b'$, the first coordinate at which they differ determines a distinct TRIT pair and cocycle in the first divergent nonadic segment of the genealogical ledger. That ledger preserves the ordered edges, not merely the final sum of their carries; hence the two histories remain distinct even if carries from later turns cancel. The direct reading (79.34) then gives

$$\beta_{\text{blk},N+1}(\Gamma_{b,1+54N}^{[54]}(x)) = \beta_{\text{blk},N}(x) \frown b. \tag{79.37}$$

The six successive trisections produce exactly 3^6 distinct extensions. Deleting the last fifty-four steps recovers x , and deleting the final block of the ledger recovers $\beta_{\text{blk},N}(x)$. This simultaneously proves bijectivity and the compatibility of $\beta_{\text{blk},N}$ with truncations.

Proposition 79.4 independently proves that Ψ_N is bijective and that

$$\text{pref}_N \circ \Psi_{N+1} = \Psi_N \circ \pi_{N+1,N}.$$

Composing this equality with the compatibility of β_{blk} and applying Ψ_N^{-1} yields

$$\pi_{N+1,N} \Psi_{N+1}^{-1} \beta_{\text{blk},N+1} = \Psi_N^{-1} \beta_{\text{blk},N} \rho_{N+1,N}^{\text{blk}},$$

which is exactly (79.36). The equality does not arise by defining an ad hoc coordinate on each branch: it results from composing two systems of constructed bijections that are compatible at every depth. $\square$

The inverse limit

$$\Omega_{\Gamma}^{\text{cal}} := \varprojlim_N (Y_N, \rho_{N+1,N}^{\text{blk}}) \quad (\text{C.40a}_0)$$

is the history limit of the TPK model constructed edge by edge in the preceding lemma. Its projection under π_{obs} preserves the configurations of the observable calendar, but the assertion does not depend on identifying this autonomous model with an ambient relation defined only by necessary conditions. The finite bijections pass to the limit and produce the cylinder homeomorphism

$$\Lambda : (\mathbb{F}_3^6)^{\mathbb{N}} \xrightarrow{\cong} \Omega_{\Gamma}^{\text{cal}}, \quad \beta_{\text{blk}} := \Lambda^{-1}. \quad (\text{C.40b}_0)$$

Denote the canonical projection by $\rho_{\infty,N}^{\text{blk}} : \Omega_{\Gamma}^{\text{cal}} \rightarrow Y_N$. Thus, Λ establishes an equivalence compatible with truncations: each ternary tape encodes a TPK history, and every prefix is extended through edges of the same generator.

79.2.3 Reversible prolongation in line coordinates

Proposition 79.7 constructs the bijections $\beta_{\text{blk},N}$ from nonadic segments of the ordered record, together with their compatibility with block deletion. The homeomorphism (C.40b₀) identifies the limit of these generated histories with the sequence of their six-trit words. We now compose that reader with the finite chart (17.22), retaining its central law.

Theorem 79.8 (Cylindrical transport of histories to ordered pairs of lines). *For every $N \geq 0$, the map*

$$\mathcal{E}_N := \mathcal{E}^{\times N} \circ \beta_{\text{blk},N} : Y_N \xrightarrow{\cong} (\mathcal{L}_{27} \times \mathcal{L}_{27})^N \quad (79.38)$$

is bijective. These bijections satisfy

$$\text{pref}_N \circ \mathcal{E}_{N+1} = \mathcal{E}_N \circ \rho_{N+1,N}^{\text{blk}} \quad (79.39)$$

and determine the homeomorphism

$$\mathcal{E}_{\infty} := \mathcal{E}^{\mathbb{N}} \circ \beta_{\text{blk}} : \Omega_{\Gamma}^{\text{cal}} \xrightarrow{\cong} (\mathcal{L}_{27} \times \mathcal{L}_{27})^{\mathbb{N}}. \quad (79.40)$$

Its inverse is $\beta_{\text{blk}}^{-1} \circ (\mathcal{E}^{-1})^{\mathbb{N}}$.

Proof. Proposition 17.7 provides the effective inverse of each factor $\mathcal{E}$, using the line table and coordinate interlacing. Consequently, $\mathcal{E}_N^{-1} = \beta_{\text{blk},N}^{-1} \circ (\mathcal{E}^{-1})^{\times N}$. For $N = 0$, both maps act on the empty word. Coordinatewise application commutes with deletion of the final factor. Using (C.40₀) gives

$$\begin{aligned} \text{pref}_N \mathcal{E}^{\times(N+1)} \beta_{\text{blk},N+1} &= \mathcal{E}^{\times N} \text{pref}_N \beta_{\text{blk},N+1} \\ &= \mathcal{E}^{\times N} \beta_{\text{blk},N} \rho_{N+1,N}^{\text{blk}}. \end{aligned}$$

This is (79.39). Applying the inverse to each factor of a sequence of line pairs determines a unique word sequence. The homeomorphism (C.40b₀) reconstructs from it a unique history in $\Omega_{\Gamma}^{\text{cal}}$. The

compositions recover the original history and the original sequence, respectively. In the cylinder topologies, a basic open set fixes finitely many coordinates. The map and its inverse transform these conditions into conditions on the same number of coordinates; both are continuous. This proves the topological assertion and compatibility of the limit with every finite truncation. $\square$

To retain the information in the complete state, let ζ denote its remaining records: frame, orientation, carry, route, memory, and boundary data. On the already constructed admissible domain, the enlarged transformation is

$$\hat{\mathcal{E}}(w, \zeta) = (\mathcal{E}(w), \zeta).$$

Its image domain retains the same compatibility conditions, expressed through the explicit inverse of the first coordinate. If $F : \mathcal{X} \rightarrow \mathcal{Y}$ is an operator between state domains, its transport is

$$F^{\mathcal{E}} = \hat{\mathcal{E}}_{\mathcal{Y}} \circ F \circ \hat{\mathcal{E}}_{\mathcal{X}}^{-1}.$$

The identity $F^{\mathcal{E}} \hat{\mathcal{E}}_{\mathcal{X}} = \hat{\mathcal{E}}_{\mathcal{Y}} F$ follows by cancelling the two inverse maps. For a relational prolongation $\mathcal{R} \subset \mathcal{X} \times \mathcal{Y}$, the corresponding definition is

$$\mathcal{R}^{\mathcal{E}} = \{(\hat{\mathcal{E}}_{\mathcal{X}}x, \hat{\mathcal{E}}_{\mathcal{Y}}y) : (x, y) \in \mathcal{R}\}.$$

The inverse transformation recovers exactly $\mathcal{R}$. If the prolongation also retains a witness ω in a domain W , its realization $\tilde{\mathcal{R}} \subset \mathcal{X} \times W \times \mathcal{Y}$ is transported by

$$(x, \omega, y) \mapsto (\hat{\mathcal{E}}_{\mathcal{X}}x, \omega, \hat{\mathcal{E}}_{\mathcal{Y}}y).$$

The inverse acts on the two endpoints and retains ω . Thus distinct witnesses remain distinct even when they share endpoints, and the recorded multiplicity is preserved. Functionality is preserved on the domains where it has already been established.

Coverage at arbitrary depth comes here from the domain $\Omega_{\Gamma}^{\text{cal}}$ and its proved reader, whereas ordered pairs of lines provide reversible coordinates for that domain. The initial census of 243 words and the global capacity of 729 prolongations per block retain their distinct roles. The diagonal, incident, and skew classification accompanies each factor; conditions selecting adjacent walks are transported as restrictions on the complete sequence.

The subcylinders are now defined as subsets of the limit, rather than as pairs of suffixes. For $x \in Y_N$ and $b \in \mathbb{F}_3^6$, let

$$\mathcal{Z}_N(x) := \{\xi \in \Omega_{\Gamma}^{\text{cal}} : \rho_{\infty, N}^{\text{blk}}(\xi) = x\}, \quad (79.41)$$

$$\mathcal{Z}_{N+1}(x; b) := \{\xi \in \Omega_{\Gamma}^{\text{cal}} : \rho_{\infty, N+1}^{\text{blk}}(\xi) = \Gamma_{b, 1+54N}^{[54]}(x)\}. \quad (79.42)$$

Theorem 79.9 (Effective partition into 729 refinement subcylinders). *For every $N \geq 0$ and every $x \in Y_N$, there is a disjoint union*

$$\mathcal{Z}_N(x) = \bigsqcup_{b \in \mathbb{F}_3^6} \mathcal{Z}_{N+1}(x; b). \quad (79.43)$$

The $3^6 = 729$ terms are nonempty clopen refinement subcylinders; each is realized by fifty-four edges of the constructed TPK.

Proof. Let $\xi \in \mathcal{Z}_N(x)$. Its level- $N + 1$ coordinate belongs to Y_{N+1} , truncates to x , and, by the bijection $\beta_{\text{blk}, N+1}$, has a unique final block $b \in \mathbb{F}_3^6$. Identity (79.37) then implies $\rho_{\infty, N+1}^{\text{blk}}(\xi) = \Gamma_{b, 1+54N}^{[54]}(x)$; therefore, ξ belongs to one of the terms on the right-hand side. The reverse inclusion follows immediately by truncation. If two terms intersected, their level- $N + 1$ coordinates would be equal and the ledger decoder would give $b = b'$; hence the union is disjoint.

Each endpoint $\Gamma_b^{[54]}(x)$ admits an infinite tail obtained by iterating the zero-block extension; consequently, every term is nonempty. Since Y_{N+1} is finite and discrete, the terms are inverse images of points under a continuous projection of the inverse limit and are therefore clopen. The very definition of $\Gamma_b^{[54]}$ establishes its fifty-four edges. $\square$

Finally, the collection $(\Phi_N)_N$ induces the homeomorphism of limits

$$\Phi_\infty : \Omega_\Gamma^{\text{cal}} \xrightarrow{\cong} \mathbb{Z}_3^6, \quad \Phi_\infty(\xi) := (\Phi_N(\rho_{\infty,N}^{\text{blk}}(\xi)))_{N \geq 0}, \quad (79.44)$$

where $\mathbb{Z}_3^6 = \varprojlim_N (\mathbb{Z}/3^N\mathbb{Z})^6$. Equation (79.36) guarantees that the right-hand side is a compatible class, and the finite bijections prove injectivity, surjectivity, and continuity in both directions. Under Φ_∞ , $\mathcal{Z}_N(x)$ is exactly the congruence class

$$\{z \in \mathbb{Z}_3^6 : z \equiv \Phi_N(x) \pmod{3^N}\},$$

and the sets $\mathcal{Z}_{N+1}(x; b)$ are its 729 refinement classes modulo 3^{N+1} . This is the effective junction between TPK extension, limit cylinders, and the Hensel tower.

For $b = (b_1, \dots, b_6) \in \mathbb{F}_3^6$, set

$$\nu(b) := \sum_{i=1}^6 [b_i]_3 3^{6-i} \in \{0, \dots, 728\}, \quad (\text{C.40c}_0)$$

where $[b_i]_3 \in \{0, 1, 2\}$ is the positional representative. The 243 visible words in the initial census are the image of the first emission; they are not to be confused with the 3^6 capacity of each extension block. Finally,

$$X_{\infty, \mathbb{Z}}^{\text{cal}} := \mathbb{Z}_{\text{HMT}} \times \Omega_\Gamma^{\text{cal}}.$$

For a prefix $s = (m; b_0, \dots, b_{N-1})$, define in the generated arithmetic of $\mathbb{Q}_{\text{HMT}}$

$$a_N(s) := m + \sum_{q < N} \frac{\nu(b_q)}{729^{q+1}}, \quad \text{Succ}(s) := \{(m; b_0, \dots, b_{N-1}, b) : b \in \mathbb{F}_3^6\}.$$

If $(m; \beta_{\text{blk}, N}(x))$ denotes the prefix obtained by prepending m to the tuple $\beta_{\text{blk}, N}(x)$, (C.40₀) gives the typed equality

$$\text{Succ}(m; \beta_{\text{blk}, N}(x)) = \{(m; \beta_{\text{blk}, N+1}(y)) : y \in \text{Ext}_N^{\text{cal}}(x)\}.$$

From their first definition, the half-open cylinders are subsets of the HMT completion:

$$I_N^{\text{HMT}}(s) := \{r \in \mathbb{R}_{\text{HMT}} : a_N(s) \leq_{\text{HMT}} r <_{\text{HMT}} a_N(s) + 729^{-N}\}.$$

They satisfy materially

$$I_N^{\text{HMT}}(s) = \bigsqcup_{y \in \text{Succ}(s)} I_{N+1}^{\text{HMT}}(y), \quad |\text{Succ}(s)| = 729, \quad \text{diam}_{\text{HMT}} I_N^{\text{HMT}}(s) = 729^{-N} \longrightarrow 0. \quad (\text{C.40a})$$

The boundary convention identifies the two representations only after they have been constructed. The partition (C.40a), nesting, and contraction produce

$$\begin{aligned} P_{\text{ar}} : X_{\infty, \mathbb{Z}}^{\text{cal}} &\twoheadrightarrow \mathbb{R}_{\text{HMT}}, \\ P_{\text{ar}}(m, \xi) &:= \text{the unique element of } \bigcap_N C_N^{\text{cl}}(m; \beta_{\text{blk}, N}(\rho_{\infty, N}^{\text{blk}}(\xi))), \\ C_N^{\text{cl}}(s) &:= \overline{I_N^{\text{HMT}}(s)}. \end{aligned} \quad (\text{C.40})$$

Therefore,

$$X_{\infty, \mathbb{Z}}^{\text{cal}} / \sim_{P_{\text{ar}}} \cong \mathbb{R}_{\text{HMT}}. \quad (\text{C.41})$$

The other representation, P_{exc} , preserves incidence, boundary, and memory toward Paley–Witt–Golay–Leech and, from there, toward V^{h} –Monster. Both arise from the same antecedent in (C.35); neither the Archimedean coordinate nor the exceptional recognition selects the seeds. The five constructions *sp*, *sol*, *gau*, *coh*, *det* remain consubstantial within $\mathcal{C}_{\text{cont}}^{\text{disc}}$ and are not disaggregated in order to carry out the cardinal step.

79.2.4 Descent of operations to the evaluation quotient

The preceding quotient identifies histories publishing the same value, not their complete registers. The following criterion characterizes exactly when an operation on histories induces an operation on that quotient without choosing a privileged history within each fiber; its formulation continues the genealogical construction established in the preceding sections.

Proposition 79.10 (Descent criterion through evaluation fibers). *Let $q : \Omega \rightarrow J$ be the evaluation of a genealogical chart onto its image. Let $\star : D \rightarrow \Omega$ be an operation with $D \subseteq \Omega^2$ saturated by the fibers of $q \times q$. There exists a unique operation $\bar{\star} : (q \times q)(D) \rightarrow J$ such that*

$$q(x \star y) = q(x) \bar{\star} q(y)$$

if and only if, for pairs in the domain,

$$q(x) = q(x'), \quad q(y) = q(y') \implies q(x \star y) = q(x' \star y').$$

Proof. If the visible operation exists, its result depends only on the two values, proving necessity. For sufficiency, define $s \bar{\star} t = q(x \star y)$, where $q(x) = s$ and $q(y) = t$. Saturation preserves the domain when representatives are changed; the stated condition preserves the result. Surjectivity of $q \times q$ onto the image of the domain proves uniqueness. $\square$

A coordinate does not automatically replace a history as an argument of all its operations: the criterion determines when that substitution is legitimate. In particular, transporting an operation through a bijection does not suffice to identify it with a native operation preceding the reading. If a chart Ω is compact and q is continuous and surjective onto a Hausdorff space J , evaluation also induces a homeomorphism $\Omega/\sim_q \rightarrow J$: it is a continuous bijection from a compact space to a Hausdorff space. It is the *quotient*, not necessarily the space of histories, that is topologically identified with J .

79.2.5 Internal reconstruction of the chart and uniform lift

Within $\mathbb{G}_{\text{HMT}}(h, m_\infty)$, the reconstruction in (C.35)–(C.41) is carried out with the same codes and operators. Denote by $\mathbb{R}_{\text{HMT}}^\mathbb{G}$ the internal HMT completion. The uniqueness of the complete Archimedean ordered field provides, **after** its construction, the internal isomorphism

$$\eta : \mathbb{R}_{\text{HMT}}^\mathbb{G} \xrightarrow{\cong} \mathbb{R}^\mathbb{G}. \quad (\text{C.41}_0)$$

For the corresponding internal interval $I_N^\mathbb{G}(s) := I_N^{\text{HMT}}(s) \cap \mathbb{R}_{\text{HMT}}^\mathbb{G}$, the isomorphism preserves endpoints and order:

$$\eta[I_N^\mathbb{G}(s)] = \{r \in \mathbb{R}^\mathbb{G} : a_N(s) \leq r < a_N(s) + 729^{-N}\}. \quad (\text{C.41}_0\text{a})$$

The surjectivity of (C.40) covers every element of that completion, not only those obtained in a finite execution.

So that “every” is not concealed in the definition of the codomain, we construct the two injections relating the line to its internal power set. Let

$$E(A) = \sum_{n \geq 0} \frac{2\mathbf{1}_A(n)}{3^{2n+2}} \quad (A \in \mathcal{P}^\mathbb{G}(\omega)). \quad (\text{C.41a})$$

This representation places 0 or 2 only in the even ternary positions; its image is contained in $[0, 1/4]$. It therefore never reaches the right endpoint of the root cylinder, and two distinct subsets have a different first even position: E is injective and is not subject to boundary ambiguity. Write $E_{\text{HMT}} := \eta^{-1} \circ E$.

Given A , begin with the root cylinder containing $E_{\text{HMT}}(A)$. If $s_N(A)$ is the prefix already chosen, write $s_N(A) = (0; b_0(A), \dots, b_{N-1}(A))$. The partition (C.40a), transported by η^{-1} , supplies a unique

half-open subcylinder $s_{N+1}(A)$ containing $E_{\text{HMT}}(A)$. By (C.40₀), that extension is also an admissible TPK extension. By induction,

$$s_{N+1}(A) \upharpoonright N = s_N(A), \quad E_{\text{HMT}}(A) \in I_N^{\text{HMT}}(s_N(A)) \quad (N < \omega). \quad (\text{C.41b}_0)$$

The recursion is a uniform formula with parameters A, h . Since both are internal, Replacement in the model of Lemma 79.3 forms the complete sequence of blocks $(b_q(A))_{q < \omega}$. Without confusing the tape with the sequence of prefixes, define

$$\xi_A := \Lambda((b_q(A))_{q < \omega}) \in \Omega_\Gamma^{\text{cal}, \mathfrak{G}}.$$

This yields the **uniform lift of the Cantor embedding**

$$Q : \mathcal{P}^\mathfrak{G}(\omega) \longrightarrow X_{\infty, \mathbb{Z}}^{\text{cal}, \mathfrak{G}}, \quad Q(A) := (0, \xi_A), \quad P_{\text{ar}}(Q(A)) = E_{\text{HMT}}(A). \quad (\text{C.41b})$$

The map Q is not called an inverse of P_{ar} : their composition expresses the embedding E_{HMT} , not the identity on the entire line. In the opposite direction, let $(q_n)_{n < \omega}$ be the generated enumeration of $\mathbb{Q}_{\text{HMT}}$, and define

$$C(r) := \{n < \omega : \eta(q_n) < r\} \quad (r \in \mathbb{R}^\mathfrak{G}).$$

The rational cut belongs to $\mathcal{P}^\mathfrak{G}(\omega)$, and C is injective by density. The injections E and C , followed by the internal Cantor–Bernstein theorem, prove

$$P_{\text{ar}}[X_{\infty, \mathbb{Z}}^{\text{cal}, \mathfrak{G}}] = \mathbb{R}_{\text{HMT}}^\mathfrak{G}, \quad \mathbb{R}_{\text{HMT}}^\mathfrak{G} \cong \mathbb{R}^\mathfrak{G} \approx \mathcal{P}^\mathfrak{G}(\omega). \quad (\text{C.41c})$$

Here an arbitrary real is used only as a universally quantified variable to prove surjectivity of an already constructed tree. It selects neither APP, TRIT, TPK, the constants, nor the transitions of that tree.

79.3 Enumeration by genealogical codes and HMT condensation

Rule (C.37) supplies derivation names from

$$(\ulcorner \varphi^\top, \sigma(p_0), \dots, \sigma(p_{k-1}) \rrcorner). \quad (\text{C.42})$$

For any object a of the hierarchy, let $\beta(a)$ be its first stage and $\sigma(a)$ its least derivation tree in the recursive order of Lemma 79.3. The term $\sigma(a)$ is not required to be a natural number when the stage is uncountable. The relation

$$a <_\mathfrak{G} b \iff (\beta(a), \sigma(a)) <_{\text{lex}} (\beta(b), \sigma(b)) \quad (\text{C.42a})$$

is the global genealogical well-order $<_\mathfrak{G}$ induced by the constructor. Repeated names are eliminated by taking the least one; neither a real nor an enumeration of the line is inserted as data.

Lemma 79.11 (Internal countability of levels below $\omega_1^\mathfrak{G}$). *For every $\beta < \omega_1^\mathfrak{G}$, the level $\mathfrak{G}_\beta(h, m_\infty)$ is countable within $\mathfrak{G}_{\text{HMT}}(h, m_\infty)$, and possesses a canonical surjection*

$$e_\beta : \omega \twoheadrightarrow \mathfrak{G}_\beta(h, m_\infty). \quad (\text{C.43})$$

Proof. The initial level has a countable code. If e_β enumerates level β , the formula codes and finite tuples of parameters in (C.42) form a countable set and enumerate the successor. If $\lambda < \omega_1^\mathfrak{G}$ is a limit, there is internally a surjection $s : \omega \twoheadrightarrow \lambda$; take the $<_\mathfrak{G}$ -least one, in the order already constructed in Lemma 79.3, and diagonalize the $e_{s(n)}$. Finally, among all surjections so obtained, define e_β as the $<_\mathfrak{G}$ -least. The recursion $\beta \mapsto e_\beta$ is a uniformly definable class of $\mathfrak{G}$; every e_β , and every restriction of the recursion to an ordinal, belongs to $\mathfrak{G}$. It uses only $h, u_{h,m}$ and the global well-order. $\square$

Lemma 79.12 (Condensation of internal reals below $\omega_1^\mathfrak{G}$). *Every real in $\mathfrak{G}_{\text{HMT}}(h, m_\infty)$ appears at a level whose index is smaller than $\omega_1^\mathfrak{G}$.*

Proof. Let $r \subseteq \omega$ be a real in $\mathfrak{G}$, and let $B := \mathfrak{G}_0(h, m_\infty)$. This base is transitive and internally countable. With the codes of natural numbers and finite pairs already fixed, $\text{Ord} \cap B = \omega + 1$. By induction,

$$\text{Ord} \cap \mathfrak{G}_\alpha = (\omega + 1) + \alpha.$$

The successor adds the ordinal formed by the ordinals of the preceding level; it cannot add a larger one, because that would not be a subset of that level. Limits are unions. Choose a limit Θ , with $\Theta > \omega^2$, such that $r \in \mathfrak{G}_\Theta$. Thus $\text{Ord} \cap \mathfrak{G}_\Theta = \Theta$, and every level $\mathfrak{G}_\xi$, with $\xi < \Theta$, belongs to $\mathfrak{G}_\Theta$. Internally form the expanded structure

$$\mathcal{A}_\Theta = \langle \mathfrak{G}_\Theta, \in, u_{h,m}, H_\Theta, <_\mathfrak{G} \upharpoonright \mathfrak{G}_\Theta \rangle$$

with the level relation as a predicate of its signature, $H_\Theta = \{ \langle \xi, x \rangle : \xi < \Theta \text{ \& } x \in \mathfrak{G}_\xi \}$, and its definable Skolem functions. Let

$$Y = \text{Hull}^{\mathcal{A}_\Theta}(B \cup \{B, r\}) < \mathcal{A}_\Theta.$$

Closure under the countable collection of Skolem functions shows within $\mathfrak{G}$ that Y is countable. Let $c : Y \rightarrow M$ be its Mostowski collapse, and let

$$\bar{\Theta} := \text{otp}(Y \cap \Theta) < \omega_1^\mathfrak{G}.$$

Since $B \subseteq Y$ is transitive, induction on membership shows that c fixes every element of B , as well as B itself. In particular, it fixes $\omega, u_{h,m}, h, m_\infty$. Since $r \subseteq \omega \subseteq Y$, it also follows that $c(r) = r$.

We prove by induction on $\xi \in Y \cap \Theta$ that

$$c^n(Y \cap \mathfrak{G}_\xi(h, m_\infty)) = \mathfrak{G}_{c(\xi)}(h, m_\infty). \quad (\text{C.43a})$$

For $\xi \in Y \cap \Theta$, the predicate H_Θ defines the set $\mathfrak{G}_\xi$ from ξ , so $\mathfrak{G}_\xi \in Y$. Relativizing each formula to that set gives $Y \cap \mathfrak{G}_\xi < \mathfrak{G}_\xi$, in the language into which Lemma 79.1 translates the HMT definitions.

The base case of (C.43a) has already been proved because the collapse fixes B . At a successor, if $\xi + 1 \in Y$, then also $\xi \in Y$ and $c(\xi + 1) = c(\xi) + 1$. For $a \in Y \cap \mathfrak{G}_{\xi+1}$, fix a finite formula defining it over $\mathfrak{G}_\xi$. The assertion

$$a = \{z \in \mathfrak{G}_\xi : \varphi^{\mathfrak{G}_\xi}(z, \bar{p})\}$$

is an ordinary formula over $\mathfrak{G}_\Theta$, with parameters $a, \mathfrak{G}_\xi$. Elementarity supplies the parameters $\bar{p}$ within $Y \cap \mathfrak{G}_\xi$. The induction hypothesis and the elementarity of that level show that $c(a)$ is the subset defined by the same formula and the collapsed parameters over $\mathfrak{G}_{c(\xi)}$. Conversely, given a formula and parameters at the collapsed level, lift the parameters to the original level. The subset defined there exists, is unique, and belongs to Y by elementarity. Its collapse is the required subset. This proves both inclusions at the successor stage without introducing a uniform truth predicate.

For a limit $\lambda \in Y \cap \Theta$, elementarity supplies, for every $x \in Y \cap \mathfrak{G}_\lambda$, a witness $\xi \in Y \cap \lambda$ with $H_\Theta(\xi, x)$. Therefore,

$$Y \cap \mathfrak{G}_\lambda = \bigcup_{\xi \in Y \cap \lambda} (Y \cap \mathfrak{G}_\xi), \quad c^n(Y \cap \lambda) = c(\lambda).$$

Continuity of the hierarchy and the induction hypothesis give (C.43a) at the limit. Cofinality of $Y \cap \lambda$ in the ambient ordinal λ is not required: its collapsed indices range over $c(\lambda)$.

Finally, every $x \in Y$ has a witness $\xi \in Y \cap \Theta$ with $H_\Theta(\xi, x)$, and $\bar{\Theta}$ is a limit. Taking unions,

$$M = \bigcup_{\eta < \bar{\Theta}} \mathfrak{G}_\eta(h, m_\infty) = \mathfrak{G}_{\bar{\Theta}}(h, m_\infty).$$

Since $c(r) = r \in M$, it follows that $r \in \mathfrak{G}_{\bar{\Theta}}$ with $\bar{\Theta} < \omega_1^\mathfrak{G}$. □

The lemma does not say that a finite-length prefix enumerates the line. It says that every generated hereditary object has a first transfinite level and a first genealogical code within that level. This is precisely the information lost by the extensional reduct when it preserves only the expressed value.

79.4 Injection of the generated line into the first uncountable ordinal of the genealogical closure

Let $\iota(r) \subseteq \omega$ be the rational cut of $r \in \mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$ relative to the enumeration of $\mathbb{Q}_{\text{HMT}}$ previously constructed from APP integers and quotients. The map $r \mapsto \iota(r)$ is definable from r and that enumeration, so $\iota(r) \in \mathfrak{G}$; Lemma 79.12 places it at a stage with countable index. Define

$$\begin{aligned}\beta(r) &:= \min\{\beta : \iota(r) \in \mathfrak{G}_{\beta+1}(h, m_\infty)\}, \\ n(r) &:= \min\{n : e_{\beta(r)+1}(n) = \iota(r)\}, \\ j_{\text{HMT}}(r) &:= \omega \cdot \beta(r) + n(r).\end{aligned}\tag{C.44}$$

Definition (C.44) is uniform in the parameters h, m_∞ . Since $\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$ is a set of $\mathfrak{G}$, Separation fixes the domain of the definition, and Replacement in $\mathfrak{G}$ forms its graph:

$$\{\langle r, \omega \cdot \beta(r) + n(r) \rangle : r \in \mathbb{R}_{\text{HMT}}^{\mathfrak{G}}\} \in \mathfrak{G}.$$

Therefore, j_{HMT} is an internal function of $\mathfrak{G}$.

By Lemmas 79.11 and 79.12, $j_{\text{HMT}}(r) < \omega_1^{\mathfrak{G}}$ for every generated real. If $j_{\text{HMT}}(r) = j_{\text{HMT}}(s)$, ordinal division by ω recovers the same pair (β, n) ; (C.43) recovers the same rational cut and, by completeness, $r = s$. Hence

$$j_{\text{HMT}} : \mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \hookrightarrow \omega_1^{\mathfrak{G}}\tag{C.45}$$

is an injection constructed from the rank and name of the HMT genealogy. Its existence uses precisely the semantic closure (C.37), the well-ordering of names from Lemma 79.3, and the condensation (C.43a); it does not receive the Continuum Hypothesis in order to choose the stage. The subsequent set-theoretic identification describes this same machinery in different notation without reversing its causal provenance.

79.5 Injection of the first uncountable ordinal into the line and cardinal equivalence

For every $\beta < \omega_1^{\mathfrak{G}}$, Lemma 79.11 allows a subset $w_\beta \subseteq \omega$ to encode a countable well-order of type β . To avoid confusing the empty order with the one-point order, both the domain $D \subseteq \omega$ and its strict order relation $R \subseteq D^2$ are encoded. Once a bijective pairing $\langle \cdot, \cdot \rangle : \omega^2 \rightarrow \omega$ is fixed, the code is

$$w(D, R) = \{2n : n \in D\} \cup \{2\langle a, b \rangle + 1 : (a, b) \in R\}.$$

The even bits recover the domain, and the odd bits recover the relation. Take the $<_{\mathfrak{G}}$ -least real encoding such a well-order. Since the property “encodes a well-order of type β ” and the global well-order are internal, Replacement forms the function $\beta \mapsto w_\beta$ within $\mathfrak{G}$. Distinct order types produce distinct codes. The ternary map, already used in (C.41a), is defined by

$$E(w) = \sum_{n \geq 0} \frac{2\mathbf{1}_w(n)}{3^{2n+2}}\tag{C.46}$$

and is injective and avoids the endpoint of the root cylinder. The lift (C.41b) ensures that $E_{\text{HMT}}(w_\beta)$ belongs to the line expressed by an HMT history. Thus

$$k_{\text{HMT}}(\beta) := E_{\text{HMT}}(w_\beta), \quad k_{\text{HMT}} : \omega_1^{\mathfrak{G}} \hookrightarrow \mathbb{R}_{\text{HMT}}^{\mathfrak{G}}.\tag{C.47}$$

The injections (C.45) and (C.47), constructed without using the equality sought, give by Cantor–Bernstein

$$|\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}| = \aleph_1^{\mathfrak{G}}.\tag{C.48}$$

Since (C.41c) identifies the complete HMT line with $\mathcal{P}^{\mathfrak{G}}(\omega)$, (C.48) rewrites exactly as

$$\mathfrak{G}_{\text{HMT}}(h, m_{\infty}) \models 2^{\aleph_0} = \aleph_1. \quad (\text{C.49})$$

The equality does not arise from 729, from an informal product of cardinality and ordinality, from the digits of a constant, or from a numerical bracket. It arises from the covering of the line by the inverse limit, the condensation of the genealogical closure, and the two explicit injections.

79.6 Full quantifier over all internal subsets

Let $A \in \mathcal{P}^{\mathfrak{G}}(\mathbb{R}_{\text{HMT}}^{\mathfrak{G}})$. If A is countable, the first alternative holds. If it is not, $j_{\text{HMT}}[A]$ is an uncountable subset of $\omega_1^{\mathfrak{G}}$. It cannot be bounded by a countable ordinal, since every initial segment of ω_1 is countable. Its increasing enumeration has order type $\omega_1^{\mathfrak{G}}$, and the inverse of j_{HMT} on its image produces an injection $\omega_1^{\mathfrak{G}} \hookrightarrow A$. The inclusion $A \subseteq \mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$, (C.48), and Cantor–Bernstein give

$$\forall A \in \mathcal{P}^{\mathfrak{G}}(\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}), \quad |A| \leq \aleph_0 \quad \text{or} \quad |A| = |\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}|. \quad (\text{C.50})$$

Here the full power set of the integral continuum generated by HMT is quantified. The bijective transport of the folding with memory carries (C.50) to any of its representations without losing parts; it does not replace the injections (C.45)–(C.47).

79.7 Representation of the genealogical closure as $L[h, m_{\infty}]$

Only after (C.49)–(C.50) is the construction in (C.36) assigned its name in the usual set-theoretic notation. Let $L[u_{h,m}]$ be the constructible hierarchy relative to the real already produced in (C.33a). Since the HMT signature is countable and all its operators are graph relations encoded by h , Lemma 79.1 eliminates its symbols uniformly. This yields the equality of classes

$$\mathfrak{G}_{\text{HMT}}(h, m_{\infty}) = L[u_{h,m}] = L[h, m_{\infty}]. \quad (\text{C.51})$$

To prove the first inclusion, strengthen the induction: for every ordinal β , both $\mathfrak{G}_{\beta}$ and the sequence $\langle \mathfrak{G}_{\xi} : \xi \leq \beta \rangle$ belong to $L[u_{h,m}]$. The base is the transitive closure of a set constructed from ω and $u_{h,m}$; the parameters h, m_{∞} are recovered from $u_{h,m}$. At a successor, the preceding level is an *internal set*, not merely a subclass of $L[u_{h,m}]$. Its satisfaction relation and the uniform elimination (C.37c) make it possible to form internally the set of its definable subsets. At a limit, transfinite recursion, Replacement, and Union in $L[u_{h,m}]$ form the preceding sequence and its union. Transitivity then gives $\mathfrak{G} \subseteq L[u_{h,m}]$.

Conversely, Lemma 79.3 supplies an internal transitive model of ZF, containing all ordinals and $u_{h,m}$ as an element. The recursion constructing $L[u_{h,m}]$ is carried out in that model: at every stage, satisfaction over the preceding level and relative definability are absolute between the two transitive domains. By induction, its levels coincide with those of the ambient construction and belong to $\mathfrak{G}$. Thus $L[u_{h,m}] \subseteq \mathfrak{G}$. Finally, (C.33a) makes $u_{h,m}$ and the pair (h, m_{∞}) primitive-recursively interdefinable, from which the second equality follows.

Equation (C.51) explicitly identifies the closure with the constructible hierarchy relative to the generated parameter. This representation does not alter causal provenance: $L[u]$ neither selects the genealogical ledger nor defines the extensions or the line; the decision of the Continuum Hypothesis follows from the exhaustivity rule and the maps proved between (C.36) and (C.50). Taken together, the composition of (C.33a), (C.36), (C.37d), and (C.49)–(C.51) and Theorem 79.13 give a complete and uniform expression of the realized universe, semantic coincidence, and cardinal closure; none of these representations replaces APP–TRIT–TPK generation.

79.8 Theorem giving an affirmative resolution of the Continuum Hypothesis

Theorem 79.13 (Decision of the continuum in HMT). *For every realized history m_∞ produced by the joint action $\text{APP} \rightarrow \text{TRIT} \rightarrow \text{TPK} \rightarrow X_\infty^{\text{enr}} \rightarrow \mathcal{C}_{\text{cont}}^{\text{disc}}$, the HMT genealogical extension (C.36) generates the internal reconstruction of the integral line (C.40), $\mathbb{R}_{\text{HMT}}^\mathfrak{G}$, constructs the injections (C.45) and (C.47), and satisfies $2^{\aleph_0} = \aleph_1$ over its entire internal power set. Therefore, HMT proves the Continuum Hypothesis affirmatively for the integral continuum that it defines and generates. The notation $L[h, m_\infty]$ is a subsequent representation of that hierarchy, not the premise deciding the result.*

79.9 Falsification criteria for the cardinal closure

The following list identifies failures that would invalidate a step of the preceding closure. Construction, semantic, and cardinality controls are distinguished:

1. there exists a level N or a surviving state without a compatible successor in (C.34);
2. $u_{h,m}$ does not hereditarily encode h, m_∞ , or some graph relation of Σ_{HMT} does not admit the uniform elimination (C.37c);
3. an object admitted by the HMT semantic rules lacks a term in (C.37a), or a term in (C.37a) expresses an object outside those rules, breaking (C.37d);
4. $\mathfrak{G}$ does not satisfy one of the axioms used, its definable well-order fails, or a purported enumeration e_β is not internal and uniform;
5. the subcylinders in (C.40a) do not form an exhaustive disjoint partition, their cylinders cease to contract, or the recursion (C.41b₀) does not produce an internal compatible history;
6. the collapse in Lemma 79.12 does not commute with a successor or limit stage, or there exists an internal real that appears in no level with index smaller than $\omega_1^\mathfrak{G}$;
7. the selection $\beta \mapsto w_\beta$ is not internal or uniform, or does not preserve order type;
8. two distinct reals have the same genealogical pair (β, n) in (C.44), two distinct ordinals produce the same code in (C.47), or one of the injections $E, C, j_{\text{HMT}}, k_{\text{HMT}}$ fails;
9. there exists an uncountable internal subset of the line whose range under j_{HMT} is bounded in $\omega_1^\mathfrak{G}$;
10. some generative stage preceding the covering (C.40a) uses the Continuum Hypothesis, $\mathbb{R}$, a target constant, the Committee on Data of the International Science Council (CODATA), the Particle Data Group (PDG), or a subsequent physical identification as a causal antecedent to define a seed, a relation $\mathcal{L}_\tau$, Ext, a coefficient, admissibility, or an extension of the tree. The subsequent lift (C.41b₀) does not constitute such an inversion: there an arbitrary already constructed real is a universal variable and selects, within the previously proved exhaustive partition, the unique subcylinder containing it.

No execution restricted to a preset number of levels replaces the universal quantifier of the inverse system; an effective failure of compatibility at some depth would falsify it. No subsequent set-theoretic notation proves (C.49) by itself. The closure jointly uses Lemma 79.3, the line–power-set identification in (C.41c), Lemmas 79.11–79.12, and the two injections constructed from the genealogy.

80 Census of residual-emitter signatures

This census specifies the finite fibers used in Section 4. Each initial condition is identified by

$$\nu = 4(9i + j) + d, \quad 0 \leq i, j \leq 8, \quad d = 0, 1, 2, 3 \longleftrightarrow N, E, S, O.$$

Euclidean division recovers $d = [\nu]_4$, $j = [\lfloor \nu/4 \rfloor]_9$ and $i = \lfloor \nu/36 \rfloor$. Thus $0 \leq \nu < 324$ enumerates each condition exactly once, and pairs of these identifiers enumerate the 104 976 conditions in Ω_{seed} .

Let $f_+(\nu)$ and $f_-(\nu)$ be the six-component signatures obtained from the emission recurrence. At every active step the cell is read before the cursor advances; the direction vectors are

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

Coordinates are reduced modulo nine and the vector is reversed after steps 27 and 54. Zero-phase steps do not move either cursor. Together with (4.2), these rules determine f_+ and f_- without a target-value catalog.

The table gives their complete images. The multiplicity m_σ counts identifiers in the fiber $f_\sigma^{-1}(a)$; $\nu_{\min}$ is its least element. Each word retains its initial zeros. The complete fiber is determined by

$$f_\sigma^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_\sigma(\nu) = a\}.$$

f_+	m_+	$\nu_{\min}$	f_-	m_-	$\nu_{\min}$
122564	18	17	000000	108	8
122898	18	24	177393	8	1
212645	18	5	177555	8	54
212988	18	0	177771	8	11
221456	18	29	339555	8	6
221889	18	12	339771	8	15
456221	18	28	339933	8	59
465898	18	21	393177	8	0
546988	18	9	393393	8	45
564122	18	16	393555	8	40
645212	18	4	555177	8	52
654889	18	33	555339	8	4
889221	18	13	555393	8	41
889654	18	32	555555	24	84
898122	18	25	555717	8	14
898465	18	20	555771	8	18
988212	18	1	555933	8	24
988546	18	8	717555	8	12
			717717	8	21
			717933	8	19
			771177	8	9
			771339	8	13
			771555	8	16
			933339	8	57
			933555	8	26
			933717	8	17

Evaluating the recurrence for the 324 identifiers gives the two partitions in the table: $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. Their fiber product yields 432 fibers of size 144, 18 of size 432, and 18 of size 1944, by injectivity of (4.3). Finally, applying (4.4) to that product of images gives 243 distinct words. This finite census belongs to the residual emitter; subsequent coinductive refinement retains its own law.

Unity preserved in its realizations

The article's development returns to its initial thesis with operations that have already been developed. APP preserves the evaluation and its quotient; TRIT distinguishes orientation and regime; the TPK composes transport, selection, and prolongation. The nonadic connection updates history during the phase return. Refinement produces the numerical readings and preserves the incidences that support their exceptional realization.

The holographic arithmetic-geometric coupling constant brings together three precise functions. Its register admits a reversible positional encoding; it enters into the closure of alpha and into marked incidence; its orbit makes it possible to identify operators with recovery bounds. The bilateral law transports this capacity through the conservation of the state and its forms.

Vacancies explain how history advances when a capacity remains fixed. The electronic section shows persistence with trajectory and register. The action sections, oriented channels, area readers, and thermal transduction convert the preceding results into calculable physical relations. The exceptional realizations preserve their carriers, incidences, and dimensional maps.

The evolutionary conservation of dimensional unity is expressed at several linked levels: normalized partition, reversible refinement, transport of forms, archive with terminal component, stable recovery, and transformation of observables. Each level has its object and its proof; the maps between them construct the continuity of the argument. The double projection remains at the centre of this organization: value and incidence originate in the same state, whose transformation preserves recoverable relations.

81 Conclusions

The holographic extreme and mean ratio is formulated through a generating structure, a reversible arithmetic evaluation, and a transport law with information recovery. The dodecaphasic register K , produced by APP-TRIT-TPK, preserves its ordering in the phase chart and connects arithmetic closure with incidence representation. Its reading

$$\kappa = \frac{234543140729659824621058914794146601}{10^{36} - 1}$$

recovers all twelve components exactly: multiplication by the denominator restores the integer and base one thousand determines its blocks. This result fixes the mathematical meaning of the holographic arithmetic-geometric coupling constant. Its value exactly represents the ordered register and participates, together with the regional coordinates of π, φ, e , in determining α_A .

The same structure permits operations to be identified. The twelve cyclic shifts of K form a basis whose Gram matrix has spectral endpoints 589609 and 6263^2 . Responses to this basis determine a linear operator with an error-amplification factor $1/\sqrt{589609}$; one vector response suffices in the class commuting with the shift. The incidence representation transports the register to its specified image with an explicit inverse. Two memory observations recover its centred part with the optimal bound $9/4$, and the uniform charge completes the register. Evaluation, incidence and identification thus compose on a single object, retaining at each step the data required by the inverse.

Agreement between the integer and analytic representations of α is established at every depth. The analytic chart extends the order-nine truncation through a recurrence with ratio $1/729$, whose initial coefficient is obtained from the previously generated pre-coordinate. Uniform convergence, the positive derivative bound and nested rational intervals identify the same simple root and the same canonical prefixes in both representations. The theorem determines their complete compatibility within this correlated construction.

The general result supporting conservation is the bilateral identity

$$(a+1)(aB-C)^*(aB-C) + a((a+1)B+C)^*((a+1)B+C) = ((a+1)^3 + a^3)I,$$

for $a > 0$ and two isometries B, C with a common domain and codomain. Cancellation of the cross terms proves the equality in arbitrary dimension. In the specialization $a = 8$, the ternary factor 73 retains its interpretation as an algebraic norm and lattice index, and satisfies $10 \cdot 73 = 729 + 1$. Refinement of admissible integer labels has an inverse by Euclidean division, with depth and carry convention specified; its lift to state bases is unitary and composes with the balance.

The law also determines what changes under observation. For unitary relative transport R , the diagonal reader transforms the observable of the common chart according to

$$\mathcal{T}(G) = \frac{72}{73}G + \frac{1}{73}R^*GR.$$

Its fixed points are exactly the observables commuting with R . The observable transported on the complete state instead preserves the initial reading through its reconstruction formula. The exterior identities extend this result to each admissible degree, retaining purely observed sectors, recorded sectors and mixed sectors. Area and volume contributions are included in a law of forms with their corresponding metrics and dimensions.

Recovery is maintained at arbitrary depth through an explicit archive policy. In the nonadic case, the terminal operator satisfies $\|R_N\|^2 \leq (729/1241)^N$. The infinite archive is therefore an isometry and its adjoint reconstructs the state with operator norm one. At every positive depth, the finite archive admits an exact inverse corrected by its Gram operator, with bound

$$\|L_N\| \leq (1 - (729/1241)^N)^{-1/2}.$$

These formulae quantify stability and separate exact inversion from the approximation obtained by truncating a recovery series. In general dynamics, memory and the surviving terminal jointly preserve the state; the stated policy proves the regime in which its entire norm transfers to the archive.

The physical realizations developed here apply these operations with explicit action, orientation, scale and representation data. In particular, for the declared symplectic protocol with two identical centred pure Gaussian inputs and finitely many modes, the antilinear component of memory fixes the symplectic spectrum of the reduction, retaining all Fock occupations. The elliptic–parabolic loop produces purity $9/11$, spectrum $(9/10)10^{-j}$, and dimensionless entropy $(10/9)\log 10 - \log 9$. Global conservation and the mixed character of the reduced observation are described within the same protocol. The electromagnetic, gravitational, thermal, and variational realizations retain the operators and normalizations used in their respective proofs.

The common construction of the continuum preserves the five correlative operations and the compatibility of their refinements. The closure of history collections in Section 79 establishes, within the genealogical universe defined there, the internal cardinal equality of its real line with that universe's $\aleph_1$. The assertion retains the domain and membership operations involved in its two injections and final identification.

The residual emitter census makes the finite basis of part of this chain explicit: it collects 44 signatures, 18 additive and 26 multiplicative, with their multiplicities and the fiber-reconstruction rule. The 648 initial-condition memberships, 324 on each sheet, are retained in the supplementary data. The seven theorems checked in Lean certify the stated arithmetic identities and their finite consequences; the operator, limit and incidence proofs are developed in the text. Census, computation and formalization thus specify their respective objects of verification.

The extension in Sections 73–77 adds a recoverable geometric realization of this continuity. The direction of K determines a positive unimodular flow, a pair of dual lattices, and a geodesic of compensating radii. The archive reconstructs the direction and controls errors in its theta and Epstein readouts. The polar structure remains constant, while spectral variation is entire. Through another construction in the same genealogy, the angular contrast is realized in the Klein chart and recovers the Planck section with its orientation; substitution into the electronic expression retains all its factors. The mass-operator decomposition distinguishes the common scale from the determinant-one congruence. These compositions extend the operational role of the coupling constant while preserving the distinct domains of conservation.

Joint selection of the constant is developed before its algebraic reconstruction: orderings of the terminal repertoire produce 19 446 registers; exceptional compatibility retains 79, and temporal compatibility determines a unique register K . The conditions of both filters and their dependence on the enriched history are retained. Section 10 connects this determination with arithmetic closure, incidence representations and operator reconstruction.

Section 65 makes the archive contribution to bilateral composition explicit, constructs the quarter-turn plane of the dodecaphasic calendar, and proves oriented action recovery through a relative commutator. Compatible amplification preserves the operator; the detector distinguishes state dilation from energy gain and the ordered archive from operation counting.

The realization in Section 42 composes the incidence inverse with the marked inscription and the local action norm to obtain the longitudinal return $L = L_* \alpha^{16} 5759/23040$. Its arrival modulus retains the full positive character and determines a unique radial operator for each character. With the explicit identification of the reduced gravitational radius, the two reciprocity identities fix the common coupling $G_a = c^3 L^2 / \hbar_a$. Residual memory, the joint variation of field, transport, and weight, and recovery of the boundary sectors retain their contiguous proofs. The compatible chronological section transports the electronic coordinate; the area scale L^2 remains common to the action sections. Constitutive restitution additionally retains the oriented collection whose endpoint evaluation determines Catalan's constant.

The finite antecedent of recovery is specified more precisely by the emitter proofs. The two signatures produce 42 possible blocks, and the third digit satisfies a check congruence. Regional junctions retain explicit seed witnesses; the ordered archive distinguishes records sharing a histogram but differing in correlation. Gauges and closure controls separate observable normalization from the conditions fixing an evaluation. The saturated boundary admits eight distributions, dual neutrality preserves a pair, and orientation selects its representative. The information received by each reader is thus specified before applying the bilateral identities, inverses, and stability bounds.

The five regional closures attain the minimum of eight in the finite problem of covering the four prescribed edges. The graph retains emission labels, and the quotient retains their incidences between phase classes. Oriented reversal makes the reader ν_{120} odd and preserves the even parity of ν_{270} ; these identities make explicit the response of memory to a change of orientation.

The nonadic phase and the quarter-turn plane recover exactly a quotient of 36 positions. Proposition 65.4 locates the ternary fibre that restores the 108 positions and retains the obstruction to splitting the clock. The memory of the enriched dynamics continues to distinguish its complete returns.

The central result is a calculable characterization of the evolutionary conservation of unity: the structure determines an evaluation, its transformations distribute information, and the inverses specify how to recover it. The coupling constant, observable laws and reconstruction bounds express this continuity through equations and operations, from the arithmetic register to the specified realizations.

References

- [1] R. P. Feynman, “Space-Time Approach to Non-Relativistic Quantum Mechanics”, *Reviews of Modern Physics* **20**, 367–387 (1948). doi:10.1103/RevModPhys.20.367.
- [2] R. P. Feynman, “The Theory of Positrons”, *Physical Review* **76**, 749–759 (1949). doi:10.1103/PhysRev.76.749.
- [3] R. P. Feynman, “The Development of the Space-Time View of Quantum Electrodynamics”, Nobel Lecture, 11 December 1965. Lecture text.
- [4] B. Musto and J. Vicary, “Quantum Latin Squares and Unitary Error Bases”, *Quantum Information and Computation* **16**, 1318–1332 (2016). doi:10.26421/QIC16.15-16-4.
- [5] G. De las Cuevas, T. Drescher and T. Netzer, “Quantum Magic Squares: Dilations and Their Limitations”, *Journal of Mathematical Physics* **61**, 111704 (2020). doi:10.1063/5.0022344.
- [6] G. De las Cuevas, T. Netzer and I. Valentiner-Branth, “Magic Squares: Latin, Semiclassical, and Quantum”, *Journal of Mathematical Physics* **64**, 022201 (2023). doi:10.1063/5.0127393.
- [7] P. A. M. Dirac, “The Lagrangian in Quantum Mechanics”, *Physikalische Zeitschrift der Sowjetunion* **3**, 64–72 (1933). Reproduction of the original article.
- [8] W. Pauli, “Exclusion Principle and Quantum Mechanics”, Nobel Lecture, 13 December 1946, in *Nobel Lectures, Physics 1942–1962*, Elsevier, 1964, pp. 27–43, especially p. 42. Lecture text.
- [9] E. Tiesinga, P. J. Mohr, D. B. Newell and B. N. Taylor, “CODATA Recommended Values of the Fundamental Physical Constants: 2018”, *Reviews of Modern Physics* **93**, 025010 (2021). doi:10.1103/RevModPhys.93.025010. Archived 2018 tables.
- [10] P. J. Mohr, D. B. Newell, B. N. Taylor and E. Tiesinga, “CODATA Recommended Values of the Fundamental Physical Constants: 2022”, *Reviews of Modern Physics* **97**, 025002 (2025). doi:10.1103/RevModPhys.97.025002. Table of constants.
- [11] I. B. Frenkel, J. Lepowsky and A. Meurman, *Vertex Operator Algebras and the Monster*, Pure and Applied Mathematics, vol. 134, Academic Press, 1988.
- [12] R. E. Borcherds, “Monstrous Moonshine and Monstrous Lie Superalgebras”, *Inventiones Mathematicae* **109**, 405–444 (1992). doi:10.1007/BF01232032.
- [13] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der mathematischen Wissenschaften 290, Springer, 1999. doi:10.1007/978-1-4757-6568-7.
- [14] Conférence générale des poids et mesures, *Resolution 1 of the 26th CGPM: On the revision of the International System of Units*, 2018. Official resolution. doi:10.59161/CGPM2018RES1E.
- [15] P. A. M. Dirac, “Quantised singularities in the electromagnetic field”, *Proceedings of the Royal Society of London. Series A* **133**, 60–72 (1931). doi:10.1098/rspa.1931.0130.
- [16] ALEPH, DELPHI, L3, OPAL and SLD Collaborations; LEP Electroweak Working Group; SLD Electroweak Group and SLD Heavy Flavour Group, “Precision electroweak measurements on the Z resonance”, *Physics Reports* **427**, 257–454 (2006). doi:10.1016/j.physrep.2005.12.006.
- [17] H.-Y. Chen, C. H. Lam and H. Shimakura, “ $\mathbb{Z}_3$ -orbifold construction of the Moonshine vertex operator algebra and some maximal 3-local subgroups of the Monster”, preprint, arXiv:1606.05961 (2016), version 2 (2017). arXiv:1606.05961.
- [18] T. Abe, C. H. Lam and H. Yamada, “A remark on $\mathbb{Z}_p$ -orbifold constructions of the Moonshine vertex operator algebra”, preprint, arXiv:1705.09022 (2017), version 4. arXiv:1705.09022.
- [19] J. H. Conway and S. P. Norton, “Monstrous Moonshine”, *Bulletin of the London Mathematical Society* **11**, 308–339 (1979). doi:10.1112/blms/11.3.308.
- [20] C. H. Bennett, “Notes on Landauer’s principle, reversible computation, and Maxwell’s Demon”, *Studies in History and Philosophy of Modern Physics* **34**, 501–510 (2003). doi:10.1016/S1355-2198(03)00039-X.
- [21] E. Noether, *Invariante Variationsprobleme*, Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse (1918), 235–257. Translation by M. A. Tavel: Invariant Variation Problems.
- [22] C. Weedbrook, S. Pirandola, R. García-Patrón, N. J. Cerf, T. C. Ralph, J. H. Shapiro and S. Lloyd, *Gaussian Quantum Information*, *Reviews of Modern Physics* **84** (2012), 621–669. arXiv:1110.3234.
- [23] J. Williamson, *On the Algebraic Problem Concerning the Normal Forms of Linear Dynamical Systems*, *American Journal of Mathematics* **58** (1936), 141–163. doi:10.2307/2371062.
- [24] A. S. Holevo, M. Sohma and O. Hirota, *Capacity of Quantum Gaussian Channels*, *Physical Review A* **59** (1999), 1820–1828. doi:10.1103/PhysRevA.59.1820.
- [25] L. Manivel, *Configurations of lines and models of Lie algebras*, arXiv:math/0507118, 2005.
- [26] S. Bochner, *Harmonic Analysis and the Theory of Probability*, University of California Press, Berkeley, 1955.
- [27] A. Postnikov, *Total positivity, Grassmannians, and networks*, preprint, 2006. <https://arxiv.org/abs/math/0609764>.